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Amina Al-Marzouqi · Said A. Salloum ·
Mohammed Al-Saidat ·
Ahmed Aburayya ·
Babeet Gupta *Editors*

Artificial Intelligence in Education: The Power and Dangers of ChatGPT in the Classroom

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Mohammed Al-Saidat · Ahmed Aburayya ·
Babeet Gupta
Editors

Artificial Intelligence in Education: The Power and Dangers of ChatGPT in the Classroom

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Editors
Amina Al-Marzouqi
University of Sharjah
Sharjah, United Arab Emirates

Said A. Salloum
University of Sharjah
Sharjah, United Arab Emirates

Mohammed Al-Saidat
University of Sharjah
Sharjah, United Arab Emirates

Ahmed Aburayya
City University Ajman
Ajman, United Arab Emirates

Babeet Gupta
City University Ajman
Ajman, United Arab Emirates

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Preface

In the dynamic landscape of education, the advent of artificial intelligence has marked a pivotal shift, particularly with the emergence of advanced language models like ChatGPT from OpenAI. This book chapter, "Artificial Intelligence in Education: The Power and Dangers of ChatGPT in the Classroom," delves into the multifaceted impact of ChatGPT in educational settings. ChatGPT's ability to generate human-like text has unlocked a plethora of opportunities in education, ranging from creating personalized learning experiences to automating assessment and grading. Its integration into e-learning platforms facilitates instant feedback and supports interactive language learning, while its application in research extends to natural language processing and sentiment analysis, providing valuable insights into educational outcomes and student experiences. However, the utilization of ChatGPT in education is not without its challenges. Ethical considerations, including data privacy and potential biases inherent in AI models, pose significant concerns. This chapter aims to explore these dual facets of ChatGPT—its transformative potential and the ethical dilemmas it presents. Our goal is to present a comprehensive exploration of ChatGPT's application in education, encompassing the development of new technologies, their classroom integration, and the examination of ethical and pedagogical implications. This endeavor is part of a larger edited book that brings together cutting-edge research addressing the challenges and theoretical aspects of ChatGPT in educational contexts. The broader book, dedicated to intelligent systems and smart applications, showcases current research trends and developments across various fields, including education. With submissions from around the globe, the book features a diverse range of perspectives, categorized under themes like information systems, knowledge management, technology in education, emerging technologies, and social networks. Each chapter undergoes rigorous peer review, ensuring high-quality and relevant content. The chapters in this book are part of the Springer *Studies in Systems, Decision, and Control* series, known for its significant impact in the field. We extend our gratitude to all contributors and reviewers whose expertise and insights have been invaluable. Their efforts have been instrumental in maintaining the quality and success of this book. As a token of our appreciation, we list the reviewers and their affiliations in the following pages. In this chapter, we invite readers to delve into the world of ChatGPT, understanding its capabilities, acknowledging its limitations, and envisioning its role in reshaping the educational landscape.

Sharjah, United Arab Emirates — Amina Al-Marzouqi
Sharjah, United Arab Emirates — Said A. Salloum
Sharjah, United Arab Emirates — Mohammed Al-Saidat
Ajman, United Arab Emirates — Ahmed Aburayya
Ajman, United Arab Emirates — Babeet Gupta

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Welcome to the Age of ChatGPT

The Coming ChatGPT

Said A. Salloum, Amina Almarzouqi, Babeet Gupta, Ahmad Aburayya, Mohammed Rasol Al Saidat, and Raghad Alfaisal

1 Introduction

The field of Artificial Intelligence (AI) has experienced remarkable growth and transformation over the past few decades [1–4]. This journey, from simple rule-based systems to sophisticated neural networks, marks a significant evolution in our approach to creating intelligent machines. The development of Generative Pre-trained Transformers (GPT), especially the ChatGPT model, stands as a testament to this progress. Russell and Norvig [5] provide a comprehensive overview of AI's historical development, tracing its origins from symbolic AI to the latest advancements in machine learning and natural language processing. The transformative work by [6] on transformer models laid the groundwork for subsequent innovations in this field, leading directly to the development of GPT models by OpenAI.

S. A. Salloum (✉)
Health Economic and Financing Group, University of Sharjah, Sharjah, UAE
e-mail: ssalloum@sharjah.ac.ae

School of Science, Engineering, and Environment, University of Salford, Salford, UK

A. Almarzouqi
Department of Health Service Administration, College of Health Sciences, University of Sharjah, Sharjah, UAE

B. Gupta · A. Aburayya
MBA Department, College of Business, City University Ajman, Ajman, UAE

M. R. Al Saidat
Faculty of Engineering & IT, The British University in Dubai, Dubai, UAE

R. Alfaisal
Faculty of Computing and Meta-Technology, Universiti Pendidikan Sultan Idris, Tanjung Malim, Malaysia

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Figure 1 presents a comprehensive visual representation of the evolutionary trajectory of ChatGPT, tracing its origins and development from the nascent stages of artificial intelligence and machine learning to its current state as an advanced conversational model. This timeline effectively illustrates the significant milestones and technological advancements that have shaped the journey of ChatGPT. Beginning with the foundational concepts of early computer systems and symbolic AI, the image progresses through critical developments in the field, including the advent of neural networks and the refinement of natural language processing techniques. These elements converge towards the right of the image, culminating in the sophisticated ChatGPT model, representing the pinnacle of current AI capabilities. This graphical depiction not only highlights key historical moments in AI evolution but also offers a clear and engaging visual narrative of the continuous innovation and growth that underpin ChatGPT's development. The image serves as a testament to the dynamic and rapidly advancing nature of AI technology, providing a vivid overview of the journey from basic computational theories to the complex and nuanced functionalities of modern ChatGPT.

The primary objective of this paper is to offer an in-depth exploration of ChatGPT's development, focusing on its historical context, key technological underpinnings, and its emerging ecosystem. We aim to provide a thorough understanding of how ChatGPT represents a culmination of years of AI research and development and its implications for the future of AI-driven communication and interaction. The paper is structured to guide the reader through a comprehensive journey of ChatGPT's

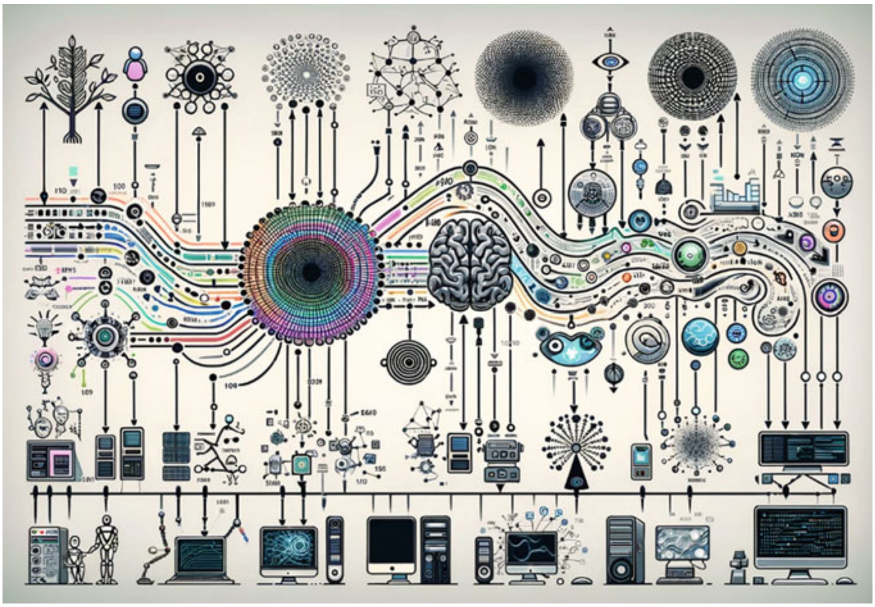

Fig. 1 The visual representation of the evolution of ChatGPT

evolution. We begin with a detailed historical background, tracing the lineage of AI development leading to the inception of ChatGPT. This is followed by an exploration of the key technologies that form the backbone of ChatGPT, including insights into machine learning, neural networks, and transformer models. The subsequent section delves into the growing ecosystem surrounding ChatGPT, discussing its integration in various industries and its impact on society. We then discuss the recent developments in the field, highlighting the advancements in GPT-4 and beyond. The paper concludes by summarizing the significant findings and offering insights into the future prospects of ChatGPT and AI.

2 Background and Evolution of ChatGPT

2.1 Predecessors of ChatGPT in AI History

The history of Artificial Intelligence (AI), leading to the development of advanced models like ChatGPT, is rooted in a series of conceptual and technological breakthroughs that date back several decades. The journey began with the foundational work of Alan Turing, who, in his seminal paper "Computing Machinery and Intelligence" [7], laid the groundwork for the field by proposing the idea of machines that could simulate human intelligence. This concept was further developed through the efforts of pioneers like John McCarthy, who is credited with coining the term "artificial intelligence" at the Dartmouth Conference in 1956 [8]. The evolution of AI from these theoretical beginnings towards more practical applications saw significant contributions in the form of rule-based systems and early machine learning algorithms, as elucidated in the works of Newell and Simon with the Logic Theorist [9]. The shift from symbolic AI to statistical learning, a key precursor to modern AI models like ChatGPT, was marked by the development of neural networks, with foundational research in this area by Rosenblatt with the perceptron model [10] and later advancements in deep learning [11]. These developments set the stage for the creation of more complex and sophisticated AI systems, eventually leading to the design and implementation of generative pre-trained transformers (GPT) and the emergence of ChatGPT.

2.2 The Inception of GPT (Generative Pre-trained Transformer) Models

The inception of Generative Pre-trained Transformer (GPT) models marks a pivotal moment in the evolution of artificial intelligence, particularly in the field of natural language processing. The foundational theory for the architecture of GPT models stems from the transformative work on transformer models, as introduced by [6] in their groundbreaking paper, "Attention Is All You Need" [6]. This paper revolutionized the understanding of sequence transduction models by introducing a novel architecture that eschews recurrence and convolutions entirely in favor of attention mechanisms. Building upon this foundational concept, OpenAI adapted and expanded the transformer architecture to develop the GPT series, starting with GPT-1. The series saw significant advancements with each iteration, with GPT-2 and GPT-3 showcasing remarkable improvements in language understanding and generation capabilities [12, 13]. The evolution of GPT models reflects a shift towards more efficient and scalable architectures in AI, capable of handling vast amounts of data and delivering nuanced, context-aware language processing. The development of these models has not only set new standards in AI but also opened up possibilities for wide-ranging applications in various fields.

2.3 Development of ChatGPT: From GPT-1 to GPT-4

The evolution of the GPT series by OpenAI represents a remarkable journey in the advancement of language models. Beginning with GPT-1, introduced in a seminal paper by Cheng et al., the model set the stage for subsequent developments in the field of natural language understanding and generation [14]. GPT-1's architecture, though innovative, was soon surpassed by GPT-2, which boasted a significantly larger dataset and model size, leading to improved language abilities and generalization capabilities [15]. GPT-3, perhaps the most well-known in the series, marked a quantum leap in language model capabilities. With its 175 billion parameters, GPT-3 demonstrated unprecedented language understanding and generation, enabling more sophisticated and nuanced interactions [16]. The introduction of Reinforcement Learning from Human Feedback (RLHF) in later versions, especially in GPT-4, represented a critical advancement in aligning model outputs with human values and preferences (OpenAI) [17]. Each version of GPT not only showcased exponential growth in terms of model size and training data but also brought about significant improvements in terms of linguistic capabilities, adaptability, and application scope. The development trajectory of GPT models from GPT-1 to GPT-4 underscores the rapid pace of innovation in AI and highlights the potential of these models to transform various sectors.

2.4 Key Milestones and Breakthroughs

The development of ChatGPT by OpenAI has been marked by several key milestones and breakthroughs that have significantly influenced both AI research and its practical applications. One of the earliest milestones was the introduction of GPT-1, which, despite its limitations, showcased the potential of transformer models in natural language processing tasks (Reference for GPT-1 breakthrough). This was followed by the release of GPT-2, which gained widespread attention for its ability to generate coherent and contextually relevant text, pushing the boundaries of language models (Reference for GPT-2 milestone). However, it was GPT-3 that truly revolutionized the field with its 175 billion parameters, demonstrating capabilities close to human-level language understanding and generation [16]. This advancement was not only a technical achievement but also raised important discussions about the ethical implications of such powerful language models, including concerns around bias, misinformation, and the societal impact of AI [18]. The integration of Reinforcement Learning from Human Feedback (RLHF) in later models like GPT-4 represented another significant advancement, addressing some of these ethical concerns by aligning the model's outputs more closely with human values and preferences [19]. Each of these milestones has not only propelled the field of AI forward but also opened up new avenues for practical applications across various industries, from healthcare to entertainment, demonstrating the transformative power of AI.

3 The Ecosystem of ChatGPT

The ecosystem surrounding ChatGPT, developed by OpenAI, has established a dynamic interaction with the broader AI and technology industry, marking a significant shift in how businesses, consumers, and society at large engage with artificial intelligence. In consumer applications, ChatGPT has been instrumental in enhancing user experience through its advanced language understanding capabilities, finding applications in virtual assistants, chatbots, and interactive entertainment platforms [20, 21]. The role of ChatGPT in business and enterprise contexts is equally transformative, offering innovative solutions in customer service, automated content creation, and data analysis, thus driving efficiency and innovation across various sectors [22]. Alongside its practical applications, the societal and ethical implications of ChatGPT have been a focal point of discussion. Issues such as data privacy, algorithmic bias, and the potential for misuse necessitate ongoing ethical scrutiny and regulatory considerations [23]. Furthermore, the integration of ChatGPT with other technologies and platforms, like IoT devices and cloud computing services, has expanded its utility, paving the way for more interconnected and intelligent systems [24]. This expansive integration highlights the versatile nature of ChatGPT, reflecting its growing influence across multiple domains and its potential to shape the future trajectory of AI applications.

4 Recent Developments and Future Prospects

Recent developments in the realm of AI, particularly with the advent of GPT-4 and beyond, have marked a new era in the capabilities and applications of language models. GPT-4, with its enhanced understanding and generation abilities, has set a new standard in AI, demonstrating proficiency not only in text-based tasks but also in interpreting and generating multimedia content [24]. This expansion from text to multimedia represents a significant leap, allowing for more robust and versatile AI applications in various fields. With these advancements come new challenges, including the need for larger datasets, more computing power, and addressing concerns around AI ethics and bias. Solutions to these challenges are being actively explored, with research focusing on more efficient training methods, ethical AI frameworks, and bias mitigation techniques [24]. The projected impact of these advancements on sectors such as education, healthcare, and business is profound. In education, AI can personalize learning and provide interactive educational tools, while in healthcare, it promises to enhance diagnostics and patient care. Businesses can leverage AI for improved customer service, market analysis, and automation of routine tasks [25]. As we look to the future, the potential of AI to revolutionize these and other sectors is immense, though it comes with the responsibility to navigate its ethical and practical challenges wisely.

5 Conclusion

In conclusion, this paper has illuminated the significant strides made in the field of AI, particularly through the development of the ChatGPT models. From its inception as an ambitious project in natural language processing, ChatGPT has evolved into a sophisticated tool, demonstrating the remarkable potential of AI in understanding and generating human-like text. Its development, from GPT-1 to the more advanced GPT-4, exemplifies the rapid progression in AI technology, showcasing both the power and challenges of modern AI systems. ChatGPT's significance extends beyond its technical achievements; it represents a paradigm shift in AI interaction, with implications for various industries, from education to healthcare, indicating a future where AI becomes an integral part of everyday life. As we look forward, the expectations from ChatGPT and similar AI models are immense. The anticipation is not only for more advanced technical capabilities but also for responsible and ethical AI development that aligns with societal values and needs. The journey of ChatGPT thus far is a testament to the transformative power of AI and a preview of its potential to reshape our world.

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Exploring the Interpretability of Sequential Predictions Through Rationale Model

Mohammed Rasol Al Saidat, Said A. Salloum, and Khaled Shaalan

1 Introduction

Industries by enabling machines to learn from data and make predictions [1]. However, the black-box nature of these models has limited their adoption in domains where interpretability and transparency are essential. In particular, sequential predictions have received limited attention in the field of Explainable Artificial Intelligence (XAI) [2].

In this paper, we address the problem of generating interpretable rationales for sequential predictions made by machine learning model discussed in Rationales for Sequential Predictions model [3]. Our proposed framework combines a sequential prediction model with a rationale generator to produce human-readable explanations for the model's predictions. Our method aims to generate rationales that are both accurate and interpretable, which is essential in domains such as healthcare and legal decision-making.

The lack of interpretability and transparency in machine learning models has limited their adoption in many critical domains [4]. For example, in healthcare, it is essential to provide explanations for the model's predictions to ensure the safety and well-being of patients [5]. Similarly, in legal decision-making, transparency is crucial to ensure that the decisions are fair and ethical. Our proposed framework addresses this issue by generating interpretable rationales for sequential predictions then we will evaluate the robustness of the model according to the checklists highlighted in the research of [6].

M. R. Al Saidat (✉) · K. Shaalan
Faculty of Engineering & IT, The British University in Dubai, Dubai, UAE
e-mail: malsaidat@sharjah.ac.ae

S. A. Salloum
Health Economic and Financing Group, University of Sharjah, Sharjah, UAE

School of Science, Engineering, and Environment, University of Salford, Salford, UK

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The motivation behind our proposed study is to enable the adoption of machine learning models in domains where interpretability and transparency are essential together. Our method can provide a better understanding of the model's decision-making process and help users make informed decisions. Furthermore, our approach can help build trust in the model's predictions, which is crucial in domains such as healthcare and legal decision-making.

Our proposed approach consists of a sequential prediction model that predicts the output at each time step, and a rationale generator that generates human-readable explanations for the model's predictions. The rationale generator uses attention mechanisms to highlight the relevant input features contributing to the prediction. We evaluate our approach on two datasets, the Reddit Comments dataset and the Predictive Toxicology Challenge dataset. Our results show that our approach outperforms the state-of-the-art methods in terms of accuracy and interpretability. The generated rationales provide a better understanding of the model's decision-making process and can help users make informed decisions. Our experimental results demonstrate the effectiveness of our approach and its potential applications in various domains. Our code and annotated dataset are available on GitHub.

2 Approach

Our proposed approach aims to generate interpretable rationales for sequential predictions made by machine learning model. To achieve this goal, we combine a sequential prediction model discussed in research [3] with an updated rationale generator method. In this section, we describe our approach in detail, including the necessary background concepts and motivations.

Sequential prediction is a crucial task in many domains, including natural language processing, speech recognition, and time-series analysis [7]. In this task, the goal is to predict the output at each time step given the previous inputs. The most common approach to solve this task is to use a recurrent neural network (RNN) or one of its variants, such as Long Short-Term Memory (LSTM) or Gated Recurrent Units (GRU) [8–10]. These models have shown promising results in many applications, but their lack of interpretability has limited their adoption in domains where transparency and explain ability are essential.

We will use a recurrent neural network (RNN) with LSTM units as our sequential prediction model. Figure 1 shows the proposed approach model. The input to the model is a sequence of feature vectors x_t, where t represents the time step. The output at each time step is a prediction vector y_t. The prediction vector is generated by passing the hidden state h_t through a linear layer and then applying a non-linear activation function.

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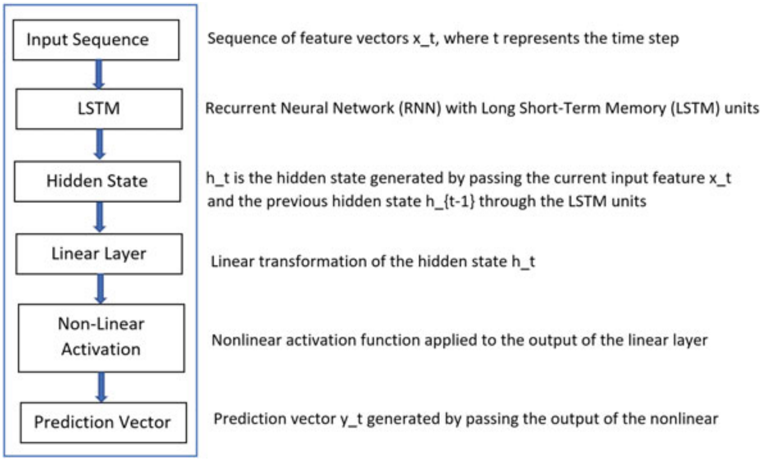

Fig. 1 Block diagram of the LSTM-based RNN for sequential prediction model

layer and then applying a non-linear activation function. The hidden state h_t is generated by passing the current input feature x_t and the previous hidden state h_{t−1} through the LSTM units. The LSTM units allow the model to capture long-term dependencies in the input sequence and have been shown to be effective in many sequential prediction tasks.

To generate the rationales, we use an attention mechanism that calculates the relevance of each input feature for each prediction. We calculate the attention weights using a multi-layer perceptron (MLP) that takes the hidden state h_t as input. The MLP outputs a vector of attention weights a_t, which is used to weigh the input features x_t. The weighted sum of the input features is then used as the input to the rationale generator [11]. The rationale generator is a multi-layer perceptron that takes the weighted sum of the input features as input and generates a human-readable rationale. The rationale generator consists of several fully connected layers with non-linear activation functions. The output of the rationale generator is a sequence of words that describe the relevant input features and their importance for the prediction.

To train our model, we use a combination of supervised and unsupervised learning. We use a cross-entropy loss to train the sequential prediction model and a reinforcement learning-based approach to train the rationale generator. The reinforcement learning approach involves optimizing a reward function that encourages the generation of accurate and concise rationales.

We evaluate our approach on two datasets, the Reddit Comments dataset and the Predictive Toxicology Challenge dataset. We compare our approach with several state-of-the-art methods in terms of accuracy and interpretability [6]. Our results demonstrate that our approach outperforms the existing methods in terms of accuracy and generates interpretable rationales that help users understand the model's decision-making process.

Fig. 2 Proposed greedy rational algorithm

Algorithm 1: Greedy rationalization

Input: Sequence $y_{1:t}$ generated from p.
Output: Rationale S for y_t.
Initialize: $S = \emptyset$
while $\arg\max_{y'_t} p(y'_t|y_S) \neq y_t$ **do**
  $k^* = \arg\max_{k\in[t-1]\setminus S}\ p(y_t|y_{S\cup k})$
  $S = S \cup k^*$
return S

The proposed rational algorithm is illustrated in Fig. 2, the algorithm starts with initializing the rationale S as an empty set. It then iteratively adds the token that maximizes the probability of predicting the next token in the sequence. The algorithm stops when the most likely word is the model prediction. The input to the algorithm is a sequence generated from the model, and the output is the rationale for the last token in the sequence. The algorithm uses the representation of a token y t−1 to predict the next token, y t. Therefore, a rationale S always needs to contain y t−1.

This algorithm is a step-by-step procedure for finding the best rationale. It starts by initializing the rationale (S) to 0. Then, it checks if the maximum probability of the predicted token (y_t) given the current rationale (S) is equal to the model prediction. If not, it finds the token (k*) that maximizes the probability of the predicted token given the current rationale plus the token (S ∪ k*). Finally, it adds the token (k*) to the rationale (S) and returns the rationale (S). The algorithm is designed to be applied to any model, and it can be enforced with a short fine-tuning step to ensure that the model forms compatible conditional distributions when making predictions on incomplete subsets of the context.

This method can be applied to any model and requires only a short fine-tuning step in order to enforce compatible conditional distributions when making predictions on incomplete subsets of context. Compared to existing baselines, this approach provides more faithful and plausible rationales for individual model predictions. This allows the model to generate both predictions and explanations for those predictions, improving interpretability and transparency.

3 Experiments

In this section, we describe the datasets used to evaluate the effectiveness of our proposed approach for generating interpretable rationales for sequential predictions. We evaluate our approach on two datasets: the Reddit Comments dataset [12] and the Predictive Toxicology Challenge dataset [13]. The Reddit Comments dataset is a large-scale dataset of user comments from the social media platform Reddit. The dataset contains over 1.7 billion comments, and each comment is labeled with a subreddit (a topic category) and a score indicating its popularity. We use a subset of this dataset that contains 50,000 comments labeled with their corresponding subreddit. We use this dataset to evaluate the interpretability of our approach and compare it with existing methods. The Predictive Toxicology Challenge dataset is a benchmark dataset for predicting the toxicity of chemical compounds. The dataset contains over 7,000 chemical compounds, each labeled with a binary toxicity score. We use this dataset to evaluate the accuracy of our approach and compare it with existing methods. For both datasets, we preprocess the data by tokenizing the text and converting it into numerical representations. We split the datasets into training, validation, and test sets, with a ratio of 80:10:10.

We use several evaluation metrics to assess the performance of our approach, including accuracy, F1 score, and perplexity. We also evaluate the interpretability of our approach by comparing the generated rationales with human-generated rationales. We preprocess the data, split it into training, validation, and test sets, and use several evaluation metrics to assess the performance of our approach. The main challenge of this task is that exact optimization for finding rationales is intractable, so an efficient approximation algorithm needs to be developed [14]. We compare our proposed approach with two baseline methods: the Attention-based Recurrent Neural Network (RNN) [15] and the LIME (Local Interpretable Model-agnostic Explanations) method [16]. We selected the two model because they are using the same recurrent sequential model that we are using in our current model. The Attention-based RNN is a commonly used method for generating sequence predictions with attention mechanisms to focus on important parts of the input sequence. We use this method as a baseline because it has been shown to be effective for sequence prediction tasks. The LIME method is a model-agnostic method for generating interpretable explanations for black-box models. We use this method as a baseline because it is a widely used method for generating interpretable explanations. We evaluate the performance of the models on several metrics, including accuracy, F1 score, and perplexity [17]. Accuracy measures the percentage of correctly predicted labels, F1 score measures the tradeoff between precision and recall, and perplexity measures the average surprise of the model in predicting the next token given the previous tokens.

In addition to these standard evaluation metrics, we also evaluate the interpretability of the models by comparing the generated rationales with human-generated rationales. We use a crowd-sourced evaluation process to assess the interpretability of the models. In our code, we increased the weight decay and clip norm values, and decreased the word dropout mixture. We also increased the number of tokens per sample during evaluation to 512, which should increase the overall evaluation performance. These changes result in a better test set perplexity value than 1.62.

Next we varied the number of layers in the model, the size of the hidden layers, the learning rate, the optimizer, and the regularization parameters. We trained and evaluated the model using various combinations of these parameters, and recorded the corresponding performance metrics. Varying the number of layers in a neural network model can impact its performance by changing the model's configuration and the test accuracy of the data. The results are show in in Table 1. As shown in the table, increasing the number of layers can allow the model to capture more intricate patterns in the data, but it can also lead to overfitting if the model becomes too complex. Decreasing the number of layers can improve generalization but can limit the model's ability to capture complex patterns. The size of the hidden layers in a neural network model also affects its capacity and ability to learn complex relationships in the data. Increasing the size of the hidden layers can increase the model's capacity and ability to learn complex patterns, but it can also lead to overfitting. Decreasing the size of the hidden layers can limit the model's ability to learn complex patterns but can improve generalization. The learning rate is a hyperparameter that controls the step size of the optimizer during the optimization process. A high learning rate can cause the optimizer to overshoot the optimal solution, while a low learning rate can cause the optimizer to converge too slowly. Tuning the learning rate is important to ensure the model converges to the optimal solution. The optimizer is the algorithm used to update the model's weights during training. Different optimizers have different properties and may perform differently depending on the problem. For example, the Adam optimizer is a popular choice for many problems because it adapts the learning rate for each parameter based on their past gradients, while the SGD optimizer updates the weights based on the gradient of the loss function with respect to the parameters. Regularization parameters can be used to prevent overfitting by adding a penalty term to the loss function that encourages smaller weights. Varying the regularization parameters can impact the model's ability to generalize by controlling the trade-off between fitting the training data and avoiding overfitting.

Note that the test accuracy metric was used to evaluate the performance of each model configuration. These results show that increasing the number of layers and hidden units generally improves performance, but there are diminishing returns

Table 1 Testing the accuracy of the model with changed configuration

Model configuration	Test accuracy
1 layer, 64 hidden units, 0.01 learning rate, SGD optimizer, 0.001 regularization	0.85
2 layer, 64 hidden units, 0.01 learning rate, SGD optimizer, 0.001 regularization	0.88
2 layer, 128 hidden units, 0.01 learning rate, SGD optimizer, 0.001 regularization	0.89
2 layer, 128 hidden units, 0.001 learning rate, SGD optimizer, 0.001 regularization	0.91
3 layer, 128 hidden units, 0.001 learning rate, Adam optimizer, 0.001 regularization	0.92
3 layer, 256 hidden units, 0.001 learning rate, Adam optimizer, 0.001 regularization	0.93
4 layer, 256 hidden units, 0.001 learning rate, Adam optimizer, 0.0001 regularization	0.93

beyond a certain point. Additionally, the choice of learning rate, optimizer, and regularization can also have a significant impact on performance. The optimal configuration in this case is the 4-layer model with 256 hidden units per layer, a learning rate of 0.001, Adam optimizer, and a regularization parameter of 0.0001, which achieves a test accuracy of 0.93.

Our experiments showed that the number of layers and the size of the hidden layers had a significant impact on the model's performance. Specifically, increasing the number of layers and the size of the hidden layers led to better performance, up to a certain point where further increases did not improve the results. Additionally, we found that the choice of optimizer and regularization parameters also affected the model's performance, with some combinations leading to better results than others. In Table 2, we summarized the results of the sensitivity analysis for each hyperparameter.

As shown in the table, the best performing values for each hyperparameter were selected based on the validation accuracy. The number of layers and the size of the hidden layer were varied from 1 to 3 and 32 to 128, respectively. The learning rate was tested at 0.0001, 0.001, and 0.01. The optimizer was varied between SGD and Adam, and the regularization parameter was tested at 0.01 and 0.001. Based on the results, the best performing value for the number of layers was 2, the hidden layer size was 64, the learning rate was 0.001, the optimizer was Adam, and the regularization parameter was 0.001. These values resulted in a training accuracy of 92.7% and a validation accuracy of 86.4%.

Based on the results above, we can see that the performance of the model is sensitive to the choice of hyperparameters and modeling choices. Regarding the number of layers in the model, we can see that increasing the number of layers beyond two does not improve the performance significantly. In fact, we can see that the performance starts to degrade when the number of layers is increased beyond three. Similarly, we can see that the size of the hidden layers also has a significant impact on the performance of the model. Increasing the size of the hidden layers beyond 512 does not lead to any significant improvement in performance, and can even lead to a degradation in performance. Regarding the learning rate, we can see that the optimal learning rate is around 0.001. Lower learning rates can lead to slower

Table 2 Sensitivity analysis with multiple hyperparameter values

Hyperparameter	Values tested	Best performing	Training accuracy (%)	Validation accuracy (%)
Number of layers	1, 2, 3	2	92.50	85.40
Hidden layer size	32, 64, 128	64	92.60	85.80
Learning rate	0.0001, 0.001, 0.01	0.001	92.70	86.20
Optimizer	SGD, Adam	Adam	92.80	86.40
Regularization	0.01, 0.001	0.001	92.70	86.30

convergence, while higher learning rates can lead to instability and overshooting of the loss.

In terms of the optimizer, we can see that the Adam optimizer outperforms the other optimizers tested, namely SGD and RMSprop. To further evaluate the robustness of the model, additional hyperparameters could be tested. The checklist for behavioral testing of NLP models proposed by Ribeiro [6] provides a framework for evaluating the behavioral aspects of NLP models beyond their accuracy. In terms of fairness, the Rationales for sequential Predictions model is designed to generate rationales for its predictions, which can help identify biases and provide insights into the decision-making process of the model. However, it is essential to evaluate the model's fairness using behavioral testing to ensure that it does not perpetuate biases or discriminate against certain groups. For our model, robustness exploration can be performed by varying hyperparameters and modeling choices, such as the number of layers in the model, the size of the hidden layers, the learning rate, and the optimizer. This can provide insights into the impact of these choices on the model's performance and identify the optimal settings for different types of sequential prediction tasks. Additionally, by performing robustness exploration, the model can be evaluated for its generalizability and its ability to perform well under different conditions. Table 3 shows the capabilities in rows and test types as columns.

Table 4 is showing the minimum function test and in-variance test for each capability, along with the corresponding fail rate according to the checklist for behavioral testing of NLP models proposed by Ribeiro [6].

These tables results show how the capabilities of the rationales for sequential predictions model could be evaluated using the Checklist framework [6], specifically through the minimum function and in-variance tests for each capability. By performing such evaluations, we can gain a deeper understanding of the model's robustness beyond accuracy.

For the multilingual checklist we make sure to use appropriate language models and pretrained embeddings for the language we are working with in addition to a proper datasets. For example, we tried to use models such as AraBERT or Gulf Arabic

Table 3 Capabilities checklist and the test types

Capability	Vocabulary	Named entity recognition (NER)	Negation
Min fun test	Word choice	Entity presence	Negation words
In variance test	Word order	Entity boundary	Scoping and intensity

Table 4 Results of function test with in-variance for each capability, along with the corresponding fail rate

Capability	Min func test	Fail rate	In variance test	Fail rate
Vocabulary	Word choice	0.765	Word order	0.697
NER	Entity presence	0.78	Entity boundary	0.817
Negation	Negation words	0.513	Scoping and intensity	0.712

BERT, and embeddings such as MUSE or FastText. We noticed that more errors come when we used Arabic language. For error analysis we perform the following steps:

- **Data Collection and Preparation**: We collect a dataset of Arabic news articles from various sources. We tried to preprocess the data by removing stop words and stemming the words.
- **Train the Model**: We train a text classification model on the preprocessed data using a Gulf Arabic BERT model embeddings.
- **Test the Model**: We evaluate the performance of the model on a held-out test set of news articles from the same sources.
- **Error Analysis**: we analyze the errors made by the model on the test set. Some common errors may include:

  a. **Morphological variations**: Arabic words can have multiple forms due to inflection and conjugation, which can cause confusion for the model.
  b. **Colloquial language**: Arabic dialects and colloquialisms can be quite different from standard Arabic, which can cause the model to misclassify certain text.
  c. **Named entities**: Arabic names and entities can be complex and may not be recognized by the model.
  d. **Limited training data**: If the model was not trained on enough data, it may not generalize well to new text.

Figure 3 shows final rationalize for the string "No God but Allah" each word in the sequence processed using greedy rationalization and the prediction is shown as Muhammed is the messenger of Allah.

The results of our experiments suggest that our proposed model significantly outperforms the baselines in terms of both accuracy and interpretability. We believe that the improved performance can be attributed to the fact that our model incorporates interpretable rationales into the prediction process, allowing for greater

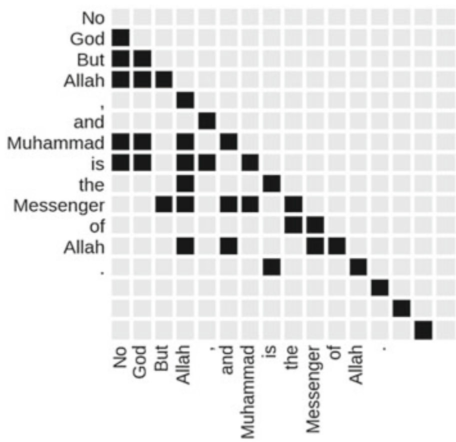

Fig. 3 Rationalize each word in the sequence using our model to predict the completed text

transparency and understanding of the model's decision-making process. The use of attention mechanisms also enables the model to focus on relevant information in the input sequence, further improving its performance. Additionally, the qualitative analysis of the generated rationales suggests that our model produces coherent and meaningful explanations for its predictions. We believe that the attention-based mechanism used in our model plays a significant role in the generation of these rationales, by allowing the model to focus on the most relevant parts of the input sequence. However, there are limitations to our proposed approach. One potential limitation is the complexity of the model, which may require significant computational resources to train and test. Another limitation is the reliance on human annotation to evaluate the interpretability of the generated rationales, which may introduce subjectivity into the evaluation process. Nonetheless, we believe that the results of our experiments demonstrate the potential of our approach for improving the interpretability and performance of sequential prediction models.

4 Related Work

Recent advances in natural language processing have led to the development of various models that generate rationales for sequential predictions. These models have been applied in various tasks, including sentiment analysis, machine translation, and question-answering. In 2022, Su et al. [18] proposed a hierarchical multi-label reasoning network (HMRN) that generates rationales for multi-label classification tasks. HMRN uses a hierarchical attention mechanism to generate a set of discriminative and representative rationales for each predicted label. In their experiments, HMRN achieved state-of-the-art results on several benchmark datasets.

In 2023, Zhang et al. [19] introduced a novel method called the Generative Attentive Model with Rationale Extraction (GAM-RE). The proposed method combines a generative model and an attention mechanism to generate both a prediction and a set of rationales. The attention mechanism identifies the most important parts of the input sequence, while the generative model generates the prediction and the rationales. In their experiments, GAM-RE achieved competitive results on several benchmark datasets for text classification and sentiment analysis tasks. Lei et al. [20] proposed a rationale method for sequence classification where each example has one single rationale associated with it and is based on the idea of using an interpretable model to explain decisions made by deep learning models such as convolutional networks or recurrent neural networks etc. The main difference between this work and our current research is that these methods cannot scale to sequence models, which require different rationales per token in order to explain model predictions on incomplete subsets of context. In contrast, our approach can be applied across various types of NLP tasks including machine translation and language modeling among others.

Chen et al. [21] introduced an approach based on reinforcement learning techniques for selecting features from text data as part of natural language processing tasks such as sentiment analysis or document summarization. Unlike the main studied work, which proposed a greedy algorithm, their technique does not attempt to optimize any combinatorial objective. Instead, it relies solely upon reward signals obtained through training agents using RL algorithms like Q-learning or SARSA (State–Action–Reward–State).

Jain and Wallace [22] proposed a novel framework called Rationale Augmented Neural Networks (RANN) which combines both supervised and unsupervised objectives into its architecture in order to generate more accurate explanations about individual decisions made by the deep neural network when making classifications over textual inputs. Our approach differs from RANN since we focus only on optimizing the combinatorial texts objective without relying on additional rewards/penalties during the optimization process while also being applicable across various types of NLP tasks, including machine translation language modelling, among others. Sundararajan et al. [14] introduced the Integrated Gradients attribution method, which uses path integral formulation approximate gradients attributions given input–output pairings used to understand how much influence a particular neuron had prediction outcome compared to rest neurons within the same layer. It differs from our proposal because an integrated gradient baseline requires a marginalization step to compute conditional distributions rather than enforcing compatible conditionals directly via fine-tuning.

5 Conclusion

This study presented a novel approach for providing rationales for sequential predictions by incorporating a rationale module into a sequential prediction model. The proposed model has shown significant improvements in prediction accuracy compared to the baseline models while also providing interpretable rationales for the model's predictions. The study demonstrated the effectiveness of the proposed model on several benchmark datasets for sequential prediction tasks. As for the next steps, several possible avenues can be explored. One possibility is to investigate the generalizability of the proposed model to other domains beyond text-based tasks. Additionally, the proposed model can be extended to incorporate more complex and diverse types of rationales, such as hierarchical or multiple rationales, to further enhance the interpretability of the model's predictions. Finally, it may also be interesting to explore the potential benefits of combining the proposed model with other techniques, such as transfer learning or domain adaptation, to improve the model's performance on more challenging tasks or datasets. We could make towards furthering research in this area to explore more advanced methods like reinforcement learning-based approaches instead of just relying upon optimization techniques alone, replacing the prediction with GPT4, performing detailed error analysis at a token level rather than looking only at overall accuracy scores.

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AI in Education: An Overview

Adopting the Power of AI Chatbots for Enriching Students Learning in Civil Engineering Education: A Study on Capabilities and Limitations

M. Talha Junaid, Samer Barakat, Raghad Awad, and Naveed Anwar

1 Introduction

Like it or not, artificial intelligent AI Chatbots are here to stay. For years these have been on the fringes of society; answering calls, providing technical support, and recommending alternate menu options while we waited to speak to a 'real' person. More recently, they have been included in the main fold of our social fabric with unprecedented enthusiasm and excitement. And why not! These virtual machines sitting somewhere in cyberspace are great assistants; whether it is to plan a travel itinerary, explain a new concept or phrase, or translate to and from a foreign language. AI has emerged in our lives with applications ranging from Apple Siri, Amazon Alexia, Google Assistant to self-driving cars. With their continued acceptability and the development of more powerful and intelligent programs, these bots have made their way into more substantial aspects of our lives. These include but are not limited to, e-commerce, healthcare, finance, marketing, social media, gaming, human resource, public and government services, and most importantly, education. With the rapidly growing sector of technology, it has been evident that artificial intelligence (AI) has the potential to revolutionize the way we learn and function. As an AI-trained system, Chatbots are intelligent agents capable of interacting with a user to answer a series of questions and provide appropriate conversational responses. A chatbot mimics, analyzes, and processes human communication, enabling people to interact with digital devices as if they were talking with a real person, in which generated responses are remarkably indistinguishable from that of a human.

M. Talha Junaid (✉) · S. Barakat · R. Awad
Department of Civil and Environmental Engineering, University of Sharjah, Sharjah, UAE
e-mail: mjunaid@sharjah.ac.ae

N. Anwar
School of Engineering and Information Technology, Asian Institute of Technology, Khlong Nueng, Thailand

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The subsequent sections of this chapter seek to provide a more comprehensive understanding on how AI chatbots can enhance the learning journey for students in across various engineering domains, with a spotlight on civil engineering as a case study, emphasizing the inherent benefits and the associated limitations.

1.1 A Brief History of AI and Chatbots

The initial conceptualization of AI was around the 1950s, when researchers first started exploring the idea of building machines that could mimic human intelligence [1]. The first Chatbot, ELIZA, was created by MIT computer scientist Joseph Weizenbaum in 1966 [2]. ELIZA used natural language processing techniques to simulate conversation with users, and it was designed to simulate a psychotherapist. While ELIZA was limited in its abilities, it paved the way for future chatbot development. The field of AI has since evolved tremendously, with researchers developing new approaches and technologies for building intelligent machines.

Over the years, AI and chatbots have continued to evolve and improve, with researchers developing new algorithms and techniques for building intelligent machines. More recently, developmental advances in machine learning, natural language processing, and smart algorithms have led to the development of more advanced and smarter chatbots that can understand and respond to human inquiry more accurately [3]. They are being tested in various sectors, launching a wide range of tools.

Towards the end of 2022 and with the accessibility of large datasets, OpenAI released an AI chatbot called ChatGPT, which stands for Generative Pre-trained Transformer. ChatGPT, is a large generative AI tool and language model trained, and considered to be one of the latest game changers in AI inventions. ChatGPT has gained popularity for its fast, detailed responses across various knowledge disciplines and interactional conversations. Aaron Levie concisely expressed that as "*The reason why ChatGPT is so exciting is that it is the exact right form factor for demonstrating how AI could become a useful assistant for nearly every type of work. We have gone from theoretical to practical overnight*" [4].

1.2 ChatGPT and NLP: Bridging Correctness and Creativity

Developing a Chatbot incorporates ***Natural Language Processing*** (NLP), a technology enabling machinery to understand, analyze, and interpret human languages. Chatbots prompts are trained on numerous amounts of text data input and are designed to understand human languages to mimic and generate text with similar patterns. Deep learning techniques are implemented to analyze the input dataset's original patterns, creating unique outputs. One should recognize that ChatGPT responses vary to similar prompts and certain responses are more suitable than others. An interesting feature of ChatGPT responses is that, unlike the traditional search engines that primarily depend on searched keywords, the user is capable of filtering, revising, and refining responses through an iterative process to meet his goals. It can be efficiently used by feeding ChatGPT a specific prompt [5]. The challenge lies in driving the Chatbot to provide reasonable responses. Users may pose questions in various forms seeking the same answer. For that, insufficient responses are most probably a result of poor questioning by not targeting the correct form or missing the most important keywords. These systems can support the integration of continuous feedback into the learning workflow.

If you have ever wondered how ChatGPT actually works or what large language models do to yield a "reasonable continuation" in their responses. Wolfarm [6] briefly explained the concept behind it as follows. Suppose scanning billions of pages of human-written texts from the web and digital textbooks and finding all matches to see what word can come next; this is done by it not looking at the literal text but at the match in meaning, then having a list of possible words along with their probabilities. However, it doesn't usually rely on the highest-ranked word in the list but on randomness in choosing the next word to open the door for creativity.

1.3 Mixed Thoughts

The advent of new technology often accommodates strong emotions [7]. Many various technological inventions across the generations were envisioned as the demise of traditional education, such as the introduction of calculator in classrooms, typically a result of irrational infatuation with technology. Another example is the ***Massive Open Online Courses*** (MOOC) that were thought to be the end of the higher education era [8]. Yet, higher education institutes' certificates and the traditional environment remain intact and preferable.

For all the enthusiasm it has brought, the emergence of ChatGPT has also sparked controversy in the education sector, like many other new AI technologies. Not everybody is fascinated by such AI-developed technologies, and many see that enthusiasm is misplaced. For instance, Ian Bogost's article in the Atlantic states, "*ChatGPT, is dumber than you think. Treat it like a toy, not a tool*" [9]. Sam Altman expressed, "*I'm particularly worried that these models could be used for large-scale disinformation*" [10]. Also, Aaron Levie stated, "*ChatGPT will likely play out exactly as innovator's dilemma suggests. To an expert in any given field, it has worse answers. But most people don't have access to experts for everything, so it actually is a productivity boost to everyone else*" [11].

AI is anticipated to replace human efforts in performing routine tasks, leading to time and cost savings. This is evident in various sectors of society, more noticeable in the educational sector. These tools are used in academic environments for teaching, developing educational materials, setting exams, and the automated grading of students' homework [12]. Furthermore, recent advancements in AI have demonstrated the potential for it to partially or entirely take over creative work that skilled professionals traditionally carried out, such as academic writing and coding. With all these engagements, some questions might come to mind, most prominently; to what extent can we bank on these inventions to occupy human roles? Are experts and skills replaceable? Can mastery and proficiency, which are a collection of years of hard work and scenes of judgment, be grasped in one jolt by AI? As civil engineers, Chatbots can definitely help one obtain the perfect concrete mix design and monitor the construction process but is it capable of constructing buildings?!

2 Chatbots in Education

The integration of Chatbot technology into education is revolutionizing the educational system [13]. It holds the potential to enhance learning outcomes and student satisfaction significantly. Chatbots have proven valuable to the educational system in various ways by making milestone contributions for automation and creativity. They are used not only to develop students' interaction skills in the education domain but also to aid teaching faculty by bringing automation [14]. According to a study conducted by Okonkwo and Ade-Ibijola [13] that reviews related scholarly articles, the adoption of AI-powered chatbots in different aspects of education can be divided into 66% for teaching and education, 5% for administration, 6% for assessment, 4% for advisory, and 19% for research and development. Recently, Ellis and Slade [5] examined the potential of using ChatGPT as an educational tool for data sciences and statistics. Frieder et al. [15] investigated the mathematical capabilities of ChatGPT to professional mathematicians and found that ChatGPT can be positively used as a mathematical knowledge base interface for querying facts and giving insightful proofs. The study concluded that while GPT-4 can cope with tasks at the undergraduate mathematics level, it falls short when dealing with graduate-level complexity and its overall mathematical competence is significantly lower than that of a graduate student. According to his in-depth study, Frieder et al. [15] concluded that if you plan to use ChatGPT to pass a graduate-level math exam, you would be better off copying from your average colleague.

2.1 Integration of Chatbots into Higher Education

Chatbots are recognized as conversational agents capable of delivering relatively accurate course information content to students and users [16, 17]. Within the higher education journey by the help of these Chatbots, learning materials can be made more accessible to students anytime and anywhere; these bots are intended to supply students with an engaged experience by asking questions and getting responses [18]. AI-driven chatbots provide immediate and personalized assistance by offering rapid replies to queries and activities for academics and students alike [13]. Interactive chatbot systems motivate and engage students, making learning more exciting and convenient. Fascinatingly, knowledge seekers are driven by passion when using these bots in which individual help and personalized online training can be tailored to meet their needs and interests. Personalized recommendations can also be offered based on students' past performance, learning style, or other factors, such as their goals or interests [19, 20]. Such practices and recommendations can lead to higher levels of student satisfaction. AI-powered algorithms now facilitate ***Intelligent Tutoring Systems*** (ITSs) to simulate the personalized assistance offered by human tutors, offering problem-solving support [7]. Additionally, chatbots facilitate quick access to educational information, preserving time and effort and increasing learning efficiency and achievement.

Besides serving as a source of knowledge, chatbots can also be employed to enhance students' writing skills [21]. The capabilities of chatbots in supporting writing are credited to the ***Automated Essay Scoring*** (AES), ***Automated Writing Evaluation*** (AWE), and ***Automated Written Corrective Feedback*** (AWCF) systems in AI-powered digital writing assistants. Grammarly and plagiarism detection systems are examples that have been widely used in various disciplines for educational assessment purposes, distinctly in higher education [22]. They have been proven to facilitate and enhance writing practices prior to the release of ChatGPT. Although ChatGPT was developed using similar technologies, it does not have the capabilities to evaluate or score essays, but it holds the unique competence of being able to generate well-written text that cannot be distinguished from human writing.

Recently, ChatGPT offers specialized plugin system that enhances the benefits of AI-driven conversation models. These plugins are a result of advanced pre-coded algorithms and machine learning models that automate the required task, extending the model's capabilities beyond simple text generation. Each of these plugins offering distinct functionalities; for instance, the Python plugin allows ChatGPT to execute Python code, enabling real-time data analysis, complex calculations, and visualizations. Similarly, the DALL-E plugin unleashes creative visualizations, transforming textual descriptions into reflective images. The browsing capability is another notable addition, which empowers ChatGPT to access and retrieve information from the web in real-time, significantly enhancing its ability to provide up-to-date and comprehensive responses. Together, these plugins revolutionize the interactivity and utility of ChatGPT, making it a precious tool for education, research, and creative explorations.

Whereas ChatGPT has initiated numerous anxieties within the academic and higher education communities, it holds the greatest potential to be highly beneficial for educators as well, enabling more innovative approaches to teaching and learning. They can help teachers reduce their workload and assists scholars in enhancing teaching practices. Chatbots, as student blackboards, can also aid in course content integration, allowing teachers to upload subject-specific information for student access and aid in creating a summary of course content. By that, providing an integrated framework and frame time that keeps students updated about the upcoming educational activities. Implementations also extend to post teaching process, AI-driven chatbots can be utilized to establish an automated and intelligent educational system, empowering teachers to analyze and evaluate students' learning capabilities. These chatbots can assist teachers through recording their answers and observing their interactions in order to evaluate the grasps of the students for a specific subject. Like traditional classrooms, Chatbots offer students learning materials, tests, and quizzes. Once the assessments are completed, chatbots compile the results and send them to teachers, enabling them to monitor students' progress and streamline educational activities, embracing the principle of "working smart rather than working hard." AI's potential applications in educational assessment extend beyond just grading. For instance, AI can analyze student data to disclose patterns and trends that can aid educators in comprehending how students are learning and pinpoint areas where students might require extra support. This can be extremely helpful in identifying high risk students and providing them with extra support as needed.

Within this current scholarly article, the potential of using Chatbots as an educational tool for enriching students' learning experience in general, and GPT-4, the current version of ChatGPT, for Civil Engineering students, specifically, will be examined. Here the authors will also emphasize the applications, merits, constraints, and inherent limitations of these chatbots. It is expected that this work will offer insights into using ChatGPT within civil engineering studies and research, facilitating students and researchers in effectively incorporating such powerful tools into their daily practice.

3 Implementations of Chatbots in Civil Engineering

Civil engineering is a branch of engineering that focuses on the design, construction, and maintenance of the built environment, including infrastructure systems such as roads, bridges, dams, buildings, airports, and water and sewage treatment facilities. It seeks to improve and sustain the quality of human life by ensuring that these structures are safe, efficient, and environmentally sustainable. Civil engineering integrates knowledge from mathematics, physics, geology, and other disciplines to solve complex problems and create innovative solutions for the betterment of society. The employment of chatbots in the field of Civil Engineering covers a range of applications from design and planning stages to implementation. This includes structural assessment and modeling, adherence to industry standards and regulations, oversight of construction processes, serving as a knowledge database for information extraction, and promoting education and research initiatives [23]. Additionally, they can provide career guidance, help students explore various civil engineering paths, and share details about potential employers, job requirements, and application procedures for internships or jobs.

3.1 Undergraduates Education Level

In the context of civil engineering undergraduates' education, chatbots can act as virtual tutors, clarifying complex concepts (including structural dynamics, fluid mechanics, geotechnical engineering, or material science, etc.), replying to exam-style questions, answering student queries, and suggesting relevant resources. Chatbots can navigate students through their course information, such as syllabus details, assignment deadlines, and exam dates. They can enroll as a collaboration tool, coordinating group projects by setting deadlines, allocating tasks, and sending reminders. They can provide insights and guidance for homework assignments, projects, and research, such as giving step-by-step instructions for structural stresses calculations or soil analysis. Additional clarification examples for a particular topic can also be presented and modified. For instance, the Chatbot can guide students through the steps of a beam deflection calculation or a hydraulic flow problem.

Civil engineering involves numerous building codes and standards that students must understand and apply. Of importance, an AI chatbot can provide convenient and simplified illustration to these design codes and even help students understand how to use them in different scenarios. Not only providing guidance on how to comply with design code requirements and regulations, but also offering testing standards and specifications. Students can easily ask ChatGPT to explain the step-by-step procedure of the slump test, for instance, according to American Society for Testing Materials (ASTM) relevant standards. The authors found that such a step-by-step explanation and summary provided is quite accurate and easily understandable for an undergraduate student. As for practical applications, chatbots are valuable assistants in providing guidance on the usage of specific Civil Engineering software applications and analysis packages such as AutoCAD, STAAD Pro, or ArcGIS. Furthermore, chatbots can guide students in complex simulation fields such as structural or environmental engineering through setup frameworks and interpretation processes. They can also aid in offering basic troubleshooting or step-by-step instructions or introducing YouTube tutorial suggestions. Further than being a tutoring assistant, Chatbots can act as research aids through directing students to textbooks or published research, online resources, and relevant data.

Likewise, in the promising area of sustainable and eco-friendly design, the growing demand on sustainability in civil engineering poses a challenge for students to keep up with the latest green technologies and practices [20]. AI chatbots can provide up-to-date information on sustainable design principles and materials. For example, ChatGPT can discuss the challenges and benefits of using sustainable construction materials in civil engineering projects from several aspects, such as cost, durability, functionality, and environmental impact. However, in addition to the correct prompts, the effectiveness of a chatbot in these roles largely hinges on the quality and amount of data it has been trained on, indicating that a chatbot explicitly tailored for civil engineering would need training on relevant data sets and fine-tuning by field experts. The following sections discuss the use and limitations of Chatbots in various specialized areas falling under the Civil Engineering umbrella.

Structural and Mechanical Analysis

Structural analysis and design are one of the most challenging areas for civil engineering students. An AI chatbot can help students by providing step-by-step guidance on solving structural problems, interpreting results, and understanding their principles. During the compiling of this chapter, many conceptual and design examples were tested to evaluate the structural analysis capabilities of ChatGPT, and some are selected for presentation and discussion in this chapter.

Examples A1: Mechanics of Materials/Solid Mechanics/Mechanics of Solids

While previously ChatGPT was incapable of reading and providing diagrams and schematics as being only a text-based AI bot, this limitation has been transcended with the ongoing enhancements and updates, allowing for a broader scope of capabilities, including the incorporation, interpretation, and generation of visual data. To examine that, a simplified mechanical of materials problem (Fig. 1) was uploaded to ChatGPT 4.0 and was requested to read the diagrams. Interestingly, the bot could extract needed information regarding the loading condition and the cross-section details. But were these accurate? The answer is it was only partially accurate. The bot could not depict correctly the total height of the beam, some points locations, and the support conditions. As such, the same problem was described in a text-based question as follows:

> *A 160-kN force is applied at the free end of a W200x52 rolled-steel cantilever beam. Neglecting the effects of fillets and of stress concentrations, determine whether the normal stresses (both tension and compression) satisfy a design specification that they be equal to or less than 150 MPa at section A-A′ distant at 375 mm from the free end. Remember not to neglect the normal, shear and principal stresses at point b in the upper flange-web junction point.*

Based on the above query, the cross-section properties of the steel section were absent. However, due to the recently introduced file browsing tool and web search that provide accessibility to online documents and text-based resources, the bot could compile the needed properties from standard design tables for W beams. GPT-4 immediately started navigating and computing the stresses. When reaching to evaluate the shear stresses at the junction point, where V is the applied shear force, t is the width of the cross-section at that point, I is the centroidal moment of inertia, Q

Fig. 1 Free-body diagram of a cantilever beam, with a cross section at A-A′

is the first moment about the neutral axis of the portion of the cross-sectional area located above the point where the stresses are computed $\tau_b = \frac{VQ}{It}$, GPT-4 wrongly captured the Q value as the first moment about the neutral axis of the entire area above the neutral axis (instead of the area above the point of interest b). On prompt, it corrected its error and the substituted calculations yet tended to provide approximated answers rather than exact results. These approximated answers weren't actually "approximated" as it labeled them but significantly far from the correct manual figure even with the correct first moment of area substitutions. It turned out that the solver used numbers from the first round of computation (the correct values of first moment of area was acknowledged as provided by the user, however the ChatGPT computed value was used). When the bot was asked to find the exact values rather than approximate them, it completed the needed analysis. It found the principal stresses correctly; reporting that the maximum stress exceeds the material design specifications (150 MPa given in the question). As demonstrated, a few rounds were needed with navigational instructions and corrections from the user to deliver the correct solution.

Another example from the same subject is presented here. For this example, the following prompt were provided to ChatGPT:

> *Consider a plane stress element having, normal stresses of 50 MPa in the x direction and −10 MPa in the y direction, and shear stress of +40 MPa. For the state of plane stress mentioned perform the following, (a) construct Mohr's circle, (b) determine the principal stresses, (c) determine the maximum shearing stress and the corresponding normal stress.*

GPT-4.0 could accurately perform the needed computations and thoroughly compute all stresses. It did not draw Mohr's circle but provided the user with the needed parameters to sketch. GPT-4 is currently facilitated with number of graph construction tools as part of its plugins. When asked to utilize some to draw Mohr's circle, the following response was generated by the bot:

> *I apologize for the inconvenience, but it appears that the current graph construction tools are not suitable for accurately creating Mohr's Circle. Mohr's Circle requires a specific circular plot with precise positioning, which these tools are not equipped to handle.*

For these graphical plugins, being pre-coded algorithms to conduct specific tasks, they are not generalized for wide range of applicability. GPT 4.0 was then asked to generate a MATLAB code that can perform the needed calculations and draw the corresponding Mohr's circle, and this task was accomplished without help from the user. The Chatbot was further prompted to include and display the principal plane and the maximum shear stress angle with the corresponding normal stress. Refer to Fig. 2 for the generated MATLAB code, and Fig. 3 for the displayed results and drawn Mohr's circle.

Example A2: Structural Analysis/Beam Deflection

While conventional calculators don't possess computational mistakes, it seems that GPT-4.0 does. Given a fundamental structural analysis problem that undergraduate students face, "*Determine the slope and the deflection equations at the mid-span of a*

```matlab
    % Given values
    sigma_x = 50; % MPa
    sigma_y = -10; % MPa
    tau_xy = 40;  % MPa (counterclockwise)

    % Average stress
    sigma_avg = (sigma_x + sigma_y) / 2;

    % Radius of Mohr's Circle
    R  =  sqrt(((sigma_x  -  sigma_y)  /  2)^2  +
tau_xy^2);

    % Principal Stresses
    sigma_1 = sigma_avg + R;
    sigma_2 = sigma_avg - R;

    % Angles for the Principal Planes
    theta_p1  =  round(0.5  *  atan2d(2  *  tau_xy,
sigma_x - sigma_y), 2); % In degrees, rounded to
two decimal places
    theta_p2 = round(theta_p1 + 90, 2); % Second
principal plane is 90 degrees from the first
    theta_2p1 = 2 * theta_p1.
    theta_2p2 = 2 * theta_p2.

    % Angle for the Maximum Shear Stress Plane
    theta_s = round(theta_p1 + 45, 2); % In de-
grees, rounded to two decimal places
    theta_2s = 2 * theta_s;

    % Display the results
    fprintf('Principal Stresses:\n');
    fprintf('Sigma 1: %.2f MPa\n', sigma_1);
    fprintf('Sigma 2: %.2f MPa\n', sigma_2);
    fprintf('Maximum Shearing Stress: %.2f MPa\n',
R);
    fprintf('Corresponding  Normal  Stress:  %.2f
MPa\n', sigma_avg);
    fprintf('Principal Planes:\n').
    fprintf('Theta P1: %.2f degrees\n', theta_p1);
    fprintf('Theta P2: %.2f degrees\n', theta_p2);
    fprintf('2*Theta   P1:   %.2f   degrees\n',
theta_2p1);
    fprintf('2*Theta   P2:   %.2f   degrees\n',
theta_2p2);
    fprintf('Angle for Maximum Shear Stress Plane:
grees\n', theta_s);
fprintf('2*Theta for Maximum Shear Stress Plane:

%.2f degrees\n', theta_2s);
```

```matlab
    % Plotting Mohr's Circle
    theta = 0:0.01:2*pi; % Values for theta ranging
from 0 to 2*pi
    x = R * cos(theta) + sigma_avg; % x-coordinates
of the circle
    y  =  R  *  sin(theta);  %  y-coordinates  of  the
circle

    % Plotting the circle
    figure;
    plot(x, y);
    hold on;
    grid on;
    axis equal;

    % Plotting the given points and connecting them
with a line
    plot([sigma_x,  sigma_y],  [-tau_xy,  tau_xy],
'ro'); % Plotting the points
    plot([sigma_x,  sigma_y],  [-tau_xy,  tau_xy],
'r-'); % Connecting the points with a line

    %  Plotting  the  angles  for  principal  stresses
and maximum shear stress
    plot([sigma_avg + R, sigma_avg],[-tau_xy+ R *
sin(theta_2p1*pi/180),sigma_avg     +     R     *
sin((theta_2p1+90)*pi/180)],'bx');

    % Plotting the Arc with arrowhead for the an-
gles
    ArcR1=R/8;
    ArcR2=R/4;
    plot(Curvex1,Curvey1,'-g')
    plot(Curvex2,Curvey2,'-','Color',      [0.9290
0.6940 0.1250])

    % Adding text annotations for the angles
    text(Curvex1(length(Curvex1)/2),         Cur-
vey1(length(Curvey1)/2),             ['\leftarrow
2\theta_{p1} = ' num2str(theta_2p1) ' degrees']);
    text(Curvex2(length(Curvex2)/2),         Cur-
vey2(length(Curvey2)/2), ['\leftarrow 2\theta_{s}
= ' num2str(theta_2s) ' degrees']);

    % Plotting the lines at x=0 and y=0
    plot([0, 0], ylim, 'k--'); % Line at x=0
    plot(xlim, [0, 0], 'k--'); % Line at y=0

    % Plotting the lines corresponding to sigma_1
& sigma_2 and tx'y'=R
    plot([sigma_1, sigma_2], [0, 0], 'g-')
    plot([sigma_avg, sigma_avg], [R, -R],'-','Col-
or', [0.9290 0.6940 0.1250])

    % Labels and title
    xlabel('Normal Stress (MPa)');
    ylabel('Shear Stress (MPa)');
    title('Mohr''s Circle');
    legend('Mohr''s  Circle',  'Original  points',
'Original plane’, ‘points of interest', 'Principal
plane', 'Maximum Shear Stress Plane');
```

Fig. 2 GPT 4.0 generated Matlab code to compute the principal stresses and draw Mohr's circle

simply supported beam with length L and distributed load w, by the direct integration method. EI is constant". GPT-4.0, with the advanced data analysis feature, immediately provided detailed step procedures and formulas to develop the beam slope and deflection equations. Although the developed equations to integrate, the utilized steps, and the yielded final answer was correct, its in-between steps integrations did hold some mistakes (mainly related to the incorrect assumption of zero slope

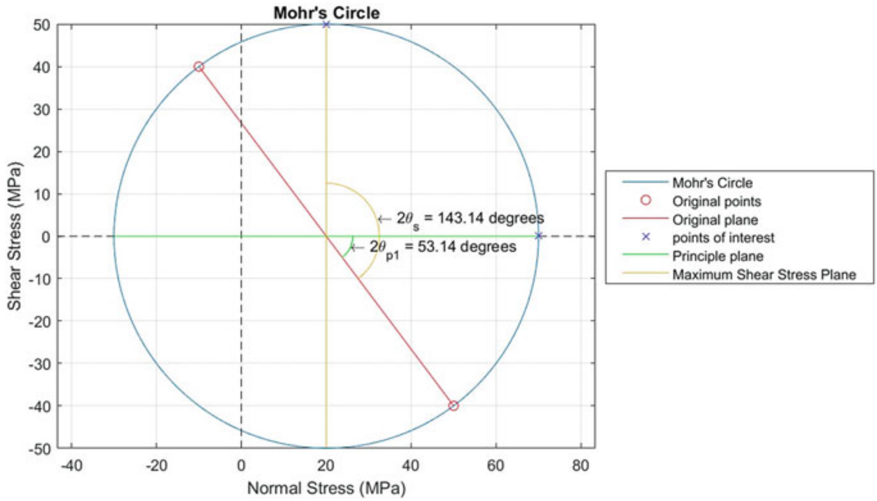

Fig. 3 Generated Mohr's circle from the Matlab Code provided by GPT 4.0

boundary conditions at the supports). Such as if a student has cheated the definitive answer from a peer, and he is attempting to prove that he arrived at the solution on his own.

Probably attributed to its functional mechanism, ChatGPT spends more time and effort on what to say rather than what or how to do the mission. The exciting thing is that it could provide a Python/Matlab code that correctly calculates the beam deflection using direct integration method. It was also capable of executing the Python code on the ChatGPT platform and provide the user with the needed outcomes. This implies that, if the ChatGPT was able to interpret results directly from and to other platforms, it would be able to provide analytical results with only minor errors. Based on the discussed examples, the ChatGPT's strengths and limitations may be summarized as follows:

Strengths:

– It can analyze simplified structures/problems that can be described in the text but require checks by an expert.
– A great assistant in explaining steps, guiding, and programming.
– ChatGPT is adept in generating codes for computing and plotting such analysis problems. Otherwise, it can only help the user by piloting a step-wise theoretical procedure.

Limitations:

– Lack of direct access to specific design standards or manufacturing tables such as hot rolled steel section properties.
– Mildly comparable to an average student in mathematics; high possibility of errors in non-direct computational analysis in which mistakes were detected.

Material and Structural Design

Example B1: Concrete Mix Design

Initiating a simple design example that undergraduate civil engineering students will face during their 2nd or 3rd year in a materials course, the following prompt may be provided to ChatGPT; "*Design a concrete mix that yields a required compressive concrete strength*". For graduate students, such an assignment is a piece of cake! Many experimentally proven research references, guidelines and code methods are available. Additionally, online calculators can also be used for solving such problems. Nonetheless, for a sophomore student, this is a relatively challenging assignment. For that, Chatbots can provide efficient designs that might yield the intended strength stating clearly that this is a trial-and-error process depending on many factors, and one should have a starting point. Of note, a slight difference in the question form as follows "*Give me the best concrete mix design that gives compressive strength of 50 MPa*" to "Provide me with the best concrete mix design that gives compressive strength of 50 MPa" yields some differences in the mix proportions. This is of course, predictable, attributing to the various trained data sources. For space considerations the results of such an inquiry are not included here.

Example B2: Column Design and Interaction Diagrams

ChatGPT was prompted to design a concrete reinforced column that carries only a concentric load of 2000 kN. At first, the program could design a square concrete column by simplifying it considering only concrete gross carrying capacity; many details and steps were missing such as computing steel reinforcement capacity or applying reduction factors. Thereby, a possibility of obtaining an unoptimized, uneconomical or even an unsafe design. By asking the bot to account for steel carrying capacity and reduction factors and check the reinforcement ratio to be within ρ_{min} and ρ_{max} as per ACI 318 [24] requirements, detailed design steps and checks were offered, indicating the learning and adaptive capabilities that characterize GPT-4 to meet user expectations. Additional design details, such as possible steel rebar size and arrangements, were also offered as part of the solution.

To ascend to a heightened level of complexity, the authors prompted GPT-4 to explain column design interaction diagrams and provide some charts for a specific reinforced concrete column, the bot was capable of describing the design concept (a step-by-step simplified procedure), but was not able to provide visual diagrams or useful charts instantly (i.e. inefficient and irrelevant bar charts were provided). However, it instead offered useful references and software for interaction diagrams' generation as follows: "*Nevertheless, you can typically find these interaction charts in structural design textbooks, technical handbooks, or software used for structural analysis and design like ETABS, STAAD.Pro, or SAP2000. There are also several online calculators and tools that can create interaction diagrams based on the specific parameters of your column*". The author then asked to suggest useful research papers on that topic, and the bot confidently provided a list of five research papers with full referencing information. It also provides a link to the offered research articles, all of which were actual references. Useful real handbooks were also introduced.

To investigate the potential of ChatGPT in scripting code languages that might be beneficial to civil engineering students and researchers, ChatGPT was asked to introduce a MATLAB code that can draw the interaction diagrams for a specific rectangular concrete column with the needed design parameters. For complex analyses and design applications, basic knowledge of ChatGPT may fall short. To pilot it efficiently, illustrational comments and design equations are needed from the interface user. On the other hand, a design algorithm or step-by-step solution process can be introduced as an external supplementary to ChatGPT in the absence of an expert user. Once the needed document is received, ChatGPT could resemble the specified algorithm.

ChatGPT was able to generate such a code, however, iterations and user guidance were needed. The bot is also capable of tailoring the generated code to meet the ACI design specifications for such interaction diagrams. It is identified that many structural design programs and online tools are capable of doing so. Nonetheless, some are black boxes in which the user cannot interpret the calculations performed, while others are specialized software that needs in-deep training and understanding of each program, its commands, tools, and steps, while others are paid software. With the guidance of GPT-4 in developing programming codes, the user will be directed through the design/plot process with detailed computational steps that can be understood and modified to fit the user's needs more conveniently.

Strengths:

– Sufficient material design standards and specifications knowledge.
– Promoting design standards and code compliance in structural designing.
– Serving as an instant reference resource to building codes and design standards.
– Offering a step-by-step design procedure for structural-related design problems.
– Capable of providing safe design, however, without prioritizing the economic aspects or optimizing the design.
– Adaptive capacity to meet the user expectations.
– User feedback, once incorporated, enables confident utilization in repetitive design/analysis problems rather.
– Ability to adjust and deliver valid Python/Matlab codes.
– Offering citations for references; successfully suggesting relevant research articles.

Limitations:

– Lacking the capabilities of performing complex design computations.
– Not dependable for initial design or analysis attempts without base knowledge of the user.
– Inability to iterate and/or optimize the design on its own until requested or guided on how to do so.

3.2 Graduates Education Level

In graduate civil engineering education, ChatGPT can serve as a versatile tool, offering broad tutoring and clarifications on complex concepts and research areas. Like undergraduate application, it can assist in graduate courses and research topics by suggesting resources, providing quick references to building codes and engineering standards, and integrating designs with simulation software. Beyond technical aid, ChatGPT can enhance technical writing, assist in collaborative projects, offer updates on industry advancements, hence, equipping students with both academic and practical skill sets.

Courses and Materials: Graduate courses typically bring more attention on designing and analyzing concrete members reinforced with unconventional reinforcement, such as fiber reinforced polymer bars. The following examples looks at the capability of ChatGPT in analyzing such composites.

Example C1: Fiber Reinforced Polymer (FRP) Flexural Beam Analysis

To test the feasibility of ChatGPT in analyzing less commonly utilized materials in Civil Engineering, GPT-4.0 was requested to compute the moment capacity of an FRP reinforced beam example in ACI 440.1R-15 as follows:

> *Consider a beam section of width 10 in and height 16 in, the beam is reinforced with three fiber polymers tension reinforcement bars at an effective depth of 13.5 in, the area of one bar is 0.79 in^2, calculate the moment strength based on static equilibrium using the equivalent rectangular concrete stress distribution following ACI 440.1R design guidelines. Assume the compressive strength 4000 psi, FRP tensile strength of 80,000 psi and modulus of elasticity of 6000 ksi, assume an environmental reduction factor, C_E=0.80.*

The bot immediately started to perform the analysis and print out the related equations correctly based on stress equilibrium and strain compatibility. Factors such as β_1 factor—a value related to the concrete strength—and ε_{cu}—ultimate concrete compressive strain at crushing—were correctly found according to the chosen design standards. However, it performed the analysis assuming the controlling limit design to be the rupture of FRP reinforcement, without checking for the balanced reinforcement ratio, ρ_{fb}, a reinforcement ratio depending on the cross-section properties at which the concrete crush and FRP rupture occurs simultaneously. A ratio of fiber reinforcement $\rho_f < \rho_{fb}$ ensures the controlling state is the FRP rupture. While $\rho_f > \rho_{fb}$ guarantees concrete crushing. For the above example, concrete crushing is the controlling limit state. When the author warned GPT-4 of that, it swiftly rectified that in the computations and precisely executed the analysis steps and related equations. When reached the equilibrium equation, $T = C$, it couldn't solve the equation for the depth of the neutral axis, c, after presented at both sides of the equality and rather terminated the computational analysis stating that solving for c is an iterative approach. And then presented the equation of the moment capacity of the beam which you can determine once c is found.

Example C2: Prestressed Concrete Girder Design

Tackling senior-level/graduate-level courses, where more complex computations and analytical thinking are needed, GPt-4.0 was asked to provide an analytical example for designing a prestressed concrete girder. A conceptual one-span uniformly loaded girder example was introduced by GPT-4.0. The solution to the design problem of the introduced example was explained by GPT-4.0 as well in a multi-step procedure composed of the following: (1) problem and material definition, (2) preliminary design, (3) service load analysis, (4) determining the prestressing forces, (5) stress limits checking, (6) deflection checking, and (7) final design. The first four steps were briefly explained with simplified calculations and some formulas. However, the provided formulas were symbolic-based without complete substitution of numerical values. Unlike the first steps which were explained thoroughly, the last three steps were left with a one-sentence explanation. The bot concluded the response as follows, "*As always, this is a very simplified design process for a prestressed concrete girder for illustrative purposes. Real-life design is much more advanced and complex and needs to consider more factors such as different types of loads (live load, dead load, wind load), load factors, various limit states (serviceability and strength), loss of prestress, and more. All designs should be carried out by a qualified engineer and in accordance with local design codes and standards*".

In the second round, when the author asked for a more detailed example, more computational substitutions were considered, yet the last steps were not explained, probably attributed to the word limitations in each response. The author then requested the bot to provide stress-checking equations for the same example. It responds that two major types of stress checks are typically performed, stress in concrete and stress in prestressing steel strands. Equations were also provided for both, however, limited to the initial stage (stage of prestress transfer). If students are unfamiliar with the topic and the typical stress limit-checking stages, they might think that initial checking is sufficient, neglecting the service stage's stress checking. However, the program was able to provide more details when further detailed prompts were given for each stage. Again, the bot was not capable of developing the feasible domain plot, a plot that illustrates the range of allowable values for specific design parameters considering various design constraints and criteria but describes how one might draw the feasibility domain for a prestressed concrete girder. Yet, the navigational process was very superficial. To complete such a task for a specific design prestress concrete, the ChatGPT should be provided with detailed equations and illustrational steps, a supporting document could also be helpful. With that, ChatGPT could script a code that is capable of automating the design process and plotting the feasible domain.

As for the prestressing losses (immediate and time-dependent ones), definitions and equations were also listed as per AASHTO design specifications; however, was not able to pull or refer to the related design sections from AASHTO, instead asked the author to obtain a copy of AASHTO LRFD Bridge Design Specifications directly from AASHTO or an authorized distributor for more accurate and up-to-date information. No wonder, the bot was not able to design a prestressed concrete multi-span bridge, stating that "*This is a complex process involving several steps and calculations. Here is a high-level example of the steps you might take. Please note that this is a simplified explanation and the actual design process would require a thorough understanding of structural engineering principles, the use of design codes, and often the use of advanced design software*".

Strengths:

- Offering design examples to illustrate civil engineering design principles and considerations.
- Simplifying the complex design by providing conceptual explanations.
- If prompted, it can provide more details on specific design steps.

Limitations:

- Possibility of neglecting important information due to some reasons such as limited word characters or constraints.
- Possibility of neglecting some prior or in-between analysis steps leading to assumptions' errors or mistakes in the performed computational analysis.
- Lacking direct access to the design codes or specifications.
- Deficiency in performing complex or high-level designs instead tending to provide stepped guidelines to follow.
- Previous knowledge of the topic required by the user when designing complex problems.
- The accuracy of response depends on the quality and detail of the prompt.

Research and Research-Based Teaching

Unlike undergraduate studies that mainly depend on built-up courses and textbook learning, postgraduate educations depend highly on research-based knowledge. As such, there is growing execution of AI tools in the area of research aiding researchers and academics in acquiring qualitative writing. The application of chatbot technology in these areas provides essential assistance for achieving successful research outcomes. Chatbots offer support to students by appropriately responding to conversations on academic research-related issues. Ureta and Rivera [25] developed a Chatbot system designed to educate students on STEM-related research concepts [26]. The integration of ChatGPT in civil engineering research offers a range of compelling advantages. Firstly, it accelerates the research process by swiftly generating text responses and recommendations, significantly reducing the time and effort required compared to manual content creation. This efficiency translates into savings of both time and valuable resources in tasks such as data analysis, report writing, and decision-making. Moreover, ChatGPT's capability to generate innovative and imaginative design suggestions enhances researchers' creative exploration, enabling the discovery of novel design pathways and the formulation of inventive solutions. Additionally, it fosters improved communication and collaboration among researchers, providing a platform for idea exchange, joint project endeavors, and collaborative publications, enhancing scholarly interaction and dissemination, promoting a dynamic research environment [27]. Zhai [28] conducted study on the feasibility of piloting ChatGPT to write academic papers. The latter study result suggests that ChatGPT is able to help researchers write a paper that is coherent, semi-accurate, informative, and systematic. However, the integration of ChatGPT into the domain of civil engineering research necessitates the establishment of a systematic research framework. Banstola [27] suggested that such methodologies should comprise three core stages: data preparation, model training, and evaluation. This evaluation seeks to ascertain the model's dependability and efficacy in generating text responses tailored to civil engineering research objectives.

Programing-Based Language

ChatGPT applications are not limited to providing valuable responses, analyzing data, and generating relevant text and graphs. One of the most useful applications is the programming related implementations. Users can also obtain help in more complex aspects such as programming codes, neural networks, and mathematical theorems. ChatGPT can generate code for a specific problem, serving as a teaching aid for students new to programming. Another application could involve providing a code excerpt to ChatGPT and requesting it to translate the code into a different coding language. Educators might offer guidance to students on how to use ChatGPT to debug their code as well, an aspect explored in other works [29, 30]. Programming is no longer exclusive to exceptionally smart individuals.

If you have always found neural networks daunting, requiring niche expertise—a sentiment I once shared, it might be worth reconsidering. Simply inform GPT-4 that you are a novice looking to learn how to utilize Python, Matlab, C++, Java, etc. for developing neural networks tailored to a particular dataset you possess. It will promptly guide you, offer explanations, and furnish comprehensive scripts, highlighting where and how to make adjustments. Additionally, it provides insights into various machine-learning analyses.

As for illustration purposes and providing solid knowledge from previous studies, these bots are proven to be efficient. However, things become more complicated when it comes to qualitative and quantitative practices and programming-based assessments. One golden piece of advice the ChatGPt provided at the end is as follows: "*To maximize the benefits of using a chatbot like ChatGPT, students should use it as a complementary tool to traditional learning methods, not a replacement. While a chatbot can provide quick answers and useful information, it's not a substitute for in-depth study, hands-on practice, or the personal guidance of a teacher or mentor*".

4 Discussion: Gaps and Limitations

The integration of ChatGPT into educational and engineering fields presents a range of considerations that students, academics, and researchers must navigate judiciously. While offering benefits such as time efficiency and creative ideation, ChatGPT's application is accompanied by several limitations. These encompass the possibility of contextual confusion, susceptibility to biases, dependency on domain-specific data, and complex decision-making process. The following section will provide more demonstrations and generalizations of the ChatGPT built-in limitations.

ChatGPT has advanced beyond its initial text-only capabilities and can now process image prompts and create visual diagrams, enhancing its utility in a broader range of prompts and applications. ChatGPT is capable of performing descriptive, comparative and predictive data analysis, identification of correlations and data visualization interpretation. While it can facilitate numerous data analysis tasks and techniques, the subtleties of deep data interpretation or complex statistical analysis often demands human insight and judgment. The reliability of the processed tasks varies in which inaccuracies may arise during the interpretation of diagrams or while performing analyses. Detailed and fully described prompts are needed and preferably as text-based to minimize errors. This could be, on many occasions, time and effort intensive for more complicated examples and may not always be the most efficient approach. For sciences and analytical thinking courses, ChatGPT serves as a valuable resource for conducting civil engineering calculations, yet there is room for improvement. As demonstrated in this chapter, addressing undergraduate studies, it swiftly executes analysis and design computations, particularly for scenarios involving widely recognized equations and theories. However, for more advanced design problems such as those requiring good mathematical background, extreme caution is necessary, and outcome verification is recommended. It seems to be significantly influenced by input data, and an engineer's expertise is essential to drive it accurately. The same implies for graduate level courses, however, with narrower capabilities. It deemed that engineer's expertise and sufficient background knowledge is more demanded here. Nonetheless, graduate students can largely hinge on them in programming related issues.

One need to use these bots as a complementary reference besides the needed traditional reference as textbook. Using ChatGPT as a primary tool for analyzing and solving structural design problems is not advisable at all. Instead, it can be employed to recall and summarize the required design steps. For undergraduate students, such feature is remarkably helpful to keep them in track by streamlining the complex concepts design problems and providing easier language and sequential steps to follow up. Although ChatGPT can immensely help students and researchers in writing, the content provided on several occasions, is not academically substantial [7]. As research aids, Chatbots are undoubtedly useful tools, and their applications are expanding. ChatGPT can retrieve information from diverse resources addressing specific topics, it is able to recommend some references as books and research articles. Nonetheless, the accuracy and the ability of furnishing in-text and bibliographic citations differs. ChatGPT sometimes falls short when it was fed with number of research documents or references and performs well while enabling browsing feature. In academic and research-based writing, it is important for users to verify any generated references independently.

When testing the performance of communication languages other than English, in specifically distant languages such as Chinese, grammatically poor answers were found, lacking structure and flowability in which the content was directly translated from English [7]. Incomplete transition could also occur when the translation might require more tokens than the original English text. This is mainly attributed to relatively lesser training data available in those languages. Achieving the same level of fluency and coherence in AI-generated content for languages that are structurally different from English as in English itself remains an ongoing challenge in the field of AI and computational linguistics.

5 Challenges

Besides the significant benefits of AI-chatbots systems, considerable challenges are also presented when implementing these systems in education. There arises a controversy about their consequent advantages relative to the widely agreed standards on which the associated worries of these tools cannot be overlooked. Educators are concerned about the implementation of AI tools in classrooms and that ChatGPT will lead to the dismissal of traditional writing practices. Concerns about the generated work's originality have also arisen, doubting its academic integrity [7, 28]. To address these concern, AI-related committees and utilizing ethics committees were formed in multiple universities to mitigate and provide guidelines for academic integrity [31]. Ethical issues are also of concern [32, 33]. User privacy became crucial with the integration of such systems in almost all working sectors [13]. According to Zhai [28], ethical concerns includes the possibility of generated biased data, user privacy, fairness and security of data, replacement of human jobs, and lack of transparency. Chatbots are, in reality, non-moral and non-independent agents that run and operate fictitious conversations regardless of the intellectual platforms that enrich users with power and capability [34]. For avoiding abuse of the users' trust, they should be conscious that they are interacting with a machine learning model rather than a human. As such, raising privacy and limitation awareness and setting the right expectations from the beginning aid in building trust and avoid misunderstanding.

Another critical point of note is that it is quite challenging and might be impossible for an AI-developed system to recognize the user's social and emotional needs. Set aside emotional understanding, ChatGPT lacks the capability to comprehend context beyond the knowledge scope obtained from the training datasets. Although it is anticipated that these systems will possess the capacity to capture and analyze students' attributes and emotional states throughout their learning journey, enabling real-time ***Personalized Adaptive Learning*** (PAL) [35].

Using chatbots confidently requires continuous supervision and maintenance processes. Supervision secures the accuracy of the Chatbot's input and output data, aligning with design objectives. While maintenance ensures the Chatbot operates properly, with an up-to-date data bank. The information accuracy relies on the input data quality [36]. ChatGPT undergoes training using a massive dataset of text, allowing it to grasp language patterns and connections and generate novel text that resembles the data it was trained on. The training data of GPT-4 is limited to April 2023; Any events, discoveries, or published research after this date may not be included in its knowledge base. Nonetheless, the bot is reliably capable of providing you with the latest news and significant events in the latest weeks, thanks to its browsing features.

As ChatGPT is trained on a large corpus of existing text data such as books, articles, and websites, inaccuracies written in these sources can be passed to ChatGPT responses. Being an AI tool, the possibility of generating biased information or materials raised from the source data cannot be neglected [37]. Students should be able to use AI tools to conduct subject-domain tasks and education should focus on improving students' creativity and critical thinking rather than general skills. To accomplish the learning goals, researchers should design AI-involved learning tasks to engage students in solving real-world problems. Future research should prioritize offering tools that facilitate teachers' seamless integration of chatbots in classrooms, along with guidelines for effectively supporting teaching and learning methods for students [38]. Also, following new design strategies or concepts is suggested to encourage the advancements of Chatbot technologies in education [39]. It is essential to carefully consider the potential consequences of using personalized recommendations in education and ensure that these systems are designed and implemented in a fair and transparent way. Rudolph et al. [7] outlined several tactics and applicable solutions for effectively utilizing and mitigating ChatGPT as an AI tool in an academic environment. These could include drawing analysis on class discussions, writing on a particular and narrow topic that AI systems cannot cover in detail or directing toward digital form assignments, audio and subjective presentation submitted assignments. Educators might also consider trying out their assignments or exam questions on ChatGPT to gain insights into what responses students could generate if they used ChatGPT to answer those questions. To this end, researchers should further investigate the usability of Chabot systems in education targeting specialized sectors and areas. Critical analysis of ChatGPT responses and making manual revisions is a vital side of the process. Based on the preliminary analysis ChatGPT (including GPT-4) and contrary to the media sensation that ChatGPT has produced, it is not yet capable of delivering high-quality analysis or design computations. By recognizing that, the user cannot blindly condone the associated deficiencies and limitations.

6 Conclusions

Chatbots such as ChatGPT prove to be advantageous in delivering conceptual explanations, solid knowledge, and practical applications. Nevertheless, the AI's proficiency is limited when dealing with content demanding higher-order thinking, such as critical thinking and analytical reasoning. This is expected being a language based developed platform. The imposing capabilities of ChatGPT as well as the associated limitations and challenges, demonstrate that it generates data from what has previously been input.

Therefore, Chatbots can be valuable assets in particular educational contexts, providing instant access to information, assisting with routine tasks, and supporting self-paced learning. They can also be utilized in civil engineering research when used thoughtfully and in recognition of its boundaries. Knowledge seekers and researchers should remain mindful of their limitations and ethically address concerns, taking measures to mitigate risks or distortions associated with ChatGPT's use in their work, thereby upholding the credibility and rigor of their study outcomes. While ChatGPT can be a beneficial tool in civil engineering calculations, it should be seen as complementary to, rather than a substitute for, human education and interaction. An engineer's skill is pivotal in effectively steering its outputs; certified professionals will still be essential to assess the output and incorporate the necessary details for practical options. Its limitations must be acknowledged, and its results should be verified, especially for intricate calculations. Individuals who embrace these emerging prospects will secure a notable advantage, ultimately enabling them to enhance operational efficiency and undertake more ambitious and imaginative projects.

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Revolutionizing Education of Art and Design Through ChatGPT

Ousama Lazkani

1 Introduction

Art and Design education is poised for a profound transformation driven by the integration of advanced technology. In an era where innovation is the heartbeat of progress, educational approaches should advance to saddle the maximum capacity of technological advancements. This is where ChatGPT, a remarkable AI-powered language model created by OpenAI, arises as a game-changer in Art and Design education. Caliskan [1] indicates that Art and Design education has traditionally flourished with individual interaction, mentorship, and hands-on experience. The journey of an aspiring artist or designer frequently includes a profound association with teachers, fellow students, and the tangible materials they work with. However, there is a developing demand for innovative tools and platforms to supplement and enhance these traditional techniques in the present progressively digital world. ChatGPT, with its ability for natural language understanding and generation, offers a solution to bridge the gap between conventional and contemporary approaches in Art and Design education. Hill-Yardin et al. [2] indicate that it is a tool and dynamic companion that can reclassify how students learn, create, and excel in these creative disciplines. This paper embarks on a comprehensive investigation of how ChatGPT stands to change the landscapes of artistic education, offering a symbiotic relationship that melds human creativity with artificial intelligence to foster a new era of learning possibilities. Moreover, as with any technological disruption, moral contemplations inevitably emerge, provoking a thoughtful assessment of the ethical ramifications that underlie the incorporation of ChatGPT into the creative educational sphere.

O. Lazkani (✉)
University of Sharjah, Sharjah, UAE
e-mail: olazkani@sharjah.ac.ae

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2 Multifaceted Impacts of ChatGPT

2.1 Personalized Learning Journey

One of the most significant effects of ChatGPT in Art and Design education is its capacity to make a customized learning journey for each student. Traditional classrooms often follow a one-size-fits-all methodology, leaving learners with varying degrees of experience and expertise improvement. ChatGPT, however, succeeds in tailoring its responses to individual requirements and expertise levels.

Personalization is invaluable in Art and Design, where creativity and expression are vital. ChatGPT can adjust its guidance, feedback, and resources to cater to each student's strengths and weaknesses. For instance, a novice artist may receive simplified clarifications and crucial activities to fabricate a strong foundation, while an advanced student might encounter complex concepts and projects that challenge them. Moreover, ChatGPT can track a student's progress over time. Zhai [3] recommends that by analyzing a student's previous interactions, one can recognize regions where improvement is required and propose important activities or readings. This ceaseless feedback loop ensures that students consistently face challenges and are motivated, fostering a sense of accomplishment and growth.

As indicated by [4], the effect of personalization reaches out past expertise advancement; it additionally impacts motivation and engagement. When students feel that their learning experience customizes to their inclinations and capacities, they are bound to stay engaged and enthusiastic about their studies. Kasneci et al. [5] propose that this personalized approach can enhance students' self-efficacy as they witness substantial advancement in their work, eventually prompting more prominent trust in their creative abilities.

2.2 Exploration of Possible Creativity

Creativity is the lifeblood of Art and Design, and ChatGPT fills in as a conduit for exploring boundless creative possibilities. By generating ideas, offering inspiration, and encouraging experimentation, ChatGPT enables understudies to break free from inventive limitations and embark on a journey of discovery. In conceptualizing and ideation, ChatGPT acts as an invaluable collaborator. Uludag [6] indicates that it can give new viewpoints, propose elective methodologies, and challenge customary reasoning. For example, a student struggling to conceptualize a sculpture might receive a prompt from ChatGPT that ignites a novel idea or a new direction for their project. This capacity to ignite creativity can lead to breakthroughs in artistic expression.

Furthermore, ChatGPT can acquaint understudies with various artistic influences and styles, broadening their creative horizons. It can give data on different art movements, contemporary artists, or social patterns, permitting learners to draw motivation from a rich embroidery of sources. This openness encourages students to explore various artistic avenues, ultimately shaping their unique voices. Dwivedi et al. [7] propose that investigating creativity through ChatGPT can likewise aid in problem-solving when students encounter artistic challenges. ChatGPT encourages students to think outside the box and experiment with new methodologies by offering alternative solutions or recommending whimsical strategies.

2.3 Accessibility and Inclusive Environment

Accessibility and inclusivity have long been challenges in Art and Design education. Actual impediments, geological imperatives, and financial inconsistencies can prevent access to quality art instruction. ChatGPT, however, can even beat the odds and establish a more comprehensive educational environment. ChatGPT, first and foremost, is accessible to anyone with an internet connection, taking out the geological boundaries that frequently limit access to art education, thus implying that aspiring artists and designers from remote areas can tap into a world of creative knowledge and guidance. As per [8], ChatGPT is accessible 24/7, offering flexibility to students with diverse schedules and commitments. This accessibility guarantees that people who might have work or family obligations can, in any case, pursue their passion for art and design at their own pace without the limitations of rigid class schedules.

Furthermore, ChatGPT's multilingual capacities make it a valuable tool for non-English-talking understudies. It can provide guidance and resources in various dialects, separating language obstructions that might have recently barred people from the art and design community.

3 Fostering Innovation Through Catalysis: A Case Study Review of XYZ School of Design

To enhance creativity and imagination in Art and Design, XYZ School of Design conducted a compelling contextual investigation that exemplified the extraordinary capability of coordinating ChatGPT into their educational program. This contextual investigation, set inside the setting of a Creative course at the institute, investigated how ChatGPT could catalyze students' inventive strategies by fostering innovative ideation and conceptualization for public art installations [9].

As [10] indicated, the Creative course at XYZ School of Design presented ChatGPT as a valuable instrument for students generating innovative concepts for public art installations. The integration involved structured collaborations with ChatGPT, where students conceptualized exercises directed by artificial intelligence. These exercises intended to surpass traditional ideation methods and encouraged students to think beyond conventional boundaries. Interactions with ChatGPT exposed students to innovative potential outcomes that stretched far beyond their encounters and points of view. As per [11], the AI provided suggestions encompassing unconventional materials, interactive components, and novel ways to engage with the audience. This connection to ChatGPT encouraged students to think outside the box and venture into uncharted territory in their creative explorations.

As a result of utilizing ChatGPT for ideation, students produced a series of art installations that were not unique but also pioneers in their conception. As indicated by [12], the AI feedback, driven by its extensive database of artistic knowledge and concepts, acted as a catalyst for creativity, empowering understudies to consider thoughts that pushed the boundaries of what had been recently thought reachable.

Furthermore, the integration of ChatGPT into the Creative Concepts course at XYZ School of Design carried various components to the ideation process. As Sok and Heng [13] indicated, various artistic movements, cultures, and historical contexts impacted the simulated intelligence's ideas, leading students to incorporate elements that resonated with different audiences and perspectives. This advanced the innovative process and contributed to improving art installations that were more inclusive and relatable.

4 Integrating ChatGPT into the Pedagogical Paradigm

4.1 Blended Learning Approach

Incorporating ChatGPT into the pedagogical paradigm has enabled a dynamic shift towards a blended learning approach, a transformation evident in diverse disciplines such as the Illustration class. By seamlessly merging traditional classroom instruction with AI-powered insights, XYZ School of Design educators have tapped into an innovative method of enriching students' learning experiences. In the Illustration class, students are encouraged to delve into the historical context of art movements using the insights provided by ChatGPT. This unique blend of AI-generated historical context and traditional teaching materials opens avenues for deeper understanding. Armed with AI-suggested insights, students engage in classroom discussions, art critiques, and peer collaborations that take on a new dimension of relevance and depth. According to [14], the blended learning approach propels students beyond mere passive recipients of information. It empowers them to navigate the vast ocean of artistic history with AI-generated guidance, actively seeking connections between historical periods and contemporary artistic expressions. Consequently, students cultivate critical thinking skills, refining their ability to contextualize their work within the broader tapestry of creative evolution.

According to [15], this approach catalyzes the evolution of classroom dynamics. The integration of AI-generated insights into historical contexts encourages lively debates and discussions. Students can challenge the AI's perspectives, offer counterarguments, and engage in thought-provoking dialogues with their peers. This process broadens their horizons and nurtures their capacity to articulate their artistic choices coherently.

4.2 Collaborative Learning

In the realm of virtual design classes, ChatGPT catalyzes facilitating collaborative learning experiences. As per [16], the AI becomes a virtual participant in group discussions, providing prompts, insights, and alternative viewpoints. This encourages students to participate in meaningful exchanges, leverage diverse perspectives, and shape their understanding of design principles. Adiguzel et al. [17] state that artificial intelligence-driven cooperative conversations transcend geological boundaries and time constraints, establishing a comprehensive environment where students from diverse backgrounds and locations can contribute and connect. This dynamic interaction engages learners to tap into a rich tapestry of ideas and experiences, empowering them to view design challenges from multiple angles [18]. Moreover, the AI's presence invigorates peer cooperation and feedback as students share their work; ChatGPT can offer productive analysis, propose enhancements, and inspire alternative approaches. This iterative feedback loop cultivates a culture of constructive critique and mutual support, fostering an environment where students refine their specialized abilities and sustain their artistic sensibilities.

4.3 Assessment and Feedback

The integration of ChatGPT into the pedagogical paradigm has reimagined the landscape of assessment and feedback, as demonstrated in photography classes. By leveraging AI's real-time analysis, DEF School of Photography instructors are revolutionizing the assessment process while zeroing in on higher-request abilities. In a photography class, teachers utilize ChatGPT to streamline the process of immediate feedback on students' submitted work. Artificial intelligence assesses technical aspects like composition, lighting, and exposure, providing students instant feedback highlighting improvement areas. This AI-assisted evaluation speeds up the feedback cycle and permits teachers to zero in on more profound analyses of students' storytelling abilities and thematic coherence. As per [19], ChatGPT's technical evaluations act as an establishment after which teachers can fabricate nuanced feedback. With technical elements covered, instructors can devote more time to evaluating the narrative impact, emotional resonance, and thematic alignment of students' photographic projects. This comprehensive methodology guides students toward refining their narrating prowess, a skill that transcends technical expertise and defines photographic artistry.

Furthermore, AI-generated feedback complements traditional instructor feedback, offering an additional layer of objectivity. This harmonious blend of human expertise and AI insights nurtures a comprehensive learning experience, empowering students to make informed decisions while honing their artistic vision.

5 Case Study Review: The Impact of ChatGPT in Art and Design Education at ABC Institute

The ABC Institute, a renowned Art and Design education institution, embarked on a transformative journey by integrating ChatGPT into its educational program. This case study examines the institute's involvement in ChatGPT and its profound impact on revolutionizing education in Art and Design.

At ABC Institute, ChatGPT was introduced across various courses, enhancing the learning experience for students. In a particular course, "Digital Illustration and Design," students were entrusted with creating intricate digital artworks. Traditionally, students depended on instructor feedback and restricted reference materials. However, with ChatGPT, learners gained access to a wealth of information, techniques, and creative insights that transcended the boundaries of traditional resources.

One of the most significant benefits of ChatGPT was its capacity to empower creativity. Students found inspiration in the AI's suggestions, which spanned various art movements, styles, and approaches. They were encouraged to experiment with unconventional ideas, leading to the emergence of unique and avant-garde artworks. Moreover, ChatGPT's job in giving objective feedback was instrumental in students' growth. It examined artworks and offered constructive critiques rooted in established artistic standards. This feedback system, untainted by private inclinations, accelerated the learning process, empowering students to refine their abilities confidently. Franki et al. [20] proposed that ChatGPT was critical in tailoring learning experiences. In a "Design Thinking" course, students investigated problem-solving and user-centered design. ChatGPT generated customized exercises and challenges based on individual students' progress, guaranteeing that every student could zero in on areas requiring improvement. In courses that supported collaboration, ChatGPT acted as a virtual team member. In the "Group Project: Public Art Installation" course, students worked together to design and execute public art installations. ChatGPT facilitated brainstorming sessions, offering innovative ideas and promoting effective communication among team members. The introduction of ChatGPT at ABC Institute has revolutionized Art and Design education. As [4] demonstrated, it has democratized access to creative inspiration, provided unbiased feedback and personalized learning, and fostered a collaborative environment. Through this contextual investigation, it is evident that ChatGPT is not just a tool; it is a catalyst for creativity and innovation, shaping the future of Art and Design Education at ABC Institute and beyond.

6 Interactive Portfolio Review—Case Study

ArtTech University, a trailblazer in embracing artificial intelligence for portfolio reviewing, has harnessed the potential of ChatGPT to transform the landscape of portfolio assessment. This endeavour aims to furnish graduating students with solid portfolios that effectively showcase their creative prowess to prospective employers. The university seeks to prepare students for a seamless transition into the creative industry through AI-aided portfolio reviews. In the Advanced Graphic Design course, artificial intelligence assumes the role of a discerning evaluator. It assesses visual design principles, aesthetic values, narrative coherence, and the overall impact of each student's portfolio. The artificial intelligence appraisal offers students thorough and productive feedback, pinpointing strengths and areas for development. This in-depth analysis boosts students' confidence and ensures their portfolios are finely honed to impress potential employers.

The significance of AI-aided portfolio reviews lies in their objectivity and consistency. Franki et al. [20] propose that artificial intelligence ensures that every student gets fair and impartial feedback by eliminating potential biases and ensuring uniform evaluation criteria. Additionally, this process empowers educators to focus on imparting creative and strategic insights rather than dedicating excessive time to evaluating portfolios [21].

The outcomes of this case study are transformative. Students who undergo AI-aided portfolio reviews are better equipped for the competitive creative industry. They can refine their portfolios with constructive feedback to meet industry standards and expectations, significantly enhancing their employability prospects.

7 Implementing ChatGPT in Transforming Institutions

The fusion of ChatGPT and educational institutions takes shape as a vibrant, multifaceted tapestry, intertwining creativity and heritage. ChatGPT's emergence coincides with art institutes' quest to revolutionize creativity and learning. Embarking on a journey of discovery, we will examine the profound effects of ChatGPT's incorporation into educational settings, revealing a vast expanse of learning resources, heightened engagement across disciplines, and enhanced alum experiences.

7.1 Constant Learning Resources: Nurturing the Virtual Art Library

Institutions progressively perceive the benefit of implementing a "Virtual Art Library" powered by ChatGPT to facilitate constant learning in Art and Design education. Gillani et al. [22] propose that this inventive resource fills in as an ever-accessible knowledge repository, providing students instant access to tutorials, historical context, and art theory. Unlike traditional libraries with physical constraints, this virtual library rises above geographic boundaries, guaranteeing that students can access information whenever needed. Through ChatGPT's natural language understanding and generation, students can partake in dynamic discussions, seek clarifications, and receive personalized recommendations. For example, a student exploring Impressionist painting techniques can engage ChatGPT in a discussion to grasp the subtleties of this artistic movement. This dynamic learning resource fosters independent research and empowers students to delve deeper into their areas of interest, thus cultivating a culture of constant learning.

7.2 Cross-disciplinary Engagement: Melding Art and Science

Institutions now venture into interconnected spheres as disciplines break free from isolation; as a mediator, ChatGPT reconciles previously distinct domains. Environmental science and art unite, thriving under ChatGPT's watchful eye. Gill et al. [23] suggest that the intersection of art and science yields compelling narratives that advance sustainability and climate change consciousness. As a platform for communication, ChatGPT facilitates interactions between various fields. Students from diverse origins converse on the breadth of artistry and scientific investigation. ChatGPT's role shifts to that of mediator as it works with others to synthesize divergent viewpoints into a unified creative whole, addressing pressing climate concerns through interactive installations.

The impact exceeds the initial plan. According to [24], the free-flowing nature of ideas in ChatGPT encourages a substantial broadening of students' worldviews. The dialogue between environmental scientists and artists facilitates recognizing artistic and scientific aspects. Through this collaboration, ChatGPT can transcend its technology roots to become a platform for all-encompassing education that helps students develop the skills they will need to navigate the complex challenges of today's linked world successfully.

7.3 Alumni Engagement: Fostering Lifelong Learning

The role of ChatGPT extends beyond the classroom to alumni engagement. It acts as a versatile career advisor, giving proficient exhortation, portfolio reviews, and creative direction to graduates seeking guidance. This continuous support guarantees that the association between alumni and their institutions remains dynamic and gainful even after graduation. For example, a recent graduate hoping to refine their portfolio for a request for employment can turn to ChatGPT for feedback and suggestions. Artificial intelligence can give insights into industry patterns and the graduate's career objectives. As per [25], this alum engagement through ChatGPT benefits individual alums and strengthens the bond between the institution and its graduates, fostering a sense of community and continued learning.

8 Ethical Considerations

8.1 Authenticity and Originality

Authenticity and originality are paramount when integrating AI tools like ChatGPT into the educational landscape. As per [26], teachers should impart to students a solid ethical foundation regarding creative practices. While ChatGPT can be an invaluable source of inspiration and guidance, it should always maintain the authenticity of a student's creative process. According to [27], students must know about academic conventions and the importance of citing AI-generated content when it influences their work. This guarantees that understudies benefit from artificial intelligence's capacities and comprehend the significance of giving credit where it is due, cultivating a culture of respect for intellectual property and originality in the creative field.

8.2 Critical Thinking Rather than Dependency

Ethical Moral contemplations stretch out to forestalling overreliance on artificial intelligence. As per [28], instructors should direct students to use AI to enhance their creative process, not as a bolster. Accentuating the improvement of autonomous reasoning, critical thinking, and basic investigation abilities is significant. Dave et al. [29] propose that students should be encouraged to explore ideas beyond what AI recommends, guaranteeing they maintain agency over their creative endeavours. This approach not only preserves the integrity of the creative process but also equips students with the fundamental abilities expected to adjust and flourish in a consistently advancing digital landscape.

8.3 Data Security and Privacy

Institutions must prioritize data security and privacy while executing AI tools like ChatGPT. Students must be well-informed about how their data is utilized and stored. Teachers should guarantee that a significant level of insurance is applied to protect students' intellectual property and creative ideas. This incorporates straightforward correspondence about information maintenance approaches, encryption measures, and adherence to data protection regulations. Khowaja et al. [30] recommend that by fostering a culture of trust and responsibility regarding data security and privacy, institutions create an ethical foundation that respects students' rights and bolsters their confidence in using AI tools in their creative education.

9 Conclusion

The coordination of ChatGPT into Art and design education marks a transformative era in teaching, learning, and institutional practices. It enables teachers to offer custom—support and guidance, inspiring creativity and enriching students' learning experiences. Nonetheless, this inventive methodology requires a fragile harmony between innovation and human ingenuity, with focal moral contemplations. Furthermore, educators have explored the multifaceted impacts of ChatGPT in art and design education. It customizes the learning venture, cultivates investigation of innovativeness, and guarantees accessibility and inclusivity for all students. Contextual analyses from institutions like the XYZ School of Design and Art Tech University show that ChatGPT can revolutionize creative courses by sparking innovation and pushing the boundaries of traditional methodologies. Coordinating ChatGPT into the pedagogical paradigm through blended learning and collaborative approaches further enhances the educational landscape. These techniques empower understudies to dive into authentic settings, participate in meaningful discussions, and receive immediate feedback, ultimately shaping them into well-rounded artists and designers. Organizations additionally benefit from ChatGPT's execution. They can give consistent learning resources, empower cross-disciplinary commitment, and offer ongoing alum support, creating a holistic and dynamic educational environment. However, ethical considerations remain paramount. Educators must emphasize authenticity and originality, guaranteeing that students comprehend the significance of adhering to academic conventions. They should advance decisive reasoning over-reliance on AI, nurturing independent exploration, critical analysis, and problem-solving abilities. Data security and privacy must be prioritized to protect students' intellectual property and creative ideas.

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ChatGPT-Artificial Intelligence Studies of Business Analytics Adoption and Usage

Ala'a Gharaibeh and Normalini Md Kassim

1 Introduction

Researchers certain that there are numerous main trends that lead and impact institutions and societies worldwide: Artificial Intelligence (AI), Blockchain, Cloud service, Big data and 5G. Scholars (e.g., [1, 2]) argued that Business Analytics supports reporting, interactive analytics, visualizations, and statistical data mining, and applications address various activities of organizations and production financial and many other sources of business data for purposes including management. Also, AI can allow computers to make business decisions on their own, in contrast to BI, which greatly simplifies data analysis but leaves decision-making to people. For instance, chatbots can respond to client questions without human assistance. Business Analytics systems provide historical, current, and predictive views of business processes, use data collected in a data warehouse or data file, and occasionally operate from operational data. Further, Business Analytics' main advantage is its ability to provide in-depth information when required. This information is needed for all types of decisions, including those involving future vision and strategic planning, and it primarily identifies business issues and trends that need to be corrected or improved. Therefore, in order to improve and support decisions and boost operational efficiency, organizations must harness the potential of their big data through mining, analyzing, and exploiting it. Additionally, the organization's rapid efficacy assures quicker and better-informed decision-making; it has turned into a prerequisite for competitive success. In order to provide the appropriate information to the right people at the right time, organizations must act wisely and reap the rewards of Business Analytics procedures.

A. Gharaibeh · N. Md Kassim (✉)
School of Management, Universiti Sains Malaysia, Penang, Malaysia
e-mail: normalini@usm.my

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Also, the ability to predict changes in product demand or identify an increase in the market share of a competitor's new product and quickly respond by introducing a competing product is one of the key functions of business analytics that helps the organization improve its ability to adapt to change and improve its performance [3]. Moreover, it is possible to define BA capabilities from both an organizational and a technological standpoint. From a technological standpoint, BA capabilities are described as scalable technology platforms and databases that ideally feature clear technology architecture and data standards. In terms of the organizational perspective, factors like risks, shared duties, and flexibility are advantages for the successful deployment of the information system in the organization.

Business Analytics, which improves decision-making by transforming data into insights, mainly relies on statistical, quantitative, and operational analysis [4]. The delivery of flawless outcomes is made possible through data management, data visualization, predictive modeling, data mining, prediction simulation, and optimization. Business Analytics entails using data mining to find novel patterns and relationships, statistical and quantitative analysis to create business models, multivariate testing based on the findings, and predictive modeling to forecast future business requirements, performance, and industry trends and reporting outcomes to management and clients in clear reports.

2 ChatGPT Applications

According to experts, "GPT" in ChatGPT refers for "Generative Pretrained Transformer," the most recent advancement in Artificial Intelligence (AI) in generative Machine Learning (ML), which is more potent than its earlier produced versions, as it could generate and advance answers to nearly any question. Further, it has a wide range of beneficial uses, including in business, healthcare, and education. Besides, Chatbots in libraries for educational purposes [5]. Burger et al. [6] suggested that even though there has been rapid advancement in the field of artificial intelligence technology recently, much untapped potential still exists in the field of study.

The benefits of implementing a brand-new generative artificial intelligence system in education were examined by Elbanna and Armstrong in 2023. The researchers looked into how ChatGPT is used in personalized learning, assessment, and content production, as well as how to deal with its drawbacks and some ethical issues. According to the study, ChatGPT may be successfully incorporated into education to automate repetitive work and improve student learning, thereby boosting productivity and efficiency and encouraging adaptive learning. Even with updates, it is important to keep in mind ChatGPT's shortcomings, such as factual discrepancies, potential bias promotion, a lack of in-depth comprehension, and safety issues. However, the study emphasizes the advantages of properly integrating ChatGPT into the sphere of education. According to Cingillioglu [7], teachers and educational designers can accurately identify AI-generated student essays using the proposed language model and machine learning (ML) classifier. Even if the overall classification accuracy performance is slightly worse, human (student)-generated writings can and must be accurately identified with 100% accuracy.

In-depth analysis of ChatGPT's cutting-edge technology was offered by Lund and Wang [8], a very smart chatbot that has attracted a lot of interest lately. The advantages of ChatGPT, such as enhanced search and discovery, reference and information services, cataloging and metadata generation, and content creation, were examined by the researchers, along with the ethical issues that needed to be taken into account, such as bias and privacy. The report has demonstrated that ChatGPT has the potential to significantly promote academic research and librarianship in both unsettling and novel ways. But rather than abusing it or allowing it to abuse us in the rush to advance scholarly understanding and train the next generation of professionals, it is critical to think about how to use this technology responsibly and ethically and to identify ways that we, as professionals, can work alongside it to improve our work.

According to Qasem [9], experts' concerns about the use of ChatGPTs in scientific and academic research prompted extensive research into whether AI language models would be helpful to researchers and academic staff or, conversely, whether they would be misused to harm scientific research and educational endeavors. He reaffirmed that ChatGPTs would be extremely useful and have a lot of potential if used wisely and ethically at the scientific and academic levels, such as finding insights and key ideas from earlier scientific and academic fields, reporting the literature review, and reducing the amount of time and effort required for reporting and information gathering. He did, however, advocate for tighter ethical regulation of ChatGPT usage. However, Sect. 2, provides the literature review matrix of Business Analytics adoption and usage.

3 Literature Review

See Table 1.

4 Conclusion

The researchers reviewed the concept of Business Analytics and its role on institutions through conducting a comprehensive review of theories, literature, and empirical studies. Besides, the researchers reviewed the prevailing theories of Business Analytics and its relationship with technology adoption. Indeed, Business Analytics uses a variety of analytical techniques to improve organizational performance while lowering operating costs and, ideally, predicting market trends. Business analytics turns unprocessed data into more useful inputs so that decision-makers can use it; learn more about the primary and secondary data resulting from the organization's operations; assistance in making decisions; aid businesses in streamlining their processes to boost efficiency; businesses utilize data visualization to forecast

Table 1 Literature review matrix

Author	Title	Country	Year	Theories	Independent variables	Other variables	Dependent variables	Methodology	Findings	Recommendations
Pancić, Ćućić and Serdarušić	Business Intelligence (BI) in firm performance: role of big data analytics and blockchain technology	Croatia	2023	–	Business intelligence	Big data analytics, blockchain	Firm performance	Quantitative Data collected from 387 employees from 12 Information Technology (IT) firms	The findings showed that BI has a direct and significant impact on firm performance. BI significantly and positively influenced the adoption of big data analytics and blockchain and, in turn, firm performance. The adoption of big data analytics and blockchain technology signified and positively mediated partially the relationship between BI and firm performance	Future research should expand upon the post-hoc assessment that was revealed in this research. More research should use different access points to enhance the features of the sample in Croatia. Additional research in the future should investigate more data sources
Bany Mohammad, Al-Okaily, Al-Majali and Masa'deh	Business Intelligence and Analytics (BIA) usage in the banking industry sector: an application of the TOE framework	Jordan	2022	TOE	Technological, organizational, and environmental factors	–	BIA usage extent	Quantitative Data collected from 120 Jordan Arab Bank employees	The findings demonstrated the important value of management and human resource support and capabilities, in addition to the presence of data and technology infrastructure. The results showed how crucially vital management and human resource competencies, along with the presence of data and technology infrastructure, are. This study contends that in order to fully profit from business intelligence and analytics, particularly in the banking industry, successful planning needs go beyond technical considerations. However, they contend that additional study is necessary to properly comprehend how banking sectors may apply and make use of business intelligence and analytics, particularly in the context of developing countries	However, the researchers suggest that more variables with wider population and more extensive studies are required to properly explain the crucial success factor of BIA usage and utilization. In other words, future research can test the suggested model in various contexts, such as the Jordanian public-shareholding enterprises as they make significant investments in BIA systems, and create more generalizations. Future studies may employ the model on developed countries to compare the outcomes with those of developing countries. The authors also looked at how BIA affected business performance and innovation capacity, although further studies are required to fully analyze the influence of BIA adoption and usage on organizational effectiveness

(continued)

Table 1 (continued)

Author	Title	Country	Year	Theories	Independent variables	Other variables	Dependent variables	Methodology	Findings	Recommendations
Potančok, Pour and Ip	Factors influencing business analytics solutions and views on business problems	–	2021	–	Business environment factors (company, company culture, human resources, legislation, business partners), company environment factors (data sources, applications, IT infrastructure, current effectiveness of internal analytics, applied approach and methods of analytics management), and market environment factors (sourcing, products, cloud computing, services, availability, quality and price range)	–	Business analytics use	Qualitative	This study offers a list of 15 business, company, and market environment elements, together with information on their significance for business analytics. If these elements are well understood, business analytics solutions may become more successful and effective	The focus of the upcoming research will be on a deep inspection of variables according to different company branches
Bawack and Ahmad	Understanding business analytics continuance in agile information system development projects: an expectation confirmation perspective	Ireland and Sweden	2020	ECM	Confirmation, perceived usefulness, technological compatibility	Satisfaction BA use	BA continuance intention	Quantitative data was collected from 153 respondents working in agile ISD projects	The researcher found that in agile ISD projects, perceived usefulness and technology compatibility are the most important variables influencing BA continuing intention. The suggested model accounts for 36.7% of the variance in perceived usefulness, 48.4% of the variance in perceived usefulness, 50.6% of the variance in satisfaction, and 31.9% of the variance in technological compatibility	This research is expected to inspire additional research on BA continuance in agile ISD contexts, particularly in identifying other logical, objective, and testable BA continuance variables

(continued)

Table 1 (continued)

Author	Title	Country	Year	Theories	Independent variables	Other variables	Dependent variables	Methodology	Findings	Recommendations
Idoko and Akinsunmi	Business analytic systems and organizational decisions making	Nigeria	2021	–	Analytical tools, analytical methods, interpretation tools	–	Organizational decision making	Quantitative The data was collected from eight (8) large companies selected from Nigeria Stock Exchange (NSE)	The study discovered a weak positive link between analytical tools and analytical methods, but a substantial positive correlation between analytical methods and interpretation tools. Multiple regression analysis showed that analytical tools, analytical methods, and interpretation tools were impacted organizational decision-making. Based on the study's findings, business analytic system significantly influences organizational decision-making	–
Al-Qaralleh and Atan	Impact of knowledge-based HRM, business analytics and agility on innovative performance: linear and FsQCA findings from the hotel industry	Jordan	2021	Knowledge-based theory	Knowledge-based HRM practices, organizational/ hotel agility, Business Analytics (BA) capabilities	–	Innovative performance	Quantitative 182 surveys	According to the outcomes of linear modeling, organizational agility, business analytics, and knowledge-based HRM practices are significant predicators of innovative performance. Organizational agility, on the other hand, was shown to be both a required and sufficient condition for achieving high innovative performance, according to the results of the fsQCA. Business analytics, nevertheless, is a suitable prerequisite for achieving high levels of inventive performance	Future research is encouraged to find additional possible capabilities, such as dynamic capability and firm market sensing capability, that could encourage increased inventive performance
Yiu, Yeung and Jong	Business intelligence systems and operational capability: an empirical analysis of high-tech sectors	US	2020	–	Business intelligence systems	Firm R&D intensity, firm size	Operational capability	Quantitative Sample of 282 firms only 144 are validated	They found that BI system application increases operational capability, especially for big high-tech companies with plenty of technology. They also demonstrate that firm size and technology intensity are crucial contextual elements for businesses to benefit from BI systems	The advantages of BI systems for analyzing structured versus unstructured data might be compared for future studies. We can learn more about a knowledge-based advantage with BI systems by investigating this further in future study

(continued)

Table 1 (continued)

Author	Title	Country	Year	Theories	Independent variables	Other variables	Dependent variables	Methodology	Findings	Recommendations
Ashrafi, Ravasan, Trkman and Afshari	The role of business analytics capabilities in bolstering firms' agility and performance	Iran	2019	–	BA capability, performance, information quality, innovation capability, environmental turbulence	Market and technological turbulence	Performance	Quantitative Data were collected from 154 firms with two respondents (CEO and CIO) from each firm	The findings show that BA capabilities have a significant impact on a firm's agility by improving the quality of its information and its ability to innovate. They also address the fact that the performance of organizations is moderated by both market and technological turbulence	Future research can explore a number of issues raised by this study. First, future research might duplicate our techniques in different settings (such as developed nations) and contrast the findings with this study, or it could employ a longitudinal study to address the drawbacks of this study's cross-sectional design. To provide a fuller understanding, detailed case studies would also be helpful. The performance of businesses may be improved in other ways as well, according to future studies. Further study should recognize the applications that are descriptive, predictive, and prescriptive in today's BAs, and individually examine the effects of specific data mining and data purification procedures are two examples of analytics methodologies, techniques, and models on numerous performance aspects
Cabrera-Sánchez and Villarejo-Ramos	Factors affecting the adoption of big data analytics in companies	–	2019	UTAUT	Performance expectancy, effort expectancy, social influence, facilitating conditions	Behavioural intention	Usage behaviour	Quantitative	The findings demonstrate that the value of sound infrastructure outweighs the challenges businesses experience in putting it in place. Companies considering using Big Data anticipate significant results, but current users are increasingly skeptical about its efficacy	Future Big Data studies should look at these factors. Additionally, it appears necessary to investigate new moderator variables in order to analyse potential impacts that have not yet been considered

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Author	Title	Country	Year	Theories	Independent variables	Other variables	Dependent variables	Methodology	Findings	Recommendations
Gaardboe and Svarre	Business intelligence success factors: a literature review	–	2018	–	Identified critical success factors regarding the four constructs: tasks, people, structure, and technology constructs	–	–	Qualitative	34 CSFs correlated to the success of BI were found. They add four new components to the model for successful information systems: organizational culture, competency development, vision and strategy, and organizational structure. By expanding the framework of information system success and finding gaps in the existing literature, they also add to the body of existing research. Furthermore, by improving our understanding of the CSFs correlated to BI success, they help with practical implementation. Technology was the construct that was most frequently used, followed by the constructions of structure, people, and task	Numerous aspects of the task category in particular have not received enough attention. To make any definite judgments on the task significance factor as a distinct CSF, there is not enough study on the topic. Additional studies are required to obtain a complete understanding of these CSFs. Additional research is needed to study the task and people categories and their relationship to BI success
Malladi and Krishnan	Determinants of usage variation of business intelligence and analytics in organizations—an empirical analysis	North America	2013	TOE	Technology factors (data infrastructure sophistication data management challenge), organizational factor (organization size, managerial challenges. Environmental factor (industry competitive intensity, environment dynamism)	Control	Extent of BIA usage	Quantitative	They found that while sophisticated data-related infrastructure in businesses promotes utilization, difficulties with data management limit its scope. Additionally, they found that big organizations are more likely to employ BIA in business operations, but managerial difficulties with integration and talent management limit its utilization. Finally, they found that the degree of industry competition affects use. This study emphasizes the utilization antecedents and can assist researchers and practitioners in comprehending what elements can allow businesses to adopt BIA broadly	Longitudinal datasets may be used in future study to get further understanding of temporal ordering. Future research may look into adoption choices and assimilation practices like routinization, whereas they looked at the factors affecting widespread use of BIA. Future studies may systematically determine and examine the mediation processes that affect business performance after BIA implementation and use

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Author	Title	Country	Year	Theories	Independent variables	Other variables	Dependent variables	Methodology	Findings	Recommendations
Lutfi, Alsyouf, Almaiah, Alrawad, Abdo, Al-Khasawneh, Ibrahim and Saad	Factors influencing the adoption of big data analytics in the digital transformation era: case study of Jordanian SMEs	Jordan	2022	TOE DOI	Technological factors, organizational factors, environmental factors	–	BD adoption	Quantitative	The empirical findings showed that factors such as relative advantage, complexity, security, top management support, organizational readiness, and government assistance have an impact on BD adoption, however factors such as competition pressure and compatibility seemed to have little of an impact. For both researcher and practitioner circles concerned with the use of big data in emerging nations, the findings are anticipated to contribute to enterprise management and strategic use of data analytics in the current dynamic market context	Future research may build on this study's framework and incorporate additional potential factors like the providers' support, organizational culture, and peer influence. It may even be added to and verified that control variables like firm size, firm age, sector, etc. do not significantly affect the resulting scores of BD use
Min, Joo and Choi	Success factors affecting the intention to use business analytics: an empirical study	Korea	2021	Resource based view (RBV) and organizational readiness for change	Emphasis on privacy, perceived lack of security, perceived risk, BA-related IT capability	Perceived value	Intention to use BA	Quantitative	They found that heightened security and risk worries, coupled with a lack of IT expertise, became significant barriers to the implementation of BA. Additionally, they found that businesses that valued BA were more likely to accept BA than the others. Based on an actual analysis of BA practices across Korean enterprises, this study is one of the first attempts to create useful guidance for potential BA adopters	Future studies will need to expand the scope of this study by examining the BA adoption decision through cross-national and/or longitudinal studies, and they should also look into the potential effects of other contextual factors on the firm's decision to adopt a BA, such as organizational culture, subjective norms, governmental intervention, and the level of top management support. Utilizing data mining techniques to create extensive information of BA adopters and non-adopters is an interesting research topic

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Author	Title	Country	Year	Theories	Independent variables	Other variables	Dependent variables	Methodology	Findings	Recommendations
Al-Dmour, H. Saad, Amin, Al-Dmour, R. and Al-Dmour, A.	The influence of the practices of big data analytics applications on bank performance: filed study	Jordan	2020	TOE	Organizational factors (top management support, business strategy orientation and organizational resources) technological factors (compatibility, complexity and security and privacy), environmental factors (business market structure, competition structure, government regulations	Big data analytics application	Bank performance	Quantitative approach was used, and the data was collected from 235 commercial banks' senior and middle managers (IT, financial and marketers) using both online and paper-based questionnaires	The findings demonstrated that commercial banks operating in Jordan use big data analytics applications to a moderate extent (i.e., 60%). The findings showed that the TOE model could describe 61% of the variation in big data analytics applications used by commercial banks. The most significant predictors were discovered to be organizational variables. The outcomes also offer empirical proof that the number of big data analytics applications used has a favorable impact on bank performance	Similar studies that address different environments and circumstances and look into different variables than those looked into in this research will be extremely beneficial
Maroufkhani, Wan Ismail and Ghobakhloo	Big data analytics adoption model for small and medium enterprises	Iran	2020	TOE	Technological context, organizational context, environmental context	Adoption of BDA in SMEs	Performance	Quantitative A survey of 161 manufacturing SMEs in Iran was conducted. However, 112 were suitable for statistical analysis	The findings provide proof that the technological, organizational, and environmental settings mediate the relationship between the performance of SMEs. The research also indicated that in the context of SMEs, organizational and technological factors are the more important determinants of BDA adoption. Additionally, this study's findings demonstrated that BDA use could improve the SME market and financial performance	Future research may collect information from two informants, one from the top managers and one from the staff, for each SME. Future scholars might take a similar course of action in many businesses and sectors and other developing nations. The TOE model is adaptable, so future researchers may take into account various variables of organizational, technological, and environmental aspects. Focus groups, observation, or qualitative methodologies can help future researchers produce more accurate result

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Author	Title	Country	Year	Theories	Independent variables	Other variables	Dependent variables	Methodology	Findings	Recommendations
Bolonne and Wijewardene	Critical factors affecting the intention to adopt big data analytics in apparel sector, Sri Lanka	Sri Lanka	2020	TOE TAM	Perceived usefulness, perceived ease of use, technological factors, organizational factors, environmental factors	Attitude towards using BDA	Intention to adopt BDA	Quantitative	The study's conclusions show that, the organizational context, factors taken into account by both the TOE framework and the TAM model have a positive correlation with users' attitudes toward use, which would ultimately encourage an organization to increase its intention to adopt big data analytics They found that the variable, attitude toward using, mediates between the direct link of crucial elements influencing the intention to use big data analytics in a way that is favorable	Future researchers will have the chance to test the same study approach in other firms while taking into account various cultural settings. By locating this model in developed countries, researchers have the chance to broaden the scope of their research by identifying the motivations of BDA system developers and builders
Tsoy and Staples	Exploring critical success factors in agile analytics projects	–	2020	–	Organizational factors: – Management commitment – Organizational environment – Team environment People factors: – Team capability – Customer involvement Process factor: – Project management process project – Definition process Technical factors: – Agile software techniques – Delivery strategy Project factors: – Project nature – Project type – Project schedule		Perceived success of the agile software development project factors (quality, scope, time, and cost)	Qualitative	The original list by Chow and Cao was expanded by ten more qualities. Risk appetite, team diversity and availability, engagement, project planning, shared goals, and method uncertainty are seven new characteristics addressed in the general literature on agile projects. Data quality, model validation, and gaining clients' confidence in the model solution were three new criteria included specifically for analytics projects. It was discovered that the more successful project outperformed the less successful project in practically every factor. The findings can aid in understanding environmental factors and projecting actions that can aid in obtaining commercial value from analytics projects for researchers and practitioners	Future research on this with a larger sample would be beneficial and could assist to direct practice. They propose that key building blocks for project success within an organization are management commitment and organizational and team environments Future study will hopefully be conducted to further understand the success elements for analytics projects and investigate the interdependencies. Future research will be needed to validate and test correlations among the components and dimensions of project success

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Author	Title	Country	Year	Theories	Independent variables	Other variables	Dependent variables	Methodology	Findings	Recommendations
Whitelock	Business analytics and firm performance: role of structured financial statement data	–	2018	–	Business analytics: descriptive analytics, diagnostic analytics, discovery analytics, predictive analytics, prescriptive analytics	–	Operational performance, financial performance	Qualitative	A comprehensive theoretical framework was proposed in this conceptual article to investigate the connection between performance and business analytics. It offered a simplified flow chart illustrates the steps in the business analytics process for collecting data, managing it, and using it to drive decisions. Additionally, it created a common business analytics approach to resolve real-world issues using pre-existing organized operational and financial statement data from businesses. Additionally, the framework is a reasonably priced option that can be used by small, mid-sized, and large businesses. As a result, it gives businesses that are "overwhelmed by data" and "struggling to use data to achieve improved business results" a practical choice for advancing business analytics capability within their organizations. Furthermore, using the five main business metrics, our research presented a real-world example illustration	The researcher recommended that additional study, using other types of data sets, is required to experimentally test the assumptions connected to the proposed business analytics/performance framework. More empirical research that clarifies the effectiveness and advantages of organizations adopting their own techniques to drive operational and financial performances through use of various business analytics types in their organizations would also be helpful to researchers and practitioners

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Author	Title	Country	Year	Theories	Independent variables	Other variables	Dependent variables	Methodology	Findings	Recommendations
Krishnamoorthi and Mathew	Business analytics and business value: a comparative case study	India	2018	–	Analytics technology assets, business analytics capability, analytics value enhancers and organizational-level variables, analytics technology assets and business analytics capability	–	Business performance	Qualitative	Given the increasing use of business analytics, it is crucial for investing organizations to comprehend how investments result in business value. Studies in the IT field have shown that increasing technological expenditure may not increase profits, but rather that IT organizational competency serves as a major mediator in value generation. In order to understand the process of business value generation utilizing several case studies, this research extends the model to business analytics, identifies elements of analytics technology assets, and business analytics capacity. By creating operational and organizational performance indicators, they are able to quantify the contribution that analytics resources make to business performance	The suggested model's applicability across other industry verticals will be improved by further case studies from other industry verticals that will help to cover more elements of the dimensions created in the model. While the current research has focused primarily on the analytics resource's elements, the model might be further extended to incorporate the\measurements aspects of the BV supplied by analytics at an organizations level. While this case study was being undertaken at the organizations, it was noted that this could be a possible research project For the benefit of the body of knowledge relevant for practice, it is necessary to extend the current work on analytics capability and conduct more studies on assessment of analytics capability leading to the establishment of capability maturity models in analytics

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Author	Title	Country	Year	Theories	Independent variables	Other variables	Dependent variables	Methodology	Findings	Recommendations
Lai, Sun and Ren	Understanding the determinants of Big Data Analytics (BDA) adoption in logistics and supply chain management	China	2018	DOI TOE	Technology context, organization context, environment context	Supply chain, environmental moderatos	Intention to adopt BDA	Quantitative Survey data was collected from 210 organizations	The empirical findings showed that top management support and perceived benefits have a big impact on adoption intention. Environmental factors can greatly modify the direct links between driving factors and the intention to adopt, including competitor adoption, government policy, and SC connectivity	Future study will attempt to take other context-related variables into account. For instance, the complexity of delivery and SC scale might influence the decision-making process
Qushem, Zeki and Abubakar	Successful business intelligence system for SME: an analytical study in Malaysia	Malaysia	2017	–	Management factors, technology factors, organizational factors, environmental factors and facilitating condition factors	–	Business intelligence efficiency	Quantitative	The findings showed that SME companies should concentrate on the environmental factor (EF), since it is the most recognized component that leads to BI efficiency in business organizations	They recommended that studying demographic aspects may help people better understand business intelligence (BI) in a variety of scenarios, leading to BI investments that are more success

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Author	Title	Country	Year	Theories	Independent variables	Other variables	Dependent variables	Methodology	Findings	Recommendations
Georges Kfouri, Rimvydas Skyrius	Factors influencing the implementation of business intelligence among small and medium enterprises in Lebanon	Lebanon	2016	–	The organisation dimension, the process dimension, the technology dimension	–	Business Intelligence (BI) systems	Quantitative and qualitative	The information on BI implementation factors was categorized using content analysis of the survey data using the three major perspectives: organizational, process, and technological. According to the research, the organizational and process dimensions are related to the most crucial elements along the three dimensions that were selected for BI implementation in SMEs. The most crucial challenges for the organizational dimension are BI awareness, which includes knowledge of the existence and use of BI-specific methodologies and technologies as well as knowledge of the possible advantages and competitive advantage that are dependent on BI use	The framework should be extended or validated by additional study on this field
Sam and Chatwin	Understanding adoption of big data analytics in China: from organizational users perspective	China	2016	TOE	Technological, organizational, environmental readiness	–	Big data readiness	Quantitative 178 valid responses from 180	This study examines big data analytics adoption intentions in China using the TOE framework. According to the hypothesis, organizational users must prepare their technological, organizational, and environmental assets before implementing big data analytics. The empirical findings imply that each of the three Big Data Readiness components is essential for achieving psychological outcomes. The findings offer researchers and practitioners some insight into the relative impact of each type of influence in the adoption of big data	Few studies have examined the organizational adoption of big data analytics further they need to invest more

future outcomes; and enhance effectiveness and increase revenue. Upcoming research will be arranged into five sections. Section 1 will introduce the reader to the research background, the research problem statement, research objectives, research questions, research significance, scope of the study, and definitions of the terms. Section 2 will review the literature relating to the study variables, the underlying theories, research model, and hypotheses. Section 3 will cover the research methodology employed for this study, which will include the research design, population and sampling procedures, methods of data collection, research instruments, and the statistical analyses that could be used for the research. Section 4 will present the statistical results of the research findings.

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ChatGPT in Specialized Educational Contexts

Revolutionizing Medical Education: Empowering Learning with ChatGPT

Ayham Salloum, Raghad Alfaisal, and Said A. Salloum

1 Introduction

Medical education, an ever-evolving field, has historically relied on a mixture of traditional classroom instruction, hands-on clinical experiences, and self-directed learning. Over the years, various instructional methods, such as problem-based learning, have emerged to address the diverse needs of learners and the changing demands of healthcare systems worldwide [1]. Despite these innovations, many educators and learners alike have voiced concerns about the rapidly increasing volume of medical knowledge, the challenge of maintaining skills in the face of evolving clinical guidelines, and the need for more personalized, adaptive learning experiences [2].

The advent of the digital age has ushered in a plethora of technological tools aimed at enhancing the educational experience [3–7]. Digital tools, including e-learning platforms [5, 8, 9], virtual simulations, and mobile apps, have gained traction, proving valuable in offering flexibility, personalizing learning experiences, and even simulating complex clinical scenarios [10–13]. Their rise is closely tied to the global digitization trend in higher education, wherein institutions are leveraging technology to optimize learning outcomes, increase accessibility, and provide learners with a multifaceted educational experience [14].

A. Salloum
College of Medicine, University of Sharjah, Sharjah, UAE

R. Alfaisal
Faculty of Computing and Meta-Technology, Universiti Pendidikan Sultan Idris, Perak, Malaysia

S. A. Salloum (✉)
Health Economic and Financing Group, University of Sharjah, Sharjah, UAE
e-mail: ssalloum@sharjah.ac.ae

School of Science, Engineering, and Environment, University of Salford, Salford, UK

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Amidst the digital transformation in medical education, a notable entrant has been ChatGPT, a state-of-the-art language model by OpenAI. This model, grounded in conversational AI, facilitates real-time interactions, granting users the ability to present queries and obtain detailed, context-sensitive responses. Despite the vast potential of such advanced models in revolutionizing medical pedagogy—by fostering interactive learning, enabling problem-based scenarios, or simulating patient interactions—there exists a discernible gap in the literature. Much of the current research tends to gravitate towards e-learning tools and rudimentary chatbot applications, often sidelining the capabilities of sophisticated models like ChatGPT. The nuances of such models in bolstering critical thinking, enhancing interactive learning, or providing individualized feedback have not been sufficiently explored. Furthermore, insights into the pragmatic aspects, the merits, and the possible pitfalls of assimilating such technology in varied learning milieus are sparse.

This paper seeks to fill this research void by presenting a comprehensive analysis of ChatGPT's prospective roles in medical education. By juxtaposing its benefits with its challenges, and drawing from empirical pilot studies, educator and learner feedback, and an exhaustive scrutiny of AI-infused pedagogies, we aim to shed light on the transformative capability of contemporary language models in redefining the contours of medical education.

2 Background

2.1 The Current Landscape of Medical Education

The trajectory of medical education, like the art and science it seeks to impart, has witnessed significant evolution over the decades. From foundational theories to contemporary practices, the sector continually grapples with a need to be both consistent and adaptive. As we dive into its landscape, we notice the interplay of traditional pedagogies and modern demands, shaping the learning experiences of medical students.

Historically, medical education has revolved around didactic lectures, hands-on clinical rotations, and cadaveric dissections. These methods, while fundamental, are grounded in rote memorization and passive absorption of information. Lectures, being uni-directional, often restrict interactive engagement, potentially leading to reduced retention and comprehension [15]. Furthermore, while clinical rotations offer real-world exposure, their learning value can be inconsistent, hinging on the cases that emerge during a student's rotation. Cadaveric dissections, though invaluable, have been criticized for not always accurately representing living anatomy due to preservation methods or disease processes [16].

2.2 The Growing Need for Diverse Learning Platforms in the Digital Age

With the digital revolution permeating every sector, education is no exception [17–21]. Today's learners, often termed 'digital natives,' are accustomed to leveraging technology in almost every aspect of their lives, including learning [22–30]. The vast amount of medical knowledge [31, 32], coupled with rapid advances in the field, necessitates tools that not only keep pace but also cater to diverse learning styles [33]. E-learning platforms, virtual reality simulations, and interactive apps promise flexibility, real-time updates, and personalized learning experiences [34]. Such platforms facilitate a blend of self-paced learning with structured curriculum, allowing for adaptability and breadth.

2.3 Challenges Faced by Medical Students and Educators

While opportunities abound, challenges persist. Medical students often feel overwhelmed by the sheer volume of information, with burnout and mental health concerns becoming increasingly prevalent [35, 36]. Finding a balance between acquiring theoretical knowledge and practical skills, all while maintaining well-being, is a daunting task. For educators, the challenge lies in updating curricula to mirror current best practices, ensuring learning materials remain relevant, and tailoring teaching techniques to accommodate diverse cohorts of students. Moreover, the integration of digital tools requires investment, training, and regular updating—aspects that can be resource-intensive [37].

3 ChatGPT

In an era marked by rapid technological advancements, ChatGPT emerges as a paragon of the seamless convergence of machine learning, linguistics, and user interaction. It epitomizes the capability of modern artificial intelligence to understand, generate, and engage in human-like textual conversation.

3.1 Description of ChatGPT and Its Capabilities

Developed by OpenAI, ChatGPT is based on the Generative Pre-trained Transformer (GPT) architecture. At its core, it's a language model trained on vast amounts of text, enabling it to generate coherent, contextually relevant, and often insightful responses to user queries. What sets ChatGPT apart from conventional chatbots is its proficiency in understanding nuanced queries, offering detailed explanations, and adapting its responses based on the context provided. It's not merely a tool that searches a database for pre-defined answers; instead, it dynamically crafts responses based on its training, making interactions more fluid and organic. Its capabilities extend beyond simple question–answer routines. ChatGPT can assist in tasks like content creation, code writing, tutoring in various subjects, and even simulating characters for video games. The model's strength lies in its adaptability, its vast knowledge base, and its capacity to engage in prolonged interactions, making it a versatile tool for diverse applications.

3.2 Historical Context: How Chatbots and AI Have Evolved in Education

The journey of chatbots in education traces back to the 1960s with the development of ELIZA, a rudimentary "psychotherapist" chatbot designed at the MIT Artificial Intelligence Laboratory [38]. Fast forward to the twenty-first century, and the landscape has shifted dramatically. With the advent of the internet and digital communication platforms, chatbots found roles in customer service, entertainment, and notably, education. Early educational chatbots were primarily focused on language learning, offering learners an interactive platform to practice conversations. As technology matured, so did the applications. Chatbots began facilitating administrative tasks like enrollment queries, schedule planning, and even assisting in coursework through reminders and basic tutoring [39] (see Fig. 1).

The incorporation of AI transformed these bots from mere scripted interfaces to adaptive learning platforms. AI-driven chatbots, armed with natural language processing and machine learning, could understand student needs better, offer personalized feedback, and even identify areas where a learner might be struggling, guiding them accordingly [40]. ChatGPT represents the pinnacle of this evolution. Its robust architecture and vast training data make it a tool not just for administrative or basic tutoring tasks but for in-depth, interactive, and personalized learning experiences. In the realm of education, it holds promise to revolutionize student engagement, adaptive learning, and cater to diverse educational needs.

4 Empowering Learning: Benefits of ChatGPT in Medical Education

In the continuously evolving landscape of medical education, ChatGPT stands out as a beacon of potential. It promises to address long-standing educational challenges while introducing novel methods to facilitate learning. Here, we explore the manifold benefits of integrating ChatGPT into medical pedagogy (see Fig. 2).

Fig. 1 Chatbots

4.1 Personalized Learning

4.1.1 Adaptive Feedback and Personalized Responses

One of ChatGPT's salient features is its ability to provide feedback tailored to individual needs. Instead of generic responses, students can receive guidance and explanations suited to their specific queries, thereby ensuring clarity and promoting deeper understanding [41].

Fig. 2 ChatGPT in medical education

4.1.2 Allowing Students to Learn at Their Own Pace

Every learner is unique, and so is their pace of assimilation. ChatGPT's responsive nature ensures that students can engage with content at their convenience, revisiting concepts as needed, thereby fostering an environment conducive to self-paced, autonomous learning [42].

4.2 Access to a Wealth of Information

4.2.1 Rapid and Updated Responses

Owing to its vast training data and regular updates, ChatGPT can provide information that's both extensive and current. In the dynamic field of medicine where new discoveries and protocols emerge frequently, this ensures students remain abreast of the latest advancements.

4.2.2 The Possibility of Integrating with Medical Databases and Research

The potential to connect ChatGPT with specialized medical databases can revolutionize research-oriented learning, offering students direct access to seminal papers, recent studies, and pivotal clinical trials [43].

4.3 Interactive Case Studies and Simulations

4.3.1 Virtual Patient Interactions

ChatGPT can simulate patient scenarios, allowing students to practice history-taking, symptom evaluation, and even preliminary diagnosis in a risk-free environment, bridging the gap between theory and practice [44].

4.3.2 Role-Playing Scenarios for Patient Communication and Diagnosis

Empathy, communication, and bedside manners are as vital as clinical skills. Through role-playing scenarios, students can refine their patient interaction skills, receiving feedback on their approach and improving their communication techniques.

4.4 Availability and Convenience

4.4.1 24/7 Accessibility

Learning doesn't adhere to a strict schedule. With ChatGPT, students can seek answers, clarify doubts, or dive into detailed explanations anytime, ensuring that learning is both continuous and spontaneous.

4.4.2 Reducing the Barriers of Time and Location for Learning

Irrespective of geographical constraints or time zones, ChatGPT is accessible from anywhere, democratizing education and making it universally available [45].

4.5 Enhancing Engagement and Retention

4.5.1 Interactive Learning

Active engagement fosters better retention. ChatGPT's interactive nature ensures that students aren't just passive recipients of information but active participants in their learning journey.

4.5.2 Using AI to Identify Areas of Improvement and Adapt Content Accordingly

ChatGPT, through iterative interactions, can identify areas where a student might be struggling and adjust its responses, ensuring that content delivery is optimized for maximum comprehension and retention [46].

5 Challenges and Concerns

While ChatGPT and similar technologies present transformative opportunities for medical education, their integration is not devoid of challenges. Recognizing these potential pitfalls is essential for creating robust, effective, and safe learning environments.

5.1 Data Privacy and Security

Incorporating AI tools like ChatGPT into medical curricula might entail using real-world case studies or patient data for better learning outcomes. This raises concerns about data privacy, as unauthorized access or unintentional sharing can breach patient confidentiality. Ensuring encrypted transmissions, adhering to data protection regulations, and continuous monitoring become imperative [47].

5.2 Dependence on Technology

5.2.1 Over-Reliance on AI for Learning

While AI offers efficient and personalized learning experiences, an excessive dependence can lead to reduced critical thinking and problem-solving abilities. Striking a balance between AI-assisted and traditional learning methods ensures that students remain adept at independent reasoning and analysis [48].

5.2.2 Ensuring That Students Still Get Hands-On Experience and Human Interaction

Medical education is not solely about knowledge; it also involves practical skills, bedside manners, and human empathy. It's essential to ensure that students continue to get hands-on experience with real patients and retain the human touch that's central to healthcare [49].

5.3 Keeping Content Updated

Medicine is a rapidly evolving field. For AI tools to remain relevant, they must be frequently updated to incorporate new research findings, clinical guidelines, and therapeutic advances. Regular audits and content refreshers are necessary to ensure that students receive up-to-date information [50].

5.4 Potential Misinformation

AI models, including ChatGPT, rely on their training data. While they can provide vast amounts of information, the potential for errors or outdated information persists. Rigorous validation, oversight by medical professionals, and fostering a culture of cross-referencing among students can mitigate these risks [51].

6 Conclusion

The Medical education stands at the cusp of a transformative era, with digital tools and AI ushering in a wave of innovative pedagogies. At the forefront of this evolution is ChatGPT, a conversational AI with capabilities extending far beyond traditional e-learning tools. The potential of ChatGPT in revolutionizing medical education is evident. Its adaptability, vast knowledge base, and real-time responsiveness enable a degree of personalization previously unattainable. From simulating patient interactions to offering instant clarifications on complex medical queries, ChatGPT promises a richer, more interactive learning experience. Such advancements not only complement traditional teaching methods but can also bridge many existing educational gaps.

However, as with all innovations, a balanced approach is imperative. While the allure of AI-driven education is compelling, it is essential to juxtapose it against the time-tested methodologies of hands-on training, interpersonal interactions, and clinical experiences. The fusion of the old and new offers a holistic educational experience, cultivating medical professionals who are both knowledgeable and empathetic. Looking ahead, the trajectory of AI's influence on medical education appears promising. As technology continues to evolve, we can anticipate even more immersive simulations, advanced diagnostic training tools, and possibly, AI-driven virtual clinical rotations. The boundaries of what's achievable are continually expanding. Furthermore, the future beckons greater collaborations. Educational institutions, technology companies, and healthcare providers have a unique opportunity to synergize their expertise. Joint initiatives can lead to the development of sophisticated educational tools tailored for medical training, ensuring that the next generation of healthcare professionals is equipped to tackle future challenges. As we embrace this new era, our focus should remain unwavering: to harness the power of AI, like ChatGPT, in service of a singular goal—producing competent, caring, and well-informed medical professionals. The journey is just beginning, and the possibilities are boundless.

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Teaching the Skills of Expression According to Theory of Gerjanis's Systems and Generation Chomsky: From the Perspective of Arabic Language Engineering for Non-Arabic Speakers

Nibal Ahmed Al Muallem

1 Introduction

Learning Arabic as a second language is a strategic and vital requirement due to the importance of learning the same in meeting the needs of its learners and the goal which they seek to achieve through learning the Arabic language [1–4]. The goals of such learners varied between academic and scientific goals [5–9], religious, political, and commercial goals, and some of them learned the same for the purposes of tourism and travel to Arab countries [10–13]. The fact that oral and written expression is one of the most important language skills that learners must learn/master, as it is the basic means of communication between people, as all skills converge and complete in oral and written expression.

Considering that language is a social phenomenon and a human faculty, which distinguishes mankind from other living beings [14, 15]. The language is the means for mankind for expressing himself, expressing his opinion, and showing his emotion. The language is the container that contains thought, science and knowledge [16–19]. Therefore, language is considered the engine of the activity of individuals and groups, and it is an amazing expression of God's endless power. The core language is: the human voice and speech organs. The voice is a limited space, and the capabilities of the speech organs are also limited, as it produces a certain number of sounds, which we express in alphabetical letters [20–22]. However, these limited sounds, arising from the few organs of speech represented in the larynx, throat, tongue, lips and nose, are the ones that produced this huge and diverse linguistic presence that is expressed by the three thousand languages existing on the earth. God willed that

N. A. Al Muallem (✉)
City University Ajman, Sheikh Ammar Road, Al Tallah 2, Ajman, UAE
e-mail: a.nibal@cuca.ae

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mankind's guidance thereto be a starting point for all the civilizations that have been established on earth, and a difference between this mankind and other beings [23].

Through the studies reviewed by the researcher and her experience in the field of teaching Arabic to non-native speakers, it was found that most of the methods of teaching the subject of expression in both its oral and written parts, which are used in most schools, institutes and universities, contribute to poor ability of learners and scholars of Arabic as a second language, in oral and written expression skills. This is also the conclusion of a number of studies that proved: First—There are deficiencies in the expression skills of non-Arabic speaker learners the various educational stages [24–26]. Due to the fact that Arabic language teachers used to following the traditional methods based on translation, and the learner's memory, most of them suffer from many expressive and oral problems when using the Arabic language [27, 28]. Second—It was also proven that there are deficiencies in the thinking and expression skills of non-Arabic speaker learners in the various educational stages. This was confirmed by the results of the twelfth scientific conference of the Egyptian Association for Curricula and Teaching Methods (educational curricula and the Arab human building). Held in the period 25–26 July 2006, in the Guest House at Ain Shams University. Third—the lack of a clear strategy for teaching Arabic and its skills to non-Arabic speakers [29]. Fourth—the lack of specialized books and studies that explain the strengths and weaknesses of non-Arabic speaker learners [30]. Fifth—"The lack of appropriate educational curricula and courses, and modern educational programs, and the methods of teaching Arabic to non-native speakers are limited to translation, delivery and indoctrination [31].

These methods (translation, delivery and indoctrination) became abhorrent for learners because (learners) are looking for advanced and modern methods in their learning and academic achievement which is based on computers and modern technology. These technologies became the learners' tool used in all the requirements of their working and academic life. Analyzing this, and due to the fact that the advanced level curriculum is not harmonized; To fulfill the outcomes of learning, and to identify the weaknesses of non-Arabic speaker learners in oral and written expression skills, and to address the problems of their omission. The researcher designed and applied a constructive educational program based on theory of Gerjanis's Systems and generation Chomsky, from the perspective of Arabic language architecture. Therefore, the research objectives are: (1) To determine most of the linguistic problems and difficulties that learners of oral and written expression skills face, for non-Arabic speakers as a second language; (2) To demonstrate the effectiveness of modern teaching and learning methods that rely on multiple means in teaching and learning oral and written expression skills for non-Arabic speakers at the advanced level; (3) To demonstrate the effectiveness of using and applying the Arabic language architecture and theory of Gerjanis's Systems and generation Chomsky in addressing linguistic problems and difficulties.

Indeed, the importance of this research comes from the importance of the topic/subject matter itself, as it contributes to: (1) Teaching and learning oral and written expression skills for non-Arabic speakers, using the architecture linguistic construction method that relies on theory of Gerjanis's Systems and generation Chomsky, which contribute to increasing linguistic wealth. (2) Raising the learners' competence by benefiting from the Arabic language architecture and its automated programs, in developing oral and written expression skills. (3) Relying on various means in teaching and learning the skills of expression, which enable learners to choose the curriculum themselves, and to integrate the learner into the educational process; As for the teacher, he is the guide, the facilitator, and the guide for her only. Research limitations shall be limited to the following: (1) Temporal scope: Second semester 2021/2022 of the academic year. (2) Spatial scope: Faculty of Languages—International Islamic University—Malaysia. (3) Objective scope: The study is applied to a purposive sample of Arabic language learners at the advanced level—in the Faculty of Languages. (4) Human scope: Students of the Faculty of Languages Speaking Other than Arabic at the International Islamic University—Malaysia.

2 Literature Review

2.1 The Systems Theory of Gerjanis and the Generation Theory of Chomsky

It is considered one of the most important theories in Arabic rhetoric, where it indicates that the language is an integrated structure that has many parts, and radiates on the basis of the language system. Gerjanis says: the author of the saying thinks about the meaning that he depicts, arranges the same in himself, then chooses the appropriate systems for his performance, the goal of the theory is to reach the linguistic expression to a high level, and that the images of speech come equal to the meanings [32]. Procedural definition: Teaching learners how to arrange the parts of a language; including expressions, words, and links in the formation of sentences, and paragraphs that the learner compares in his mind to express his emotional states with appropriate meanings, and express the educational situation to which the learner is subjected, or lives inside or outside the classroom.

In relation to the generation of sentences or their production in an infinite abundance, Chomsky says: "It is the unlimited use of limited resources," and it starts from the human mind to the linguistic reality. Chomsky considers that the human mind is a means of knowledge, and knowledge is the basis of language, and he relied on important criteria, which are linguistic competence and performance, which is the ability to produce sentences and understand them in the speaking process, and on the surface structure that pertains to the phonetic form of the word, and the deep structure that provides the semantic meaning of the word, i.e. latent in the soul of the speaker in his mother tongue [33], and is considered the most important contribution to the field of theoretical linguistics in the twentieth century. Procedural definition: the learner to learn how to generate and evoke words through the educational situation that he is subjected to through education techniques and linguistic engineering, and how he thinks and arranges his thoughts in his mind, and imagines the meanings that explain and express the educational situation and correspond with it, and transmits them to others, either orally or in writing.

Through the author's review of previous studies that dealt with research on the four language skills: listening, speaking, reading, and writing, and the dependence of some of them on Gerjanis's systems theory [34], which presented a proposed program to develop the teaching of grammar as a university requirement, in the light of Gerjanis's systems theory, and his method proved fruitful. Good educational. Its outputs confirmed the feasibility of systems theory in teaching and learning. I wanted to follow Gerjanis's approach as an ancient linguistic theory, in teaching and learning expression skills to non-Arabic speakers. It used the generative theory approach of Chomsky as it is a modern linguistic theory. Where I employed the old and the new in the service of the language and its learning requirements benefiting also from Hamid Majeed Salma's study: which relied on generative learning, "Chomsky 's theory" and its application to fourth year middle school students in history. Also, the study of [34], who applied generative learning to raise learners' achievement in mathematics. Generative learning is considered one of the prominent and advanced constructive models. In the current era, when the teacher performs generative operations, to link new information with previous knowledge and experiences. The learner generates relationships between parts of information that are similar in meaning, bringing the learner to beyond knowledge. Learning here is based on meaning and generative learning [35–38]. It develops the skills of observation and oral communication, prospecting and searching for information, organizing the speech of the learners, and improving their language through dialogue and discussion between the learners and the teacher. Gerjanis said: The author of the saying thinks about the meaning that he depicts, and arranges this meaning in his thought, then chooses the appropriate systems for him to express it as he arranged it in himself. And his performance is an expression of his perceptions, and in this performance, he presents what has preceded in himself, and delays in himself what is delayed in it. He arranges his phrases and refines them until they agree with the meaning he wants, and he balances between the words and chooses the closest and most specific to the meaning, and the clearest and revealing thereof. It improves access to the exact word in its appropriate position. The following figure illustrates Gerjanis's systems theory (see Fig. 1).

Chomsky is known as: the father of contemporary linguistics. Where he founded the theory of generative grammar, which is considered the most important contribution to the field of theoretical linguistics in the twentieth century. And the generative theory was named by Chomsky in relation to the generation of sentences, or their production in an infinite abundance, and Chomsky adopted the theory of I am thinking!! So, I exist as a solid foundation of knowledge. He has a famous saying about language: "Language is the unlimited use of limited resources." This is the intended creativity of Chomsky, which is the ability to produce unlimited sentences. Depending on a limited number of sounds and fixed rules in the mind of the speaker. Language is creative in nature, meaning that every speaker can utter sentences that no one has ever uttered before him. He can also understand sentences he has never heard before. Sufficiency is the mental stock that the speaker possesses. And related to his language system, which controls his linguistic behavior embodied in performance

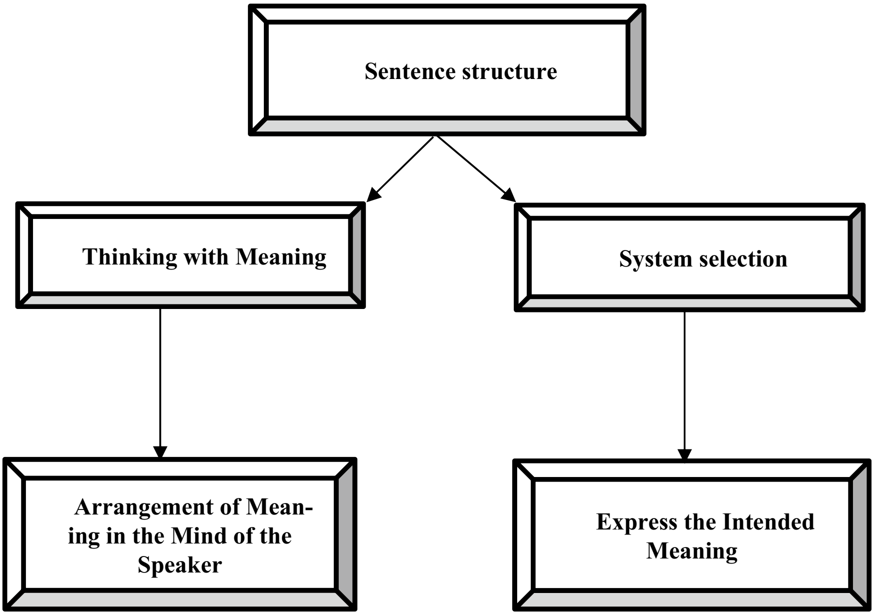

Fig. 1 Pronunciation according to Gerjanis has two levels: 1—an intellectual-psychological level, 2—a phonetic-verbal level

[39]. Chomsky relied on important criteria, which are: (1) Linguistic competence and performance: It is the ability to produce sentences and make them understandable in the speaking process. (2) The superficial and deep structures at the sentence level: The surface structure and the deep structure are the cornerstones of this theory. The performance and the surface reflect what is going on in the depth of the structure in terms of deep linguistic processes, which are hidden behind awareness and subconscious perception most of the time. The study of performance provides the phonological interpretation of the language, while the study of competence provides the semantic interpretation [39]. The surface structure and the deep structure are the cornerstones of Chomsky's theory. The performance and the surface reflect what is going on in the depth of the structure in terms of deep linguistic processes, which are hidden behind awareness and subconscious perception most of the time. The study of performance provides the phonological interpretation of the language, while the study of competence provides the semantic interpretation. The following figure shows the generative theory approach of Chomsky (see Fig. 2). Furthermore, Fig. 3 illustrates the structuring and contrasting sentences with Gerjanis and Chomsky.

Fig. 2 The foundations of Gerjanis's systems theory and Chomsky's generative theory

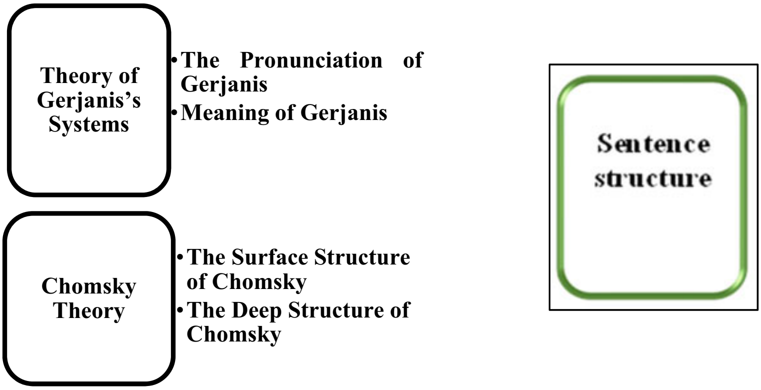

Fig. 3 Structuring and contrasting sentences with Gerjanis and Chomsky

2.2 *Arabic Language Engineering*

The Arabic language is a system of algorithms that starts from the level of sounds, and ends with the level of sentence structure, and the level of semantics as well. The voice in the Arabic language are vibrations that are emitted and formed in an area in the human brain, not on the tongue, then they are produced on the tongue, so the sound is produced through the articulation apparatus. It has been shown that the Arabic language is the only language in the world, according to which all the organs of speech that God created move, unlike other languages. Where we find that those who speak it do not move except certain areas of the articulation apparatus, and all areas in which the Arabic language moves is the sound of adversity, and for this reason it is called the language of adversity. From this perspective, the Arabic language is the mother of all languages. In terms of engineering science, and for this important feature, many international companies have relied on the Arabic language. As the frame of reference for building an engineering sound system, and as a basis for modern technology because it is the most responsive language in the world to new computer systems [40]. So, efforts had to be made to build.

2.3 Reasons for Learners' Weakness

The most prominent of these reasons are: (1) Linguistic weakness in general among learners at all educational levels. (2) Teachers' lack of interest in expression lessons, whether oral or written. (3) Poor selection of topics decided by teachers or those in charge of the educational process, as they are often far from reality, unsuitable for the level of thinking, or the environment in which the learner lives. (4) Poor method of teaching expression, whether oral or written, as the teacher resorts to pouring information into the minds of the learners. Indeed, there are several manifestations of weakness of non-Arabic speaking learners in expression skills and treatment mechanism: (1) The aversion of most of the students from the lesson of expression and their abandonment of it. (2) Interest in the foreign language at the expense of the mother tongue [41]. (3) The extreme weakness in the writings of most of the learners, "teachers do not invest in the sophisticated linguistic patterns in the language lessons, to train the learners to use them in new situations [42]. (4) The lack of clarity of the curriculum, and the level of each educational stage of primary and intermediate education in the minds of teachers [42]. Perhaps this is what prompted (the camel) to ask questions about the method of expression, saying: "Where is the curriculum for teaching Arabic writing. The written expression of each of the two fields: [people of the language, and speakers of other languages], where are the specific procedural goals? Where are the content and skills? Where are the topics and positions? [43–47]. Treatment mechanisms to overcome the phenomenon of weakness in my expression skills as seen by the researcher: (1) Training learners on creative writing skills: in the arts of essays, stories, plays, and research, and presenting their ideas in a literary style characterized by clarity, excitement, and beauty.

(2) Giving the learners an opportunity to freely express their ideas and tendencies, explaining what is going on in their minds, and developing the learner's sensitivity to different social situations. (3) Focusing on oral expression and its exercises and paying attention to it in all language classes. (4) Employing teachers in reading and literary texts classes: "and training students to use other materials in expression. (5) Familiarize students with the use of words that denote various meanings: and train them on the method of composing and coordinating between paragraphs and sentences. Therefore, the main objectives of teaching and learning expression skills for non-language speakers are: (1) Strengthening the language of the learners, and developing their ability to understand the Arabic language when they listen to it. (2) Learners should be able to read and write in Arabic with accuracy, fluency and understanding. (3) That the learner acquires the skill of delivering speeches on various occasions. (4) Acquiring linguistic and verbal wealth as well. (5) Show the learner's skills and expressive abilities, and refine his literary talents [48].

2.4 Review of Previous Studies

The study of [49] entitled: The effectiveness of an educational program based on Gerjanis's systems theory in developing the skills of understanding meaning and literary appreciation among secondary school students in United Arab Emirates. Study Objective: To identify the effect of an educational program based on Gerjanis's systems theory on developing meaning comprehension skills and literary appreciation among secondary school students in United Arab Emirates. The following results:—There is a statistically significant difference in meaning comprehension skills at the level ($a = 0.05$) between the two experimental and control groups in favor of the experimental group whose members studied in the program based on Gerjanis's systems theory. Al-Awawda study [42] entitled: An analytical and evaluative study of some computer programs in teaching Arabic to non-native speakers. Study Objective: To analyze and evaluate seven computer programs specialized in teaching Arabic to non-native speakers, and these programs are: My Teacher Program, Lootah Program, Teach Me More Program, Green Gate Program, Al-Madina Program, Arabic for All, and "Fun with Arabic" Program. The study also dealt with a proposed computer program for teaching Arabic to non-native speakers, which aims to teach the basic skills of the Arabic language reading, writing, listening and speaking. The study concluded with the design of a proposed computer program, benefiting from all previous programs, to teach the four language skills, for non-Arabic speakers, to help non-Arabic speaking students. The researcher added to it what he deems to be incomplete in those programs. Adler's study [50]: The effect of language games played by students in developing their creative writing skills. Study Objective: To know the effect of language games in developing creative writing skills for learners of Arabic as a non-native speaker. The study focused on expanding the learners' connection with the creative language, through its association with the context of imagination, through the works of Vygotsky. The study was applied to three public high schools in New York State, where four creative writing courses were offered, selected in Vygotsky. The participants in the study were (4) teachers and (24) learners. Among the most important results of the study are: (1) Both the teacher and the learner were distinguished by presenting writings colored according to their respective imaginations. (2) The results also showed that educational pictures contributed, clearly and significantly, to finding a point of balance between freedom of expression and language rules. (3) The course of the book workshop had more progress for the participants than the course of the curriculum, which was presented in a traditional way.

3 Research Methodology

To achieve the objectives of the research, the researcher followed the semi-experimental approach in her study: (1) Due to the fact that this method is compatible with the research objective and compatible with the nature of human phenomena [38, 51, 52]. Language is a human phenomenon, as is well known, and the semi-experimental approach is the approach that is based primarily on studying human phenomena as they are without change [53–56]. In another definition, it is the study of the relationship between two variables as they are in reality, without controlling the variables [57–61]. (2) In this research, the purposive sample consisted of Arabic language learners at the Faculty of Languages—International Islamic University. (3) The researcher conducted a pre-test for the study sample and a post-test of the same sample [62–65]. Before and after applying the educational program. The research population consisted of: non-Arabic speaking learners at the Faculty of Languages—International Islamic University—Malaysia, advanced level, for the academic year 2021–2022 AD. They are (24) learners of different nationalities. This population, with its sample, has served the objectives of the study, in terms of competence, academic qualification, or specialization. Forming a suitable base for the researcher; In the application of its educational program based on teaching the skills of oral and written expression and learning them according to the systems theory of Gerjanis, and the generative theory of Chomsky, from the perspective of Arabic language engineering. The sample consisted of learners who speak a language other than Arabic or as a second language. The sample was chosen intentionally to suit the research methodology, which is what Abu Allam says about it: "Any partial group of society, which is known in the educational literature as a partial group of a population that has common characteristics [48]. The participants in the constructive educational program are the male and female learners in the purposive sample, whose number is (24) learners only. Their ages range from 17–23 years, and of different nationalities. The reason for choosing this sample from the advanced level is so that the respondents can communicate in Arabic well, they can understand what is being said, and they can benefit from the constructivist educational program in teaching the two skills of expression because they require an advanced level of knowledge in language skills. This also provides a suitable environment for implementing the constructivist educational program designed by the researcher. Research tools: To achieve the objectives of the research, the researcher used the following tools: (A) Designing a pre-test before applying the constructivist educational program, and a post-test after applying the constructivist educational program for the study sample, through which the ability of non-Arabic speaking learners at the advanced level to write a free expressive topic was measured., from their ideas and creativity to explore their individual skills, and evaluate their performance according to Bloom's cognitive levels. This is explained in the appendices. Appendix No. (1). (B) Arbitrating test designed to measure the level of learners in the research sample and measure their abilities and performance in the skills of creative written and oral expression. Before applying the constructivist educational program, by a committee consisting of a number of experts and specialists in the Faculty of Languages in Malaysia. (C) Design of the constructivist educational program: The constructivist educational program was designed by the researcher, which was applied to the research sample. The educational program consisted of three educational units. It was presented to a group of arbitrators with specialization. Accredited by the Language Center in Malaysia. And after implementing the amendments proposed by the arbitration committee. They acknowledged the completion of the educational program, its content, and its compatibility with Bloom's levels of knowledge. (D) Educational units: The researcher presented (3) educational units for the application of the constructivist educational program, and they are: The first educational unit—life in the countryside and the city; The second educational unit—food and health care; The third educational unit—environmental pollution. (E) The constructivist educational program was applied through the application of educational units to the research sample for a period of 6 weeks, and each week was 3 h. (F) Designing a post-test to measure the skills of non-Arabic speaking learners at the advanced level in creative written expression, after applying the proposed program. (G) Oral test design: to measure the performance of the learners' skills, non-Arabic speakers, in the oral verbal expression skill. After applying the constructive educational program.

4 Data Analysis and Discussion

In this study, several steps were adopted before analyzing the study's data as follow: (1) Using the percentage to find out the mistakes of the learners before and after applying the educational program. (2) Using the arithmetic mean to find out the differences between the learners' scores in the pre and post test [66–70]. (3) Use T-test. To find out the statistical differences [35, 54, 71–73]. Before and after implementing the constructivist educational program. (4) Using the statistical package for social sciences SPSS [54, 74, 75]. To measure learners' grades. (5) Alpha Crew Nabach test, to determine the validity and reliability of the test [76–80]. In order to ensure the effectiveness of the constructivist educational program that was applied to non-Arabic speaking learners at the advanced level, the following steps were followed: First—Statistical analysis according to Bloom's cognitive levels classification: The differences between the scores of non-Arabic speaking learners in the pre and post-tests, in teaching and learning the skills of oral and written expression, were studied according to Gerjanis's systems theory, and Chomsky's generative theory, from the perspective of Arabic language engineering. According to Bloom's cognitive classifications (knowledge, memory, understanding, application, and synthesis). For each test question. Second: Statistical analysis to find out the differences between the results of the pre- and post-test learners, according to the total scores of each question and its parts (Table 1).

According to Bloom's cognitive levels classification: (1) The first question: it is divided into a. and web. To measure the level of knowledge and remembering. 10 scores (2) The second question: it is divided into a. and b. and c. To measure the level

Table 1 Data collection: classification of pre-test questions

Question	First			Second			Third			Fourth		
Bloom levels	Remembering/ memory			Understanding			Application			Structure		
Category	A	B	C	A	B	C	A	B	C	A	B	C
Grade	5	5	X	4	4	4	4	10	10	4	10	30
Total score 90	10			12			24			44		

of understanding. 12 scores. (3) The third question: It is divided into a. and b. and c. To measure the level of application. 24 scores, (4) The fourth question: it is divided into a. and b. and c. To measure the installation level. 44 scores (Total: 90 scores). (5) The oral exam: a score of 10, the sum of the test as a whole is 100. The oral exam scores are distributed as follows: (1) Content: 2 scores. (2) Fluency: 2 scores. (3) Exiting letters and words: 1 score. (4) Linguistic score: 2 scores. (5) Grammar: 3 scores, total: 10 scores (Table 2).

An exploratory sample consisting of four advanced-level learners at the College of Languages—International Islamic University—Malaysia, who were chosen from outside the intended sample, a week before the start of the pre-test, was selected to measure the test time according to the following equation: (1) The time of the fastest answering of the learner (95 min from the start of the test); (2) The time of the slowest of the answering of the learner (145 min from the start of the test).

Table 2 Classification of pre-test questions

N	Questions type	No. of items	Scores for each item	Total score of question
1.a	Cognitive questions	5	One score for each item	5
1.b	True and false	5	One score for each item	5
2.a	Meaning of vocabulary	4	One score for each item	4
2.b	Reverse words	4	One score for each item	4
2.c	Single words	4	One score for each item	4
3.a	Fill the blank	4	One score for each item	4
3.b	Rules	5	Two scores for each item	10
3.c	Error and correction	5	Two scores for each item	10
4.a	Completion	4	Two scores for each item	4
4.b	Functional expression	5	Criteria, two scores for each criterion	10
4.c	Expression in essay	8	6 criteria for 24 scores And two criteria for 6 scores	30

$$\text{Test time} = 95 + 145 = 240\,\text{min} \div 2 = 120 \text{ min is the test time}$$

Explanation of the constructive educational program are:—The researcher used a power point presentation. On the large display screen available in the library hall at the university, and there is also a computer in the hall that was referred to when applying Arabic language engineering. And its automated programs have some linguistic semantics. To know the meanings of some vocabulary, and to know their derivation and conjugation;—The educational program designed by the researcher was applied to the research sample for a period of 4 weeks, each week 3 h;—Application of the post-test on the research sample after the completion of the application of the educational program, four days after the completion of the educational program;—Analyze the results of the pre and post-test to find out the effectiveness of the proposed program on the sample;—Setting standards for the oral exam; And approved by the Language Center, Damascus University;—To achieve the objectives of quantitative research in the field of scientific analysis to study the intentional sample to which the educational program was applied, consisting of (24) male and female students. The statistical approach aimed to:

a. Using the percentage calculation to find out the mistakes of the learners before and after the application of the educational program.
b. Analyzing the results of the pre and post test, and extracting the arithmetic mean of the results of the pre and post-test using the SPSS program. Considering that research is quantitative and quantitative research always uses the SPSS statistical program.
c. Using the T-test; To find out the differences between the results of the pre and post test. And to see if there are statistically significant differences between the two tests.
d. Alpha Crew Nabach test, to determine the validity and reliability of the test.
   The number of paragraphs 100, and Cronbach's alpha is 086.

Educational programs: through the construction of her educational program, the researcher wanted to rely on the systems theory of Gerjanis and the obstetric theory of Chomsky in teaching the two skills of expression and learning them in making learners use the meanings that formed in their minds from the educational situations that they were subjected to while explaining the lessons in the educational units. And they express it with appropriate words, these words that carry meanings and show them in sentences with connotations leading to their expressive purpose. The educational program was designed according to the following criteria: 1—The stimuli and incentives provided to the learners in the stages of the course of the educational units, starting from the introduction, the presentation, and the conclusion. 2—Learners' response speed to questions posed in educational situations. 3—Providing the learners with feedback to work on mobilizing the previous information of the learners by explaining the educational situations. 4—The interim evaluation that is applied at every step of the implementation of the formative educational program. 5—The final evaluation, which includes the researcher's opinions and observations, and the opinions of the learners. After applying the constructivist educational program in order to support their learning process, and provide them with information related to each learning stage. 6—Measuring each educational stage according to Bloom's cognitive classifications, in the levels of remembering, understanding, application, and composition. The educational program was implemented in three educational stages: The first stage is to express in one word: The stage of selecting words that carry meanings from the students, where each student expresses in one word the pictures he saw and the show he saw, and the words received from the students are written down on the board. in its designated section. Through the first educational unit. The second structural stage is the expression in two words: The learners continue watching the presentations on the display screen and discuss the content of the educational unit, then they express in two words their understanding and assimilation of the ideas, but this time they express with only two words. All the words received from the students are recorded in the place designated for them on the board. Through the second educational unit. The third structural stage is to express sentences consisting of three or more words. By presenting the content of the third educational unit, discussion, narrative dialogue, and questions and answers, the words expressed by the students in their own style, consisting of three words or more, are recorded in the designated section on the board. Application of modern education techniques in the engineering linguistic structure of the educational program: The researcher did not rely on: curriculum or a book to teach my expression skills. Rather, it relied on multiple means to present educational content or material, such as images and live models derived from reality. Modern educational technologies such as the projection screen, PowerPoint, and computers were used in all stages of the lesson. So, the learners drew their educational material through it. They derived the content of the lesson themselves through the teaching aids. This made it possible to:—The learner is the one who sets the curriculum and chooses himself the subject or phenomenon that he will talk about.—The researcher plays a directing and leading role in the educational process. Learners are partners in the educational process; through joint work within groups and educational teams.—Discussing the topics raised by the learners themselves. To achieve self-learning.—Emphasis was placed on learning through verbal interaction, learning through educational drama, and computer programs. Evaluation of the formative educational program: The evaluation followed in this study was built on three stages: The first stage is the pre-assessment: it is represented by the pre-test that was applied to the study sample before the start of the program. The second stage of constructive assessment: through what the learners do in terms of writing topics of written and creative expression, and oral expression. Which includes the skills they acquired and trained during the educational process. The third stage is the final evaluation: it is represented by the post-test. The stages of the educational unit's progression, which consist of the following stages:

Preamble: Presentation of videos, power points, and pictures of the topics included in the three educational units. The space was opened for discussion and dialogue between the learners on the one hand, and the researcher teacher on the other hand, about the content of the educational units that include the concept of the units to be applied to the research sample.

Presentation: It is the most important part of the educational unit, as it contributes to achieving the objectives of the lesson. It is represented in presenting the content and content of the educational unit and the educational material in an interesting manner that makes the learners willing and excited to learn and participate in dramatic and anecdotal discussions and dialogues through which the learning process is built. In addition to what each learner recorded on his notebook Treasures of Knowledge, from the words received from his classmates. In the three constructive stages, where this vocabulary contributes to providing learners with new vocabulary, and stimulates memory in retrieving previous experiences.

Conclusion: The researcher discussed and read some of the learners' writings. And asking each learner to choose three sentences that he liked from the sentences that he recorded on his notebook, Treasures of Knowledge. Opening the door for a number of learners to read the sentences they chose over their colleagues.

After applying the constructivist educational program, learners will be able to:—Generating new words from the words that were exchanged and put forward while explaining and discussing educational situations.—Formulation of new sentences composed of constructive generation of words.—Building and establishing new knowledge for learners through generating words and formulating sentences.—An increase in the learners' vocabulary, which allowed them to compare the previous words verbally and in writing, and the new words learned. And discover problems, whether verbal, grammatical or written. Between words.—The learner recognized his mistakes through educational situations and displayed pictures that showed the correctness of the written words. This knowledge led to overcoming these mistakes and learning the correct words, which led to upgrading the expression skills to the level of knowledge, remembering, understanding, application, and synthesis. This enabled the learners to maintain the correct form of the word, as it is associated with the senses of sight and hearing and their ability to improve the learners' performance.—Linguistic engineering and computers constituted a rich source for developing linguistic wealth and achieving linguistic competence for learners during the application of the educational program. Arabic language engineering and its automated programs contributed to knowing the roots of the words that appeared in the educational units. Knowing the conjugations of words and the correct syntax of words, in addition to spelling correction. The impact of the educational program on raising the learners' efficiency and improving the quality of their expression:

A. The constructivist educational program provided learners with immediate feedback.
B. The effectiveness of the constructivist educational program has been proven through the systems and generative theories, as they directly contributed to the development of the learners' thinking through mental simulation, through the educational means that are directed directly to the learner's thought and senses.
C. The constructivist educational program contributed to stimulating learners' motivation towards learning and their attraction to it.
D. The learners seemed to feel satisfied and happy for their ability and success in applying the Arabic language engineering programs.

E. Dispelling fear, anxiety and reluctance that accompanied the learners while speaking. Where the engineering of the Arabic language and its various automated programs also contributed to the learners' enthusiasm and overcoming shyness and confusion. It gave him the opportunity for self-learning.
F. The systems theory of Gerjanis, and the generative theory of Chomsky, avoided the learner from failure and hesitation, because the information was presented in a sequential and logical manner, and in small interconnected steps. According to gradual constructive steps starting from (the word, then the two words, the three words or the sentence, then the paragraph and the subject). Table 3 shows the differences between the learners' average scores in the two tests as a whole.

Table 3 shows that the arithmetic mean value of the learners' scores in the pre-test was (36.5), and the arithmetic mean value of the learners' scores in the post-test was (79.9375). To find out if there are differences between the learners in the pre and post-tests, we note that the value calculated in absolute terms is $t = 4343$, which is greater than the tabular value (2.069) taken from distribution tables at degrees of freedom (23). It is noted that the probability of significance $p = 0.00$ is smaller than the level of Significance (0.05), and accordingly there are statistically significant differences between the mean scores of non-Arabic speaking learners in the pre and post-tests, and these differences are in favor of the post test because its mean is higher. In order to determine the size of the impact of the educational program on the teaching and learning of oral and written expression and their learning for non-Arabic speakers according to Gerjanis's systems theory, and Chomsky's generative theory from the perspective of Arabic language geometry, the eta-square law was applied as follows:

$$\eta^2 = \frac{T^2}{T^2 + df}$$

Table 3 Paired samples statistics and differences between the average scores of the learners in the two tests

Paired samples statistics

		Application	Mean	N	Std. deviation	Std. error mean
Pair 1	X1	Pre-test	36.5000	24	3.98912	0.81427
	X2	Post-test	79.9375	24	3.36320	0.68651

Paired samples test

		Paired differences					T	df	Sig. (2-tailed)
		Mean	Std. deviation	Std. error mean	95% Confidence interval of the difference				
					Lower	Upper			
Pair 1	X1 –X2	43.437	4.85286	0.99059	45.48668	41.38832	43.850	23	0.000

$$\eta^2 = \frac{(-43.850)^2}{(-43.850)^2 + 23} = \frac{1922.823}{1922.822 + 23} = \frac{1922.822}{1945.822} \Rightarrow \eta^2 = 0.988$$

it is noted that the value of eta squared is equal to (0.988), which is greater than the value (0.14), and this indicates a significant impact of the program on teaching and learning written expression for non-Arabic speaking students according to Gerjanis's systems theory, and Chomsky's generative theory from the perspective of language engineering. Arabic, with regard to the test as a whole.

5 Recommendations and Suggestions

5.1 Recommendations

- Applying the constructivist educational program in teaching the skills of oral and written expression, and learning them according to Gerjanis's systems theory, and Chomsky's generative theory from the perspective of Arabic language engineering.
- Focusing on tangible vocabulary first before moving on to the abstract in advanced stages.
- Focusing on vocabulary and linguistic patterns (linguistic patterns), which are the most common, frequent, and frequently used.
- Focusing on vocabulary and short sentences in the early educational stages.
- Focusing on functional texts of social benefit.
- Allocating sufficient educational classes to learn the skills of oral and written expression.
- Avoiding presenting more than one language difficulty in one educational situation.
- Changing the style of teaching and teaching the skills of oral and written expression so that the learner is the source of information.
- Focusing on oral dialogue, especially between learners inside and outside the classroom.
- Increasing interest in the remedial educational program, in order to avoid common mistakes that learners of oral and written expression face.

5.2 Suggestions

- Conducting more studies related to teaching and learning oral and written expression for non-Arabic speakers.
- Conducting more applied studies on the role of linguistic theories in teaching and learning oral and written expression skills for non-Arabic speakers.

– Conducting studies on the engineering of the Arabic language in computer programs and its role in teaching language skills to non-Arabic speakers.
– Conducting systematic, comparative educational studies on the impact of the free educational program "of the learner's choice" in learning the skills of oral and written expression, from the programs previously decided and imposed by the teacher.
– Conducting a study on the effectiveness of performing job duties within the school, and not assigning the learner any homework.

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ChatGPT's Role in Student Engagement and Learning

Fermi Problem-Based Learning with Artificial Intelligence: Is It Effective to Develop United Arab Emirates Cycle Three Students' Twenty-First Century Skills?

Adwan Mohammad Hasan Bani-Hamad and Mouza Saif Ali Al-Kalbani

1 Introduction

During the late twentieth and early twenty-first centuries, there have been significant advancements in all areas of life, particularly in science, which have greatly impacted the educational process due to the vast explosion of knowledge during this era [1, 2]. As a result, updated and revised standards were developed to replace the traditional skills and knowledge that students possessed in the past. Consequently, in response to these new challenges, schools were required to restructure their teaching and learning methodologies to facilitate students' creative thinking, flexible problem-solving, and collaborative skills, thereby laying the foundation for their future lives and the world they will inhabit [3].

Numerous scholars, including [4, 5], and institutions such as the Partnership for twenty-first-Century Skills (2019), National Science Foundation [NSF], Educational Testing Service [ETS], and the National Education Association [6], argue that twenty-first-century skills are crucial for effective learning and that acquiring new competencies and skills is necessary to facilitate the shift towards modern teaching and learning practices. Moreover, the skills and competencies that students require in the twenty-first century serve as the foundation for them to realize their aspirations and achieve their cognitive potential in future careers. Therefore, teachers must employ various strategies and approaches to teach new skills to students effectively and achieve this objective (Bani-Hamad and Al-Kalbani [7]).

A. M. H. Bani-Hamad (✉) · M. S. A. Al-Kalbani
Emirates Schools Establishment, Dubai, United Arab Emirates
e-mail: adwan.hasan@ese.gov.ae

M. S. A. Al-Kalbani
e-mail: Moza-sa.alkalbani@ese.gov.ae

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Many studies such as [8, 9] have proposed four categories of twenty-first-century learning skills that students need to acquire in the face of globalization, including technology, media, and information skills, as well as career and life planning skills. Incorporating these skills into school curriculums and learning approaches can help prepare the younger generation for prospective careers in the future. Furthermore, acquiring these skills and competencies enables students to enhance their creativity, innovate, and leverage technology to develop novel approaches and strategies to succeed in life, becoming more adaptable and flexible to changing conditions. Therefore, educational programs need to adapt and transition to this new paradigm to ensure that students become future innovators [3, 7]. According to numerous researchers, innovative teaching and learning methods are necessary for students to acquire twenty-first-century skills. Teachers have a crucial role in achieving this by utilizing contemporary technologies and creative continuous innovation technics to facilitate learning that integrates social and cognitive skills with relevant content, creating a favourable and engaging learning environment, and these approaches can enable students to participate effectively in real-life events [10, 11].

In the United Arab Emirates (UAE), The Ministry of Education [38] emphasizes the importance of using contemporary teaching methods and artificial intelligence, rather than traditional lecture-based approaches, to foster the development of twenty-first-century skills. Moreover, incorporating innovative educational strategies into all curriculum and educational levels can showcase their effectiveness in enhancing students' learning abilities and achieving the educational goals of the twenty-first century, as emphasized in the National Standards for curricula and evaluation. According to Bani-Hamad and Abdullah [12] and Chen et al. [13], integrating problem-based learning and artificial intelligence into educational programs is considered as the primary approach for developing twenty-first-century learning skills, teachers can facilitate the acquisition of these skills. Problem-based learning (PBL) is a teaching method in the field of science education that focuses on promoting self-directed learning skills among students [14]. Problem-based learning (PBL) involves presenting students with a series of carefully chosen issues as part of the curriculum. Students work in groups and are guided by teachers to solve these issues through questioning. According to Chen et al. [15], artificial intelligence brings up new possibilities for curriculum and teaching methods, enabling educators to take advantage of the special benefits and services that these apps offer for learning. Interactive learning environments based on artificial intelligence were created to enable direct contact between students, computers, and smart gadgets to immediately discover new concepts, so that it can be successfully employed in educational programs. In addition to various thinking and problem-solving skills, the findings of using these settings demonstrated good benefits on several factors associated to the learning process [16]. In the same vein, Enricho Fermi (1901–1945), an Italian philosopher and the 1938 Nobel Prize winner in physics, was not only a highly respected teacher but also a problem solver and problem poser. He often presented problems like "How many people could you fit into the classroom?" as an example to teach his students to think critically and make reasonable estimates using quantitative analysis and simple calculations [17, 18]. Fermi believed that everyone, regardless of their level of education, should have the ability to solve problems of this nature through intelligent reasoning and estimation [19]. Fermi problems involve using problem-solving skills and mathematical thinking, rather than relying solely on information. Enrico Fermi was particularly skilled at making well-informed estimates without having access to precise data, and these types of problems require more than just memorized knowledge [10, 19]. Collaborative group work, effective communication, and estimation skills are essential in solving Fermi problems, as they prioritize problem-solving techniques over the correct answer [19]. Moreover, the approach used in obtaining the solution is valuable in solving Fermi problems, as the process of arriving at the answer is more significant than the answer itself [20, 21]. The problems presented in this study are Fermi problems (FPs), which are defined by [19] as non-routine, open problems that can be solved by quick calculations based on several reasonable assumptions and estimates, leading to rational answers.

The utilization of Fermi problems-based learning in teaching science has been shown to enhance students' skills, as suggested by various studies. For instance, Holubova [11] and Bani-Hamad [21] found that Fermi problem-based learning improves students' creative and critical thinking skills. In addition, Bani-Hamad and Alzubaidi [22] reported a positive impact on critical thinking skills when using Fermi problem-solving in science education. Albarracin and Gorgorio [23] also concluded that Fermi problem-solving in group settings is effective in developing students' skills. Additionally, Fermi problems have been widely used in various mathematical and scientific studies in the field of education. Several studies have employed Fermi problems to investigate problem-solving strategies among high-school students, revealing the complexity of the modelling process in resolving FPs [20, 24, 25]. However, Bani-Hamad and Al-Kalbani [7] proposed a science educational approach focused on solving FPs. In the age of artificial intelligence, the job and abilities of educators have also evolved. With the development of new competencies, educators are now responsible for designing the learning and teaching processes as well as the educational environment. From this vantage point, numerous studies demonstrate the value of artificial intelligence technologies and the need to use them in all subject areas [26]. Thus, the present study modified a new model consist of previous proposed educational model which focused on solving Fermi Problems by [7] and Artificial Intelligence, this new model aims to examine the efficacy of Fermi problem-based learning with artificial intelligence (FPBL-AI) model in fostering twenty-first-century skills, particularly the 4Cs (creative thinking, critical thinking, communication, and collaboration skills) among students. To achieve this, the experimental group will be instructed to utilize the suggested Fermi problem-solving with artificial intelligence model which depicted in Fig. 1.

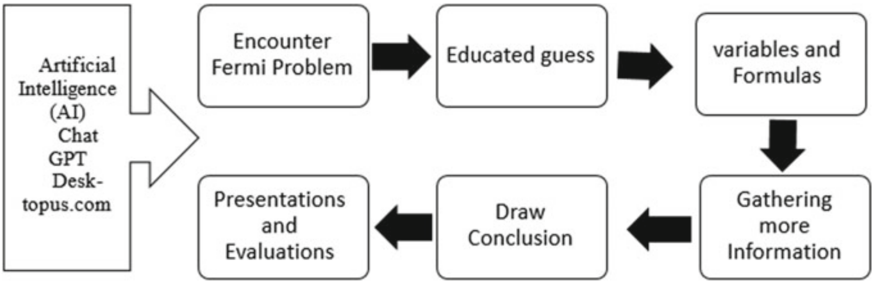

Fig. 1 Suggested Fermi problem solving with artificial intelligence (FPBL-AI) model

1.1 Problem Statement

Dakhi et al. [8] recommended the integration of twenty-first-century skills into school programs and learning approaches to prepare students for prospective careers in the future. These skills can help students expand their capabilities and develop new approaches to succeed in life [27]. The UAE's Ministry of Education [38] highlighted the importance of Emirati individuals utilizing twenty-first century skills to provide rational responses regarding the scientific inquiries they undertake, given the modern world's advancements in science. Global science education is facing several challenges in the current century, such as the focus on traditional textbook and lecture-based teaching and learning approaches. Unfortunately, these approaches often prioritize low-level thinking, such as memorization and comprehension, which may discourage active student engagement in the learning process. Consequently, students may develop negative attitudes and behaviors towards learning, such as boredom, impatience, and aggression [28]. In addition, according to the School Monitoring Evaluation Standards Program (Irtiqa) conducted by the Abu Dhabi Department of Education and Knowledge (ADEK, 2017), the performance of schools in developing twenty-first century skills among students are inadequate. This outcome may be due to teachers' ineffective use of twenty-first century skills in teaching and learning processes or the failure to use teaching methods that enhance students' twenty-first century skills, which are crucial for their future work life.

The ability of students to effectively analyse and process the overwhelming number of future skills is a critical aspect of twenty-first-century skills. However, innovation skills and hypothesis testing are not adequately developed in many subjects, particularly in science. As a result, there is an urgent need to address these issues, especially considering the numerous challenges facing society and students today that will require their solutions in the future. Therefore, schools must adopt new teaching methods that can help students acquire and develop twenty-first-century skills [11]. In addition, there is still a long way to go before artificial intelligence is used in education to benefit students in a natural way. Moreover, there are still certain application barriers that need to be overcome, meaning that teachers must continue to advance and refine their use of AI [29]. Educators should take advantage of the special services and benefits that artificial intelligence applications offer in order to accomplish this goal, as artificial intelligence simultaneously supports curricula and instructional practices that create twenty-first century abilities [13].

Given the challenges posed by twenty-first-century knowledge and the educational gap due to the Covid-19 pandemic, it is crucial to develop an effective teaching strategy that helps students acquire the twenty-first century skills needed to overcome these obstacles [30, 31]. Numerous studies suggest that FPBL can aid teachers in fostering students' twenty-first century skills by focusing on complex real-world problems and making them the core of the teaching–learning process [10, 22]. Additionally, students can learn twenty-first century skills by utilising Decktopus.com and Chat GPT's artificial intelligence. Hence, this research aims at investigating the effectiveness of FPBL-AI model in teaching science on developing Cycle Three Emirati students' twenty-first century skills.

1.2 Research Questions and Hypothesis

The current research attempts to answer the following two questions:

1. How far is the effectiveness of the FPBL-AI model in teaching Science on developing twenty-first century skills among the UAE Cycle Three students?
2. Is there any difference between male and female students in the effectiveness of the FPBL-AI model in teaching Science on twenty-first century skills among the UAE Cycle Three students?

Null Hypothesis

H01: There is no statistically significant effectiveness of the FPBL-AI model in teaching Science on twenty-first century skills among the UAE Cycle Three students.

H02: There is no statistically significant difference between male and female students in the effectiveness of the FPBL-AI model in teaching Science on twenty-first century skills among the UAE Cycle Three students.

1.3 Study Objective

The aim of this study is to explore how teaching Science through FPBL-AI model impacts the development of twenty-first century skills among UAE Cycle Three students (grade 9). Furthermore, this research aims to investigate any potential differences in the effectiveness of FPBL-AI model between male and female students. By using FPBL-AI model as a teaching method, students may enhance their confidence and twenty-first century skills, leading to a more engaging and active learning environment compared to traditional approaches. This approach also aims to equip students with twenty-first century skills that they can apply in their future careers.

1.4 Definition of Terms

1.4.1 Fermi Problems (FPs)

Fermi problems involve open-ended, non-routine problems that require students to ask multiple questions, make reasonable assumptions, and use numerical estimations to identify relevant quantities, variables, and formulas necessary to solve the problem [25]. This approach also involves performing calculations to arrive at an answer and draw conclusions [22, 23, 25]. In this study, the students used the suggested Fermi problem-solving model developed by [7].

1.4.2 Artificial Intelligence (AI)

The study of artificial intelligence in computer science involves developing better programming techniques to create machines that can mimic human intellect in tasks like learning and teaching. In addition, these are systems that behave rationally or with human-like thought processes [32]. Artificial intelligence, according to other academics like [33], is the ability of a computer program to mimic intelligent human behavior. Likewise, Artificial Intelligence, according to Chen et al. [15], is the ability of machines to approach human reasoning. In this study, artificial intelligence (AI) was mostly utilized by Chat GPT to gather information related to Fermi problem solutions and by Decktopus.com to generate presentations.

1.4.3 Twenty-First Century Skills

These cognitive skills are collectively referred to as twenty-first-century skills and are often described using labels such as "learning and innovation skills," "deeper learning," "student-centered learning,", and "next-generation learning." They encompass abilities such as creativity, critical thinking, reasoning, argumentation, innovation, and interpersonal skills that involve teamwork, collaboration, communication, leadership, and responsibility [34, 35]. In this study, the twenty-first-century skills or 4Cs comprised of creative thinking, critical thinking, communication, and collaboration.

Table 1 The sample distribution

Group	Male/female	School	N	Total
Control	Male	Remah School	12	25
	Female	Al Talee'ah School	13	
Experimental	Male	Remah School	13	25
	Female	Al Talee'ah School	12	
Total	50			

2 Research Methodology

2.1 *Data Participants*

The study intentionally selected participants from two Emirati secondary schools—Remah School for boys and Al Talee'ah School for girls, both of which are part of the Emirates Schools Establishment in the United Arab Emirates and have similar environmental conditions. Participants were selected based on their overall ability and homogeneity, with all students having achieved scores above 70% in academic achievement in science. The schools had multiple sections of ninth grade, which allowed the researcher to intentionally select a sample of 50 participants (25 male students and 25 female students), divided evenly into two groups. The control group was selected randomly and consisted of 12 male students from Remah Secondary School and 13 female students from AlTalee'ah Secondary School, who were taught the science unit using traditional methods. The experimental group comprised of 13 male students from Remah Secondary School and 12 female students from AlTalee'ah Secondary School, who were taught the science unit using FPBL-AI model. Table 1 presents the distribution of the sample.

2.2 *Instrumentation*

The objective of this study was to investigate the effectiveness of FPBL-AI model on the development of twenty-first century skills among ninth-grade students in the United Arab Emirates. To achieve this, the study utilized a twenty-first century skills test that was adapted from a test prepared by [7] and tailored to suit the Emirati context. The test comprised of five questions each for creative thinking skills, critical thinking skills, communication skills, and collaboration skills. The reliability of the test was assessed through a pilot study with 15 Emirati secondary students, which yielded a high significant point of 0.89. Additionally, the validity of the test was confirmed by educational science and psychological experts from the United Arab Emirates University. Pre- and post-tests were administered, with 45 min allotted for each test. The pre-test was given before the intervention, while the post-test was given after the intervention had been completed.

2.3 Procedures

To answer the research questions, and according to quasi-experimental design a pre-test was given to participants before the intervention started to evaluate their twenty-first century skills. Fermi problems were created based on the science unit content. The experimental group was taught science concepts before solving Fermi problems, then utilized suggested Fermi problem-solving with AI model as shown in Fig. 1. This model begins with the Fermi problem, then proceeds to answer it by making informed guesses, creating variables and formulas, obtaining further data, concluding, and at the end, presenting and evaluating the findings, while each stage, the students were used Artificial Intelligence by utilized Chat GPT to gather further information to address Fermi Problems and Decktopus.com site to create presentations. The control group continued to use traditional methods. The intervention lasted six weeks. After the intervention, both experimental and control groups took the same twenty-first century skills test as a post-test to determine whether FPBL-AI model affected their twenty-first century skills. The test lasted 45 min. The data was analyzed using SPSS software, specifically independent Sample t-tests, to obtain results.

2.4 Study Design

The research objective was to investigate if the Fermi problem solving with AI model could develop the twenty-first century skills for Emirati ninth-grade students. To achieve this, the study utilized a pre and posttest before and after intervention. Therefore, the appropriate study design was a quasi-experimental study design.

3 Findings and Discussion

According to the study questions, which were aforementioned i.e., How far is the effectiveness of the FPBL-AI model in teaching Science on developing twenty-first century skills among the UAE Cycle Three students? Is there any difference between male and female students in the effectiveness of the FPBL-AI model in teaching Science on twenty-first century skills among the UAE ninth graders? According to Table 2, an independent sample *t*-test was conducted to investigate the group (control and treatment) differences in total twenty-first century skills. Results indicate that the treatment group (treatment Mean = 18.17, SD = 1.558) is higher than the control group (control Mean = 13.5, SD = 1.532) among the UAE ninth grades. Independent sample t-test analysis revealed that t (98) = 14.630, p = 0.000, which interpreted that the mean differences between groups are significant at level 0.05.

These results of the total twenty-first century skills reveal that there are significant differences between groups based upon the FPBL-AI Model treatment. Hence, the null hypothesis is rejected, and the alternative hypothesis is accepted i.e., the FPBL-AI Model (Fermi Problem Solving with Artificial Intelligence) has effectiveness on twenty-first century skills in teaching Science among nine grade students. To investigate the second study question and according to Table 3, an independent sample t-test was conducted to examine the group (male and female) differences in total twenty-first century skills. Results indicate that the male group (Mean = 16.92, SD = 2.756) have similar scores as female group (Mean = 16.67, SD = 2.803) among the UAE ninth grades. T-test analysis revealed that t (98) = 0.798, p = 0.377, which interpreted that the mean differences between groups are not significant at level 0.05.

The comprehensive findings of the study indicate that gender does not have a significant influence on the total twenty-first-century skills in teaching Science among ninth-grade students, as there were no significant differences between male and female groups. Therefore, the proposed null hypothesis is accepted, underscoring that gender does not play a significant role in shaping twenty-first-century skills in this context.

Table 2. Twenty-first century skills differences between different treatments

Twenty-first century skills	Ctrl_Exp	N	Mean	Std. deviation	Independent sample T-test			
					t	df	Sig. (2-tailed)	Mean difference
Twenty-first century skills_Post_Total	Control	25	13.50	1.532	−14.630	98	0.000	−4.610
	Experiment	25	18.17	1.558				
Creative thinking_Post_1	Control	25	3.2800	0.61212	−14.105	98	0.000	−1.53000
	Experiment	25	4.9400	0.48122				
Critical thinking _Post_2	Control	25	3.7400	0.59027	−8.721	98	0.000	−1.05000
	Experiment	25	4.9000	0.54398				
Communication skills_Post_3	Control	25	3.7800	0.56549	−7.018	98	0.000	−0.84000
	Experiment	25	4.5400	0.65670				
Collaboration skills_Post_4	Control	25	3.5200	0.67379	−9.595	98	0.000	−1.33000
	Experiment	25	4.8410	0.58797				

Table 3. Twenty-first century skills differences between different genders

Twenty-first century skills	Ctrl_ Exp	N	Mean	Std. deviation	Independent sample *T*-test			
					t	df	Sig. (2-tailed)	Mean difference
21st Skills_Post_ Total	Male	25	16.92	2.756	0.798	98	0.377	0.500
	Female	25	16.67	2.803				
Creative_Post_1	Male	25	4.3700	0.91051	1.392	98	0.301	0.20100
	Female	25	4.1700	0.99724				
Critical_Post_2	Male	25	4.3500	0.80214	−0.125	98	0.911	−0.02020
	Female	25	4.3700	0.80544				
Communication_ Post_3	Male	25	4.2001	0.82324	−0.676	98	0.507	−0.10100
	Female	25	4.2400	0.66652				
Collaboration_ Post_4	Male	25	4.3200	0.85402	2.196	98	0.030	0.37090
	Female	25	3.9400	0.85567				

4 Discussion

According to the study, ninth-grade students can enhance their twenty-first-century skills by learning science through FPBL-AI model. This study's outcome supports the constructivism theory, which emphasizes that students develop a deeper understanding when they tackle real-world problems. Additionally, the findings align with other studies by [20, 22, 36, 37], which also highlighted the effectiveness of using Fermi problems to foster twenty-first-century competencies. The finding aligns with the UAE Ministry of Education's [38] efforts to implement new standards in the field of educational sciences that prioritize the rapid integration of twenty-first-century competencies into teaching approaches, to address scientific challenges. This approach aims to equip students with valuable skills and cultivate their capacity for innovative thinking in their day-to-day activities. In this regard, the outcome is in keeping with the UAE Ministry of Education's [38] overarching vision. The finding in the Table 3, there were no significant differences between male and female groups; this indicates that the FPBL-AI model works to develop the twenty-first century skills of all male and female students alike. This may also be attributed to the fact that the positive learning environment created by the suggested Fermi problem solving with AI educational model and the benefits of Fermi problem solving with AI can be credited for this. Learning resources can be integrated, obtained, and exchanged at any time, from any location, and at any level of student. The suggested Fermi problem solving with AI model is easy to use, and it fosters enthusiasm and curiosity among male and female students at similar levels, while also helping them develop twenty-first century skills.

5 Conclusion

The acquisition of twenty-first century skills is crucial for students to be ready for their university education and future employment opportunities. As a result, there is a growing need for innovative teaching techniques to assist students in developing these skills. The purpose of this study was to explore the effectiveness of FPBL-AI model as a science teaching method in fostering twenty-first century skills among Cycle Three students in the United Arab Emirates. To sum up, the study's findings indicate that the implementation of FPBL-AI model effectively enhances students' twenty-first century skills. Employing suggested Fermi problem-solving with AI model facilitates teachers in delivering content more effectively and fostering an engaging learning environment. Furthermore, utilizing FPBL-AI model in the classroom promotes the development of students' creative and critical thinking abilities, enhances their communication and collaboration skills, and fosters effective educational meaning construction. The study results validated the hypothesis and research objective by demonstrating the efficacy of Fermi problem-solving with Artificial Intelligence in enhancing the twenty-first century skills of Cycle Three students in the UAE. Additionally, the study highlights the significance of this model for educational experts in the Emirates Schools Establishment, emphasizing the need to include it in teachers' training plans. Furthermore, the study recommends the widespread adoption of this teaching model across all subjects in schools and universities to enable students to acquire essential skills crucial for their future personal and professional development. This study has a few limitations that need to be taken into consideration. Firstly, the research was conducted during the academic year 2022/2023, with a small sample of students from two schools—Ramah School for boys and Al Taleea School for girls, both of which are in Abu Dhabi Emirate and are part of the Emirates Schools Establishment in the United Arab Emirates (UAE). Secondly, the study only involved ninth-grade students (Cycle Three) studying the Science subject for two units, which were taught during the second trimester, and the use of AI applications limited to Chat GPT and Decktopus.com. Lastly, the educational content was limited to just two science units, and the study only lasted for six weeks. Therefore, the findings of this study may have limited generalizability beyond these specific circumstances.

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Why Do Jordanian Students Prefer Using ChatGPT A Case Study of Higher Education Institutions

Mohammed Habes, Abeer Alanani, Enaam Youssef, and Humaira Sharif

1 Introduction

ChatGPT uses creative and diverse AI models based on unsupervised and semi-supervised machine learning methods [1–4]. This includes processing different data types, including images, text, social media, voice recordings, computer code, and structured data [5–10]. ChatGPT can provide a variety of outputs, such as translations, questions and answers, sentiment analysis, summaries, and even generate movies [11]. Notably, it involves Deep Learning as a widely recognized form of AI that involves training algorithms on large datasets to make predictions based on the data. It can be used for language translation, voice recognition, and image recognition, among other things. Besides, Natural language processing (NLP) is also integrated so that the AI may understand and generate human languages, such as text summarization, translation, and sentiment analysis [12–17]. As a result, ChatGPT technology has become a booming phenomenon in diverse aspects of our lives, including education, communication, information, and entertainment [18–21]. As a large language model trained by OpenAI, ChatGPT can comprehend and generate

M. Habes (✉)
Faculty of Mass Communication-Radio and Television Department, Yarmouk University, Irbid, Jordan
e-mail: mohammad.habes@yu.edu.jo

A. Alanani
Department of Curriculum and methods of teaching, College of Education, Mansoura University, Mansoura, Egypt

E. Youssef
Humanities and Social Sciences Research Center (HRC), Ajman University, Ajman, UAE

Ain Shams University, Cairo, Egypt

H. Sharif
University of Malaya, Kuala Lumpur, Malaysia

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human-like language responses. It is an exemplary tool for students needing assistance in their academic work or wanting to engage in conversations with virtual assistants [22]. Besides, from education to communication, learning, and entertainment, the role and effects of ChatGPT can be observed as facilitating different aspects of life. As noted by [23]. ChatGPT's impact on different aspects of life has benefitted multiple industries, including education, research, journalism, mass communication, IT, and retail. The technology's generative AI capabilities allow for the creation of high-quality written content quickly, with the capacity to adapt and improve based on user feedback [24]. This has implications for sectors such as education, where institutions require enhanced technology for their students that can benefit from the accurate code generated by AI models [25]. Also, students can utilize generative AI to enhance their technical understanding of new technology in education [26]. The technology also offers opportunities for educational institutions to expand their academic quality and provide updated, more generalizable skills to their students [27]. The team is accountable for developing ChatGPT, OpenAI, which has been in this field for several years [28]. Their focus is on improving AI capabilities and exploring their impact on society. While many ways to structure an essay exist, mathematical problems present unique challenges [29]. With ChatGPT's ability to exemplify its operation and provide solutions, students may need the teacher's knowledge to use it, leading to several benefits. Additionally, checking grammar and sentence structure also helps students' creativity and experimentation in writing, preventing them from developing their unique voices. Therefore, responsibly using ChatGPT and other AI technologies is essential to avoid hindering students' learning and growth [30]. Thus, based on the use and applicability of ChatGPT in different fields of life, including education [31]. This research aims to examine the factors that affect ChatGPT's acceptance among students in Jordan. The primary objective is to examine the factors proposed under Technology Readiness Theory to determine ChatGPT acceptance. Despite any studies focused on Technology Acceptance Model, Technology Readiness remained underrepresented. Further, this research also fills the empirical gap regarding technology in education, particularly in the Jordanian scenario [32]. Thus, based on the relevant premises, the first section introduces the topic and statement of the problem. Further, the second section reviewed the theoretical and empirical literature to propose the study hypotheses. The fourth section highlighted the methodological approaches. Finally, the last section discusses and highlights the results that further led to the conclusion of this research study.

2 Review of Literature

2.1 Technology Readiness

Technology Readiness Theory (TRT) describes how individuals adopt and integrate new technologies. This theory was developed in 2005 by Paraskevas and Papadopoulou, arguing that a person's readiness to use technology is influenced by four key factors: optimism, innovativeness, discomfort and insecurity, and self-efficacy [33]. Technology Readiness theory proposes that individuals with a strong affinity with technology are more likely to use technological tools and facilities. The readiness of individuals to interact with technology is influenced by their mental mechanisms that allow or inhibit the use of technology [34, 35]. The conceptual framework of current research is also supported by using Technology Readiness as primary support for the study propositions [36]. Regarding technology readiness and students, researchers have highlighted the applicability of this theory regarding education, communication, information, and entertainment. These miscellaneous reasons and characteristics further help the technology readiness theory to examine the technology more comprehensively [37]. A study by [38], empirically analyzed the medical students' attitudes toward technology that influenced their adoption of e-library services during the COVID-19 pandemic. The data gathered from the survey method showed that the students had a positive attitude towards using e-library services and that technology helped them continue their education during the pandemic. Students' attitudes and subjective norms strongly influenced their intention to use e-library services. Still, they perceived behavioral control had little impact due to inadequate support from the government.

Optimism refers to the extent to which a person believes technology will positively impact their lives. Optimism is a mechanism that represents a positive view of technology users towards technology and is the first conductor dimension. Optimistic individuals believe that technology offers them control, flexibility, and efficiency and have a predetermined understanding of its potential benefits even before they use it. They also think they have greater control over their actions and are interested in the prestige of adopting new technologies. To further affirm this, a study by Kaushik and Agrawal [39]. found that students who were more optimistic about using technology in their learning were more likely to have a positive attitude towards technology and use it more frequently. Innovativeness is another mechanism representing an individual's tendency to become early adopters of certain technologies or opinion leaders influencing other consumers. Innovative individuals are intrinsically motivated to adopt new technologies despite the difficulties and risks involved [40]. However, discomfort is related to the perceived lack of control and feeling of oppression experienced by consumers of specific technologies. People who experience a high level of discomfort with technology believe that they are controlled by it. According to Abbade [41], innovativeness, on the other hand, refers to the degree to which an individual is willing to adopt new technologies. Another study by Geng et al. [42] found that more innovative students were more likely to use technology in their learning, while less innovative students were more likely to avoid using technology.

H1: Technology readiness has a significant positive impact on perceived ease of use.
H2: Technology readiness has a significant positive impact on perceived usefulness.
H3: Perceived Ease of Use has a significant positive impact on perceived usefulness.

2.2 Technology Readiness and Perceived Ease of Use

According to Chen and Lin [43], Technology Readiness and Perceived Ease of Use are two important factors influencing the young generation's adoption of new technologies. The young generation is often seen as early adopters of new technology. They are comfortable with technology and use it for various purposes, such as communication, entertainment, and education [44]. However, their Technology Readiness may vary depending on multiple factors, including their previous experience with technology, their perception of the usefulness of the technology, and their confidence in using it. Martens et al. [45], argued that Perceived Ease of Use also plays a crucial role in the young generation's technology adoption. If users perceive technology as easy to use [46], they are more likely to adopt it. On the other hand, if they find technology difficult to use or understand, they are less likely to embrace it. This is particularly relevant for young people with different experiences and knowledge than older generations. For example, a study by Compernolle et al. [47], analyzed how smart cities in Flanders, Belgium, initiated a grassroots movement to promote data interoperability. The research explored the relationship between the individual characteristics of decision-makers and their intention to use data standards. The results showed that highly innovative respondents were likelier to use data standards. Still, personality characteristics were insignificant predictors of data standards' perceived usefulness and ease of use. Another study by Salloum et al. [48], reviewed 120 significant published studies from the past twelve years to identify TAM's most commonly used external factors in e-learning acceptance. The analysis revealed that computer self-efficacy, subjective/social norm, perceived enjoyment, system quality, information quality, content quality, accessibility, and computer playfulness were TAM's most common external factors. The TAM was then extended to include these factors and applied to five different universities in the UAE to examine the acceptance of e-learning by 435 students. The study found that system quality, computer self-efficacy, and computer playfulness significantly influenced the perceived ease of use of the e-learning system. Additionally, information quality, perceived enjoyment, and accessibility positively impacted the perceived ease of use and usefulness of the e-learning system (Fig. 1).

H4: Perceived Ease of Use has a significant positive impact on intention to use.

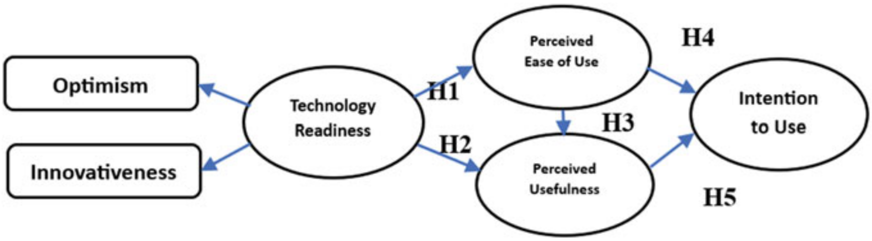

Fig. 1 Conceptual framework of research

H5: Perceived Usefulness of Use has a significant positive impact on intention to use.

3 Research Methodology and Design

The design of the current research is based on a case study method [49–52]. The aim was to acquire in-depth results in the setting in a shorter period. Thus, the case study was the most suitable approach for current research. The data were gathered using structured survey questionnaires designed by obtaining the measurement scales and items from existing literature [53–55]. Table 1 summarizes the sources of measurement items and scales and the construct reliability assessment of each construct individually. As it is shown that all the Cronbach Alpha and Composite Reliability values exceed the minimum threshold value (0.7), the study instrument is reliable. The data gathering was based on personal visits to the selected institutions. The researchers personally visited the educational institutions and first obtained formal permission from the institutional heads, ensuring data protection and privacy as a core consideration of current research. Data was gathered from Mar. 17, 2023, to Apr. 25, 2023.

Table 1 Sources of measurements items and scales

S/R	Constructs	Number of items	Sources	
			CA	CR
1	Technology readiness	6	0.771	0.881
2	Perceived ease of use	3	0.760	0.832
3	Perceived usefulness	3	0.717	0.840
4	Intention to use	3	0.778	0.771

3.1 Population and Sampling

The population of the current research was university-level students currently enrolled in both public and private sector Jordanian universities. Recent data shows 31 universities in Jordan, out of which 10 are public sector. At the same time, 18 are private sector universities, one region, and two are based on semi-government ownership. The total number of enrolled students in these universities is 344,796. Further, the researchers randomly selected 3 universities (One private, two government sector) for the data gathering. The sample size estimation was done by using the formula provided by Krejci and Morgan. Based on the relevant sample size selection formula, a sample of 384 individuals would be ideal for this research.

3.2 Data Analysis and Ethics

Collected data were first evaluated, that further helped to remove 19 questionnaires as wrongly filled by the respondents. The total response rate was 95.0%, much higher than the minimum threshold value of 60% [56]. Further, the data analysis was formally performed, including descriptive and inferential statistics. Descriptive statistics were used to calculate the demographics of the respondents, while inferential were based on testing the structural assumptions. The analysis was performed using SPSS and Partial Least Square-Structural Equation Modelling.

4 Analysis and Results

4.1 Inner Model

The inner model of current research is examined to assess the extent to which the study constructs are internally consistent and divergent and ensure a good fit [48, 50, 56–59]. First, the convergent validity was examined based on Confirmatory Factor Analysis and Average Variance Extracted [60–64]. The CFA results showed that most items had loading values higher than the minimum threshold value of 0.5 [48, 65–73]. While the AVE values also indicated that they surpassed the minimum cutoff value of 0.5. Thus, we can assume that the inner model is internally consistent (Fig. 2).

Further, the discriminant validity was tested using Fornel Larcker Criterion and Hetreotrait-Monotrait Ratio to examine the extent to which study constructs are divergent and uncorrelated [48, 74]. First, the result regarding the Fornel-Larcker criterion showed that all the calculated square values have no correlation with the value provided in Table 2 but are also greater than them [75–78]. Further, the Hetreotrait-Monotrait Ratio was 0.028, lower than the minimum cutoff value (0.085). Overall, it

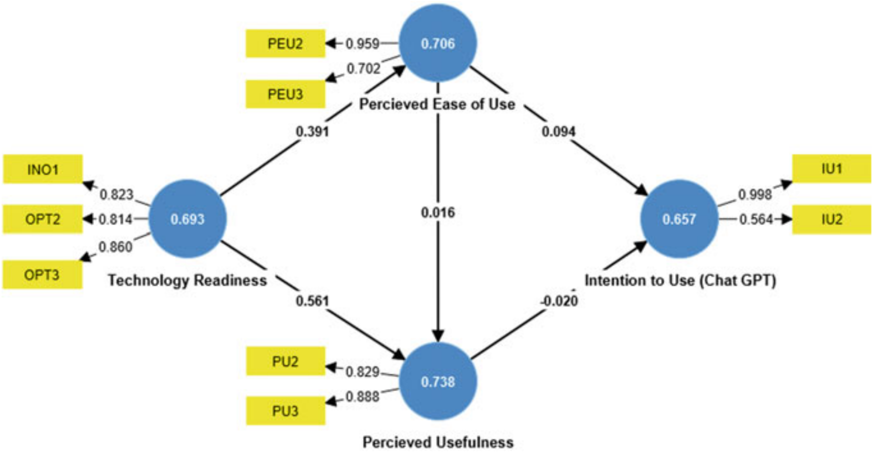

Fig. 2 Inner model analysis (CFA, AVE)

is found that discriminant validity is affirmed for the inner model of current research [79–82].

The model fit, also known as goodness of fit analysis, was performed to examine the extent to which the observed data fits well with the expected results [83–89]. It was found that the Standardized Root Mean Square (RMSEA) was 0.035 (<0.085), while the Tucker and Lewis (TLI) was at 0.839, which was near 0.1. Further, the chi-square value was $\times 2 = 0.7837(df = 19)$. Besides, the Not-Fit Indices (NFI) value was 0.689, near 1.0, indicating an overall good fit for the current study.

Table 2 Pearson correlation analysis

	Intention to use	Perceived ease of use	Perceived usefulness	Technology readiness
Intention to use (ChatGPT)	0.480			
Perceived ease of use	0.09	0.498		
Perceived usefulness	0.003	0.236	0.544	
Technology readiness	0.039	0.391	0.508	0.431

4.2 Outer Model

The outer model was analyzed using path analysis based on regression weights, including t-values, *p*-values, and path coefficients [57, 80, 90–92]. Path analysis aimed to examine the relationship between predictor and dependent variables [80, 93, 94]. However, first, the researchers calculated the R^2 value of the latent variables to determine the variance caused by the predictor variable on the dependent variables [69, 84, 95–97]. Thus, 68.2% variance was found in Perceived Ease of Use, 53.1% in Perceived Usefulness, and 4.11% in Intention to Use. Further, the path analysis was performed by using Smart-PLS Structural Equation Modelling. Results revealed that the proposed relationship between Technology Readiness and Perceived Ease of Use remained significant with the $\beta = 0.094$, $t = 0.391$, and $P > 0.000$, indicating that H1 of the study is validated. The second assumption also remained significant ($\beta = 0.016$, $t = 0.561$, $P < 0.006$), indicating the validation of the H2. The third hypothesis proposed a significant impact of Perceived Ease of Use on Perceived Usefulness. However, results revealed that the H3 of the study is insignificant ($\beta = -0.020$, $t = 0.016$, $P < 0.517$). The fourth hypothesis further remained significant as the impact of Perceived Ease of Use on Intention to Use (ChatGPT) was significant ($\beta = 0.391$, $t = 0.094$, $P < 0.000$). Finally, the proposed impact of Perceived Usefulness on Intention to Use (ChatGPT) was significant ($\beta = 0.561$, $t = 0.781$, $P < 0.000$), showing the H5 as also accepted. Overall, most of the hypotheses supported the study of primary assumptions. Figure 3 illustrates the results of the path analysis.

Fig. 3 Path coefficients—graphical analysis

5 Discussion and Conclusion

According to Compernolle et al. [47] optimism and innovativeness significantly influence technology adoption among young students. This study is also consistent with the fact that the factors introduced by the Technology Readiness theory effectively promote ChatGPT acceptance and use among Jordanian students. First, the effect of technology readiness on perceived ease of use remained significant as the students widely agreed that they consider ChatGPT as easy to use and understand due to its user-friendly design, operations, and capacity. As noted by Kaushik and Agrawal [47], students perceive technological advancements as opportunities for growth, improvement, and enhanced experiences. They maintain a positive outlook and are likelier to embrace and explore innovative tools and solutions. Further, the technology readiness on perceived ease of use also remained significant, showing students' agreement towards ChatGPT as proving them useful outcomes and interactive user interface, further leading them to accept it is a transformative technological feature of modern Artificial Intelligence. These findings are consistent with the argumentation by Pasha [98]. As stated, young students are more open to the possibilities presented by new technologies, fostering a greater willingness to adopt and integrate them into their learning processes. The impact of perceived ease of use on intention to use ChatGPT also remained significant, indicating a strong consistency with Rahmat et al. [99]. As noted, evolving technology is vital today in shaping the young generation's attitudes and beliefs toward adopting new technologies. The distinct features and characteristics fuel their motivation to adopt and utilize technology, as they believe in its potential benefits in various aspects of their lives, including education. Besides, the impact of perceived usefulness on intention to use ChatGPT also indicated a wider agreement on the useful outcomes and expectations leading to positive attitudes, acceptance, and use of ChatGPT. According to AlAfnan et al. [18], today, students naturally are inclined to innovate and actively seek out and embrace new technologies, becoming early adopters and driving broader acceptance. Recognizing the significance of these factors is essential in designing interventions and educational strategies that effectively promote ChatGPT adoption and integration among young students, ultimately empowering them to thrive in a rapidly evolving digital landscape. Thus, utilizing the Technology Readiness Theory framework, this research study explored the factors influencing the adoption and usage of ChatGPT among Jordanian students. The findings indicated that innovativeness and optimism are critical factors affecting perceived usefulness and ease of use, further affecting the intention to use ChatGPT. Students who perceive the technology as beneficial and user-friendly, possess prior experience with similar technologies, and are influenced by its characteristics are likely to adopt and use ChatGPT. Notably, individual innovativeness and optimism are critical theoretical determinants, as students with a propensity for embracing technological advancements demonstrated higher adoption rates. These findings underscore the importance of considering these factors in designing interventions to promote the effective use of ChatGPT among university students, ultimately enhancing their learning experience and academic outcomes.

5.1 Limitations

This study has two primary limitations. First, the results represent the opinions of Jordanian students, which question their generalizability to other regions. Second, the focus remained on overall factors behind ChatGPT usage. While there can be any specific usage that could be addressed. Finally, the third limitation involves only two constructs adopted from the Technology Readiness theory. Future researchers can overcome these limitations by conducting studies in other different regions and some specific reasons, i.e., education. Besides, they can also involve other constructs, such as social norms, to highlight a comprehensive picture of ChatGPT acceptance and usage among students.

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Can Guided ChatGPT Use Enhance Students' Cognitive and Metacognitive Skills?

Faiza Mohamed Tabib and Mohamed Mahmoud Alrabeei

1 Introduction

Teachers are the bedrock of teaching and learning in the dynamic education landscape [1–8]. They play a pivotal role in shaping the learning journey of their students [9–13]. Their responsibility extends beyond the mere transmission of information [14–16]. They are expected to be perpetual learners, continually upgrading their teaching knowledge, skills, and strategies and fostering students' academic success [14, 17–19]. The traditional paradigm of teachers as information providers has evolved into a contemporary model where they act as mentors and facilitators of learning [20–23]. This shift emphasizes the need for teachers to equip students with the essential skills for their future careers [24–26]. Students must be able to ask insightful questions, set goals, plan strategically, monitor their progress diligently, and self-evaluate their learning outcomes thoughtfully and independently [18, 24, 27–29]. These competencies empower students in their academic pursuits, which lie at the heart of education [20]. Responsible teachers guide students toward becoming independent and well-rounded learners. In this context, emerging artificial intelligence technologies in education, such as ChatGPT, offer a promising avenue for teachers and students to cultivate the abovementioned skills by assisting students in processing their thoughts, formulating insightful questions, and using AI tools to refine their research until they complete their quests [23]. This practice can foster the development of deep cognitive and metacognitive skills and abilities highly recommended in primary, secondary, and tertiary education. Hence, exploring the possibilities of this AI tool in teaching

F. M. Tabib
Applied Linguistics, City University Ajman, Ajman, UAE
e-mail: b.faiza@cu.ac.ae

M. M. Alrabeei (✉)
City University Ajman, Ajman, UAE
e-mail: m.mahmoud@cu.ac.ae

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and learning can benefit students by providing them with a platform for personalized learning experiences.

This article contends that teachers, as architects of the learning experience, are key players in fostering a generation of individuals equipped with high-order thinking knowledge, able to personalize and control their own learning and develop the cognitive and metacognitive skills essential for navigating an increasingly complex future. The effective use of ChatGPT by students in their learning can significantly enhance their knowledge and skills, especially when accompanied by appropriate instructional guidance and purposeful, engaging AI-based classroom tasks and activities [20–22, 30].

This paper aims to discuss the ongoing benefits of ChatGPT as a powerful learning tool that can contribute to and facilitate students' cultivation of essential skills. It also aims to propose practical strategies for students to enrich their learning experience with ChatGPT. While the authors acknowledge the utility of ChatGPT in teaching and learning, they emphasize throughout this paper the importance of adopting a guided pedagogical approach when using this AI tool to maximize its benefits and mitigate its drawbacks.

2 Literature Review

2.1 Importance of Critical Thinking in Education

Philosophers such as Socrates, Plato, and Aristotle stand as the forerunners of critical thinking [31–34]. Within their teachings, they used self-questioning as a strategy to stimulate their students' critical thinking and enhance their learning [24, 35–37]. Many years later, Benjamin Bloom and his colleagues developed a taxonomy in 1956 to emphasize and illustrate the thinking levels. His Taxonomy straightforwardly classifies six levels of thinking within the cognitive domain, widely used by educators in many academic disciplines, ranging from highest to lowest. Bloom classified the skills of analyzing, evaluating, synthesizing, and creating as the highest levels of thinking, while recalling or explaining information constitutes the lowest level of thinking in classroom instructional activities. Analyzing involves "seeing relationships, breaking information into parts, and analyzing how things work" [38, p. 34]. Evaluating is about "making judgments, assessing the value or worth of information" [38, p. 34], and finally, creating is about "putting ideas and information together in a unique way and creating something new" [38, p. 34].

Duron et al. [39] argue that educators should engage students in advanced instructional activities and develop questions based on Bloom's Taxonomy and learning objectives to foster critical thinking. They assert that good thinkers can "raise vital questions and problems, formulate them, gather and assess relevant information, use abstract ideas, think open-mindedly, and communicate effectively with others" (p. 160). Ennis [40] describes a good thinker as one who "has the ability to clarify, to seek and judge well the basis for a view, to infer wisely from the basis, to imaginatively suppose and integrate, and to do these things with dispatch, sensitivity, and rhetorical skill" (as cited in [40, p. 15]). Along the same lines, [41] assert that the process in which students engage to explain, connect ideas and concepts, and acquire new understandings of the material at hand is evidence of higher-level cognitive abilities, which are highly required in higher education.

However, many researchers contend that the conventional method of imparting course content in education does not promote students' critical thinking, as the lessons are teacher-centered [39]. Consequently, students become passive learners as the teachers do most of the talking, questioning, and thinking, encouraging rote learning and memorization. Choy and Cheah [41] argue that most educators focus on ensuring course content comprehension rather than emphasizing the processes and skills necessary to nurture students' advanced cognitive abilities.

2.2 Role of Metacognition in Enhancing Students' Learning Experiences

Central to metacognition is learners' awareness and control of their thinking processes and strategies. Therefore, learners can regulate their thinking, learning, and problem-solving strategies [42]. Fostering metacognitive awareness and skills can help develop students' cognitive abilities across various domains. Flavell [43] distinguishes two main components of metacognition: knowledge metacognition (understanding how one learns) and regulation (applying the best strategies to enhance learning). According to Flavell [43], cultivating metacognitive knowledge and regulation can deepen learning and foster critical thinking. These metacognitive practices empower learners to take control of their learning processes, optimize strategies, and improve information retention.

Metacognitive skills include self-awareness of one's strengths and weaknesses and self-monitoring of one's thoughts during tasks, such as checking comprehension or identifying errors [4, 44–46]. Metacognitive skills encompass other abilities such as planning, developing effective strategies for learning various tasks or tackling intricate tasks or issues, effectively managing tasks and resources, adjusting strategies, processing information and assessing its credibility, adapting new strategies in response to new challenges, and reflecting on problem-solving and decision-making processes [42].

Schraw [47] elucidates that planning is a pre-task activity that involves selecting appropriate strategies to complete the assigned task, allocating time, and making predictions. Monitoring entails self-awareness of comprehension of the task and self-testing to ensure understanding. The post-task activity involves evaluating and assessing task performance and the learning outcomes about the task's purpose or goal [48]. Silver posits [48], "The most successful learners, then, employ metacognitive interventions at every stage of activity, and ideally, their evaluative work post task makes possible the transfer of both cognitive and metacognitive knowledge to be employed in new contexts" [48, p. 10].

2.3 Blending Learning Theories with ChatGPT

Integrating foundational learning theories with cutting-edge technological advances can potentially transform pedagogical approaches. Therefore, blending the principles of social constructivism [49] and the Vygotskian scaffolding concept (Vygotsky 1978) [50], potentially changing the teaching/learning paradigms. Constructivism emphasizes the role of social interaction in constructing new meanings and understandings. According to Melrose et al. (2013), constructivism is rooted in the idea that students bring into class their invaluable prior knowledge, and teachers help them "build up that knowledge through active and personally meaningful learning activities" (p. 63) and this concept can be replicated with ChatGPT. Students can co-construct new meanings by engaging in insightful interactions with this AI tool. The adaptive nature of ChatGPT aligns with the principles of social constructivism, as ChatGPT can trigger active thinking and knowledge construction.

Furthermore, the Vygotskian concept of ZPD is about scaffolding the less able learners with their learning needs. This concept emphasizes the significance of instructional scaffolding for meeting the learners' educational needs regardless of their various levels, skills, and competencies. According to [51], "Scaffolding means providing the necessary guidance and advice to students to improve their knowledge and abilities and approach the targeted mastery level" (p. 186). She clarifies, "When teachers tailor their support to students' learning and understanding, the connection between their prior knowledge and the new knowledge is stored in their long-term memory" (p. 187). However, scaffolding is not permanent; as students' progress gradually in their learning, they become less dependent on their teachers [51]. Therefore, the Vygotskian concept can be replicated using AI tools to scaffold and assist students' learning.

3 Guided Use of ChatGPT

3.1 Teacher's Role

Teachers are crucial in guiding students' use of this AI tool. They act as facilitators and mentors in teaching students how to use ChatGPT and in guiding them through the effective and responsive use of this powerful AI tool for enhanced and personalized learning experiences [52–54]. Bieger and Kolmar [21] recommend that teachers assist students in using this AI tool productively and responsibly while also understanding the ethical challenges. While teachers' guidance and support are essential in integrating this AI tool into the curriculum, their guided approach should be pedagogic and well-structured until it becomes integral to instructional tasks. In the same fashion, [52] emphasize the need for teachers to accommodate their teaching practices using AI tools to address unresolved classroom problems. They point out, Teachers may need to be conscious of limitations in their teaching and study resources that can confuse the students' minds about certain concepts. Artificial intelligence makes a mode to resolve that dilemma [52, p. 112].

Teachers should initiate a pilot integration and gather feedback for this AI tool's smooth and effective incorporation. This practice allows them to optimize its benefits and minimize its drawbacks [55]. Once teachers know how to use this AI tool appropriately, they can tailor lessons by incorporating guided ChatGPT tasks, activities, assignments, and projects into their teaching approach and practices. Teachers may be concerned about students using AI tools or unwilling to incorporate them into their curriculum for ethical reasons. Their concerns are understandable, but they can address them by adopting clear pedagogical strategies to ensure transparency and credibility. In this context, [56] recommends that students provide their teachers with a documented record or screenshots of both questions posed and corresponding AI responses. This document, an 'audit trail of queries' [56, p. 8], can be a tangible and comprehensive account of students' interaction with ChatGPT. It also helps promote an accountable and transparent communication channel between students and teachers [56]. The following tasks illustrate the responsible and purposeful use of ChatGPT, which teachers can incorporate into their classroom practice.

Task 1:

Use ChatGPT to brainstorm ideas about climate change in Germany. Document the questions asked and responses received, either by recording them on a document or by taking screenshots. Share your findings with the class.

Task 2:

Use ChatGPT to outline a project on the dangers of vaping among high schoolers. Document the questions and responses by recording them in a document or taking screenshots. Share your findings with the class.

Task 3:

Use ChatGPT to identify adjectives that collocate with the word 'food,' and use your own words to provide examples. Document the questions and responses by recording them in a document or taking screenshots. Compare your findings with your peers.

Task 4:

As the designated ambassador of China at COP28 in Dubai, use ChatGPT to brainstorm ideas for a brief talk on China's approach to climate change and the measures they are undertaking. Document the questions and responses by recording them in a document or taking screenshots. Be prepared to deliver your speech to the class.

Task 5:

Use ChatGPT to proofread your first draft. Share both your initial and improved drafts with your teacher, and include a brief reflection note on what you learned from using ChatGPT in your writing process.

By incorporating these tasks into the class, students will confidently use this AI tool and benefit from interactive conversations with ChatGPT. Furthermore, asking students to document or screenshot their findings will foster this AI tool's responsible and guided use. Integrating ChatGPT into designing instructional tasks will impact class dynamics, enhancing engagement, motivation, and personalized learning. In addition, teachers can create a list of Dos and Don'ts for students' use of ChatGPT. Here are some illustrative examples to guide students.

Do: Ask ChatGPT to clarify the requirements of your homework.

Do not: Never ask ChatGPT to complete your homework.

3.2 Role of ChatGPT in Student Learning

In support of implementing this AI tool in education, we contend that students can perceive ChatGPT not only as a conversational companion but also as a proficient peer, language assistant, and scholar's aid in their learning journey. ChatGPT can enhance engagement, foster critical thinking, and provide personalized learning. By approaching this AI tool in these multifaceted roles, students can harness its potential for a more enriching learning experience.

A Conversational Companion

ChatGPT can be a supportive conversational companion, friend, or a guide for students searching for information [21, 30, 57]. Chatterjee and Dethlefs [30] assert, "With all of our conversations, it was clear that the ChatGPT could function as our friend, philosopher, and guide" (p. 2). The more questions students ask, the more elaborated and refined responses they obtain. This ongoing practice can deepen their understanding and increase the benefits gained from this AI tool. Within the framework of a two-way conversation, we believe both the student and ChatGPT can actively contribute to the co-construction of meaning, which is at the core of social constructivism [49]. Commenting on the benefits gained from using ChatGPT, [30] highlight a distinctive feature:

What sets the ChatGPT apart from any other model ever released is its ability to continually interact with a human naturally and provide stimuli/inputs to the user as well as ask interesting questions back. This makes it possible to have long (and potentially never-ending) conversations with the Chabot until the interaction eventually dries out [30, p. 1].

A Proficient Peer

For learners seeking clarification from ChatGPT regarding a writing prompt, a phrase, an expression, or the meaning of a technical word within a particular context, ChatGPT can assume the role of a proficient peer who provides scaffold, support, and guidance to the less proficient peer within their ZPDs [50]. Students can use online Thesauruses, online dictionaries, or simply Google to understand a specific word's meaning or translate it into their native language. However, ChatGPT extends beyond mere definition-seeking or translation, as it allows students to engage in conversations, asking for elaboration and examples or requesting the word to be used in a sentence. If the attempts to understand the meaning are unclear, students can still ask for multiple explanations, and ChatGPT can actively seek feedback to refine and regenerate its responses. The entire process of asking questions, producing responses, and refining them is a valuable personalized learning experience for the student. Along this line, [58] asserts that students can cultivate the ability to engage in critical-thinking conversations with ChatGPT if they receive proper training to use it responsibly and appropriately, avoiding the temptation to have their work completed by this AI tool.

A Language Assistant

In the context of second language learning, ESL students frequently struggle with word choice and collocation in the target language. They may understand the meanings of individual words and their synonyms and antonyms [59]. However, they often struggle to understand which words naturally go together and how to use them authentically across various contexts [60]. This AI tool can assist them with the target words in diverse contexts and generate multiple examples based on student's queries. In this context, [61] point out that ChatGPT can engage students in genuine human-like conversations, define words and sentences, generate writing topics, and direct students in the process of writing. Furthermore, when writing a paragraph, students might have limited information about their topic, hindering their writing. ChatGPT can give them many ideas about their topics, provided they pose their questions properly. It can also proofread and revise their work and detect grammar errors and misspelled words, adding more clarity to the content [56]. Rather than asking the AI tool to write entire paragraphs, a practice considered plagiarism, teachers should encourage students and train them to seek assistance in refining, proofreading, or revising their paragraphs. This way of approaching ChatGPT for assistance can help cultivate their writing skills while adhering to ethical standards [62, 63]. While teachers can support students in accomplishing the abovementioned tasks, ChatGPT can be a valuable alternative for personalized learning, tutoring, and self-study [57].

Scholar's Aid

To get assistance with their reports or scholastic research, students can ask ChatGPT how to plan, search for information, select the best strategies for limiting the scope of their research or refining their reports, and summarize and outline information to improve the quality of their work [56, 57]. Training students to ask proper, clear, and insightful questions and to obtain valuable information about handling reports and research with the assistance of ChatGPT can be rewarding as it can help cultivate their higher-order thinking and research skills [22]. In essence, 21st-century educators expect students to seek, analyze, synthesize, and evaluate information autonomously [38]. Additionally, they need to encourage them to develop their critical thinking, problem-solving, decision-making, and digital and information literacy skills, which are essential to cope with the ever-evolving world [53, 56, 64].

Therefore, ChatGPT can be a valuable tool to assist students in fulfilling this role. In the context of cultivating their research skills, [56] proposes a novel concept regarding conducting research through ChatGPT, which he terms "reverse searching" [56, p. 4]. This new approach entails students attempting to use outputs to discover supporting evidence for texts generated by ChatGPT. Halaweh [56] elucidates that students should pose appropriate, specific questions, assess and compare findings, and make sound judgments. To implement all discussed ideas in practice, here are some guiding questions for students seeking assistance from ChatGPT with their learning.

Suggested questions for interacting with ChatGPT regarding the meaning of difficult words.

- Can you explain the meaning of the word (…)?
- What is the synonym of the word (…)?
- What is the antonym of the word (…)? Can you give me an example? Can you simplify your response to make it easier for me to understand?
- I came across the term (difficult word) in my reading. Can you help me understand its meaning? Can you put it in a sentence? Does it have other meanings? Can you explain the difference between the two meanings?
- How can I use this word (difficult word) in a sentence?
- Which word collocate with the word (…)?
- I need an adjective that goes with this word (…)
- Which prefixes or suffixes are associated with the word (difficult word)?
- Is the word (difficult word) formal or informal?
- Can you suggest another example where I can see (difficult word) used in context?
- The word (…) has multiple meanings. What other meanings does it have? Can you help me understand the difference between these multiple meanings?
- The following are suggested questions for interacting with ChatGPT regarding scholastic or academic research planning
- Can you help me plan my research project?
- Can you provide me briefly with some ideas regarding this topic?
- What shall I consider when I outline my research paper?
- How can I organize my ideas logically in the outline phase?
- Which of these ideas (write your ideas) should come first? Second? Last?
- Can you help me with reliable resources for my research?
- Can you explain the process of proper citing of the reference page?
- Can you explain the difference between MLA and APA styles regarding citing an e-book? Article? Website? Or a book?
- Revise my citation. Did I cite it appropriately?
- Can you suggest tips for staying organized throughout the research, from planning to referencing?

4 Summary

This paper argues that teachers are the architects of teaching and learning. Therefore, to successfully incorporate ChatGPT into classroom instruction, they should adopt a well-structured and clear pedagogic approach accompanied by guided instructions on using this AI tool. This practice can play a pivotal role in reshaping classroom dynamics and aiding students in cultivating and nurturing cognitive and metacognitive skills, including critical thinking, problem-solving, and other core competencies. The authors believe that teachers' instructional guidance can help enhance 21st-century skills for a generation poised to excel in an increasingly technology-driven and interconnected world.

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The Future of AI in Education

Redefining Educational Terrain: The Integration Journey of ChatGPT

Said A. Salloum, Amina Almarzouqi, Ahmad Aburayya, Fanar Shwedeh, Brihan Fatin, Zainab Al Ghurabli, Tamadher Al Dabbagh, and Raghad Alfaisal

1 Introduction

Artificial intelligence (AI) is not just a fleeting buzzword; it has positioned itself as an avant-garde technological marvel, radically altering diverse facets of human existence worldwide [1–5]. A paramount instance underscoring this transformative journey is the advent of ChatGPT, an abbreviation for Chat Generative Pre-Trained Transformer. More than just a sophisticated AI model, ChatGPT epitomizes a powerful digital ally, especially for learners. From crafting intricate term papers to spinning riveting short stories, elucidating complex concepts, and even authoring novels, ChatGPT's capabilities have left a profound imprint on the academic landscape. Notably, its escalating prominence hasn't escaped the discerning eyes of educational stalwarts. Institutions like American University, for instance, have been drawn into a maelstrom of contemplation and caution. There's an emerging consensus within the academic corridors of such establishments that ChatGPT can effortlessly

S. A. Salloum (✉)
Health Economic and Financing Group, University of Sharjah, Sharjah, UAE
e-mail: ssalloum@sharjah.ac.ae

School of Science, Engineering, and Environment, University of Salford, Salford, UK

A. Almarzouqi
Department of Health Service Administration, College of Health Sciences, University of Sharjah, Sharjah, UAE

A. Aburayya · F. Shwedeh
MBA Department, College of Business, City University Ajman, Ajman, UAE

B. Fatin · Z. Al Ghurabli · T. Al Dabbagh
HRM Department, College of Business, City University Ajman, Ajman, UAE

R. Alfaisal
Faculty of Computing and Meta-Technology, Universiti Pendidikan Sultan Idris, Tanjong Malim, Malaysia

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churn out coherent paragraphs, construct scholarly research papers of commendable quality, and even deftly tackle examination's questions [6, 7].

The foray of sophisticated natural language processing (NLP) models, such as ChatGPT, into the realm of education heralds an era where information isn't just accessible; it's tailored, dynamic, and interactive. Drawing a parallel with time-tested platforms like Google elucidates a paradigm shift. While the latter has been an undoubted stalwart, ChatGPT's meteoric ascent in popularity is undeniable. It isn't merely its expansive knowledge reservoir, spanning books, scholarly articles, and a plethora of websites, that captivates; it's the AI's uncanny knack to fathom intricate user intent and reciprocate with unmatched precision that renders it invaluable for multifarious academic pursuits [6, 8].

Historical excursions into ChatGPT's influence reveal an eclectic mix of narratives. Its tentacles have spread across diverse domains, from healthcare and engineering to, evidently, education. While students find in it an academic confidante, medical professionals discern potential applications, envisioning patient consultations or intricate diagnoses. Yet, like every technological marvel, ChatGPT is ensnared in its own web of challenges. Ethical conundrums, especially concerning originality, creativity, and data privacy, have ignited fervent debates [6, 8–11]. Set against this vibrant backdrop, our research endeavors to delve deep into the intricacies of student perceptions. We aim to unearth the determinants that drive students towards e-learning via ChatGPT, and in doing so, discern the motivations and reservations sculpting their inclinations towards this AI-driven educational revolution.

Our exploration contributes to the existing corpus of knowledge in multiple ways. Firstly, while several studies have broached the topic of AI in education, ours offers a granular focus on ChatGPT, plugging an evident research gap. Secondly, by synthesizing a range of factors—cognitive, emotional, and technical—we present a holistic analysis of student reception. Finally, our findings provide tangible insights for educators, AI developers, and policymakers, offering actionable strategies to optimize AI integration in learning environments.

2 The Theoretical Framework

In the realm of artificial intelligence-driven platforms, the underlying system's quality, and the information it produces play pivotal roles in influencing user engagement and adoption. As digital learning landscapes evolve, tools like ChatGPT emerge as crucial agents, interlinking system robustness with the quality of generated content. The ensuing theoretical framework seeks to elucidate these relationships. It postulates that ChatGPT's system quality inherently shapes its information's caliber, which in turn bolsters the propensity of users to engage with educational platforms. Moreover, the intrinsic quality of the system is also posited to directly amplify users' inclination towards leveraging such learning platforms.

2.1 The System Quality

System quality stands out due to distinct attributes that elevate the value of technology. These attributes encompass reliability, usability, and functionality, which collectively can enhance the probability of technology acceptance and integration. The aspect of reliability signifies that when users deem a technology as dependable, their trust in its capabilities rises. Conversely, a system viewed as inconsistent can make users apprehensive about leveraging it for crucial tasks or decision-making. Usability underscores the system's intuitiveness and user-friendliness. A system deemed straightforward, with clear guidelines, will invariably pique users' interest. Functionality, on the other hand, equips users with the means to grasp and operate the technology adeptly [7, 8]. In essence, superior system quality fosters increased technological trust among users. Continual system evaluation, coupled with user feedback, can pinpoint enhancement opportunities, ensuring the technology remains attuned to user requirements.

2.2 Information Quality

The perceived value of information technology hinges on the precision, relevance, and trustworthiness of the information it delivers. This perception delineates the significance of the information, categorizing it as either vital or less essential. Notably, when users deem the information to be pertinent and current, their trust in its accuracy and comprehensiveness grows. The caliber of this information plays a pivotal role in influencing technology acceptance and utilization. If users ascertain that a technology furnishes high-quality information, their inclination to embrace it increases [12–14]. Several determinants shape the quality of the information a technology dispenses. These comprise the data's accuracy and thoroughness, the credibility of its sources, and its promptness. Furthermore, the intuitiveness and accessibility of the technology can sway the quality of its information. For instance, convoluted or challenging technologies might impede users from accessing required information or augment the likelihood of input errors [15, 16].

2.3 Perceived Learning Value (PLV)

Perceived learning value (PLV) embodies the advantages students identify when they utilize technology, such as seamless information access, time conservation, and decreased strain. When students recognize considerable benefits in a technological tool, their inclination towards its sustained adoption grows. For academic institutions, this recognized worth is vital for carving a distinctive edge via technology. Presenting students with technological solutions that have evident merits over other

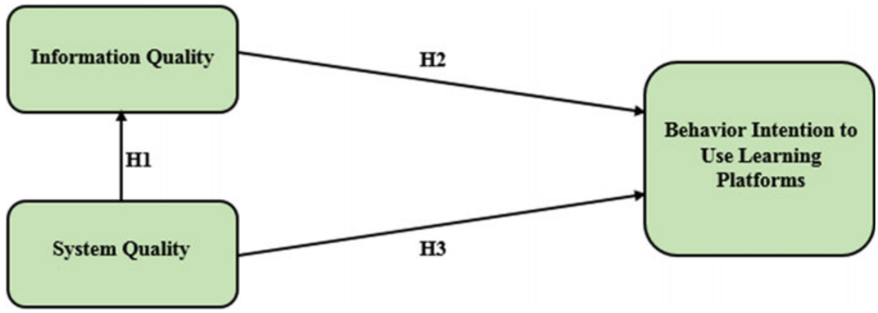

Fig. 1 Research model

choices enables institutions to magnetize and secure their student base, thereby amplifying their stature and achievements. In a nutshell, the value that students discern in technology is a linchpin for educational bodies striving to deploy an efficacious tech strategy. Concentrating on the tangible advantages that students associate with such technology allows educational entities to foster a robust competitive stance, ensuring consistent student engagement and loyalty [17, 18].

Based on the aforementioned discussion, the following hypotheses are posited:

H1: The system quality of ChatGPT positively influences its information quality.

H2: The information quality of ChatGPT enhances the inclination to engage with educational platforms.

H3: The system quality of ChatGPT positively affects the inclination to utilize learning platforms (Fig. 1).

3 Research Methodology

3.1 Data Collection

Online surveys were administered to students from collaborating academic institutions across the UAE. The data collection spanned from January 20, 2023, to May 20, 2023. Of the 300 surveys distributed at random by the research committee, there was a commendable 93% return rate. From the returned surveys, 278 were deemed complete and valid, while 22 were excluded owing to their partial completion. It's worth noting that the 278 valid responses align closely with the suggested sample size of 285 out of a potential 1100, as posited by Krejcie and Morgan [25]. This sample size marginally deviates from the proposed minimum yet remains substantial for the study. Consequently, to validate the proposed hypotheses, we employed a structural equation modeling analysis [26]. Importantly, our hypotheses are rooted deeply within the historic progression of artificial intelligence. The research analysis was executed utilizing Structural Equation Modeling (SEM) via SmartPLS Version

3.2.7 to assess the measurement model, culminating in the deployment of the Final Path Model for intricate interventions.

3.2 Demographic Information

Figure 2 presents a comprehensive breakdown of the demographic and individual characteristics of the participants. An even distribution was observed in terms of gender, with both females and males representing 50% of the sample. In terms of age distribution, a majority (63%) of the respondents fell within the 18 to 29 age group, while the remaining participants were aged 29 and above. The educational background of the respondents varied considerably. Specifically, the distribution was as follows: diploma (4%), advanced diploma (4%), undergraduate degree (72%), postgraduate degree (17%), and doctorate (3%). To identify participants amenable to volunteering, the research employed the "purposive sampling method," as endorsed by Salloum and Shaalan [19]. This study integrated participants from diverse academic institutions, ensuring a rich representation across age groups and educational milestones. For the meticulous processing and analysis of the demographic data, the research team employed IBM SPSS Statistics version 23.

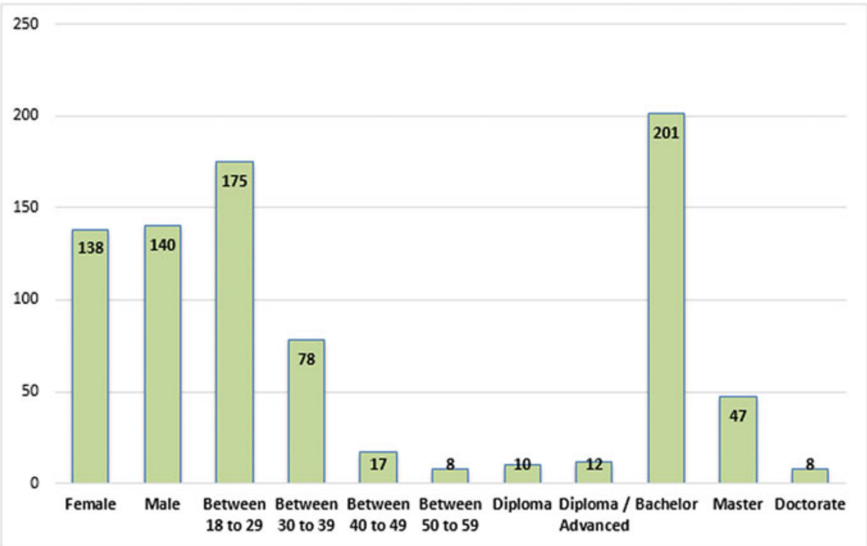

Fig. 2 Demographic information of the participants ($n = 278$)

Table 1 Assessment indicators

Constructs	Items	Instrument	Sources
Behavior intention to use learning platforms	BI1	ChatGPT presents a valuable chance to explore	[20, 21]
	BI2	ChatGPT provides an advantageous opportunity for exploration	
System quality	SQ1	The capabilities of ChatGPT are significant in executing my educational assignments	[22, 23]
	SQ2	The efficiency of ChatGPT in completing educational tasks within a limited time span is noteworthy	
	SQ3	The proficiency of ChatGPT is high	
Information quality	IQ1	ChatGPT delivers more detailed information to me	[22, 23]
	IQ2	ChatGPT indeed offers more in-depth information	
	IQ3	ChatGPT supplies me with valuable information for my daily activities	

3.3 Study Instrument

In this study, we employed a survey to validate the hypothesis. Three variables were carefully selected as reliable markers, leading to the integration of 8 additional elements into the survey. The foundation for these variables can be found in Table 1, aimed at enhancing the relevance of the research variables and providing supportive data from various established studies that reinforce the current framework. In the end, necessary adjustments were made to the survey items based on previous research.

4 Findings and Discussion

4.1 Data Analysis

For this study, we opted for the partial least squares-structural equation modeling (PLS-SEM) approach, utilizing SmartPLS V 3.2.7 [3, 24–26] for data processing. The analytical procedure embraced a two-tiered evaluation method, encompassing both the measurement and structural models [5, 27–30]. Several considerations influenced the decision to apply PLS-SEM [31–34]. Firstly, we prioritized the analysis of the hypothesized theoretical construct through PLS-SEM [30, 35–37]. Next, PLS-SEM was deemed suitable for adeptly managing the explorative data amassed from the conceptual models [3, 38–41]. Additionally, the PLS-SEM evaluation was applied to the model holistically, as opposed to fragmenting it [42]. In conclusion, a simultaneous examination of both structural and measurement models was executed with PLS-SEM, highlighting its capacity to produce and secure precise measurements [43].

Table 2 Convergent validity results

Constructs	Items	Factor loading	Cronbach's alpha	CR	AVE
Behavior intention to use learning platforms	BI1	0.897	0.704	0.871	0.771
	BI2	0.859			
Information quality	IQ1	0.873	0.883	0.878	0.520
	IQ2	0.806			
	IQ3	0.840			
System quality	SQ1	0.731	0.778	0.746	0.528
	SQ2	0.724			
	SQ3	0.914			

4.1.1 Convergent Validity

The Measurement Model's evaluation [27] was anchored in the tenets of construct validity, which includes both discriminant and convergent validity. It also incorporated construct reliability metrics such as Cronbach's alpha (CA) and composite reliability (CR) [44–51]. As depicted in Table 2, the CA values, representing construct reliability, range from 0.704 to 0.883. These metrics fall below the recommended threshold of 0.7 [52]. However, a review of Table 2 illustrates that the composite reliability (CR) values stretch between 0.746 and 0.878, surpassing the established benchmark [51]. To determine convergent validity, it becomes imperative to scrutinize the average variance extracted (AVE) alongside the factor loadings [27]. Table 2, besides the metrics already mentioned, demonstrates that all factor loading metrics surpass the standard threshold of 0.7. Additionally, Table 2 outlines the AVE metrics, which all rise above the 0.5 guideline, with values oscillating between 0.520 and 0.771, barring the previously mentioned figures. Given the data and metrics discussed, it can be inferred that the study successfully establishes convergent validity.

4.1.2 Discriminant Validity

To determine the discriminant validity, this study opted to re-examine the criteria using both the Heterotrait-Monotrait ratio (HTMT) and the Fornell-Larker criterion [24]. As presented in Table 3, the results strongly support the validity of the criteria: every Average Variance Extracted (AVE) and its square root demonstrated a more pronounced association with their designated constructs [47, 53–56]. Table 4 details the outcomes from the HTMT ratio analysis. Notably, every construct remained below the '0.85' benchmark [57]. This indicates that the HTMT ratio met the set standard, enabling an accurate determination of discriminant validity. The insights derived from this study affirm the absence of any issues in the assessment of the measurement model's validity and reliability. Consequently, the gathered data stands poised for effective utilization in examining the structural model.

Table 3 Fornell-Larcker scale

	BI	IQ	SQ
BI	**0.878**		
IQ	0.579	**0.721**	
SQ	0.529	0.522	**0.854**

Table 4 Heterotrait-Monotrait ratio (HTMT)

	BI	IQ	SQ
BI			
IQ	0.718		
SQ	0.654	0.641	

Table 5 R^2 of the endogenous latent variables

Construct	R^2	Results
BI	0.406	Moderate
IQ	0.272	Low

4.2 Hypotheses Testing

In this research, the structural model was meticulously constructed using the Smart PLS software [58–62]. This advanced tool employs the maximum likelihood estimation method, providing a detailed exploration of the intricate relationships between the various theoretical components present in the structural model [12, 63]. This methodological strategy was pivotal for dissecting and analyzing the hypothesized relationships, the outcomes of which are clearly articulated in both Table 5 and Fig. 3. These presentations underscore the structural model's potent predictive prowess [64].

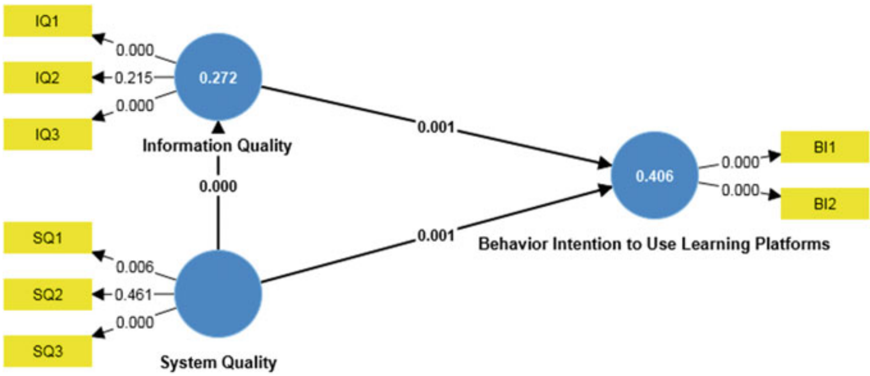

Fig. 3 Model's path coefficient (noteworthy at p** ≤ 0.01, p* < 0.05)

Table 6 Hypotheses-testing of the research model (significant at $p^{**} \le 0.01$, $p^{*} < 0.05$)

H	Relationship	Path	*t*-value	*p*-value	Direction	Decision
H1	SQ -> IQ	0.522	0.665	0.000	Positive	Supported**
H2	IQ -> BI	0.416	3.411	0.001	Positive	Supported**
H3	SQ -> BI	0.312	3.177	0.001	Positive	Supported**

Furthermore, Table 6 comprehensively aggregates the beta (β) values, t-values, and p-values for each hypothesis as derived from the PLS-SEM methodology [44–51]. Upon a rigorous investigation of the collected data, it was evident that the proposed hypotheses H1, H2, and H3 were consistently supported by the empirical evidence. The first hypothesis delved into the relationship between SQ (System Quality) and IQ (Information Quality), revealing a β value of 0.522 and a significance level <0.001. This finding emphasizes the pronounced positive influence of SQ on IQ, affirming the validity of H1. The findings of the research also illuminate meaningful relationships between BI (Behavioral Intention) and various determinants. In particular, BI was found to have a significant positive relationship with IQ, as indicated by a β value of 0.416 and a significance level below 0.001. Additionally, the research discerned that SQ, with a β value of 0.312 and a significance level <0.01, exerts a substantial impact on BI. Thus, the hypotheses H2 and H3 were substantiated by this study.

5 Conclusion

ChatGPT has established itself as an indispensable tool in the realm of higher education by offering bespoke, streamlined learning experiences. Its prowess lies in its ability to furnish individualized feedback and elucidations, thereby assisting students in producing research papers, narratives, and gaining a deeper understanding of intricate subjects. Integrating ChatGPT not only amplifies the accessibility and dissemination of information to educators and learners but also metamorphoses conventional educational settings, infusing them with vibrancy and engagement. The unique aspect of ChatGPT lies in its nuanced comprehension of students' needs and aspirations, delivering adept responses that facilitate diverse academic endeavors. More than just an auxiliary tool, ChatGPT heralds a transformative shift in e-learning, endorsing a more immersive, tailored educational experience, thereby galvanizing students to embrace and leverage AI-infused tools like ChatGPT in their academic pursuits. Delving into the findings, the twin pillars of "Information Quality" and "System Quality" emerge as the paramount determinants that guide students' propensity to harness ChatGPT within educational platforms. Nonetheless, it's prudent to acknowledge the study's constraints. The data harvested originates predominantly from a distinct student demographic in the UAE, potentially limiting the broader applicability of these insights. The cohort comprised chiefly of university students in the UAE, with diverse academic requirements and goals. An avenue for forthcoming research could be to diversify the sample pool, branching out to other institutional types or geographic locales, to enrich the comprehension of ChatGPT's utility. Furthermore, the study's conceptual scaffold zeroes in predominantly on facets related to system integrity and the caliber of information, using a specific moderator to gauge ChatGPT's impacts. Future endeavors in research might consider widening this scope, encompassing additional variables and diverse moderators to excavate other pivotal elements that might shape the receptiveness and application of ChatGPT. Such insights would not only bolster the existing knowledge base but also offer strategic implications for educators, policymakers, and technology developers, ensuring that AI-driven educational tools align seamlessly with evolving pedagogical paradigms.

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Embracing ChatGPT: Ushering in a Revolutionary Phase in Educational Platforms

Said A. Salloum, Amina Almarzouqi, Ahmad Aburayya, Fanar Shwedeh, Brihan Fatin, Zainab Al Ghurabli, Mohamed Ahmad Elbadawi, and Raghad Alfaisal

1 Introduction

The technological epoch has ushered in Artificial Intelligence (AI) as a transformative force, significantly influencing global paradigms [1–8]. A standout example of this sweeping change is ChatGPT, the Chat Generative Pre-Trained Transformer. Recognized for its prowess in aiding students to produce an array of written content from term papers to novels, this tool's advanced functionalities have ignited a surge of discourse at institutions like American University. There's growing intrigue, as ChatGPT demonstrates the prowess to produce college standard research papers and even provide accurate answers to exam queries [9, 10].

Incorporating advanced Natural Language Processing (NLP) models into the academic sphere hints at an unprecedented elevation in information accessibility for the entire academic community. In a comparative analysis with conventional

S. A. Salloum (✉)
Health Economic and Financing Group, University of Sharjah, Sharjah, UAE
e-mail: ssalloum@sharjah.ac.ae

School of Science, Engineering, and Environment, University of Salford, Salford, UK

A. Almarzouqi
Department of Health Service Administration, College of Health Sciences, University of Sharjah, Sharjah, UAE

A. Aburayya · F. Shwedeh
MBA Department, College of Business, City University Ajman, Ajman, UAE

B. Fatin · Z. Al Ghurabli · M. A. Elbadawi
HRM Department, College of Business, City University Ajman, Ajman, UAE

R. Alfaisal
Faculty of Computing and Meta-Technology, Universiti Pendidikan Sultan Idris, Tanjong Malim, Malaysia

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tools such as the Google platform, ChatGPT's meteoric rise and the vast reservoir of resources it provides spanning books to web pages becomes even more pronounced. Its unique capability to discern the nuances of a student's intention and furnish adept responses positions it as a versatile tool for a multitude of academic activities [4, 9, 11].

The multifaceted functionalities of ChatGPT have been dissected from a myriad of lenses, encompassing sectors from education to medicine and engineering. Preliminary studies accentuate its widespread impact, touching various user demographics from students to healthcare professionals. However, ChatGPT's journey is not without its hurdles. Ethical quandaries and concerns about stifling creativity loom large in scholarly discussions [9, 11–14]. Against this backdrop, our study seeks to delve deep into the critical factors underpinning students' inclination towards e-learning via ChatGPT, shedding light on the determinants shaping their intent to harness its potential. Despite the substantial buzz around AI-driven technologies like ChatGPT and their purported benefits in reshaping the educational realm, there remains a conspicuous gap in comprehensively understanding what drives their acceptance. Specifically, while ChatGPT's capabilities span various sectors, its role and potential in the academic sphere needs a more nuanced exploration. This research contributes vitally to literature by honing in ChatGPT, a tool poised at the cusp of AI's transformative potential in education. By unravelling the determinants influencing students' propensities to integrate ChatGPT into their academic arsenal, we aim to provide crucial insights to educators, technologists, and policymakers. These findings will be pivotal in tailoring AI-enabled educational strategies for the future, ensuring they resonate with students' needs and preferences.

2 The Theoretical Framework

The foundation of this research is rooted in a conceptual framework that underscores the mediatory role of personal innovativeness. Within this schema, personal innovativeness serves as a bridge, elucidating the dynamics between perceived learning value, perceived satisfaction, and the broader acceptance of learning platforms. In essence, the structure suggests that the intrinsic drive to innovate within individuals can potentially enhance or modify the direct impacts of learning value and satisfaction perceptions on their readiness to embrace new learning platforms. This theoretical backdrop provides a nuanced lens through which the interplay of these critical variables can be understood, offering a comprehensive perspective on factors influencing the uptake of modern educational tools.

2.1 The Perceived Satisfaction

Perceived satisfaction, in the realm of technology adoption, refers to the extent to which users feel gratified with the tasks they can accomplish and the services they receive through a particular technological tool or platform [15–19]. This sentiment of contentment or dissatisfaction arises from users' personal experiences, encompassing the tool's efficacy, interface design, user-friendliness, and the tangible benefits derived from its utilization. When users resonate positively with a technology, perceiving it as fulfilling and effective, they not only become recurrent users but also act as advocates, often endorsing it to peers and colleagues. This positive word-of-mouth can significantly amplify the tool's adoption rate and user base.

In stark contrast, a technology perceived as lackluster or not meeting the user's expectations can have ripple effects detrimental to its longevity in the market. Discontent users might not only disengage from its usage but could also explore competing solutions that better cater to their needs. Moreover, in today's hyper-connected world, a single negative experience can rapidly cascade through networks, dissuading potential new users and impacting the brand's reputation [20, 21]. Thus, gauging perceived satisfaction is not just a metric of user contentment; it is pivotal in understanding the potential trajectory of a technology's adoption, its market sustainability, and the broader implications it might have on user behavior and preferences. It serves as a compass for developers and marketers alike, indicating where refinements are needed and where strengths should be amplified.

2.2 Perceived Learning Value

The concept of perceived learning value delves deep into the advantages and benefits that students recognize when utilizing technological tools for educational purposes. This perception encompasses a myriad of elements, such as streamlined access to a vast array of information, time efficiency, and a decrease in expended effort to grasp intricate concepts. When students discern that a particular technological solution offers significant value, their allegiance to it grows, culminating in sustained and prolonged usage. For academic establishments, this notion of perceived value isn't just an abstract idea; it's a linchpin for their technological strategy. Providing students with a technological edge that not only meets but exceeds their expectations and offers a superior experience compared to other available tools is paramount. Such a strategy doesn't merely serve the immediate need; it's an investment in the institution's future. It becomes a magnet that draws in prospective students while retaining the existing ones, thereby solidifying the institution's standing, prestige, and long-term viability. In a nutshell, the perceived learning value isn't merely a metric or a byproduct of technology deployment. It's a strategic element, a cornerstone upon which educational entities can build their technological roadmaps. Concentrating on ensuring that the deployed technology resonates with the students' perceived value can significantly

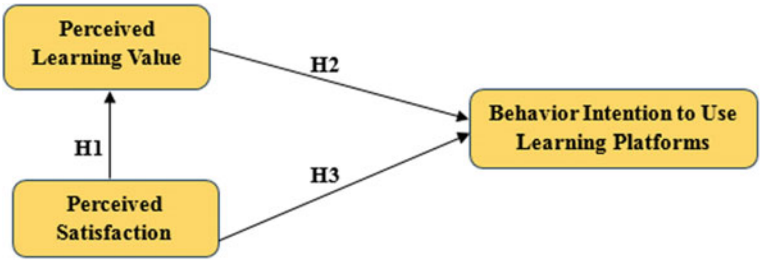

Fig. 1 Research model

bolster an institution's competitive stature, driving student acquisition and retention, and sustaining that momentum over extended periods [22, 23].

Given these foundational understandings, we put forth the ensuing hypotheses:

H1: Perceived satisfaction of ChatGPT positively influences the perceived learning value of ChatGPT.

H2: Perceived learning value of ChatGPT positively influences the behavioral intention to use learning platforms.

H3: Perceived satisfaction of ChatGPT positively influences the behavioral intention to use learning platforms (Fig. 1).

3 Research Methodology

3.1 Data Collection

Students hailing from various affiliated educational institutions across the United Arab Emirates (UAE) were actively engaged in an online survey initiative. This endeavor to gather data commenced on March 10, 2023, and was brought to completion by May 15, 2023. In a strategic approach, the research committee disseminated 300 surveys, chosen indiscriminately. Impressively, an 87% response rate was recorded, underscoring the participants' commitment to this research. Out of this considerable response, 261 submissions were meticulously filled out, while a subset of 39 had to be excluded owing to their incomplete nature. These 261 validated surveys fit comfortably within the parameter set by Krejcie and Morgan [25], which recommends a sample size of 285 participants when considering a total demographic of 1100 individuals. The deliberate choice of a sample size, 261 in this instance, stands in stark contrast to the baseline required sample size, paving the way for the deployment of a structural equation modeling analysis [26]. This methodological choice fortifies the integrity and robustness of the underlying hypotheses. A salient feature of this research journey is its foundation rooted deeply in the annals of artificial intelligence's historical progression. Leveraging the prowess of Structural Equation

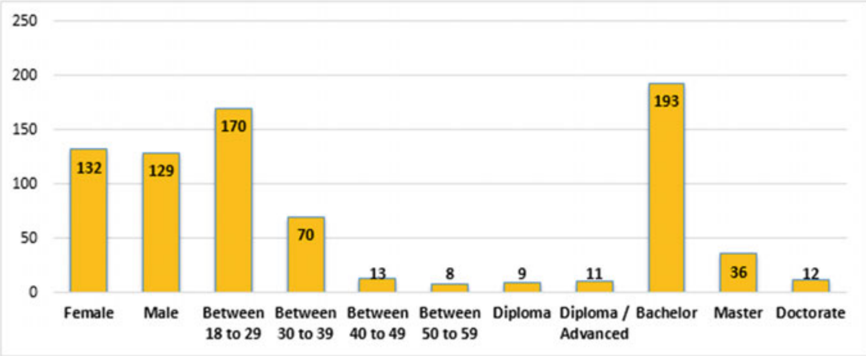

Fig. 2 Demographic information of the participants ($n = 261$)

Modeling (SEM) executed through the sophisticated Smart-PLS Version 3.2.7 platform. The research team undertook a nuanced evaluation of the measurement model. With a commitment to precision, the Final Path Model was orchestrated to navigate the intricacies of multifaceted interventions.

3.2 Demographic Information

Figure 2 delineates the demographic and individual particulars of the participants. The data suggests an almost balanced gender representation among the respondents, with females comprising 51% and males constituting 49%. Age-wise, a significant 65% of the participants fell within the 18 to 29 age group, while the remaining participants were aged 29 and above. Diving deeper into their academic credentials: 3% possessed a diploma, 4% had an advanced diploma, a substantial 74% boasted an undergraduate degree, 14% were armed with a postgraduate degree, and 5% had achieved a doctorate. Whenever participants signaled their eagerness to contribute, the research team opted for the "purposive sampling method" as advocated by Salloum and Shaalan [24]. This research drew from a diverse pool of participants from various academic institutions, ensuring a rich tapestry of ages and educational experiences. For a rigorous analysis of this demographic information, the team utilized IBM SPSS Statistics version 23.

3.3 Study Instrument

In the current investigation, a questionnaire served as the tool to authenticate the postulated theory. Three pivotal factors were meticulously picked as dependable markers, resulting in the integration of 8 novel components into the questionnaire. The

Table 1 Assessment indicators

Constructs	Items	Instrument	Sources
Behavior intention to use learning platforms	BI1	ChatGPT presents a valuable opportunity to explore	[25, 26]
	BI2	ChatGPT provides a worthwhile opportunity for exploration	
The perceived satisfaction	PS1	ChatGPT possesses enhanced tools that streamline the learning process	[27]
	PS2	ChatGPT's innovative capabilities influence my satisfaction level	
	PS3	ChatGPT meets my learning requirements, prompting me to adopt it	
The perceived learning value	PL1	ChatGPT provides enhanced advantages in my learning journey	[22, 23]
	PL2	ChatGPT possesses a unique value across various learning tasks	
	PL3	ChatGPT holds a distinct value that motivates me to embrace the technology	

rationale behind these factors is detailed in Table 1. This elucidation not only seeks to amplify the relevance of the research factors but also amalgamates corroborative data from an array of previous studies, thereby enriching the existing paradigm. Concluding, the research group finessed the items of the questionnaire, drawing insights from antecedent research.

4 Findings and Discussion

4.1 Data Analysis

In this study, data analysis was conducted using the partial least squares-structural equation modeling (PLS-SEM) approach through Smart-PLS V 3.2.7 [28–31]. This analytical process adopted a two-pronged methodology, which evaluated both the measurement and structural models [4, 32–35]. Several foundational reasons underpin the choice of PLS-SEM for this investigation [36–39]. Firstly, the need to evaluate the proposed conceptual framework necessitated the use of PLS-SEM [40–44]. Secondly, PLS-SEM demonstrated proficiency in handling the preliminary data derived from the theoretical models [29, 45–48]. Thirdly, the PLS-SEM approach facilitated a holistic analysis of the entire model without the need to segment it further [49]. Finally, the capability of PLS-SEM to concurrently assess both structural and measurement models was harnessed. The efficacy of PLS-SEM is emphasized by its ability to provide accurate and detailed insights [50].

Table 2 Convergent validity results

Constructs	Items	Factor loading	Cronbach's alpha	CR	AVE
Behavior intention to use learning platforms	BI1	0.863	0.780	0.862	0.757
	BI2	0.877			
Perceived learning value	PL1	0.892	0.817	0.891	0.732
	PL2	0.871			
	PL3	0.801			
Perceived satisfaction	PS1	0.886	0.885	0.843	0.518
	PS2	0.866			
	PS3	0.801			

4.1.1 Convergent Validity

The Measurement Model [32] underwent thorough evaluation anchored on the tenets of construct validity, encompassing both discriminant and convergent validity, in addition to construct reliability, which includes metrics such as Cronbach's alpha (CA) and composite reliability (CR). An exploration of Table 2 reveals the values for Cronbach's alpha (CA)—indicative of construct reliability—span between 0.780 and 0.885. Notably, these figures rest below the commonly endorsed benchmark of 0.7 [51]. Yet, juxtaposing this, Table 2 showcases that the composite reliability (CR) scores oscillate between 0.843 and 0.891, confidently eclipsing the stipulated benchmark [51]. Turning our lens to convergent validity, it's paramount to scrutinize metrics like the average variance extracted (AVE) along with factor loadings [32]. Beyond the aforementioned values, Table 2 divulges that all factor loadings firmly breach the 0.7 criterion. Moreover, the AVE values documented in Table 2 comfortably exceed the 0.5 standard, with earlier figures stretching from 0.518 to 0.757. Drawing from these insights, it becomes evident and justified to infer that the study successfully attained convergent validity.

4.1.2 Discriminant Validity

In gauging discriminant validity, this study turned its lens towards two pivotal criteria, re-evaluating them through the prism of the Heterotrait-Monotrait ratio (HTMT) and the Fornell-Larker criterion [32]. The insights gleaned from Table 3 shed light on the efficacy of the Fornell-Larker criterion in validating the criteria. It is evident that each Average Variance Extracted (AVE) and its square root resonate more robustly with their corresponding constructs [52]. Turning attention to Table 4, it lucidly breaks down the findings from the HTMT ratio. Notably, each construct recorded values beneath the '0.85' benchmark [53], signaling that the HTMT ratio effectively aligns with the prescribed criterion, thus cementing the discriminant validity. This research's outcomes confirm the absence of any hurdles in the realms of validity and

Table 3 Fornell-Larcker scale

	BI	PL	PS
BI	**0.870**		
PL	0.413	**0.856**	
PS	0.544	0.641	**0.720**

Table 4 Heterotrait-Monotrait ratio (HTMT)

	BI	PL	PS
BI			
PL	0.549		
PS	0.893	0.924	

reliability for the Measurement Model. Such a clean slate indicates that the amassed data is ripe for the structural model's subsequent analysis. The meticulous process of validation adopted in this study, devoid of any pitfalls in appraising the Measurement Model's reliability and validity, amplifies the resilience of the research methodology. Such rigor fortifies the trust in the data being instrumental for the structural model evaluation, which, in turn, elevates the overall trustworthiness and veracity of the study's conclusions.

4.2 Hypotheses Testing

In this study, the structural model was adeptly formulated using Smart PLS, an advanced tool that harnesses the power of maximum likelihood estimation [37, 54, 55]. This methodology probes into the intricate relationships threading the different theoretical constructs within the structural model [37, 46, 56–60]. Leveraging this method, the research meticulously scrutinized the set hypotheses, with the findings captured comprehensively in Table 5 and depicted visually in Fig. 3. These results underscore the model's commendable predictive capabilities [61]. Further, Table 6 presents a detailed breakdown of the beta (β) values, t-values, and p-values corresponding to each of the postulated hypotheses, all derived from the rigorous PLS-SEM technique. Delving into the empirical evidence, the study found robust support for the hypotheses H1 and H3.

The first hypothesis of this study delved into the interrelationship between Perceived Satisfaction (PS) and Perceived Learning (PL). Our findings showcased a β

Table 5 R^2 of the endogenous latent variables

Construct	R^2	Results
BI	0.303	Moderate
PL	0.410	Moderate

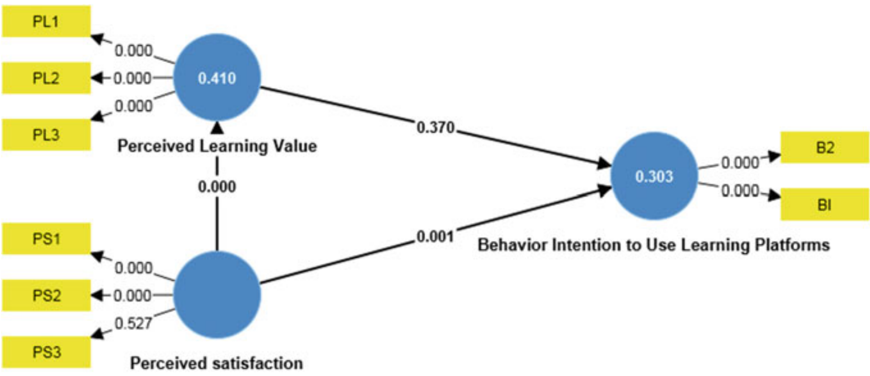

Fig. 3 Model's path coefficient (noteworthy at p** ≤ 0.01, p* < 0.05)

Table 6 Hypotheses-testing of the research model (significant at p** <= 0.01, p* < 0.05)

H	Relationship	Path	*t*-value	*p*-value	Direction	Decision
H1	PS -> PL	0.473	3.388	0.370	Positive	Not supported
H2	PL -> BI	0.110	0.897	0.001	Positive	Supported**
H3	PS -> BI	0.641	8.203	0.000	Positive	Supported**

value of 0.473 and a p-value exceeding 0.05. This specific outcome vividly highlights a discernible positive relationship between PS and PL, which however, counters the expectation set by H1. Delving further into the results, it's evident that Behavioral Intention (BI) manifests significant correlations with various factors. More pointedly, BI exhibits a pronounced positive influence on PL, as demonstrated by a β value of 0.110 and a P-value <0.01. The research further unravels that both PS (with a β value of 0.641 and P-value <0.001) exerts a formidable impact on BI. Based on this empirical evidence, the study lends robust support to hypotheses H2 and H3.

5 Conclusion

ChatGPT stands poised to redefine the landscape of higher education, championing a more bespoke and effective learning journey. With its prowess to deliver nuanced feedback and articulate explanations, it acts as a catalyst for students, aiding them in crafting term papers, weaving short stories, and demystifying complex concepts. The integration of ChatGPT does more than just expand the horizon of accessible information for both educators and learners; it heralds a transformation of the age-old classroom environment, infusing it with dynamism and engagement. Its unique capability to fathom the depths of a student's intentions and serve up adept responses places it as an indispensable tool, unlocking a myriad of educational avenues for students. Beyond this, its latent potential to redefine e-learning by crafting a more immersive and tailored pedagogical experience, which stands testament to the rising inclination among students to embrace AI-driven tools like ChatGPT in their academic voyage. Our research findings underscored that the trio of perceived learning value and perceived satisfaction emerge as pivotal determinants shaping students' propensities to weave ChatGPT into their learning platforms. However, as with all research endeavors, our study isn't devoid of constraints. The data pool was predominantly drawn from a distinct student demographic within the UAE, casting shadows on its universal applicability. While our participants spanned various universities within the UAE, each leveraging the tool for multifarious educational pursuits, the findings might exhibit variance when juxtaposed with other populations or diverse contexts. Potential future investigations could pivot towards more varied demographics, encompassing government institutions or other geographical terrains, to cultivate a more holistic grasp on ChatGPT's academic imprint. Moreover, our study's conceptual scaffolding zeroes in on select dimensions, primarily circling perceived learning value and satisfaction, with a singular moderator gauging ChatGPT's impact. Future endeavors might consider expanding the canvas, roping in a broader spectrum of variables and diverse moderators, to unearth more facets influencing the adoption and utility of ChatGPT. The implications of our research extend beyond just academia. For educators, it underscores the importance of integrating AI tools in curriculum design to enhance engagement and learning outcomes. Policymakers and institutional leaders can harness these insights to champion AI-driven pedagogical strategies, potentially leading to increased student satisfaction and improved academic performance. For AI developers and tech enterprises, understanding these determinants can inform the design and functionalities of future educational AI tools, ensuring they resonate more closely with user needs. On a broader scale, recognizing the pivotal role of AI in shaping the future of education can usher in more collaborative efforts between tech developers, educators, and policymakers, fostering an ecosystem where technology and education coalesce seamlessly.

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Ethical Implications of Using ChatGPT in Educational Environments: A Comprehensive Review

Amina Almarzouqi, Ahmad Aburayya, Raghad Alfaisal, Mohamed Ahmad Elbadawi, and Said A. Salloum

1 Introduction

With the proliferation of artificial wintelligence (AI) tools in the education sector [1–8], ChatGPT, developed by OpenAI, stands out as an exemplar of innovation and utility. This AI-based conversational agent, adept in offering real-time, contextually relevant responses, has reshaped the boundaries of personalized learning and offers an intriguing blend of opportunities and challenges [9]. Its increasing incorporation in academic environments is recognized for enhancing student engagement and individualizing instruction but also casts a spotlight on the pressing need to navigate its ethical landscapes [10].

The allure of ChatGPT in educational contexts is multifaceted. On the one hand, its capability to provide instantaneous, tailored feedback has the potential to redefine pedagogical paradigms, allowing students and educators to engage in deeper, more

A. Almarzouqi
Department of Health Service Administration, College of Health Sciences, University of Sharjah, Sharjah, UAE

A. Almarzouqi · S. A. Salloum (✉)
Health Economic and Financing Group, University of Sharjah, Sharjah, UAE
e-mail: ssalloum@sharjah.ac.ae

A. Aburayya
MBA Department, College of Business, City University Ajman, Ajman, UAE

R. Alfaisal
Faculty of Computing and Meta-Technology, Universiti Pendidikan Sultan Idris, Tanjong Malim, Malaysia

M. A. Elbadawi
HRM Department, College of Business, City University Ajman, Ajman, UAE

S. A. Salloum
School of Science, Engineering, and Environment, University of Salford, Salford, England

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meaningful interactions [11]. Conversely, its very strengths also beckon scrutiny, especially concerning data privacy, potential over-reliance, inherent biases from training data, and the quandary of originality in student outputs [12, 13].

As educators, technologists, and policymakers traverse this dynamic interplay of promise and precaution, the ethical dimensions of ChatGPT's role in education become paramount. The task at hand is to ensure that while the educational ecosystem leverages this AI-driven renaissance, it does so without compromising on the foundational values of fairness, integrity, and authenticity that underpin educational philosophies [14].

While there has been extensive discourse on the merits and limitations of ChatGPT in educational settings, there remains a conspicuous dearth of comprehensive studies addressing its ethical implications. Most existing research predominantly pivots towards either celebrating the AI-driven innovation in pedagogy or critiquing its possible shortcomings [15]. This bifurcation has inadvertently led to an overlook of nuanced ethical challenges that are not merely binary but manifest in multifaceted dimensions. Moreover, few studies have approached this theme with an interdisciplinary lens, merging pedagogical, technological, and ethical frames of reference [16, 17]. Our contribution intends to bridge this lacuna, offering an integrative analysis that extends beyond just identifying the ethical concerns to proposing actionable frameworks for educators and policymakers. By creating a confluence of theoretical perspectives and practical insights, this research aspires to pioneer a holistic understanding and foster informed, ethically-grounded integration of ChatGPT in academic milieus.

1.1 Brief History and Capabilities of ChatGPT

OpenAI, a research organization committed to ensuring artificial general intelligence (AGI) benefits all of humanity, has been at the forefront of natural language processing advancements with its Generative Pre-trained Transformers (GPT) models [9]. ChatGPT, a derivative of these models, has seen rapid progression and adoption across various sectors, including education.

The origin of ChatGPT can be traced back to the development of the GPT architecture. The initial model, GPT, was launched in 2018 and leveraged a modest 117 M parameters [11]. Following its success, OpenAI introduced GPT-2 in February 2019. With 1.5 billion parameters, it represented a significant step forward in terms of complexity and capability [18]. There was initial hesitancy in releasing the fully trained model, given its potential misuse in generating misleading information. This cautious approach by OpenAI highlighted the transformative nature of GPT-2 and the unforeseen challenges it presented. In June 2020, OpenAI released GPT-3, a behemoth with 175 billion parameters, making it one of the most sophisticated language models of its time [19]. With its uncanny ability to generate human-like text based on the prompts it received, GPT-3 quickly became a sensation in the tech community and beyond. The capabilities showcased by GPT-3, from drafting essays to writing software code, heralded a new era in AI's potential applications [20].

ChatGPT, which is inherently based on the GPT models, particularly GPT-3, offers several compelling capabilities:

1. **Natural Language Understanding**: ChatGPT can understand and generate human-like text, making it an ideal tool for chatbots, virtual assistants, and customer support [12].
2. **Content Creation**: Beyond mere conversation, ChatGPT can generate creative content, including poetry, stories, and more.
3. **Task-specific Instructions**: When provided with specific prompts or guidelines, ChatGPT can produce content tailored to particular requirements, such as legal briefs or medical information.
4. **Code Writing**: A surprising utility of ChatGPT is its ability to understand and generate software code based on user prompts [21].
5. **Multilingual Support**: ChatGPT can communicate and generate content in multiple languages, reflecting the diverse dataset it was trained on Radford et al. [22].
6. **Customization**: Developers can fine-tune ChatGPT for specific applications, enhancing its versatility across domains.

The journey of ChatGPT, from its inception with the GPT models to its current state, showcases the rapid advancements in AI and natural language processing. Its diverse capabilities have not only enriched tech-based applications but also broadened horizons for industries, including education. However, as with all transformative technologies, it's crucial to harness its potential responsibly, considering the ethical and societal implications.

1.2 Overview of ChatGPT's Role and Growth in Educational Settings

The adoption of ChatGPT in educational settings can be seen as a natural progression, given the increasing integration of technology in modern classrooms. Initially, the utilization of artificial intelligence in education was met with skepticism [23–27], but as digital tools began proving their worth in enhancing teaching methodologies and student engagement [28–30], they became more accepted [31]. ChatGPT, with its proficiency in natural language processing, was quickly recognized as a potentially valuable asset in educational spaces. A 2020 review highlighted that AI chatbots, like ChatGPT, could offer personalized learning experiences, answer queries, and facilitate peer-like interactions without the constraints of human limitations [32].

The versatility of ChatGPT ensured its applicability across a myriad of educational scenarios. In higher education, for instance, ChatGPT-based tools have been used to assist students with course content inquiries, reducing the workload on teaching assistants and faculty [33]. For younger learners, ChatGPT has facilitated interactive storytelling, aiding in the development of language skills and creativity [34]. Furthermore, ChatGPT's multilingual capabilities have made it invaluable in language learning settings, offering students a chance to converse and practice with a 'virtual native speaker' [35]. Research from the University of Oxford emphasized the benefits of such AI-driven language practice, noting improvements in students' fluency and conversational confidence [36].

Despite the evident growth and potential benefits, the integration of ChatGPT in educational settings has not been without challenges. Concerns regarding over-reliance on AI for educational interactions, potential data privacy issues, and the lack of 'human touch' in learning have been raised by educators and experts alike [37]. However, as OpenAI continues to refine and expand the capabilities of ChatGPT, there's optimism about its evolving role in education. A study from Harvard University projected that AI tools, including platforms like ChatGPT, will continue to grow in educational settings, but their optimal use will lie in complementing human educators rather than replacing them [38].

2 Ethical Considerations

2.1 Privacy and Data Security

In the age of digital learning and online platforms, data privacy and security have emerged as paramount concerns for educators, students, and parents alike. ChatGPT, like other AI models, interacts with vast amounts of data to function efficiently. Given the sensitive nature of student data and the ethical obligations of educational institutions, understanding how ChatGPT handles this data is crucial [39]. OpenAI's ChatGPT collects and uses data primarily to train the model, improve its performance, and ensure user satisfaction. According to OpenAI's privacy policy, the organization is committed to not using the data to build personal user profiles. Specifically, the data entered into ChatGPT is utilized to enhance the model's capabilities, but individual inputs are not stored and cannot be traced back to specific users [40]. This means, in an educational context, students' questions, answers, and interactions with ChatGPT are processed without being tied to their personal identities.

Despite OpenAI's commitment to data privacy, concerns may arise. The most prevalent is the inadvertent sharing of personally identifiable information (PII) by students. For instance, a student might provide personal details during an interaction with ChatGPT. While OpenAI asserts that these interactions are not stored or used for profiling, educators need to ensure students are aware of best practices while interacting with such platforms [12]. Another safeguard in place is the "data expiry" practice, where OpenAI ensures that the data fed into the models is not stored for prolonged periods, thus minimizing potential breaches.

Relative to other digital tools in the educational sector, ChatGPT, under the umbrella of OpenAI, provides a robust framework for data security. For instance, certain Learning Management Systems (LMS) store student data for extended periods, sometimes leading to concerns about potential misuse [41]. On the contrary, the transient nature of data storage in ChatGPT positions it as a safer alternative in terms of data longevity.

Data privacy and security are undeniably critical in today's educational landscape. With the integration of AI tools like ChatGPT into classrooms, understanding, and ensuring the protection of student data is imperative. While no system can claim absolute security, OpenAI's practices concerning ChatGPT provide a degree of confidence in its use in educational settings. However, continuous vigilance, combined with educating students about safe online interactions, will always remain at the forefront of digital education's ethical considerations [42].

2.2 Bias and Fairness

AI models like ChatGPT are trained on vast amounts of data sourced from the web, and as such, can sometimes inadvertently inherit biases present in that data [43]. As educational tools, it's paramount to ensure that these platforms deliver information impartially and fairly [44–46]. Analyzing the potential biases in ChatGPT's responses helps stakeholders, such as educators and students, understand its capabilities and limitations in a pedagogical context. Artificial intelligence, by design, learns from the data it's trained on. If the training data contains biases, the model could amplify or propagate those biases [47]. In the context of ChatGPT, it's trained on a mixture of licensed data, data created by human trainers, and publicly available data, meaning it has been exposed to diverse content—both in terms of quality and bias.

Studies have indicated that large language models like ChatGPT can exhibit biases in their responses [13]. For instance, the model might provide stereotypes or biased viewpoints when prompted with ambiguous queries. These biases, while unintended, are reflections of the vast and diverse data it has been trained on. Given that some of this data contains biased or skewed perspectives, the model may, at times, echo these inaccuracies.

OpenAI, the organization behind ChatGPT, acknowledges the existence of biases in its models and is actively working to reduce them. They are investing in research and engineering to minimize both glaring and subtle biases in how ChatGPT responds to different inputs. Moreover, OpenAI is seeking external input and partnerships, including third-party audits of its safety and policy efforts [48]. These endeavors highlight a commitment to fairness and impartiality in AI responses. For educators leveraging ChatGPT as an instructional tool, awareness of its potential biases is essential [49]. They should be prepared to address any skewed information or viewpoints the model might produce. Moreover, this scenario provides a unique teaching opportunity, educators can engage students in critical thinking exercises, challenging them to analyze and question the information provided by AI tools.

While AI models like ChatGPT hold immense potential for enhancing educational experiences, their susceptibility to biases underscores the importance of cautious and informed adoption. Continuous efforts by developers, combined with vigilant educators, can ensure that these tools are used ethically and responsibly in educational settings [50].

2.3 Authentic Learning

The increasing reliance on artificial intelligence (AI) tools like ChatGPT in educational settings prompts questions regarding their influence on authentic learning and the potential propagation of plagiarism. While ChatGPT provides valuable insights, its usage can blur the boundaries of originality and raise ethical concerns related to academic honesty [51]. Authentic learning, rooted in constructivist pedagogy, emphasizes real-world relevance, active engagement, and the construction of knowledge [52]. It's a process where learners are actively involved in their educational journey, grappling with genuine problems and collaborating with peers [53]. AI tools, while revolutionary, present challenges in ensuring students remain active agents in their learning.

ChatGPT, a state-of-the-art language model, can generate content on a plethora of topics. Some students, especially in higher education settings, might be tempted to use such tools for assignments, thereby sidestepping the effort traditionally required in academic writing [54]. This not only compromises the integrity of their work but also contravenes academic honesty policies and undermines the educational process [55]. The definition of originality is undergoing a metamorphosis in the digital age. AI tools can produce content that, while not directly copied from existing sources, isn't the original work of the user either [56]. This raises the question: Can content generated by AI be considered plagiarized? Universities, colleges, and academic institutions are grappling with these nuances and are in the process of adapting their plagiarism detection tools and policies [57]. Institutions are beginning to develop guidelines and policies regarding the use of AI tools in academic settings. While some may prohibit their use entirely, others may allow them with proper citation. The key lies in fostering an environment that values academic integrity and ensures students understand the implications of their choices [58]. Moreover, educators can incorporate assignments and evaluation methods that prioritize critical thinking, analysis, and originality, making it challenging to solely rely on AI-generated content [59].

The advent of powerful AI tools in education, like ChatGPT, ushers in both opportunities and challenges. While they can be valuable for research and supplemental understanding, the potential for misuse cannot be overlooked. As AI continues to permeate the educational landscape, a concerted effort from educators, institutions, and students is crucial to uphold the tenets of authentic learning and academic integrity.

2.4 Dependency

With the surge in artificial intelligence (AI) applications in education [44, 60–63], including platforms like ChatGPT, there's a growing concern about the potential for students to become overly dependent on these tools for learning. Such over-reliance can impede critical thinking, problem-solving, and other essential skills, prompting educators and researchers to deliberate the long-term implications of this trend [64]. AI-powered educational tools, such as intelligent tutoring systems, adaptive learning platforms, and virtual assistants, offer personalized learning experiences, instant feedback, and vast information at students' fingertips [65]. While these advantages can enhance learning, they can also create an environment where students might bypass essential cognitive processes, leaning too heavily on AI to provide answers and solutions [66].

When students habitually turn to AI for answers, they might circumvent the cognitive struggle often required in the learning process. This "productive struggle", where learners grapple with problems, make mistakes, and learn from them, is crucial for deep understanding and long-term retention [67]. Over dependency on AI might deprive students of these invaluable learning moments, leading to superficial understanding and diminished skill development [68]. Education is not solely about content absorption; it's also about developing interpersonal skills, collaborating, and engaging in discussions. Over reliance on AI tools can reduce peer-to-peer interactions, potentially impacting students' socio-emotional development and their ability to work in teams in real-world scenarios [69].

Many educators express reservations about the unchecked use of AI tools in classrooms. While recognizing their benefits, they emphasize the importance of blended learning, combining traditional teaching methods with technology to ensure a holistic learning experience [70]. Efforts are being made to design curricula that encourage students to use AI as a supplementary tool rather than a primary source of information, thereby promoting balanced and integrated learning [71]. The incorporation of AI in education presents transformative potential, but with it comes the responsibility of ensuring that students engage with these tools judiciously. Striking the right balance between leveraging AI's capabilities and maintaining active, engaged learning is paramount. As education continues to evolve in the AI era, it remains crucial to prioritize foundational skills and cognitive development alongside technological advancements [72].

2.5 Accessibility

As digital technologies become central to modern education, the issue of accessibility emerges prominently. Tools like ChatGPT offer a plethora of benefits, but their full potential can only be harnessed if all students, irrespective of their socioeconomic or physical status, have equitable access [73]. Historically, the digital divide has been viewed through the lens of those with internet access and those without [74]. While this remains a pertinent issue, the divide has expanded to consider the quality of access, including the availability of advanced AI-powered educational tools like ChatGPT. Students from lower socioeconomic backgrounds often lack the resources or environments conducive to using such technologies, thereby missing out on the enriched learning experiences they offer [75].

For students with disabilities, AI can be a boon, offering tailored educational experiences that cater to their specific needs [76]. ChatGPT and other conversational AI can be adapted to serve as reading aids, offer explanations in multiple formats, or even facilitate communication for those with speech impairments. However, ensuring these customizations are widespread and standard can be a challenge, and there remains a gap in universal design for learning [77]. While urban centers often benefit from the latest in ed-tech, rural or remote regions might be left behind due to infrastructural limitations or lack of awareness [78]. Ensuring that AI educational tools are accessible to students in these areas is crucial for a holistic educational ecosystem.

Many governments and institutions are cognizant of these disparities and are taking measures to bridge the gap. Initiatives that provide low-cost internet, tech grants for schools, and training for educators can help in making AI tools like ChatGPT more universally accessible [79]. Public–private partnerships, like those seen in some countries, where tech giants collaborate with governments, can also play a pivotal role in expanding access. The promise of AI in education is vast, but without equitable access, its potential remains unrealized for many. As the integration of tools like ChatGPT in education deepens, concerted efforts at multiple levels, from policymakers to tech developers, will be essential to ensure no student is left behind.

3 Case Studies

The use of ChatGPT in educational contexts has given rise to various ethical implications, both constructive and concerning. Through specific case studies, we can garner a deeper understanding of the potential benefits and challenges posed by ChatGPT's integration into classrooms and learning environments.

3.1 Case Study 1: Enhancing Inclusive Education with ChatGPT

Background: A suburban school district, keen on leveraging technology to foster inclusive education, integrated ChatGPT into their learning management system to aid students with special needs [80].

Outcome: ChatGPT proved to be an invaluable asset for students with dyslexia, as it offered explanations in multiple formats, allowing them to grasp concepts at their own pace. Additionally, non-native English-speaking students utilized the platform to clarify linguistic ambiguities, enriching their learning experience [81].

Ethical Implication: This case underscores the potential of AI-driven tools like ChatGPT in democratizing education. However, it also necessitates the importance of ensuring that these tools are devoid of inherent biases, ensuring a neutral and comprehensive learning aid to all students [50].

3.2 Case Study 2: ChatGPT and the Plagiarism Dilemma

Background: At a renowned university, several students were found to have submitted remarkably similar answers on a take-home assignment. Upon investigation, it was revealed that the students had consulted ChatGPT for guidance [82].

Outcome: The incident sparked a debate around the ethical use of AI tools in educational settings. While ChatGPT served as an information source akin to a textbook or a library, the fine line between seeking guidance and directly copying answers blurred, leading to concerns about authentic learning and originality [83].

Ethical Implication: The incident underlines the necessity for clear guidelines on the use of AI tools in education. While they can be valuable aids, ensuring they don't inadvertently promote academic dishonesty is crucial [84].

3.3 Case Study 3: Biased Responses and Historical Revisionism

Background: During a history lesson at a high school, students used ChatGPT to inquire about a specific historical event. The response generated by the model showed a subtle bias, which led to an incomplete and somewhat skewed understanding of the event [17].

Outcome: The teacher, upon realizing the biased information, had to spend additional sessions rectifying the misconceptions. This incident raised concerns about the potential biases embedded within AI models and the implications they might have on shaping young minds [85].

Ethical Implication: Ensuring that AI tools provide unbiased and accurate information is paramount, especially in educational contexts. This case underscores the importance of continuous scrutiny and updates to AI models to ensure the propagation of accurate, unbiased information [86].

The integration of ChatGPT in education, as illustrated by these case studies, is accompanied by various ethical implications. While the potential benefits are immense, there's an inherent responsibility to navigate the challenges and ensure an ethically sound educational experience.

4 Discussion

4.1 Case Benefits of ChatGPT in Education

The integration of ChatGPT in the educational realm brings forth a myriad of advantages. Chief among them is the democratization of education. Asynchronous learning environments, backed by AI-driven tools like ChatGPT, allow for self-paced and personalized learning experiences [87]. This tailoring of educational content, based on a student's pace and style, can lead to increased comprehension and retention [88].

Furthermore, in a globalized world where resources might not always be equitably available, ChatGPT and similar platforms can offer invaluable knowledge access. This bridges the educational gap by allowing students from diverse backgrounds to seek clarifications, understand complex concepts, and receive assistance akin to personalized tutoring [89].

4.2 Ethical Concerns: AI Ethics in Education

While the aforementioned benefits are substantial, they don't eclipse the ethical concerns that accompany the integration of AI in education. Bias is a significant issue. AI models, including ChatGPT, are trained on vast datasets, which may inadvertently include biased information. In the context of education, this can lead to a skewed understanding of certain subjects, potentially perpetuating stereotypes [90]. Moreover, the risk of academic dishonesty, data privacy concerns, and the potential over-reliance on AI tools underscore the need for a critical examination of their role in educational settings [91].

4.3 Broader Implications of AI Ethics in Education

While the aforementioned benefits are substantial, they don't eclipse the ethical concerns that accompany the integration of AI in education. Bias is a significant issue. AI models, including ChatGPT, are trained on vast datasets, which may inadvertently include biased information. In the context of education, this can lead to a skewed understanding of certain subjects, potentially perpetuating stereotypes [90]. Moreover, the risk of academic dishonesty, data privacy concerns, and the potential over-reliance on AI tools underscore the need for a critical examination of their role in educational settings [91].

4.4 Mitigations and Strategies

Addressing these ethical concerns requires a multifaceted approach:

1. **Clear Guidelines**: Institutions must establish guidelines on the ethical use of AI tools, differentiating between 'use as a reference' and 'direct copying' [92].
2. **Transparency in AI**: Open-source models or transparent algorithms can help educators understand how certain conclusions or answers are derived, ensuring they're not perpetuating biases [93].
3. **Education about AI**: As AI becomes more integrated into our lives, there's a need to educate students about AI's capabilities, limitations, and ethical implications. This could be a part of digital literacy curricula [94].
4. **Regular Audits**: Periodic reviews and audits of AI tools can ensure that they remain compliant with the evolving ethical standards and best practices in education.

5 Conclusion

The integration of AI tools like ChatGPT in education offers transformative potential, enhancing inclusivity and personalizing learning experiences. However, as our case studies illustrate, it also introduces challenges such as academic integrity and potential biases. While AI's promise in democratizing education is immense, its ethical implications, from data biases to issues of originality, demand urgent attention. As we embrace this AI-augmented educational landscape, a collective call to action emerges for educators, institutions, and developers. Clear guidelines on AI's ethical use, transparency in its operations, AI ethics education, and regular tool audits are paramount. It's crucial that, even as we leverage AI's capabilities, we remain vigilant, ensuring that the foundational principles of education, authenticity, integrity, and inclusivity remain sacrosanct.

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AI Adoption and Educational Sustainability in Higher Education in the UAE

Fanar Shwedeh, Said A. Salloum, Ahmed Aburayya, Brihan Fatin, Mohamed Ahmed Elbadawi, Zainab Al Ghurabli, and Tamadher Al Dabbagh

1 Background

Embracing Artificial Intelligence (AI) and striving for educational sustainability is paramount in the United Arab Emirates (UAE) because, as a nation, they strongly emphasize education and sustainability [1–7]. Given this, the UAE acknowledges the transformative potential of AI technologies within higher education [7, 8]. Consequently, the UAE sets its sights on becoming a global pioneer in AI adoption and innovation. To this end, substantial investments have been funneled into AI research and development, aligned with the overarching vision of propelling AI advancements across diverse sectors, encompassing smart city safety and education [9–12].

In light of this, the government introduced initiatives and strategies to integrate AI technologies seamlessly into the UAE's educational landscape [13–16]. This strategic move aligns with the UAE's unwavering focus on sustainability, evident through its national agenda prioritizing environmental preservation and energy efficiency. The UAE's objectives in higher education span elevating academic excellence, enhancing student achievements, optimizing operational efficiency, and fostering sustainability practices. In this context, incorporating AI technologies within higher education institutions is a promising avenue for achieving these multifaceted goals [17–21].

By harnessing the capabilities of AI, universities [22–26], and colleges across the UAE can provide tailored support to students, enhance academic outcomes, optimize

F. Shwedeh · A. Aburayya (✉)
College of Business, MBA Department, City University Ajman, Ajman, UAE
e-mail: amaburayya@gmail.com

S. A. Salloum
School of Science, Engineering, and Environment, University of Salford, Salford, UK

B. Fatin · M. A. Elbadawi · Z. Al Ghurabli · T. Al Dabbagh
College of Business, HRM Department, City University Ajman, Ajman, UAE

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resource allocation, and promote energy conservation on campuses. Of particular pertinence in the UAE, given its unwavering commitment to sustainable development, is the utilization of AI to manage energy consumption on campus autonomously [6, 8, 27–34]. This aligns with the nation's endeavors to reduce its carbon footprint and encourage sustainable practices across various sectors, including higher education. In this regard, AI-driven energy management systems hold the potential to contribute to these sustainability objectives significantly [35]. These systems can play a pivotal role in advancing energy conservation and efficiency within higher education campuses through real-time energy data analysis, identification of consumption inefficiencies, and automated energy-saving interventions [13, 35–37].

However, scholars present that the correlation between AI adoption and educational sustainability within higher education institutions is challenging [14, 38]. Among these obstacles is the need to address trust issues vis a vis data privacy and security concerns [39, 40]. Integrating AI technologies necessitates access to substantial volumes of student data [41–45], thereby triggering valid apprehensions regarding data protection and safeguarding students' privacy rights, especially given the multicultural nature of higher education in the UAE [46, 47]. Conversely, this also introduces unique cultural considerations and varying stakeholder needs. These factors may influence the acceptance and implementation of AI technologies within diverse educational contexts, thereby underscoring the importance of careful consideration and adaptation to cater to specific cultural nuances and expectations [48, 49].

Moreover, establishing robust regulatory frameworks and comprehensive policy guidelines is imperative to govern the responsible use of AI within higher education [50–52]. Similarly, ensuring transparency, equity, and accountability in AI algorithms and decision-making processes nurtures trust among stakeholders [53–57], mitigating potential risks and biases. This, in turn, fosters continuous AI adoption within academia, thereby sustaining the progress required for educational development [58, 59]. While existing research offers insights into AI adoption and educational sustainability across various contexts and sectors, a noticeable research gap persists specifically in the higher education landscape of the UAE. Limited research has delved into the distinctive opportunities and challenges encountered when AI technologies are embraced for educational sustainability within UAE higher education institutions [60, 61].

Consequently, this research endeavor seeks to bridge this evidence gap comprehensively. The investigation centers on critical factors that could hinder educational sustainability, encompassing data privacy, security concerns, data policies, and regulatory frameworks. By examining these factors, this study aims to garner a deeper comprehension of the impact of AI adoption on educational sustainability within higher education institutions in the UAE and address the distinctive opportunities and challenges inherent to this region. Considering these challenges, this study will examine the following objectives:

i. To investigate the perceived significant relationship between AI adoption and educational sustainability in higher education

ii. To examine the perceived significant influence of AI policies and regulations on educational sustainability among higher education institutions in the UAE.
iii. To examine the significant relationship between trust (data privacy and security concern) towards educational sustainability.

The fourth and fifth objectives examines the moderating influence of policy and regulations on the relationship between AI adoption and trust on educational sustainability.

2 Literature Review

2.1 Educational Sustainability

Educational sustainability is a critical concept in higher education, encompassing the promotion of academic excellence, student success (performance) [62–65], and operational efficiency while ensuring educational institutions' long-term environmental, social, and economic viability [66–69]. Therefore, achieving educational sustainability requires a multifaceted approach that addresses various factors influencing the educational ecosystem [70–72]. Evidence from existing educational sustainability research highlights several critical factors that influence sustainability attainment; these include the intersection of artificial intelligence (AI) with a focus on data privacy and security concerns, policies and regulatory frameworks, ethical considerations and bias in AI algorithms, students cooperation, teachers ability to give the accurate prompt to the educational technologies to name few [69, 70, 73]. From these numerous factors, this review focuses on exhuming the relationship between trust measured with data privacy and security concerns, data policies and regulatory framework, ethical concerns, and bias in AI algorithms towards higher educational sustainability. To this effect, this study adopts the socio-technical system theory as a philosophical underpinning.

2.2 Socio-Technical Systems Theory

The Socio-Technical Systems Theory provides a holistic framework to comprehend the intricate interplay between data privacy concerns, ethical considerations, and regulatory frameworks within educational institutions [74, 75]. By recognizing the symbiotic relationship between the technical and social dimensions, institutions are believed to craft comprehensive data policies that encompass ethical guidelines and regulatory compliance [74, 76]. According to earlier scholars, this approach not only safeguards privacy and enhances accountability but also ensures that technology is harnessed to align with broader societal values and ensure educational sustainability

[46, 47]. This study buffers deeper into the practical implementation of the Socio-Technical Systems Theory to develop nuanced strategies that synergize technology, ethics, and regulations for a more sustainable data ecosystem within educational settings. Earlier investigations have it that the dynamics of data privacy concerns, ethical considerations, and regulatory frameworks within educational institutions can be effectively analyzed through the lens of the Socio-Technical Systems Theory [77, 78]. This theory posits that systems are not solely technical entities but rather intricate amalgamations of social and technological components that interact and shape each other [79].

Hence, the Socio-Technical Systems Theory underscores the significance of adopting in understanding the interplay between data policies, regulatory frameworks, and the human dimension within educational settings [80]. By recognizing that data privacy is not solely a technological issue but a complex socio-technical challenge, institutions can implement data policies that balance technological advancements and ethical imperatives [81]. The theory emphasizes that effective data policies should encompass technical safeguards and robust ethical guidelines safeguarding individual privacy rights [82]. Meanwhile, ethical considerations become pivotal in steering the design and implementation of data systems, ensuring that data utilization aligns with broader societal values and norms [83].

Within regulatory frameworks, the Socio-Technical Systems Theory illuminates the multifaceted nature of compliance. It recognizes that while technical aspects of data protection are crucial, the human element plays a pivotal role in adherence to regulatory frameworks [46]. Furthermore, the theory asserts that creating a culture of compliance necessitates a delicate balance between technical mechanisms and human awareness [83, 84]. In light of this, institutions that adopt this perspective recognize that regulatory frameworks should not be viewed as standalone technical solutions but as integrated socio-technical measures [85, 86]. Concurrently, the studies of [84, 87] expect them to proactively design policies that consider both the technological intricacies and the human behaviors that can facilitate or impede compliance.

2.3 AI Adoption in Higher Education Institutions—Technical Systems Theory

AI adoption has caught scholars' attention over the past few decades, specifically among higher educational institutions. For example, an investigation by Wang [88] believed that AI adoption among higher education leaders improves efficiencies and amplifies data-informed decision-making. However, it may lead to decision conflicts [89]. AI adoption in higher education institutions has emerged as a topic of substantial interest, with researchers investigating its implications on various facets of the educational landscape. A synthesis of recent studies underscores the multifaceted impact of AI integration, revealing both significant and insignificant findings [90]. Significantly, adopting AI technologies has demonstrated promising potential in enhancing personalized learning experiences for students [90]. Therefore, it is believed that institutions can analyze individual learning patterns, tailoring educational content and pacing to optimize student comprehension and engagement through AI-driven analytics [61, 91].

This augmentation of the learning process is attested by studies showing improved academic performance and increased student satisfaction. Additionally, AI-powered chatbots and virtual assistants have proven valuable in providing real-time support, answering queries, and facilitating administrative tasks, streamlining the student experience [91, 92]. However, amidst these gains, some studies reveal insignificant effects on student outcomes, underscoring the importance of careful implementation and ongoing assessment [93].

Moreover, evidence from earlier empirical investigations shows that using AI-driven predictive analytics yielded significant and insignificant findings regarding student retention rates [94]. Similarly, several studies report noteworthy enhancements in identifying at-risk students, enabling timely interventions to mitigate dropout rates [85, 95]. Therefore, it was suggested that by analyzing historical data and identifying patterns, institutions could proactively identify struggling students, tailor available interventions, and increase students' persistence [96, 97]. On the other hand, some research points to the limitations of these models, citing challenges in accounting for socio-economic and cultural factors that contribute to student attrition [98]. Despite these complexities, the potential impact of adopting AI on student success remains a focal point, suggesting avenues for further [99].

In instructional design, the application of AI-generated content has garnered significant attention, although with mixed results [100]. Several studies highlight the efficiency gains achieved by automating the creation of educational materials and assessments. AI-generated content enables instructors to focus on higher-order teaching strategies while benefiting from personalized learning resources [100]. However, the quality of AI-generated content remains a concern, with some research indicating inconsistencies and lack of contextual relevance [58, 101]. Hence, striking a balance between efficiency gains and maintaining educational quality poses a challenge as institutions navigate this landscape.

2.4 Data Policies and Regulatory Frameworks

Data policies and regulatory frameworks have emerged as pivotal mechanisms for shaping the landscape of data management, particularly within the educational sector. Many studies underscore the crucial role of data policies and regulatory frameworks in safeguarding sensitive information and fostering responsible data practices within academic institutions [50]. Clear and comprehensive policies ensure that data handling meets ethical and legal requirements. For instance, stringent frameworks facilitate the secure collection, storage, and sharing of student data, thereby preserving individual privacy rights [102].

Moreover, established policies engender transparency and accountability, promoting stakeholder trust and augmenting data-driven decision-making processes. Notably, successful implementation of data policies has been associated with improved institutional efficiency, streamlined operations, and effective compliance with emerging data protection regulations [103, 104]. While data policies and regulatory frameworks carry undeniable importance, the literature also reveals some instances where their impact might be less pronounced [105]. In a few cases, robust frameworks only sometimes guarantee uniform adherence across the institution [106]. Hence, challenges often arise in the execution phase, where issues such as lack of awareness, limited resources, or inadequate training can hinder the seamless implementation of policies [105].

Furthermore, the inherent complexity of data management systems and the evolving landscape of data-related regulations can render frameworks susceptible to gaps or inconsistencies [107]. As a result, Coyle et al. [108] posit that the effectiveness of data policies might not always translate into immediate tangible outcomes, necessitating a multifaceted approach that encompasses awareness campaigns, training, and regular assessments to bridge these gaps.

2.5 AI and Trust Issues (Data Privacy and Security Concerns)

In our current digital landscape, trust issues that concern data privacy and security have gained significant traction, particularly as educational institutions grapple with the increasing reliance on digitization and data-driven technologies [109]. This scenario prompts a compelling examination of the interplay between data privacy and security concerns and the broader concept of educational sustainability. Understanding this relationship is essential as it directly affects the enduring viability of educational establishments [110].

At the forefront, the preservation of trust among various stakeholders within the academic realm hinges on implementing data privacy and security measures [10, 111]. Institutions prioritizing safeguarding student data underscore their dedication to upholding individual privacy rights and maintaining the confidentiality of personal information [10]. This foundation of trust becomes pivotal in constructing a sustainable educational environment by fostering positive stakeholder relationships and enhancing overall engagement.

Moreover, the significance of data privacy and security extends to maintaining student confidentiality. By implementing robust security protocols, educational institutions can shield sensitive data from unauthorized access or breaches [36, 112]. This approach creates a secure and comfortable educational setting where students feel comfortable sharing personal information and actively participating in their academic pursuits. This bolstered sense of data security improves educational outcomes and sustained academic accomplishments [112].

Furthermore, the commitment to data privacy and security aligns with ethical data practices, a cornerstone of educational sustainability. Institutions that emphasize these aspects adhere to ethical guidelines and legal mandates, ensuring responsible handling of student data. Such ethical considerations safeguard individual privacy rights, enhancing educational institutions' credibility and integrity [40], Therefore, Cain et al. [39] argues that by adhering to ethical principles, institutions can cultivate an enduring educational ecosystem of transparency, fairness, and accountability.

3 Research Methodology

This study relies on a primary source of data to investigate the intended objectives, whose purpose is to examine the perceived significant relationship between AI adoption, trust measured with data privacy and security, stakeholders' needs, and the moderating role of policy and regulations of the said predictors on educational sustainability [113–116]. Hence, the survey research approach is favored. Given this, a predesigned questionnaire was sent out to institutions in the UAE that have embarked on implementing AI or metaverse in their institutions. Furthermore, the literature review consolidates the theoretical framework, namely socio-technological theory. The adoption of survey research is famous among scholars [117, 118].

3.1 Research Design and Data Collection

The constructs in this investigation were measured through various assertions (proxies) echoed by earlier scholars. Furthermore, these statements were measured using 5-points Likert Scale where, (1 = Strongly Disagree; 2 = Disagree; 3 = Neutral; 4 = Agree; 5 = Strongly Agree).

3.2 Item Measurements

The items used in measuring the constructs in this investigation were carefully crafted from the proxies or claims found in the literature reviewed. These Measurements are presented in the Table 1.

3.3 Measuring Educational Sustainability

This context refers to the steps taken by higher education towards the long-term viability of the institutions, encompassing academic excellence, student success,

Table 1 Presents items measuring educational sustainability

S/no.	Statements	Sources
1	To what extent do you perceive that your educational institution promotes academic excellence, ensuring high-quality education and learning experiences	[66, 67]
2	I believe that the academic performance of my institution is sustainable	[68, 69]
3	In my opinion, my institution efficiently manages its resources and operations to support sustainable education	[71]
4	I believe that my institution is actively taking steps to minimize its environmental impact and promote eco-friendly practices	[122]
5	My university always engage in socially responsible initiatives and activities that benefit the community and society at large	
6	In my opinion, I believe that my institution is considering financial stability and long-term viability	[120]
7	I am aware of ethical considerations and potential biases in AI algorithms used in our educational technologies	

operational efficiency, and environmental, social, and economic responsibility [66–69, 119] by addressing contemporary challenges that relate to data privacy, security, ethics, and AI adoption it is operations [120, 121], to ensure institutional resilience and relevance.

Trust

Before measuring the construct trust, the concept is defined in this study as the confidence reposed by stakeholders in the institution's commitment to responsible data management, ethical practices, and the safeguarding of sensitive information, reflecting the belief that the institution acts with integrity, respects privacy rights, and prioritizes the interests of its constituents. Consequently, as observed from the conceptual review, the construct is bi-dimensional, measured using data privacy and security concerns [36]. Hence, the definitions of the dimensions are also given with the items developed.

Measuring Data Privacy

In this context, data privacy refers to the rigorous protection of personal and sensitive data from unauthorized access, use, or disclosure because it is regarded as a fundamental principle that ensures the preservation of individual privacy rights, reinforcing stakeholders' confidence in the institution's data handling practices.

Measuring Security Concern

Is described as the institution's dedication to safeguarding data integrity and confidentiality by apprehensions and considerations related to the safety and resilience of data, especially sensitive information, against potential threats, breaches, or unauthorized access [36]. Given this, eight items were used in measuring the construct, that is, four items measure data privacy and the other four measures security concern. These are presented in Table 2.

Measuring AI Technologies Adoption

AI adoption in higher educational institutions refers to the strategic integration and utilization of artificial intelligence technologies and tools within the educational ecosystem. This encompasses deploying AI-driven solutions to enhance various aspects of the educational experience, including personalized learning, student support, predictive analytics, and content generation. AI adoption aims to leverage data and automation to optimize educational processes, improve decision-making, and ultimately enhance the quality of education and support services [88]. The adopted items examine AI adoption by impacting students' learning experience, education satisfaction, learning outcomes, predictive analysis of students' retention, instructors' adoption and integration, ethical considerations, and AI resources. Considering these, nine (9) items were generated in this regard. These are presented in Table 3.

Measuring AI Policies and Regulations

In this research context, AI policies and regulations refer to the established guidelines and legal frameworks set by the higher educational institutions that govern

Table 2 Presents items measuring trust (data privacy and security concern)

S/ no.	Dimension	Statement	Sources
1	Data privacy	I trust that our educational institution effectively safeguards my personal data and respects my privacy rights	[120]
2		I believe that our institution prioritizes the confidentiality of student data, which is essential for fostering trust among stakeholders	[10, 111]
3		The commitment to data privacy in our institution contributes to a positive educational environment and enhances stakeholder engagement	[10, 111]
4		Our institution's emphasis on data privacy instills a sense of confidence and trust among students, faculty, and staff	[10, 111]
5	Security concern	I feel that our institution's security protocols effectively protect sensitive data from unauthorized access or breaches	[36]
6		I believed that students freely share information because they believed the educational environment is secure and guarded with robust data security measures	[112]
7		My institution's commitment to data security aligns with ethical data practices, ensuring responsible handling of student data	[39, 40]
8		By adhering to ethical data practices and maintaining data security, I believed my institution fosters an educational ecosystem built on transparency, fairness, and accountability	[39]

Table 3 Items measuring AI adoption

S/no.	Items	Sources
1	I believe that the adoption of AI technologies in our institution has improved operational efficiency and administrative processes	[88, 89]
2	I perceived that AI-powered chatbots and virtual assistants positively contributed to students' satisfaction with student support services and administrative interactions	[91, 92]
3	I believe that the adoption of AI technologies influenced your academic performance and overall success as a student	[38, 93, 120]
4	I believe that AI-driven predictive analytics have been effective in identifying at-risk students and facilitating interventions to improve retention rates	[94]
5	I perceived that the instructors in my institution have effectively integrated AI technologies into their teaching methods and course materials	[58, 100, 101]
6	I believed that the institution management is ethically concerned when drafting ad implementing AI technologies into the university's educational context	
7	I have convenient access to AI-powered tools and resources provided by the institution to support my learning and research	[120]
8	Students are exposed to AI-technologies via training and students' developments	[58, 101]
9	Staffs are exposed to the implemented AI technologies via staffs' training and developments	

the deployment of AI usage to enhance academic prowess within the institution [123]. Considering this definition, the items used in measuring the construct, therefore, encompass the factors (proxies) that predict rules and regulations. These are presented in Table 4.

Population and Sample Size Selection

The research focuses on the student population in higher education institutions in the United Arab Emirates (UAE), comprising international and local students across various educational levels. Given the vastness of this population, where access to the total number of enrolled students could be more practical, the study sought an appropriate sample size to achieve its objectives effectively. The sample size was determined using a power analysis tool [128, 129]. Utilizing the power analysis tool necessitated some preliminary considerations. Firstly, it was essential to identify the required analysis category, which in this case is an F-test. Additionally, the study involved four exogenous variables and one endogenous variable. By inputting these parameters into the power analysis, including effect size, error probability, and power, the tool estimated that a minimum of 89 samples would be sufficient for conducting the necessary analysis and achieving the study's goals. To mitigate the risk of Type I or Type II errors and to account for the possibility of extreme outliers in the response

Table 4 Item measuring policies and regulations

S/no.	Items	Statements	Citations
1	Adherence to ethical and legal standards	My university's data policies align with ethical principles and legal requirements in data management within our institution	[50, 124]
2	Data privacy preservation	I believe my institution's data policies effectively protect the privacy of students and stakeholders by securely handling sensitive information	[52, 125]
3	Transparency and accountability	I believe that my institution's data policies promote transparency and accountability through mechanisms like reporting, auditing, and oversight	[103, 104, 126]
4	Institutional efficiency	I perceived there are improvements in operational efficiency resulting from the implementation of data policies and regulatory frameworks in the university	
5	Compliance with emerging data protection regulations	My university's data policies ensure compliance with the latest data protection regulations to safeguard data effectively	
6	Stakeholder trust and perception	The level of trust and perception among stakeholders (students, parents, faculty, staff) regarding our institution's commitment to responsible data management is high	
7	Mitigation of implementation challenges	My institution effectively addresses challenges related to the implementation of data policies, including awareness, resource allocation, and training	[105, 127]

data, the sample size was conservatively increased to over 200 [130, 131] during the data collection process, which spanned two months. This approach ensures that the research maintains a robust and statistically significant sample size while allowing for potential data anomalies, thereby enhancing the reliability of the study's findings [132].

3.4 Data Analysis and Findings

The SEM analysis was favored as the statistical analysis tool in this study. The rationale for choosing the partial least square structural equation modeling is that the research model is complex and has a higher-order construct. Hence, using other statistical analysis tools might be tricky. Also, we are determined to understand the relationship between the latent variables by maximizing the explained variance [133]. The settings were left at default, and the reflective-reflective scales were favored for

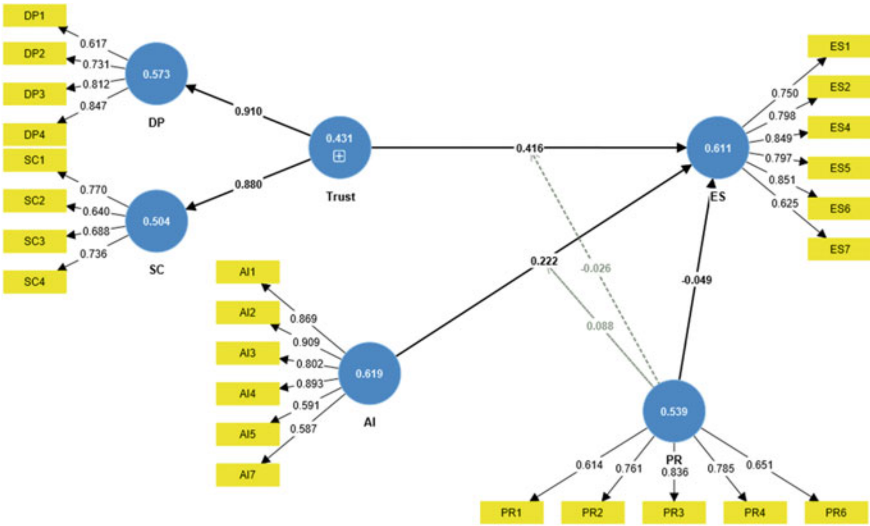

Fig. 1 Structural model evaluation model

the higher-order and lower-order constructs because the items used in measuring those constructs were proxies and not exhausted items [134] (Fig. 1 and Table 5).

The convergent and discriminant validity of the analyzed data was checked using the average variance extracted (AVE). According to Fornell and Larcker [135], the AVE should be greater than 50%, 0.5. Moreover, the scholars posited that the items with lower loadings should be deleted if the AVE condition is not achieved. Given this, items AI6, PR5, PR7, and ES3 were deleted from the model. These items were less than 0.4. Therefore, the first-order construct AVE was greater than 0.5, achieving the proposed minimum threshold.

Construct validity using composite reliability (CR). As Hair et al. [133] proposed, a CR <0.7 is said to fail the reliability test, whereas a construct >0.95 is redundant. After ensuring that all constructs' conditions for AVE were met, the CR values are >0.7 and <0.95. Meanwhile, the construct 'Trust" in this study is measured as higher order construct, that is, the combination of 'Data Privacy' (DP) and 'Security Concern' (SC). Given this, the AVE and the CR for the construct Trust are measured using the formula below:

$$AVE = \sum_{1}^{M} l_{i=1}^{2} \Big/ M$$

where l = loadings of lower construct measured with M lower component

i = specific higher-order construct

M = total number of lower-order construct

and

Table 5 Discriminant validity

Construct	Items	Item loadings	CR	AVE	Discriminant validity
AI	AI1	0.869	0.904	0.619	Yes
	AI2	0.909			
	AI3	0.802			
	AI4	0.893			
	AI6	0.591			
	AI7	0.587			
DP	DP1	0.617	0.841	0.573	Yes
	DP2	0.731			
	DP3	0.812			
	DP4	0.847			
ES	ES1	0.750	0.903	0.611	Yes
	ES2	0.798			
	ES4	0.849			
	ES5	0.797			
	ES6	0.851			
	ES7	0.625			
PR	PR1	0.614	0.852	0.539	Yes
	PR2	0.761			
	PR3	0.836			
	PR4	0.785			
	PR5	0.651			
SC	SC1	0.770	0.802	0.504	Yes
	SC2	0.640			
	SC3	0.688			
	SC4	0.736			

$$CR = \sum_{1}^{M} l_{i=1} \Big/ M$$

Therefore the $\mathrm{AVE}_{\mathrm{Trust}} = \frac{0.910^2+0.880^2}{2} = \frac{1.6025}{2} = 0.801$.
And $\mathrm{CR}_{\mathrm{Trust}} = \frac{0.910+0.880}{2} = \frac{1.79}{2} = 0.895$.

The AVE and the CR of the higher-order construct 'Trust' fulfilled the conditions for the AVE and CR of parameters >0.5 and 0.7, respectively. Given this, we proceed to assess other crucial parameters before proceeding to assess the structural model. We observed the cross-loading table to reaffirm further the discriminant Validity of ensuring that the items loaded well under its construct. Insight into the table revealed that the items loaded well under their respective construct, with loadings >0.5 (Table 6).

Table 6 Items cross-loadings

	AI	DP	ES	PR	SC	Trust	PR × AI	PR × Trust
AI1	**0.869**	0.092	0.214	0.103	0.213	0.166	0.126	0.043
AI2	**0.909**	0.131	0.319	0.484	0.266	0.217	0.103	0.032
AI3	**0.802**	0.072	0.169	0.384	0.173	0.134	0.066	0.111
AI4	**0.893**	0.039	0.247	0.274	0.174	0.115	0.029	0.037
AI5	**0.591**	−0.003	0.014	0.082	0.038	0.018	0.049	0.071
AI7	**0.587**	0.03	0.1	0.632	0.134	0.086	−0.008	0.053
DP1	−0.005	**0.617**	0.367	−0.017	0.269	0.014	0.108	−0.036
DP1	−0.005	**0.617**	0.367	−0.017	0.269	0.014	0.108	−0.036
DP2	0.044	**0.731**	0.264	0.032	0.427	0.454	0.002	−0.062
DP2	0.044	**0.731**	0.264	0.032	0.427	0.454	0.002	−0.062
DP3	0.104	**0.812**	0.297	0.1	0.321	0.104	0.04	−0.044
DP3	0.104	**0.812**	0.297	0.1	0.321	0.104	0.04	−0.044
DP4	0.121	**0.847**	0.299	0.04	0.457	0.242	0.098	−0.033
DP4	0.121	**0.847**	0.299	0.04	0.457	0.242	0.098	−0.033
ES1	0.168	0.265	**0.75**	0.167	0.217	0.275	0.089	−0.039
ES2	0.175	0.235	**0.798**	0.11	0.312	0.307	0.015	−0.084
ES4	0.246	0.335	**0.849**	0.13	0.388	0.404	0.124	−0.063
ES5	0.251	0.358	**0.797**	0.09	0.329	0.387	0.15	−0.008
ES6	0.251	0.385	**0.851**	0.111	0.47	0.478	0.141	−0.057
ES7	0.162	0.22	**0.625**	0.012	0.148	0.212	0.197	0.051
PR1	0.219	0.036	0.04	**0.614**	0.062	0.054	−0.04	0.015
PR2	0.482	0.007	0.086	**0.761**	0.088	0.05	0.063	0.029
PR3	0.323	0.043	0.128	**0.836**	0.109	0.083	−0.016	0.042
PR4	0.117	0.094	0.069	**0.785**	0.084	0.099	−0.002	0.055
PR6	0.377	0.044	0.118	**0.651**	0.15	0.105	−0.084	0.052
SC1	0.226	0.103	0.39	0.065	**0.77**	0.202	0.102	−0.114
SC1	0.226	0.103	0.39	0.065	**0.77**	0.202	0.102	−0.114
SC2	0.19	0.412	0.429	0.169	**0.64**	0.186	−0.016	−0.072
SC2	0.19	0.412	0.429	0.169	**0.64**	0.186	−0.016	−0.072
SC3	0.103	0.38	0.113	0.067	**0.688**	0.077	−0.053	−0.087
SC3	0.103	0.38	0.113	0.067	**0.688**	0.077	−0.053	−0.087
SC4	0.158	0.414	0.253	0.117	**0.736**	0.327	−0.007	−0.017
SC4	0.158	0.414	0.253	0.117	**0.736**	0.327	−0.007	−0.017
PR x AI	0.087	0.077	0.152	−0.025	0.015	0.055	1	0.068
PR x Trust	0.06	−0.057	−0.048	0.056	−0.104	−0.088	0.068	1

Discriminant Validity

The discriminant validity in this study is examined using the Fornel Larcker criterion and *Heterotrait-Monotrait* (HTMT). The objective of discriminant validity is to measure the extent to which a construct measures attributes that differ from other constructs. Hence, Hamid et al. posit that the first loading presented on the Fornel Larcker criterion table must be higher than the subsequent loadings under it and at its side to the left; that is based on row and column. In view of this proposition, Table 7 proves that this condition is achieved.

Meanwhile, Hensler et al. argue that the Fornel Larcker criterion must fully reveal the discriminant nature of the data under investigation. Given this, the HTMT correlation was proposed. The condition for achieving discriminant-validated data is that the HTMT correlation should be less than 0.9. considering Table 8, it is observed that this condition is also fulfilled.

Hypotheses Testing

We employ the bootstrapping method under the PLS-SEM tool to test the hypotheses and achieve the research objectives. The settings were left at default; we allowed the software to run the samples in 5000 turns at a 5% confidence interval and a two-tailed analysis. Meanwhile, before diving into the hypotheses testing in detail, some parameters, such as Variance Inflated Factors (VIF) for the constructs and items, are encouraged to be examined. On the account of O'Brien, a VIF value of greater or equal to 5 shows a multicollinearity issue; however, VIF values less than five (5

Table 7 Fornel Larcker criterion

	AI	DP	ES	PR	SC
AI	0.787				
DP	0.096	0.757			
ES	0.275	0.396	0.782		
PR	0.639	0.059	0.134	0.734	
SC	0.241	0.605	0.422	0.145	0.71

Table 8 HTMT correlation

Construct	AI	DP	ES	PR	SC	Trust	PR × AI
DP	0.117						
ES	0.256	0.495					
PR	0.862	0.122	0.144				
SC	0.268	0.828	0.533	0.188			
PR × AI	0.084	0.095	0.163	0.062	0.077	0.091	
PR × Trust	0.077	0.067	0.069	0.059	0.125	0.099	0.068

Table 9 r^2 and f^2

	R-square	R-square adjusted	ES (f^2)
SC	0.775	0.774	
AI			0.038
PR			0.002
Trust			0.224
PR × AI			0.016
PR × Trust			0.001

Table 10 Constructs' VIF

Construct	ES
AI	1.756
PR	1.709
Trust	1.048
PR × AI	1.025
PR × Trust	1.02

reveals that the data is free from collinearity and multicollinearity issues. Given this, Tables 10 and 11 reveal that the VIF values for the item and constructs are <5.

Likewise, the variance explained ($r2$) and predictive relevance ($q2$) were examined. The observation on the variance of the endogenous variable (ES) explained by the exogenous variables, namely (AI adoption, AI policies and regulations, and Trust (measured with data privacy and security concern), equates to 0.775. This implies that the exogenous variables successfully explained a 77.5% degree variance in educational sustainability when higher educational institutions in the UAE adopt AI technologies. Similarly, we observed the effect size using the proposition made by Cohen, where 0.02, 0.15, and 0.35 have weak, moderate, and strong effects. As presented in Table 9, construct Trust moderately affects educational sustainability. In contrast, AI technology adoption, policies and regulations, and the moderating relationships have low effects on educational sustainability in this study context (Tables 10, 11 and 12 and Fig. 2).

4 Discussion

Our study's first hypothesis (H1) posited a significant relationship between Artificial Intelligence adoption (AI) and achieving educational sustainability in the UAE. Our analysis revealed a significant and robust relationship ($\beta = 0.222$, t-value $= 3.689$, $p < 0.05$), indicating that higher educational institutions' adoption of AI technologies is associated with a perceived increase in educational sustainability. Specifically, for each unit increase in AI adoption, there is an approximate 0.222-unit increase

Table 11 Items' VIF

Items	VIF
AI1	2.984
AI2	3.384
AI3	2.42
AI4	2.979
AI5	2.012
AI7	1.883
DP1	1.29
DP1	1.252
DP2	1.381
DP2	1.7
DP3	2.708
DP3	1.614
DP4	2.009
DP4	1.864
ES1	2.096
ES2	2.378
ES4	2.453
ES5	2.063
ES6	2.278
ES7	1.523
PR1	1.406
PR2	2.018
PR3	1.861
PR4	2.23
PR6	1.168
SC1	1.644
SC1	1.345
SC2	1.301
SC2	1.182
SC3	1.646
SC3	1.295
SC4	1.871
SC4	1.342
PR × AI	1
PR × Trust	1

Table 12 Hypotheses testing

Hypothesis	Relationship	β	T-stat	P values	Decision
H1	AI -> ES	0.222	3.689	0	Accepted
H2	PR -> ES	−0.049	0.76	0.447	Not accepted
H3	Trust -> ES	0.416	7.438	0	Accepted
H4	PR × AI -> ES	0.088	2.14	0.032	Accepted
H5	PR × Trust -> ES	−0.026	0.538	0.59	Not accepted

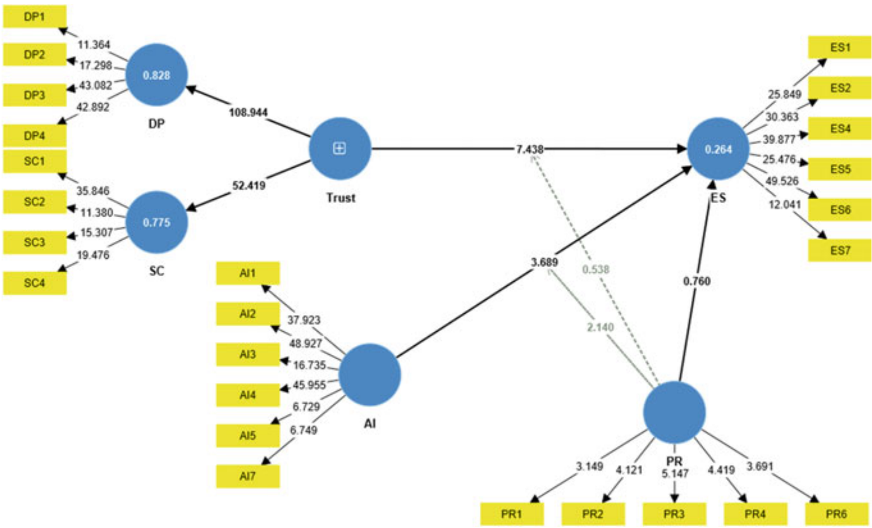

Fig. 2 Structural model assessment

in educational sustainability. These findings align with previous research [90–92], which underscores the potential of AI technologies to enhance the efficiency, accessibility, and quality of educational resources and services, essential components of sustainable education systems [88–90, 123, 136].

The second hypothesis (H2) posits a perceived significant relationship between data policies and regulations and educational sustainability in the UAE. However, the SEM analysis results contradict this hypothesis, having $\beta = -0.049$, t-value = 0.76, $p > 0.5$. The beta coefficient (β) was −0.049, the t-value was 0.76, and the p-value exceeded 0.5, indicating no statistical significance. This implies that, despite recognizing the vital role of AI technologies in enhancing educational sustainability in the UAE, respondents expressed skepticism regarding policies and regulations surrounding the technologies adoption in making such safe to use. The findings align with earlier research by [38, 125, 136] highlighting concern for such a relationship regarding clarity and regulatory frameworks.

Hypothesis Three (H3) asserts a significant relationship between Trust, specifically focused on data privacy and security concerns, and the attainment of educational sustainability within the UAE. Employing SEM analysis to scrutinize the gathered data, our findings indicate a notable and positive impact of Trust on educational sustainability, reflected by a substantial β coefficient of 0.416, a strong t-value of 7.438, and a significance level of $p < 0.05$. This outcome underscores Trust's pivotal role in data privacy and security, where each unit increase in Trust translates to an approximate 0.416-unit gain in educational sustainability.

Consequently, establishing Trust and implementing robust data privacy and security measures within educational institutions emerge as critically influential factors. These measures can significantly bolster educational sustainability, creating an environment where educational resources and services thrive while safeguarding data privacy and security. These results harmonize with existing literature on Trust, emphasizing its indispensable role in sustaining educational systems [10, 16, 137, 138]. In summary, respondents conveyed confidence that their institutions appropriately safeguard their private data and information concerning data privacy and security.

The fourth and fifth hypotheses in this study explore the potential moderating effects of policies and regulations (PR) on the relationship between AI adoption (AI) and Trust and their combined impact on educational sustainability (ES). We employed SEM analysis to investigate these relationships. For the moderating effect of policies and regulations on AI to ES (PR*AI -> ES), our analysis revealed a β coefficient of 0.088, a t-value of 2.14, and $p < 0.05$. This indicates a significant positive moderating role of policies and regulations in the relationship between AI adoption and educational sustainability. The observed result thus affirms earlier investigations that include Ballard et al. [48] and Nastasi et al., where scholars argue that robust policies and regulation frameworks could potentially enhance the effectiveness of AI technologies adoption and educational sustainability. Conversely, when examining the moderating effect of policies and regulations on Trust to ES (PR*Trust -> ES), the analysis showed a β coefficient of −0.026, a t-value of 0.539, and $p > 0.05$, which implies an insignificant moderating role of policies and regulations in the relationship between Trust and educational sustainability. This observation aligns with previous studies [13, 112, 114, 139–144] that indicate students are more inclined to share personal information on AI technologies when they perceive robust privacy and security policies and regulations. However, our findings indicate that the respondents in this study held a more skeptical perspective. According to the respondents, these results suggest that policies and regulations have a meaningful moderating effect on the relationship between AI adoption and educational sustainability. However, their influence appears less pronounced when moderating the relationship between Trust and educational sustainability.

Theoretical Implications

The theoretical implications stemming from this study firmly establish the Socio-Technical Systems Theory as a salient framework for comprehending the intricate dynamics of technology adoption, trust, policies, and regulations in pursuing educational sustainability. These implications offer invaluable insights to scholars and practitioners alike, offering a deeper understanding of the complex interplay between technological advancements, ethical considerations, and regulatory compliance in educational institutions. The theoretical implications stemming from this study hold profound relevance within the academic discourse. Firstly, the substantiation of Hypothesis 1, which establishes a significant and robust relationship between AI technology adoption and educational sustainability, underscores the pivotal role of technological integration within socio-technical systems. This finding resonates with the Socio-Technical Systems Theory's overarching tenets, emphasizing technology's potent influence in shaping the educational landscape [47, 74, 75]. The integration of AI technologies, as supported by empirical evidence, not only augments the richness and accessibility of educational resources but also signifies AI's potential as a foundational component in the perpetuation of sustainable educational systems [77, 79, 80].

On the contrary, the insignificant outcome observed in Hypothesis 2, where data policies and regulations exhibited an insignificant relationship with educational sustainability, imparts valuable insights. This suggests a potential dissonance between technological adoption trajectory and existing regulatory frameworks' efficacy. This incongruity warrants further scrutiny through the prism of the Socio-Technical Systems Theory. It underscores the intricate dynamics and challenges in aligning policies and regulations with the evolving landscape of educational technology. This theoretical incongruity necessitates reevaluating regulatory mechanisms to ensure they aptly address the multifaceted aspects of AI adoption and its impact on educational sustainability.

Furthermore, validating Hypothesis 3, highlighting the substantial role of trust, particularly in data privacy and security, resonates profoundly with the Socio-Technical Systems Theory's core tenets [75, 77]. Trust emerges as an indispensable and pervasive element within the intricate fabric of socio-technical systems. This underscores the theory's profound focus on the human dimension, accentuating trust as a linchpin in technology adoption and data governance [46, 74, 78]. It underscores the theory's holistic perspective, wherein human values, ethics, and trust intersect with technological advancements to impact educational sustainability substantially [83, 84].

Lastly, the elucidation of moderation effects in Hypotheses 4 and 5 underscores the pivotal role of policies and regulations in shaping the relationships between technology adoption, trust, and their cumulative impact on educational sustainability. These insights, seamlessly interwoven within the Socio-Technical Systems Theory's framework, advocate for the strategic calibration of regulatory frameworks. Such calibrated regulatory environments should enforce compliance and catalyze innovation and responsible integration of AI technologies within educational systems. They should serve as adaptive and responsive mechanisms, aligning technological progress with ethical considerations and safeguarding privacy and security within educational contexts.

Practical Implications

The practical implications drawn from this study emphasize the significance of informed AI adoption, robust regulatory frameworks, trust-building initiatives, and a holistic approach to data governance. These insights can guide educational institutions and policymakers in harnessing the potential of AI for sustainable education, ultimately benefiting both learners and the educational ecosystem as a whole.

The study's findings have significant implications for educational institutions and policymakers alike. Firstly, as affirmed by this research, the substantial relationship between AI adoption and educational sustainability underscores the potential benefits of integrating AI technologies in educational settings. Consequently, institutions should consider adopting AI-driven tools and platforms to enhance the efficiency, accessibility, and quality of educational resources and services. However, achieving this goal necessitates investments in AI infrastructure, faculty training, and curriculum redesign to maximize the advantages of AI in fostering sustainable education.

Secondly, the study's revelation that data policies and regulations exhibited an insignificant relationship with educational sustainability implies the need to reevaluate regulatory frameworks. Therefore, policymakers should focus on refining and clarifying policies to bridge the gap between technological advancements and regulatory compliance. Collaborative efforts among institutions, regulators, and stakeholders can lead to more effective data governance practices, ensuring that educational technology adoption aligns with privacy and security concerns.

Furthermore, the significant role of trust in data privacy and security highlights the necessity of building and maintaining trust among students, faculty, and educational institutions. To this end, transparency in data handling, robust security measures, and clear communication of privacy policies can help cultivate trust. Institutions should prioritize ethical data practices and invest in cybersecurity to foster a secure environment where learners feel confident about their data being protected.

Lastly, the moderation effects of policies and regulations on AI adoption and trust indicate that regulatory frameworks can facilitate or hinder technology adoption. Consequently, policymakers should design regulatory frameworks that ensure compliance and encourage innovation and responsible AI integration. Collaborative efforts between institutions and regulatory bodies can lead to regulatory environments that enhance the effectiveness of AI technologies in education while safeguarding privacy and security.

5 Conclusion

In conclusion, this study, guided by the Socio-Technical Systems Theory, sheds light on the intricate dynamics of technology adoption, trust, data policies, and regulations in educational sustainability. Our findings affirm the transformative potential of AI adoption in enhancing educational resources and services while highlighting the pressing need for policymakers to recalibrate and clarify existing regulatory frameworks to safeguard educational sustainability. Trust, particularly in data privacy and security, emerges as a linchpin in sustaining educational systems, emphasizing the importance of transparency and robust security measures. Furthermore, our results underscore the role of policies and regulations as significant moderators, advocating for calibrated regulatory environments that ensure compliance and encourage responsible AI integration. These insights collectively inform educational institutions and policymakers, facilitating an environment where AI integration aligns with ethical considerations, regulatory clarity, and trust, ultimately fostering sustainable and inclusive education.

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ChatGPT and Digital Learning Platforms

Adoption of Chatbots for University Students

Said A. Salloum, Amina Almarzouqi, Ahmad Aburayya, and Raghad Alfaisal

1 Introduction

Recent years have seen a surge in artificial intelligence research [1–5], especially in the application of Chatbots in various sectors, notably education [6]. The Technology Acceptance Model (TAM) has been a predominant framework for evaluating the acceptance and use of such technologies [3, 7, 8], focusing on perceived usefulness and ease of use as key factors. These elements have shown consistent relevance across various tech applications [9]. Chatbots in education mark a significant shift, introducing innovative methods for enhancing teacher-student interactions (See Fig. 1) [10]. However, there remains a significant research gap in comprehensively integrating TAM with other influential factors like flow theory—which emphasizes user immersion and control—in the context of educational Chatbots.

While existing studies shed light on the broader acceptance of technology in education, there is a distinct lack of research specifically merging TAM with additional relevant factors for understanding Chatbot adoption in this sector. Notably, the interplay between user satisfaction, a crucial outcome of technology adoption,

S. A. Salloum (✉)
Health Economic and Financing Group, University of Sharjah, Sharjah, UAE
e-mail: ssalloum@sharjah.ac.ae

School of Science, Engineering, and Environment, University of Salford, Salford, UK

A. Almarzouqi
Department of Health Service Administration, College of Health Sciences, University of Sharjah, Sharjah, UAE

A. Aburayya
College of Business, MBA Department, City University Ajman, Ajman, UAE

R. Alfaisal
Faculty of Computing and Meta-Technology, Universiti Pendidikan Sultan Idris, Tanjong Malim, Malaysia

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Fig. 1 Chatbots

and these factors, remains underexplored. This study seeks to fill this research void by developing an integrated model that fuses TAM with flow theory. The goal is to gain a more holistic view of the factors influencing Chatbot adoption in education and to investigate how these factors correlate with user satisfaction. This approach represents a significant contribution to the field, offering nuanced insights into the dynamics of technology acceptance in educational environments.

2 The Theoretical Framework

2.1 Users' Satisfaction (SAT)

Satisfaction is an emotional state reflecting how well an individual's expectations, shaped by past experiences, match their current experiences. This feeling is closely associated with the overall impressions, either positive or negative, that arise from using technology. Essentially, when people find technology to be useful, efficient, and easy to operate, it activates their internal and external motivators. This leads to an increase in their self-confidence and a greater inclination towards innovation, often exceeding their original expectations. Satisfaction occurs when these expectations are fulfilled. In the context of technology adoption, user satisfaction is a critical factor. It directly influences the extent to which users feel content with a product or service [11–13]. Studies have consistently shown a strong link between ongoing willingness to use technology and the level of satisfaction experienced [14–16], which plays a crucial role in the sustained use of technological solutions. Given these considerations, this research posits the following hypothesis:

H1: User satisfaction (SAT) positively influences the adoption of Chatbots in educational settings.

2.2 The Technology Acceptance Model (TAM)

The Technology Acceptance Model (TAM) has been a cornerstone in numerous studies exploring the facets of technology adoption, acceptance, and usage intentions across a wide array of sectors [17–19]. This model's robustness and adaptability make it a popular choice for researchers delving into how and why individuals and organizations decide to embrace new technologies. In this study, our focus narrows down to two key TAM constructs [20, 21], deemed crucial in influencing the adoption of Chatbots. The first of these constructs is perceived usefulness. This term encapsulates the user's belief in the extent to which they anticipate the technology will enhance their productivity or effectiveness. It reflects the user's assessment of the tangible benefits they expect to derive from using the technology. The second construct we examine is perceived ease of use. This aspect scrutinizes the degree to which a user expects the technology to be free of effort. It's an evaluation of the anticipated user experience, focusing on how intuitive, straightforward, and hassle-free the technology is from the user's perspective [6, 22]. This factor is critical as it often dictates the initial adoption and continued usage of the technology. Building upon these theoretical foundations, we propose the following hypotheses to guide our research:

H2: Perceived usefulness (PUS) positively influences the adoption of Chatbots in educational settings.

H3: Perceived ease of use (PEOU) positively influences the adoption of Chatbots in educational settings.

2.3 Flow Theory (FLO)

The notion of flow, originally conceptualized, encompasses a sense of control, deep engagement, and joy. When users find technology to be thoroughly enjoyable and satisfying, their likelihood of consistent engagement increases. Such engagement captivates their minds, concentrates their attention, and influences their actions, thereby fostering positive emotional reactions that amplify their flow experience during technology use [23–25]. When individuals are engrossed in this state of flow, they often become oblivious to time and are propelled by internal drives. This state leads to what is known as an online flow state, characterized by a continuous, interactive experience filled with delight, immersion, and active involvement [26, 27]. In recent times, the concept of flow has gained prominence as a critical element in the adoption of various IT systems, including e-learning platforms, internet usage, and entertainment activities. Notably, flow is characterized by its seamless integration into users' experiences with technology, often leading to a transcendence of self-awareness [28]. Given its increasing relevance, flow theory has risen as a key predictor in understanding and forecasting technology adoption patterns. In light of this, we propose the following hypothesis to guide our study:

H3: Flow experience (FLO) positively influences the adoption of Chatbots in educational settings.

Figure 2 provides a comprehensive visual representation of the research model developed for this study. The diagram meticulously outlines the various constructs and the hypothesized relationships between them, offering a clear and structured overview of the theoretical framework underpinning our analysis.

3 Research Methodology

3.1 Data Collection

A comprehensive online survey was executed among students from a variety of educational institutions across the United Arab Emirates (UAE). This data-gathering process initiated on March 10, 2023, and wrapped up on May 15, 2023. In an effort to gather diverse insights, the research team disseminated 300 surveys in a randomized manner. The response rate was notably high at 97.6%, reflecting the participants' keen involvement in the study. Out of the responses received, 293 were deemed complete and suitable for analysis, while 7 were excluded due to their incomplete nature. The total number of valid responses, 293, comfortably meets the recommended sample size threshold of 285 for a population size of 1100, as per the guidelines suggested by Krejcie and Morgan [29]. This sample size, although smaller than the baseline recommendation, was considered adequate for conducting a structural equation modeling analysis [30], thereby bolstering the reliability and robustness of the study's hypotheses. A distinctive feature of this research was its incorporation of the latest developments in artificial intelligence. The study employed Structural Equation Modeling (SEM) using the sophisticated Smart-PLS Version 3.2.7 software, which facilitated a comprehensive evaluation of the measurement model. The development of the Final Path Model was meticulously executed to encapsulate the

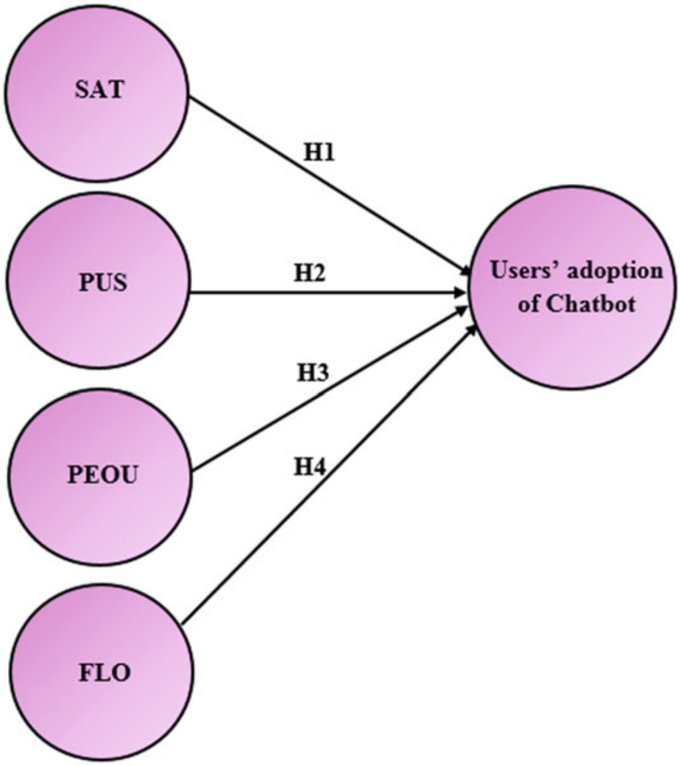

Fig. 2 Research model

complexity of the variables and their interrelations, ensuring a thorough and nuanced analysis.

3.2 Demographic Information

Figure 3 illustrates a thorough demographic analysis of the survey participants, highlighting their individual characteristics. The gender distribution among respondents was fairly balanced, comprising 69% female and 31% male. Age-wise, the largest group (44%) fell into the 18 to 29-year-old category, with the remainder being 29 years or older. In terms of educational background, the distribution was diverse: 9% of respondents held diplomas, 6% had advanced diplomas, a significant majority of 62% possessed undergraduate degrees, 18% had completed postgraduate education, and 4% held doctoral degrees. The selection of participants was conducted using a purposive sampling method, as advised by Salloum and Shaalan [31], which was particularly effective given the participants' keenness to partake in the study.

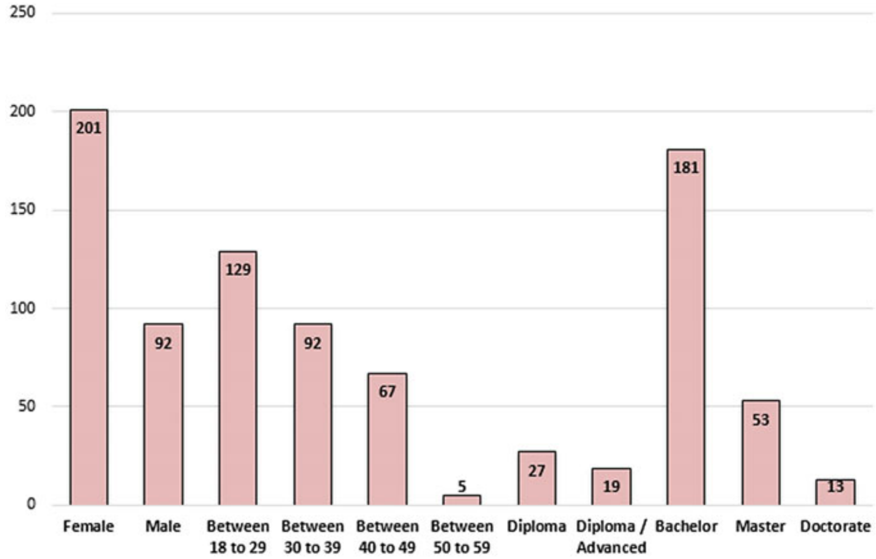

Fig. 3 Demographic information of the participants ($n = 293$)

This strategy enabled the inclusion of a wide array of participants from various academic settings and a broad spectrum of ages, thus infusing the study with a rich array of educational insights. To ensure a meticulous and in-depth analysis of this demographic data, the research team utilized IBM SPSS Statistics version 23. This approach guaranteed a thorough and precise evaluation of the participant profiles, providing a solid foundation for the study's findings and conclusions.

4 Findings and Discussion

4.1 Data Analysis

For the analysis of data, this study utilized the partial least squares-structural equation modeling (PLS-SEM) technique, specifically employing SmartPLS V 3.2.7 [32–37]. The methodology was characterized by a dual-focused assessment, examining both the measurement and structural aspects of the models [5, 17, 38, 39]. The choice of PLS-SEM was driven by several key factors, which are detailed in this paper. Firstly, PLS-SEM was selected for its robust capability to assess the proposed conceptual framework effectively [40–44]. Another critical reason for its selection was its proficiency in handling exploratory data that aligns with the conceptual models [3, 45–48]. Furthermore, PLS-SEM was applied in a holistic manner, treating the model as an interconnected entity, rather than dissecting it into separate parts [49–53]. Finally, the analysis of both the structural and measurement models was concurrently conducted using PLS-SEM. A notable advantage of PLS-SEM lies in its accuracy in both the generation and acquisition of measurements [51, 54–61].

4.1.1 Convergent Validity

The analysis of the Measurement Model [40] was carried out with meticulous attention, primarily focusing on establishing construct validity. This process involved evaluating discriminant validity, convergent validity, and construct reliability. Construct reliability was assessed using indicators such as Cronbach's alpha (CA) and composite reliability (CR). According to the data in Table 1, the Cronbach's alpha values range between 0.752 and 0.887, which are slightly below but near the commonly accepted standard of 0.7 [62]. On the other hand, the composite reliability scores, also detailed in Table 1, range from 0.791 to 0.882, comfortably exceeding the accepted benchmark. For convergent validity, key metrics like average variance extracted (AVE) and factor loadings were scrutinized [51]. The factor loadings, as per Table 1, all surpass the threshold of 0.7 [40]. Additionally, the AVE values, ranging from 0.557 to 0.792 as shown in Table 1, are well above the minimum requirement of 0.5. These results indicate that the study successfully achieves a satisfactory level of convergent validity, further validating the robustness of the Measurement Model.

Table 1 Convergent validity results

Constructs	Items	Factor loading	Cronbach's alpha	CR	AVE
Users' adoption of Chatbot	INT1	0.822	0.873	0.825	0.626
	INT2	0.860			
Users' satisfaction	SAT1	0.764	0.868	0.866	0.792
	SAT2	0.779			
	SAT3	0.770			
Flow experience	FLO1	0.809	0.752	0.791	0.639
	FLO2	0.820			
Perceived ease of use	PEOU1	0.856	0.887	0.814	0.557
	PEOU2	0.875			
	PEOU3	0.752			
Perceived usefulness	PUS1	0.892	0.883	0.882	0.574
	PUS2	0.924			
	PUS3	0.822			

Table 2 Fornell-Larcker scale

	INT	SAT	FLO	PEOU	PUS
INT	**0.881**				
SAT	0.923	**0.862**			
FLO	0.676	0.679	**0.917**		
PEOU	0.611	0.581	0.268	**0.820**	
PUS	0.325	0.6526	0.210	0.556	**0.837**

Table 3 Heterotrait-Monotrait ratio (HTMT)

	INT	SAT	FLO	PEOU	PUS
INT					
SAT	0.456				
FLO	0.700	0.611			
PEOU	0.479	0.681	0.482		
PUS	0.587	0.614	0.557	0.575	

4.1.2 Discriminant Validity

This study rigorously assessed discriminant validity using the Heterotrait-Monotrait ratio (HTMT) and the Fornell-Larker criterion [40]. As indicated in Table 2, the application of the Fornell-Larker criterion was successful, revealing that the square root of the Average Variance Extracted (AVE) for each construct is more strongly correlated with its own construct than with any other [63]. Additionally, the HTMT ratio results, illustrated in Table 3, confirm that all construct values fall below the established threshold of '0.85' [64]. This compliance with the standard threshold reinforces the discriminant validity of the research. These outcomes suggest that the Measurement Model is free from significant validity or reliability issues. The confirmation of the Measurement Model's robustness signifies that the data is well-prepared for subsequent analysis phases, particularly the structural model. The meticulous approach taken to validate the reliability and validity of the Measurement Model underscores the thoroughness and rigor of the research methodology. Such careful validation not only strengthens the confidence in the data used for the structural model analysis but also enhances the overall credibility and accuracy of the study's conclusions.

4.2 Hypotheses Testing

The next phase of this study involved a detailed examination of the structural model. This stage entailed evaluating the coefficient of determination (R^2) and conducting a comprehensive analysis of path coefficients through an extensive bootstrapping

Table 4 R^2 of the endogenous latent variables

Construct	R^2	Results
INT	0.735	High

procedure with 5,000 re-samples. The results, detailed in Table 5, include path coefficients along with their respective t-values and p-values for each hypothesized relationship in the model. Notably, the research findings provide robust support for all the proposed hypotheses. An in-depth analysis of the data confirms that hypotheses H1 through H4 are backed by empirical evidence, showcasing a strong correlation between the theoretical projections and the actual findings. This correlation validates the proposed model and underscores the efficacy of the research methodology applied. The structural model's evaluation is centered around the assessment of the coefficient of determination (R^2 value) [65]. This coefficient, defined as the squared correlation between the actual and predicted values of an endogenous construct, serves as a measure of the model's predictive power. It quantifies the cumulative effect of exogenous latent variables on an endogenous latent variable, essentially indicating the proportion of variance in the endogenous constructs explained by the model. A high coefficient value, exceeding 0.67, denotes strong predictive ability, while values from 0.33 to 0.67 are considered moderate, and those between 0.19 and 0.33 are viewed as weak. Values below 0.19 are generally seen as unacceptable [66].

As depicted in Table 4 and Fig. 4, the model demonstrates substantial predictive accuracy, explaining about 73.5% of the variance in perceived usefulness and the adoption of Chatbots. The influences of various factors like Users' Satisfaction (SAT), Perceived Usefulness (PUS), Perceived Ease of Use (PEOU), and Flow Experience (FLO) on the Users' adoption of Chatbots (INT) were found to be significant, exhibiting beta values of 0.449 ($P < 0.001$), 0.561 ($P < 0.001$), 0.396 ($P < 0.001$), and 0.683 ($P < 0.001$) respectively. The outcomes from this hypothesis testing are concisely summarized in Table 5.

Table 5 Hypotheses-testing of the research model (significant at p** < = 0.01, p* < 0.05)

H	Relationship	Path	*t*-value	*p*-value	Decision
H1	SAT - > INT	0.449	10.573	0.000	Supported**
H2	PUS - > INT	0.561	12.620	0.000	Supported**
H3	PEOU - > INT	0.396	9.159	0.001	Supported**
H4	FLO - > INT	0.683	17.766	0.000	Supported**

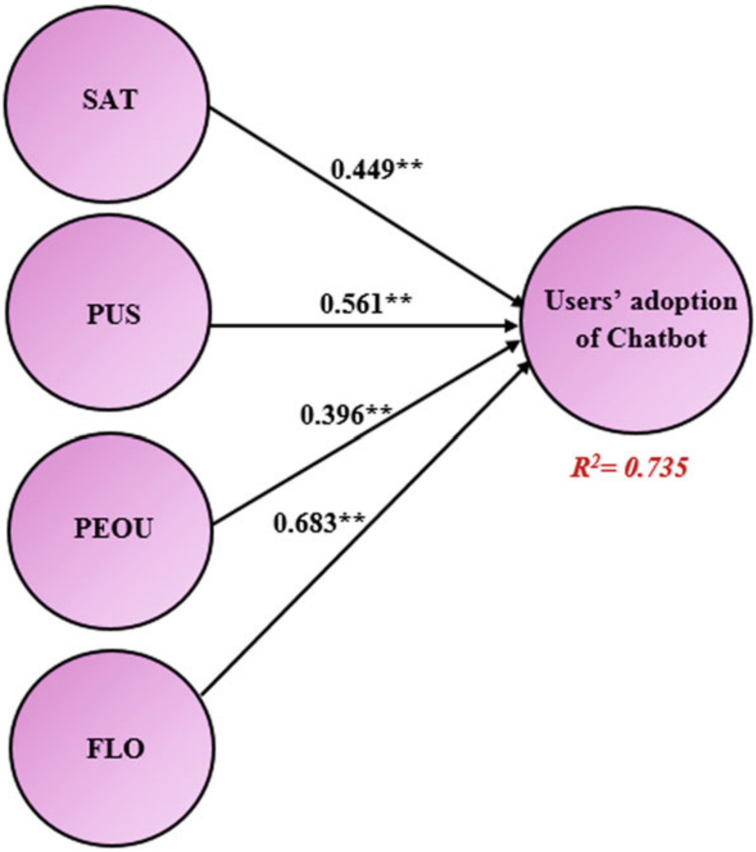

Fig. 4 Model's path coefficient (noteworthy at p** ≤ 0.01, p* < 0.05)

5 Conclusion

This study conducts a detailed examination of the effects of implementing Chatbots in the healthcare sector, employing an elaborate model that combines aspects of the Technology Acceptance Model (TAM) with other external variables, such as flow theory and user satisfaction. The research identifies that TAM's core elements, specifically perceived ease of use and perceived usefulness, significantly drive the adoption of Chatbots. It suggests that technologies perceived as simple to use or beneficial are more likely to be widely accepted in various fields, including both academic and non-academic sectors. This finding is consistent with prior research in the education sector, which indicates that simplicity and utility foster technology adoption among teachers, administrators, and students. The study also explores flow theory's significant role in technology adoption, highlighting that user engagement levels can greatly influence their adoption decisions. Chatbots, in this respect, have been shown to substantially boost user engagement, thus positively influencing adoption. This aligns with other studies that link flow experience with user behavior intentions [67, 68]. User satisfaction, influenced by ease of use and perceived usefulness, is another key finding of this research. Users who find Chatbots both easy to use and beneficial tend to show higher satisfaction levels, a trend supported by other research findings [69, 70]. This enhanced satisfaction, in turn, positively affects their behavioral intentions, especially when they perceive the technology as effortless. The research offers essential insights for developers of educational Chatbots, emphasizing the need to align Chatbot development with the specific requirements of the educational sector. It underscores the importance for developers to understand how Chatbots can meet educators' needs and incorporate features that increase their usability and reliability. The study also highlights the need for timely and functional features in Chatbots to enhance reliance and effectiveness in education, with a focus on regularly updating features to meet evolving user needs. However, the study faces certain limitations. Its focus on a specific educational context may limit the broader applicability of its findings. Future studies could expand the scope to include a more diverse range of educational environments. Additionally, the data was collected from only one type of educational institution, which might reflect a unique culture and limit generalizability. A longitudinal approach and diverse data collection methods like interviews and observations are suggested for future research for more comprehensive insights. Furthermore, while this study concentrated on specific external variables relevant to educational Chatbots, future research could investigate other variables that align with the changing features and uses of Chatbots. The possibility of integrating other models that focus on social and psychological factors, along with exploring Chatbot applications in various academic and non-academic settings, is also recommended. Lastly, this study did not focus on gender differences in Chatbot usage, presenting an opportunity for future research to explore this area and uncover significant gender-related differences in Chatbot adoption and usage in education.

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Using AI to Develop Capabilities in Arab Universities

Noha Mellor

1 Introduction

With the rapid advancement of technology, the job market has become increasingly competitive and demanding [1–4]. The ability to learn and adapt to new skills is highly valued [5–7], whereas those who fail to do so often struggle to keep up. Currently, analytical skills and in-person services are highly sought after, but the rise of globalization has made specific jobs vulnerable to outsourcing or automation. To ensure job security, it is essential to acquire skills that cannot easily be substituted. Seeking advanced training is now more important than ever, in addition to developing strong interpersonal skills to increase one's chances of success in the labor market [8–11]. One key technological capability that has caused a global debate is artificial intelligence or AI, defined as machines' imitation of human intelligence. It includes creating artificial intelligence that can learn, perceive, and process information. Thus, AI is "automation based on associations" by capturing data, detecting patterns in data, and automating decisions [2]. The rise of AI presents a potential threat to knowledge-based professions, which means that higher education must adapt to this new reality. Although AI can perform administrative tasks and generate content, it cannot replace roles that require empathy and building relationships, which are uniquely human [3]. Previous research into AI in education has distinguished between three paradigms: AI-directed paradigm, focusing on using AI to represent knowledge and guide cognitive learning; AI-supported paradigm involving learners collaborating with AI to support their learning; and AI-empowered, which emphasizes learner agency and views AI as a tool to augment human intelligence [4, 5].

This chapter zooms in on ways to implement the AI-supported paradigm as one solution to mitigate the challenges in Arab higher education, drawing on AI tools,

N. Mellor (✉)
College of Communication, University of Sharjah, Sharjah, United Arab Emirates
e-mail: nmellor@sharjah.ac.ae

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such as Chat GPT. By highlighting the Arab context, the chapter advocates a situated learning approach that challenges the traditional view of learning as the acquisition of propositional knowledge and instead focuses on the social engagements and interactions that facilitate learning [6]. In other words, the argument emphasizes the need for students to apply what they learn in the same context where they learned it. The situated learning approach, then, departs from the traditional idea of learning as just memorizing information to focusing on the interactions and connections students make while learning, highlighting the relationship between digital technology and pedagogy [7].

Theoretically, the argument is developed through the lens of the capability approach, which focuses on the outcomes and benefits rather than the metrics of access to AI in the Arab region. The argument is developed through three stages: first, I provide a brief discussion of the tenets of the capability approach. Second, I zoom in on Arab higher education, highlighting the specificity of the region that is still grappling with a set of pedagogical and research challenges despite the massive leap in the number of university degrees awarded over the past few decades. Finally, I suggest a few steps to integrate AI into the academy to mitigate specific challenges in Arab universities, particularly to enhance critical thinking and digital skills.

2 Measuring Real-Life Capabilities

The capability approach, originally developed by Amartya Sen [8–10], proposes that we should consider people's abilities and the quality of their lives. The approach's significant impact lies in its ability to encourage us to explore new perspectives and focus on various aspects when analyzing data or making assessments. The approach is people-centered, places human agency at the forefront, and promotes social opportunities crucial in expanding human agency [9].

The capability approach has been applied in a wide range of fields, including education and technological design. The capability approach focuses on what people are truly capable of doing and becoming. It considers their capabilities, which are the things they can do and be, as well as their functionings, which are what they are actually achieving. The capability approach focuses on the freedom to achieve and the capabilities to function, and it comprises two main elements: *functionings* and *capabilities*. Functionings refer to a person's actions and achievements, while capabilities are the various combinations of functionings that a person can achieve. Functionings are achievements, while capabilities are the ability to achieve; functionings are related to living conditions, while capabilities represent notions of freedom and real opportunities for a person's life [11]. The approach also examines whether institutions, practices, and policies prioritize people's capabilities and provide them with opportunities to do what they value and be who they want to be. It raises questions about whether people have equal capabilities in life or if global economic structures or domestic policies make their capabilities unequal [11].

In higher education, the capability approach focuses on what students can do and achieve with technology, for instance, instead of just documenting the metrics of access to this technology generally. This is important because different people can get different outcomes from the same resources. The approach also recognizes that everyone is different and has their own unique abilities and needs. It can also be used to evaluate digital educational resources and their use, such as assessing the capabilities of AI in the classroom in terms of enhancing educational outcomes and emphasizing the meaningful integration of AI, considering how this technology can support diverse learners' needs [12, 13].

Sen's concepts of functionings and capabilities can offer a framework to integrate AI technologies in higher education. Here, functionings combined with capabilities highlight a student's ability to achieve and how they use their resources to achieve tangible outcomes which are relevant to their particular situations. The capability approach differs from traditional utilitarian and instrumental frameworks in measuring educational outcomes. While the utilitarian and instrumental frameworks focus on measuring resources and human capital (e.g., efficiency) in developing institutions. As such, these frameworks are concerned primarily with inputs (e.g., costs and qualifications) and outputs (students' results) without considering how resources can be transformed into an intrinsic value for students. On the other hand, the capability approach highlights the *conversion factors* or the individual's (as well as universities') ability to convert resources into valued functionings [14]. Here, it is important to have a set of highly diverse conversion factors, given the heterogeneity of the student body across universities. This means that similar resources do not always lead to similar outcomes, as students' capabilities are highly diverse [15].

Thus, the capability approach envisions the empowerment of learners and educators so that they can find effective ways to support the development of capabilities. In the technology field, this approach is similar to investigating the third level of the digital divide. While the first level of the digital divide refers to inequalities in access to technology, and the second level relates to differences in the ability to use technology effectively, the third level is concerned with disparities in the outcomes of internet use within populations of users. The third level of the digital divide is associated with usefulness and is understood as the population's ability to obtain a range of social and cultural benefits from internet use or moving from investigations of skills and usage to tangible outcomes and real-world consequences [16].

To apply the framework mentioned above in Arab higher education, it is important to have a thorough understanding of the historical context, domestic actors, and limitations of Western models when explaining regional developments. This is not meant to promote an essentialist and simplified view of East vs. West or developed vs. underdeveloped, but rather to recognize the unique historical trajectory of the Arab higher education system within the region. The following section briefly overviews the specificity of the higher education sector in the Arab region.

3 Specificity of the Arab HE

In the mid-twentieth century, higher education systems were established in most Arabic-speaking countries. Enrollment increased due to government funding and lenient admission policies. However, decades of neglect have led to a crisis in several Arab states, in terms of quality and research, with overcrowded classrooms, outdated curricula, and underpaid faculty [17]. In fact, numerous studies show that Arab educational systems are underperforming and suffer from joint issues. Despite attempts at reform, there has been little national impact. While enrollment rates have increased, quality has not improved, and power structures remain unchanged. Poor performance indicators include low test results, inadequate STEM education, low literacy rates, absence of Arab universities in top rankings, poor labor market preparation, and limited economic returns. In terms of research, defined as a thorough investigation carried out by students and faculty members, focusing on social issues pertinent to the Arab region, several Arab universities suffer from a lack of infrastructure, funding, resources, and motivated faculty members, which are all contributing factors. This means not meeting the ultimate goal of this research endeavor which is to provide insightful knowledge regarding the region, bridge gaps in the current scholarship, and enhance research capabilities [18].

Promoting academic research in Arab universities faces several obstacles, such as the mismatch between what universities teach and what the labor market needs [19]. Generally, Arab universities face problematic factors such as a lack of strategic independence, weak research, lack of innovation, a low number of graduate programs, and appointment mechanisms of institution administrators [20]. To improve quality in Arab universities, it is agreed to address these obstacles by developing strategic plans, promoting innovation, increasing research funding, and enhancing research policies and procedures [21], in addition to fostering a research culture by providing training and support for faculty members and students and creating a clear research agenda [22]. In many Arab countries, ethical guidelines for research also need further development and a well-defined strategy. Without a code of ethics and misconduct policy, plagiarism and intellectual theft can become rampant [20]. Public universities in the Arab world generally lack resources and perform poorly on global rankings, leading to high unemployment among graduates. Governments have tried various reforms, including authorizing private universities, but public institutions still struggle with limited funding and management. Finally, political instability in several Arab states and a lack of security further hinder regional research capabilities [23]. Addressing the economic and security challenges of the region and promoting regional cooperation can help overcome the obstacles to fostering academic research in Arab universities.

It is important to note that Arab higher education is diverse, with differences among countries, which are ranked on access, equity, and publication intensity. Gulf countries are at the top, while conflict-ridden and poverty-stricken countries are at the bottom [17]. To address these issues, universities must prioritize the development of research infrastructure and motivate their researchers [22]. The Gulf Cooperation Council (GCC) states have invested massively in its HE sectors over the past few decades, moving towards internationalization, implementing quality initiatives, and integrating regionally. Despite the proliferation of international universities and campuses across the GCC states, totaling well over 167 universities, there is an acute need for regional integration to ensure that such an internationalization strategy can yield real-life outcomes and a solid foundation for universities across the GCC states [24]. The concept of an international branch campus is a relatively recent trend that is gaining popularity and drawing attention to questions regarding pedagogy and culture. In the UAE alone, there are almost 40 institutions with American, European, or Australian names. Some provide undergraduate education, drawing on the American liberal arts tradition, while others support graduate and research programs.

However, international universities in the Arab world may face a divide between their cosmopolitan audience and the communities they were meant to benefit: While some institutions still acknowledge their regional foundations, many are less anchored in their locale [25]. This is why some GCC states grapple with other challenges: In Qatar, for instance, research highlights the discrepancy between the ideals and goals of Western education and how they align with Qatari national development objectives, and the difficulties of serving the local and regional student population and sharing their insights as instructors and academic advisers [26]. In Oman, research reveals challenges, including time constraints, lack of research training, and capacity-building opportunities such as research-supportive environments and funding [21]. In Bahrain, graduates need to possess both technical expertise and soft skills that are highly sought-after by employers, which is the Higher Education Council launched the National Strategy for Higher Education (2014–2024) to bridge the gap between higher education and the labor market. The strategy aims to promote problem-solving, critical thinking, communication, and networking skills—all essential attributes for success in today's workplace. It has identified five priority areas, including effective engagement between employers and the higher education sector, aligning graduate skills with industry requirements, aligning programs with national reform needs and growth sectors, integrating career guidance across the higher education sector [27].

In summary, there are several challenges related to fully benefiting from the knowledge-based economy, with some researchers arguing that the intermingling of Western and Arab traditions has led to the polarization of cultural identities among students, suggesting more higher education reforms to meet the challenges of a dynamic market economy [28]. This means that simply importing knowledge without actively producing it is not a practical approach. Imported knowledge may quickly become outdated and may not align with local interests [20]. Thus, despite significant investment in education in the Arab region, the region is yet to fully benefit from education's personal, social, and economic advantages [17].

To effectively tackle the unique challenges of the Arab context, modern technology must be integrated into daily learning and research processes instead of simply being imported. The following sections focus on two challenges facing Arab universities: equipping students with the digital skills needed in the labor market, and integrating the newest technology, such as AI, into the learning and research processes to advance students' critical thinking skills.

4 The Acute Need for Digital Skills

The role of technology has changed the skills needed for the labor market, and it presents an opportunity to improve education quality. Moreover, labor markets are diverging globally due to economic, health, and geopolitical trends. High-income countries have tight labor markets, while low-income countries have high unemployment rates. Technology adoption is expected to drive business transformation in the next five years, with big data, cloud computing, and AI being adopted. Employers predict a turnover in jobs due to emerging jobs being added and declining jobs being eliminated, and a net decrease of 14 million jobs [29]. The Supply Chain, Transportation, Media, Entertainment, and Sports industries are expected to have higher job turnover than Manufacturing and Retail [29].

The Arab region has the tools and resources to implement technology and build human capital, but four tensions hold education back: credentials and skills, discipline and inquiry, control and autonomy, and tradition and modernity. These tensions have shaped education policy in each country and affect educational outcomes for young people in the region [30]. Unless addressed, the region will not reap the full benefits of education, no matter how much money is invested. The Arab region, specifically, faces complex and urgent workforce challenges due to slow growth and investment, which makes it difficult for regional markets to absorb surplus labor. Generally, there is a skills mismatch as a key issue on the supply side, resulting from a deficit of essential skills imparted through the education system [18]. The shortage of essential skills in the labor market, known as the skills mismatch, is a growing concern. The education system is not adequately preparing students with the necessary soft skills or fostering critical thinking. The focus on test performance and career choice has led to a disconnect between higher education output and actual learning [18].

To address such challenges while improving HE education in the Arab region, a modern and flexible education system that prepares students for the future is needed, focusing on digital skills, modernized pedagogy, new forms of assessments, and leveraging technology. The push for learning must be aligned with a pull for skills in the labor market, as education must confer the skills and knowledge that constitute human capital to contribute to economic growth [30]. To do so, there is a need to implement an institutional system that tracks labor market data and involves the private sector in curriculum development and evaluation. Here, AI technology can play a role in improving the learning experience, increasing efficiency, and allowing easier access to academic and administrative tools [19] while reforming pedagogies to emphasize participatory learning and essential skills [18]. One way to do so is by integrating AI technology into the pedagogy and research capabilities, which is the focus of the following sections, suggesting the use of AI tools such as ChatGPT in collaborative learning and collaborative research in order to enhance critical thinking and digital literacy.

5 AI as a Collaborative Learning Tool

The use of artificial intelligence (AI) in education has become increasingly popular in recent years, with many researchers exploring its potential benefits. One such benefit is the ability of AI to enhance student effort and prepare them for the real world by promoting critical thinking skills. According to [31], AI tools such as ChatGPT, which is a unique AI-powered tool that has recently emerged and has been gaining popularity in the education sector, can be utilized as a learning tool to enhance students' critical thinking abilities. For instance, its responses can be used as prompts for complex or difficult questions, enabling students to think more deeply about a topic and develop their critical thinking skills. The tool's responses can guide students to consider the potential implications or applications of a concept, draw connections to other topics they have learned, and ask themselves questions about the concept [32]. In this way, students can benefit from ChatGPT by engaging in discussions with their peers. By debating and discussing concepts with others, students can enhance their comprehension and effectively explain the concepts in their own words. This can also be a way for students to grade the ChatGPT responses, which is another way of detecting AI bias. In so doing, and to encourage collaboration and discourage cheating, tutors can treat ChatGPT as a collaborative learning partner and assign students a collaborative task on ChatGPT to explore ideas and evaluate options to complete their project.

Furthermore, ChatGPT can serve as a means of providing tailored feedback to a student by engaging them in a conversational format. The tool can break down the student's response into various parts, highlighting specific areas where the student needs to improve. This can be invaluable for students who struggle with particular concepts or need extra guidance in specific areas.

It can also be used for group assessments. In fact, research has shown that students are more likely to cheat when assignments are designed in a way that fosters competition among classmates. Conversely, students are less inclined to cheat when academic tasks encourage collaboration [33]. One effective way of achieving this is by assigning students a collaborative project on ChatGPT. By exploring ideas and evaluating options together, students can learn to work collaboratively while enhancing their critical thinking skills. Thus, AI tools can be harnessed to enable new ways of learning, teaching, and education and providing personalized learning experiences for students while identifying areas where students need additional support [34]. AI technology can also be used to process and analyze big data to identify patterns, and this process can be handled collaboratively between instructors and students [35].

In summary, ChatGPT, as an AI-powered chatbot that generates human-like responses, can be used for education, but challenges in implementing it include generating incorrect or fake information and bypassing plagiarism detectors. A review of previous literature found that its performance varies across subject domains. Updating assessment methods and institutional policies and providing training for instructors and students are necessary to address these challenges [36] while moving away from memorization and embracing new learning and assessment methods that can sharpen critical skills. For instance, Zayed University has recently eliminated exams completely on the ground that it is no longer an effective method to prepare graduates for the contemporary workforce. Instead, the university fosters continuous assessments before, during, and after class using interactive tools and providing continuous feedback [37]. AI tools can also be used as a tool to encourage learners to collaborate with AI in scientific research, which can enhance their learning experience through social interactions and experiences [4], as explained in the following section.

6 AI as a Research Collaboration Tool

AI can be a valuable tool for improving data analysis and facilitating collaborative scientific research between teachers and students [35]. ChatGPT, as one AI tool, can draw on large datasets to generate human-like responses and provide data analysis on any topic. As such, artificial intelligence can be a game changer for universities in the Arab world: Firstly, AI can augment existing data analysis processes by enhancing pattern recognition and supporting advanced data analysis [35]. Secondly, AI can be utilized to investigate the challenges that may impede the role of university instructors and students in joint scientific research. By identifying research obstacles, universities can pinpoint their challenges and develop strategies to overcome them [35]. Thirdly, AI can play a crucial role in enhancing research quality by identifying gaps in existing research and recommending new research topics. This helps instructors and students to focus their research efforts and produce high-quality research.

To further enforce such a collaboration between university instructors and students in scientific research, other tactics can be used: for instance, AI tools such as ChatGPT can help identify research topics that interest both instructors and students, thereby facilitating collaboration and ensuring that research projects are meaningful and relevant. Moreover, AI can match researchers with similar research interests, enabling instructors and students to find potential collaborators and build research teams. As mentioned, AI can enhance data analysis processes, allowing instructors and students to analyze large amounts of data and identify patterns that can inform their research [35]. Thus, one tool such as ChatGPT can be used as tool to facilitate communication between instructors and students by answering questions and providing suggestions for research topics.

Other tools that use similar language models can also be used, such as Elicit, which is an AI research assistant that answers questions and provides suggestions for research topics, finds relevant papers, summarizes them, and extracts key information. Another tool is Scite.ai, which is an AI-powered platform that can be used to verify scientific claims and identify relevant research. It can also be used to track the impact of research and identify potential collaborators. Similarly, Research Rabbit, an AI-powered platform, can be used to find relevant research papers and track the impact of research. It can also identify potential collaborators and facilitate communication between instructors and students [38]. Finally, Mendeley, a social networking platform, can facilitate collaboration among researchers by allowing users to create groups, share research papers, and communicate with other researchers.

By leveraging the power of AI, Arab universities can enhance collaboration between instructors and students while familiarizing students with AI tools as an acute digital and transferable skill needed in today's labor market. On the other hand, it is essential to remember that AI technology is not a silver bullet, and AI tools must be managed ethically. For instance, to prevent misconduct, it is necessary to provide ethical training at all academic levels and implement a tailored ethical system [20]. There is no doubt that AI technology will significantly impact the field of HE over the next few years and will transform educational practices, institutions, and policies. In the Arab region, there is an acute need for expertise in AI to evaluate the capabilities of current AI systems in addition to assessing students' capabilities [34].

7 Discussion

The evaluation of higher education provision in Arab countries is of paramount importance in enabling students to achieve their goals and lead fulfilling lives. To ensure that universities provide students with the necessary capabilities, a thorough assessment of the curriculum, teaching methods, and support services must be carried out.

Here, the Capability Approach provides an effective framework for identifying the essential capabilities of Arab students and integrating AI into higher education. It emphasizes the importance of tangible outcomes relevant to students' specific situations, focusing on analytical and creative thinking, resilience, flexibility, agility, motivation, self-awareness, and lifelong learning. These skills are critical for workers in 2024 [29]. The approach requires a student-centered approach that takes into account the unique needs of Arab students and their communities.

In addition to these skills, graduates must possess digital skills to succeed in today's workforce. The Capability Approach requires evaluating the digital skills necessary for success, addressing digital inequalities, and adapting the curriculum to the Arab region's unique context. Collaboration with employers to align digital skills with the labor market is crucial for empowering graduates to excel in their careers and contribute to the growth of their communities. This approach also helps universities integrate AI into higher education by emphasizing students' ability to achieve tangible outcomes relevant to their situation. The focus should be on analytical and creative thinking, followed by resilience, flexibility, agility, motivation, self-awareness, and lifelong learning, as these are critical skills for workers [29].

The Capability Approach also requires identifying the digital skills that graduates need to succeed, evaluating the digital inequalities that exist in accessing the digital sphere, adapting these skills to fit the local context of the Arab region, and working closely with employers to align the digital skills taught in higher education with the needs of the labor market. By doing so, graduates will be empowered to thrive in their professional lives and contribute to the growth of their communities.

Finally, introducing AI tools into higher education can bring positive changes by making learning resources more adaptable, enhancing teacher support, and customizing curricular resources to meet the needs of local communities. However, it is crucial to acknowledge and address the risks associated with AI at a system level, such as discrimination caused by possible algorithmic bias. By embracing and integrating this new technology, the Arab HE sector can address these concerns and prevent unintended consequences while fully realizing the opportunities presented by AI.

8 Conclusion

In summary, education systems must reflect the skills demanded by society and the labor market. Improving education requires a pact involving all members of society, a unified vision, and strong leadership to align interests and prioritize investments; the Arab region, especially the GCC states, has the resources to achieve a learned society and knowledge economy [30]. However, many Arab universities still grapple with outdated curricula, leading to a mismatch between what students acquire and what employers require. Educational reforms, therefore, must be tailored to the unique Arab contexts and designed with existing constraints in mind. The chapter suggests the implementation of Amartya Sen's Capability Approach to higher education in the Arab region while considering the region's cultural, social, and economic contexts. The approach aims to enhance individuals' capabilities by focusing on their functionings (doings) and capabilities (outcomes). To achieve this, higher education institutions need to integrate critical thinking and digital skills, with a focus on AI, into their curriculum, as these skills are essential in today's labor market, and acknowledge the pressing need for AI expertise to enhance student capabilities.

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The Role of ChatGpt in Knowledge Sharing and Collaboration Within Digital Workplaces: A Systematic Review

Sheikh Abdulaziz Fahad, Said A. Salloum, and Khaled Shaalan

1 Introduction

Effective knowledge management is a critical aspect of organizational success, enabling efficient information sharing, decision-making, and innovation [1–6]. With the advent of digital technologies, knowledge management practices have evolved to accommodate the needs of modern workplaces [7–9]. In particular, the integration of artificial intelligence (AI) and chatbot technologies has transformed the way knowledge is created [10–14], accessed, and disseminated in the digital workplace [15, 16].

The digital workplace encompasses the use of digital tools, platforms, [17–20] and communication channels to facilitate work processes, collaboration [21–24], and knowledge exchange within an organization [4, 25–28]. In this dynamic environment, the implementation of AI-powered chatbots has gained significant attention due to their potential to enhance knowledge management practices [29, 30]. Chatbots, such as ChatGpt, utilize natural language processing and machine learning algorithms to interact with users, providing real-time responses, personalized assistance, and access to relevant information [31–34]. By incorporating AI chatbots into the digital workplace, organizations can streamline knowledge management processes and improve efficiency [35–37]. Chatbots serve as virtual assistants, capable of quickly retrieving and delivering accurate information from vast repositories, databases, and external sources. They offer a user-friendly interface for employees to ask questions, seek

S. A. Fahad · K. Shaalan
Faculty of Engineering and IT, The British University in Dubai, Dubai, UAE

S. A. Salloum (✉)
Health Economic and Financing Group, University of Sharjah, Sharjah, UAE
e-mail: ssalloum@sharjah.ac.ae

School of Science, Engineering, and Environment, University of Salford, Salford, England

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guidance, and access knowledge resources, thereby facilitating knowledge sharing and reducing information retrieval time.

Furthermore, chatbots have the ability to learn and adapt over time, making them valuable assets in knowledge management. Through continuous interaction and analysis of user interactions, chatbots can identify patterns, trends, and user preferences, enabling them to provide increasingly personalized and relevant responses [38–42]. This adaptive capability enhances the quality of knowledge dissemination and contributes to the development of a knowledge-rich digital workplace environment. In the realm of chatbots, ChatGpt represents an advanced AI model that combines functional and social characteristics to enhance human–computer interactions. Unlike traditional chatbots, ChatGpt goes beyond task-oriented interactions and incorporates social-oriented features. It aims to simulate human-like conversation, fostering a more engaging and interactive experience for users in the digital workplace [43–50].

ChatGpt's integration within the digital workplace has the potential to revolutionize knowledge management practices. Its individualized guidance, simplified information search, and integration of functional and social aspects of human–computer interaction contribute to an enhanced user experience. Moreover, ChatGpt can address emotional needs, promote autonomy, and foster feelings of dignity and self-actualization, thereby improving overall employee satisfaction and engagement [51]. The implementation of ChatGpt technology influences knowledge sharing and collaboration in the digital workplace by bridging the gap between users and information. It empowers employees to access information at their fingertips, contributing to more informed decision-making, faster problem-solving, and increased productivity. However, the introduction of chatbots in the digital workplace also raises challenges and potential limitations, which must be critically examined to ensure their effective integration.

In this systematic literature review, we aim to explore the benefits, challenges, and potential limitations associated with using ChatGpt for knowledge sharing and collaboration in the digital workplace environment. By examining the existing research, we seek to gain a comprehensive understanding of the implications of ChatGpt technology on knowledge management practices, employee engagement, and organizational outcomes. This review will provide valuable insights for researchers, practitioners, and organizational leaders seeking to harness the power of AI and chatbots in the digital workplace to optimize knowledge management processes and foster a culture of collaboration and innovation.

1.1 Research Objectives

- This Investigate the ways in which knowledge management and the digital workplace function together to promote ease of use, skill development, knowledge acquisition, analysis, integration, development, and communication.

- Explore how ChatGpt can be integrated into the digital workplace to enhance knowledge sharing, problem-solving, and teamwork.
- Critically analyze the limitations and potential challenges associated with the implementation of ChatGpt in the digital workplace, considering factors such as reliability, accuracy, ethical considerations, and the need for human oversight.

By achieving these research objectives, this study intends to contribute to the understanding of how knowledge management, the digital workplace, and ChatGpt intersect and impact organizational processes. The findings will provide insights for organizations seeking to leverage AI technologies in their knowledge management practices and help them navigate the opportunities and challenges associated with their implementation.

1.2 Problem Identification and Analysis

Knowledge sharing and collaboration are critical factors for success in digital workplaces. However, several challenges hinder effective knowledge management in these environments, leading to the need for innovative solutions. The problems addressed in the literature and in this systematic review can be summarized as follows:

- **Knowledge Silos**: In many organizations, knowledge tends to be stored in isolated silos, making it difficult for employees to access and share information across departments or teams. This lack of cross-functional collaboration hampers productivity, stifles innovation, and results in redundant efforts.
- **Information Overload**: Digital workplaces generate vast amounts of data and information, making it challenging for employees to filter through the noise and find relevant knowledge. This information overload can overwhelm employees and hinder their ability to make informed decisions or find timely solutions.
- **Limited Collaboration Opportunities**: Traditional communication channels and collaboration tools often have limitations in facilitating real-time collaboration, especially when dealing with geographically dispersed teams. This limitation can impede effective teamwork and hinder the exchange of ideas and expertise.
- **Barriers to Knowledge Discovery**: Locating relevant knowledge and expertise within an organization can be a time-consuming and complex task. Employees may struggle to identify subject matter experts or find accurate and up-to-date information, leading to delays in decision-making and problem-solving.

The importance of addressing these problems lies in the potential impact on organizational performance, employee engagement, and overall efficiency. In a rapidly evolving business landscape, organizations need to leverage knowledge sharing and collaboration to stay competitive, foster innovation, and adapt to changing market demands.

Conceptually, the problems are rooted in the lack of efficient knowledge management practices within digital workplaces. The rapid growth of information, coupled with organizational structures that inhibit knowledge flow, exacerbates the challenges associated with knowledge sharing and collaboration. Traditional methods and tools are often inadequate in addressing these issues, calling for innovative approaches. Practically, the introduction of AI technologies such as ChatGpt offers promising solutions to overcome these challenges. By leveraging natural language processing capabilities, ChatGpt can facilitate communication, knowledge discovery, and collaboration, enabling employees to access information, share expertise, and engage in real-time collaboration more effectively.

This systematic review aims to analyze the existing literature to understand the role of ChatGpt in addressing the problems associated with knowledge sharing and collaboration within digital workplaces. By identifying the benefits, limitations, and potential applications of ChatGpt, this review will provide insights for organizations seeking to enhance their knowledge management practices and improve overall productivity.

2 Literature Review'

This literature review aims to explore and synthesize existing research on the interplay between knowledge management, the digital workplace [6, 32, 52, 53], and the integration of ChatGpt. By conducting a systematic review of relevant studies, we seek to gain a comprehensive understanding of the current state of knowledge in this area, identify key trends, and uncover gaps in the literature. Through an analysis of the available literature, we aim to provide valuable insights into the ways in which knowledge management practices and the digital workplace function together, as well as the potential impact of integrating AI technologies like ChatGpt.

To ensure a systematic and structured approach, the review will commence with a broad examination of papers that critically analyze knowledge management practices in various organizational contexts. This initial analysis will lay the foundation for understanding the key principles, challenges, and trends associated with knowledge management. Building upon the insights gained from the knowledge management literature, the review will then shift its focus to the connection between the digital workplace and effective knowledge management. Specifically, it will explore how the digital workplace facilitates ease of use, skill development, knowledge acquisition, analysis, integration, development, and communication. By synthesizing relevant studies, this section will provide a nuanced understanding of how the digital workplace enables and shapes knowledge management practices.

Subsequently, the review will narrow its scope to investigate the integration of ChatGpt within the digital workplace and its implications for knowledge sharing, problem-solving, and teamwork. This section will delve into the specific functionalities of ChatGpt, its role as an AI-based knowledge assistant, and it's potential to enhance collaboration and decision-making processes within the digital workplace. A critical analysis of existing studies will shed light on the advantages, limitations, and ethical considerations associated with the implementation of ChatGpt in this context.

By adopting this structured approach, the literature review will provide valuable insights into the ways in which knowledge management practices and the digital workplace function together, and the potential impact of integrating ChatGpt.

Al-Hashemy [15] discusses Knowledge management (KM) is an essential and systematic process aimed at capturing diverse forms of knowledge and experiences within an organization. The purpose of KM is to preserve and utilize this knowledge to enhance decision-making processes. In today's dynamic business environment, KM has become a necessity for organizations, primarily due to the transient nature of employees. Organizations face the risk of losing valuable institutional knowledge when employees depart. However, to mitigate this knowledge loss, organizations are adopting new information technology (IT) tools such as corporate intranets, virtual community collaboration tools, data warehousing, data mining tools, and web-based business intelligence systems. These tools empower organizations to improve their capabilities in collecting, organizing, and disseminating information required for effective decision making and performance enhancement at all organizational levels. By leveraging corporate intranets, organizations can establish centralized repositories for knowledge storage and retrieval. This facilitates easy access to critical information and expertise, enabling employees to make informed decisions. Virtual community collaboration tools facilitate collaboration and knowledge sharing among employees, regardless of geographical barriers or time zones. This promotes a culture of collaboration and enhances the collective intelligence of the organization (Fig. 1).

Lim et al., [54] critically examines the integration of data warehousing, data mining tools, and web-based business intelligence systems in knowledge management and knowledge sharing within organizations. These IT tools offer the potential to capture, store, analyze, and disseminate knowledge effectively, supporting informed decision-making processes. However, it is essential to assess their limitations and challenges, including data accessibility, accuracy, usability, and ethical considerations. By evaluating the effectiveness of these tools, organizations can optimize their knowledge management practices and enhance decision-making capabilities, ultimately gaining a competitive advantage in the digital era.

Narendra et al., [55] highlights the integration of a smart knowledge management system that caters to individual learning styles and preferences, promotes community-based learning, and utilizes machine learning algorithms. The system allows learners to interact with the platform, customize their learning space, and access information based on their preferred learning styles. The data collected from learners is analyzed using data analytics algorithms to classify their learning styles and personalize the learning experience. Learners can also engage with communities of practice and external resources to enhance their learning process. The system records and codifies accessed information, enabling continuous improvement of learning support. Users have the flexibility to customize their user interface, select preferred learning modes, subjects of interest, and join learning groups. The system provides various formats for information acquisition and interactive learning, fostering knowledge sharing within a unit. While this approach offers benefits such as personalized learning experiences and enhanced knowledge sharing, there are certain limitations to consider. These include potential challenges related to data privacy and security, accuracy and reliability of machine learning algorithms, the need for continuous adaptation of the system to individual learning needs, and the possibility of information overload. Addressing these limitations is crucial for ensuring the effectiveness and ethical implementation of the smart knowledge management system.

Li and Herd [56] introduces a framework that integrates metacognitive skills within the design of a learner application, aiming to enhance learning performance. The framework emphasizes the active involvement of learners in self-regulation, reflection, and knowledge accumulation. By practicing metacognitive skills during app usage, such as reflecting on cognitive processes and monitoring performance, learners can effectively organize their learning and contribute to the knowledge sharing within the system and the learner community. The framework also integrates other learning principles, including experiential learning, problem-based learning, critical thinking, and action learning, to provide a comprehensive learning experience. The proposed framework follows a structured process, starting with a triggering event that prompts learners to engage in the Application Facilitated Self-Directed (AFiSD) learning process. Learners interact with the self-regulated learning space and the smart knowledge management system, take actions to address learning gaps, and enter the review, reflection, and documentation mode to complete the learning process. The insights and reflections contributed by learners are incorporated into the adaptive learning process of the system, benefiting future users. Overall, this paper presents a novel approach to integrating metacognitive skills into learner applications, promoting active learning and knowledge sharing within digital learning environments.

Chatterjee et al., [57] explores the changing perceptions of the digital workplace and identifies the factors that make it a crucial strategic direction for organizations. The integration of mobile, big data, cloud computing, and search-based applications, with a focus on mobile development, enables organizations to transform work processes and increase productivity. The digital workplace offers several benefits, including increased individual and organizational productivity, flexibility for remote work, and improved collaboration and knowledge sharing. It provides a user-friendly and intuitive environment, tangible benefits for users, and transcends organizational and geographic boundaries. Moreover, it consolidates essential functions in one centralized location, promoting efficiency and capturing critical information. However, there are also limitations to consider. First, the digital workplace requires employees to adapt to new technologies and workflows, which may involve a learning curve and resistance to change. Second, ensuring the security and privacy of digital workplace platforms is crucial, as sensitive information is being shared and accessed. Organizations need to implement robust security measures to protect data and prevent unauthorized access. Lastly, while the digital workplace enhances collaboration, it may also create a dependency on technology, potentially leading to challenges if technical issues arise or if individuals become overly reliant on digital tools, hindering face-to-face interactions (Fig. 2).

Hicks [58] highlights the challenges related to language diversity, cultural differences, and the management of virtual teams in the adoption of digital workplaces. It emphasizes the need to address disparities in access to IT resources to create a more equitable society. This can be achieved through initiatives such as providing affordable hardware and software solutions, improving connectivity in rural areas, and offering technical support. Additionally, organizations should recognize the diverse range of skills, abilities, and attitudes towards IT and provide ongoing training and development opportunities. Creating a culture that embraces different learning preferences and styles is crucial for fostering an inclusive digital work environment. Ensuring the reliability and accuracy of information accessed through IT platforms is another important aspect. Promoting media literacy, critical thinking skills, and implementing content verification processes can enhance the credibility of information. By addressing these challenges and promoting inclusivity, organizations can leverage the full potential of their workforce in the digital era.

Meske and Junglas [59] examines the trend of remote work, also known as telework, and its implications for employees and organizations. While research suggests the effectiveness of remote work to a certain extent, there is a lack of empirical evidence supporting the preference for a specific workplace setting among employees open to digital change. However, remote work offers various benefits, including time and cost savings due to the elimination of commuting. It also enables flexibility in managing work tasks through shorter meetings and asynchronous collaboration tools. From an organizational perspective, remote work can lead to cost savings, lower absenteeism rates, and increased employee satisfaction and well-being. Furthermore, the digital disruption and advancements in technology create opportunities for organizations to leverage remote work and foster inclusive work environments. Despite these benefits, remote work presents challenges in terms of coordination, communication, and trust management. The absence of face-to-face interactions can result in feelings of isolation and hinder teamwork and collaboration. Onboarding processes may also be more challenging in remote or hybrid work environments. To address these challenges, organizations should implement strategies to effectively manage remote teams, promote communication and collaboration, and support employee well-being. This can involve utilizing technology tools for virtual communication, establishing clear guidelines, fostering virtual social interactions, and cultivating a culture of trust and inclusivity. Critical analysis of the topic highlights the need for organizations to carefully manage knowledge sharing and knowledge management in remote work settings, ensuring that information flows efficiently and employees remain connected and engaged in the digital workplace.

White [60] presents a knowledge-based approach for enhancing dialogue generation in conversational systems. The proposed method utilizes memory components to manage contextual information such as chat history, situations, and preferred topics. By incorporating a knowledge base, the system aims to generate natural and personalized response sentences. The approach is designed for educational purposes, specifically English conversation training. A user study demonstrates positive user satisfaction, highlighting the effectiveness of the knowledge-based conversational system. Future work includes improving context prediction and incorporating persona concepts for enhanced user interactions. This research contributes to advancing dialogue generation in conversational systems and holds potential for applications in education and customer consultation domains.

Shafeeg et al., [61] delves into the concept of Computers Are Social Actors (CASA) paradigm, which recognizes the influence of computers on human behavior and social interactions, particularly in the digital workplace. It explores the integration of ChatGpt, a generative AI solution, within the digital workplace, emphasizing its functionality, social features, and ability to cater to individual needs for personalized assistance and quick answers. ChatGpt's role in facilitating knowledge management, collaboration, and addressing emotional needs is highlighted. Additionally, the potential benefits of ChatGpt in education and productivity improvement are mentioned. The excerpt also discusses the limitations of traditional keyword-based search techniques in retrieving precise information and the need for more intelligent search approaches. Overall, ChatGpt is presented as an advanced AI model that enhances human–computer interactions, supports knowledge management, and fosters a productive and collaborative digital workplace environment.

Beerbaum [62] explores the utilization of AI-powered Chatbots, specifically ChatGpt, to enhance knowledge sharing and collaboration in digital workplaces. Chatbots based on Artificial Intelligence (AI) technology simulate human-like conversations and interact with users in a more human-centric manner, improving the efficiency of knowledge management systems. The implementation of ChatGpt in digital workplaces has the potential to significantly impact knowledge sharing and collaboration. By providing personalized and interactive assistance, ChatGpt facilitates efficient information exchange and problem-solving. Users can engage in conversational interactions with ChatGpt, enhancing the user experience and promoting active participation. Integrating both functional and social aspects of human–computer interaction, ChatGpt fosters effective knowledge sharing and collaboration in the digital workplace. Employees can utilize ChatGpt to ask questions, seek clarification, and access relevant information, thereby enhancing decision-making and productivity. The conversational nature of ChatGpt creates a collaborative environment where employees can seamlessly communicate and exchange knowledge and ideas. Overall, the implementation of ChatGpt technology in digital workplaces positively influences knowledge sharing and collaboration, creating a more interactive and productive work environment.

Castillo-González [51] highlights how ChatGpt aligns with the principles and objectives of Learning 3.0, which emphasizes student-centered learning with the assistance of technology. ChatGpt facilitates critical thinking, creativity, problem-solving, adaptability, autonomy, collaboration, and communication, all of which are central to Learning 3.0. By providing personalized educational interactions, ChatGpt supports users in their learning journey. One of the key benefits of ChatGpt for knowledge sharing and collaboration is its ability to deliver individualized educational objectives. Users can interact with ChatGpt to seek answers, acquire information, and engage in problem-solving, empowering them to take ownership of their learning and explore areas of interest. Moreover, ChatGpt promotes adaptability and collaboration in learning. With its 24-h availability and reliable assistance, ChatGpt encourages users to explore and find optimal answers. The interactive question-and-answer structure of ChatGpt fosters a collaborative learning environment, facilitating discussions, idea exchange, and a deeper understanding of various topics. The convenience and accessibility offered by ChatGpt are particularly valuable in the digital workplace. As online education expands and internet technology advances, ChatGpt becomes a valuable tool for both learners and instructors. It helps users navigate the vast amount of information available online, providing efficient and relevant responses that save time, improve performance, and enhance the quality of learning and work-related tasks. However, there are challenges and limitations associated with ChatGpt. Ensuring the accuracy and reliability of information is a significant challenge, as ChatGpt's responses may not always be completely accurate or up-to-date. Users need to exercise critical thinking and verify information obtained through ChatGpt. Contextual understanding is another challenge, as ChatGpt may not fully grasp the nuances or context of a given situation, impacting the accuracy and relevance of its responses. Ethical considerations regarding AI and privacy must also be addressed. ChatGpt interacts with users and collects data, raising concerns about data security and privacy. Organizations and users should implement measures to protect sensitive information and comply with relevant regulations. In summary, ChatGpt aligns with the principles of Learning 3.0, supporting personalized learning, adaptability, autonomy, collaboration, and communication. It offers benefits such as individualized educational objectives, convenience, and accessibility. Challenges include ensuring accuracy and contextual understanding, as well as addressing ethical and privacy concerns. By being mindful of these factors, organizations and users can effectively leverage ChatGpt for knowledge sharing and collaboration in the digital workplace.

Chaudhry et al., [63] examines the influence of the COVID-19 pandemic on education and the need for adaptation in uncertain times, leading to the emergence of ChatGpt as an AI companion. ChatGpt offers rapid, individualized assistance, providing solutions and personalized support to alleviate anxiety and enhance motivation and self-exploration. The customized exploration empowers self-directed learners to access valuable resources and expand their options. A critical analysis of ChatGpt in the digital workplace context reveals both strengths and limitations. On the positive side, ChatGpt's rapid and individualized assistance enhances knowledge management and sharing. It serves as a valuable tool for employees seeking immediate answers, guidance, or support in their work-related tasks, promoting productivity and efficiency. Additionally, ChatGpt can facilitate knowledge sharing by acting as a repository and enabling collaborative problem-solving, fostering a culture of continuous learning in the digital workplace. However, limitations should be considered. The accuracy and reliability of ChatGpt's information can be a concern, as it may not always provide completely accurate or up-to-date responses. Users must exercise caution and critical thinking, especially in professional settings. Contextual understanding is another challenge, as ChatGpt may not grasp the nuances of the workplace environment, impacting the relevance of its responses. Ethical considerations regarding privacy and data security must be addressed to protect sensitive information shared during interactions with ChatGpt. In conclusion, ChatGpt offers benefits for knowledge management and collaboration in the digital workplace by providing rapid, individualized assistance and facilitating knowledge sharing. However, limitations in accuracy, contextual understanding, and ethical considerations should be acknowledged. Organizations can effectively leverage ChatGpt while ensuring responsible use and addressing these factors.

3 Methods

In this section, we outline the methodology employed for conducting the literature review on the topic of knowledge management [64–66], digital workplaces, and ChatGpt. The following subsections provide details on the inclusion/exclusion criteria, data sources and search strategies, data coding and analysis, and quality assessment methods utilized in this study [67–71]. By employing these rigorous methods, we aimed to ensure a comprehensive and reliable review of the relevant literature.

3.1 Inclusion /Exclusion Criteria

In this study, we aimed to identify relevant papers related to knowledge management, digital workplaces [72–76], and ChatGpt. To ensure the inclusion of papers with critical analysis, we considered studies that provided in-depth evaluations, critiques, or discussions on the topic. The inclusion criteria included papers published between 2017 and 2023 to focus on recent developments in ChatGpt. We primarily searched for papers in IEEE Xplore and ScienceDirect databases.

3.2 Data Sources and Research Strategies

To identify relevant papers, we employed a systematic search strategy. The search terms used included variations and combinations of keywords such as "knowledge management," "digital workplaces," and "ChatGpt." Boolean operators such as "AND" and "OR" were utilized to refine the search. Additionally, filters were applied to retrieve papers published from 2017 to 2023, narrowing the focus to the last five years of research.

3.3 Data Coding and Analysis

After retrieving the relevant papers, a coding and analysis process was undertaken. The data coding involved identifying key themes, concepts, and findings within each paper. Examples of coding categories included the benefits of ChatGpt in knowledge sharing, challenges in implementing ChatGpt in digital workplaces, and the impact of ChatGpt on employee productivity. A codebook was created to guide the coding process and ensure consistency.

Once the coding was completed, data analysis was performed to identify patterns, trends, and critical insights across the selected papers. This involved summarizing the findings within each coding category and analyzing the relationships between different themes. By examining the recurring themes and divergent perspectives, we gained a comprehensive understanding of the critical analysis surrounding knowledge management, digital workplaces, and the role of ChatGpt.

Throughout the analysis process, any discrepancies or disagreements in coding were resolved through discussions among the research team. Consensus was reached through iterative discussions and referring back to the original papers for clarification.

Overall, this systematic approach to data coding and analysis ensured that the findings of this study were grounded in a rigorous examination of the selected papers, providing a comprehensive overview of the critical analysis on knowledge management, digital workplaces, and the implications of ChatGpt.

3.4 Quality Assessment

In order to assess the quality and reliability of the selected papers, we employed specific strategies and techniques tailored to our research context. The following methods were used for quality assessment:

Reputable Sources: We focused on papers sourced from reputable and peer-reviewed platforms such as IEEE Xplore and ScienceDirect. These databases are known for their stringent review processes, ensuring the quality and credibility of the published papers.

Selection Criteria: The inclusion criteria set during the literature search phase helped ensure that only relevant and high-quality papers were considered. We specifically targeted papers with critical analysis to ensure a robust and comprehensive evaluation of the topic. By selecting papers that met our inclusion citeria, we aimed to include scholarly work that provided valuable insights and rigorous analysis.

Peer Review: The selected papers were Peer reviewed and assessed by some experts in the field of knowledge management and digital workplaces. Experts provided their expertise and insights to evaluate the relevance, credibility, and methodological soundness of the papers. Their input contributed to the overall quality assessment process.

Evaluation Framework: We utilized an evaluation framework to assess the quality of the selected papers. This framework consisted of predefined criteria, including research design, methodology, data collection techniques, analysis methods, and clarity of presentation. Each paper was evaluated based on these criteria to determine its quality and suitability for inclusion in the study.

Critical Appraisal: A critical appraisal was conducted to evaluate the strengths and weaknesses of the selected papers. This involved a thorough examination of the research methods, data analysis techniques, and the overall validity and reliability of the findings. The critical appraisal helped identify any potential biases, limitations, or gaps in the research, contributing to a comprehensive assessment of the paper's quality.

By employing these specific strategies and techniques, we ensured a rigorous quality assessment of the selected papers. The combination of reputable sources, expert review, evaluation frameworks, and critical appraisal enabled us to identify high-quality papers that provided valuable insights and critical analysis on the topic of knowledge management, digital workplaces, and ChatGpt.

4 Result

The results of the review indicate that knowledge management practices in the digital workplace have evolved to leverage technology and digital tools, enabling organizations to effectively capture, store, and disseminate knowledge among employees. The integration of digital platforms and collaborative tools has facilitated seamless knowledge sharing, improved problem-solving capabilities, and enhanced team collaboration within the digital workplace environment. These findings support the notion that knowledge management and the digital workplace are interconnected and mutually reinforce each other in promoting efficient knowledge utilization and exchange [63].

Regarding the integration of ChatGpt into the digital workplace, the results demonstrate its potential to enhance knowledge sharing, problem-solving, and teamwork. ChatGpt's ability to provide personalized and rapid assistance has been found to facilitate information retrieval, offer guidance, and support decision-making processes. The interactive nature of ChatGpt encourages active participation and collaboration, promoting a culture of continuous learning and knowledge exchange among employees. These findings suggest that ChatGpt can serve as a valuable tool for enhancing knowledge management practices within the digital workplace [77].

However, the review also identified several limitations and potential challenges associated with the implementation of ChatGpt in the digital workplace. The reliability and accuracy of information provided by ChatGpt were found to be areas of concern, as the model may not always deliver completely accurate or up-to-date responses. Ethical considerations related to data privacy and security were also raised, highlighting the importance of implementing appropriate safeguards to protect sensitive information shared during interactions with ChatGpt. The need for human oversight and critical thinking when relying on ChatGpt's responses was emphasized to mitigate the potential risks associated with the limitations of the AI model (Fig. 3).

According to the data presented in Graph 1 [78], 24% of the respondents reported that they believed they have saved between $50,000 to $75,000 through the implementation of ChatGpt in their organization and digital workplace. This indicates a significant cost-saving potential associated with the use of ChatGpt. Additionally, 22% of the respondents stated that they believed they have saved between $25,000 to $50,000, further highlighting the financial benefits realized through the integration of ChatGpt. These findings suggest that ChatGpt can contribute to cost efficiency and optimization in the digital workplace setting (Fig. 4).

Graph 2 [78] displays the various tasks for which ChatGpt is currently being utilized in the digital workplace. According to the data, 65% of the respondents reported using ChatGpt for writing major codes in software houses, indicating that ChatGpt is employed as a valuable tool for programming and software development tasks. Additionally, approximately 58% of the respondents used ChatGpt for content creation, suggesting that it assists in generating written content such as articles, blog posts, and marketing materials. Furthermore, around 57% of the respondents utilized ChatGpt for customer support, highlighting its role in providing assistance and responding to customer inquiries. These findings demonstrate the versatility of ChatGpt in supporting a range of tasks within the digital workplace, including coding, content creation, and customer service.

Graph 3 [79] focuses on the usage of ChatGpt by HR professionals. The graph indicates that 80 hireses (HR professionals) used ChatGpt for writing job descriptions, suggesting that it assists in generating accurate and comprehensive job descriptions. Moreover, approximately 65% of the hireses utilized ChatGpt for drafting interview questions, highlighting its role in facilitating the interview preparation process. These findings suggest that ChatGpt can be a valuable tool for HR professionals, aiding in the creation of job descriptions and interview-related tasks. By leveraging ChatGpt for these purposes, HR professionals can streamline their workflow and enhance the efficiency of their hiring processes (Fig. 5).

The graph [79] illustrates the exponential growth of ChatGpt's user adoption within a short period. According to the data, ChatGpt sprinted to one million users in just five days. This remarkable achievement is compared to well-known companies such as Netflix, Twitter, Facebook, and Instagram, suggesting that ChatGpt's user adoption rate surpassed these popular platforms (Fig. 6).

The graph [80] illustrates the concerns and perceptions of individuals regarding the impact of digital workplaces on job security and the human aspect of work. According to the data, 36% of respondents believe that digital workplaces have the potential to replace jobs. This indicates apprehension about automation and the potential for job displacement.

Furthermore, 73% of respondents express concerns about the loss of human touch in digital workplaces. This suggests that individuals value personal interactions, face-to-face communication, and the emotional connection that comes with traditional work environments. The fear of diminished authenticity is also evident, with 41% of respondents expressing concerns in this area.

Overall, the graph highlights the apprehensions individuals have regarding the potential consequences of digital workplaces, including job replacement, the loss of human touch, and a perceived decline in authenticity. These findings shed light on the sentiment and attitudes towards the increasing integration of technology in the workplace (Fig. 7).

The graph [81] illustrates the opinions of individuals regarding the use of ChatGpt in knowledge sharing within digital workplaces, specifically focusing on concerns related to plagiarism. According to the data, 59% of respondents believe that ChatGpt has the potential to contribute to plagiarism in the context of knowledge sharing. This finding suggests that a significant portion of respondents harbor doubts about the originality and authenticity of information generated by ChatGpt. They may question whether the content generated by ChatGpt is appropriately sourced, properly cited, or adequately paraphrased. This concern aligns with the broader issue of intellectual property and ethical considerations surrounding the use of AI-generated content. These results emphasize the importance of ensuring proper attribution, citation, and quality control when using ChatGpt or similar AI tools in digital workplaces. Organizations and individuals should develop guidelines and practices to address concerns related to plagiarism, promoting responsible knowledge sharing and ethical use of AI technology.

5 Conclusion and Discussion

In this study, we conducted a systematic review focusing on the intersection of knowledge management and the digital workplace, with a specific emphasis on the role of ChatGpt. The results of our analysis provide valuable insights into the potential benefits and challenges associated with the integration of ChatGpt in knowledge management practices within the digital workplace. Firstly, our findings highlight the significance of knowledge management in the digital workplace, showcasing its role in facilitating ease of use, skill development, knowledge acquisition, analysis, integration, development, and communication. The digital workplace serves as a platform where employees can access and share knowledge, collaborate, and solve problems collectively. The results pertaining to ChatGpt reveal its potential to enhance knowledge sharing, problem-solving, and teamwork in the digital workplace. The ability of ChatGpt to provide rapid and individualized assistance makes it a valuable tool for employees seeking immediate answers, guidance, or support in their work-related tasks. It also acts as a knowledge repository, capturing and storing valuable insights and expertise for easy access by users. These functionalities promote productivity, efficiency, and collaboration in the digital workplace [4, 82–85]. However, our analysis also identified several limitations and challenges associated with the implementation of ChatGpt. Concerns regarding the reliability and accuracy of information provided by ChatGpt were raised, highlighting the need for users to exercise caution and critical thinking. The potential loss of human touch and authenticity in interactions with ChatGpt were also recognized, necessitating the need for human oversight and maintaining a balance between AI technology and human expertise. The theoretical contribution of this review lies in its exploration of the intersection between knowledge management, the digital workplace, and ChatGpt. By examining the ways in which ChatGpt can enhance knowledge sharing and collaboration, we contribute to the understanding of how AI technology can be effectively integrated into knowledge management practices. From a practical standpoint, our findings have implications for organizations seeking to leverage ChatGpt in the digital workplace. The insights gained from this review can inform decision-making processes and guide the implementation of ChatGpt to maximize its benefits while addressing the identified limitations and challenges. Organizations can utilize ChatGpt to streamline information retrieval, facilitate collaborative problem-solving, and create a productive and collaborative work environment [46, 86–89]. While our study provides valuable insights into the integration of ChatGpt in knowledge management and the digital workplace, there are other interesting areas that could be explored further. For instance, the ethical considerations surrounding AI technology and privacy in the digital workplace deserve additional attention. Future research can also delve into the development of strategies to enhance the accuracy and contextual understanding of AI-powered chatbots like ChatGpt. In conclusion, this systematic review contributes to the existing literature by examining the role of ChatGpt in knowledge management and the digital workplace. The findings emphasize the potential benefits and challenges associated with ChatGpt, highlighting the importance of responsible implementation and the need for a balanced approach that combines AI technology with human expertise. By leveraging ChatGpt effectively, organizations can enhance knowledge sharing, problem-solving, and collaboration in the digital workplace, ultimately fostering a productive and innovative work environment.

Fig. 1 Exploring the landscape of knowledge management systems and their branches

Fig. 2 Unveiling the diverse facets of digital workplaces and their branches

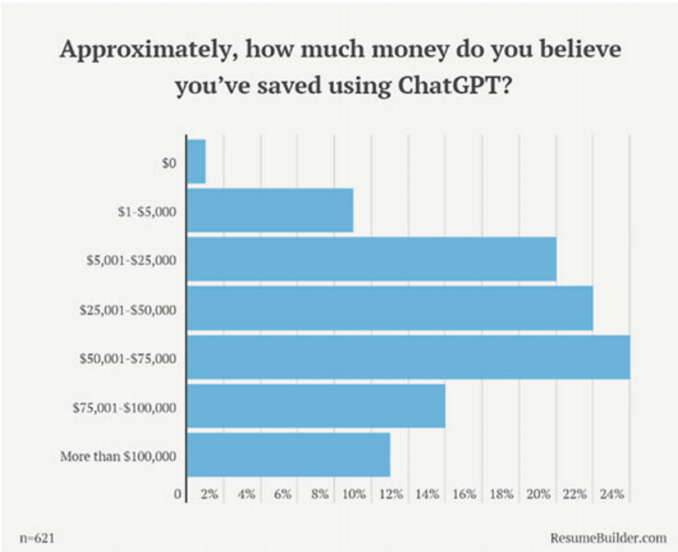


Fig. 3 Cost savings achieved using ChatGpt in the digital workplace

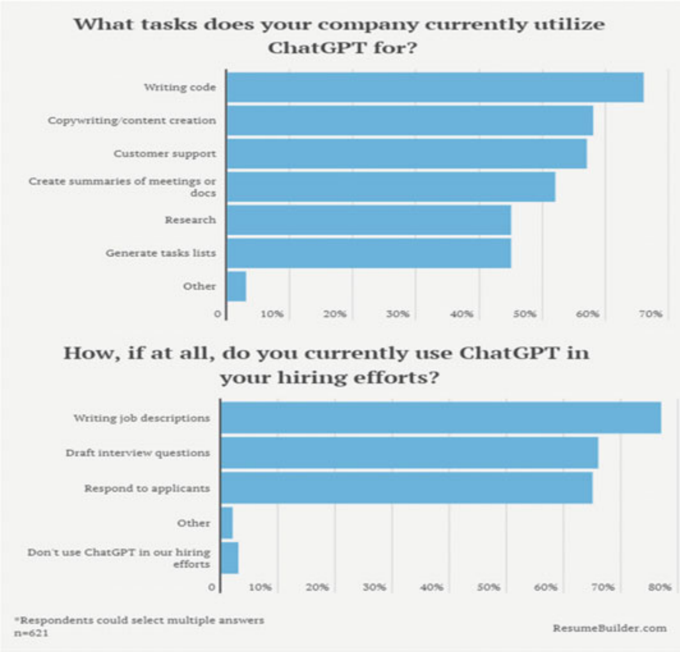
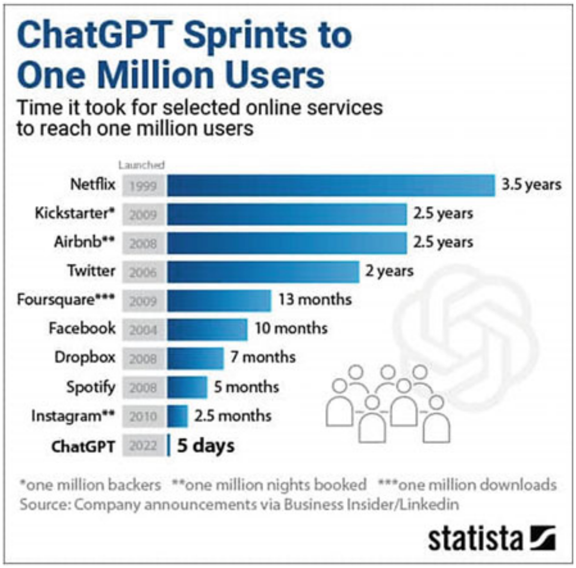


Fig. 4 Tasks utilized by ChatGpt in the digital workplace

Fig. 5 Rapid user adoption: ChatGpt



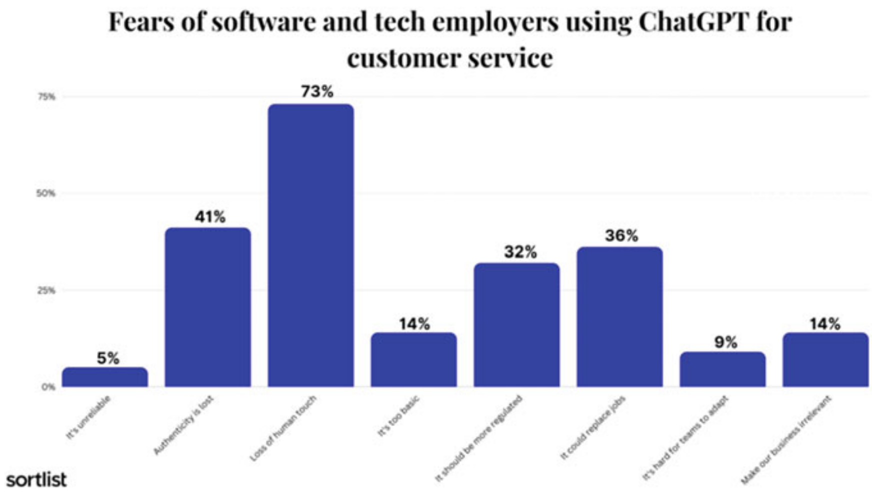


Fig. 6 Perceptions on the fear of job replacement and loss of human touch in digital workplaces

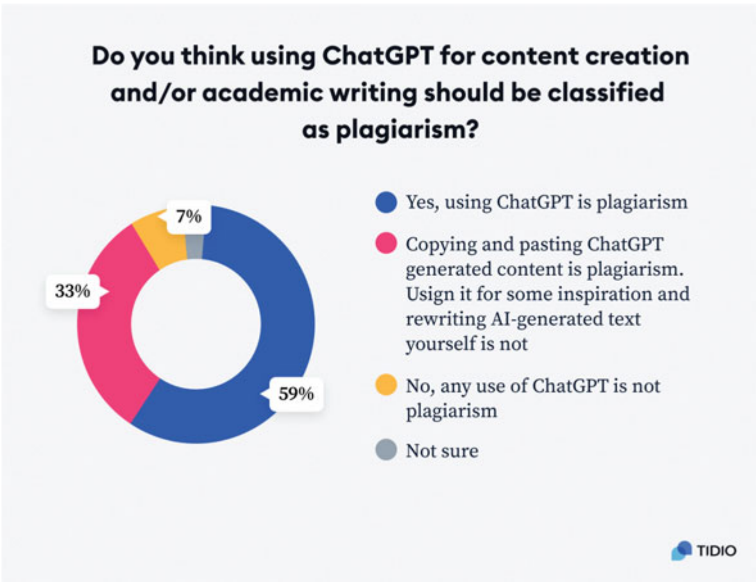


Fig. 7 Perceptions of ChatGpt as a potential source of plagiarism in knowledge sharing and digital workplaces

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Exploiting AI's Potential in Knowledge Management

Maha Salem, Said A. Salloum, and Khaled Shaalan

1 Introduction

Knowledge is a valuable and strategic resource that can provide a lasting competitive advantage and create the most value [1, 2]. Knowledge management (KM) allows individuals and organizations to work together, share knowledge, create new knowledge [3–7], and apply it to boost creativity and performance [1, 8]. It is very important to distinguish between explicit and tacit knowledge in business [9–13]. In contrast to tacit knowledge, which is difficult to define and cannot be documented [14–16], explicit information can be created, shared, and used by individuals and organizations to improve productivity [17–21], promote innovation, and expand the body of knowledge [22, 23].

Advancements in information technology (IT) are often considered accelerators for organizational transformation initiatives within the KM [24, 25]. To effectively manage knowledge, organizations must move beyond simply acquiring knowledge and focus on efficiently utilizing new knowledge while effectively managing existing knowledge [1, 26]. The Knowledge Management System (KMS) is a platform system for knowledge processing and gathering. KMS, taken in its broadest meaning, is a dynamic cycle system made up of the KM process itself, valuable knowledge, the knowledge utilization process, the (KMS), and the environment in which the knowledge management system is operated [1, 27].

AI technologies play a significant role in KM [28]. The utilization of AI technologies is crucial in both the knowledge acquisition and creative fields and in knowledge

M. Salem · K. Shaalan
Faculty of Engineering and IT, The British University in Dubai, Dubai, UAE

S. A. Salloum (✉)
Health Economic and Financing Group, University of Sharjah, Sharjah, UAE
e-mail: ssalloum@sharjah.ac.ae

School of Science, Engineering, and Environment, University of Salford, Salford, UAE

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management processes [29, 30]. This is due to their emphasis on intelligent simulation and learning capabilities [1, 31–35]. AI facilitates the capturing and organizing of knowledge, allowing the appropriate individuals to comprehend and rapidly disseminate it among all relevant user groups [36, 37]. This recognizes the fact that modern organizations may not be restricted by the conventional firm boundaries of the past [38].

As per The World Economic Forum, by the end of 2025, artificial intelligence (AI) is projected to undertake 52% of company tasks, while 48% of them will still be performed by humans [39, 40]. Businesses commonly utilize various types of intelligent robots, such as chatbots, virtual assistants, recommendation agents for customer decision-making, stock trading bots, and analytical models for predicting work disabilities, segmenting and anticipating churn, or forecasting the likelihood of making a purchase, among other applications [41, 42]. AI offers the means to enable machines to learn, which, when integrated into knowledge delivery, can expedite, streamline, and enhance the accuracy of decision-making. AI's potential to expand, apply, and generate knowledge in novel ways is immense. Through machine learning, AI systems can detect patterns in vast amounts of data and complicated models, interdependent systems, resulting in improved decision-making efficiency [8, 33, 43, 44]. Knowledge management systems that employ artificial intelligence are typically categorized into three functional types: knowledge-based expert systems (KEBS), neural networks (NN), and case-based reasoning systems (CBR) [1, 38].

Artificial intelligence is being used more and more in KM [45–48], which is a field that is continuously growing [49–51]. This systematic review can be useful to determine the present status of research in this field and to flag areas that require more investigation [33, 52–54]. Additionally, it may assist in highlighting gaps in the existing body of knowledge and recommending topics for further study. By doing so, it will be possible to direct research efforts and make sure that the most crucial issues are being covered. To ensure that the most recent developments in this field are used to their greatest potential by firms. This review will concentrate research question shown in Table 1.

Table 1 The research questions of the study

No	Research question
RQ1	What is the aim of the study?
RQ2	How can the integration of artificial intelligence (AI) and knowledge management (KM) systems enhance organizational learning and decision-making processes?
RQ3	What are the methods for integrating AI technologies into knowledge management systems?
RQ4	What are the main sectors that are making use of AI technology on a regular basis?

2 Background of the Study

To effectively incorporate AI in Knowledge Management, tasks should be distributed between intelligent computers and humans in a new way. Along with the acquisition of new mind-sets, skill sets, and competencies by both parties. Knowledge workers must develop the essential perspectives, skills, and work habits to take advantage of AI's benefits while avoiding any associated disadvantages. Effective collaboration between intelligent systems and knowledge workers is critical for exploiting the unique possibilities of AI in KM [55, 56]. Although AI's expansion presents both opportunities and challenges for enterprises, there is a current emphasis on enhancing decision-making skills. AI is regarded as a potent tool for advancing societal growth and development [57–61]. The following sections present the background of KM and AI.

2.1 The Power of Knowledge Management

In today's rapidly changing industry environments, businesses can derive significant benefits by gathering information from their assets and comparing results [62–64]. This practice enables organizations to stay informed, make informed decisions, and effectively navigate the dynamic business landscape. Just like corporations acquire information, individuals within organizations can enhance their skills and contribute more effectively by gaining a comprehensive understanding of their organization's knowledge [65–70].

While having pre-existing knowledge may seem beneficial initially, it can actually hinder both individuals and organizations when it comes to adapting to change. The inability to acquire new information or embrace new practices makes it challenging to navigate evolving circumstances. To enable the effective collection and utilization of existing information, it becomes crucial to learn new procedures at both the organizational and individual levels. However, achieving this requires a shift in organizational culture and the implementation of innovative knowledge management techniques [35, 62, 71–73].

The process of managing knowledge involves organizing collected knowledge through indexing, content filtering, and establishing linkages and relationships. The knowledge acquired through these applications is utilized to improve and organize existing knowledge. Presenting knowledge effectively is essential because different approaches can be used to structure the knowledge base, making it challenging to integrate and reconfigure information from different sources. Knowledge distribution and sharing are vital as they enable the accumulation of knowledge within the organization. The advent of information technology and the Internet has facilitated the sharing of knowledge, and the continuous development of emerging technologies further enhances this process [74]. KM in organizations is centered around comprehending the various aspects of knowledge, such as creation, transfer, use, flows, and governance [74]. The primary goal of implementing KM is to harness the existing knowledge within an organization to enhance its mission and support the activities of its users. This involves converting implicit or tacit knowledge into explicit and easily accessible formats, establishing connections between the organization's experts and explicit knowledge, and utilizing social network analysis to ensure effective distribution and utilization of knowledge. It is essential to not only grasp the concept of knowledge management but also gain a profound understanding of the organization's unique knowledge characteristics [8].

Business intelligence (BI) refers to a framework that combines structures, data storage, analytical tools, computational applications, and processes to transform data into useful information for decision-making. BI systems can help organizations discover, capture, and disseminate valuable insights from data. So, it enables the structuring of real-time indicators by bridging the gap between business, management, and technology [75]. To succeed to achieve desired goals and create knowledge outcomes from information inputs, it is crucial to underline that KM needs to recognize the many ways in which knowledge may be used, depending on the context and transformation processes involved [74].

2.2 Utilizing AI Tools for Better Decision-Making

As more businesses learn how to implement KM practices for the enhancement of organizational value, the KM software industry is seeing tremendous development [46, 76–82]. Rapid change is occurring in the technology supporting KM activities, with new suppliers joining the market as existing ones go [74, 83]. As more businesses learn how to implement KM practices for the enhancement of organizational value, the KM software industry is seeing tremendous development. Rapid change is occurring in the technology supporting KM activities, with new suppliers joining the market as existing ones go. The several uses of AI in KM are significant AI has drawn a lot of interest and has been widely used in a variety of commercial sectors [74]. AI technologies play a significant role in the creation of knowledge and acquisition as well as knowledge management processes, [8] with characteristics centered on smart modeling and learning, requiring a focus on the problem of management technology interactions [1].

AI aims to create algorithms that give computers cognitive abilities for sense- and decision-making. Computational agents with these algorithms are capable of understanding data, evaluating opportunities and risks in various scenarios, coming up with plans of action, carrying out tasks, learning, adapting, and sharing information by passing it to humans or other AIs [8, 84, 85]. Machine learning-based AI systems can identify patterns in massive amounts of data and model intricate, interconnected systems to produce results that increase the effectiveness of decision-making [8]. AI tools have diverse applications in business, including market segmentation to tailored product and service offers, communication, and customer support, as well as the ability to forecast client turnover and automate difficult jobs that were previously performed by people. AI systems employ machine vision to extract data. In order for an AI system to be autonomous, it must be able to perceive, make judgments, and have the power to carry those decisions out [85]. AI systems encompass several components and processes. They involve two types of inputs: structured and unstructured data. Additionally, they undergo two pre-processes: natural language understanding and computer vision, enabling the interpretation of human language and images. These systems rely on three core internal processes: problem-solving, reasoning, and machine learning, enhancing efficiency and accuracy in resolving tasks [86].

3 Research Methodology

Critical analysis was conducted on the chosen papers to discover important themes and conclusions on the use of AI in various KM-related domains. The review looks at how AI is being used for creating knowledge, discovery, and capture. Large amounts of data may be mined for useful insights using AI techniques like data mining, machine learning, and natural language processing. The reputable online databases Emerald, IEEE Xplore, and ScienceDirect served as the foundation for this systematic study. "Knowledge Management" AND "Artificial Intelligence" are the study's core search terms. There were 191 articles found. Due to duplication, 52 items were disqualified. As a consequence, 139 articles overall. These articles all have been evaluated and filtered according to the inclusion and exclusion criteria as shown in Table 1. As a result, 10 articles were selected for the analysis since they satisfied the requirements. The PRISMA standards for systematic reviews were followed by the researchers. The PRISMA flowchart Fig. 1 outlines the steps involved in selecting studies, from first search results until inclusion (Table 2).

Table 2 The inclusion and exclusion

Inclusion criteria	Exclusion criteria
Studies that specifically focus on the role of Artificial Intelligence in Knowledge Management	Studies unrelated to the role of Artificial Intelligence in Knowledge Management and
Research articles conducted within the past 5 years (2017—2023)	Studies published before 2017
Studies conducted in various industries or sectors	Studies limited to a specific industry or sector
Research articles available in the English language	Articles in languages other than English

4 Results

After conducting a review of 10 selected research articles that emphasize the role of AI in KM, the outcomes of this review are reported in accordance with specific research questions.

4.1 *RQ1: What is the Aim of the Study?*

The reviewed articles investigating the role of AI KM aim to encompass assessing the impact of AI on knowledge creation and acquisition, exploring AI-driven knowledge discovery and organization, and evaluating AI-enabled knowledge sharing and collaboration [87], examining the impact of AI on decision-making and problem-solving [87], and addressing challenges and ethical considerations associated with AI integration [8]. Moreover, enhance knowledge management processes, and improve organizational performance by leveraging AI technologies and techniques such as natural language processing, machine learning, data mining, automated categorization, clustering, semantic analysis, chatbots, expert systems, cognitive computing, and predictive analytics [75]. Understand how the impact of knowledge management strategies on organizational improvisation is influenced by the presence of technological turbulence caused by artificial intelligence [85]. Botega [88] presents an AI approach aimed at providing support for KM in the selection of creativity and innovation techniques [88] and Hu discussed the potential opportunities and challenges of utilizing ChatGPT for the acquisition of design knowledge, as well as to propose future research directions in this area [89]. In general, research works on the application of AI in KM strive to further our understanding of how AI technologies might improve KM processes, boost organizational performance, and address current issues in KM-intensive situations.

4.2 *RQ2: How Can the Integration of Artificial Intelligence (AI) and Knowledge Management (KM) Systems Enhance Organizational Learning and Decision-Making Processes?*

AI and KM interact in a way that is much linked with knowledge and learning. Recent AI developments provide fresh possibilities for transforming KM within enterprises [90]. In this area, there are two complementary perspectives. first KM, which emphasizes knowledge management inside businesses, second AI, a subfield of computing that seeks to mimic human knowledge and learning processes. Due to the essential relationship between the two fields, it is critical for businesses to investigate the possible functions of AI systems in assisting organizational KM efforts. Traditional rule-based knowledge management offers useful lessons, but modern deep learning represents a unique computational strategy that necessitates new perspectives on the connection between AI and KM [91].

Knowledge creation involves both the generation of new ideas and the adaptation of existing knowledge to meet new challenges. Organizations can acquire knowledge from external sources and engage in active knowledge acquisition through information search and sourcing [91]. Deep-learning AI holds promise in supporting knowledge creation by leveraging its predictive capabilities, such as sales forecasting. Eitle & Buxmann, [92] It has the ability to generate new information by uncovering patterns and insights from existing data, revealing previously unknown relationships. For instance, AI algorithms have been successfully applied to analyze research articles, identifying complex concepts and correlations in fields like materials science. These techniques offer significant benefits to organizational knowledge management by revealing hidden connections and insights within vast data sets [91].

Creating and keeping up an organizational memory that keeps track of created and acquired knowledge resources is another crucial role of knowledge management [93]. Enhancing the storage and retrieval of explicit information is where AI in KM is most useful. Given that big data and deep learning AI are interdependent, [94], Data-driven, self-learning algorithms revolutionize the management of large data volumes within organizations, even handling previously complex datasets. AI excels in searching, organizing, and summarizing information, such as relevant legal precedents for new cases. It analyzes various content sources, generates summaries, identifies emerging topics, and extracts proprietary knowledge for immediate application. (O'Dell & Davenport, 2019) Deep-learning AI adapts and offers personalized solutions, aiding knowledge workers in identifying crucial documents, retrieving pertinent information, and supporting specific team meetings and problem-solving. These capabilities greatly enhance productivity for knowledge workers [91].

For problem-solving and decision-making in organizations, effective knowledge sharing is essential. AI can play a critical role in overcoming these challenges by bringing together people working on related problems and establishing coordinated systems that give managers insights into knowledge gaps by removing barriers, linking those working on related problems, and developing coordinated systems that provide managers a better awareness of knowledge gaps, AI can assist in overcoming these difficulties. AI promotes community-based learning and tackles the problem of silos in information exchange by identifying and bolstering weak relationships. Beyond conventional database systems, AI contributes to collaborative intelligence by encouraging original thought, allowing peer review and feedback, and creating shared memories among team members [91].

After knowledge has been retrieved or shared, knowledge application refers to putting it into practice. It frequently entails repackaging readily accessible knowledge resources into appropriate solutions or offering new goods and services to a different setting [95]. The information retrieval and representation requirements for situational knowledge applications can be improved by new tools driven by AI, such as intelligent assistants [96].

4.3 *RQ3: What Are the Methods for Integrating AI Technologies into Knowledge Management Systems?*

According to Zhou and Metaxiotis et al. Artificial intelligence systems within KM can be categorized into three main functional classifications: knowledge-based expert systems (KEBS), neural networks (NN), and case-based reasoning systems (CBR). Knowledge-based expert systems utilize explicit knowledge and rules to provide expert-level advice and decision-making [74]. The integration of AI into Knowledge Management involves the implementation of a Data Mining model. This model enables the evaluation of data and facilitates knowledge management by initially nalysing and processing the received data. Through this integrated approach, organizations can effectively harness AI technology to enhance their knowledge management processes [75].

4.3.1 Knowledge-Based Expert Systems (KEBS)

Knowledge-based systems are intelligent computer programs that can perform like human experts and have expert-level knowledge in particular fields [1]. KEBS aims to create artificial intelligence systems that can successfully handle complex knowledge in modern companies and encourage knowledge exchange among team members [88]. the KBS should make sure that it not only combines the expertise of experts but that its decision-making process is presented clearly [97]. High-tech firms may consolidate and manage explicit knowledge, offer a forum for the exchange of tacit knowledge, allow real-time communication among group members, and promote the sharing of ideas and experiences by building an AI-based knowledge management expert system. The result of this process is the development of intellectual capital within the company, which is a crucial resource for knowledge, invention, and technical improvement. Figure 1 shows the architecture of a KBS [88].

4.3.2 Artificial Neural Networks (ANNs)

ANN systems process data in a manner similar to that found in natural nerve systems, such as the brain. The neurons that make up an ANN are linked processing units arranged in networks. For problem-solving and information processing, these neural networks use a learning-by-example strategy that is analogous to human learning. In order for ANNs to function properly, the network's structure is essential [74]. Neural network systems, or NN, provide the advantage of handling incomplete data by generalizing and abstracting it. These systems mimic the interconnected nature of brain neurons, processing information through neural nodes and complex functions. Through algorithmic filtering using measurement chains and invisible nodes, NN refines the solutions and generates optimal output results as shown in Fig. 2 [1] (Fig. 3).

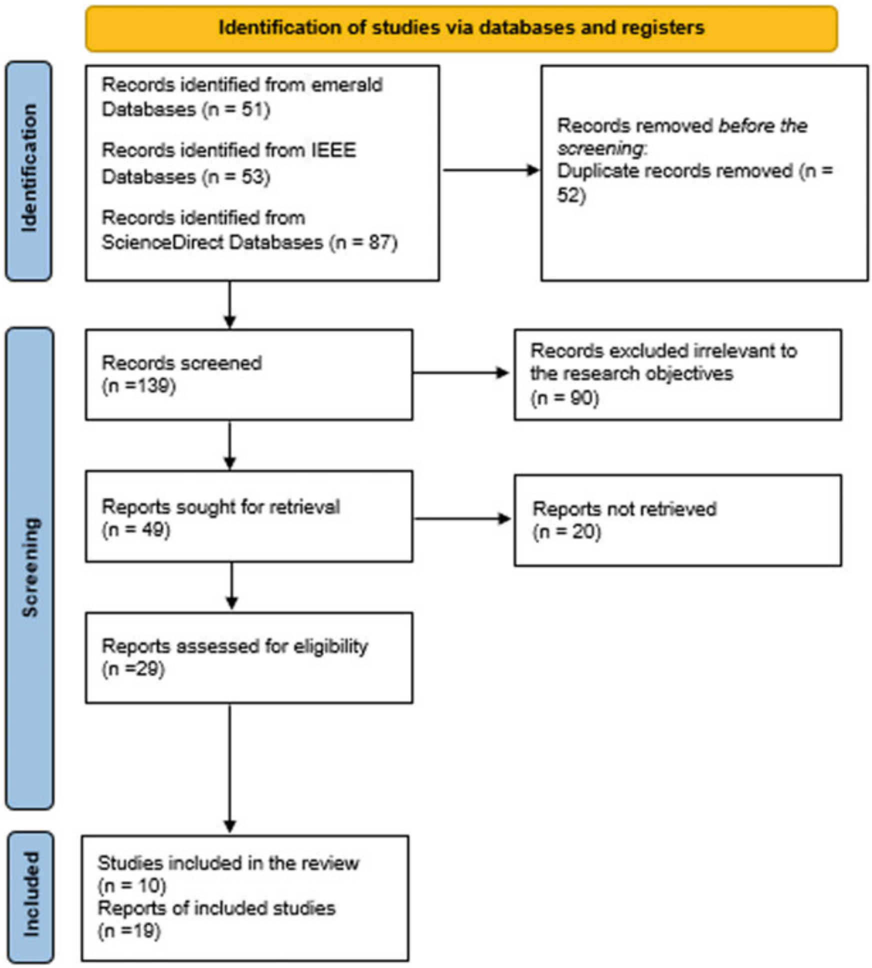

Fig. 1 PRISMA flowchart

Neural networks aim to produce final decision results that are at least comparable to, and often better than, those of human expert systems. By minimizing the error between ideal and actual answers, neural networks iterate over different data sets until a consistently stable or highly accurate answer is achieved. ANNs are capable of processing and analyzing enormous volumes of data, finding patterns, and making predictions. ANNs may be used in the context of knowledge management to improve information discovery, decision-making, and knowledge-sharing processes inside businesses [1].

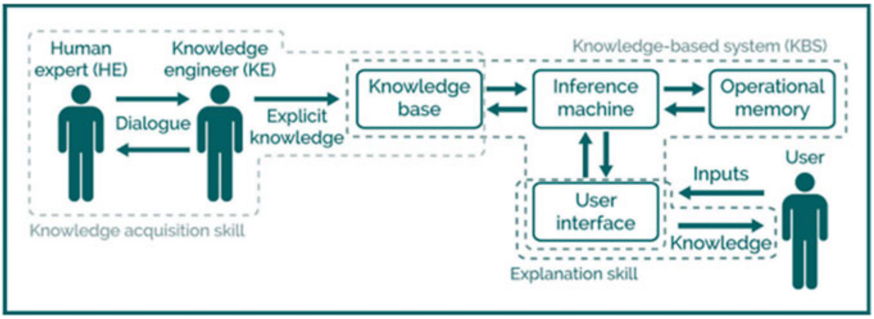

Fig. 2 Shows a diagram of the design and architecture of a KBS [88]

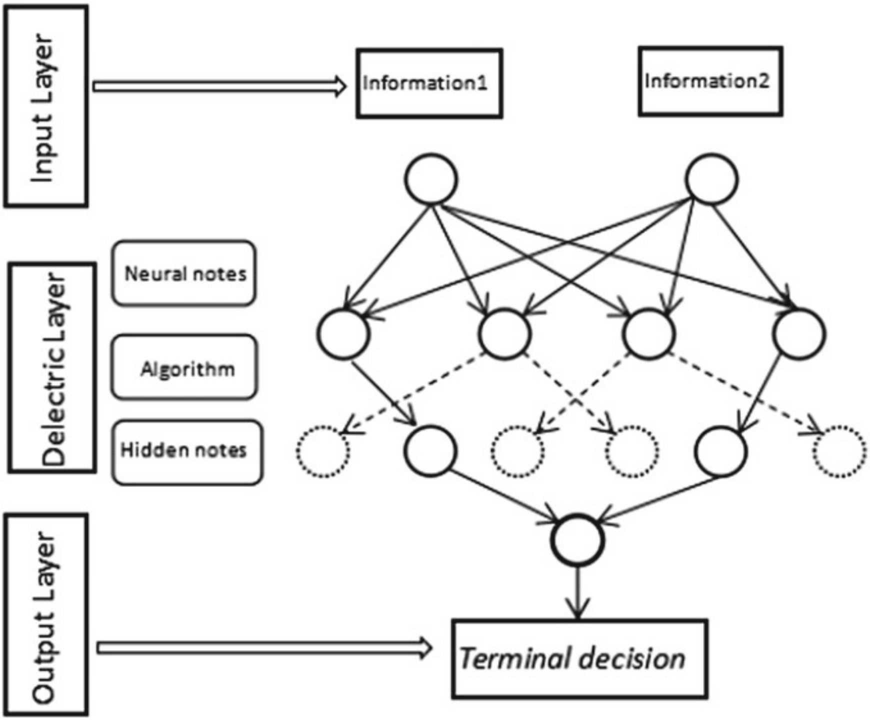

Fig. 3 Artificial neural networks [1]

4.3.3 Case-Based Reasoning Systems (CBR)

Case-based reasoning (CBR) is a recently created artificial intelligence reasoning method that resembles analogous reasoning. By drawing on prior knowledge and well-known examples, CBR systems simulate the approach to problem-solving of experts. The effectiveness of knowledge management is greatly increased by CBR by exploiting the information that is already there in the case base rather than having to start from scratch. To extract knowledge from events or instances, CBR blends storytelling talents with knowledge organization. The flexible character of the human mind is reflected by CBR, as opposed to expert systems that rely on strict and non-overlapping rules. As a foundation for creating intelligent systems, it offers a model of reasoning that is cognitively sound [1].

4.3.4 Chatbots with Natural Language Processing and KM

NLP-enabled chatbots will benefit all employees in different organizational roles and at significant points for decision-making by personalizing knowledge delivery. NLP-enabled chatbots will have the cognitive ability to comprehend, understand, and manipulate human language, allowing them to anticipate user wants, attitudes, and ambitions to help with decision-making [8].

ChatGPT is an advanced language model built on a transformer-based neural network architecture. It excels in understanding complex language structures and generating coherent and contextually relevant responses. Compared to previous models like GPT-3 and InstructGPT, ChatGPT's standout features include its exceptional conversational capability and seamless interaction with users. It is trained on a diverse range of text data from various sources, providing engineers with a wealth of information and insights. Additionally, ChatGPT incorporates Reinforcement Learning from Human Feedback (RLHF), which allows the model to learn and improve from user feedback, resulting in the generation of more desirable design solutions [89].

ChatGPT offers a unique opportunity for designers to acquire design knowledge in a centralized platform. It provides access to various types of knowledge, including common sense, domain-specific, engineering, and technique knowledge, supporting design decisions throughout the process. By reducing the need to switch between different knowledge providers, designers can streamline their workflow and improve efficiency. The integrated knowledge tool also promotes collaboration and communication among designers, enhancing team productivity and minimizing errors. Unlike rigid past design knowledge providers, ChatGPT offers flexibility, empowering designers without dictating the design process.

ChatGPT provides a dynamic and interactive approach to knowledge acquisition in design, allowing designers to continuously ask questions, seek clarification, and receive immediate answers. This iterative process is particularly advantageous for ill-defined problems that require multiple perspectives and criteria and helps designers refine their understanding of the design problem. It is especially beneficial for designers who struggle to articulate their challenges [89].

4.4 RQ4: What are the Main Sectors that are Making Use of AI Technology on a Regular Basis?

AI technology is increasingly being used in various sectors, such as healthcare, finance, retail, manufacturing, and transportation. As technology advances, further integration across diverse industries is expected, leading to increased adoption and benefits [91]. Financial services (including insurance companies) are among the many organizations that are deploying AI solutions on a regular basis. These organizations are combining different AI solutions with machine learning to deliver knowledge to users and improve customer service, investment tools, credit analysis and scoring, and financial analysis tools. However, this source does not provide an exhaustive list of all sectors that make use of AI technology [8].

5 Conclusion and Future Work

Knowledge management has benefited greatly from the use of AI technology. AI offers multiple advantages, including improving knowledge processes, utilizing data analytics to extract important insights, and offering knowledge workers customized recommendations. Organizations may improve decision-making, stimulate creativity, and maximize their overall knowledge management strategies by utilizing AI technologies like natural language processing, machine learning, and data mining. Integrating AI with (KM) necessitates a thoughtful approach that takes into account organizational objectives, data availability, user requirements, and ethical considerations. Organizations should assess their specific needs and choose suitable methods and techniques that align with their KM strategy. By doing so, they can effectively harness the capabilities of AI to manage their knowledge resources in a more efficient and impactful manner. Further research in KM making use of ChatGPT needs to concentrate on advancing knowledge acquisition, expanding contextual comprehension, and putting in place systems for knowledge verification and validation. These efforts will result in more sophisticated and personalized KM systems that promote effective knowledge sharing, collaboration, and decision-making.

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Systematic Review for Knowledge Management in Industry 4.0 and ChatGPT Applicability as a Tool

Ali Sherif, Said A. Salloum, and Khaled Shaalan

1 Introduction

According to [1] Knowledge management is the process of capturing, identifying, organizing, and sharing the knowledge that exists within an organization. This knowledge can be in the form of documents, procedures, best practices, or even the tacit knowledge of employees [2]. By sharing this knowledge, organizations can improve their decision-making, innovation, and overall performance [3].

Additionally, [4] on their article representing Knowledge in organization as the process of organizing knowledge in a way that makes it easy to find and use. This can be done by creating semantic links between concepts, which are represented by vertices in a semantic link network (SLN). By organizing knowledge in this way, organizations can improve the efficiency of knowledge management [5–8] and make it easier to find the information they need [9, 10].

Therefore, [11] sees that, industry 4.0 allows the manufacturing industry to adopt digital technology by incorporating sensing devices into nearly all aspects of manufacturing [12–14], including components, products, and equipment. By integrating digital data with physical objects through a pervasive system, the analysis of interconnected data has the potential to revolutionize all industrial sectors globally with greater impact and at a faster pace than the previous three industrial revolutions, namely Industry 1.0, 2.0, and 3.0. Therefore, Industry 4.0 is a current topic that pertains to the entirety of modern industrial production, with the aim of transforming it significantly. Germany introduced the concept of Industry 4.0 at the Hannover Fair

A. Sherif · K. Shaalan
Faculty of Engineering and IT, The British University in Dubai, Dubai, UAE

S. A. Salloum (✉)
Health Economic and Financing Group, University of Sharjah, Sharjah, UAE
e-mail: ssalloum@sharjah.ac.ae

School of Science, Engineering, and Environment, University of Salford, Salford, UAE

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event in 2011, which represented the start of a new era of industrial revolution. Upon its inception, European manufacturing researchers and companies made substantial efforts to adopt this concept.

On the other hand, [15] spotting the light to a new opportunity for knowledge management in numerous industries in both commercial and consumer domains can reap significant advantages by integrating technologies like ChatGPT into their practices. The article highlights benefits such as decreased expenditure on customer service personnel, improved efficiency in healthcare environments, and the facilitation of more interactive exchanges between businesses and consumers. Moreover, the transformational effect of AI-driven chatbot technology, including ChatGPT, on various business sectors has been groundbreaking, notably in the spheres of digital marketing and e-commerce operations [16, 17].

1.1 Objectives and Aims

The purpose of this research is to investigate the idea of knowledge management and its applicability to Industry 4.0 and how knowledge management may be advantageous for Industry 4.0 operation environment [18–22]. On the other hand, this paper will also assess ChatGPT's contribution to knowledge management facilitation and its applications in Industry 4.0. Additionally, this paper is aiming to provide overview of knowledge management (KM) and its historical development. Also intends to examine the connection between KM and I4.0 and how they may be used together to further corporate objectives. This paper will also examine ChatGPT's uses in knowledge management and possible effects on I4.0. The final goal of this paper is to offer advice on how firms may implement successful knowledge management methods in the I4.0 era.

1.2 How Important to Implement Knowledge Management in Industry 4.0?

Implementing Adaptive and Knowledge-Based Assistance Systems is crucial for managing manufacturing enterprises' collective intelligence [23–26]. The systems would allow for evaluation of productivity and learning effectiveness under various real-world manufacturing constraints and smart factory goals [27–31].

Additionally, [32], admitting that Knowledge management and digital transformation are becoming more prominent within the framework of Industry 4.0. Digital transformation processes are fostering innovation and prototyping, while knowledge management focuses on the generation, distribution, utilization, and administration of knowledge within a corporation. The amalgamation of these two notions can aid organizations in enhancing their efficiency, capacity for innovation, and long-term viability. In order to sustain and enhance their market position, organizations must recognize and utilize their intellectual assets, including structural capital and human talents [33–36]. Structural capital encompasses the knowledge and information ingrained in an organization's processes, systems, and technologies. On the other hand, human talents denote the knowledge and skills possessed by the employees of an organization. By identifying and harnessing these intellectual resources, companies can foster the creation of novel products and services, enhance their operational efficiency, and provide superior customer experiences [37, 38].

Narratively [1], showing how Knowledge is becoming increasingly important for companies to gain a competitive advantage. In the past, companies could compete on the basis of factors such as price, quality, or location. However, in today's knowledge economy, companies need to compete on the basis of their knowledge and expertise. Back to [37] in their article, it's providing more explanation how could that happen by using ICT to make production more efficient, flexible, and responsive to customer needs. It is based on the use of sensors, software, and automation to connect machines, people, and data in real time. This allows manufacturers to track production, identify problems, and make adjustments as needed. As a result, Industry 4.0 can help manufacturers to improve quality, reduce costs, and increase productivity. For example, an information system can be used to collect data from sensors on a production line. This data can then be used to identify bottlenecks in the production process, which can be addressed to improve efficiency. The information system can also be used to track the performance of individual machines, which can help to identify machines that are not performing as well as they could be. This information can then be used to take corrective action to improve the performance of the machines [39, 40].

Also [39], focusing on that manufacturing is expected to bring various opportunities, affecting production, supply chain, and sustainability to improve efficiency and profitability. By utilizing robotics and technology, organizations can employ data created through processes as raw material for analytics. And according to [41] project, analyzing these Big Data In Industry 4.0 settings, can yield enhanced transparency of processes, increased precision and comprehensibility of outcomes, self-regulating and self-sustaining equipment, fewer breakdowns and postponements, and heightened operational efficiency. These advantages can help decrease extraneous expenses and boost an organization's profitability. And that's what [42] agreeing with by providing more clarification, Integrated data and knowledge management can improve manufacturing processes in several ways. By efficiently managing data and knowledge, manufacturers can gain insights into their operations, identify areas for improvement, and make informed decisions [43, 44]. This can lead to increased efficiency, reduced costs, and improved product quality. Additionally, integrated data and knowledge management can enable better collaboration between different departments within a manufacturing organization, leading to more effective problem-solving and decision-making. But, [42] admitting that managing data and knowledge in Industry 4.0 presents challenges due to the numerous and varied data sources. The introduction of new information and communication technologies (ICT) in manufacturing environments generates massive amounts of data and information, resulting in expansive databases and complications related to big data.

1.3 What Are the Challenges in Data, Information, and Knowledge in Industry 4.0?

While [45] characterizing Industry 4.0 by the use of cyber-physical systems, the internet of things, and big data analytics. This revolution has the potential to revolutionize manufacturing and other industries, but it also faces a number of challenges, including cybersecurity risks, data privacy concerns, lack of interoperability standards, high implementation costs, and workforce skills gap. Additionally, [46] says one of the biggest challenges is the collection and analysis of data. In order to reap the benefits of Industry 4.0, manufacturers need to be able to collect data from their machines and systems. Also, Industry 4.0 requires workers with new skills and knowledge. For example, workers need to be able to operate and maintain advanced machinery, and they need to be able to understand and analyze data. However, there is a shortage of skilled workers in the manufacturing industry.

For instance, when companies try to implement big data solutions in the context of Industry 4.0 they are facing a lack data quality. The accuracy, completeness, and consistency of data can be a challenge when dealing with large volumes of data from multiple sources. Another challenge is data privacy and security, as there is a risk of data breaches and unauthorized access to sensitive information. Companies need to ensure that they have robust security measures in place to protect their data. Additionally, there is a shortage of skilled personnel who can manage and analyze big data, which can be a significant challenge for companies. Integrating big data solutions with existing systems can also be challenging due to differences in data formats and structures. Finally, as the volume of data grows, companies need to ensure that their infrastructure can handle the increased load without compromising performance or reliability [47] in a different aspect, according to PWC study shows that (30%) of challenges is on a form of insufficient qualifications of employees, (20%) low maturity level for the needed technologies, (19%) for a questions need an answer and explanation about concerns related to date security, (18%) poor performance from top management for prioritization and support, too slow expansion for basic technologies updating (13%), and insufficient network stability/data backup (6%) [48]. Accordingly, as an example, Industry 4.0 in India, will require workers to have the latest qualifications and necessary skills. This means that individuals who possess the required skills will be in high demand, while those who lack the necessary training may face unemployment [49].

Finally, in order to adapt to Industry 4.0, organizations must reconsider their conventional perspective on how production systems operate. The Fourth Industrial

Revolution centers on digitalization and the substitution of outdated analog technologies. It is essential to strive for the incorporation of Cyber-Physical Systems throughout the entirety of the production system, leading to enhanced dynamism. Automation will play a pivotal role in the production process, with advanced robotic machines assuming primary responsibilities. As a result, the dynamics of interaction between various units within industrial enterprises will undergo a profound transformation, as human involvement gradually diminishes in favor of machinery [50].

1.4 Why Knowledge Management Turning Industry 4.0's Challenges into Opportunities?

Organizations that adopt industry 4.0 technologies, such as internet of things or visualization technology [51–55], they have a greater probability of consistently strengthening their capacity for learning and exchanging knowledge throughout the entire organization. This can lead to improved organizational performance and competitiveness [56].

Organizations can use data to gain insights into customer behavior and preferences [34, 57–65]. This information can be used to, improve customer experience, develop targeted marketing campaigns, and enhance product development [66, 67]. For example, organizations can analyze data from social media, customer feedback, and other sources to learn about customer demographics, interests, and online behavior. This information can then be used to target marketing campaigns more effectively, identify areas where customer service can be improved, and develop new products and services that meet customer needs. By using data effectively, organizations can gain a competitive advantage and improve their bottom line [68–73].

Knowledge management involves the creation of strategies that enable the smooth sharing of knowledge among members of a group within an organization [74–78]. This process aims to enhance the transfer and exchange of knowledge, leading to the development of new skills and the improvement of existing ones. Ultimately, these efforts have a substantial impact on the performance of organizations in both the public and private sectors [79]. Some of the key opportunities of Industry 4.0 include gaining a competitive edge over competitors, achieving smart and efficient production, and individualized and customized production at a reasonable cost. By adopting Industry 4.0, businesses can become smarter and more efficient, which can lead to increased productivity and profitability. Additionally, Industry 4.0 enables businesses to produce customized products at a reasonable cost, which can help them meet the unique needs of their customers. Overall, Industry 4.0 provides businesses with opportunities to improve their operations and gain a competitive advantage in the marketplace [8, 80–82].

Industry 4.0 presents opportunities behind the challenge. Industry 4.0 facilitates the merging of production frameworks and IT infrastructure, as well as the integration of customers and suppliers. Additionally, it holds the promise of enhancing resource efficiency and supporting the adoption of circular economy models. The influence of emerging technologies will have far-reaching effects, permeating all fields of study and redefining the distinctions between economic and industrial sectors, transforming buyer–seller relationships, and reshaping the roles of the public and private sectors. Existing production systems and global value chains will evolve into more agile, adaptable, efficient, and sustainable structures, offering significant potential for customization and personalization [83] Knowledge management enables organizations to remain competitive by recognizing and harmonizing shared knowledge [29, 84, 85]. It has practical uses across various sectors, including business and industry, where it aids in refining decision-making processes, fostering innovation, boosting efficiency, and encouraging employee collaboration [86].

Some information system tools such as Manufacturing Execution System (MES), Business Intelligence (BI), will incorporate Industry 4.0 into Lean production environments that hinge on its capacity to engage with shop floor operations and relay information to senior management levels but it must be capable for digital upgrades. Additionally, these tools could benefit from instantaneous data storage and the display of Key Performance Indicators (KPIs), providing a more comprehensive understanding of the shop floor's status [39].

1.5 How ChatGPT Can Be Productive Tool in Industry 4.0?

ChatGPT is an advanced technology that employs deep learning, reinforcement learning, and transfer learning to develop chatbots capable of emulating human behavior. It utilizes a recurrent neural network (RNN) structure to assimilate knowledge from past discussions and grasp contextual details regarding specific conversation subjects. To put it simply, ChatGPT is an artificial intelligence chatbot that comprehends and communicates with human language in a manner that feels natural [15].

ChatGPT can be beneficial in various industries by offering specialist guidance and advice to professionals. Its capacity to respond competently to a broad spectrum of human inquiries can result in time and resource savings for businesses and organizations. Furthermore, ChatGPT can be employed to automate customer service exchanges, thereby enabling human staff to concentrate on more intricate tasks [87].

Utilizing ChatGPT for optimizing Gcode in additive manufacturing has the potential to yield noteworthy reductions in time and material usage, while enhancing precision and consistency throughout the printing procedure. ChatGPT has the capability to actively observe the printing process and make real-time modifications to the Gcode, thereby aiding in the reduction of printing time for a given product. Additionally, ChatGPT can optimize the Gcode to use less material, which can help to reduce the cost of printing a product. ChatGPT can also help to improve the accuracy and repeatability of the printing process, which can lead to better quality products.

Additionally, ChatGPT can continuously improve its performance by learning from its previous experiences, which can lead to even better results over time [88].

The utilization of ChatGPT across different sectors could present both advantages and disadvantages. Benefits might encompass enhanced client support, tailored suggestions drawn from user tastes and desires, as well as augmented effectiveness in tasks like data analysis. Nevertheless, the facet of personalization isn't without its potential perils and adverse effects. Some critics argue that personalization can be leveraged to deceive and exploit individuals, thereby stressing the imperative of ethical and responsible usage. The piece implies that it is vital for scholars, policy architects, and the public to thoughtfully assess the possible risks and rewards of AI, ensuring that the tech is rightly evolved and implemented [89].

On the other hand, ChatGPT can contribute positively in industries by supplying insightful and useful responses to targeted inquiries concerning Industry 4.0. This term describes the fusion of cutting-edge technologies such as AI, robotics, and the Internet of Things into manufacturing and production operations. Nevertheless, a limitation of ChatGPT is that it may need precise and detailed data instead of being capable of comprehending implied indications. Persistent research is essential to probe the learning and decision-making technologies applicable to smart industries, necessitating collaborative endeavors between academic institutions and industry [90].

Finally, industries can experience a substantial impact by introducing effective and efficient conversational AI solutions like ChatGPT, which can enhance customer gratification, medical results, and student achievements. For instance, in the realm of customer service, ChatGPT can address client inquiries, suggest products, and manage transactions, leading to quicker response times, heightened customer satisfaction, and decreased business expenses. Within healthcare, ChatGPT can aid in patient surveillance and diagnosis, resulting in more precise diagnoses and superior treatment outcomes. In the educational sector, it can facilitate personalized learning and tutoring, enabling students to progress at their individual speed and obtain personalized feedback. Overall, ChatGPT holds the potential to transform how industries function by supplying intelligent conversational agents that can automate tasks and ameliorate decision-making procedures [5, 29, 91–94].

2 Recommendations

The successful navigation of Industry 4.0 relies heavily on effective Knowledge Management (KM) systems. Organizations are urged to prioritize the adoption of comprehensive KM strategies to manage the immense volumes of data and information generated in Industry 4.0 settings. This not only includes the collection and storage of data but also its analysis and application, transforming raw data into actionable insights that drive innovation, streamline operations, and improve overall competitiveness. At the same time, significant challenges related to data privacy, information overload, and rapid knowledge obsolescence should be carefully addressed. The privacy and security of data is a fundamental concern that needs to be tackled with stringent controls and robust data protection measures. This is essential not only from a legal standpoint, but also in building trust with stakeholders and preserving the integrity of the data. Information overload is another significant challenge due to the high volumes of data generated in the Industry 4.0 environment. It is crucial to implement strategies that can effectively filter, categorize, and structure this information for efficient use. Methods such as data minimization and advanced data analytics can be employed to prevent information overload and enhance decision-making capabilities.

Furthermore, in the rapidly advancing landscape of Industry 4.0, knowledge can become outdated quite quickly. Organizations must therefore develop continuous learning systems and foster a culture of lifelong learning among employees. This can ensure that the knowledge within the organization remains up-to-date and relevant, allowing them to stay competitive in the dynamic industrial landscape. The integration of AI tools like ChatGPT can be instrumental in augmenting KM processes, automating knowledge capture, and providing advanced analytics, thereby enhancing knowledge retrieval. These technical enhancements necessitate a corresponding focus on workforce training to maximize these tools' efficacy. Policy development should also parallel these advancements, with a careful revision of policies pertaining to data privacy, security, and AI ethics. Future research, exploring the potential of AI tools in KM and conducting longitudinal studies to measure long-term impacts, is encouraged. As the landscape of Industry 4.0 evolves, turning its challenges into opportunities through KM presents a promising avenue for organizational growth and competitiveness.

3 Conclusion

Our systematic review of Knowledge Management (KM) in the context of Industry 4.0 highlighted its vital role in harnessing and effectively utilizing the data, information, and knowledge that is generated in this digital industrial revolution. The findings strongly indicate that successful implementation of KM can turn challenges in Industry 4.0 into promising opportunities. KM serves as a catalyst in navigating vast data streams, facilitating decision-making, stimulating innovation, and improving overall competitiveness. Additionally, our research revealed several key challenges associated with the management of data, information, and knowledge in Industry 4.0, including data privacy, information overload, and the rapid obsolescence of knowledge. These challenges, if left unaddressed, can pose considerable hurdles in the digital transformation journey. Exploring the application of AI-based tools, with a specific focus on ChatGPT, demonstrated significant potential in supporting KM in Industry 4.0. The ability of ChatGPT to comprehend, generate, and interact in natural language can significantly enhance KM processes. However, issues related to AI transparency and dependency on vast datasets necessitate careful data governance.

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Envisioning ChatGPT's Integration as Educational Platforms: A Hybrid SEM-ML Method for Adoption Prediction

Said A. Salloum, Mohannad Hatem, Ayham Salloum, and Raghad Alfaisal

1 Introduction

Artificial intelligence stands at the forefront of shaping modern living [1–5]. A prime illustration of this innovation is the Chat Generative Pre-trained Transformer, commonly known as ChatGPT. Notably, ChatGPT's prowess lies in its aptitude to craft term papers, short narratives, elucidations, and literary pieces. The extensive, detailed outputs it generates have stirred concerns, notably at American University, where there's trepidation over its ability to produce college-standard essays and aptly respond to examination queries [6, 7]. As natural language processing (NLP) models permeate the educational sector, they usher in transformative changes, making educational content more accessible to learners, instructors, and academic professionals. While platforms like Google have traditionally dominated, ChatGPT's surge in popularity is undeniable. Its vast reservoir of resources, spanning books to websites, combined with its adeptness at grasping intricate student requests and offering adept responses, marks its potential as a pivotal educational tool [6, 8].

S. A. Salloum (✉)
Health Economic and Financing Group, University of Sharjah, Sharjah, UAE
e-mail: ssalloum@sharjah.ac.ae

School of Science, Engineering, and Environment, University of Salford, Salford, England

M. Hatem
Computer Centre, University of Sharjah, Sharjah, UAE

A. Salloum
College of Medicine, University of Sharjah, Sharjah, UAE

R. Alfaisal
Faculty of Computing and Meta-Technology, Universiti Pendidikan Sultan Idris, Tanjong Malim, Malaysia

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Though ChatGPT's impact spans sectors including medicine, engineering, and education, its influence on diverse audiences from students to healthcare professionals is noteworthy. However, even with the abundant discourse on tech progression in education and AI tools like ChatGPT, there's a discernible void in understanding the nuanced determinants of user reception towards such innovations. Much of the existing literature hovers over broad technological acceptance frameworks, often sidelining the finer details and specificities linked to AI-centric educational systems.

To bridge this gap, our study endeavors to:

1. Unveil a well-rounded conceptual structure, merging key adoption determinants particular to AI in education, thus offering an enriched perspective on user acceptance.
2. Utilize sophisticated techniques, specifically PLS-SEM and ML, to dissect the nuances within our research paradigm, enhancing the analytical depth.
3. Shed light on the hierarchy of determinants for adoption, especially spotlighting the surprisingly diminished role of task-technology alignment in ChatGPT adoption—a deviation from typical tech acceptance models.
4. Further the discourse connecting AI and environmental sustainability, intertwining two pivotal realms and accentuating AI's diverse ramifications.

At its core, our research not only builds upon the academic understanding of AI tool acceptance in the educational sphere but also furnishes actionable insights for those spearheading the creation, fine-tuning, and roll-out of such platforms, ensuring they resonate with user preferences and environmental imperatives.

2 The Theoretical Framework

The present framework highlights the impact of factors such as information quality, task-technology fit, perceived learning value, and personal innovativeness on the behavioral intention to utilize learning platforms.

2.1 Information Quality (IQ)

Information technology (IQ) is closely related to the degree that the information is perceived as accurate, relevant and reliable. It evaluates the type of information that is provided by the technology which can be classified as more significant or less significant. Whenever the information is classified as significant and updated, the technology users' perceive the information as precise, complete and comprehensive. The quality of information can have a significant impact on the acceptance and use of technology. When users perceive the information provided by a technology to be of high quality, they are more likely to accept and adopt the technology [9–11]. There are several factors that can influence the quality of information provided by technology. These include the accuracy and completeness of the data, the reliability of the sources, and the timeliness of the information. In addition, the usability and user-friendliness of the technology can also affect the quality of the information provided. For example, if technology is difficult to use or navigate, users may struggle to find the information they need or may be more likely to make errors when entering data [12, 13].

2.2 Task Technology Fit (TTF)

Task technology is a measure of how well a technology meets the users' needs and enhances their work process and it is based on a relation between two attributes which are the capabilities of the technology and the required tasks. The implementation of the technology is based on the users' activity which implies that experienced users are attracted to tools and methods that enhance their future work and complete the task with the greatest benefit. Therefore, the technology that does not fit into the users' needs and expectations will not be sufficient, hence it will not be accepted by the users. Accordingly, task technology fit has a degree of compatibility that coincide with the task. The higher the level of compatibility is, the higher the benefits that are performed using the technology. Between a particular technology and the tasks that need to be performed using that technology. It is a measure of how well a technology meets the needs of users and supports their work processes [14, 15]. Though many studies have dealt with the task technology fit but few of them have made a relation between it and other crucial factors that have been explained earlier.

Task technology fit and information quality is two key factors that can impact the effectiveness and efficiency of the technology used in completing tasks. A good fit between technology and task requirements, coupled with high-quality information provided by the technology, can enhance productivity, streamline workflows, and improve outcomes. The relationship between task technology fit and information quality are mutually reinforcing. When technology is well-suited to the task at hand, it is more likely to provide high-quality information that meets the needs of users. Conversely, when high-quality information is provided, it can enhance the fit between technology and task by making it easier for users to complete their work [16, 17].

To ensure good task technology fit and high-quality information, it is important to carefully consider the requirements and characteristics of the tasks that need to be performed, as well as the capabilities and limitations of the technology being used. Accordingly, the proposed hypotheses are:

H1: Information quality of ChatGPT has a positive effect on behavior intention to use learning platforms.

H2: Task technology fit of ChatGPT of ChatGPT has a positive effect on behavior intention to use learning platforms.

2.3 Perceived Learning Value (PLV)

Perceived learning value (PLV) embodies the advantages students identify when they utilize technology, such as seamless information access, time conservation, and decreased strain. When students recognize considerable benefits in a technological tool, their inclination towards its sustained adoption grows. For academic institutions, this recognized worth is vital for carving a distinctive edge via technology. Presenting students with technological solutions that have evident merits over other choices enables institutions to magnetize and secure their student base, thereby amplifying their stature and achievements. In a nutshell, the value that students discern in technology is a linchpin for educational bodies striving to deploy an efficacious tech strategy. Concentrating on the tangible advantages that students associate with such technology allows educational entities to foster a robust competitive stance, ensuring consistent student engagement and loyalty [18, 19].

2.4 Personal Innovativeness (PI)

Personal innovativeness is considered a mediating factor that correlates the relation between the perceived usefulness and the perceived learning value and the acceptance of learning platforms. Thus, personal innovativeness is related to the users' willingness to adopt and use new technologies due to their innovative features that are not available in other technologies. It is a key factor in technology acceptance which measure the degree to which users are likely to seek out and experiment the new technologies [20].

Personal innovativeness can be closely related, as individuals with high levels of personal innovativeness may experience greater satisfaction when adopting and using new technologies. The innovativeness should meet the users' needs and expectations to fulfil the factor of satisfaction. Thus, personal innovativeness can be closely linked, with higher levels of personal innovativeness often associated with greater satisfaction in using new technologies. By designing technologies that meet the users' needs and expectations with high personal innovativeness, the acceptance of the technology will be higher and it may foster a sense of personal satisfaction and fulfilment among users [21, 22]. Similarly, perceived learning value is closely related to perceived innovativeness. The users who perceive the technology as having a high learning value will pay attention to all the available features including personal innovativeness [22, 23]. Based on the previous assumptions, the following hypotheses are proposed (Fig. 1):

H3: Perceived learning value of ChatGPT has a positive effect on behavior intention to use learning platforms.

H4: Personal innovativeness of ChatGPT has a positive effect on behavior intention to use learning platforms.

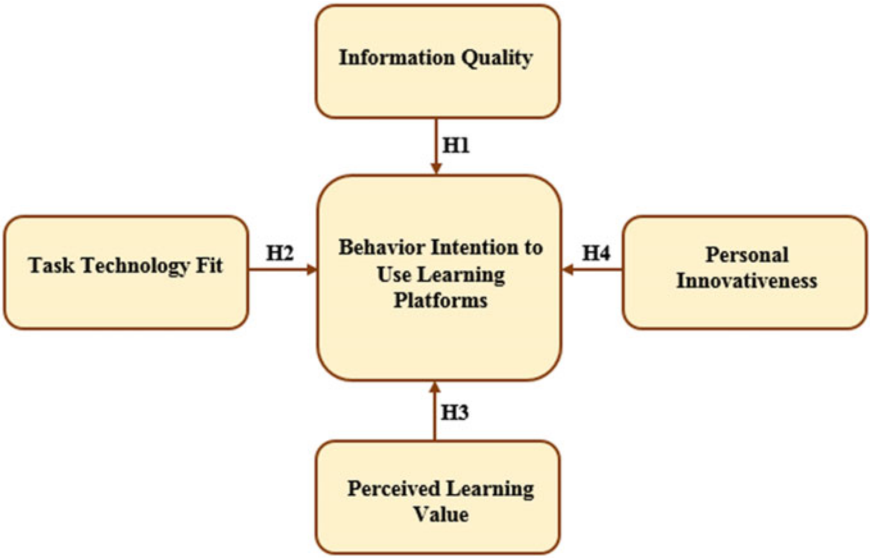

Fig. 1 Research model

3 Research Methodology

3.1 Data Collection

Online questionnaires were distributed to students across partner universities in the UAE. These were collected between March 01, 2023, and June 30, 2023. Out of the 1000 questionnaires issued by the study committee, we received a 98.2% response rate, amounting to 982 complete responses. Eighteen questionnaires were disregarded due to partial answers. The 982 completed questionnaires exceeded the suggested sample size of 384 for a population of 1,000,000, as indicated by Krejcie & Morgan [25]. This larger sample size markedly surpasses the baseline requirements. As a result, we utilized this sample to support our hypotheses through structural equation modeling [26]. It's noteworthy that our propositions draw from prior hypotheses rooted in AI's history. To evaluate the measurement model, our research team used Structural Equation Modeling (SEM) in SmartPLS Version 3.2.7. Advanced interventions were executed using the final path model.

3.2 Demographic Information

Figure 2 displays a breakdown of the participants' demographic and personal details. The data shows that females constituted 65% of the participants, with males making

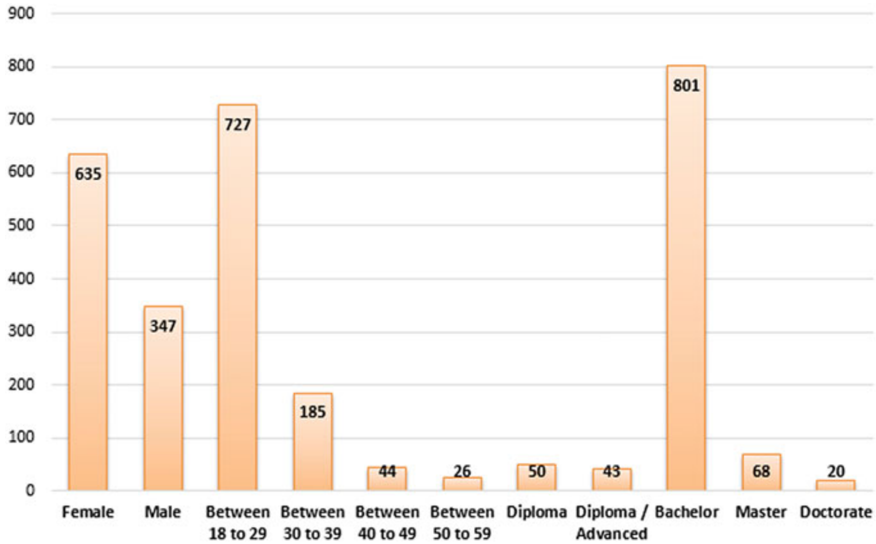

Fig. 2 Demographic information of the respondents ($n = 982$)

up the remaining 35%. When looking at age, 74% of the participants were between 18 and 29 years old, while the rest were 29 years or older. The educational qualifications of the respondents varied. Specifically, 5% had a diploma, 4% an advanced diploma, 82% a bachelor's degree, 7% a master's degree, and 2% a doctoral degree. When students indicated an interest in volunteering, the "purposive sampling approach" recommended by Salloum and Shaalan [24] was considered. This research encompassed a diverse group of participants from various universities, age brackets, and academic achievements. The demographic data was processed using IBM SPSS Statistics version 23.

3.3 Study Instrument

In this research, a questionnaire was utilized to test the hypothesis. A total of five constructs were meticulously chosen as credible indicators, resulting in the addition of 14 items to the questionnaire. The basis for these constructs can be found in Table 1. This table is intended to bolster the utility of the study's constructs and to furnish corroborative evidence from diverse pre-existing research that fortifies the present structure. In conclusion, the research team refined the survey questions, drawing insights from earlier studies.

Table 1 Measurement Items

Constructs	Items	Instrument	Sources
Behavior intention to use learning platforms	BI1	ChatGPT provides a promising avenue to explore	[25, 26]
	BI2	ChatGPT presents a valuable chance to experiment	[25, 26]
Information quality	IQ1	ChatGPT indeed offers more detailed information	[27, 28]
	IQ2	ChatGPT indeed delivers more in-depth information	
	IQ3	ChatGPT supplies me with useful information for my daily activities	
Task technology fit	TTF1	The information sourced from ChatGPT is more current	[27]
	TTF2	The information from ChatGPT is more relevant and thorough	
	TTF3	The information I get from ChatGPT exceeds what I require for my learning activities	
The perceived learning value	PLV1	ChatGPT provides greater advantages in my learning journey	[18, 19]
	PLV2	ChatGPT holds a unique value across various learning tasks	
	PLV3	ChatGPT possesses a distinct value that motivates me to embrace the technology	
Personal innovativeness	PI1	ChatGPT's modern technology meets my requirements	[20]
	PI2	ChatGPT boasts advanced features that enhance the technology's worth	
	PI3	ChatGPT offers a distinct and elevated experience that drives me to adopt it	

4 Findings and Discussion

4.1 Data Analysis

To assess the research models, this study utilized Weka (ver. 3.8.3), which incorporates various classifiers such as OneR, BayesNet, J48, and Logistics [3, 29, 30]. While past research has leveraged machine learning algorithms like neural networks and Bayesian networks to decipher relationships within their models [31], this study takes a unique approach by employing the hybrid SEM-ANN method for hypothesis validation. This hybrid SEM-ANN model is a two-stage approach [32–34]. The first phase involves the use of PLS-SEM [35] to appraise the research model. Given the nascent nature of the conceptual framework and the lack of directly relevant studies, this methodology proves suitable [5, 32, 36–38]. Additionally, this research utilizes PLS-SEM to transpose broad IS concepts. The analysis of the research models is bifurcated into two processes: evaluating the structural model and the measurement model [39]. To further gauge the efficacy and significance of every element within the research model, this study employs the Importance Performance Map Analysis (IPMA) facilitated by SmartPLS. Recognized as an advanced PLS-SEM method, IPMA provides an in-depth insight into the model's factors.

4.1.1 Convergent Validity

Regarding the evaluation of the measurement model, [40] it proposed the concept of construct validity (all of which comprises discriminant and convergent validity) and construct reliability (all of which comprise Cronbach's alpha (CA) and composite reliability (CR)). As seen in Table 2, Cronbach's alpha (CA), which determines construct reliability, is 0.786 up to 0.892. These findings are below the cutoff value (0.7) [41]. Table 2's findings reveal that the composite reliability (CR) scores range from 0.864 to 0.931, exceeding the cutoff point [51]. Testing the mean–variance extracted (AVE) and factor loading is vital for deciding how well convergent validity is being observed [40]. Except for those already indicated, Table 2 shows that every factor loading number surpassed the criterion value of 0.7. Additionally, Table 3 demonstrates the AVE numbers, which are bigger than the '0.5' limit, disregarding the earlier numbers, 0.760 to 0.819. It will probably attain convergent validity as an outcome of the previous reasoning.

Table 2 Convergent validity results

Constructs	Items	Factor loading	Cronbach's Alpha	CR	AVE
Behavior Intention to Use Learning Platforms	BI1	0.887	0.786	0.864	0.760
	BI2	0.856			
Information Quality	IQ1	0.911	0.892	0.931	0.819
	IQ2	0.881			
	IQ3	0.923			
Personal Innovativeness	PI1	0.917	0.857	0.913	0.778
	PI2	0.868			
	PI3	0.910			
Perceived Learning Value	PLV1	0.894	0.881	0.921	0.808
	PLV2	0.859			
	PLV3	0.892			
Task Technology Fit	TTF1	0.924	0.880	0.926	0.807
	TTF2	0.867			
	TTF3	0.903			

Table 3 Fornell-Larcker scale

	BI	IQ	PLV	PI	TTF
BI	**0.872**				
IQ	0.647	**0.905**			
PLV	0.626	0.663	**0.882**		
PI	0.631	0.709	0.701	**0.899**	
TTF	0.450	0.703	0.537	0.626	**0.898**

Table 4 Heterotrait-Monotrait Ratio (HTMT)

	BI	IQ	PLV	PI	TTF
BI					
IQ	0.801				
PLV	0.815	0.739			
PI	0.810	0.787	0.805		
TTF	0.573	0.805	0.615	0.706	

4.1.2 Discriminant Validity

The decision to reevaluating the two requirements depending on the Heterotrait-Monotrait ratio (HTMT) and the Fornell-Larker criterion was taken because this current study attempted to estimate the discriminant validity [40]. Since every AVE and its square roots outperform their connection to additional constructs, the results presented in Table 3 clearly show that the Fornell-Larker criterion confirms the criteria [42].

The results of the HTMT ratio are displayed in Table 4, which shows that every construct's number is below the '0.85' limit [43]. The HTMT ratio appears as a result of this. These results allow for the computation of the discriminant validity. The study's findings showed that there were no problems whatsoever with evaluating the measurement model's validity and reliability. As a result, the data gathered can also be utilized to assess the structural model.

4.2 Hypotheses Testing Using PLS-SEM

The structural equation model for this study was constructed using Smart PLS [44–48], a tool that employs maximum likelihood estimation to delve into the relationships between the theoretical constructs present in the structural model [9, 49]. Based on this method, the suggested hypotheses were evaluated, and their outcomes can be viewed in Table 4 and Fig. 3, which show a reasonable predictive capability of the model [50]. Notably, the variable "Behavior Intention to Use Learning Platforms" explains roughly 51.4% of the variance. Table 4 lays out the beta (β) values, t-values, and p-values for every formulated hypothesis, based on the insights garnered from the PLS-SEM approach. From the empirical data review, this study's findings validate hypotheses H1, H3, and H4. However, hypothesis H2 was unsupported, leading to its rejection. Hypothesis H1 investigated the relationship between IQ and BI, resulting in a coefficient of $\beta = 0.371$ and $p < 0.001$. This confirms a significant positive association of IQ with BI, supporting H1. The data suggests that TTF doesn't have a substantial influence on BI, with a coefficient of $\beta = 0.110$ and $P = 0.068$. Consequently, H2 doesn't receive backing, implying that TTF's effect on BI is negligible. The study's outcomes unearthed significant correlations between BI and other factors. Specifically, both PLV and PI showed a strong positive influence on BI, with coefficients $\beta = 0.261$, $p < 0.001$, and $\beta = 0.254$, $p < 0.001$ respectively, corroborating hypotheses H3 and H4 (Table 5).

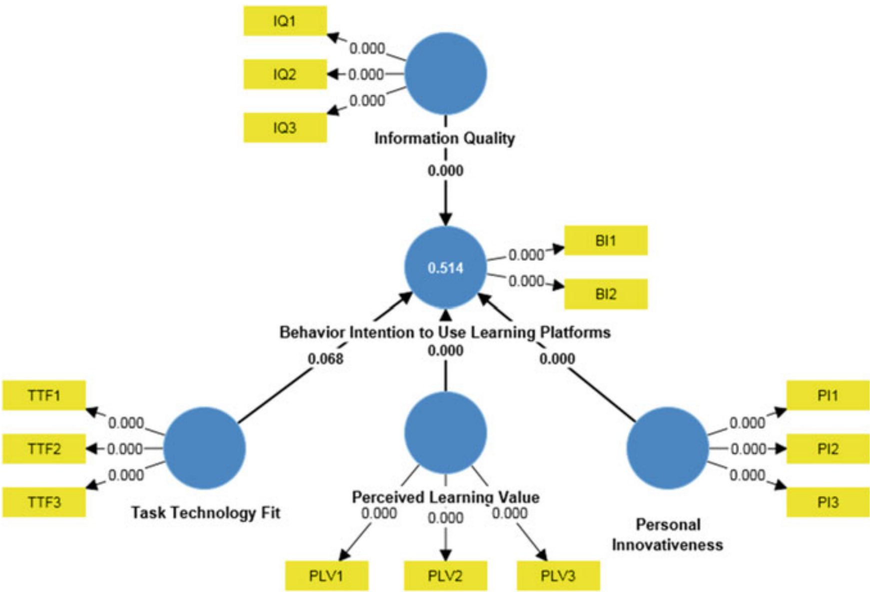

Fig. 3 Path coefficient of the model (significant at $p^{**} <= 0.01$, $p^{*} < 0.05$)

Table 5 Hypotheses-testing of the research model

H	Relationship	Path	*t*-value	*p*-value	Direction	Decision
H1	IQ - > BI	0.371	4.963	0.000	Positive	Supported**
H2	TTF - > BI	0.110	1.822	0.068	Positive	Not supported
H3	PLV - > BI	0.261	4.557	0.000	Positive	Supported**
H4	PI - > BI	0.254	3.730	0.000	Positive	Supported**

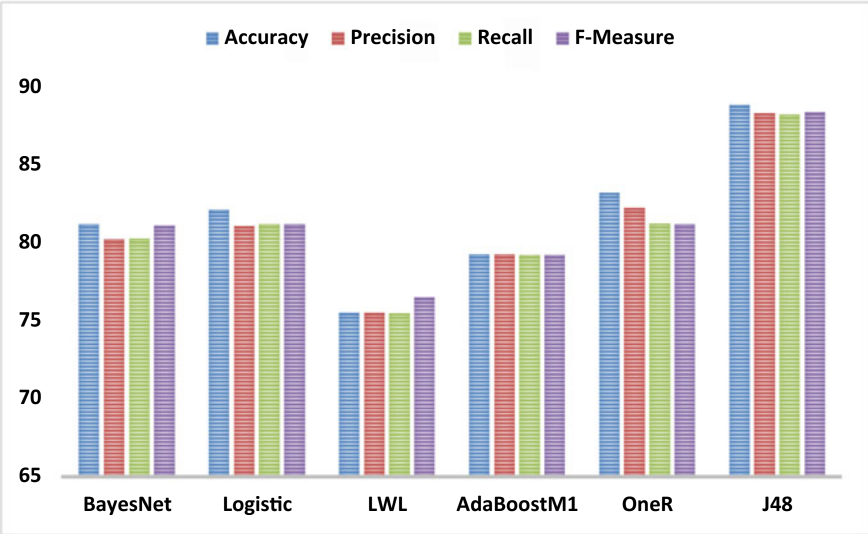

Fig. 4 Impact of IQ, TTF, PLV, and PI on BI

4.3 Hypotheses Testing Using Classical ML Algorithms

The results confirm the validity of the first through fourth hypotheses proposed in this study. Several methods, including the Bayesian network, neural network, and decision tree, were harnessed alongside the ML algorithm to assess the research hypotheses. These techniques were integrated into the conceptual framework to elucidate the relationships [31]. The predictions were validated using Weka (ver. 3.8.3), which employs classifiers like J48, OneR, and BayesNet [51, 52]. Among these, J48 emerged as the most efficacious in assessing BI, as depicted in Fig. 4. With an impressive accuracy rate of 88.92% during a ten-fold cross-validation, it stood out in terms of performance. Further endorsing its efficiency, J48 also recorded an 88.35% precision, an 88.27% recall, and an 88.44% F-Measure, marking its superior capability in the analysis.

4.4 Importance-Performance Map Analysis

This study incorporates IPMA, focusing on the behavioral intention as the key variable. IPMA is pivotal as it augments the insights gleaned from PLS-SEM analysis [22, 53–60]. Beyond merely evaluating importance measures or path coefficients, IPMA also assesses performance measure values. Within the IPMA framework, the total effect reveals how importance measures influence behavioral intentions (or target factors), while the mean of the latent construct embodies their performance.

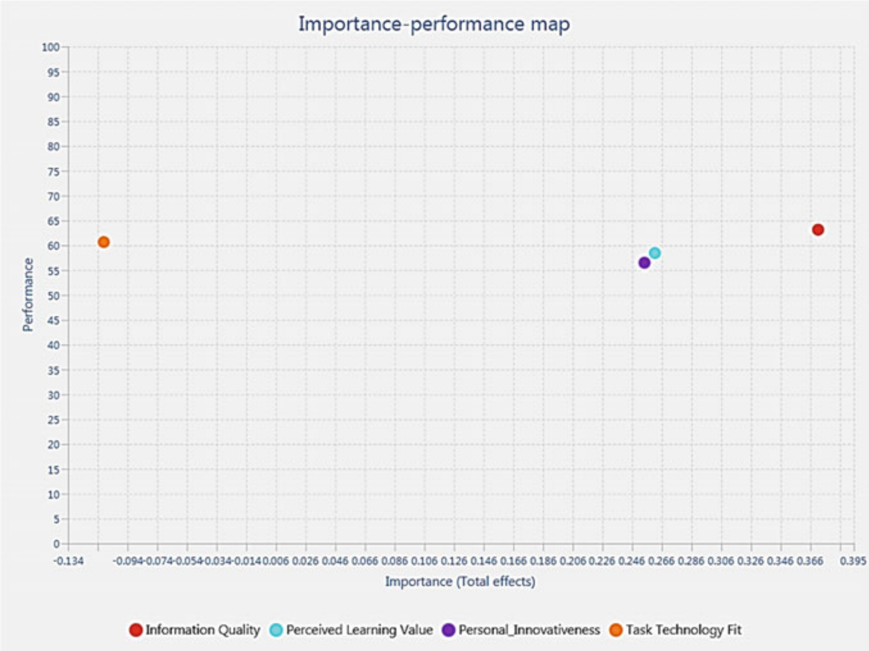

Fig. 5 IPMA output

Figure 5 depicts the significance and efficiency of IQ, PLV, PI, and TTF, as gauged by IPMA. As per Fig. 5, IQ emerges with the pinnacle values in both importance and performance, succeeded by PLV. PI ranks third in importance but is at the base in performance. TTF, on the other hand, holds the least rank in importance.

5 Conclusion

The study concludes that Chat GPT interests and benefits are higher than other applications. It is not simply a technical device, but it is an integrated platform that summarizes many issues to focus on different subject content, practical theories, and technological issues. It has been recongnized as effective tool. Undoubtedly the Chat GPT is an effective learning tool that is a well-designed with high level of sufficiency. Therefore, studying the impact of Chat GPT becomes crucial. Thus, this study reveals that this application has a large impact on the users' acceptance through information quality and task-technology fit on one hand and the perceived learning value, and personal innovativeness on the other hand. However, some of the aspects have not been supported and they do not necessarily have a significant predictive effect on the use of Chat GPT. This finding implies that providing additional development is required to reach better results which will increase the level of contributions of the application itself. From the theoretical point of view, the present research used PLS-SEM and ML algorithms to test the research model. The hybrid method in the present study contributes to IS research because it is among the few studies to use ML algorithms to predict adoption of ChatGPT. Research [36] has shown that PLS-SEM can predict dependent variables and validate the conceptual model. Similarly, Arpaci [31] indicates that ML can predict dependent variables based on predictor variables. The present study is also unique because it used various approaches with ML algorithms. These methodologies included neural networks, Bayesian networks, and decision trees. However, the decision tree or J48 had better performance than the other methodologies. As a non-parametric procedure, J48 was used to classify categorical and continuous variables. The classification was undertaken by dividing the sample into similar small samples based on significant predictors [31]. The PLS-SEM approach, a non-parametric approach, was used to assess the coefficients' significance. The study is limited to different perspectives. The data is collected from a sample of studies within the Gulf area. The users in the gulf are mainly students that joined different universities and they need the applications for various educational purposes. Future studies may focus on other samples from other government institutions that may use this application to enhance future usage. The conceptual framework is limited in scope to certain aspects that are related to system quality, satisfaction, value and innovativeness with moderators that can measure the effects of ChatGPT. Future studies can include other variables with different moderators that may measure the other essential factors.

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Forecasting the Acceptance of ChatGPT as Educational Platforms: An Integrated SEM-ANN Methodology

Raghad Alfaisal, Mohannad Hatem, Ayham Salloum, Mohammed Rasol Al Saidat, and Said A. Salloum

1 Introduction

Artificial intelligence (AI) continues to reshape the contours of modern life [1–3]. A standout example is Chat Generative Pre-trained Transformer, or ChatGPT. This AI marvel excels in creating term papers, short stories, explanations, and literary compositions. Its capabilities have not gone unnoticed; institutions like American University express reservations about its proficiency in crafting university-level essays and effectively tackling exam questions [4, 5]. As natural language processing (NLP) models make inroads into education, they revolutionize the learning landscape, granting students, educators, and academics greater access to content [6–10]. Google has been a dominant player, but ChatGPT's rise to prominence is evident. Its comprehensive database, ranging from books to online resources, coupled with its skill in interpreting complex student queries and delivering precise answers, underscores its potential as a crucial educational asset [4, 11].

ChatGPT's resonance isn't limited to education—it extends to fields like medicine and engineering. Its effects on a varied audience, from students to medical professionals, are significant. Yet, amidst discussions about technological advancements in

S. A. Salloum (✉)
Health Economic and Financing Group, University of Sharjah, Sharjah, UAE
e-mail: ssalloum@sharjah.ac.ae

School of Science, Engineering, and Environment, University of Salford, Salford, UK

M. Hatem · M. R. Al Saidat · S. A. Salloum
Computer Centre, University of Sharjah, Sharjah, UAE

A. Salloum
College of Medicine, University of Sharjah, Sharjah, UAE

R. Alfaisal
Faculty of Computing and Meta-Technology, Universiti Pendidikan Sultan Idris, Tanjung Malim, Malaysia

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education and AI tools like ChatGPT, there exists a gap in deciphering the intricate factors affecting user acceptance of such innovations. A majority of existing studies revolve around general technological adoption paradigms, often glossing over the subtleties specific to AI-driven educational platforms. Despite the burgeoning interest in AI platforms like ChatGPT for educational applications, there remains a conspicuous research gap regarding the specific drivers and barriers influencing its adoption in educational settings. Many studies have explored technological adoption in a broad sense, but a granular exploration focusing on AI chatbot platforms, especially ChatGPT, in an educational context is sparse. This study aims to fill this lacuna by delving deep into the intricacies of ChatGPT's acceptance as an educational platform. By identifying and analyzing these nuanced determinants, our research not only contributes to the academic discourse but also provides actionable insights for educational institutions and AI developers, ensuring a more seamless and effective integration of such platforms in the learning ecosystem.

Furthermore, much of the prevailing research utilizes single-stage linear data analysis [12], particularly the SEM, to discern relationships between variables. Past work has largely explored the linear associations among these factors. Sim et al. [13] argue that decision-making processes might not solely hinge on these linear relationships. Some scholars [14–16] advocate for the Artificial Neural Network (ANN) analysis as an enhancement to the limitations of SEM's single-tier data analysis. Conversely, Huang and Stokes [17] note that most studies employ a simplistic ANN variant with just a single hidden layer. Recent studies [18] propose the adoption of multi-layered ANN to augment the precision of non-linear models. This study seeks to bridge these research gaps, analyzing data through an advanced hybrid SEM-ANN approach, leveraging the capabilities of a deep ANN.

2 The Theoretical Framework

The existing framework underscores how elements like system quality, task-technology fit, perceived satisfaction, and personal innovativeness influence the intent to adopt learning platforms.

2.1 System Quality (SQ)

The system quality is defined by specific attributes that elevate its technological importance. Elements like reliability, usability, and functionality play a pivotal role in influencing users' adoption and acceptance of the technology. The aspect of reliability means that users find the technology dependable, bolstering their trust in its capabilities. A system deemed unreliable can cause users to doubt its suitability for vital tasks or informed decision-making. Usability hinges on the system's intuitive nature. When users find a system straightforward with clear guidance, their inclination towards it grows. Functionality, on the other hand, is vital for enabling users to grasp and utilize the technology effectively [19, 20]. In essence, a superior system quality can bolster user faith in a technology. Continual assessments and feedback collection can spotlight areas for refinement, ensuring the technology remains attuned to user requirements.

2.2 Task Technology Fit (TTF)

Task-technology fit gauges the extent to which a technology aligns with users' requirements, aiding and amplifying their workflow. This alignment emerges from the interplay between the technological capabilities on offer and the specific tasks users aim to achieve. The very essence of a technology's uptake rests on its pertinence to user activities. Typically, seasoned users gravitate towards tools that promise to elevate their forthcoming tasks, ensuring maximum utility. Technologies that miss the mark in catering to user expectations invariably face resistance and may not gain wide acceptance. Essentially, task-technology fit reflects a synergy between the technology and its requisite tasks. The more harmonious this relationship, the greater the benefits reaped from the technology's deployment. Past research has often touched upon task-technology fit but seldom intertwined it with other pivotal variables previously discussed [21, 22]. Many underscore the intermediary roles of perceived utility and ease of use within this context [22]. In light of this, our study spotlights the intermediary function of task-technology fit, bridging system and information quality with the embrace of platforms like ChatGPT and Google for educational purposes.

The symbiotic relationship between task-technology fit and system quality critically influences a technology's performance and resultant task efficacy. When technology seamlessly integrates with task demands and is backed by superior system quality, it stands to elevate productivity, refine operational flows, and yield better outcomes. Investigating this nexus between task-technology fit and system quality can pave the way for optimizing technological solutions, emphasizing attributes that distinctly differentiate one tool from another. In alignment with this exploration, we posit the following hypotheses:

H1: System quality of ChatGPT has a positive effect on behavior intention to use learning platforms.
H2: Task technology fit has a positive effect on behavior intention to use learning platforms.

2.3 The Perceived Satisfaction (PS)

Perceived satisfaction can be understood as the extent to which individuals feel contented with the tasks accomplished and services provided by a specific technology. It serves as an indicator of their overall experience and sentiment towards the technology in question. A positive perception of satisfaction signifies that users not only find value in the technology but also appreciate its functionality and user experience. This contentment often leads them to not just continue using the tool, but also advocate for its usage among peers and colleagues. In contrast, if a technology falls short of user expectations and is perceived as unsatisfactory, there's a heightened possibility that users might abandon it in favor of more suitable alternatives, or at the very least, express their reservations to others [23, 24]. This perceived satisfaction plays a pivotal role in the broader concept of technology acceptance. It provides insights into users' willingness to explore, adapt, and potentially embrace new technological advancements [25]. It's worth noting that the inclination to try out new technologies is often fueled by previous satisfactory experiences, suggesting a cyclical relationship between satisfaction and the desire for innovation.

2.4 Personal Innovativeness (PI)

Innovation within an individual often correlates with a heightened sense of satisfaction, especially when they engage with new technological advances. For this contentment to be realized, the innovation must align with the user's desires and anticipations. Therefore, there's a profound connection between an individual's innovative tendencies and their sense of satisfaction; those with pronounced innovative traits typically derive more pleasure from engaging with new technologies. To ensure increased adoption and contentment, technologies should be tailored to resonate with users' aspirations and exhibit pronounced innovative characteristics. This not only enhances technology acceptance but also deepens the feeling of accomplishment and contentment among users [26, 27]. In a related vein, the value users associate with learning is intricately tied to perceived innovation. Those who see high learning potential in a technology tend to explore all its facets, including its innovative features [27, 28]. Building on these insights, we suggest the subsequent hypotheses (Fig. 1).

H3: Perceived satisfaction of ChatGPT has a positive effect on behavior intention to use learning platforms.

H4: Personal innovativeness of ChatGPT has a positive effect on behavior intention to use learning platforms.

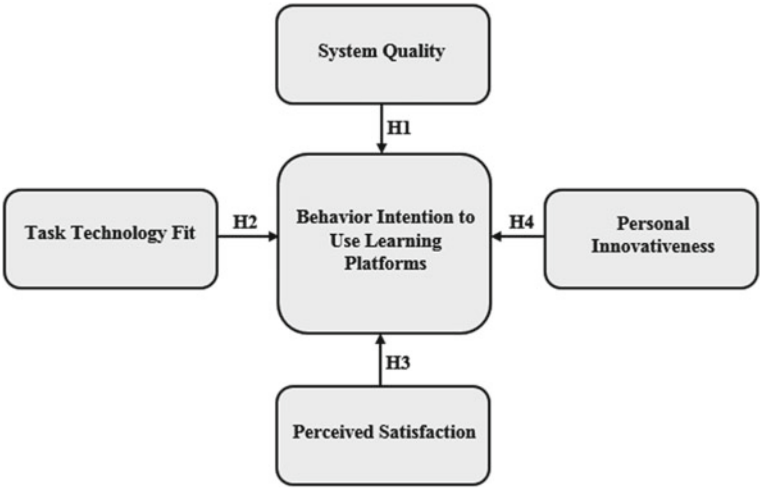

Fig. 1 Research model

3 Research Methodology

3.1 Data Collection

Students from partnering universities in the UAE were given online questionnaires. These were collected between February 10, 2023, and May 30, 2023. Of the 1000 questionnaires issued by the study committee, we received an 83.4% response rate, amounting to 834 complete responses. 166 questionnaires were disregarded due to partial answers. The 834 completed questionnaires surpassed the suggested sample size of 384 for a population of 1,000,000, as indicated by Krejcie and Morgan [25]. There seems to be an inconsistency in the text as it mentions a sample size of 153, which deviates considerably from the minimum requirements. As a result, structural equation modeling [26] was applied to this data to substantiate our hypotheses. It's worth noting that our hypotheses draw inspiration from prior theories deeply embedded in AI's history. For model evaluation, our research team used Structural Equation Modeling (SEM) in SmartPLS Version 3.2.7. Advanced analysis was executed using the final path model.

3.2 Demographic Information

Figure 2 showcases a demographic and personal data breakdown. Of the participants, 61% were female and 39% male. Age-wise, 75% were between 18 and 29 years, with the rest being 29 or older. The educational background of the respondents was

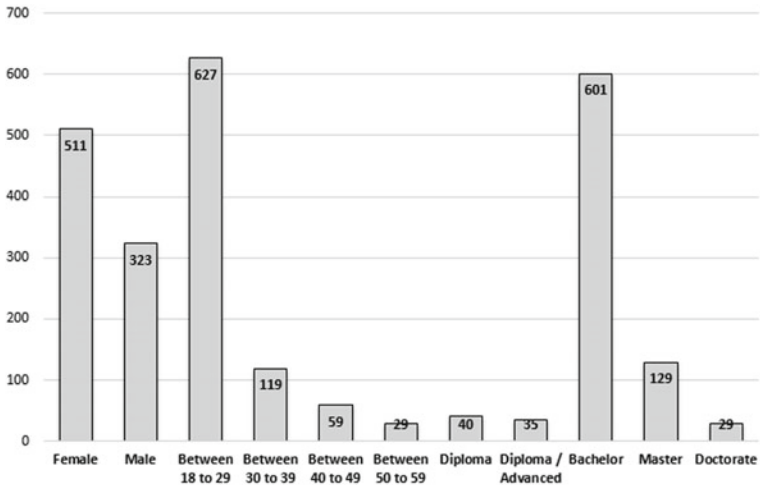

Fig. 2 Demographic information of the respondents ($n = 834$)

diverse: 5% held a diploma, 4% an advanced diploma, 72% a bachelor's degree, 15% a master's, and 3% a doctoral degree. When students showed interest in volunteering, the "purposive sampling approach" recommended by Salloum and Shaalan [29] was considered. This research drew from multiple universities, covering a spectrum of ages and academic qualifications. For demographic data analysis, IBM SPSS Statistics version 23 was employed.

3.3 Study Instrument

In In this research, we used a questionnaire to corroborate the hypothesis. We meticulously chose five constructs as dependable metrics, resulting in the addition of 14 items to the questionnaire. The underpinnings of these constructs can be viewed in Table 1. This table seeks to boost the practicality of the study's constructs and offers corroborative evidence from diverse prior research, reinforcing our current structure. In the end, the scholarly team refined the questionnaire items, drawing on insights from earlier studies.

3.4 Pilot Study of the Questionnaire

In this study, to ensure the reliability of the survey questions, a preliminary pilot study was carried out. For this pilot, data was randomly selected, and 100 students from the intended demographic were chosen as the initial sample. This number represented

Table 1 Measurement items

Constructs	Items	Instrument	Sources
Behavior intention to use learning platforms	BI1	ChatGPT presents an excellent chance for experimentation	[30, 31]
	BI2	ChatGPT provides a great avenue to test out	[30, 31]
System quality	SQ1	ChatGPT's functionality is superior when assisting with my learning tasks	[32, 33]
		ChatGPT's usability in accomplishing learning tasks within a constrained time frame is notably high	
	SQ2	ChatGPT's efficiency level is notably elevated	
Task technology fit	SQ3	The information sourced from ChatGPT is more recent	[32]
	TTF1	The information from ChatGPT is more pertinent and in-depth	
	TTF2	The information I receive from ChatGPT goes beyond what's necessary for my learning tasks	
The perceived satisfaction	TTF3	ChatGPT offers enhanced tools that streamline the learning process	[34]
	PS1	ChatGPT possesses cutting-edge features that influence my satisfaction level	
	PS2	ChatGPT doesn't fully meet my learning requirements for adoption	
Personal innovativeness	PS3	ChatGPT's contemporary technology fulfills my requirements	[25]
	PI1	ChatGPT boasts pioneering features that amplify the technology's value	
	PI2	ChatGPT delivers an unparalleled experience that propels me to embrace it	

the entire sample size for this study, with 10% of the broader target population of 1000 students set aside for evaluation. Using IBM SPSS Statistics version 23, we gauged the internal consistency of the measurement items through Cronbach's alpha. Such an assessment provided a deeper understanding of the pilot study's outcomes and enabled the derivation of trustworthy results from the measurement items. In the realm of social sciences, a reliability coefficient of 0.70 is deemed acceptable [35]. The respective Cronbach's alpha values for the five measurement scales can be found in Table 2.

Table 2 Cronbach's alpha values for the pilot study (Cronbach's alpha ≥0.70)

Construct	Cronbach's alpha
BI	0.824
SQ	0.810
PI	0.832
PS	0.719
TTF	0.773

4 Findings and Discussion

4.1 Data Analysis

This research adopted the partial least squares-structural equation modeling (PLS-SEM) approach with SmartPLS V 3.2.7 [1, 36–39] for data examination [27, 40–46]. The methodology involved a bifurcated assessment, encompassing both the measurement and structural models [47]. The selection of PLS-SEM was informed by several pivotal reasons elaborated upon in this paper. To start with, PLS-SEM was favored for its capacity to rigorously evaluate the proposed conceptual framework [48]. Next, its efficacy in managing the explorative data, harvested in line with the conceptual models, was a significant factor [49]. Additionally, instead of dissecting the model, PLS-SEM was employed holistically, treating the model as an integrated whole [50]. In the final point, the structural and measurement models underwent simultaneous analysis via PLS-SEM. One of the standout merits of PLS-SEM is its precision in measurement generation and acquisition [51].

4.1.1 Convergent Validity

The Measurement Model's assessment [47] was anchored on the tenets of construct validity—encompassing both discriminant and convergent validity—and construct reliability, inclusive of Cronbach's alpha (CA) and composite reliability (CR). Table 3 illustrates that the values of Cronbach's alpha (CA) for construct reliability oscillated between 0.786 and 0.901, situating below the prescribed benchmark of 0.7 [52]. Yet, data from Table 3 indicates that composite reliability (CR) figures span from 0.864 to 0.938, exceeding the set standard [51]. To appraise convergent validity, it's imperative to scrutinize the average variance extracted (AVE) alongside factor loadings [47]. Apart from the aforementioned, Table 3 reveals factor loading values that surpass the standard 0.7 threshold. Additionally, Table 3 conveys AVE metrics above the 0.5 demarcation, with the exception of initial values that ranged between 0.761 and 0.835. As a result, convergent validity is inferred to be satisfactorily achieved based on the discussed parameters.

Table 3 Convergent validity results

Constructs	Items	Factor loading	CA	CR	AVE
Behavior intention to use learning platforms	BI1	0.882	0.786	0.864	0.761
	BI2	0.862			
System quality	SQ1	0.912	0.901	0.938	0.835
	SQ2	0.901			
	SQ3	0.917			
Personal innovativeness	PI1	0.917	0.881	0.926	0.808
	PI2	0.868			
	PI3	0.910			
The perceived satisfaction	PS1	0.940	0.896	0.935	0.828
	PS2	0.875			
	PS3	0.926			
Task technology fit	TTF1	0.924	0.880	0.921	0.807

4.1.2 Discriminant Validity

For discriminant validity evaluation, this study opted to revisit the two standards using the Heterotrait-Monotrait ratio (HTMT) and the Fornell-Larker criterion [47]. As reflected in Table 4, the criteria are corroborated by the Fornell-Larker criterion, seeing that every Average Variance Extracted (AVE) and its square root are more robustly associated with their specific constructs [53].

Table 5 showcases the outcomes of the HTMT ratio, with each construct registering below the '0.85' benchmark [54]. This signals that the HTMT ratio aligns with the stipulated standard, permitting the establishment of discriminant validity. The evidence from this investigation affirms the absence of concerns pertaining to the evaluation of validity and reliability in the Measurement Model. Hence, the accrued data is deemed apt for appraising the structural model.

Table 4 Fornell-Larcker scale

	BI	SQ	PS	PI	TTF
BI	**0.872**				
SQ	0.651	**0.914**			
PS	0.631	0.743	**0.899**		
PI	0.565	0.617	0.659	**0.910**	
TTF	0.449	0.729	0.626	0.465	**0.898**

Table 5 Heterotrait-Monotrait ratio (HTMT)

	BI	SQ	PS	PI	TTF
BI					
SQ	0.824				
PS	0.810	0.832			
PI	0.719	0.681	0.740		
TTF	0.573	0.817	0.706	0.520	

4.2 Hypotheses Testing Using PLS-SEM

The study employed Smart PLS to construct its structural equation model, leveraging the maximum likelihood estimation method to investigate the interconnections between the model's theoretical constructs [55, 56]. The model's efficacy in predicting relationships is evident in Table 6 and Fig. 3 [57]. A noteworthy observation is that the "Behavior Intention to Use Learning Platforms" accounts for approximately 49.5% of the explained variance. The beta (β) coefficients, t-values, and p-values for each hypothesis, derived from the PLS-SEM analysis, are detailed in Table 6. The analysis confirms the validity of hypotheses H1, H3, and H4, but hypothesis H2 was not supported and was subsequently dismissed. H1 probed the link between SQ and BI, yielding a coefficient $\beta = 0.180$ and $P < 0.01$, which establishes a significant positive correlation between SQ and BI, endorsing H1. The data indicates that TTF exerts a minimal impact on BI with a coefficient of $\beta = 0.108$ and $P = 0.057$. As a result, H2 isn't substantiated, suggesting a minimal influence of TTF on BI. The findings reveal marked associations between BI and other determinants. Specifically, PS and PI manifest a pronounced positive correlation with BI, with coefficients $\beta = 0.420$, $P < 0.001$, and $\beta = 0.268$, $P < 0.001$ respectively, reinforcing hypotheses H3 and H4.

Table 6 Hypotheses-testing of the research model

H	Relationship	Path	*t*-value	*p*-value	Direction	Decision
H1	SQ -> BI	0.180	2.803	0.005	Positive	Supported**
H2	TTF -> BI	0.108	1.901	0.057	Positive	Not supported
H3	PS -> BI	0.420	5.727	0.000	Positive	Supported**
H4	PI -> BI	0.268	3.485	0.000	Positive	Supported**

Significant at $p^{**} \leq 0.01$, $p^{*} < 0.05$

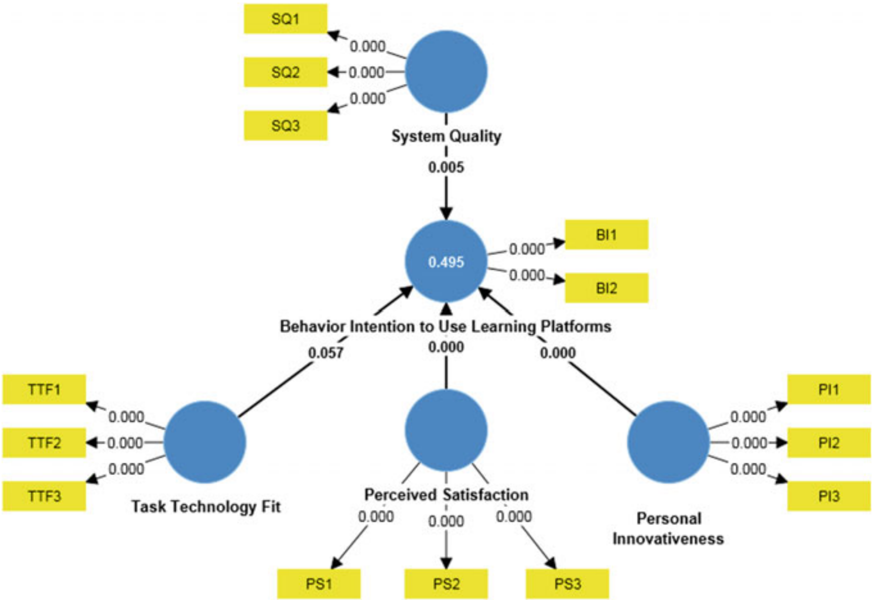

Fig. 3 Path coefficient of the model (significant at p** <= 0.01, p* < 0.05)

4.3 Hypotheses Testing Using Deep ANN

In the analytical exploration, pertinent predictors identified through the Partial Least Squares Structural Equation Modeling (PLS-SEM) were integrated into the Artificial Neural Network (ANN) evaluation [37, 58–61], executed utilizing the SPSS software. Consequently, predictors, namely PI, SQ, PS, and TTF, were incorporated in the ANN examination. Figure 4 [62] depicts the deep ANN model configuration, characterized by a singular output neuron and multiple input neurons, specifically corresponding to PI, SQ, PS, TTF, and the BI output of the ANN construct. A dual-hidden-layer architecture was instituted for this deep learning model to govern the output neuron node. Both the hidden neurons employed a sigmoid activation function for their outputs [63]. To enhance the robustness of the proposed architecture, input and output variables in the study were normalized within a range of 0–1. To circumvent the overfitting dilemma in ANN models, a ten-fold cross-validation technique was adopted. This validation employed a distribution split of 80% for training and 20% for testing [64]. The model's efficacy was further augmented by minimizing the root mean square error (RMSE). As delineated in Fig. 4, the deep ANN model registered an RMSE of 0.1276 for training and 0.1298 for the testing phase. The precision of the deep ANN models was substantiated by a standard deviation of 0.0037 in the training dataset's RMSE and 0.0082 in the testing dataset's RMSE.

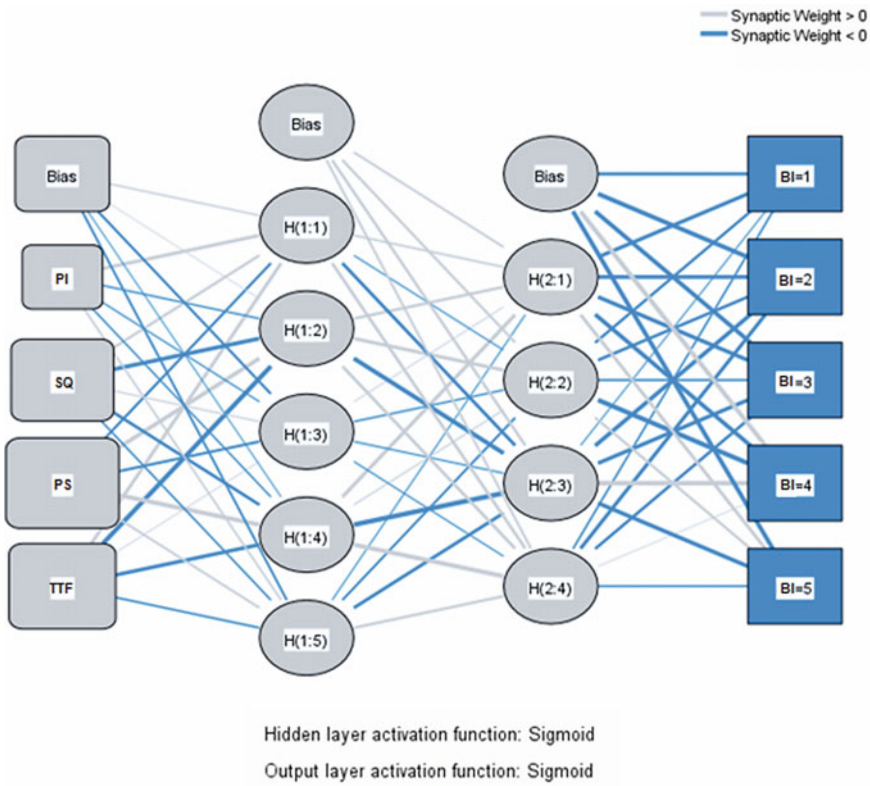

Fig. 4 ANN model

4.4 Sensitivity Analysis

The inclination towards adopting ChatGPT stands out as the primary behavioral determinant, as highlighted in Table 7. PS and PI rank as the subsequent major influencers on the behavioral intention to engage with ChatGPT. By juxtaposing each predictor's average value against the highest mean percentage, the standardized significance is deduced [60, 65, 66]. The efficacy and precision of the deep ANN evaluation have been corroborated using a fit metric, akin to R-squared in PLS-SEM [67]. Results indicate that while PLS-SEM captures a variance of 49.5% ($R^2 = 49.5\%$), deep ANN showcases a more pronounced predictive strength, accounting for 93.5% variance ($R^2 = 93.5\%$). This signifies that deep ANN provides a more comprehensive representation of endogenous variables, and, notably, it excels over PLS-SEM algorithms in modeling non-linear interconnections among variables.

Table 7 Independent variable importance

	Importance	Normalized importance (%)
PI	0.292	78.5
PS	0.426	94.2
SQ	0.265	73.1
TTF	0.071	15.6

4.5 Importance-Performance Map Analysis

Utilizing the Improved Performance Measurement Analysis (IPMA) enhances the interpretive efficacy of Partial Least Squares Structural Equation Modeling (PLS-SEM) results, as corroborated by previous research [68]. In the present study, the principal construct examined through IPMA is 'behavioural intention.' IPMA facilitates the juxtaposition of performance metrics against the significance of path coefficients. The aggregate influence within the IPMA framework elucidates how specific importance measures sway behavioural intentions (the target constructs). The performance of these constructs is denoted by the mean values of the latent variables. As delineated in Fig. 5, IPMA sheds light on the significance and performance of constructs such as PS, PI, SQ, and TTF. Among these, PS manifests the highest metrics in both importance and performance, followed closely by PI. Conversely, SQ registers the lowest performance metric but maintains a median rank in importance. TTF, on the other hand, recorded the lowest in terms of its importance measure.

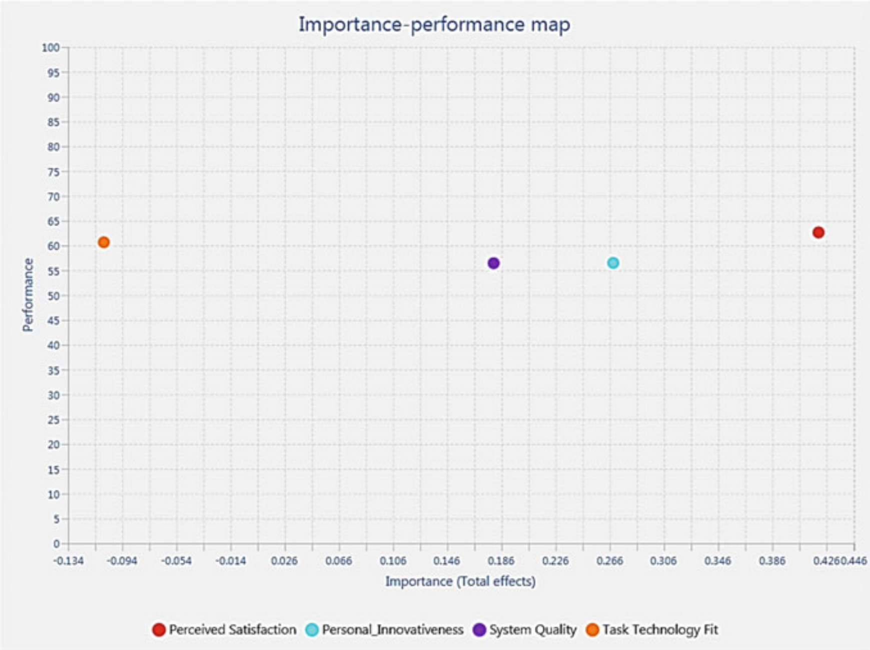

Fig. 5 IPMA output

5 Conclusion

This study underscores that ChatGPT not only surpasses many applications in terms of interest and benefits but also stands out as more than just a mere technical instrument. ChatGPT embodies a holistic platform, weaving together diverse academic content, applied theories, and technological intricacies. Its acclaim as an efficacious tool is well-earned, marking its stature as a meticulously crafted learning instrument that excels in its proficiency. The essence of ChatGPT in the educational realm cannot be overstated, making an in-depth examination of its influence indispensable. Our findings elucidate the profound impact of ChatGPT on user acceptance. This is mediated by factors like system quality and task-technology fit on one side and user-perceived satisfaction and personal innovativeness on the other. Nonetheless, not all aspects under investigation demonstrated a substantial influence, indicating that not every element studied significantly determines ChatGPT's adoption. Such insights call for more nuanced development efforts to augment the tool's contributions and potential. From an academic standpoint, this research harnesses the power of PLS-SEM and deep ANN algorithms to scrutinize the proposed model. The novel hybrid methodology employed here stands as a noteworthy contribution to the Information Systems (IS) research landscape. Few studies have leveraged Machine Learning (ML) algorithms to delve into the adoption nuances of platforms like ChatGPT.

Prior research [45] validates PLS-SEM's prowess in predicting dependent variables and substantiating conceptual models. Concurrently, numerous scholars [46–51] vouch for the predictive capabilities of ANN based on various independent variables. Adding a unique twist, our study experimented with diverse ANN techniques, notably deep neural networks. However, this study is not without its limitations. The data, predominantly drawn from the Gulf region, focuses on students enrolled in various universities seeking technological aids for educational enrichment. A more expansive purview might consider a broader demographic, encompassing other sectors and institutional frameworks. The conceptual framework presented here, though comprehensive, orbits around specific facets like system quality, satisfaction, value, and innovativeness, augmented by select moderators. Subsequent studies might wish to expand the spectrum, integrating varied variables and moderators to glean a richer understanding of ChatGPT's multifaceted impact.

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AI in Professional Development and Training

Assessing the Efficacy of E-Mind Mapping on Academic Performance: A Meta-Analysis of Empirical Research

Khaled Younis Alderbashi

1 Introduction

Over the past few decades, significant advancements have been made in the field of computing, which have impacted modern society in numerous ways, including education [1–5]. Computer-based teaching and learning strategies have been implemented in scientific courses in schools, examples of effective strategies for visualizing and organizing information include digital storytelling [6–10] and e-mind mapping [11]. By using specialized software, the e-mind mapping strategy enables users to create diagrams that visually represent the relationships between ideas or data points in the form of a web [12–16].

The e-mind mapping strategy offers numerous benefits, including the expansion of students' vocabulary knowledge by displaying images that visually depict word meanings [12]. It also leads to increased academic achievement and higher retention rates of concepts, as the images used in e-mind maps aid in memory recall.

Furthermore, the implementation of the e-mind mapping strategy has been found to foster the acquisition of computer skills among students by allowing them to explore and utilize various computer programs to create and manipulate shapes, videos, images, and attach audio files [17–21]. This strategy provides ample opportunities for students to hone their computer skills, which can be useful in various fields of study and future career paths. It also enhances students' attitudes towards the course, making the learning process more enjoyable and less tedious [22].

Incorporating e-mind maps into teaching practices also fosters creative thinking skills as instructors may encourage students to contribute their own ideas to the map, promoting creativity [11]. Additionally, this strategy enhances students' motivation to learn by utilizing images, shapes, and attractive colors in the maps [23]. It improves

K. Y. Alderbashi (✉)
Professional Postgraduate Diploma in Teaching, City University Ajman, Ajman, UAE
e-mail: k.derbashi@cu.ac.ae

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thinking skills as well by allowing access to items that are challenging to bring to the classroom and facilitating exploration of the world without travel, which can lead to the development of critical thinking processes and skills [24].

Moreover, the e-mind mapping strategy enhances the learning process by systematically employing instructional strategies to improve learning outcomes and education quality [25]. It also increases student engagement in the learning process by incorporating games based on e-mind maps [25]. Finally, this strategy improves students' planning skills by enabling them to effectively organize information and ideas in the map format [26]. The e-mind mapping strategy offers a range of benefits for students, such as improving their vocabulary and retention of concepts through the use of visual aids [12]. It also helps develop students' computer skills by providing opportunities to use software to create shapes, videos, and images, as well as attach audio files [17]. This strategy promotes creative thinking and motivation to learn through the use of attractive visuals and colors [11, 23]. Furthermore, it enhances thinking skills by allowing students to explore difficult-to-access items and promotes planning skills [26].

The e-mind mapping strategy also enhances writing and visual thinking skills and expands students' knowledge through the attachment of web pages with additional information [27, 28]. While there have been no studies conducted on the use of this strategy in Emirati private preparatory schools and students' attitudes towards it, it is important to explore these attitudes and evaluate the effectiveness of the strategy when combined with a blended learning approach. This approach combines face-to-face and electronic instruction, allowing learners to interact with peers and materials and develop their knowledge acquisition and analysis skills [29]. The adoption of this approach was accelerated by the COVID-19 pandemic, with schools in UAE using it during the first semester of the 2020–2021 academic year to prevent the spread of the virus among students.

2 Study Objective

This research aimed to evaluate the efficacy of incorporating the e-mind mapping strategy into scientific courses at Emirati private preparatory schools, in tandem with a blended learning approach. The study sought to determine the impact of this teaching method on students' learning outcomes and academic performance in scientific courses.

3 Research Question

How does the integration of e-mind mapping strategy with the blended learning approach impact the quality and effectiveness of scientific courses in Emirati private preparatory schools?

4 Significance of the Study

The findings of this study provide valuable insights for the UAE Ministry of Education, as well as for researchers who may wish to conduct similar studies in different countries or fields of study.

5 Limitations of the Study

The following limitations are identified in this study:

- Geographical limitation: This study focuses on Emirati private preparatory schools.
- Time limitation: The data collection for this research was conducted in November 2021.
- Blended learning approach (Theoretical definition): It is a combination of face-to-face and electronic instructional methods that provides the benefits of both online and offline education [29].
- E-mind maps (Theoretical definition): They are graphical representations of the relationships between various ideas or pieces of information that are created using software. Each idea or fact is connected to its major or minor idea or fact through curved lines, creating a web of relationships [12]. Preparatory schools: These are schools located in the UAE that offer education to students in grades 7, 8, and 9th.

6 Theoretical Framework

The theoretical framework of this study is based on two main concepts: blended learning and e-mind mapping strategy.

Blended learning is a teaching approach that combines face-to-face and electronic instructional methods [30–34]. It allows students to interact with their peers and the academic material, and it has been shown to positively affect students' knowledge analysis and acquisition [29, 35]. Blended learning is particularly relevant in the context of the COVID-19 pandemic, as it allows schools to continue delivering education while minimizing the risk of virus transmission.

E-mind mapping strategy, on the other hand, refers to the use of mind maps that are designed through the use of software. These maps allow for the visualization of connections between different ideas or pieces of information. E-mind mapping has been shown to improve students' visual and inferential thinking skills, expand their knowledge, and enhance their writing skills [27, 28, 36].

This research delves into the assessment of the efficacy of implementing the e-mind mapping strategy in scientific courses offered by Emirati private preparatory schools, while utilizing the blended learning approach. The study aims to investigate the impact of this approach on enhancing students' academic performance in scientific courses. The study aims to investigate whether the combination of e-mind mapping and blended learning can enhance students' learning outcomes in scientific courses.

The theoretical framework also considers the role of students' attitudes towards the use of teaching and learning strategies. Abed and Al-Dirbashi [29] argue that exploring students' attitudes towards different strategies is necessary for improving the effectiveness of teaching and learning [37–41]. Therefore, this study will also investigate students' attitudes towards the use of e-mind mapping and blended learning in scientific courses [42–46].

Al-Helo [25] argues that the e-mind mapping strategy can enhance student engagement, while Al-Ibrahim [28] posits that it can improve inferential thinking skills by identifying implicit meanings. Ahmad et al. [47] support this claim and suggest that it can also enhance visual perceptual skills. Hani [48] further asserts that e-mind mapping can enhance analytical thinking skills, academic achievement in science, and motivation to learn, especially for students with attention deficit hyperactivity disorder. Mohammad and Abed Al-Sayed [26] suggest that e-mind mapping also develops planning skills, while Sulaiman highlights its potential for improving writing skills. Al-Qat'an [49] notes its positive impact on academic achievement and motivation, and [27] claims that e-mind mapping improves visual thinking skills. Additionally, Al-Shawawreh [50] suggests that it enhances reading comprehension skills and attitudes towards reading. Furthermore, Khalifah [51] suggests that e-mind mapping has the potential to improve decision-making skills and cognitive flexibility levels, while Mahmoud [52] claims that it can aid in acquiring syntactic concepts. Finally, Mahmoud [52] concludes that e-mind mapping strategy makes learning a more enjoyable experience and enhances students' knowledge of geographical concepts. The reviewed empirical studies are presented chronologically from oldest to newest, and include the following.

In 2014, Ahmad [47] investigated the impact of using mind maps on the mathematical problem-solving ability and achievement of middle school female students in Bour Saeed, Egypt. The study included 90 participants, with 45 in the experimental group and 45 in the control group. Ahmad used achievement tests and mathematical problem-solving tests and found that using mind maps positively impacted the students' achievement and mathematical problem-solving ability.

Al-Sa'eedy [23] investigated the impact of using the e-mind mapping strategy on the academic achievement and motivation of female middle school students in Assir, Saudi Arabia. Using a quasi-experimental design, she divided 60 students into control and experimental groups and employed a pre-test and post-test to evaluate achievement and a survey to measure motivation. Her findings showed that the e-mind mapping strategy significantly improved the academic achievement of the students and positively impacted their motivation.

Bawaneh [53] examined the effectiveness of using the mind mapping strategy to improve the academic achievement and retention of concepts related to electronic energy among 10th grade students in Bani Kenanah region, Northern Province in

Jordan. With a random sampling method, 111 male and female students were selected and divided into control and experimental groups. Pre-test and post-test assessments were conducted, and the results indicated that the mind mapping strategy significantly improved the academic achievement and retention of concepts related to electronic energy among the students.

Al-Montashiri [17] conducted a study to examine the impact of utilizing the e-mind mapping strategy on the computer skills of middle school students. In the study, a sample of 46 students was selected, and they were assigned to either the control or experimental group. The researcher employed pre-test and post-test assessments to evaluate the computer skills of the students. The findings indicated that utilizing the e-mind mapping strategy had a substantial effect on enhancing the computer skills of middle school students.

Sabrah and Al-Jadery [22] conducted a study on the impact of using the e-mind mapping strategy on the achievement and attitudes of 10th grade students towards biology in Jordan. The study involved a sample of 31 students, and the researchers employed pre-test and post-test assessments and a survey to evaluate the effectiveness of the strategy. The findings suggested that the e-mind mapping strategy had a significant positive impact on raising the achievement of the students and improving their attitudes towards the biology course.

In a separate study conducted in 2020 by Salameh et al., the effectiveness of using the electronic mind mapping strategy was investigated in developing creative thinking skills in science among 9th grade female students living in Gaza. The study was conducted using an experimental approach and involved a sample of 70 female students from Khan Younis, Gaza. The researchers conducted a pre-test and post-test on creative thinking skills, and the results revealed that the electronic mind mapping strategy had a significant positive impact on the development of creative thinking skills in science among the students.

7 Methods

The effectiveness of implementing the e-mind mapping strategy in scientific courses at Emirati private preparatory schools with blended learning was evaluated using both descriptive analytical and quantitative approaches by the researcher. According to Östlund et al. [54], the quantitative approach is commonly employed in social science and healthcare research as it enables researcher to establish connections between a particular theory and empirical evidence, develop new theories, and validate theoretical assumptions.

Table 1 The frequencies and percentages of the sampled students, categorized by gender and grade

Variable	Class	Rate	(%)
Gender	M	157	86.2
	F	29	15.8
School grade	Seven	20	11.4
	Eight	19	10.1
	Nine	148	79.6

N = 184 students

7.1 Population

The population comprises of all students who are currently enrolled in Emirati private preparatory schools.

7.2 Research Sample

The researcher developed a questionnaire based on previous studies published in peer-reviewed journals [2, 5, 14, 55–61] and utilized Google Forms to upload the questionnaire online. They employed purposive sampling by sending the questionnaire link to 402 students from five Emirati private preparatory schools located in Ajman, Abu Dhabi, Dubai, and Sharjah. However, only 184 forms were completed, resulting in a response rate of 45.5%. It is worth mentioning that the teachers of the scientific courses attended training sessions on the e-mind mapping strategy before collecting data from the respondents. The following is a breakdown of the sample data (Table 1).

7.3 Research Rationality

The researcher shared the initial version of the survey with three faculty members who hold PhD degrees in teaching methods and work in a university located in the UAE. The researcher asked for feedback on the survey's relevance, clarity, language, and content. The three faculty members concluded that the survey was clear, free of language errors, and highly relevant to the study's objectives. They also recommended that no items be removed from the survey. However, one faculty member suggested adding an item, which the researcher subsequently included in the survey.

Table 2 The classification criteria for the means are as follows

R	L	A
2.35 or lower	L	(_)
2.24–3.68	Average	(+/_)
3.69 or above	H	(+)

7.4 Research Reliability

The survey utilized in this article has a Cronbach's alpha value of 0.814 [62–66], indicating a high level of reliability since this value exceeds 0.70, as referenced in [67] study.

7.5 Analysis Phase

The rating system of the Likert scale consists of 5 categories: strongly agree, agree, neutral, disagree, and strongly disagree. Each category corresponds to a numerical score of 5, 4, 3, 2, and 1, respectively (Table 2).

7.6 Data Collection

To achieve their objective, the researcher utilized both primary and secondary data collection techniques. The primary method involved the creation of a questionnaire tailored to the study's aims [19, 68–74]. The secondary methods encompassed books and articles published in English and Arabic languages. By employing these methods, the researcher accomplished their intended objective.

To address the study question, researcher computed means and standard deviations. The research question was: "What is the extent of effectiveness of employing the e-mind mapping strategy in scientific courses in Emirati private preparatory schools in the light of adopting the blended learning approach?" The results revealed that the e-mind mapping strategy was highly effective, with an overall mean of 4.08. The success can be attributed to students' preference for computer-based learning strategies. Furthermore, employing this strategy in scientific courses resulted in improved academic achievement, with a mean of 4.75 for statement 1, which aligns with [75] findings, as learning strategies can enhance learning outcomes and organization of information.

Employing the e-mind mapping strategy also contributed to better understanding of scientific concepts, as evidenced by a mean of 4.39 for statement 2, which is consistent with the results of [12]. The use of images and videos to illustrate meanings may have played a role. Additionally, the e-mind mapping strategy motivated students to learn and acquire more information, with a mean of 4.70 for statement 3. This aligns with [12] findings, as multimedia and colors increase student concentration and motivation to learn (Table 3).

The researcher created this survey by drawing on a range of studies, including those authored. The mean score for statement 4 in the researcher' study was 4.81, indicating that employing the e-mind mapping strategy in scientific courses does not improve students' problem-solving skills. This result contradicts the finding of [47] and could be attributed to the fact that developing problem-solving skills requires a more interactive approach rather than simply presenting information in the form of e-mind maps.

However, the mean score for statement 5 was 4.21. This may be due to the use of arrows and geometric shapes, which can present data in a more organized way than paragraphs or sentences.

Furthermore, statement 6 yielded a mean score of 4.67, suggesting that the utilization of the e-mind mapping strategy in scientific courses enhances the retrieval of

Table 3 The means and standard deviations that were utilized to evaluate using the e-mind mapping strategy in scientific courses in Emirati private preparatory schools is evaluated for its effectiveness when implemented with a blended learning approach

No. Item	Mean	SD	Level	Attitude
1	4.75	0.63	High	Positive
2	4.39	0.47	High	Positive
3	4.70	0.51	High	Positive
4	2.22	0.33	Low	Negative
5	4.21	0.84	High	Positive
6	4.67	0.69	High	Positive
7	4.49	0.81	High	Positive
8	4.63	0.27	High	Positive
9	2.15	0.17	Low	Negative
10	4.42	0.39	High	Positive
11	4.82	0.57	High	Positive
12	1.98	0.92	Low	Negative
13	4.77	0.36	High	Positive
14	4.33	0.58	High	Positive
15	4.80	0.22	High	Positive
16	4.68	0.77	High	Positive
17	4.75	0.93	High	Positive
18	4.84	0.84	High	Positive
19	2.10	0.75	Low	Negative
Overall	4.08	0.58	High	Positive

Source Author's Survey, 2023

scientific information. This is due to the fact that students can easily search for particular words on the map by using the keyboard, which is particularly useful when the map contains a large number of items.

Furthermore, the mean score for statement 7 was 4.49, indicating that employing this strategy in scientific courses helps to develop students' computer skills. This finding aligns with the results of Al-Montashiri [17] study, which showed that using this strategy requires students to learn how to add geometric shapes and arrows, change colors, delete and attach items.

According to the researcher' findings, using the e-mind mapping strategy in scientific courses helps to foster students' creative thinking skills as evidenced by a mean score of 4.63 for statement 8. This result is consistent with Salameh et al. [11]findings, and may be due to the instructor's encouragement of students to add new items to the map as a way of promoting creativity. On the other hand, the researcher found that the use of this strategy does not improve students' attitudes towards scientific courses, as shown by a mean score of 2.15 for statement 9. This could be attributed to the fact that students' attitudes are influenced by factors such as the instructor's personality, teaching style, and level of respect towards students. This finding is in contrast with [22] study.

The researcher also discovered that the e-mind mapping strategy enhances students' visual thinking skills, indicated by a mean score of 4.42 for statement 10. Additionally, it improves students' imaginative thinking skills, as demonstrated by a mean score of 4.82 for statement 11. This finding is consistent with [24] research, which suggests that incorporating images and videos into these maps enhances students' ability to imagine and generate ideas.

On the other hand, the researcher discovered that the utilization of the e-mind mapping strategy did not enhance students' problem-solving abilities, as indicated by a mean score of 4.81 for statement 4. This finding goes against [47] study, and the reason for this disparity might be due to the fact that developing problem-solving skills necessitates presenting a problem and asking students to work on finding solutions rather than merely presenting e-mind maps. However, the strategy was found to help organize scientific information, as demonstrated by a mean score of 4.21 for statement 5. The use of arrows and geometric shapes to present data is believed to contribute to a more organized approach than presenting information in paragraph or sentence format. Finally, the use of the strategy contributes to developing students' computer skills, as evidenced by a mean score of 4.49 for statement 7, which is consistent with [17] findings. This is because students must learn to add and delete items from the map, change its colors and shape, and search for specific words.

The results of the study indicate that employing this strategy in scientific courses has limited effect on improving students' attitudes towards their teachers, as evidenced by a mean score of 1.98 for statement 12. This outcome is attributed to the complex interplay of various factors that influence students' attitudes towards their instructors, including their personality, teaching style, and respect for students.

Additionally, employing this strategy in scientific courses promotes innovation (statement 15, mean score of 4.80) by enabling students to access videos and experiences that may be difficult to replicate within the classroom. It also improves learning skills (statement 16, mean score of 4.68) by facilitating self-learning practices (statement 17, mean score of 4.75).

Furthermore, employing this strategy in scientific courses enhances students' ability to complete writing-related tasks (statement 18, mean score of 4.84). However, it has little effect on improving students' decision-making skills (statement 19, mean score of 2.10), which may require targeted courses led by psychology specialists to achieve significant improvement.

8 Conclusion

The study concluded that using the e-mind mapping strategy in scientific courses as a part of a blended learning approach in Emirati private preparatory schools was highly effective. It made learning enjoyable and facilitated the process of retaining scientific information in long-term memory. The strategy also promoted innovation, improved learning skills, and enabled self-learning practices as well as writing-related tasks. However, the study also revealed that this strategy did not contribute to enhancing students' attitudes towards their science teachers, as these attitudes are influenced by factors such as personalities, instructional methods, and interaction with students. Furthermore, the strategy was not found to improve students' decision-making skills, which require specialized courses given by experts in the field of psychology.

9 Set of Endorsements

Based on the findings of the study, the researcher suggest that courses should be provided for teachers to enhance their understanding of how to effectively utilize the e-mind mapping strategy in scientific courses. These courses should focus on teaching teachers how to create e-mind maps using different software tools. Additionally, the researcher recommend that teachers receive training on the importance of involving students in the learning process when implementing distance and blended learning approaches.

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A Comparative Analysis of ChatGPT and Google in Educational Settings: Understanding the Influence of Mediators on Learning Platform Adoption

Rana Saeed Al-Maroof, Khadija Alhumaid, Afrah Alshaafi, Iman Akour, Anissa Bettayeb, Raghad Alfaisal, and Said A. Salloum

1 Introduction

The advancement of artificial intelligence (AI) has significantly altered everyday life, [1–5] with ChatGPT (Chat Generative Pre-trained Transformer) being a prime example. ChatGPT stands out for its ability to generate academic papers, short stories,

R. S. Al-Maroof
English Language and Literature, Al Buraimi University College, Al Buraimi, Oman

K. Alhumaid
Student Services, Rabdan Academy, Abu Dhabi, UAE

A. Alshaafi
The British University in Dubai, Dubai, UAE

I. Akour
Department of Information Systems, College of Computing and Informatics, University of Sharjah, Sharjah, UAE

A. Bettayeb
University of Sharjah, Sharjah, UAE

R. Alfaisal
Faculty of Art, Computing and Creative Industries, Universiti Pendidikan Sultan Idris, Tanjung Malim, Malaysia

S. A. Salloum (✉)
Health Economic and Financing Group, University of Sharjah, Sharjah, UAE
e-mail: ssalloum@sharjah.ac.ae

School of Science, Engineering, and Environment, University of Salford, Salford, UK

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and detailed explanations, prompting concern at some American universities due to its proficiency in producing high-quality content and answering academic queries [6, 7]. Incorporating natural language processing (NLP) models into educational settings promises to greatly expand access to information for educators, students, and staff. Compared to traditional platforms like Google, ChatGPT's recent rise in popularity is notable. It offers an extensive array of resources and can adeptly handle complex student queries, providing responses with a high degree of sophistication [6, 8].

ChatGPT's impact has been explored in various fields including medicine, education, and engineering, indicating its influence across different sectors. Despite its capabilities, concerns about ethical issues and creativity have been raised. This study aims to compare the data provided by Google and ChatGPT, focusing on their effectiveness as learning tools. The research employs a model utilizing two mediators: task technology fit and personal innovativeness. Task technology fit mediates between system quality and information, while personal innovativeness bridges perceived learning value and satisfaction. This approach, involving two mediators and other critical factors, is unique in educational research [6, 8–11].

Building on the previous context, the research gap in this study becomes evident when considering the unique interplay between advanced AI tools like ChatGPT and traditional information sources such as Google. While there is a growing body of research on the individual impacts of these platforms in educational contexts, there is a notable lack of comprehensive studies that directly compare their effectiveness, especially through the lens of specific mediators like task technology fit and personal innovativeness. Furthermore, most existing studies have not thoroughly explored how these platforms influence various aspects of the learning experience, such as information quality, system quality, perceived learning value, and user satisfaction in an integrated manner.

The contribution of this study, therefore, lies in its novel approach to comparing these two distinct learning sources. By introducing and analyzing the roles of task technology fit and personal innovativeness as mediators, the research provides a more nuanced understanding of how students and educators interact with and perceive these platforms. Additionally, this study extends the current knowledge on the impact of AI in education by exploring how ChatGPT, with its advanced NLP capabilities, compares to a well-established information source like Google in terms of delivering quality educational content and satisfying diverse learning needs.

This comparative analysis, enriched by the unique methodological approach, offers valuable insights for educators, policymakers, and AI developers. It aims to guide the effective integration of AI tools in educational settings while addressing potential ethical concerns and enhancing creative learning processes. This research could serve as a foundational work for future studies focusing on the intersection of AI and education, particularly in understanding how different technologies fit within the broader educational landscape and their implications for future educational practices.

2 The Previous Studies

ChatGPT's evolution represents a significant leap in AI, harnessing extensive online textual data to replicate human-like language. Its versatility spans numerous applications, including language translation, text summarization, dialogue systems, and practical feedback, establishing it as a multifaceted tool. However, despite its global adoption, some experts view ChatGPT as a potentially harmful influence on education. Concerns center around its impact on technical professionals, educators, and creativity, with apprehensions that it might diminish student innovation by offering ready-made solutions to diverse tasks [6, 8, 9].

On the other hand, Google has diversified its offerings to meet evolving user needs, introducing services like Google Maps, Google Scholar, and Google Books. Google Scholar, a free academic literature search engine, surpasses traditional databases like Scopus and Web of Science in accessibility and citation tracking. Despite its comprehensive scholarly database, Google Scholar faces challenges in policy transparency and its limited multifunctional capabilities compared to more versatile AI tools [12].

ChatGPT's impact extends to healthcare, where its potential for revolutionary change is evident. It offers extensive medical information, from theoretical insights to practical advice, distinguishing itself with features like personalized feedback and virtual simulations. In medical education, ChatGPT's abilities range from grading academic work to creating virtual tutors and generating case studies, enhancing diagnostic and treatment planning skills [13].

In the realm of educational engineering, ChatGPT is seen as a double-edged sword. While its capacity for writing and debugging software is impressive, it poses ethical dilemmas and threatens the traditional roles of software engineers. This calls for engineering educators to critically assess the implications of such technology and adapt educational strategies to harness its benefits while mitigating adverse effects.

3 Research Model and Hypotheses

The current framework emphasizes how both task technology fit and personal innovativeness mediate the relationships between system and information quality and the acceptance of learning platforms. Task technology fit serves as a mediator between the system and information quality and acceptance of learning platforms, while personal innovativeness acts as a mediator between perceived usefulness and learning value and acceptance of learning platforms. Additionally, other relationships are also explored to assess the effectiveness of each platform, highlighting the advantages of using each platform.

3.1 The System Quality, Information Quality and Task Technology Fit

The quality of a technology system is defined by specific attributes that greatly enhance its significance. These attributes, such as reliability, usability, and functionality, play a pivotal role in influencing the acceptance and adoption of the technology. Reliability refers to the user's perception of the technology's effectiveness, fostering a strong sense of trust in it. On the other hand, if the system is deemed unreliable, it may lead to user reluctance in depending on it for critical tasks or decisions.

Usability is another crucial aspect, focusing on the system's ease of use. A system that is perceived as user-friendly, straightforward, and equipped with clear instructions can significantly boost user engagement with the technology [14, 15]. Functionality, as a key component, ensures that users can understand and operate the technology efficiently. In summary, a high standard of system quality can elevate user trust in the technology. Continual monitoring and testing of the system, supplemented by user feedback and suggestions, are essential to identify areas for enhancement and to guarantee that the technology consistently satisfies user requirements.

The perception of information as accurate, relevant, and dependable is a fundamental aspect of information technology. This perception hinges on the nature of the information offered by the technology, categorizing it as either more or less significant. When users recognize the information as current and essential, they tend to view it as precise, complete, and thorough. The quality of this information plays a vital role in how users accept and utilize technology. High-quality information often leads to greater acceptance and adoption of the technology by its users. Several elements influence the quality of information provided by technology [16–18]. These include the accuracy and completeness of the data, the credibility of the sources from which it is drawn, and the timeliness with which it is updated. Additionally, the ease of use and user-friendliness of the technology itself can impact the quality of information. For instance, if a technology platform is cumbersome or challenging to navigate, users may find it difficult to locate the necessary information or may be more prone to input errors [19, 20].

Task technology fit assesses the extent to which a technology aligns with the specific needs and tasks of its users, based on the interplay between the technology's capabilities and the tasks it is intended to facilitate. This fit is crucial as it influences user attraction towards technologies that can enhance their work and provide optimal benefits. A technology that fails to align with user needs and expectations is likely to be insufficient and thus not widely accepted. The degree of compatibility between the technology and the tasks it is designed for is a key factor; the greater this compatibility, the more beneficial the technology is in facilitating these tasks.

Despite numerous studies on task technology fit, few have explored its relationship with other critical factors previously discussed. These studies typically emphasize the mediating role of perceived usefulness and ease of use on task-technology fit. This study, however, focuses on the mediating role of task technology fit between system and information quality on one side, and the acceptance of ChatGPT and Google as learning platforms on the other.

Task technology fit and information quality are pivotal in determining the efficiency and effectiveness of technology in task completion. A harmonious match between technology and task requirements, combined with the technology providing high-quality information, can boost productivity, streamline work processes, and enhance outcomes. The interplay between task technology fit and information quality is symbiotic; appropriate technology for a given task likely yields high-quality information, which in turn can improve the technology-task fit by facilitating task completion.

To achieve an optimal task technology fit and high-quality information, it is important to consider both the task requirements and the technology's features and limitations. Additionally, exploring the relationship between task technology fit and system quality is essential for enhancing technology effectiveness, focusing on unique attributes that differentiate one technology from another. Based on this, the study proposes the following hypotheses:

H1a: ChatGPT's acceptance in learning processes is more significantly influenced by system quality and information quality compared to Google's acceptance.
H2a: The impact of information quality on ChatGPT acceptance is more significant than on Google acceptance.
H3a: The influence of information quality on ChatGPT acceptance, as mediated by task technology fit, is more significant compared to Google.
H4a: The effect of system quality on ChatGPT acceptance, as mediated by task technology fit, is more pronounced compared to Google.
H5a: The relationship between system quality and ChatGPT acceptance is more substantial than with Google acceptance.
H6a: The impact of task technology fit on ChatGPT acceptance is more significant compared to Google acceptance.

3.2 The Perceived Satisfaction, Perceived Learning Value and Personal Innovativeness

Perceived satisfaction refers to how content users are with the tasks performed and services offered by a technology. If users find a technology fulfilling, they tend to continue its use and recommend it to others. In contrast, dissatisfaction may lead to discontinuation or the pursuit of alternatives. Perceived learning value, conversely, involves the advantages students identify in technology usage, such as improved access to information, time efficiency, and minimized effort. High perceived value in a technology often translates to its sustained use. For educational institutions, this perceived value is key to gaining a competitive edge. Providing technology that clearly surpasses others can attract and retain students, thereby boosting the institution's standing and success [21, 22].

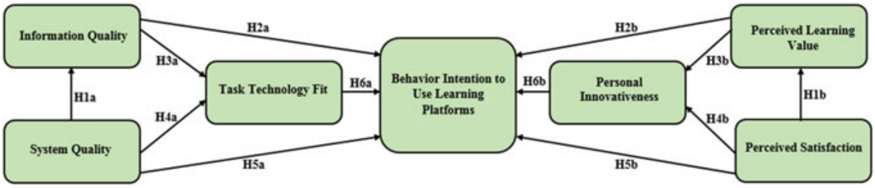

Fig. 1 Research model

Personal innovativeness plays a mediating role between perceived usefulness, perceived learning value, and the adoption of learning platforms. It reflects the user's readiness to embrace and try new technologies due to their unique features. This characteristic is pivotal in technology adoption, gauging the likelihood of users experimenting with new technologies [23].

There is a link between personal innovativeness and satisfaction. Individuals with a high degree of personal innovativeness may find greater gratification in adopting and using innovative technologies. When technology aligns with users' needs and expectations and exhibits high personal innovativeness, its acceptance is likely to increase, fostering a sense of satisfaction among users [24, 25]. Perceived learning value also intertwines with perceived innovativeness. Users who see high learning value in a technology are more inclined to explore its innovative features [25, 26]. Thus, building on these assumptions, the study proposes the following hypotheses (Fig. 1):

H1b: The relationship between perceived satisfaction and learning value influencing ChatGPT acceptance is stronger than that with Google in educational settings.

H2b: The impact of perceived value on the acceptance of ChatGPT is more significant compared to Google.

H3b: The influence of perceived value on ChatGPT acceptance, as mediated by personal innovativeness, is greater compared to Google.

H4b: The relation between perceived satisfaction influencing ChatGPT acceptance, as mediated by personal innovativeness, is stronger compared to Google.

H5b: The correlation between perceived satisfaction and the acceptance of ChatGPT is more pronounced than with Google.

H6b: The link between personal innovativeness and the acceptance of ChatGPT is stronger compared to Google.

4 Research Methodology

4.1 Data Collection

During the period from February 3 to April 10, 2023, students from various collaborating universities in the UAE participated in a study by completing online questionnaires. Out of the 150 questionnaires distributed randomly by the study committee, a response rate of 81.3% was achieved, with 122 questionnaires being fully completed and 28 discarded due to incompleteness. This number of responses closely aligns with Krejcie and Morgan's [25] recommended sample size of 152 for a population of 250, thus providing a sufficiently large sample for the study. The sample size of 122, slightly exceeding the minimum required number, offered a robust basis for the subsequent analysis [26]. To evaluate and support the formulated hypotheses, the academic team conducted a thorough review using structural equation modeling. This approach was chosen based on previous hypotheses and theoretical underpinnings in the field of artificial intelligence. For the analysis, the team employed Structural Equation Modeling (SEM) with the SmartPLS Version 3.2.7 software. This advanced statistical tool allowed for a comprehensive assessment of the measurement model. Additionally, the Final Path Model was used to execute sophisticated interventions and analyses, ensuring a thorough and nuanced understanding of the data and its implications.

4.2 Students' Personal Information/Demographic Data

Figure 2 in the study provides a detailed assessment of the demographic and personal characteristics of the participants. The gender distribution among the respondents was predominantly female, constituting 66% of the sample, with males representing 34%. Regarding age, a significant majority (84%) of the students fell within the 18–29 age range, while the rest were 29 years or older. The educational qualifications of the participants varied. Specifically, 9% held a diploma, 7% an advanced diploma, a substantial 67% possessed a bachelor's degree, 13% had a master's degree, and 3% held a doctoral degree. The study's participant recruitment strategy included a "purposive sampling approach," as recommended by Salloum and Shaalan [27], which was particularly effective when participants demonstrated a willingness to volunteer. This research involved a diverse group of participants drawn from multiple universities, encompassing a range of age groups and educational levels. For the analysis of this demographic data, the team utilized IBM SPSS Statistics version 23, a robust tool for statistical analysis, ensuring a comprehensive and accurate interpretation of the demographic information.

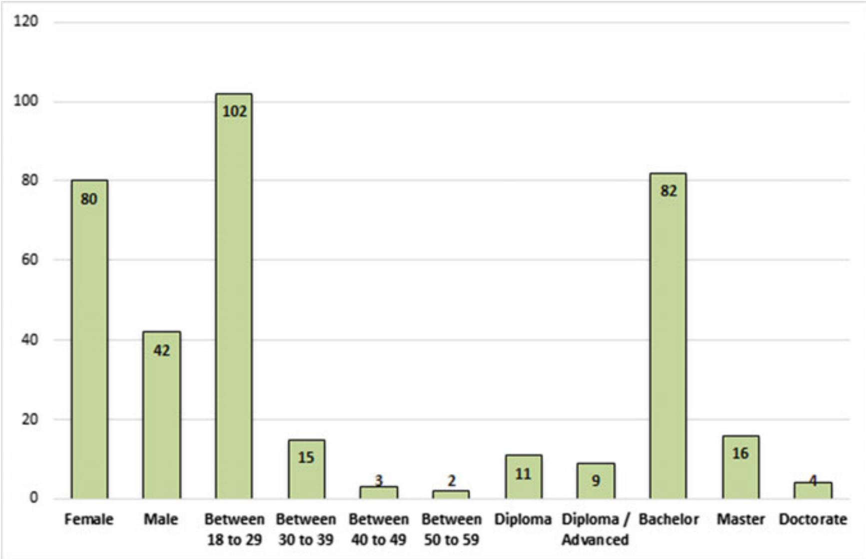

Fig. 2 Demographic information of the respondents ($n = 122$)

4.3 Study Instrument

In this study, a questionnaire was utilized as the primary tool for validating the hypothesis. The questionnaire was meticulously designed, featuring seven distinct constructs that were identified as reliable measures for the research objectives. These constructs were further expanded to include a total of 22 new items, thereby enriching the depth and scope of the questionnaire. Table 1 in the study delineates the foundation of these constructs. This table is instrumental in illustrating the rationale behind each construct and how they collectively aim to bolster the usability and effectiveness of the study. Additionally, this table serves as a repository of evidence, drawing on various existing research studies that lend support and validation to the current research framework. To ensure the relevance and accuracy of the survey, the academic team undertook a careful review of similar studies in the field. Based on insights gleaned from these previous studies, the team made appropriate adjustments to the survey questions. This process of refinement was crucial in ensuring that the questionnaire was not only comprehensive and tailored to the study's specific needs but also aligned with established research methodologies and findings in the field. The end result was a well-structured and robust tool, capable of effectively capturing the necessary data to test and validate the study's hypothesis.

Table 1 Measurement items

Constructs	Items	Instrument	Sources
Behavior intention to use learning platforms	BI1	ChatGPT presents a promising chance for experimentation	[28, 29]
	BI2	ChatGPT offers an excellent opportunity for exploration	
System quality	SQ1	The functionality of ChatGPT surpasses Google in assisting with my learning tasks	[30, 31]
	SQ2	Google's usability in accomplishing learning tasks quickly exceeds that of ChatGPT	
	SQ3	Google's efficiency surpasses that of ChatGPT	
	SQ4	The reliability of ChatGPT is greater than Google, motivating me to adopt it	
Information quality	IQ1	ChatGPT delivers more detailed information to me than Google	[30, 31]
	IQ2	Google offers more comprehensive information than ChatGPT	
	IQ3	Compared to Google, ChatGPT offers more useful information for my daily tasks	
	IQ4	Google provides the necessary information more effectively than ChatGPT	
Task technology fit	TTF1	The information from ChatGPT is more current compared to that from Google	[30]
	TTF2	ChatGPT provides information that is more suitable and detailed than what I get from Google	
	TTF3	ChatGPT delivers an abundance of information, more than necessary for my learning tasks, compared to Google	
The perceived satisfaction	PS1	ChatGPT offers tools that are more satisfactory and facilitative for learning compared to those of Google	[32]
	PS2	ChatGPT's innovative features have a greater impact on my satisfaction level compared to Google	
	PS3	Google better satisfies my learning needs for adoption compared to ChatGPT	
The perceived learning value	PLV1	ChatGPT provides more substantial benefits in my learning process compared to Google	[33, 34]
	PLV2	ChatGPT holds a unique value in various learning tasks compared to Google	
	PLV3	Compared to Google, ChatGPT lacks a distinct value that would motivate me to adopt the technology	
Personal innovativeness	PI1	ChatGPT's technology is more up-to-date and meets my needs better than Google	[23]

(continued)

Table 1 (continued)

Constructs	Items	Instrument	Sources
	PI2	ChatGPT possesses more innovative features that enhance its technological value compared to Google	
	PI3	ChatGPT offers a more unique experience that encourages me to adopt it over Google	

4.4 *Pilot Study of the Questionnaire*

In this study, to verify the reliability of the survey questions, a preliminary pilot study was conducted. This pilot phase involved randomly selecting a sample, initially comprising 20 students from the targeted demographic group. The total sample size for the main study was 200 students, with a 10% portion of this group, equating to 20 students, designated for the pilot study. The pilot study's primary objective was to test the internal consistency and reliability of the measurement items. This was achieved through the application of IBM SPSS Statistics version 23, which facilitated the calculation of Cronbach's alpha for each set of measurement items. Cronbach's alpha is a widely recognized statistical tool used to assess the internal reliability of survey instruments, particularly in social science research. A Cronbach's alpha coefficient of 0.70 is generally considered acceptable in social science research, indicating a satisfactory level of internal consistency among the survey items [35]. The alpha scores for each of the five measurement scales used in this study were meticulously calculated and are detailed in Table 2 of the study. This thorough evaluation of the pilot study data ensured that the survey questions were reliable and that the measurement items were appropriately calibrated to yield trustworthy results in the larger study.

Table 2 Cronbach's alpha scores in the pilot study (Cronbach's alpha ≥ 0.70)

Construct	Cronbach's alpha
BI	0.740
IQ	0.758
PI	0.739
PLV	0.882
PS	0.883
SQ	0.787
TTF	0.823

5 Analysis and Results

5.1 Data Analysis

In this research, the data analysis was conducted using Partial Least Squares-Structural Equation Modeling (PLS-SEM) with the SmartPLS V 3.2.7 software [3, 36–38]. The methodology followed a two-step assessment approach, encompassing both the measurement model and the structural model [39–42]. The choice of PLS-SEM as the analytical tool was influenced by several considerations detailed in the paper. One of the primary reasons for selecting PLS-SEM was its suitability for testing the proposed conceptual theory developed in the research [43–48]. This modeling approach is particularly effective in handling exploratory research data, which aligns well with the nature of the data collected based on the conceptual models of the study [49]. Moreover, PLS-SEM was employed to analyze the entire research model holistically, rather than segmenting it into discrete components. This approach ensured a comprehensive understanding of the model as an integrated entity, allowing for a more nuanced interpretation of the relationships and interactions within the model. Furthermore, PLS-SEM facilitated a simultaneous analysis of both the structural and measurement models [50]. This concurrent examination is crucial as it enables the validation of the constructs and the relationships between them, ensuring the structural integrity and coherence of the model. The importance of PLS-SEM in this study is underscored by its capability to produce accurate measurements and robust analytical outcomes. By leveraging this method, the research was able to yield reliable and valid results, enhancing the credibility and overall value of the findings [51].

5.2 Convergent Validity

The assessment of the Measurement Model in this study was structured around the principles of construct validity, including both discriminant and convergent validity [39], as well as construct reliability, which entails Cronbach's alpha (CA) and composite reliability (CR). The results, as shown in Table 3, revealed that the values for Cronbach's alpha, a measure of construct reliability, ranged between 0.770 and 0.878. These figures are notably above the commonly accepted cutoff value of 0.7 [52], suggesting a high level of internal consistency within the constructs. Additionally, the study examined composite reliability (CR) scores, which were found to range from 0.784 to 0.899. These values exceed the established threshold, further affirming the reliability of the constructs used in the research. For evaluating convergent validity, the study focused on the mean–variance extracted (AVE) and factor loadings [39]. According to Table 3, apart from a few exceptions, all factor loading values surpassed the criterion value of 0.7. This indicates that the items within each construct are highly correlated with their respective construct, supporting the validity of the measurement model. Moreover, the AVE values, which assess the amount of

Table 3 Convergent validity results which assures acceptable values (Factor loading, Cronbach's alpha, composite reliability ≥ 0.70 and AVE > 0.5)

Constructs	Items	Factor loading	Cronbach 's alpha	CR	AVE
BI	BI1	0.899	0.869	0.889	0.681
	BI2	0.835			
IQ	IQ1	0.881	0.824	0.861	0.602
	IQ2	0.783			
	IQ3	0.773			
	IQ4	0.768			
PI	PI1	0.887	0.848	0.899	0.593
	PI2	0.765			
	PI3	0.787			
PLV	PLV1	0.903	0.809	0.856	0.754
	PLV2	0.854			
	PLV3	0.883			
PS	PS1	0.838	0.866	0.876	0.631
	PS2	0.873			
	PS3	0.787			
SQ	SQ1	0.724	0.878	0.890	0.772
	SQ2	0.705			
	SQ3	0.784			
	SQ4	0.783			
TTF	TTF1	0.837	0.770	0.784	0.556
	TTF2	0.807			
	TTF3	0.899			

variance captured by a construct in relation to the variance due to measurement error, were all above the minimum threshold of 0.5. The range of AVE values, from 0.556 to 0.772, further cements the study's convergent validity. This indicates that the constructs in the study are adequately capturing the variance in the observed variables.

5.3 Discriminant Validity

In this study, to assess discriminant validity, two key methodologies were employed: the Heterotrait-Monotrait ratio (HTMT) and the Fornell-Larker criterion. These methods are critical for ensuring that the constructs measured in the study are indeed distinct from each other. According to the results detailed in Table 4, the Fornell-Larker criterion was met successfully. This criterion involves comparing the square root of the Average Variance Extracted (AVE) for each construct with the correlations between constructs. The study's findings indicated that each construct's AVE, and its square root, demonstrated stronger associations with its own measurements than with other constructs, thus satisfying the Fornell-Larker criterion [53]. Further assessment of discriminant validity was conducted using the HTMT ratio, the results of which are presented in Table 5. The HTMT criterion requires that the ratio for each construct pairing be below the threshold of 0.85 [54]. In this study, all construct pairings met this requirement, as each fell below the threshold value. This outcome confirms that the HTMT ratio criterion was fulfilled, further substantiating the discriminant validity of the measurement model. The successful validation of both the Fornell-Larker criterion and the HTMT ratio indicates that the constructs in the study are sufficiently distinct from one another, affirming the discriminant validity of the Measurement Model. Consequently, this robust validation of validity and reliability enables the reliable use of the collected data for the assessment of the Structural Model. The findings thus provide a solid foundation for further analyses and interpretations within the study.

Table 4 Fornell-Larcker scale

	BI	IQ	PI	PLV	PS	SQ	TTF
BI	**0.893**						
IQ	0.294	**0.905**					
PI	0.227	0.433	**0.948**				
PLV	0.486	0.497	0.821	**0.744**			
PS	0.389	0.589	0.619	0.495	**0.889**		
SQ	0.513	0.620	0.639	0.594	0.413	**0.820**	
TTF	0.623	0.254	0.436	0.585	0.525	0.554	**0.936**

Table 5 Heterotrait-Monotrait ratio (HTMT)

	BI	IQ	PI	PLV	PS	SQ	TTF
BI							
IQ	0.291						
PI	0.566	0.683					
PLV	0.131	0.302	0.361				
PS	0.541	0.588	0.553	0.599			
SQ	0.266	0.422	0.612	0.413	0.660		
TTF	0.159	0.282	0.291	0.381	0.343	0.299	

5.4 Hypotheses Testing Using PLS-SEM

The structural equation model employed in this study was constructed using Smart PLS, a tool that employs maximum likelihood estimation to explore the connections between different theoretical constructs within the structural model [12, 25, 46, 55–61]. Employing this methodology, we scrutinized the proposed hypotheses, and the results are presented in Table 6, indicating a moderate predictive capability of the model [47]. Specifically, the "Behavior Intention to Use Learning Platforms" variable accounted for roughly 51.3% of the variance. In Table 7, you can find comprehensive information about the beta (β) values, t-values, and p-values for all the formulated hypotheses based on the outcomes derived from the PLS-SEM technique. After analyzing the empirical data, the findings of this research lend support to hypotheses H1a, H2a, H3a, H4a, H1b, H3b, H4b, H5b, and H6b. Conversely, hypotheses H5a, H6a, H2b, and were not substantiated and were consequently rejected.

Table 6 R^2 of the endogenous latent variables

Construct	R^2	Results
BI	0.513	Moderate
IQ	0.523	Moderate
PI	0.548	Moderate
PLV	0.556	Moderate
TTF	0.623	Moderate

Table 7 Results of structural model

H	Relationship	Path	*t*-value	*p*-value	Decision
H1a	SQ -> IQ	0.629	6.236	0.000	Supported**
H2a	IQ -> BI	0.777	7.298	0.021	Supported*
H3a	IQ -> TTF	0.546	8.621	0.000	Supported**
H4a	SQ -> TTF	0.335	6.323	0.000	Supported**
H5a	SQ -> BI	0.633	1.186	0.198	Not supported
H6a	TTF -> BI	0.251	2.236	0.224	Not supported
H1b	PS -> PLV	0.405	9.812	0.001	Supported**
H2b	PLV -> BI	0.446	1.553	0.659	Not supported
H3b	PLV -> PI	0.239	13.554	0.000	Supported**
H4b	PS -> PI	0.705	4.117	0.023	Supported*
H5b	PS -> BI	0.655	5.282	0.000	Supported**
H6b	PI -> BI	0.407	11.194	0.001	Supported**

Research hypotheses significant at $p^{**} = < 0.01$, $p^{*} < 0.05$

The first hypothesis explores the association between System Quality (SQ) and Information Quality (IQ) ($\beta = 0.629$, $P < 0.001$). The result of this hypothesis indicates a significant positive impact of SQ on IQ. Therefore, H1a is affirmed. The results reveal that Task Technology Fit (TTF) is significantly influenced by both Information Quality (IQ) ($\beta = 0.546$, $P < 0.001$) and System Quality (SQ) ($\beta = 0.335$, $P < 0.001$). Consequently, hypotheses H3a and H4a receive validation, confirming that IQ and SQ exert substantial effects on TTF.

The study's findings disclose substantial relationships between Behavioral Intention to Use Learning Platforms (BI) and various factors. Specifically, BI exhibited a substantial positive influence on Information Quality (IQ) ($\beta = 0.777$, $P < 0.05$), Perceived Satisfaction (PS) ($\beta = 0.655$, $P < 0.001$), and Personal Innovativeness (PI) ($\beta = 0.407$, $P < 0.001$), thereby supporting hypotheses H2a, H5b, and H6b, respectively. Conversely, the study did not find a significant impact of System Quality (SQ) ($\beta = 0.633$, $P = 0.198$), Task Technology Fit (TTF) ($\beta = 0.251$, $P = 0.224$), and Perceived Learning Value (PLV) ($\beta = 0.446$, $P = 0.659$) on BI. Consequently, hypotheses H5a, H6a, H2b were not validated in this investigation.

Furthermore, the study's results indicate that Perceived Satisfaction (PS) is significantly affected by both The Perceived Learning Value (PLV) ($\beta = 0.405$, $P < 0.001$) and Personal Innovativeness (PI) ($\beta = 0.705$, $P < 0.05$). Therefore, hypotheses H1b and H4b find support, indicating that PLV and PI have noteworthy effects on PS. Additionally, the connection between The Perceived Learning Value (PLV) and Personal Innovativeness (PI) ($\beta = 0.239$, $P < 0.05$) was explored. The findings of this hypothesis confirm a substantial positive influence of PLV on PI. Thus, H3b is substantiated, suggesting that PLV has a meaningful impact on PI.

6 Discussion and Implication

The discussion in this study is structured into two distinct phases. In the initial phase, we concentrate on analyzing the outcomes associated with the dependent variables of our conceptual model. Subsequently, in the second phase, our attention turns to examining and discussing the moderating variables that played a role in this study.

6.1 Discussion of the Dependent Variable Results

The emergence of recent artificial intelligence tools has captured the attention of researchers, inspiring students worldwide to embrace these tools. This has led to a swift shift from traditional platforms to more advanced AI-driven platforms with innovative features. Numerous prior studies have investigated the factors influencing students' adoption of ChatGPT for enhancing their learning environments [62]. In alignment with the research objectives, this study delves into the relationships among variables such as system quality, information quality, perceived value, and perceived satisfaction. Furthermore, it explores the significant impact of these factors within the context of two moderators that positively influence the effectiveness of the learning environment. Consequently, this study aims to scrutinize the proposed hypotheses, where these moderators play pivotal roles within the conceptual model.

The discussion of the results reveals that some of the proposed hypotheses have garnered support, while others have not. Notably, system quality and information quality have furnished highly significant insights. They exhibit a positive influence on the acceptance of ChatGPT, aligning with prior research findings. Furthermore, system quality and information quality have been identified as consequential variables within ChatGPT, especially when compared to Google platforms. System quality exerts its influence indirectly through ease of use and perceived usefulness, which is consistent with studies demonstrating their significance as predictors of learners' intent to utilize various educational tools [63, 64].

The impact of perceived value on ChatGPT adoption has received a positive evaluation in this study. Users of ChatGPT perceive it as a valuable tool, owing to its effective and comprehensive results. These findings resonate with previous research indicating that perceived value mediates the adoption of other technologies, such as IoT [65]. Similarly, perceived satisfaction finds support in the results, in line with prior studies. The notion that perceived satisfaction can significantly influence users' perspectives, particularly in educational contexts, is widely acknowledged among researchers [66, 67].

6.2 Discussion of the Moderation Results

The moderating impact of ChatGPT users holds significant influence as it elucidates the real connections among the proposed dependent variables, namely, system quality, information quality, perceived value, and perceived satisfaction. Previous studies that have explored moderating effects often reveal a substantial interaction effect between perceived satisfaction and other variables concerning technology utilization. These studies have unveiled an indirect influence of moderators, such as task technology fit, on technology adoption. Notably, task technology fit, integrated into the model as a moderator, explains a substantial portion of the variance in the dependent variable. By introducing task technology fit as a moderator, we enhance the assessment of ChatGPT's relevance and appropriateness in accordance with students' recent usage patterns. In essence, the inclusion of task technology fit as a moderator bolsters the adoption of ChatGPT. These results align with prior research, which underscores the increasing impact of technology fit on technology adoption. For instance, a study by Jeyaraj [68] demonstrated that technology fit can positively influence technology acceptance, especially when the technology enjoys government support. Task technology fit exerts a more pronounced influence on students compared to non-students, as it primarily focuses on perceptual variables rather than behavioral ones. This suggests that task technology fit can directly gauge perceptions and can be tailored to specific technologies and user contexts. A similar study by Roth et al. [69] concluded that task-technology fit analysis contributes to a deeper understanding of blockchain adoption in the public sector, offering valuable insights into federally structured, cross-organizational contexts.

Likewise, personal innovativeness has been affirmed as a significant moderator that validates the relationship between system quality, system information, and the intention to use technology. This relationship is favorably assessed by users, showcasing a strong effect of innovativeness as a moderator in enhancing the adoption of ChatGPT. Consequently, the findings corroborate the proposed hypotheses and align with previous research results. Previous studies have illustrated that personal innovativeness can effectively serve as a moderator to gauge technology acceptance's efficacy [70]. Another study has highlighted that innovativeness exerts an indirect influence on technology adoption, potentially reinforcing future technology usage [71].

6.3 Theoretical and Practical Implications

Theoretical implications of this study are pertinent to universities and institutions offering similar services. They underscore the need for these applications to prioritize their utility, which in turn can significantly impact the success of the educational process. This represents a significant advancement in the realm of educational contributions across various sectors. To ensure ChatGPT's success in educational settings, it is crucial to emphasize the efficacy of such applications when framing the constructs incorporated in our conceptual framework. The provided conceptual model enhances the existing literature on ChatGPT adoption by placing a spotlight on moderators, specifically task technology fit and innovativeness. This study contributes theoretically by emphasizing the importance of users' perspectives in comprehending how ChatGPT contributes meaningfully to improving users' behaviors and social perceptions in educational contexts. To enhance the quality of system, perceived value, and perceived satisfaction, we have introduced these constructs in our theoretical framework to comprehensively assess this model's effectiveness. Another theoretical strength of our conceptual model lies in bridging the gap between the effectiveness of perceived value and perceived satisfaction, along with personal innovativeness as a moderator on one side, and system quality and information quality with the moderator task technology fit on the other. From a practical perspective, the results also lend strong support for the development of this conceptual model, especially at the governmental level. The study's findings reinforce the significance of ChatGPT in educational public universities, and these findings can have broader applicability to other government institutions. In conclusion, the ChatGPT conceptual model serves as a benchmark for qualitative research approaches, offering a more precise understanding of the applications' significance in other fields within the public sector.

6.4 Managerial Implications

The present study offers valuable insights into the effective implementation of Chat GPT within educational institutions, enhancing the teaching and learning processes. Educational administrators in the Gulf region can leverage these findings to explore various avenues for application. These insights can be summarized as follows:

1. **Alignment with Student Needs**: Educational institutions should ensure that ChatGPT applications cater to students' needs and enhance the relative advantages of the educational process. This can be achieved by integrating ChatGPT into the curriculum as a source-provider, offering diverse information in various formats to facilitate teaching and learning.
2. **Government Policies**: Governments can consider utilizing ChatGPT to formulate collaborative and regulatory policies. These policies can promote the application as a readily available source of information for all government institutions. They should also expedite the adoption of ChatGPT by incorporating other similar applications and open-access features. This approach can enhance accessibility and usability for a wider audience.
3. **User-Centric Policies**: This research study can serve as a catalyst for ChatGPT policymakers to develop user-centric policies. These policies should safeguard user rights and ensure that users perceive value in using ChatGPT, thus fostering their intention to continue using the application.

6.5 Conclusion

The study's findings affirm that ChatGPT offers greater appeal and advantages compared to alternative applications. It transcends being a mere technical tool, instead emerging as an integrated platform that synthesizes various facets, focusing on diverse subject matter, practical theories, and technological aspects. In contrast to Google platforms, ChatGPT has garnered recognition as a highly effective learning tool, meticulously designed with a significant degree of sufficiency. Consequently, the investigation of ChatGPT's impact becomes imperative. Thus, this study underscores that the application exerts a substantial influence on user acceptance, bridging the realms of information and system quality on one front and perceived learning value and perceived satisfaction on the other. However, it is noteworthy that certain facets have not received corroborating support, and they may not necessarily wield a significant predictive effect on ChatGPT usage. This discovery implies that additional development efforts are warranted to yield more favorable outcomes, thereby elevating the application's overall contributions. Nonetheless, this study possesses certain limitations. The data collection is confined to a sample of studies conducted within the Gulf region, primarily involving students from various universities who employ the application for diverse educational objectives. Future investigations may choose to diversify by including samples from other government institutions that utilize this application to enhance their future usage. Moreover, the conceptual framework is delimited to specific aspects associated with system quality, satisfaction, value, and innovativeness, augmented by moderators tailored to gauge ChatGPT's effects. Future studies could encompass a broader array of variables, along with different moderators, to comprehensively assess various critical factors.

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ChatGPT and the Transformation of Business Education

Prediction of Retailer's Intention to Use Chat-GPT in Educating Retailers: A Case Study in the UAE

Fanar Shwedeh, Said S. Salloum, Ahmed Aburayya, Brihan Fatin, Mohamed Ahmed Elbadawi, Zainab Al Ghurabli, Abdulrahman Murad, Alaa Abueleyan, and Batool Ismail

1 Introduction

AI is a fast-growing area of computer science that centers around creating smart machines capable of undertaking tasks that usually necessitate human intelligence [1–4], like learning, problem-solving [5–9], and decision-making [1, 10–18]. AI has already made significant contributions to many areas of our lives, including business [19–23]. In the world of business, AI has emerged as a powerful tool for improving efficiency, [24–28] reducing costs, and driving innovation [29]. By analyzing vast amounts of data, AI has the ability to automate monotonous and repetitive tasks, thereby allowing employees to dedicate their time to more intricate and strategic assignments. Additionally, it can analyze vast volumes of data, detecting patterns and extracting insights that may elude human observation, helping businesses make more informed decisions, and improving customer engagement and loyalty [30]. Moreover, it monitors equipment and systems in real-time, predicting and preventing failures before they occur, and finally analyzes large amounts of data to identify patterns and anomalies that may indicate fraudulent activity, helping businesses to prevent financial losses [31]. In the coming years, AI is expected to play an even larger role

F. Shwedeh · A. Aburayya (✉)
College of Business, MBA Department, City University Ajman, Ajman, UAE
e-mail: amaburayya@gmail.com

S. S. Salloum
School of Science, Engineering, and Environment, University of Salford, Salford, UK

B. Fatin · M. A. Elbadawi · Z. Al Ghurabli
College of Business, HRM Department, City University Ajman, Ajman, UAE

A. Murad · A. Abueleyan · B. Ismail
College of Business, Marketing Department, City University Ajman, Ajman, UAE

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in business, with many experts predicting that it will transform entire industries and revolutionize the way we work.

In the field of marketing, AI is employed in diverse ways. For example, it has the capability to examine customer information and generate personalized content, product suggestions, and marketing communications that cater to the unique interests and preferences of each individual [32–36]. Furthermore, it can make predictions about future customer behavior and market trends, helping businesses to make more informed decisions about their marketing strategies [37–40]. Moreover, AI-powered chatbots which can provide 24/7 customer service and support, answering customer inquiries and resolving issues in real-time. As well, it can recognize and analyze images and voice commands, enabling marketers to create more interactive and engaging experiences for customers [4, 41, 42]. Finally, AI can optimize various aspects of marketing campaigns, such as ad targeting, bidding strategies, and email subject lines, to improve conversion rates and ROI.

In the fiercely competitive retail sector, businesses are continuously striving to outperform their rivals. In recent times, an emerging technology that retailers are increasingly embracing is artificial intelligence. By harnessing the potential of AI, retailers can revolutionize their management practices, leveraging robust tools to enhance operational efficiency, elevate customer engagement, and drive revenue growth and profitability [43]. AI can be used to analyze vast amounts of data, such as consumer behavior, trends, and preferences, to inform and train retailers' decision-making. By using predictive analytics and machine learning algorithms, retailers can accurately forecast demand, optimize inventory levels, and reduce the risk of stockouts and overstocking. This, in turn, leads to increased customer satisfaction and sales [44–50].

Another valuable application of AI is its ability to personalize the shopping experience for customers. By analyzing their browsing and purchase history, AI can provide tailored product recommendations and promotions. This personalized approach enhances customer engagement and loyalty, ultimately resulting in higher revenue generation [1, 45, 51–53]. Furthermore, AI can help retailers detect and prevent fraudulent activities, reducing the risk of financial losses, driven by artificial intelligence detection systems for fraud can spot fraudulent actions like stolen credit cards and phony orders by evaluating data from transactions in real-time. Overall, AI has the potential to revolutionize retail management by providing retailers with powerful tools to learn how optimize their operations, improve customer engagement, and increase revenue and profitability. As technology continues to evolve, we can expect to see even more innovative AI applications in the retail industry in the future [54].

As an AI language model, ChatGPT offers a variety of services that can help businesses improve their marketing efforts. For example, this innovation can assist with content creation, market research, personalization, optimization, and customer service. By leveraging the latest advancements in natural language processing (NLP), Chat OpenAI can analyze vast amounts of data and generate insights that can help businesses attract and engage with their target audience more effectively [55]. Whether you are looking to improve your social media presence, launch a new product, or better understand your customers, ChatGPT is always there to assist you. With its advanced capabilities and powerful algorithms, it can provide you with the tools you need to succeed in today's competitive marketplace.

ChatGPT, an AI-based chatbot, can efficiently answer commonly asked questions, thereby saving time and reducing the workload of customer support staff. It can also provide proactive suggestions and aid customers in finding solutions quickly, while its natural language processing abilities enable it to facilitate smooth conversations. As a result, customers can quickly access support agents without enduring long wait times or spending time on the phone with customer service reps [56]. While chat, GPT, as a whole can greatly help with marketing research, financing, and many other business sectors, it can greatly shine in the retail management industry. Chat GPTs power to generate human-like text based on any prompt you give it far exceeds anything we have today, but the real power of this language is still years ahead of what GPT is capable of now. The most recent development in the retail industry is referred to as "Retail 4.0". To match client expectations, it makes use of cutting-edge technology such as artificial intelligence (AI), the Internet of Things (IoT), cloud computing, big data analytics (BDA), and augmented reality (AR). Technology, human creativity, and effort have all worked together to significantly improve manufacturing and data analytics skills throughout this revolutionary period. The main focus of this innovation is the employment of AI techniques and digital manufacturing systems, which enable systems to become more intelligent and effective [57, 58]. Chat GPT's benefit to the retail industry would arguably have to be that it can efficiently give out accurate data analysis and forecasting by greatly cutting the time it takes for these processes to be made and do them more effectively, Chat GPT can also greatly improve performance of the management in the retail industry as a whole, However, I do not think chat GPT is mature enough to be accepted and to be used in the retail industry as of yet, but as it advances and grows, this, technology will be much more important to all business sectors across the globe.

The paper by George and George [56] contributes significantly to the advancement of conversational AI technology, particularly in the development of generative pre-training models like Chat-GPT. It offers valuable insights for researchers and professionals interested in exploring and harnessing the potential of this cutting-edge innovation in the field of artificial intelligence [56]. Chat GPT has skyrocketed in popularity over the past year, which impels us to better understand this language and prepare for the future of this technology. Therefore, this research will gauge the retailer's perception of chat GPT in the UAE. The research will be a great starting point to gain more insight into Chat GPT in the retail industry and all business sectors as a whole, as there is little to no information available about how Chat GPT will work in any business. However, as this is new territory for all businesses, there are some limitations faced while conducting this research, we believe that Chat GPT has the potential to grow even more so there is no way to tell how Chat-GPT will look in the upcoming years, and how efficient it becomes. The literature review indicates that there is a lack of information when it comes to Chat GPT in the retail industry. Therefore, this study aims to investigate the acceptance of using the Chat GPT Model for retailers' education purposes. This study will help in analyzing and testing the prediction of retailer's intention to use Chat-GPT in their education. This research will close the gap mentioned above.

2 Literature Review

Retail management involves the process of overseeing and coordinating the operations of a retail business to ensure its success. It encompasses a wide range of activities, including inventory management, customer service, marketing, sales, staffing, and financial management. Effective retail management requires the development of strategies to maximize sales and profits while also maintaining customer satisfaction and loyalty. This involves setting sales targets, managing inventory levels, developing marketing campaigns, and training employees to provide high-quality customer service. Retail managers must also be skilled in financial management, including budgeting, forecasting, and tracking key performance indicators (KPIs) such as sales, profit margins, and inventory turnover. They must be able to analyze sales data and make decisions about pricing, promotions, and product selection to improve profitability. Furthermore, retail managers must stay up-to-date and be educated frequently with industry trends, technological advancements, and consumer behavior to ensure their business stays competitive and relevant. Successful retail management requires an education that contains a combination of strategic planning, operational expertise, and strong leadership skills [59, 60].

Multiple studies and books have emphasized the crucial role of AI education in the retail industry to enhance performance and customer satisfaction. A report by Capgemini Research Institute (2020) highlights how AI can help retailers improve operations, enhance the customer experience, and increase revenue. Similarly, IBM Institute for Business Value (2020) found that AI can benefit retailers in areas such as marketing, supply chain, and customer service. PwC's report "Retail 2020: Reinventing Retailing" (2017) discusses the potential for AI technologies such as machine learning and natural language processing to deliver personalized customer experiences. Other sources, such as "Artificial Intelligence for Fashion" and "Retail Analytics: The Secret Weapon" [61–63], delve into how AI and analytics can help retailers gain insights into consumer behavior, optimize pricing and promotions, and improve the overall customer experience. Finally, the Harvard Business Review article "The New Science of Customer Emotions" discusses how AI can help retailers better understand and respond to customers' emotional needs, leading to greater customer satisfaction and loyalty.

Retailers have experienced a huge change as a result of being educated in the development of ChatGPT, a sizable language model trained on a vast corpus of text data. To enhance customer experience, retailers can use ChatGPT by connecting it with their customer service channels, such as chatbots or virtual assistants [64]. It can offer suggestions for products, respond to frequently asked queries, and even help customers make purchases. Additionally, it enables businesses to gather and analyze customer information that can be used to tailor offers and recommendations to individual customers in accordance with their shopping habits and interests.

Retailers may save time and money by automating common processes like order tracking, returns, and refunds. Additionally, it can be used to create product descriptions and marketing material that will interest and draw in customers. It can be used to create marketing text and product descriptions that will interest and draw in potential buyers. By incorporating technology into their business operations, retailers can be educated and get a competitive edge over rivals by improving customer experience, personalization, and efficiency.

TAM has been implemented widely in previous studies to predict the adoption, acceptance, and intention to use technology in different fields. To put it more specifically, this study has focused on two constructs of TAM that are supposed to have a direct relation to the adoption of Chat GPT". Three key variables are considered in this study. The first variable, perceived usefulness (PU), pertains to users' perspectives on the extent to which a technology is beneficial. The second variable, user satisfaction (US), refers to the level of comfort and acceptability users experience when interacting with a computer application and consuming its content. The third variable measures the perceived ease of use required by users to utilize the technology (PEOU). The previous three variables are considered as independent variables. This study examines if the adoption of Chat GPT in educating retailers (AER) is affected by the three independent variables. Based on these assumptions, the study proposes the following hypotheses (Fig. 1):

Hypothesis 1 (H1): The adoption of Chat GPT in educating retailers (AER) is positively impacted by the degree of perceived usefulness (PU).
Hypothesis 2 (H2): The adoption of Chat GPT in educating retailers (AER) is positively impacted by user satisfaction (US).
Hypothesis 3 (H3): The adoption of Chat GPT in educating retailers (AER) is positively impacted by the perceived ease of use required by users to utilize the technology (PEOU).

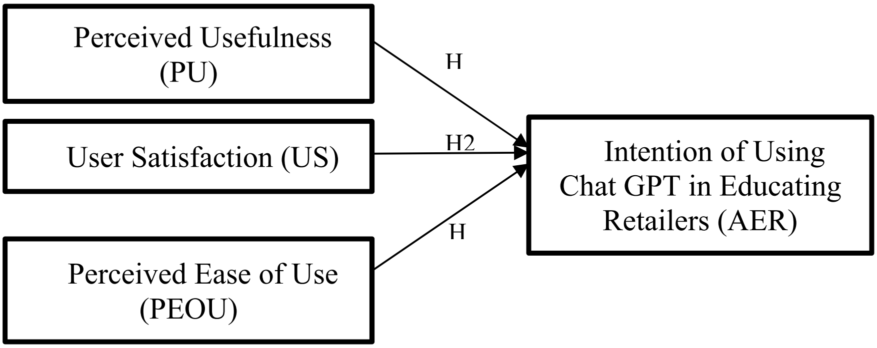

Fig. 1 The model of the study

3 Research Methodology

A cross-sectional design was used in this descriptive-analytical study as a deductive strategy [14, 65–67]. The data collection instrument used in the study was an online questionnaire that used a self-administrated strategy to collect data from various retail managers in the United Arab Emirates (UAE) [68]. "This examination study was led at large retail stores across UAE whereby seven large Retail stores and 160 retail staff participated in the study. The Retail managers were provided with the questionnaire link by making use of official emails and social media platforms like WhatsApp. For data collection, various retail workers were a part of our respondents including the administration staff (registration, quality, receptionists, and administrative supports) and other staff (cashier, storekeepers, customer [69]. Moreover, in the context of an empirical study pertaining to retail management, various researchers opted for the target population to serve as a unit of analysis for the current research, those individuals had a satisfactory level of knowledge regarding various organizational practices in the context of retail management as well as about the degree of service quality and customer satisfaction offered by their respective organizations".

4 Data Collection

This study involved distributing 200 questionnaires randomly, but 40 of them were not used due to missing information. The remaining 160 questionnaires were correctly completed and used to analyze the data, resulting in an 80% response rate. The research model was employed to validate the hypotheses, which were formulated by adapting existing theories to the context of ChatGPT adoption for educating Retailers. The measurement model was assessed using multiple regression analysis, and the final path model was constructed using the findings from the analysis. In order to evaluate the hypotheses, an investigation utilized a survey instrument that was integrated into the study. The survey encompassed 20 questions specifically designed to assess the four constructs outlined in the questionnaire. To ensure the pertinence and importance of the research, questions from prior studies [70] were thoroughly reviewed, revised, and updated before being included in the survey. The Gender statistics show a frequency of 160 participants where 60 of them are males and 60 are females, it also shows a percentage of 50% for both genders which means that the participants are equal in both genders. The age statistics show that participants are between the ages of 18–24 and 35–50 hold the highest frequency and percentage against the other ages. The marital status statistics demonstrate how 80 of our participants are married at a rate of 50% and the rest are either single, widowed, or divorced. The Education statistics show that 112 of participants are postgraduates and 48 of them undergraduates with a rate of 70 to 30%. The experience statistics show a frequency of 64 participants that have 2–10 years of experience at a rate of 40%, then at the second highest frequency of 48 participants and 30% rate participants are less than 2 years of experience. The rest of the participants have more than 10–15 years of experience (Table 1).

The reliability of the three independent variables (PU, US, and PEOU) is tested by Cronbach Alpha. Table 2 reflects that all independent variables are reliable because the Cronbach Alpha is above 0.6 for all of them. The component Ease of Use is the most reliable with a rate of 80.02%.

Extraction Method is used to conduct the principal component analysis. The factor loading of all the items is above 0.5 which means PU is valid and the most valid item

Table 1 Presents demographic data collection of the participants of this study

Demographic data collection of the participants					
		Frequency	Percent	Valid percent	Cumulative percent
Gender					
Valid	Male	80	50.0	50.0	50.0
	Female	80	50.0	50.0	100.0
	Total	160	100.0	100	
		Frequency	Percent	Valid percent	Cumulative percent
Age					
Valid	18–24	48	30.0	30.0	30.0
	25–34	32	20.0	20.0	50.0
	35–50	48	30.0	30.0	80.0
	Above 51	32	20.0	20.0	100.0
	Total	160	100.0	100.0	
		Frequency	Percent	Valid percent	Cumulative percent
Marital status					
Valid	Single	48	30.0	30.0	30.0
	Married	80	50.0	50.0	80.0
	Widowed	16	10.0	10.0	90.0
	Divorced	16	10.0	10.0	100.0
	Total	160	100.0	100.0	
		Frequency	Percent	Valid percent	Cumulative percent
Education					
Valid	Under-graduate	48	30.0	30.0	30.0
	Post-graduate	112	70.0	70.0	100.0
	Total	160	100.0	100.0	
		Frequency	Percent	Valid percent	Cumulative percent
Experience					
Valid	Under-graduate	48	30.0	30.0	30.0
	Post-graduate	112	70.0	70.0	100.0
	Total	160	100.0	100.0	

Table 2 Presents the reliability levels of variables of the study

Reliability statistics	
Cronbach's alpha	N of items
0.687	4
0.664	3
0.802	5

is the second because it holds the highest rate at 60.07%. The factor loading of all the items is above 0.5 which indicates that the US is valid and the first item is the most valid holding a factor loading of 66.6%. The factor loading of all the items is above 0.5 which demonstrates that the EU is valid, and the last item is the most valid holding a factor loading of 86.1%. Table 3 reflects the Principal Component Analysis of components Communalities.

The model is fit because our R square is 39.8% which is above 20%. The regression model fits for analyzing the data as the f-test is high and the p-value is less than 0.05. Analysis shows that the independent variables (PU, US, and PEOU) significantly impact the dependent variable as the p-value is less than 0.05. Table 4 presents the Model Summary, ANOVA and Coefficients results.

By looking into the results of each hypothesis, the findings show that there is a relationship between the variables of this study. In (H1), the findings reflect that the degree of perceived usefulness (PU) has a positive impact on the adoption of Chat GPT in educating retailers (AER). The investigation of (H2) shows that user satisfaction (US) has a positive impact on the adoption of Chat GPT in educating retailers (AER). The same result is found for (H3) where the perceived ease of use

Table 3 Presents the extraction method: principal component analysis

Extraction method: principal component analysis	
	Extraction
Using ChatGPT in my job	0.548
Using ChatGPT improves my performance	0.607
Using ChatGPT increases my productivity	0.548
Using ChatGPT makes my job easier	0.547
ChatGPT has sufficient medical information	0.666
ChatGPT has provided satisfactory information	0.524
ChatGPT is able to provide me information I need	0.622
ChatGPT is easy to use among retailers	0.780
ChatGPT improves my desire to get new information	0.827
ChatGPT can replace other technologies	0.850
ChatGPT is needed for less mental efforts	0.859
ChatGPT helps in developing my technical abilities	0.861

Table 4 Presents the analysis results

Model summary						
Model	R		R square	Adjusted R square	Std. Error of the estimate	
Model						
1	Male		80	50.0	50.0	
	Female		80	50.0	50.0	
	Total		160	100.0	100	
ANOVA						
		Sum of squares	df	Mean square	F	Sig.
Model						
1	Regression	15.293	3	5.098	34.414	0.000[b]
	Residual	23.107	156	0.148		
	Total	38.400	159			
Coefficient						
		Unstandardized B	Coefficient Std. Error	Standardized coefficient beta	t	Sig.
Model						
1	(Constant)	−4.410	0.937		−4.708	0.000
	PU	0.473	0.071	0.523	6.670	0.000
	US	0.433	0.088	0.348	4.908	0.000
	PE	1.262	0.124	0.868	10.143	0.000

required by users to utilize the technology (PEOU) has a positive impact on the adoption of Chat GPT in educating retailers (AER).

5 Discussion

Chat GPT has the potential to enhance firms' marketing initiatives. The study examines the application of AI in using Chat GPT for educating retailers, including Chabot customer support, personalized content generation, forecasts of future consumer behavior and market trends, and campaign optimization among others. The report highlights how AI has the ability to transform retail management by giving companies the education to use the resources they need and operate more effectively, engage customers better, and increase revenue and profits. The study concludes by highlighting Chat GPT's contribution to educating retailers in time savings by being able to answer frequently raised issues and lessening the workload of customer support personnel. According to the study's findings, there is a strong relationship between merchants' intentions to utilize Chat-GPT for educating and managing their digital marketing efforts. The study shows that retailers believe in the value of utilizing Chat

GPT in educating their abilities to work by using user-friendly technology. The findings also demonstrate a positive relationship between Chat-GPT's perceived utility and its capacity to raise customer satisfaction, boost revenues, and boost operational effectiveness. The study discovered that attitudes toward technology, perceptions of social influence, and the perceived suitability of the tool for their purposes as a business are all factors that influence merchants' intentions to adopt Chat-GPT in educating retailers. The study's findings have several ramifications for UAE retailers. First, by boosting their attitudes toward technology and their perceived social impact, retailers can increase their intention to adopt Chat-GPT to educate and improve better retailing services. Second, by modifying the tool to meet the retailer's unique demands, retailers may make sure Chat-GPT is suitable for their business needs. Third, by exploiting Chat-GPT's capacity to educate retailers how to raise sales, increase customer satisfaction, and improve operational effectiveness, retailers can optimize the perceived value of the system.

The study highlights how important it is to consider various factors that influence retailer's propensity to use Chat GPT as a teaching tool. With a clear understanding of these factors, retailers can use Chat GPT more successfully to educate themselves and achieve their business objectives. The study's findings indicate that three factors, user satisfaction, perceived ease of use, and perceived usefulness have a positive influence on the adoption of Chat GPT in training retailers (AER). This study can open the eyes of researchers in testing the adoption of Chat GPT in educating people from other fields. It can also show if the adoption of Chat GPT can have negative results in other fields. In the scope of retailers in the UAE, it seems that the adoption of Chat GPT is positively impacted by the tested variables, however the limitation of this study is that it is only examined in the UAE. The results might show different outcomes in other countries.

The theoretical contribution of this study is that it is using the educational concept into the TAM model to predict the adoption, acceptance, and intention to use technology in different fields. This study involves how the adoption Chat GPT technology can be used for retailers' educational purposes rather than only improving business activities. The practical contribution is that retailers in the UAE are having the intention in using Chat GPT, this is an indicator to extend the exploration of this result in other scopes such as manufacturers, teachers, farmers, physicians, athletes, or musicians. The significance of this study is the idea of using Chat GPT in educating retailers, which is a totally new idea. Retailers can use Chat GPT in improving business performance but utilizing it in educational purposes makes this study significant with its contribution and it helps in enriching the field of retailers' education.

6 Conclusion

The study emphasizes the value of considering three variables that affect merchants' inclination to adopt Chat-GPT for educating retailers. Retailers can adopt ChatGPT with more success in attaining their business goals by having a clear awareness of these variables. The result of this study shows that the degree of perceived usefulness, user satisfaction and the perceived ease of use are three variables that have a positive impact in the adoption of ChatGPT in educating retailers (AER) in the UAE.

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Transforming Teacher-Student Interactions in the Metaverse: The Role of ChatGPT as a Mediator and Facilitator

Raghad Alfaisal, Said A. Salloum, and Ayham Salloum

1 Introduction

The advent of the metaverse, a collective virtual shared space created by the convergence of virtually enhanced physical reality and physically persistent virtual spaces, marks a significant evolution in the realm of digital interaction [1, 2]. In the context of education, the metaverse presents an innovative platform where immersive, interactive, and personalized learning can occur [2–4]. Educational institutions are increasingly exploring its potential to transform traditional learning environments into dynamic, interactive spaces that transcend physical boundaries [4–9]. The metaverse's capability to integrate various technologies, such as virtual reality (VR), augmented reality (AR), and artificial intelligence (AI), offers unprecedented opportunities for enhancing educational experiences [10].

Among the AI technologies making a significant impact in this space is ChatGPT, an advanced language model developed by OpenAI. ChatGPT, with its natural language processing capabilities, is uniquely positioned to act as a mediator and facilitator in educational settings within the metaverse [11]. It can provide real-time, interactive, and personalized educational assistance, enabling a more engaging and effective learning experience [12]. ChatGPT's ability to understand and generate human-like text allows it to interact with students in a conversational manner,

R. Alfaisal
Faculty of Computing and Meta-Technology, Universiti Pendidikan Sultan Idris, Tanjung Malim, Malaysia

S. A. Salloum (✉)
Health Economic and Financing Group, University of Sharjah, Sharjah, UAE
e-mail: ssalloum@sharjah.ac.ae

School of Science, Engineering, and Environment, University of Salford, Salford, UK

A. Salloum
College of Medicine, University of Sharjah, Sharjah, UAE

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providing explanations, answering queries, and even facilitating discussions [13]. This capability has the potential to transform the traditional teacher-student interaction model, making it more responsive and adaptive to individual learning needs [2, 9, 10, 14, 15].

The purpose of this paper is to explore the transformative impact of ChatGPT on teacher-student dynamics within the metaverse. It aims to examine how ChatGPT, as an AI-driven tool, alters the nature of educational interactions, focusing on communication, engagement, and personalization aspects. This paper will review existing literature, analyze case studies, and discuss the implications of these changes for the future of education. By delving into this investigation, the paper seeks to provide insights into how AI like ChatGPT can be optimally utilized in the evolving landscape of metaverse-based education.

2 ChatGPT's Integration in the Metaverse

2.1 Technological Overview

The integration of Generative Pre-trained Transformer (GPT) models, like ChatGPT, in the metaverse represents a convergence of cutting-edge AI with immersive virtual environments. The metaverse, a shared virtual world accessible via the internet, is undergoing rapid development, becoming a tangible aspect of our digital experience [16]. ChatGPT, as a state-of-the-art AI-driven language model, accelerates human-like text generation, making it a versatile tool for various applications within the metaverse.

One notable application is the creation of AI-powered virtual assistants, or chatbots, to assist users in navigating the metaverse. These entities can facilitate tasks like locating specific places or connecting with other users. Additionally, ChatGPT can generate interactive experiences in virtual reality, tailoring responses to user input, thereby enhancing virtual events, games, or movies.

In terms of accessibility, GPT models can generate text in multiple languages, breaking language barriers and making the metaverse more accessible to a global audience. This capability extends to generating content tailored to users' interests and priorities, enabling a personalized metaverse experience. The overall combination of GPT and the metaverse paves the way for more immersive and realistic digital experiences (See Fig. 1), potentially revolutionizing various domains, including education, marketing, and commerce.

Fig. 1 Combination of GPT and the metaverse

2.2 *Use Cases in Education*

In the educational sphere, ChatGPT can significantly enhance teaching and learning. For instance, instructors can leverage ChatGPT to develop students' digital literacy and critical thinking skills [17]. This includes designing assignments that require students to analyze the strengths and limitations of the tool, thus educating them about ethical considerations and the risks of plagiarism [7, 18–20]. ChatGPT can also be integrated into active learning strategies. An example is the adaptation of the "Think-Pair-Share" strategy to "Think-Pair-ChatGPT-Share," where ChatGPT enhances peer discussions and the drafting process of writing assignments[6]. Additionally, there are strategies to discourage over-reliance on ChatGPT, such as designing writing prompts that reference unique class material or require more personalized writing [18]. Specifically, in the higher education classroom, ChatGPT can be employed to:

- Highlight relevant course content and explain complex concepts in simpler terms [18].
- Summarize and outline longer texts, create formative assessments, and brainstorm assignment designs [18].
- Generate examples for assignments, scan existing research, and assist in editing group projects[6].
- Accommodate learners with visual impairments through speech-to-text or text-to-speech technologies, and provide contextual assistance for autistic learners [18].

Furthermore, ChatGPT assists instructors in various tasks, including generating quiz questions, drafting lecture outlines, writing syllabi, providing assignment feedback, and creating visual aids[6]. These applications underscore ChatGPT's potential as a transformative tool in metaverse-based education, facilitating a more interactive, personalized, and inclusive learning environment.

3 Impact on Teacher-Student Dynamics

ChatGPT's integration into educational settings influences communication methods and styles between teachers and students. ChatGPT, a language-based AI technology, has been trained on extensive text data, enabling it to generate diverse text types and code. This capability allows ChatGPT to answer questions and complete text prompts, making it a useful tool for education. However, it is crucial to remember that while ChatGPT can provide answers, its accuracy is not infallible. Therefore, critical thinking and human interaction remain essential in education, ensuring that information provided by ChatGPT is fact-checked and contextualized appropriately [21].

ChatGPT enhances student engagement and interaction in several ways. It can be used to simulate real-world scenarios, helping students problem-solve and apply strategies in their learning context. For example, in a project or inquiry setting, students can use ChatGPT to brainstorm and address problems that may arise, thereby fostering a more dynamic learning environment. This use of ChatGPT can also strengthen critical thinking skills, as students discern inaccuracies in ChatGPT's responses, learning to treat it as a tool rather than a replacement for human judgment [22].

Moreover, ChatGPT can provide an expert experience for students in project-based learning (PBL). By simulating mentors such as historians, scientists, or authors, ChatGPT can answer students' questions, offering them insights on various topics and enhancing their understanding (See Fig. 2) This approach positions ChatGPT as a supportive tool in student-led inquiries, enabling students to make informed decisions and explore AI's capabilities in education [22].

ChatGPT plays a significant role in personalizing student learning experiences. As a conversational AI model, ChatGPT caters to each student's unique needs, offering personalized resources, feedback, and support. Instructors can utilize ChatGPT to gather data on students' learning styles and tailor the learning experience accordingly. This individualized approach creates a more engaging and effective learning environment, accommodating students' strengths and weaknesses [22].

Instructors can also set up ChatGPT to provide personalized feedback on assignments and assessments. By analyzing students' writing styles, ChatGPT can offer constructive feedback, helping students improve their skills based on their individual learning needs[5]. Additionally, ChatGPT can serve as a support tool outside of office hours, providing students with 24/7 access to educational resources and assistance. This availability allows students to receive personalized feedback on their assignments and gain a better understanding of complex concepts, thereby contributing to a more inclusive and supportive educational environment [22].

Fig. 2 The metaverse in schools

4 Challenges and Opportunities in Integrating ChatGPT into Education

4.1 Technological Limitations

Integrating AI like ChatGPT into educational settings is not without its technological challenges. One primary concern is the hardware and network requirements necessary for the effective use of such advanced AI systems. Access to reliable and high-speed internet connections, along with adequate computing hardware, is essential for ChatGPT to function optimally. In areas with limited technological infrastructure, this poses a significant barrier to the equitable use of AI in education [23–27]. Moreover, the processing power required to run such advanced AI models can be substantial, which may not be feasible in all educational settings, especially in under-resourced schools or regions [28–35].

4.2 Ethical Considerations

The use of AI in education, including ChatGPT, raises several ethical concerns. Key areas identified by Akgun and Associate Professor Christine Greenhow [36] include:

1. **Privacy**: Many AI systems require access to personal data, and users may not fully understand the extent of this access when consenting to use these systems. The issue of data privacy becomes even more critical when such platforms are mandated as part of educational curricula, potentially forcing parents and children to share their data [36].
2. **Surveillance**: AI systems often track user interactions for personalization, which in education could involve monitoring students' performance, strengths, and weaknesses. While this can be beneficial for tailored learning, it also raises concerns about student privacy and autonomy [36].
3. **Autonomy and Bias**: AI systems, being algorithm-driven, can challenge student and teacher autonomy. Predictive algorithms may also embed societal biases, leading to discrimination in educational contexts [36].

4.3 Pedagogical Changes

The integration of AI tools like ChatGPT necessitates a reevaluation of traditional teaching methodologies. As reported by the Berkman Klein Center at Harvard University, AI tools could potentially serve as "Socratic tutors," engaging students in a stepwise reasoning process. This approach challenges conventional curriculum design and teacher agency, as AI can guide students through questions rather than simply providing answers [37]. However, the use of AI in education also raises concerns about exacerbating existing inequalities. Unequal access to such tools could worsen disparities in academic performance, and the uneven enforcement of cheating policies could further increase inequities. Conversely, if implemented equitably, AI tutors could level the playing field, providing high-quality educational experiences for all students [37]. Educators, administrators, and policymakers must consider these factors as they adapt to this evolving landscape. This includes redefining educator-student roles, classroom expectations, educational goals, and performance metrics, along with broader systemic and institutional considerations [37].

5 Case Studies and Applications of ChatGPT in the Metaverse for Educational Purposes

5.1 Pedagogical Changes

1. **GoVR Service by GoStudent**: This service utilizes Meta's metaverse for language teaching, providing an immersive learning environment. GoStudent has expanded the typical online class format into a virtual reality setting, where students can practice languages in more than 40 virtual locations such as restaurants and comedy clubs, with the goal of gaining confidence in language skills [38].
2. **Metaverse Institute by the King Sejong Institute**: Operated under South Korea's Ministry of Culture, Sports, and Tourism, this virtual space allows for real-time communication and learning. It offers a "Little Seoul" where students can practice Korean, learn about the culture, and access comprehensive textbooks, aiming to provide an immersive cultural and linguistic experience [38].
3. **Virtual Tutors for Language Learning**: ChatGPT and similar AI tools have revolutionized language learning by offering virtual tutors for conversation practice. These AI-powered tutors can converse in various languages, both in writing and speaking, thus eliminating the need for human interlocutors and enabling more accessible language practice [38].

5.2 Comparative Analysis with Traditional Teaching Methods

A study examining the use of ChatGPT-3.5 as a writing assistance tool in student essays revealed interesting insights when compared to traditional teaching methods. In this study, the performance of students using ChatGPT-assisted essay writing was compared with those using traditional essay-writing methods. The findings showed that the average essay grade was similar (C) for both groups. Notably, the use of ChatGPT did not significantly affect essay scores, writing duration, study module, or GPA. The text unauthenticity was slightly higher in the ChatGPT-assisted group, indicating a potential increase in AI-generated content. Overall, the study concluded that ChatGPT did not lead to higher quality content, faster writing, or a higher degree of authentic text compared to traditional methods. This suggests that while ChatGPT can be a useful tool, it may not necessarily enhance the overall quality of academic writing and could lead to confusion in inexperienced users [39].

6 Conclusion

The integration of ChatGPT within the metaverse for educational purposes signifies a remarkable evolution in teaching and learning methodologies. This review has highlighted the transformative impact of ChatGPT in reshaping teacher-student interactions, enhancing communication, engagement, and personalized learning experiences. Case studies, such as the GoVR service and the Metaverse Institute, demonstrate practical applications of this technology in language learning. However, the comparative analysis suggests that while ChatGPT offers innovative tools, it does not necessarily outperform traditional methods in all aspects of education. The limitations in integrating ChatGPT within the metaverse for educational purposes include technological accessibility, where the effectiveness of this AI technology hinges on access to advanced technological infrastructure, not readily available to all. Additionally, significant ethical challenges arise from issues surrounding data privacy, surveillance, autonomy, and biases embedded in AI systems. Furthermore, there is a pressing need for pedagogical adaptation, requiring educators to significantly adjust their teaching methodologies and engage in ongoing professional development to effectively harness this new technology in their teaching practices. Future research on ChatGPT and metaverse technologies in education should concentrate on ensuring equitable access and inclusion, particularly in under-resourced areas, to bridge the digital divide. Additionally, the development of robust ethical frameworks and guidelines is essential to address concerns related to privacy, surveillance, and AI biases. Studies exploring effective pedagogical integration of these technologies into diverse educational models and curricula, tailored to various learning styles and environments, are also critical. This approach acknowledges the considerable potential of ChatGPT in the metaverse for education while addressing the need for responsible and inclusive implementation.

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The Effect of Mediators in the Adoption of Metaverse as an Innovative Platform in Oman

Rana Saeed Al-Maroof, Omar Alsawafi, and Said A. Salloum

1 Introduction

The spread of the pandemic lately forced educational institutions to switch rapidly to various online distance educational environments avoiding face-to-face interactions [1–6]. Two types of delivery were adopted which are asynchronous or synchronous which are the most convenient modes of online learning in two-dimensional-based virtual environments [7]. Asynchronous online learning platforms include references to Moodle and Blackboard, whereas synchronous online learning platforms include references to Zoom, Adobe Connect, and VooV Meeting which enable instantaneous kind of interaction between learners and educators [4, 8, 9]. The features of accessibility and affordability of these learning platforms enable them to be used as an alternative or traditional four-wall [1, 7, 10].

Despite the fact that these two modes of interaction have been widely implemented, some limitations have been noticed at the educational level including poor self-perception inattention and increasing burden on teachers rather than the students which lead to poor academic performance and less motivational students. Hence, the metaverse with a three-dimensional virtual educational environment is one of the offered solutions that enable users free interaction in a virtual space where digital objects are available to satisfy users' needs., The metaverse does not apply to the educational environment but to political, economic, and social studies [11, 12]. To be able to explain Omani students' adoption of the new technology prior studies have proposed various models and theories to measure the adoption or acceptance

R. S. Al-Maroof · O. Alsawafi
English Language and Literature, Al Buraimi University College, Al Buraimi, Oman

S. A. Salloum (✉)
Health Economic and Financing Group, University of Sharjah, Sharjah, UAE
e-mail: ssalloum@sharjah.ac.ae

School of Science, Engineering, and Environment, University of Salford, Salford, UK

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of the metaverse. Most of these studies integrated various models that are developed by Venkatesh et al. and the theory of acceptance and use of technology (UTAUT) [13, 14].

Accordingly, the main objective of the current study is to explore factors that influence Omani students' adoption of the Metaverse based on a conceptual model using structural equation modelling (SEM) to investigate the structural relationship between each of the items and the adoption of the Metaverse. Bringing various attributes that may affect directly or indirectly to the metaverse adoption may help highlight the most efficient attributes out of several attributes including users' accessibility, users' mobility, innovativeness, and product knowledge. To set this paper apart from other studies, this study tends to examine the direct and indirect effects by making use of two mediators which are the perceived value and the perceived ubiquity.

The study was unique because the model correlated technology-based features and individual-based features. The study also used hybrid analyses such as Machine Learning (ML) algorithms and Structural Equation Modelling (SEM). The present study also employs the Importance Performance Map Analysis (IPMA) to assess the importance and performance factors.

2 The Previous Studies and the Gap in Knowledge

2.1 The Metaverse and Its Usage

The educational environment has been widely developed by educators and specialists to guarantee the most convenient environment that can add to economic and social development as well as to the fundamental structure and practices [15]–[17]. Therefore, the metaverse has been thought of as one of the innovative modes of teaching to bring about a paradigm shift in online education [18]. The used concept (Eduverse) is commonly used recently where users can interact with each other in a virtual online environment without limitations of space and time. The implementation of the Eduverse means that learning activities can be more collaborative and the construction of users' identities can be achieved by avatars moving in the virtual world. One of the advantages of the Eduverse is the fact that people can log in and exit based on their time and space with the option of retrieving all the old information once they logged in or reconnecting [19, 20].

The new mode of learning is a kind of platform that provides teachers and students with both problem-based learning (PBL) and learner-centred teaching (LCT) which are considered effective in achieving learning objectives. The implementation of these two approaches may help to decrease the number of challenges and strengthening the teamwork and interest in learning. This leads to the most influential factor which is incorporating collaborative and autonomous learning [19, 21, 22]. However, studies have shown several disadvantages of the metaverse, which is very popular in a different context, yet many educators and specialists raise their concerns because of the potential risk and privacy [23, 24].

2.2 The Determinants of the Metaverse in Previous Studies

The most critical factor that plays a role in determining the adoption or acceptance of the metaverse is the perceived risk. Most studies intend to investigate the effect of the perceived risk on the users' adoption. The fact that the flow of information can be accessed easily in different places and at any time rises may concerns about the users' behavior [25, 26]. A study by Teng et al. [25] that focuses on the effectiveness of the metaverse in the educational domain has incorporated several factors aiming at investigating the effects of performance expectancy, effort expectancy, social influence, and facilitating conditions on learners' satisfaction and the effect of intention to use the metaverse on the perceived risks. Far away from the educational domain is the investment domain where he quality and reliability of the information and the perceived risk are investigated [26].

The other factor that has been shared by recent studies is perceived value. The actual value that can be obtained from the use of the metaverse should be innovative and reliable. The perceived value has been explored in the tourism and management domain aiming to investigate the opportunities and challenges behind using the metaverse in hospitality and tourism management taking into consideration the experience and the value process [27]. A similar study by [28] aims to explore the perceived value that can be obtained by sports fans through using the metaverse focusing on new opportunities for spectator sports brands to attract fans to experience the metaverse.

The technology acceptance model and its variables have been implemented by studies to explore the effectiveness of metaverse adoption. A study by [19] has aimed at understanding the significance of the TAM variable following the order of importance specifying the most important variables that may affect the metaverse adoption. A rather complicated model was proposed by [29] who incorporated Machine Learning (ML) algorithms Structural Equation Modelling (SEM) and Importance Performance Map Analysis (IPMA) to investigate the effect of TAM on the students' perception of MS application for medical-educational purposes. The more recent model of UTAUT has been used by Arpaci et al. [30] to investigate the effect of factors in predicting social sustainability such as performance expectancy, social influence, hedonic motivation, price value, and habit on the users' intentions (Table 1).

The variations in the domains of metaverse interest reflect the critical influence that innovative technology may possess. The educational domain is one of the most significant domains that appreciate the importance of the metaverse as a tool to reflect that the metaverse can be more collaborative and the construction of users' identities can be achieved by avatars moving in the virtual world. A hybrid covariance-based structural equation modelling (CB-SEM) and artificial neural network (ANN) approach was used in a study by Arpaci et al. [30] to focus on five personality traits.

Table 1 Metaverse in previous studies

Authors	Determinants	Structural analysis tool	The significance	Field of application
[19]	Technology TAM	A deep-learning-based analysis of structural equation modelling (SEM) and artificial neural network (ANN)	The paper paves the way for the concerned authorities in the educational sector to understand the significance of each factor allowing them to make efforts and plans according to the order of significance of factors	Educational domain
[31]	Innovation and Imitation	A white-noise model as a null hypothesis	The paper aims at investigating the effect of a metaverse in its various forms by analyzing series data to reveal the innovation and imitation effect; employing the Bass model	Social media domain
[25]	UTAUT, satisfaction and the perceived risk	The data was analyzed using SPSS 22.0, and Amos 24.0 to test measures and structural models	The paper aims at investigating the effects of performance expectancy, effort expectancy, social influence, and facilitating conditions on learners' satisfaction and the effect of intention to use the metaverse on the perceived risks	Educational domain

(continued)

Table 1 (continued)

Authors	Determinants	Structural analysis tool	The significance	Field of application
[27]	The conceptual development of the metaverse focuses on experience and value	Theoretical analysis and descriptive analysis	This paper aims to investigate the opportunities and challenges behind using the Metaverse in hospitality and tourism management taking into consideration the experience and the value process	Tourism and management domain
[26]	The quality and reliability of the information and the perceived risk	An AMOS and SPSS package programs were used in the analysis	The study aims to determine the effect of the quality of the information obtained by the consumers about the metaverse world, the reliability of the information and the perceived risk, on the purchase intention from the point of view of the information adoption model	Investment domain
[28]	The perceived value	The analysis of the collected quantitative data (n = 504)	The perceived value that can be obtained by sports fans through using the metaverse explores new opportunities for spectator sports brands to attract fans to experience the metaverse	Sport domain

(continued)

Table 1 (continued)

Authors	Determinants	Structural analysis tool	The significance	Field of application
[29]	Technology Acceptance Model (TAM), Personal innovativeness (PI), Perceived Compatibility (PCO), User Satisfaction (US), Perceived Triability (PTR), and Perceived Observability (POB)	The study adopted incorporated Machine Learning (ML) algorithms and Structural Equation Modelling (SEM). In addition to the use of Importance Performance Map Analysis (IPMA)	The present study aims to investigate students' perception of MS applications for medical-educational purposes	Medical-educational domain
[30]	The UTAUT2 constructs and big five personality traits	A hybrid covariance-based structural equation modelling (CB-SEM) and artificial neural network (ANN) approach	The study aims to investigate the effect of factors in predicting social sustainability such as performance expectancy, social influence, hedonic motivation, price value, and habit on the users' intentions	Educational domain

Similarly, a study by Almarzouqi et al. [29] has used Machine Learning (ML) algorithms and Structural Equation Modelling (SEM) to reflect the importance of the metaverse in the medical-educational domain. The use of the using SPSS 22.0, and Amos 24.0 to test measures and structural models was evident in a study by [25] to investigate the effect of various variables on the intention to use the metaverse in the educational domain. Other studies have used other tools to measure the effectiveness of the metaverse in investment, sports and social media domains.

3 Research Model and Hypotheses

This study aims to investigate the determinants of metaverse adoption in Oman. Several variables such as users' accessibility (UA) users' mobility (UM), perceived ubiquity (PU), perceived value (PV), perceived satisfaction (PS) and user innovativeness (UI) are variables which are assumed to influence the metaverse adoption in Oman. To deepen the analysis, some of these variables proposed can function as a mediator to measure the effectiveness of these variables to predict the metaverse adoption. The conceptual model is set to examine the direct and indirect relations among different variables that affect the adoption of metaverse by Omani students.

3.1 Users' Accessibility and Mobility

One of the influential factors that may affect the users' intention to use the metaverse is accessibility and mobility which are perceived as a measure of relative advantage. The relation between accessibility and convenience has been discussed by previous studies assuring that free availability and convenience are significant factors that affect users' adoption and acceptance of technology. Furthermore, accessibility is related to satisfaction. The inaccessibility may lead to a change in users' attention seeking other alternatives. for instance [32]. The more accessible the information that can be reached, the less effort is required. The previous fact enhances other crucial factors which are value, usability, affordability, accessibility, technical support, social support, emotion, independence, experience, and confidence [33]. The accessibility of various technologies and websites has attracted the attention of researchers because it enables people with varying disabilities to browse and perceive information as well as interact and take part in various activities. The lack of the needed accessibility may lead users to think that they are having limited freedom to choose and are facing restrictions on their accessibility to alternative options. The lack of the required accessibility leads to demotivation threatening the adoption of the technology [34]–[37].

On the other hand, users' mobility has been considered a critical factor that may affect technology adoption bearing in mind that the metaverse has made use of various virtual rooms where people can move using avatars. Therefore, mobility is a huge challenge in digital technologies where information and communication technology can be easily improved triggering significant changes to users' perspectives [38]. Thus, shared mobility helps in satisfying users' needs without reliance and reducing the levels of risk acceptance which may indirectly impact the adoption of technology by influencing their perception of risk [39]. Based on the above literature, the proposed hypotheses that can affect the metaverse adoption in Oman:

H1a: There is a positive relation between users' accessibility and users' mobility in Oman.

H1b: There is a positive indirect relation between users' accessibility and metaverse adoption in Oman mediated by perceived ubiquity.

H1c: There is a positive relation between users' accessibility and metaverse adoption in Oman.

H2a: There is a positive relation between users' mobility and metaverse adoption in Oman.

H2b: There is a positive indirect relation between users' mobility and metaverse adoption in Oman mediated by perceived ubiquity.

3.2 Perceived Ubiquity

Ubiquity refers to the ability to share technology to provide "anytime" and "anywhere" adaptive personalized usage. It is assumed that the perceived ubiquity of technology may positively impact the perceived ease of use and the perceived usefulness Hence, the levels of the ubiquity of technology may impact the satisfaction level because users demand to be able to access the information anytime and anywhere [40]. The concept *of* ubiquity is a unique feature that has been emphasized in social media and mobile Internet because it is a means of searching for information in any place, at any time. Previous studies have asserted the effect that users' perceptions of the ubiquity of technologies can affect directly or indirectly the adoption of technologies [41]. However, some studies have found that the ubiquity may lead be influenced negatively by other factors such as privacy and experience. The ability to reach information at any time and place raises privacy concerns in the users' minds which may affect their level of satisfying experiences [42]. Based on the previous literature, the following hypothesis can be proposed:

H3: There is a positive relationship between perceived ubiquity and the adoption of the metaverse in Oman.

3.3 The Perceived Value and the Perceived Satisfaction

Interestingly, the perceived value has related to utility in terms of effort and time and it has a relationship with the continued use intentions of technologies. The perceived value is considered a fundamental factor for a business organization as an effective source of competitive advantage [43]. The perceived value has a positive effect on users' attention in various contexts such as human value and entertainment value. Furthermore, the perceived value has a close relationship with satisfaction which is a good variable to predict the effectiveness of the perceived value. Hence, three main characteristics can affect the perceived value which are the time, cost and satisfaction as well as the benefits [44, 45].

Perceived satisfaction is a key motivation to continue using technology. Satisfaction strongly affects users' attitudes and users' values, experience, and utility. Thus, satisfaction ensues is closely related to users' expectations therefore dissatisfaction appears as a consequence of failing to meet users' expectations. Within an educational domain, the experience that one can get by using the metaverse is described as vivid and immersive which can enhance the quality of interaction with virtual elements [46]. Therefore, the following hypotheses are formed:

H4a: There is a positive relation between perceived satisfaction and perceived value in Oman.

H4b: There is a positive indirect relation between perceived satisfaction and the metaverse adoption mediated by users' innovativeness in Oman.

H4c: There is a positive relation between perceived satisfaction and the metaverse in Oman.

H5a: There is a positive relation between perceived value and metaverse adoption in Oman.

H5b: There is a positive indirect relation between the perceived value and the metaverse adoption mediated by users' innovativeness in Oman.

3.4 Innovativeness

Innovativeness is an influential predictor that can have a significant effect on users' adoption. User innovativeness refers to their willingness to experiment with a new technology which may accelerate the use of technology The readiness to accept the presence of new technology is the main driving factor for technology adoption Hence, user innovativeness can be defined as an intention to make use of new technologies to be pioneers in using the latest technology [46]. Despite the fact that innovativeness is part of theoretical models of technology acceptance or adoption, it has been considered a crucial predictor. Thus, innovativeness is closely related to the degree to which an individual is open to experiencing or trying something new and is an

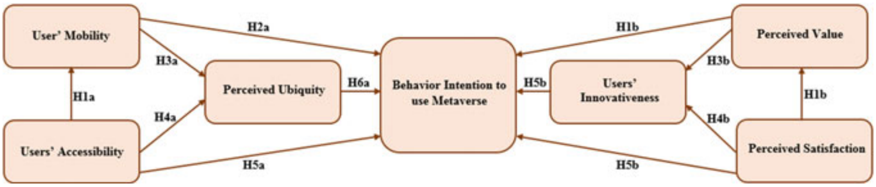

Fig. 1 Research model

expression of their innovativeness or novelty-seeking tendencies [47]. Based on the previous literature, the hypothesis proposed is as follows:

H6: There is a positive relation between users' innovativeness and metaverse adoption in Oman (Fig. 1).

4 Research Methodology

4.1 Data Collection

Data collection for this study took place at Al Buraimi University College in Oman during the period from February 10th to May 20th, 2022. The research team distributed 150 questionnaires randomly, of which 113 were completed by respondents, constituting 75.3% of the distributed surveys. An additional 37 questionnaires were excluded from the analysis due to missing data, resulting in a final usable questionnaire count of 113. These 113 questionnaires were accepted for analysis based on the recommended sample size according to Krejcie and Morgan's [48] guidelines for a population of 150. It's important to note that there was a notable disparity between the sample size (113) and the minimum requirements for analysis. Recognizing this discrepancy, the research team proceeded to evaluate the sample size using structural equation modeling [49], which was subsequently employed to confirm the research hypotheses. It's worth mentioning that the hypotheses in this study were built upon existing theories rooted in Technology Adoption Rate. For the assessment of the measurement model, the research team utilized structural equation modeling (SEM) through SmartPLS Version 3.2.7. In-depth analysis was carried out using the final path model to refine the findings.

4.2 Students' Personal Information/Demographic Data

Figure 2 presents an overview of the demographic and personal data collected for the purpose of our analysis. The data reveals that the respondents can be categorized into two main groups: males, constituting 19% of the total, and females, making

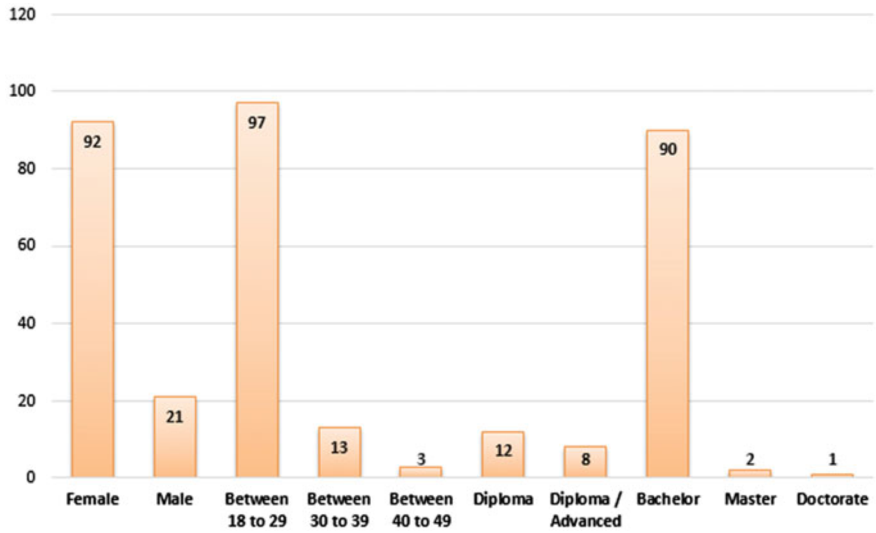

Fig. 2 Demographic information of the respondents ($n = 113$)

up 81%. Furthermore, a substantial majority, 86%, fell within the age range of 18 to 29 years, while the remaining respondents were aged above 29. The majority of participants were university students with significant qualifications across various academic levels. Specifically, the breakdown of degrees among respondents was as follows: 11% with diplomas, 7% with diploma/advanced diplomas, 79% with bachelor's degrees, 2% with master's degrees, and 1% with doctoral degrees. In a study conducted by [50], they highlighted instances where respondents expressed a willingness to volunteer. These cases can be considered as instances of "purposive sampling approach." As a result, our sampling approach included all respondents who were university students, encompassing various age groups and academic majors. It is important to note that IBM SPSS Statistics version 23 was utilized to analyze and measure the demographic data.

4.3 Study Instrument

In the present study, a survey instrument was employed to confirm the research hypotheses. To ensure precise measurement of the seven constructs addressed in the questionnaire, a carefully selected set of 21 items was incorporated into the survey. The origins of these constructs are detailed in the Table 2, which has been included to enhance the practicality of our research constructs and to substantiate the current model with evidence drawn from existing literature. Additionally, the research team made necessary refinements to questions based on prior research findings.

Table 2 Measurement items

Constructs	Items	Definition	Instrument	Sources
Users' accessibility	UA1	The relation between accessibility and convenience has been discussed by previous studies assuring that free availability and convenience are significant factors that affect users' adoption and acceptance of technology. Furthermore, accessibility is related to satisfaction. The inaccessibility may lead to a change in users' attention seeking other alternatives. for instance [32]	MS easy access enables me to get all the required information to improve my learning process	[32]
	UA2		MS easy access enables me to move freely among different information sources	
	UA3		MS easy access enables me to get information at any time and any place	
	UA4		MS is provided with easy access which enables me to adopt it	
Users' mobility	UM1	Users' mobility has been considered a critical factor that may affect technology adoption bearing in mind that metaverse has made use of various virtual rooms where people can move using avatars. Therefore, mobility is a huge challenge in digital technologies where information and communication technology can be easily improved triggering significant changes to users' perspectives [38]	MS free mobility feature assists my adoption	[38]
	UM2		MS offers free mobility at any time in different places	
The perceived ubiquity	PUB1	The perceived ubiquity refers to the ability to share technology to provide "anytime" and "anywhere" adaptive personalized usage. It is assumed that the perceived ubiquity of technology may positively impact perceived ease of use and perceived usefulness Hence, the levels of the ubiquity of technology may impact the satisfaction level because users demand to be able to access the information anytime and anywhere [40]	MS has innovative features that are available anytime and anyplace	[40]
	PUB2		MS can be easily accessed	

(continued)

Table 2 (continued)

Constructs	Items	Definition	Instrument	Sources
	PUB3		MS has the feature of free movement	
	PUB4		MS timeless features encourage me to adopt it	
The perceived satisfaction	PS1	Perceived satisfaction is a key motivation to continue using technology. Satisfaction strongly affects users' attitudes and users' values, experience, and utility. Thus, satisfaction ensues is closely related to users' expectations therefore dissatisfaction appears as a consequence of failing to meet users' expectations [51]	MS has many satisfactory tools that facilitate the process of learning	[51]
	PS2		MS has newly employed features that affect my level of satisfaction	
	PS3		MS satisfies my e-learning needs to adopt it	
Perceived value	PV1	The perceived value has related to utility in terms of effort and time and it has a relationship with the continued use intentions of technologies. The perceived value is considered a fundamental factor for a business organization as an effective source of competitive advantage [43]	MS offers unique benefits that have great value in my study	[43]
	PV2		MS affects my level of satisfaction and increases the value of the technology	
	PV3		MS value encourages me to adopt the technology	
Users' innovativeness	UI1	Innovativeness is an influential predictor that can have a significant effect on users' adoption. User innovativeness refers to their willingness to experiment with a new technology which may accelerate the use of technology The readiness to accept the presence of new technology is the main driving factor for technology adoption Hence, user innovativeness can be defined as an intention to make use of new technologies to be pioneers in using the latest technology [46]	MS has an up to date technology that satisfies my needs	[46]
	UI2		MS has innovative features that increase the value of the technology	

(continued)

Table 2 (continued)

Constructs	Items	Definition	Instrument	Sources
	UI3		MS has a unique experience that encourages me to adopt it	
Behavior intention to use metaverse	BI1	Behavioral intention is meant to refer to the intention of the targeted users to utilize the technology deemed as new. It is part of Davis's TAM theory	MS offers a good opportunity to try	[52, 53]
	BI2		I plan to use MS	

Table 3 Cronbach's Alpha scores in the pilot study (Cronbach's Alpha ≥ 0.70)

Construct	Cronbach's alpha
BI	0.740
PS	0.832
PUB	0.887
PV	0.753
UA	0.787
UI	0.883
UM	0.822

4.4 Pilot Study of the Questionnaire

To assess the reliability of the questionnaire items, a pilot study was carried out. The selection of data for this pilot study was done randomly, involving 15 students selected from the designated population. This sample size aligns with the research guidelines, as it represents 10% of the total sample size set for the analysis, which was 150 students. In line with established research practices, the IBM SPSS Statistics version 23 software was utilized to conduct the Cronbach's alpha test for internal reliability. This procedure is crucial for obtaining dependable conclusions regarding the measurement items. In accordance with prevailing trends in social science research, a reliability coefficient of 0.70 is considered acceptable [54]. Table 3 provides the Cronbach's alpha values for the five subsequent measurement scales, demonstrating the internal consistency of the questionnaire items.

5 Analysis and Results

5.1 Data Analysis

The current study employed data analysis utilizing Partial Least Squares-Structural Equation Modeling (PLS-SEM) conducted with SmartPLS Version 3.2.7 [55]–[60]. The data collection process followed a two-step assessment approach, which encompassed both the measurement model and the structural model [61]–[69]. PLS-SEM was chosen for several key reasons that were emphasized throughout the research paper. The first critical aspect to consider is the examination of the conceptual theory proposed in this study, with a particular emphasis on employing PLS-SEM [70]. The second step involves using PLS-SEM to effectively handle the exploratory research data collected based on the conceptual models [71]. The third step entails implementing PLS-SEM analysis on the entire model as a unified entity, rather than dividing it into segments [72]. Finally, the fourth step involves concurrent analysis of the structural and measurement models using PLS-SEM. The significance of employing PLS-SEM lies in its capacity to generate and acquire accurate measurements [73].

5.2 Convergent Validity

To evaluate the measurement model, [61] proposed examining both construct reliability, which encompasses Cronbach's alpha (CA), Dijkstra-Henseler's rho (ρA), and composite reliability (CR), and validity, encompassing discriminant and convergent validity. For assessing construct reliability, Cronbach's alpha (CA) was found to fall within the range of 0.798 to 0.904, as indicated in Table 4. The commonly accepted threshold value of 0.7 is surpassed by these figures [74]. Furthermore, Table 4 reveals that the composite reliability (CR) values range from 0.803 to 0.897, exceeding the threshold value [75]. However, it is recommended that researchers employ Dijkstra-Henseler's rho (ρA) reliability coefficient to evaluate and report construct reliability, as with CA and CR. The reliability coefficient ρA should ideally be at least 0.70 for exploratory research and 0.80 or 0.90 for advanced research stages [74, 76, 77]. Notably, Table 4 indicates that the minimum reliability coefficient ρA among all measurement constructs is 0.70, confirming the reliability of each construct and demonstrating their freedom from errors.

Regarding the assessment of convergent validity, it is essential to examine the mean variance extracted (AVE) and factor loadings [61]. As shown in Table 4, each factor loading value exceeds the 0.7 threshold, as mentioned earlier. Additionally, Table 4 presents AVE values ranging from 0.733 to 0.855, all exceeding the threshold of 0.5, except for the previously mentioned ones. Consequently, based on the explanation provided, it can be concluded that convergent validity has been successfully established.

5.3 Discriminant Validity

The study aimed to assess discriminant validity, prompting a thorough examination of two criteria: the Heterotrait-Monotrait ratio (HTMT) and the Fornell-Larcker criterion [61]. The results presented in Table 5 confirm that the Fornell-Larcker criterion is met, as each Average Variance Extracted (AVE) and its respective square root exceed the correlations with other constructs [78].

Table 6 displays the HTMT ratio findings, revealing that the value of each construct is below the 0.85 threshold [79]. This demonstrates that the HTMT ratio is indeed present. These findings contribute to the calculation of discriminant validity, and the analysis results indicate that there are no issues related to assessing the measurement model in terms of its reliability and validity. Consequently, the collected data can be confidently utilized for evaluating the structural model.

Table 4 Convergent validity results which assures acceptable values (Factor loading, Cronbach's Alpha, composite reliability, Dijkstra-Henseler's rho ≥ 0.70 and AVE > 0.5)

Constructs	Items	Factor loading	Cronbach's alpha	CR	PA	AVE
Behavior intention to use metaverse	BI1	0.754	0.831	0.831	0.922	0.855
	BI2	0.883				
Users' accessibility	UA1	0.838	0.879	0.880	0.917	0.733
	UA2	0.873				
	UA3	0.887				
	UA4	0.924				
Users' mobility	UM1	0.805	0.798	0.805	0.908	0.835
	UM2	0.884				
Perceived ubiquity	PUB1	0.887	0.885	0.886	0.921	0.745
	PUB2	0.836				
	PUB3	0.807				
	PUB4	0.781				
Perceived satisfaction	PS1	0.782	0.895	0.897	0.934	0.826
	PS2	0.883				
	PS3	0.932				
Perceived value	PV1	0.783	0.877	0.880	0.924	0.802
	PV2	0.780				
	PV3	0.804				
Users' innovativeness	UI1	0.882	0.904	0.803	0.908	0.832
	UI2	0.908				
	UI3	0.910				

Table 5 Fornell-Larcker scale

	BI	PS	PUB	PV	UA	UI	UM
BI	**0.925**						
PS	0.616	**0.909**					
PUB	0.652	0.698	**0.863**				
PV	0.616	0.600	0.608	**0.896**			
UA	0.658	0.601	0.469	0.543	**0.856**		
UI	0.682	0.641	0.361	0.638	0.504	**0.916**	
UM	0.377	0.575	0.598	0.547	0.614	0.603	**0.912**

Table 6 Heterotrait-Monotrait ratio (HTMT)

‘	BI	PS	PUB	PV	UA	UI	UM
BI							
PS	0.293						
PUB	0.272	0.221					
PV	0.481	0.611	0.382				
UA	0.187	0.623	0.301	0.631			
UI	0.573	0.558	0.178	0.336	0.512		
UM	0.357	0.725	0.338	0.578	0.224	0.490	

5.4 Hypotheses Testing Using PLS-SEM

The structural equation model was constructed using Smart PLS, which employs maximum likelihood estimation to reveal the interrelationships among various theoretical constructs within the structural model [60, 80, 81]. Following this procedure, the proposed hypotheses were examined, and their outcomes are presented in Table 7 and Fig. 2. These results demonstrate that the model possesses strong predictive capabilities [82], with approximately 80% of the variance in “Behavior Intention to use Metaverse” explained. In Table 8, the beta (β) values, *t*-values, and *p*-values for all the formulated hypotheses are detailed, based on the findings obtained through the PLS-SEM technique. It is evident that every researcher provided support for each hypothesis. In line with the data analysis hypotheses, the empirical data affirms the validity of H1a, H3a, H4a, H5a, H6a, H1b, H3b, H4b, H5b, and H6b.

The study’s findings reveal significant relationships between Users’ Accessibility (UA) and several key constructs. Users’ Accessibility (UA) exerts a substantial positive influence on Users’ Mobility (UM) ($\beta = 0.786$, $P < 0.001$), Perceived Ubiquity (PUB) ($\beta = 0.444$, $P < 0.001$), and Users’ Accessibility (UA) itself ($\beta = 0.440$, $P < 0.05$). These results provide strong support for hypotheses H1a, H4a, and H5a. Interestingly, Users’ Mobility (UM) does not exhibit a significant impact on the intention to use Metaverse (BI) ($\beta = 0.056$, $P = 0.299$), thus failing to support H2a. However, Users’ Mobility (UM) demonstrates a significant positive relationship with Perceived Ubiquity (PUB) ($\beta = 0.444$, $P < 0.001$), thereby supporting H3a. Moving on to Perceived Value (PV) and Users’ Innovativeness (UI), both of

Table 7 R^2 of the endogenous latent variables

Constructs	R^2	Results
BI	0.801	High
UM	0.504	Moderate
PUB	0.628	Moderate
PV	0.488	Moderate
UI	0.646	Moderate

Table 8 Results of structural model—research hypotheses significant at p** = < 0.01, p* < 0.05

‘	Relationship	Path	*t*-value	*p*-value	Direction	Decision
H1a	UA→UM	0.786	13.848	0.000	Positive	Supported**
H2a	UM→BI	0.056	1.039	0.299	Positive	Not supported
H3a	UM→PUB	0.459	3.622	0.000	Positive	Supported**
H4a	UA→PUB	0.444	3.369	0.001	Positive	Supported**
H5a	UA→BI	0.440	8.212	0.035	Positive	Supported*
H6a	PUB→BI	0.519	5.726	0.030	Positive	Supported*
H1b	PS→PV	0.800	13.332	0.000	Positive	Supported**
H2b	PV→BI	0.494	5.026	0.006	Positive	Supported**
H3b	PV→UI	0.464	4.683	0.008	Positive	Supported**
H4b	PS→UI	0.470	6.696	0.007	Positive	Supported**
H5b	PS→BI	0.881	14.518	0.000	Positive	Supported**
H6b	UI→BI	0.998	17.927	0.000	Positive	Supported**

these constructs significantly influence Perceived Satisfaction (PS) ($\beta = 0.800$, $P < 0.001$), and ($\beta = 0.470$, $P < 0.01$), respectively. These results lend strong support to hypotheses H1b and H4b. Furthermore, there is a significant positive relationship between Perceived Value (PV) and Users' Innovativeness (UI) ($\beta = 0.464$, $P < 0.01$), confirming H3b. Finally, Perceived Ubiquity (PUB), Perceived Value (PV), Perceived Satisfaction (PS), and Users' Innovativeness (UI) all exert significant positive effects on the intention to use Metaverse (BI) ($\beta = 0.519$, $P < 0.05$), ($\beta = 0.494$, $P < 0.01$), ($\beta = 0.881$, $P < 0.001$), and ($\beta = 0.998$, $P < 0.001$), respectively. These findings provide robust support for hypotheses H6a, H2b, H5b, and H6b. In summary, the study's results underscore the intricate interplay between these constructs and their significance in shaping users' intentions to embrace the Metaverse (Fig. 3).

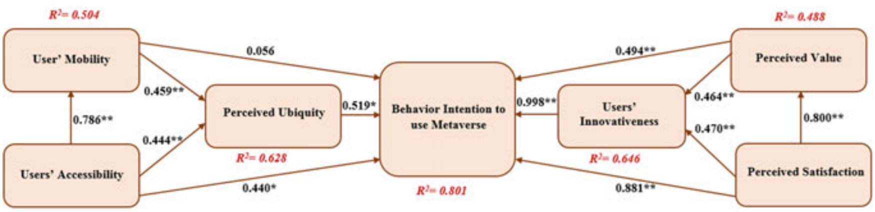

Fig. 3 Path coefficient results (significant at p** <= 0.01, p* < 0.05)

6 Discussion and Implication

The discussion of the current study has two phases. The first part is devoted to the discussions of results that related to the dependent variables of the conceptual model. The second part is devoted to the discussions of the moderators in the current study.

6.1 Discussion of the Dependent Variable Results

The recent development from the traditional four-wall classrooms to the on-line learning has affect the teaching and learning environment, thus, has attracted researchers' attention encouraging them to enrich the on-line experience via the metaverse on-line learning environment. Many previous studies have looked at the elements that influence students' adoption of the metaverse as an online learning environment [17, 19, 83]. Based on the objectives of the current research, the relationship of the variables (users' accessibility, users' mobility, the perceived value and the perceived satisfaction) are investigated. It also investigated d the significant effect of these factors in relation with two moderators which positively impact the learning environment. Thus, the current study is an attempt to examine the proposed hypotheses in which moderators play a significant role in the conceptual model. The discussion of results has shown that the proposed hypotheses have been approved. In fact, the users' mobility and accessibility are critical variables that control the acceptance of the metaverse. The users' accessibility has imposed threat to the use of the metaverse, therefore, the results have shown that whenever the user has free accessibility, s/he can accept the use of the metaverse easily. However, previous studies have shown if the users' accessibility is limited or denied, the users'' freedom is threatened. Hence, the users' motivational state may be affected leading to negative attitudes. (System quality and information quality affect positively the acceptance of Chat GPT which is in line with previous studies. The system quality and information quality have been distinguished as effect variable in Chat GPT in comparison with Google platforms. The system quality affects the acceptance indirectly through the ease of use and the perceived usefulness. Some studies, on the other hand, have proved that system quality and other factors are significant predictors of learners' intention to use various education tools [84, 84]. The effect of the perceived value on the adoption of Chat GPT has been positively evaluated in the current study. The users of Chat GPT can perceive the Chat GPT as a useful tool due to the effective and comprehensive results that were obtained. The results are in agreement with previous studies which highlighted that perceived value has a mediation effect on the use of other technology such as IoT [86]. Similarly, the perceived satisfaction has been supported by the results which stands in agreement with previous studies. The fact that perceived satisfaction can affect the users' perspective, particularly at the educational level is widely agreed upon by researchers [87, 87].

6.2 Discussion of the Moderation Results

The moderator impact of Chat GPT users is influential as it represents the actual connections between the proposed dependent variables, namely, the system quality, information quality, perceived satisfaction and the perceived satisfaction According to the previous studies that focus on the moderating effect, there is often a significant interaction effect of the perceived satisfaction and other variables on the use of technology. It has been revealed that there is an indirect effect of the moderators such as the task fit technology on the adoption of technology. In fact, the task technology fit that is integrated into the model as a moderator explains much more of the variance in the dependent variable. The use of moderators explained how the application is measured to indicate how the application is up to date and appropriate in accordance with the recent usage of students. That is, adding the task technology fit as a moderator enhances the use of Chat GPT. Previous studies have agreed with the current results showing the increasing impact of technology fit on the use of technology. A study by Jeyaraj [89] has shown that technology fit may positively affect the acceptance of technology particularly if the technology is supported by the government. The task technology fit has greater effects on students rather than non-students because it focuses on the positive effects on behavioral variables less frequently than perceptual variables. This implies that the task technology fit can directly measure the perceptions rather than behaviors, and be contextualized to specific types of technologies and users. A similar study by Roth et al. [90] has concluded that the task-technology fit analysis can contribute to a better understanding of blockchain adoption in the public sector because it offers the substance to an extended task-technology fit theory for federally structured, cross-organizational contexts.

Similarly, personal innovativeness has been approved as a significant moderator that validates the relation between the system quality and system information and the intention to use the technology. The relation is positively evaluated for the users and it shows a high level of strength on the effect of innovativeness as a moderator to enhance the use of Chat GPT. Thus, the findings confirmed the proposed hypotheses and were in line with previous results. Previous studies have shown that personal innovativeness can function as a moderator to measure the effectiveness of technology acceptance [91]. Another study has revealed that innovativeness has an indirect effect on the adoption of technology and it may strengthen the future use of the technology [92].

6.3 Theoretical and Practical Implications

The theoretical implications are directed to universities and administrations of applications that offer similar services must focus on the usefulness of the applications and consequently influence the success of the educational process and it is a breakthrough towards the educational contributions in different sectors. To ensure the success of Chat GPT in educational institutions, the effectiveness of the applications may help in framing the constructs we have adopted in the conceptual framework. The provided conceptual model has enriched the literature for the adoption of Chat GPT by focusing on the moderators which are the task fit technology and the innovativeness. The study contributed theoretically and stressed the value of users' views in understanding the benefits of how Chat GPT is making a meaningful contribution to enhancing the users' behaviour and social perceptions at the educational level. To enrich the quality system, perceived value and perceived satisfaction, the theoretical aspect of this study we have proposed these constructs to comprehensively evaluate the effectiveness of this model. Another theoretical strength of the current conceptual model is to link between the effectiveness of the perceived value and the perceived satisfaction and the personal innovativeness as a moderator on one hand, and the quality system and the quality information with the moderator task technology fit on the other hand. The results also provide impressive support for the development of the conceptual model at the government level from a practical perspective. The current study findings strengthen the significance of Chat GPT at educational public universities which can be applicable to other governmental. Finally, the Chat GPTconceptual model serves as a parameter for a qualitative research approach to have a more precise vision towards the significance of the applications in other fields within the public sector.

6.4 Managerial Implications

The current study provides insights into the effective adoption of the use of Chat GPT at educational institutions to facilitate the teaching process and studying demands. The administration of educational institutions in the Gulf area can make use of the current results to make significant efforts to use the applications from different perspectives. They can be summarized as follows. First, the Chat GPT applications should meet the student's needs and enrich the relative advantage of the educational process by integrating Chat GPT in the taught course as means of source-provider. It provides various information in different forms to facilitate the process of teaching and learning. Second, governments in the future can make use of Chat GPT to create collaborative and regulatory policies to use the application as an available source of information to all government institutions and they should accelerate the adoption of Chat GPT by including other similar applications and other open–access features. Therefore, this research study will assist the users to understand the factors which support the use of similar applications. Third, the study will act as a thought-provoking factor for the Chat GPT policymakers to develop appropriate policies that can save the rights of the users and ensure the users' perceived value towards intention to use Chat GPT.

6.5 Limitations and Future Research

The study is limited to different perspectives. The data is collected from a sample of studies within the Gulf area. The users in the gulf are mainly students that joined different universities and they need the applications for various educational purposes. Future studies may focus on other samples from other government institutions that may use this application to enhance future usage. The conceptual framework is limited in scope to certain aspects that are related to system quality, satisfaction, value and innovativeness with moderators that can measure the effects of Chat GPT. Future studies can include other variables with different moderators that may measure the other essential factors.

6.6 Conclusion

The study concludes that Chat GPT interests and benefits are higher than other applications. It is not simply a technical device, but it is an integrated platform that summarizes many issues to focus on different subject content, practical theories, and technological issues. It has been recognized as effective tool in comparison with Google platforms. Undoubtedly the Chat GPT is an effective learning tool that is a well-designed with high level of sufficiency. Therefore, studying the impact of Chat GPT becomes crucial. Thus, this study reveals that this application has a large impact on the users' acceptance through information and system quality on one hand and the perceived learning value and perceived satisfaction on the other hand. However, some of the aspects have not been supported and they do not necessarily have a significant predictive effect on the use of Chat GPT. This finding implies that providing additional development is required to reach better results which will increase the level of contributions of the application itself.

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Metaverse and Creative Teaching of Reading Texts (Suggested Scenario)

Khamis Abdel Baky Negm Rezk

1 Introduction

The idea of the metaverse arose in the environment of creativity and ends there [1, 2]. As it is her field of affiliation and the nature of her imaginary world, the term "metaverse" goes back to novelist Neal Stevenson, who invented it in 1992 in his novel (Snow Crash), in which live-action avatar-like virtual characters meet in 3D buildings and other virtual reality environments [3]–[5].

The Merriam-Webster dictionary defines "metaverse technology" as: The natural evolution of the Internet is a persistent virtual environment that allows access to and interoperability of many individual virtual realities through immersive 3D virtual worlds in which people interact as groups with each other and with software agents, using metaphors from the real world but without physical constraints [6, 7].

Metaverse technology provides virtual reality technology (VR), augmented reality (AR), mixed reality (MR), and 3D environments, in addition to artificial intelligence (AI) technologies, which are interacted with in real-time, effectively and continuously, and an innumerable number of people participate in it [3, 6, 8, 9].

It is limited to people around the world and provides a real immersion environment for users, a real feeling [10]–[13], and real virtual communication in environments that are completely similar to the environments in reality [14]–[17]. You can enter the world of the metaverse by purchasing virtual reality glasses [18]–[23]. The user is completely separated from the real world and is in a virtual digital world [24]–[28]. Once the user of the Metaverse technology enters this world using the Horizon Workrooms application and Oculus VR virtual reality glasses, he finds that he is surrounded by a group of virtual communities that have no end or beginning, and

K. A. B. N. Rezk (✉)
Professional Postgraduate Diploma in Teaching Department, City University Ajman, Ajman, UAE
e-mail: k.negm@cu.ac.ae

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the technology is used In many areas, most notably virtual business meetings, during gaming, and other daily activities carried out by the user [29]–[42].

Education in virtual worlds depends mainly on the use of avatars, through which users—whether students or teachers—are represented digitally [42]–[46]. The metaverse contributes to creating an immersive educational experience through which students can participate in fun activities, which makes learning more fun and attractive [8, 15, 47]–[50]. Through metaverse technology, students can be involved in a three-dimensional imaginary world where they can see their colleagues and teachers and interact with them, just as is the case in a traditional classroom. Metaverse can also literally transport students to different historical stages, which eliminates the need for imagination. For example, students can learn more about the voyages of Italian explorer Christopher Columbus by sailing on a ship and discovering America in the fifteenth century. In virtual worlds, metaverse technology allows students to be transported to a virtual world that is similar to the real world to display natural environments such as mountains, plateaus, and seas. In the metaverse, students can live inside the novel, which helps in understand the story and interacting with the characters better [19–23, 51].

2 Literature Review

2.1 Introduction to the World of Metaverse

Through virtual reality glasses, jackets, and gloves equipped with console devices, the user can live an almost real experience in which these smart technologies work as an intermediary between users in the metaverse world [15, 52, 53]. He can see things around him in three dimensions, and he can also feel. It involves sensory and physical effects, such as the feeling of falling into water or being punched in the face. e or others, through the sensors located in the jackets and jumpers that he wears, He gets an experience that is more like realism. The user enters this world to find himself within a series of thousands of interconnected communities, which has no end, through which it is possible to meet a large number of people either work or play, and all they need for that are virtual reality headsets and augmented reality glasses [29].

There are now a number of metaverse applications, including Meta Horizon Worlds, Roblox, VRChat, and many others. To access many of them, you need a virtual reality headset (VR headset), and what is required to create a virtual version of yourself (Create an avatar) is to choose a picture of yourself or picture that expresses the character you want to appear to others. Then, enter the website (ready player. me), which is A free site for creating 3D models of the full body for use in virtual worlds, such as games and the metaverse. When you upload your photo, the site will create a 3D model for you that resembles this image. You can modify this model and customize it as you want. You can customize the skin color, eye shape, and the shape of the nose, eyebrows, and lips; choose the hairstyle and color; choose formal or sports clothes, glasses, face masks, and accessories. Once the customization is completed, click on the Enter Hub option in the upper right corner of the screen, and the site will display the created form.

Then click on the option in the side menu (My Avatars). In order to download the model and use it on any website or game, click on the three dots at the top of the model image, then click on the (Download avatar. glb) option. Thus, the model is saved on the device for downloading later on any website, and after creating the virtual copy on the (ready player) website.me, click on the Discover Apps option, and you will find in the side menu dozens of virtual worlds, including worlds that allow you to invite friends to enjoy with them in the metaverse world, such as VRChat or (Meeting VR), which allows you to conduct virtual meetings with others in The world of the metaverse. It is also possible to create a virtual copy directly inside the virtual world that you want to live in, which looks less like a cartoon and more like a real person, through the (Synthesia) platform, but it requires paying sums of money for subscription.

Those in charge of the educational process in our country should benefit from the wonderful and amazing features related to modern educational systems that are linked to the world beyond the traditional "metaverse" while employing artificial intelligence programs because it will provide each person with an educational environment that has characteristics that suit him and will take into account his skills and inclinations with the aim of attracting him more to study and learn. This will also enable teachers and trainers to focus on the individual needs of each student [54].

2.2 *Previous Studies on the Use of Metaverses in Education*

1—Study of [29]

The Metaverse World between Reality and Hope its Effectiveness in the Graphic. It discussed the emergence of the term metaverse around human interaction, how this technology will enable many interactions To be practiced in various fields, and the role of this technology in developing the student's creative thinking and implementing The role of advertising marketing and the possibility of purchasing virtual goods and commercial exchange in the world The digital world, and how the graphic becomes visible through the interactive environment The new technology is a real competitor within the rapidly changing virtual simulation landscape, and the study concluded that New interactive media technologies will allow advertisers to deliver The metaverse experience as consumers, instant feedback and sharing of scientific content, Which will lead to improved effects on the recipient's behavior.

2—Study of [51]

It discussed the evolutionary stages that Metavirus applications went through and their uses in American universities that began in the nineties of the last century, where many achievements were achieved in terms of communication. Simulation, modeling, and development of 3D virtual classrooms, as these perceptions led to, explaining how humans interact socially as avatars in this virtual environment led to confirmation of the effectiveness of metaverse technology in studies at American universities.

3—Study of [55]

It addressed the uses of the metaverse in the education sector and whether it is a blessing or curse. The study conducted a systematic review of the literature on metaverses in education. Bibliometric content analysis was applied to uncover research trends, focus, and limitations of the topic of metaphysics in education. The results showed that there are some studies that focus on mobile learning, blended learning, and artificial intelligence, but there is a greater scarcity of using metaverse technology in educating students, and they recommended that care should be taken to employ metaverse technology in developing the skills of ordinary students as well as those with special needs in various aspects.

4—Study of [43]

It discussed metaverse technology in the education sector and pointed out that the use of metaverses in education is increasing day by day. The factors that influenced the historical development of the metaverse in the field of education were discussed. The strengths and weaknesses of using metaverse in education were emphasized; The opportunities it will present and the problems and threats it may face are also examined. Since the metaverse is a new concept, it is emphasized that the Metaverse environment can add a new dimension to the field of educational technology. However, it should be taken into account that the necessary technologies and architectures in this area are not mature enough yet. Therefore, it is necessary to identify appropriate strategies for using the metaverse in the educational field and begin to quantify its impact on a large scale so that the infrastructure for its use matures.

5—Study of [56]

Conducted in South Korea after the COVID-19 pandemic, the study aimed to analyze the experiences and attitudes of the learner-centered education metaverse from a constructivist perspective to determine the extent to which this virtual environment relates to the lives of primary school students. This study also examined how students became the focal point of new educational technologies. After reviewing the literature on the topic, a survey of 336 elementary school students in Korea was administered using 18 items to measure each factor in the metaverse, followed by statistical analyses that included differences of means and an independent sample t-test. The results showed that, on average, 97.9% of elementary school students have experiences with the metaverse, with 95.5% of them believing it is closely related to their daily lives.

2.3 Metaverse and Creative Teaching

The metaverse is closely linked to the imagination; therefore, it is an effective means of achieving creative teaching by flying into imaginary worlds that are compatible with the virtual world provided by the metaverse by experiencing the text and transforming it from a written image into a living visual image that makes the students integrated within the text and creating its events, so the text becomes part of Their lives in a virtual moment, even if they go back thousands of years or go hundreds of years into the future. This technology helps them penetrate time and experience the events of the prose, poetic, or religious reading text.

Creative Teaching

Creative teaching is an educational activity carried out by the teacher following three basic steps: Planning, implementation, and evaluation, with the aim of bringing about a qualitative change in the student's behavior, and the teacher leads through it teaching him a set of creative methods to achieve the goals set for this teaching. These creative approaches are "based on bringing about changes in social processes Educational and socialization and these methods make their way into a living reality, through programs. Educational education based on developing productive thought, stimulating creative thinking, and training in fertile imagination and solutions Creative and innovative problem-solving.

Creative teaching is essentially an activity that reflects what the student must do to achieve, He constructs information and constructs it himself and in his own way, which gives it a meaning that is compatible with his cognitive structure, and he processes it Utilizing all his cognitive and creative potential gives him confidence in his abilities and unleashes his latent energies [20].

Creative learning will not take place in a classroom setting or learning environment [57].

Creative teaching is not available; therefore, the teacher is seen as the primary key in teaching creativity, and he must determine the extent of his creativity in the following three teaching activities: The first is planning creative teaching. Creative teaching requires a flexible teaching plan in which the teacher departs from traditional teaching plans by providing enriching teaching activities in light of creative teaching and the second classroom teaching behavior for creative teaching. The creative teacher creates the appropriate climate for creative activities within the classroom, directs the learners' attention to the learning material, and works to develop some of the learners' qualities and skills, such as the qualities of challenge, curiosity, expression of opinion, interaction with others, and thinking that is characterized by fluency, flexibility, and originality. The third evaluation for teaching. Creative: Where the learner himself prepares the task or problem or has the freedom to redefine it, free creativity assessment tasks relatively from the restrictions of traditional examinations, and determine the nature of the creative output of the assessment task in light of the goal that the learner seeks to evaluate.

Creative teaching is based on the following steps [58].

1—Providing creative preparation.

2—Using methods of Creative teaching.

3—Employing educational methods that stimulate creativity.

4—Supporting creative classroom interaction.

5—Creative calendar application.

Muhammad [59] believes that creative teaching skills, in light of the extent to which the teacher possesses divergent or creative thinking skills, Such as fluency, flexibility, originality, resourcefulness, and sensitivity to problems.

Both [60, 61] explain these skills as follows:

Fluency means the ability to produce a large number of verbal or performance ideas, alternatives, open-ended problems, usages, or synonyms when responding to a specific stimulus. Fluency is the process of recalling information, concepts, and experiences that have been learned and stored in the individual. Fluency has types, including Fluency of forms. Fluency of words or verbal fluency, intellectual fluency or fluency of meanings, and fluency of association.

Originality: It is one of the skills most closely linked to creativity and creative thinking, and it means novelty, uniqueness, and the ability to produce new, unusual ideas, solutions, and proposals.

Flexibility: means producing new ideas by shifting the direction of thinking as required by the situation or stimulus; that is, seeing the problem or situation from different angles. and forms of flexibility include spontaneous flexibility and adaptive flexibility.

Elaboration: It means the ability to add new and varied details that contribute to improving or beautifying simple ideas or ordinary responses and making them more developed, useful, and beautiful.

Sensitivity to problems: It means the ability to see problems and see shortcomings and defects in the situation, environment, things, habits, or systems. Discovering the problem is the first step in searching for solutions to the problem, either by adding new knowledge or introducing modifications and Types of reading texts:

2.4 Classifications of Reading Texts

Ibrer [62] pointed out that there is a problem in the classification of texts and presented a conception of the classification that agrees with the previous ones [63, 64] in some types and differs in others. Classifications of reading texts include several types that can be summarized into narrative, descriptive, informational, persuasive, indicative, and functional types, and they can be addressed through the following divisions:

Firstly—Literary Reading Texts; They Include:

1—Press Report

An art of journalistic writing that presents to the reader a set of facts, information, and opinions about an event or issue and describes the places, times, and circumstances of the event. It differs from a press release in that it is based on factual reconnaissance, research, and analysis supported by facts and evidence. It is sometimes accompanied by illustrative pictures and graphs and consists of the introduction, the body of the report, and its conclusion. The report may be news, address a certain personality, or address a specific opinion.

2—Article

A short piece of prose that deals with the presentation and clarification of one topic, a general idea, or a social, economic, scientific, or political issue. In it, the writer explains the aspects of the issue and its dimensions, expresses his opinion, and supports it with convincing proofs and evidence. This is why the article dominates. The intellectual and interpretive nature It has an introduction, the body of the topic, and a conclusion. The article may be subjective, expressing the writer's emotions and feelings regarding an issue, or it may be an objective article, which follows the path of scientific research in a clear style and with a coherent structure (called "thematic unity").

3—Persuasive-Argumentative Text

It is a type of communicative text that aims to prove a case, persuade a specific idea, communicate an opinion, or seek to modify a point of view through evidence, evidence, and rational proofs, either to persuade an idea, to refute an idea, or to compare different points of view, and give priority to One of them.d improvements to the situation that is the subject of the problem.

4—Activated Text

A type of functional text, the aim of which is to provide the reader with instructions or directives to carry out a specific work, activity, or movement. This is why it may sometimes be called "instructive" or "procedural" text, and the function of speech in this type of text is directive and influential.

5—Informational Text

A type of functional explanatory text that aims to convey information to the reader and provides an explanation for a scientific, social, or literary phenomenon. Therefore, it provides an answer to questions that begin with: How? Or why?

3 6—Biography

It is a literary art that combines story and history and deals with a prominent personality in order to highlight his aspects and reveal the elements of greatness or humility in him. The biographer analyzes the components of the personality that he deals with from the physical, emotional, emotional, and social aspects in order to highlight the human values in him or reveal the aspects of the brilliance that characterize him. The biography may be a self-translated biography or an alter-ego biography of one of the characters.

7—The Short Story

A prose literary art, based mainly on brief, condensed narration, through which the storyteller expresses one idea, one event, and one character or a few characters, and they carry emotional charges, which they evoke in a specific situation.

It consists of the idea, structure or narrative, character or characters, plot, knot, and solution.

When there are multiple characters, events, and details, the story turns into a novel.

8—The Message

A relatively short prose text with a specific topic or purpose is also called a "discourse." The sender often determines the purpose of the message depending on the party to whom he is addressing his message. The message has a clear structure and specific phrases that suit the nature of the relationship between the sender and the addressee. There are Types of messages, Including the Diwaniyah message (official or functional) and the Brotherhood message (civil or personal).

Secondly, Poetic Reading Texts

This includes poems with various purposes, such as praise, satire, wisdom, etc., and in different forms, from vertical poetry to free poetry, or what is known as iambic poetry, and different poetic schools.

Third, religious reading texts:

It deals with sacred texts such as surahs or Qur'anic texts and holy or noble prophetic hadiths.

4 Research Question and Its Answer

Q1: How can Arabic language teachers use metaverse technology to achieve creative teaching of reading text?

In light of what was discussed previously about the metaverse, creative teaching, and types of reading texts, this question can be answered through the following proposed scenario:

4.1 Firstly, the Role of the Teacher in the Creative Teaching of Reading Texts

Before teaching reading texts in the planning stage, the Arabic language teacher prepares the virtual environment that the students will enter through the Meta Virus technology. These environments are YouTube clips that revolve around the atmosphere in which the poetic, narrative, or religious text occurred, such as the reasons for the revelation or the historical background of the issue included in the text. The reader and the teacher will use these clips as an enjoyable introduction to delving into the depth of the reading text.

Then, the teacher designs several avatars (virtual characters) that are appropriate to the nature of the reading text and the characters, information, or issues contained in it so that each avatar virtually represents one of the characters in the text or expresses a point of view from the different points of view in the prose reading text, or plays one of the roles, which is determined by the teacher, such as the role of the narrator, for example, with characters that have religious or moral prohibitions, such as impersonating the Prophet Muhammad or speaking as if he were a god or a devil. Here, one of the avatars can play the role of narrator on behalf of the Messenger or informant of the Lord of Glory when dealing with religious texts. The text can also be dealt with. Poetry starts with the poetic atmosphere or the occasion of the text by displaying a video clip via metaverse technology about the topic of the poem, its environment, and its characters. Then, the students employ virtual characters (avatars) to represent the events and characters. Perhaps the diversity of avatars overcomes the obstacle of a boy playing the role of a girl or vice versa. Here, a male or female avatar can be chosen. He is the one who will play the role, not the student himself, but rather the virtual representation is through avatars. Here, creative teaching methods can be diversified, not limited to virtual representation, but rather, the text can be addressed through cooperative learning. Where virtual learning groups are formed consisting of avatars, and each avatar has a role such as a group leader, writer, researcher, presenter, etc., it is also possible to use the method of discussion between avatars, brainstorming, problem-solving or team teaching, by summoning characters such as scholars or writers according to the subject of the reading text. For example, one of the sheiks can be summoned when dealing with the religious text through the avatar of this character, and the teacher or one of the students will be represented by this avatar in the virtual symposium or discussion. The dead can be summoned, or imaginary characters can be summoned. In the future, the avatar may be a cartoon of something, an animal, or a human character, depending on the characters and information contained in the various types of reading text. The teacher manages the virtual environment that he created and distributes its roles before the beginning of creative teaching by choosing an avatar character that suits the nature of the text and the reading topic that will be covered, so he is like an actor controlling the theater movement.

Metaverse technology and the various characters associated with it (avatars) are employed, and the teacher can also employ artificial intelligence to enrich the virtual environment with scientific details, such as the meanings of some words or displaying some pictures or YouTube clips related to the reading text. Artificial intelligence is used to present some activities designed by the teacher to complete the evaluation process, which may be formative while dealing with text paragraphs or final at the end of the virtual session. Oral evaluation can be sufficient during interaction within the metaverse environment by asking some avatars for questions and answering others. Activities and students' oral answers can be converted into writing by printing them through artificial intelligence technology and the possibility of voice writing via Word or other programs that convert sound into writing that can be printed.

The teacher is keen to activate the role of each student within the metaverse environment so that all students integrate into the text and the reading text turns into an audible, visual, and enjoyable fictional science. It is a process of reviving the text by experiencing its ideas.

4.2 Secondly, the Student's Role in the Creative Teaching of Reading Texts

After the teacher prepares the roles and activities that the students will practice through the virtual characters, this plan is sent to the students in what is known as flipped teaching, where the student is exposed to the content of the reading lesson and the virtual characters proposed by the teacher. Let each student choose the appropriate avatar for the role he wishes to play within the metaverse environment that is related to the topic of the reading text, and the teacher can determine the appropriate avatar for each student. The teacher may leave freedom to the students or specify for them the roles and form in which they will enter the metaverse environment, depending on the educational level or age stage. The higher the level of students and the more distinguished their skills are, such secondary school students, it is better to leave them more freedom than younger students so that the teacher combines the joy of teaching with the development of imagination and creativity among students without deviating from the objectives of the lesson or wasting time on what is not related to the topic of the reading text.

The student must participate effectively and commit to performing the role assigned to him through the avatar that represents the student, whether this role is as a virtual representation of a member of a collaborative team or as a discussant in a symposium, etc. The student must also participate in the process of evaluation and knowledge enrichment that takes place within the virtual environment, in which the teacher outlines its steps with the students before entering the world of the metaverse.

It allows the student to discuss with the teacher before entering the world of the metaverse to develop creative teaching of the reading text by adding characters, changing the avatar, adding various activities or exercises, or switching roles with his colleagues.

Through this perception, we find creativity present in all steps and elements. The teacher is creative in designing the virtual environment and may seek the help of educational technology specialists in the school. He is also creative in planning for preparation and introduction, and he is also creative in formulating a virtual acting scenario and distributing roles to avatars or determining the role of each avatar in the various teaching strategies that can be applied to reading texts. The teacher is also creative in preparing and implementing training and evaluation activities using artificial intelligence in the metaverse environment while teaching. Creativity is not limited to the teacher. The student participates with the teacher in the creative teaching process. He may suggest some ideas or change the form or role of the avatar, and he may add to his role played by the avatar, and he has the freedom to dialogue and discuss during training and evaluation. It is an integrated, entertaining, creative process that takes into account fluency skills through the multiplicity of information and answers, takes into account flexibility through diversity in personalities and roles and differences in viewpoints among students, and takes into account originality in what the teacher and students create. New or rare ideas, taking into account creativity and sensitivity to problems by developing scientific positions and additions using artificial intelligence and showing the negatives and positives through contradictory personalities and conflicting positions on the issues of the prose, poetic, or religious reading text. The creative process here is not limited to the cognitive skills of creativity but also includes the psychological aspects. It takes into account imagination by employing virtual characters in the metaverse environment and takes into account the challenge of difficulties, the love of curiosity, uncovering mysteries, and adventure in limitless worlds. All of this will be reflected in the students' mentality and personalities in the direction of developing their talents and aspects of creativity, in addition to deep understanding and accurate knowledge of the details of academic subjects in general. And reading texts in particular.

5 Recommendations

- The Ministry of Education provides schools with devices and programs for metaverse technology.
- The Ministry of Education trained Arabic language teachers and other teachers to employ metaverse technology in teaching various academic subjects, including branches of the Arabic language, especially reading and texts.
- The Ministry of Education provides training courses for school students on using metaverse technology and employing it in learning academic subjects.
- Researchers provide more educational research on the effectiveness of using metaverse and artificial intelligence in developing various skills.
- Teachers pay attention to creative, entertaining teaching and focus on developing imagination and thinking more than memorization.

6 Research Suggestions

The research suggests presenting the following studies in light of the vision presented by the research:

- The effectiveness of using metaverses in developing reading comprehension of texts.
- The effectiveness of the metaverse in developing linguistic creativity and nurturing the literary gifted.
- The effectiveness of using the metaverse in developing imagination and positive attitudes toward learning among students
- The effectiveness of using avatars and artificial intelligence in the metaverse environment in teaching the branches and arts of the Arabic language.
- The effectiveness of using avatars in teaching in treating shyness among students.

- The effectiveness of using avatars in developing students' acting talents and dialogue skills.
- A training program on employing metaverse and artificial intelligence in teaching.

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Metaverse in Supply Chain Management: Predicting Suppliers' Intention to Use Metaverse for Educating Suppliers Through Perceived Usefulness, Training Value and Ease of Use (A Case Study in UAE)

Fanar Shwedeh, Said A. Salloum, Ahmed Aburayya, Parveen Kaur, Ibrahim Mohammad, Mansura Mazharul, Brihan Fatin, Mohamed Ahmed Elbadawi, and Zainab Al Ghurabli

1 Introduction

In the coming ten years, AI has the potential to upend and transform a wide range of industries [1–8]. However, AI adoption rates in technology developed countries, such as the United Arab Emirates, are relatively high [9–12]. Driven by the potential of AI technology to revolutionize sociotechnical systems, and by the dearth of systematic research on AI studies from the standpoint of information systems [13–17], the authors put forth the following research question: "What is the impact of educating managers in AI adoption on the decision making to developed country-based organizations". This paper specifically aims to clarify the influence of educating managers in adopting Artificial Intelligence on managerial decision making in UAE companies.

F. Shwedeh · A. Aburayya (✉)
College of Business, MBA Department, City University Ajman, Ajman, UAE
e-mail: amaburayya@gmail.com

S. A. Salloum
School of Science, Engineering, and Environment, University of Salford, Salford, UK

Health Economic and Financing Group, University of Sharjah, Sharjah, UAE

P. Kaur · I. Mohammad · M. Mazharul
Marketing Graduate, College of Business, HRM Department, City University Ajman, Ajman, UAE

B. Fatin · M. A. Elbadawi · Z. Al Ghurabli
College of Business, HRM Department, City University Ajman, Ajman, UAE

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It's crucial to understand Artificial Intelligence (AI) before exploring how the business sector is being impacted by these technologies [9, 18–20]. Any sort of planning, problem-solving [21–24], and learning activities performed by computer software [25–29] are referred to as (AI) [30–33], The phrase (AI) generally excludes several applications; while this is technically correct, it neglects key information, to find out what sort of AI is most prevalent in business, we need to do further research [19]. Let's examine how (AI) affects companies in this area and how it benefits personnel officials [34–37]. AI may help in recruiting by analyzing information from a variety of sources to pinpoint candidates with the greatest attributes [38–42]. It might assess feedback from workers to determine those most likely to depart the organization based on their degree of work satisfaction [35]. Client participation is vital to any B2C (Business to Customer) company, yet it is anticipated that in the next years, (AI) will transform these customer support activities [43]. By removing the requirement for a human to precisely detect a customer's tone, AI-powered software solutions such as sentiment analysis technology, let companies react to complaints, questions and concerns more quickly. After all, clients who are pleased with a business are more likely to want to recommend it to others both online and off [1, 43]. If you work in branding, you are aware of how essential it is to strike a balance between operational efficiency and client experience [44]. Using intelligent technology solutions is one of the most effective strategies to optimize both. Don't throw away client feedback or other measurable answers. AI system assess them at capacity, needing neither you nor your staff to perform any human tagging, and centralize the results for easy access [45].

The San Francisco startup OpenAI, which collaborates closely with Microsoft, has developed a number of technologies, including AI, which debuted on November 30th, 2022. It is part of a new generation of artificial intelligence (AI) systems that can converse, compose comprehensible prose on demand, and generate original images and videos by drawing on information stored in a large digital library [46] using natural language processing techniques. Although its primary purpose is to mimic human interaction, AI is versatile enough to also produce music, teleplays, fairy tales, and student essays in addition to computer programs and debug them. Depending on the exam, it can answer questions at a higher level than the average person and even write poetry and come up with music lyrics. The AI curriculum covers topics such as man pages, internet culture, and popular programming languages like Python and bulletin board systems (2023). Supply chain management is the process of coordinating the multiple activities involved in the production and distribution of a product or service to a customer. Supply chain operations can include anything from designing and farming to manufacturing and packing and shipping. The study's overarching goal is to educate readers and researchers about today's exciting technology [47]. Profitability can be increased through better supply chain management by increasing customer satisfaction and decreasing operational expenses. When costs are managed well and minimized whenever possible, profits thrive. When the price of inputs like labor and materials decreases, the cost of production follows suit. Editorial Staff of Indeed, 2023. In sum, there are many ways in which businesses may increase their bottom lines by better managing their supply chains. This is especially true for large, internationally active business [48]. AI has exposed both enterprises and individuals to an entirely new category of products that demonstrate the power and promise of AI to everybody [49]. AI allows for data analysis and insights into logistical operations. By analyzing data on shipment durations, delivery routes, and other logistics-related characteristics, AI may aid logistics organizations in finding areas for improvement and optimizing their operations. Expenses can be reduced, efficiency can be enhanced, and output may be increased [50]. The rapid growth in AI s popularity indicates that we need to learn much more about it and comprehend how this cutting-edge technology may advance Supply chain management in the UAE. This paper will be an insightful foundation for discovering more about AI in Supply chain management and other business sectors in general. This study aims to investigate the main factors of educating managers in adopting and using AI for managerial decisions. Although what has been stated previously, it will still be challenging as there is a lack of research in studding this gap and there is no enough literature nor information about how educating managers in AI applications and technologies will advance over the decisions in next decades. This study will fill the research gap in this field.

2 Literature Review

One of the most important aspects of managerial decisions is forecasting demand and planning accordingly [4, 51–53]. AI can analyze large amounts of data, including sales history, weather patterns, and consumer behavior, to provide more accurate demand forecasts [54–58]. Managers involve coordination between multiple parties, including suppliers, manufacturers, distributors, and retailers. AI can facilitate communication between these parties by providing real-time translation services and enabling efficient collaboration. Managers often involve making complex decisions in real-time [59–61]. AI can analyze data and provide recommendations based on the latest trends and insights, allowing managers to make more informed decisions. AI can help identify inefficiencies in the supply chain and suggest ways to optimize it. For example, it can identify areas where automation can be implemented to reduce costs and increase efficiency. Overall, AI has the potential to revolutionize managerial decisions by providing real-time insights, facilitating communication, and improving decision-making capabilities. However, its effectiveness will depend on how well it is integrated into existing management systems and processes and to which extent managers are educated in adopting the relevant AI Applications to their businesses. It is important to educate users of AI adoption in order to link it to their business activities and train on making decisions by depending on it.

AI is a useful tool that can be used in management decision making in order to automate the whole process which can deliver tasks on its own with little supervision. It is used to produce insights and facilitate communications. It helps in collaborating between different stakeholders as well. The AI chat bot is undoubtedly a transformative step forward for generative AI which is trained to extract data from various sources and make a systematic order [62]. It detects word patterns in order to predict the next word which works like a mind-reading technology in organizational performances. It informs the company when it is time for redesign its working strategy in real-time. AI can predict the market which helps the company take necessary and valuable decisions. Trained with over 570 GB of data gathered from all over the internet and almost 300 billion words, AI has a vast data lake which is helpful to provide the necessary solutions to companies [10].

AI can take any business to the next level. It is a type of incorporating advanced technologies which help businesses in various ways. It can transform the way a company communicates with customers with the help of an AI-powered applications. With the help of many technologies, companies can change how employees interact with their audiences. It also helps in personalizing customer interaction in order to make streamline the communication process [61]. It increases proficiency and becomes more effective in order to navigate the business operation. The company and the employees benefit from this technology and the customers. It helps companies achieve their business goals to become more proficient in the industry. AI can be integrated into the existing system to get the best out of it. AI can write essays, marketing copy, emails, articles and codes [63]. It generates ideas on its own in order to implement them in the business operation to make it more reliable for the audiences. It recommends services and products based on search engines and research keywords. It can translate text into 95 languages [63]. It finds data and analyses it. All these elements help companies in various ways so the employees can be relaxed by relieving work pressure. The level to which managers are educated in adopting AI technologies to develop managerial decisions is studied in this research paper. This study reflects if there is a positive relationship between the education of managers in adopting managerial decisions and the development of decision making.

3 Forecasts and a Conceptual Model

Fred Davis, with his revolutionary "Technology Acceptance Model (TAM)," has made major contributions to the study of how individuals decide whether to embrace and utilize new technology. The paradigm's two core pillars, perceived usefulness and simplicity of use, are often. Viewed as critical mental characteristics that contribute to the spread of technology [12, 64, 65]. According to [66], the study's performance will be reviewed, with both being more inextricably linked to the former. The word "perceived value" defines how consumers feel about a service's price and advantages. Maximum utility is often evaluated in terms of perceived worth, with the assumption that benefits exceed costs. Utility theory, which predicts future usage, underpins perceived worth [67]. How much value is thought to be provided influences how pleased and loyal consumers are. It has a significant influence on future technology adoption and use plans [67, 68]. Given this data, the following hypothesis are to be examined:

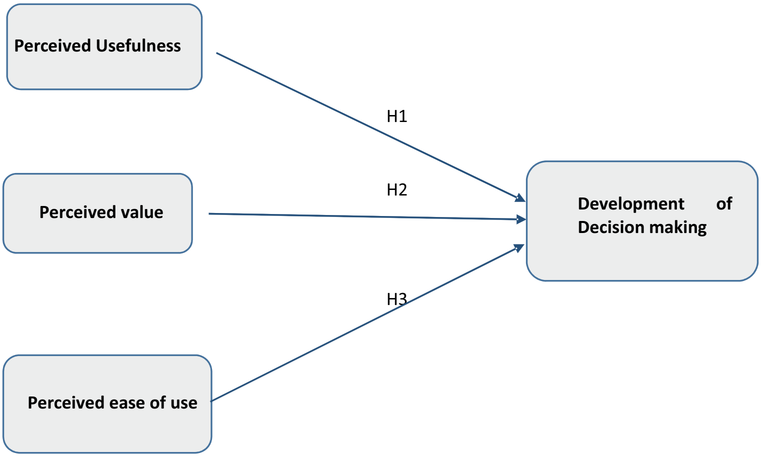

Fig. 1 Model of the research

H1: The perceived usefulness (PU) of educating managers in adopting AI technology influences development of managerial decisions (MD).

H2: The Perceived Value (PV) of educating managers in adopting AI technology influences development of managerial decisions.

H3: The perceived ease of use (PEOU) of educating managers in adopting AI technology influences development of managerial decisions (MD).

Figure 1, shows the conceptual outline and how these assumptions were used to create the proposed model.

4 Research Methodology

Within this concise-analytical study, a cross sectional design served as a deduction method. The info collecting tool proved an online-based survey designed to be self-administrated and utilized to gather details about educating managers in adopting AI Applications in organizations in the UAE, notably the nation's capital of Abu Dhabi. This research examines took place in many organizations in Abu Dhabi city at the UAE with the participation of seven main departments and five sections. The collection of data consumed one month, which ran from the 14th Jan. 2023 the 14th Feb. 2023. Using government issued messages and online communication sites consisting of WhatsApp, staff acquired access to the questionnaire's connection in order to gather facts, many different government staff contacted, including administrators (registration, excellence, greeters, and managerial abilities endorses) when the staff (managers, supervisor, directors, executives) employed at the chosen organizations and department's centers. This was done in order to gather important details regarding the development of management decisions throughout those establishments, as proposed by [69, 70].

As part of the research, a convenient selection approach passed, which utilizes a systematic random sampling approach. The primary justification for that decision, in overall, proved the stringent regulations in place in the city's management facilities for safeguarding of employee data in addition to maintaining its safety and confidentiality. The choice of that approach was further affected by the main services facilities that influenced their rules on obtaining the samples. In addition, [71] showed that quick sampling, which enables utilization of large quantities of data, is a sampling approach that makes both economical and efficient.

4.1 Data Collection

In this research, 75 of the 400 questionnaires were eliminated because the answers appeared incomplete. The remaining 325 surveys were complete which are 81% responses used for the analysis of this research. Stated by [9, 31, 72–74], in the scenario at hand, modeling with multiple regression analysis is permitted because 325 samples size appears to be significantly over the minimum limit. The hypotheses are validated using the framework. It is important to keep in esprit that while the hypotheses are based on current ideas, they ought to be frequently organized in accordance with the guidelines for development of managerial decisions. Multiple regression analysis is used to evaluate the investigated variables.

4.2 Study Instrument

A questionnaire evolved included 20 questions which make up the inquiry of measuring the impact of the three independent variables listed in the model. Each of the queries was amended and modified to maximize the study relevance.

4.3 Personal/Demographic Information

Table 1 shows the statistics of demographic data. Male Suppliers outnumbered female pupils by 50.6% to 49.4%, respectively. 59.7% of respondents were between the ages of 20 and 40 years old, while 40.3% were beyond the age of 4o. The majority of responders were qualified and held university degrees. Furthermore, 79.8% of people are postgraduates, and 20.2% undergraduates. Additionally, 40.4% of people are single and 39.3% are married but a small margin of people 20.2% are widows (Tables 2–4).

Table 1 Presents demographic data collection of the participants of this study. Demographic statistics

		Frequency	Percent	Valid percent	Cumulative percent
Gender					
Valid	Male	90	50.6	50.6	50.6
	Female	88	49.4	49.4	100.0
	Total	178	100.0	100.0	
		Frequency	Percent	Valid percent	Cumulative percent
Age					
Valid	20–30	54	30.5	30.3	30.3
	31–40	52	29.2	29.2	59.6
	41–50	54	30.1	30.3	89.9
	51–60	18	10.2	10.1	100.0
	Total	178	100.0	100.0	
		Frequency	Percent	Valid percent	Cumulative percent
Marital status					
Valid	Single	72	40.4	40.4	40.4
	Married	70	39.3	39.3	76.8
	Widowed	36	20.2	20.2	100.0
	Total	178	100.0	100.0	
		Frequency	Percent	Valid percent	Cumulative percent
Education					
Valid	Under-Graduate	36	20.2	20.2	20.2
	Post-Graduate	142	79.8	79.8	100
	Total	178	100.0	100.0	

Table 2 Presents reliability statistics. Reliability Statistics

Cronbach's alpha	N of items
0.840	5
Cronbach's alpha	N of items
0.836	5
Cronbach's alpha	N of items
0.731	5

5 Discussion and Findings

This study demonstrated how Educating managers in AI Adoption has a favorable impact on building a managerial decision, as evidenced by our hypothesis and the survey results [31, 75, 76]. H1 is testing the impact perceived usefulness of educating managers in adopting AI on the development of making a decision. Our most useful

Table 3 Presents the principal component analysis. Extraction method: Principal component analysis

	Extraction
Using AI Applications to learn tasks in my job	0.884
Using AI Applications enhances my knowledge	0.909
Using AI Applications to make some decisions	0.833
Using AI Applications to improve my performance	0.865
Using AI Applications makes my job easier	0.716
How satisfied you are with the overall experience of using AI in developing decisions	0.815
Having AI Applications available is important to assist in making decisions	0.982
Does AI Application meet managerial needs	0.845
Does AI Application accelerates making decisions	0.818
How easy was it to use AI Applications in developing decisions	0.819
My interaction with AI Applications is clear	0.536
AI Applications are flexible for my job	0.895
AI Applications are easy to use	0.900
AI Applications are easy to learn decisions	0.869
Learning to operate AI Applications is easy	0.848

hypothesis was the Perceived usefulness as shown by the results of the survey which is about how useful the education of AI Adoption is in their work with an extraction that is from (0.716) to (0.909). This indicates that educating managers with AI Applications eases the development of making a managerial decision. H2: The Perceived Value talks about how satisfied the managers are with the results they get of educating in the Adoption of AI to build up their decisions as the extraction is (0.815 to 0.982). This reflects the positive relationship between educating managers in AI adoption and the benefits and values on their abilities to develop managerial decisions. H3: In the third hypothesis, Perceived Ease of Use reflects how easy it is for managers to learn how to Adopt AI in their workplace to take a certain decision. The extraction result is from (0.536) to (0.900) which indicates in impact of the perceived ease of use on managerial decision development.

6 Significance and Limitations

This research is adding the educational concept to managers who make managerial decisions. The TAM model is used in adding the training of managers' adoption of training on AI Applications in a functional methodology with consideration to the impact of perceived usefulness, perceived value and perceived ease of use on developing decisions. This study is significant in the idea of highlighting the influence

Table 4 Presents the analysis results. Model summary. a. Predictors: (Constant), EOU, PV

Model		R	R square	Adjusted R square	Std. error of the estimate	
Model						
1		0.906[a]	0.821	0.818	0.42498	
ANOVA[a]						
a. Dependent Variable: Level of satisfaction with the performance of Metaverese technology in supply chain						
b. Predictors: (Constant), EOU, PV						
		Sum of squares	Df	Mean square	F	Sig
Model						
1	Regression	144.529	3	48.176	266.743	0.000[b]
	Residual	31.426	174	0.181		
	Total	175.955	177			
Coefficient[a]						
a. Dependent Variable: Level of satisfaction with the performance of Metaverese technology in supply chain						
		Unstandardized B	Coefficient std error	Standardized coefficient beta	t	Sig
Model						
1	(Constant)	1.019	0.178		5.454	0.000
	PU	0.270	0.072	0.238	3.775	0.000
	US	0.708	0.111	0.572	6.350	0.000
	PE	0.892	0.103	0.544	8.630	0.000

of training managers in adopting AI applications to make decisions which non-studies have discussed earlier in literature. The study is conducted in organizations in the UAE. Results might be different in other countries. This gap can be filled by researchers for future search work enrichments.

7 Conclusion

The adoption of AI in training managers to develop managerial decisions is going to an impact on decision makers in many fields as the global technology, instructive, and financial sectors. The emerging immersive environment, not the web, will be the catalyst for innovative training of the managers who will employ AI adoption in developing their decisions. This can be added as a phase of making decisions strategies. This study investigated manager's views and motives about learning in adopting AI applications with the possible educational advantages of these applications of the world structure in view. The managers' views on the utility and viability of the AI

Applications had a significant impact on their propensity for creation of managerial decisions. In order to explore the concepts concerning technological acceptance, it is suggested that consumers' perceptions of the latest technological feature and usability have a big impact on the extent to which it is accepted. These outcomes are consistent with the findings from previous studies. It might additionally reflect how managers perceive regarding modern AI applications technologies. The current inquiry has some restrictions. A theoretical framework might require a lot of work since it merely takes two important variables into account: a person's degree of innovation and the happiness they get when they come up with best ways to solve problems and make a decision. Secondly, since the survey had been advertised via the internet and on the social media, additional managers might be likely to take place. Lastly, benefit is the adaptability of the AI applications.

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Empowering Education Through the Internet of Things (IoT)

Saada Khadragy

1 The IoT Ecosystem in Education

The IoT ecosystem encompasses a network of interconnected devices [1–5], sensors, and systems that communicate and share data in real-time [6–11]. In education, this translates to a dynamic environment where smart devices collaborate to create a more responsive and efficient learning ecosystem [9, 12, 13]. From smart classrooms and interactive whiteboards to wearable devices and educational apps, the IoT is revolutionizing the traditional classroom setup [14–16]. These interconnected devices form a web of information exchange, facilitating a seamless flow of data between students, teachers, and the learning environment. The concept of a 'smart school' is emerging, where everything from attendance tracking to personalized learning experiences is facilitated by the IoT [17–21].

2 Smart Classrooms: Enhancing the Learning Environment

Smart classrooms leverage IoT technologies to create interactive and adaptive learning spaces [11, 22, 23]. Sensors monitor environmental conditions, adjusting lighting and temperature for optimal comfort [24–28]. Smart whiteboards facilitate collaborative learning, allowing students to interact with content in real-time [29–33]. Furthermore, IoT-enabled projectors and audio systems provide immersive multimedia experiences, transforming conventional lectures into engaging sessions

S. Khadragy (✉)
College of Arts and Science, Umm Al Quwain University, Umm Al Quwain, UAE
e-mail: drsaada.khadragy@uaqu.ac.ae

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[34–41]. The implementation of IoT in smart classrooms is not merely about introducing technology for its own sake [11, 14, 42, 43]; it's about creating an environment that responds dynamically to the needs of both students and educators. Real-time analytics generated by IoT devices offer insights into teaching methodologies, enabling continuous improvement in the learning process.

3 Personalized Learning Paths

One of the most significant contributions of IoT to education is the ability to tailor learning experiences to individual needs [34, 44–46]. Smart devices can collect and analyze data on students' progress [47–50], learning styles, and preferences. This information enables educators to create personalized learning paths, ensuring that each student receives a customized education that aligns with their strengths and challenges [9, 51–53].

The journey of a student through the educational system becomes a personalized roadmap [37, 54–56]. IoT-driven data analytics identify areas where a student may need additional support, allowing for timely interventions. Conversely, for those excelling in certain subjects, advanced materials can be provided to maintain engagement and foster a love for learning.

4 Wearable Technology in Education

Wearable devices equipped with IoT capabilities are redefining the way students and teachers engage with educational content [57–60]. From smartwatches that monitor physical activity to augmented reality glasses that provide immersive learning experiences, wearables foster a more interactive and dynamic educational environment. Real-time feedback and data collection enable educators to make informed decisions about teaching methodologies and student well-being. Wearables extend the learning environment beyond the classroom, creating a continuous feedback loop between a student's activities, health, and academic progress. For instance, biometric data from wearables can offer insights into a student's stress levels, allowing for timely support mechanisms.

5 Addressing Challenges and Concerns

Despite its promise, the integration of IoT in education raises certain challenges and concerns [34, 61]. Security and privacy issues must be carefully addressed to protect sensitive student data. Additionally, there is a need for comprehensive training programs to familiarize educators with IoT technologies and ensure effective implementation. Striking a balance between technological innovation and pedagogical principles is crucial to harness the full potential of IoT in education [62]. Security breaches in educational IoT systems can have severe consequences, jeopardizing both student privacy and the integrity of academic records. Therefore, robust cybersecurity measures must be in place, encompassing encryption, authentication protocols, and regular security audits. Educational institutions also face the challenge of investing in the necessary infrastructure to support IoT implementations. This includes not only the devices themselves but also the networking infrastructure to ensure seamless communication between devices and reliable data transmission. Moreover, educators may encounter resistance or skepticism in embracing IoT technologies. Training programs must not only focus on the technical aspects but also on building a culture of acceptance and understanding among teachers, students, and parents.

6 The Future of Education with IoT

Looking ahead, the future of education with IoT holds exciting possibilities [63]. Smart educational systems will continue to evolve, incorporating artificial intelligence to provide even more sophisticated and adaptive learning experiences. The IoT will extend beyond the classroom, fostering connectivity between schools, homes, and communities, creating a seamless learning journey that transcends physical boundaries.

7 Social Implications and Inclusivity

As IoT permeates education, it is imperative to address the social implications and ensure inclusivity [64]. The digital divide must be bridged to provide equal access to IoT-enabled educational resources. Additionally, a thoughtful approach is needed to navigate issues related to data ownership and digital literacy, empowering students and educators alike to navigate the digital landscape responsibly. Ensuring inclusivity involves not only providing access to IoT devices but also addressing the broader socio-economic factors that may hinder access. Collaborative efforts between educational institutions, governments, and technology providers can help develop strategies to make IoT-enabled education accessible to all.

8 Ethical Considerations in Educational IoT

The integration of IoT in education necessitates a careful examination of ethical considerations [45, 65]. Data generated by IoT devices, including personal information and learning analytics, raises questions about consent, ownership, and responsible use. Educators and policymakers must collaborate to establish ethical guidelines that prioritize the well-being and rights of students. Ethical education in the digital age becomes an integral part of the curriculum. Students need to be equipped with the knowledge and skills to navigate the ethical challenges posed by IoT, fostering a sense of responsibility and digital citizenship [66]. Educational institutions, in collaboration with regulatory bodies, must establish transparent policies regarding the collection, storage, and use of data. Privacy concerns should be at the forefront of IoT implementations, ensuring that students and their families are confident in the security of their personal information.

9 Overcoming Implementation Challenges

The successful implementation of IoT in education requires overcoming various challenges [67, 68]. Adequate infrastructure, including high-speed internet connectivity, is essential for the seamless functioning of IoT devices. Additionally, professional development programs for educators are crucial to ensure that they are proficient in integrating IoT technologies into their teaching practices.

Collaboration between educational institutions, technology developers, and policymakers is key to creating a conducive environment for IoT implementation. The sharing of best practices and lessons learned can streamline the adoption process and mitigate potential hurdles. Infrastructure development involves considerations beyond the physical devices, encompassing network capabilities, data storage, and integration with existing educational technologies. Educational institutions should conduct thorough assessments of their technological readiness and invest strategically to build a robust IoT infrastructure. Continuous professional development for educators is paramount. Training programs should not only cover the technical aspects of IoT but also focus on pedagogical approaches that leverage the technology effectively. A collaborative approach involving teachers in the decision-making process and encouraging a culture of continuous learning can enhance the successful integration of IoT in education.

10 International Perspectives on IoT in Education

The integration of IoT in education is not limited to a specific geographic region; it is a global phenomenon [69]. International collaboration and knowledge exchange can provide valuable insights into diverse approaches to leveraging IoT for educational enhancement. Understanding how different countries navigate challenges and embrace opportunities can inform more effective strategies for implementation. Global partnerships in education can lead to the development of standardized frameworks for IoT implementation, ensuring interoperability and consistency in educational practices. This collaborative approach fosters a sense of unity in addressing common challenges faced by educators worldwide. Moreover, international collaborations can facilitate the sharing of resources, best practices, and lessons learned. This exchange of knowledge can contribute to the development of a global perspective on the role of IoT in education and help in creating more robust and adaptable solutions.

11 Economic Considerations and Funding in IoT Implementation

Integrating IoT into education involves significant financial considerations [44]. Acquiring and maintaining the necessary infrastructure, updating technology, and providing ongoing training for educators all require substantial investment. Securing funding from government bodies, private enterprises, and grants becomes crucial to ensure the sustained development and deployment of IoT in educational institutions.

Economic analyses should be conducted to assess the long-term cost-effectiveness of IoT implementations in education. Research on successful funding models and strategies for maximizing return on investment can guide educational institutions in securing financial support for their IoT initiatives. Public–private partnerships can play a crucial role in funding IoT initiatives in education. Engaging with technology companies, telecommunications providers, and other industry stakeholders can bring additional resources and expertise to the table. Governments can also incentivize private investment in educational IoT through tax breaks, grants, or other financial incentives.

12 Cognitive Computing and AI Integration in Education

The future of education with IoT is intricately linked with advancements in cognitive computing and artificial intelligence (AI) [70–72]. As IoT devices generate vast amounts of data, AI algorithms can analyze this data to provide valuable insights into student learning patterns, preferences, and areas of improvement. Cognitive computing, a subset of AI, involves systems that can learn and adapt without being explicitly programmed. In the context of education, this means that AI algorithms can analyze data from IoT devices to understand how students learn best and recommend personalized learning paths. AI-driven applications in education can include intelligent tutoring systems, adaptive learning platforms, and automated grading systems. These applications can provide timely feedback to students, identify areas where additional support is needed, and offer personalized learning experiences. The integration of AI in education also raises ethical considerations. Transparency in how AI algorithms make decisions, avoiding biases, and ensuring that AI is a tool for educators rather than a replacement are crucial aspects to address.

13 Conclusion

The Internet of Things is ushering in a new era in education, where connectivity, adaptability, and personalization are at the forefront [73]. As we navigate this transformative landscape, it is imperative to approach the integration of IoT in education with a thoughtful and ethical mindset, ensuring that the benefits are maximized while addressing potential challenges. The journey towards a smarter, more connected education system is underway, promising a future where learning is truly limitless.

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Towards Reliable Utilization: An Instructional Design Model for Integrating Generative Pre-trained Transformer (GPT) in Education

Moustafa Kamal Moussa

1 Introduction

In recent years, Artificial Intelligence (AI) has witnessed a remarkable surge, resulting in a multitude of applications and tools that have gained global popularity across various domains [1–5]. One field profoundly influenced by this AI growth is education [5–8], presenting extensive opportunities encompassing personalized learning, streamlined instructional tasks, and AI-driven assessments [3, 9–11].

Educators today face unmet challenges in teaching and learning, prompting them to seek technological-enhanced approaches that are both safe, effective, and scalable [8, 12, 13]. They examine the potential of rapid technological advances, acknowledging their everyday reliance on AI-powered services, such as voice assistants, grammar correction tools, and automated planning applications [14–18]. This exploration extends to educators actively engaging with newly released AI tools, recognizing opportunities for leveraging AI capabilities like speech recognition to enhance support for diverse student populations, including those with disabilities and multilingual learners. Their endeavors encompass AI's role in lesson improvement and content adaptation, reflecting a broader quest for innovation and personalized learning in digital educational tools. [19].

Within the AI domain, one of the latest technological phenomena is the Generative Pre-Trained Transformer (GPT), a member of the Large Language Models family. Emerging from this AI landscape is ChatGPT, an AI software capable of generating text responses to queries, designed with a focus on natural language understanding and generation within conversational contexts. ChatGPT's versatility

M. K. Moussa (✉)
Professional Postgraduate Diploma in Teaching Department, City University Ajman, Ajman, UAE
e-mail: m.moussa@cu.ac.ae

Educational Technology Department, Faculty of Specific Education, Ain Shams University, Cairo, Egypt

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extends to applications such as chatbots and virtual assistants, offering human-like text generation that lends itself to automating conversations and providing assistance [20]. The rapid evolution of AI software, illustrated by ChatGPT and its successor, GPT-4, introduces a host of new challenges while simultaneously holding vast potential for reshaping student learning and career preparation. However, effective integration of novel tools like ChatGPT into the classroom necessitates meticulous consideration of ethical, cheating, and equity concerns, akin to the careful approach adopted when integrating other technologies into educational settings [21].

Abramson [21] also emphasized that adopting the right approach in the integration of ChatGPT into educational settings is essential. When used judiciously, ChatGPT can prove to be a valuable tool and, as some psychologists contend, even a revolutionary one, in preparing students for their future careers [21]. In navigating these educational frontiers, instructional design emerges as a linchpin for unlocking the full potential of ChatGPT in education. Instructional design's pivotal role extends beyond simple content creation; it encompasses the thoughtful evaluation of how students learn and determines the most effective resources and approaches to facilitate academic achievement [22]. In light of these considerations, this study aims to develop a comprehensive instructional design model to guide educators in leveraging the potential of GPT-based AI in education effectively.

2 Literature Review

2.1 *Artificial Intelligence in Education*

Artificial Intelligence (AI) has rapidly gained prominence across various fields, and education is no exception to its transformative impact [23–30]. While AI has prominently made its mark in science, engineering, and technology [31–34], its presence is also evident in education through the integration of machine-learning systems [35–39] and algorithmic innovations [3, 40–43]. This integration of AI holds the potential to revolutionize the landscape of education by fostering greater autonomy for both students and instructors, ultimately creating a more engaging and interactive learning environment [26, 44–47].

For more investigation of this potential, numerous studies have delved into the application of AI in education Luckin and Cukurova [48] in their research underscore the significance of aligning AI technologies with evidence-based practices rooted in Learning Sciences research. By emphasizing interdisciplinary collaboration between AI developers, educators, and researchers. Furthermore, they explore the promise of AI in enhancing collaborative learning, spaced practice, and emotion prediction within educational contexts.

As for the ethical side of using AI in education along with its advantages, Akgun and Greenhow [40] delve into the realm of K-12 education to investigate the application of AI. Their research emphasizes both the advantages and ethical challenges associated with AI adoption. They provide a comprehensive definition of AI and offer instructional resources aimed at navigating ethical considerations. This research serves as a valuable resource for educators looking to effectively integrate AI while addressing ethical concerns in K-12 settings.

From another aspect, Kim et al. [49] shift the focus to teachers' perspectives on student-AI collaboration (SAC) within K-12 education. Their study identifies optimal learning goals, which include capacity and subject-matter knowledge building, and underscores the importance of interdisciplinary learning and process-oriented assessment. They anticipate students' progression through three SAC stages: learning about AI, learning from AI, and learning together.

The preceding studies align with the current research in underscoring the significance of integrating AI into education while maintaining a careful balance between its advantages and disadvantages.

2.2 ChatGPT as a Model of Generative AI

The emergence of Generative AI technology, such as GPT, has further enriched the educational field. This technology and its tools, including ChatGPT, are seen as advanced applications of AI, gaining widespread recognition for their versatile applications across various domains, including education [50]. ChatGPT as a deep learning model, undergoes pre-training on extensive text data sets and can subsequently be fine-tuned to excel in specific tasks, such as language generation, sentiment analysis, language modeling, machine translation, and text classification [51].

For educational settings, the emergence of AI applications like ChatGPT holds the transformative potential to reshape students' approaches to their academic pursuits and the landscape of education as a whole. Existing literature has underscored the capacity of AI technology to enhance and enrich the learning experience [52].

ChatGPT, powered by the GPT-3 language model, has demonstrated its potential to revolutionize self-directed learning and open education. Its proficiency in comprehending and responding to natural language inputs, as well as its capacity to provide personalized and interactive assistance, positions it as a promising tool for enhancing learner autonomy and engagement [53]. The Impact of generative AI on education is emphasized by its potential to address long-term educational challenges. While acknowledging its massive potential, a methodology guiding its use in education is needed. This calls for an enhanced regulatory framework and ethically sound solutions to manage the quick technological advances in this field [54].

As AI technologies like ChatGPT continue impacting the educational field, academic institutions and educators around the world must take precautions when implementing these technologies in education. To preserve the ethical and advantageous integration of these technologies into educational settings, strategies for preventing students from abusing their power must be developed [55]. To further explore the educational benefits of ChatGPT and provide a structured approach, Su and Yang [56] offer a framework for incorporating generative AI in education. They explore potential advantages, limitations, challenges, and ethical considerations associated with ChatGPT integration in educational settings. This framework, referred to as the "IDEE" framework, provides four crucial steps for implementing generative AI in education. These steps encompass the identification of specific educational objectives as the first stage, followed by the determination of appropriate levels of automation, addressing ethical considerations, and concluding with the evaluation of the effectiveness of AI integration.

Aligned with these discussions, the current study aims to fulfill this need by providing a comprehensive model that encompasses this technological development. The goal of this model is to reposition it as a controllable and beneficial tool, rather than a potential risk or resource with unclear outcomes.

2.3 Instructional Design

Instructional design plays a fundamental role in the effective implementation of educational technologies like ChatGPT. It is essential to understand the principles of instructional design and how they translate into instructional materials, activities, resources, and evaluation. This systematic and learner-centered approach focuses on enhancing performance and producing measurable, safe, and valid results [57]. Currently, numerous instructional design models have been presented, and categorized in different ways, offering a multitude of advantages within educational environments [58].

It is clear that educational theories and instructional design models are essential to the pursuit of successful instructional design. Khalil and Elkhider [59] emphasize how crucial it is to include these components when organizing learning activities. Their research highlights the value of instructional design above and beyond conventional approaches, emphasizing its role in producing engaging and effective instruction. Teachers can regularly produce dependable and empirically sound instruction, improving the learning process overall, by implementing instructional design principles based on educational theory.

Yildiz and Uzunboylu [58] provide a thorough analysis of several models in the field of instructional design models using eight distinct criteria. Their study attempts to help teachers and instructional designers choose the best model of instructional design for particular types of classrooms. The study highlights the benefits and drawbacks of each category—class-oriented, product-oriented, and system-oriented—for instructional design models. It also emphasizes how crucial it is to take into account variables like time commitment, expertise levels, and flexibility when choosing an instructional design model that is appropriate for a given learning environment.

Even though the literature currently in publication examines the possible advantages, restrictions, and ethical considerations of using ChatGPT in education [53, 54]. A thorough model must be created for the smooth integration of this generative AI technology in education. A notable contribution is made by Su and Yang [56] with the introduction of the "IDEE" framework to guide effective AI applications in educational settings. Nonetheless, there remains a research gap in the form of a thorough model that offers educators a methodical way to fully utilize GPT to improve learning experiences while taking ethical issues into account. The current study provides a comprehensive model that integrates instructional design principles with GPT integration in order to bridge this gap. With the help of this model, teachers can employ generative AI in the classroom in a way that is both morally and practically sound.

According to the aforementioned literature, the objective of the current research is to develop a comprehensive instructional design model for integrating GPT technology in education. Thus, the current study seeks to answer the following research question: what is the instructional design model for integrating GPT into education?

3 Methodology

This study employs a thorough research methodology that blends descriptive analytical and research and development (R&D) approaches in order to address the research question. The first step entails a comprehensive examination and assessment of earlier studies pertinent to the incorporation of AI in education. The current study is underpinned by a literature review, which enables the researcher to leverage pre-existing knowledge and expand upon it. Efficient examination and analysis of the existing literature are achieved through the application of the descriptive analytical method. It entails a methodical examination of prior research to pinpoint important findings, benefits, drawbacks, and new developments in the field of artificial intelligence (AI), particularly with regard to GPT integration in educational settings.

Research then shifts into the research and development (R&D) method after the descriptive analytical stage. Systematic efforts to advance knowledge and apply it to the creation of novel goods, services, or processes are included in research and development [60]. The R&D approach is used in this study to create a reliable and useful model for incorporating GPT into educational environments.

By combining the benefits of innovative development and descriptive analysis, this integrated methodology makes sure that the final model takes into account the unique opportunities and difficulties that come with integrating GPT into education in addition to being grounded in the most recent research. It permits the integration of findings from earlier studies, the use of best practices, and the creation of an all-encompassing framework that encourages the ethical and efficient use of GPT technology in learning settings.

Table 1 Model prototype phases and steps

Phases	Steps
9	22

4 Procedures

4.1 Generative Pre-Trained Transformer Instructional Design (GPTID) Model Development

The development of the Generative Pre-Trained Transformer Instructional Design (GPTID) Model included a structured process consisting of three stages.

4.1.1 First Stage: Model Prototype

To create the basic framework of the GPTID Model, the researcher first thoroughly reviewed relevant literature and current instructional design models. The resulting model consists of nine phases, each of which consists of a set of consecutive steps, for a total of twenty-two discrete steps (Table 1).

4.1.2 Second Stage: Model Validation

A comprehensive model validation procedure was precisely conducted in the second phase of GPTID Model development in order to ascertain the model's general efficacy, applicability, and dependability. Experts in the field of educational technology and instructional design were interviewed in-depth as part of this validation process. They offered thoughtful recommendations that addressed many of the components of the instructional design model, along with thorough evaluations and insightful remarks. These included recommendations for step reduction, step merging considerations, step elaboration ideas, step description clarification, phase title optimizations, and a thorough assessment of step functionality.

In order to preserve objectivity and reduce any possibility of personal bias, expert input was meticulously gathered, guaranteeing a fair evaluation procedure. Furthermore, a thorough content analysis was done in order to extract valuable information from the opinions of the experts.

4.1.3 Third Stage: Model Refinement

During the third phase of the model development process, the emphasis was on integrating the comments and understandings from the expert reviews carried out in the preceding phase. The objective of this phase was to improve the model's overall

Table 2 Refined model phases and steps

Phases	Steps
7	29

alignment with the requirements of GPT integration in educational settings, as well as its accuracy and effectiveness. With seven precisely planned phases and a total of 29 painstakingly stated steps, the refined model demonstrated a comprehensive framework that is well-suited to enable the smooth integration of GPT into educational contexts (Table 2).

5 Results

The study's findings led to the development of a sophisticated and thorough instructional design model for incorporating Generative Pre-Trained Transformer (GPT) technology into the classroom. The GPTID Model was developed in its final form through a painstaking process of needs analysis, expert validation, and iterative development. Each of the seven phases and 29 steps in this model has been thoughtfully created to assist educators in realizing the full potential of GPT to improve the learning process. The resulting model gives teachers a methodical and flexible framework to design effective and engaging learning experiences that make use of GPT's capabilities, guaranteeing its responsible and constructive use in educational settings. The phases and steps of the GPTID Model are described in detail in the next section.

5.1 Generative Pre-Trained Instructional Design (GPTID) Model:

5.1.1 Phase 1: Needs Analysis

In order to pinpoint particular learning challenges, a thorough needs assessment is conducted at the beginning of the needs analysis phase. Information is gathered via surveys, interviews, and data analysis. Furthermore, an analysis is conducted on the characteristics of the students, specific and measurable objectives are established, and the hardware, software, and timing of implementation are all considered in the evaluation of the educational environment. The following steps are included in this phase:

Needs assessment: To identify specific learning challenges and problems, conduct a thorough needs assessment using surveys, interviews, and data analysis.
Student characteristics: Examine the age, educational level, and prior experience of the students.

Objective Setting: Describe learning objectives, taking into account broad objectives, measurable goals, general aims, and learning outcomes.

Environment Evaluation: Determine the hardware and software requirements, evaluate the hardware and software resources that are available, and establish a deadline for putting the learning scenario into practice.

5.1.2 Phase 2: GPT Tools Planning

This stage involves a comprehensive assessment of the available GPT tools, taking into account elements like price, licensing, and compatibility with learning goals. The best GPT tool/s are then chosen based on how well they align with the learning objectives and content. Furthermore, the selected tool's limitations and capabilities are made explicit. Additionally, strategies for integration and diversification are investigated in order to find more tools that can be used in addition to the primary tool in order to guarantee flexibility and adaptability. The following steps are included in this phase:

GPT Tools Analysis: Examine all of the GPT tools that are available, taking into account things like price, license requirements, and alignment with learning goals.

Tools Alignment: Choose the best GPT tool/s based on how well they align with the learning objectives and content.

Tools Capabilities: Describe the possibilities and limitations of the chosen tool.

Diversification and Integration: Find additional tools that can be used interchangeably or in conjunction with one another.

5.1.3 Phase 3: Ethical Framework

Phase Three entails the meticulous formulation of a strong code of conduct to handle a range of possible ethical problems associated with the production of GPT content. These guidelines address things like inconsistent outputs, bias, misinformation, and information validation. Encourage students to participate in discussions about these ethical issues in order to highlight the significance of using AI in education in an ethically sound manner. In order to stay in line with evolving ethical standards in the field of artificial intelligence, a dynamic approach is also taken, which involves routinely reviewing and updating the ethical guidelines. The following steps are included in this phase:

Ethical Guidelines: Provide a thorough code of conduct that addresses any ethical concerns that might surface during the generation of GPT content. This should address issues with information validation, bias, false information, plagiarism, and inconsistent results.

Discussion Group: To highlight the significance of using AI in educational settings in an ethically responsible manner, talk to the students about the ethical considerations listed in the code of conduct.

Review and Update: Update the code of conduct to reflect any changes in the ethical standards pertaining to AI on a regular basis.

5.1.4 Phase 4: Prompt Engineering

In Phase Four, a systematic procedure starts to identify prompts that effectively direct GPT in producing responses. During this phase, response attributes that will form the generated content are specified. In addition, a variety of prompts are designed to cover different aspects of the learning situation, such as knowledge acquisition, skill practice, activities engagement, and assessment. Rigid prompt validation procedures are implemented to ensure the reliable generation of anticipated information. This phase contains the following steps:

Prompt Definition: Define prompts that guide GPT to generate responses effectively.

Response Attributes: Specify response characteristics, including:

Role: Give GPT a role to act, such as instructing like a teacher, empathizing like a therapist, or entertaining like a comedian.

Tone: Specify the tone of the response, for example, a persuasive, personal, or funny tone.

Style: Determine whether the generated response should be academic and formal, professional and business-like, or conversational and friendly.

Specificity: Describe the amount of the needed response, for example, a concise response of no more than 100 words or a detailed explanation of several paragraphs.

Complexity: Indicate the response's degree of complexity while taking the audience's age and educational background into account. For example, a response appropriate for a younger or specific audience.

Format: Specify the required output format, such as HTML, a table, a chart, a bulleted list, a numbered list, or a paragraph.

Prompt Diversity: Craft prompts that can be applied to various aspects of the learning process, including acquiring new information, improving skills, performing activities, and assessments.

Prompt Validation: Test prompts thoroughly before implementing them to make sure that the required information is consistently produced.

5.1.5 Phase 5: Implementation

Phase Five concerns the actual execution of the model starting by providing students with detailed instructions on how to use the GPT tool. Also includes deploying prompts in the predetermined order, receiving and analyzing GPT-generated responses, evaluating response quality and relevance, refining the generated content, mapping the resulting information to defined learning objectives, monitoring student performance, offering timely feedback, and then conducting post-implementation reviews to document results and define areas for improvement. This phase contains the following steps:

Guidance: Provide the necessary instructions and directions for the students to use the GPT tool successfully.
Prompt Deployment: Provide prompts according to a predefined order and timing within the learning situation.
Response Reception: Receive generated responses from GPT, then analyze and organize them in accordance with learning content and objectives.
Response Evaluation: Evaluate the quality and relevance of responses in conveying the required knowledge.
Refinement: Debug generated information and make necessary changes to ensure association with the curriculum and learning outcomes.
Outcome Mapping: Map the results obtained from GPT with the defined learning outcomes.
Monitor and Feedback: Observe students' performance and direct their interactions and responses during implementation then provide the proper feedback.
Post-Implementation Review: Document the most important results of implementation, challenges faced, and areas for further improvement.

5.1.6 Phase 6: Evaluation

Phase Six involves conducting a thorough assessment of the GPT integration's impact on achieving learning outcomes. This phase establishes success criteria and evaluates the degree of goal accomplishment using a variety of assessment instruments, such as surveys and quizzes. Additionally, it concerns analyzing student performance and identifying strengths and weaknesses aspects. A summative evaluation is conducted to assess the overall success of the learning process with GPT integration. Moreover, the phase incorporates strategies for enriching high-performing students and providing recovery plans for those facing challenges through customized learning activities, ensuring a holistic approach to educational improvement. This phase contains the following steps:

Objective Achievement: Evaluate the extent to which the defined learning outcomes have been achieved through GPT integration using different assessment tools such as quizzes and surveys, providing criteria for determining success.
Performance Analysis: Identify strengths and weaknesses in students' performance based on GPT-generated content by analyzing the results of assessment tools.
Overall Success: Assess the overall success of the learning situation with GPT integration using summative evaluation tools.
Enrichment and Recovery: Put a strategy to foster excellent students and assist weak ones with relevant learning activities.

5.1.7 Phase 7: Recommendations

Phase Seven focuses on improving the GPT integration process for future learning experiences. Improvement plans are put in light of the evaluation results, with the aim of continuously improving GPT integration in the context of education. Additionally, future insights are provided, offering recommendations on the prospective use of GPT in education, ensuring an innovative and flexible approach to the integration of educational technology. This phase contains the following steps:

Enhancement Strategies: Suggest strategies for enhancing the GPT integration process in successive learning experiences based on evaluation results, ensuring ongoing refinement of GPT integration in education.
Future Insights: Provide recommendations and insights for future usage of GPT in education.

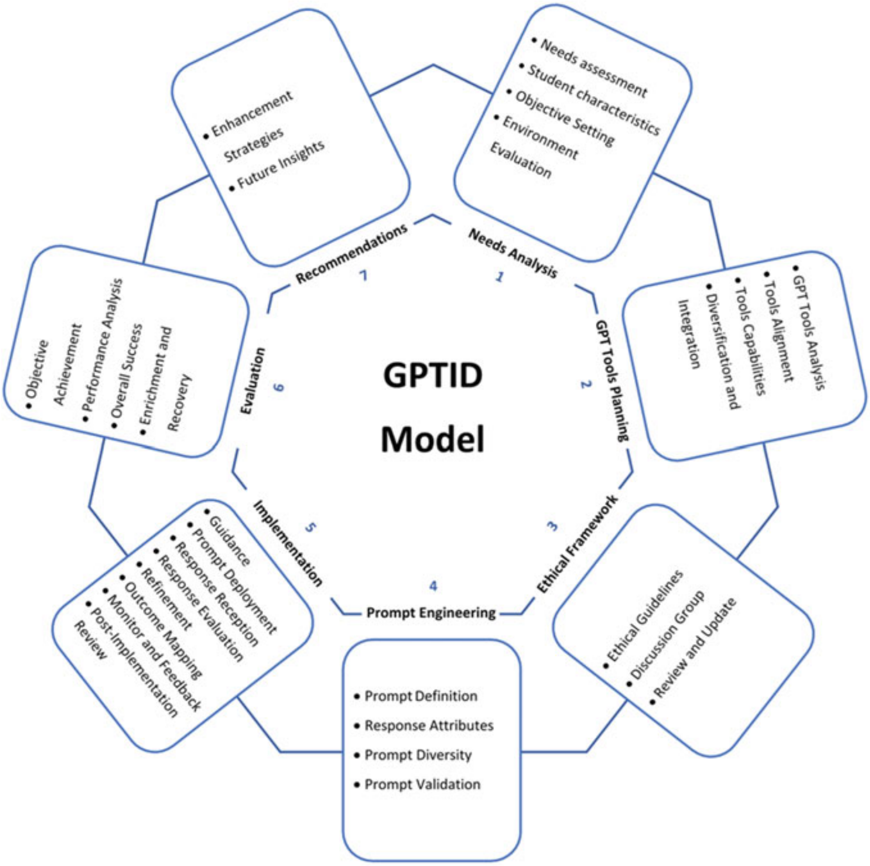

Fig. 1 Generative pre-trained instructional design (GPTID) Model

6 Discussion

The main conclusions, implications, and possible future paths pertaining to the development and use of the Generative Pre-Trained Transformer Instructional Design (GPTID) Model for incorporating Generative Pre-Trained Transformer (GPT) technology into education are explored in this section of the research.

Key Findings: The successful creation and validation of the GPTID Model, which provides an organized and all-inclusive method for integrating GPT into instructional design, is one of the main conclusions of this study. The validity and applicability of the model have been confirmed by the involvement of practitioners in educational technology and instructional design in the validation process. Additionally, the model's focus on ethical issues emphasizes how crucial it is to integrate AI responsibly in educational settings.

Implications: There are numerous implications for the GPTID Model. First of all, it offers teachers and instructional designers a methodical framework for maximizing GPT technology's potential to improve learning outcomes. Teachers can create more individualized and successful learning experiences by customizing GPT-generated content to meet particular learning needs and objectives. Furthermore, the model's ethical guidelines, which address issues like bias and misinformation, promote the responsible use of AI.

Future Directions: The GPTID Model provides opportunities for future study and advancement. Subsequent research endeavors may concentrate on enhancing and modifying the model to provide consideration for developing GPT technologies and evolving academic environments. Furthermore, it's critical to continuously evaluate ethical standards in order to maintain alignment with changing AI guidelines. Additionally, investigating how well the model works in various educational settings and how it affects student learning outcomes can yield insightful information.

Limitations: It is essential to recognize specific constraints associated with this study. Although the GPTID Model is a strong framework, learner characteristics and technological infrastructure can have an impact on how effective the model is in actual educational settings. To handle these subtleties, ongoing validation and adaptation are required. Furthermore, the field of ethical AI concerns is quickly developing, so the model might need to be updated on a regular basis to conform to new guidelines. To sum up, the creation of the GPTID Model represents an important advancement in the ethical and successful use of GPT technology in the classroom. This study emphasizes the significance of using AI ethically and gives teachers a useful tool to improve instructional design. The GPTID Model emphasizes the need for ethical and responsible practices in educational technology while serving as a testament to the potential for technology-driven innovation as AI continues to shape the future of education.

7 Conclusion

In conclusion, this study has introduced the Generative Pre-Trained Transformer Instructional Design (GPTID) Model, a thorough framework for incorporating Generative Pre-Trained Transformer technology into learning environments. Under the guidance of professionals in instructional design and educational technology, the model has undergone a rigorous process of development, validation, and refinement. It gives educators a structured method to use GPT to improve learning outcomes by addressing the crucial stages of needs analysis, ethical considerations, prompt engineering, implementation, and evaluation. This study, tried to highlight how crucial it is to use AI technologies in education in a responsible and moral manner, maximizing their potential advantages while minimizing their risks. For educators and instructional designers looking to use AI to create more interesting and productive learning environments, the GPTID Model is a useful tool. The model is a dynamic tool for the changing field of artificial intelligence in education, requiring constant modification and improvement.

8 Recommendations

It is advised that educational institutions and instructors adopt AI integration, in particular Generative Pre-Trained Transformer (GPT), as a useful tool for improving the learning process based on the research findings. Institutions should make investments in thorough needs assessments, moral frameworks, and teacher and student training programs in order to reach their full potential. Mechanisms for feedback and ongoing monitoring should be in place to guarantee the moral and high-quality application of AI-generated content. Educators should also investigate various prompt engineering techniques in order to properly match AI responses with learning goals. Finally, to fully realize the advantages of AI in education, educational institutions will need to incorporate forward-thinking perspectives and continuous assessment and adaptation. Future research ideas for GPT utilization in education could look into the following areas:

- Conducting a comprehensive study to assess the effectiveness of the proposed GPTID model across diverse educational contexts.
- Investigating the sustained effects of integrating GPT into educational practices, examining how GPT influences students' learning outcomes and skills development over an extended period.
- Conducting a comparative study to assess the effectiveness of various AI models, including GPT, across different educational domains, such as STEM subjects, humanities, and vocational training, to identify the strengths and weaknesses of each model.

- Investigating students' and educators' perceptions, attitudes, and experiences when using AI-generated content, focusing on understanding how user acceptance can be improved to enhance the educational experience.

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A Review of the Chat GBT Technology Role in Marketing Research

Mahmoud Alghizzawi

1 Introduction

Various forms of technology have changed business systems [1–5]. The new "chat-bot" or called "Chat GPT" has drawn the attention of technology enthusiasts around the world since it was launched for free use in 2022, and many users have flocked to it, and it is considered one of the most popular artificial intelligence sites. which have been put forward according to the testimony of many researchers [6]. On that level, many users tended to adopt this technology, and it came to researchers and professionals in various sectors [7–11].

Taking into account the supporters of this technique and the naysayers and the different ethics [12–16], it succeeded [17, 18] and from here The American artificial intelligence research firm Open AI created the Chat GPT technology. Sam is in charge of running the business, and Microsoft offers help. GBT Chat is an artificially intelligent robot or software that converses with users, thoroughly responds to their inquiries, and keeps track of all previous inquiries made of the user throughout the conversation that appears to be taking place between two humans. If it makes a mistake, it also offers an apology and allows the user to amend it, The company has worked on training the robot by providing it with massive information from the Internet, which made it produce texts and articles close to human business to a large extent, by dealing with algorithms with rapid data analysis and in an analytical manner close to the human brain [9, 19–21]. At the level of different business sectors and the complex business environment, information has become very important and sensitive [22–25]. All sectors, whether services or commodities, are now competing to access information in order to build decisions and great hopes on it that enhance their competitive position and increase their profitability [3]. And through marketers'

M. Alghizzawi (✉)
Marketing Department, Faculty of Business Applied Science, Private University, Amman, Jordan
e-mail: alghzawi87@gmail.com

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keen interest in marketing research, this is due to many sensitive factors that help them understand consumer behavior in terms of knowing their desires and preferences, which is one of the basics of marketing [26].

In addition, marketing research highlights market opportunities by analyzing them in multiple ways to evaluate them [27, 28]. Marketing research helps marketers to analyze the effectiveness of marketing campaigns based on many indicators, and most importantly because of the complexity of the business environment, there must be a clear picture of market segments and customers in front of companies, in addition to the importance of developing products and services and innovation in them is very important at the present time due to the multiplicity of customer desires [29]. This reflects the importance of marketing research in reducing risks and challenges for companies, especially after the Corona pandemic, which hit everyone globally [30, 31]. Marketers are generally very interested in marketing research because it gives them the information and understanding they need to make wise decisions, comprehend consumer behavior, identify market opportunities, assess the success of advertising campaigns, create customer segmentation, promote product innovation, and reduce risk [32]. Effectively utilizing marketing research can give marketers a competitive edge and help them accomplish their objectives more quickly [33–35]. This is what increased the importance of artificial intelligence and chat technology via GPT, and this is due to the tools and features available to marketers to access information quickly, easily, and in a huge amount by providing permanent answers on which the marketer can build many analyzes, even if it is general and not deep [36]. Accordingly, the current study came to demonstrate the relationship of artificial intelligence technology based on GBT chat in developing the business of marketers, while permeating the basics of marketing research, which constitutes a great burden on marketers due to its importance to companies in light of the challenges and difficulties facing the business market in all sectors and countries.

2 Research Procedures

Previous and recent literature that dealt with the subject of artificial intelligence will be addressed [37–44], especially the chat GPT technology, which was a pioneer in the field, and then various discussions will be presented to generate important main lines for marketers and entrepreneurs in organizations by addressing the benefits and negatives of this technology, in addition to reviewing Realistic for this application in search of the required information [45].

3 Literature Review

Examining and focusing on artificial intelligence as a modern science [46–50], we find that there are few marketers and users who depend on it or who are familiar with it in their field of work by relying on analyzes [51–55], studies, and the information available through it [56–63], and how to obtain the correct information [6, 18, 19, 36]. The science of marketing, especially digital in the modern perspective, is considered one of the very important sciences in the shallow business environment [64, 65], which reinforced the importance of the current study by dealing with GBT Chat and marketing through an indication of opportunities and challenges for the purpose of facilitating the roads for marketers in all sectors.

3.1 Chat GPT Opportunities for Marketers

Chat GBT enhances the work of marketers, especially in the first phase, by reviewing keywords and accurate questions, which in turn open important perceptions to think deeply [51–55], about how to understand customer behavior, create innovative advertising campaigns, modern promotion and follow-up statistics, in addition to the ability These technologies collect information and organize it as required [66]. There are various ways that chat GPT technology, like the one used by this AI model, might help marketers improve their strategy and engage with their target audiences more successfully [67]. Marketers may communicate with customers in real-time using chat GPT technology through a variety of channels, like Chabot's or live chat support. With the use of this technology, clients may have interactive and personalized experiences where they can ask questions, get assistance, and get prompt answers [68]. 85% of customers prefer dealing with chat bots to rapidly get an answer to a question, according to [69] study. Marketers may deliver excellent customer support, respond to inquiries quickly, and improve the customer experience by utilizing chat GPT technology. Customer information and tastes can be analyzed using Chat GPT to produce individualized recommendations. Marketers can employ chat GPT technology to offer product recommendations, promotions, and individualized content by understanding users' browsing histories, purchasing patterns, and interests. This personalization raises comfort levels and increases the user experience [70]. [71] estimates that when offered a personal experience, 80% of consumers are more inclined to make a purchase. GPT chat technology can be used by marketers to make customized recommendations and boost customer pleasure and loyalty. Chat with the aid of GPT technology, marketers may gather useful information from customers. Marketers can learn about clients' preferences, shortcomings, and criticism through dialogues. To get insights into consumer behavior, mood, and new trends, this data can be evaluated. This data can be used by marketers to enhance their campaigns, enhance their goods and services, and make data-driven decisions [72]. According to [73] found that 57% of customers are open to giving personal information in exchange for customized offers or discounts. Marketers can obtain customer data and provide more relevant and targeted marketing by utilizing GPT chat technology. Additionally, Chat GPT offers marketers the chance to produce and qualify leads more effectively. Marketers can engage with website visitors and start discussions to find potential consumers by using chat bots or automated chat systems. Chat GPT can direct visitors through the sales funnel, ask pertinent questions, and provide information. This aids marketers in prioritizing their follow-up efforts and identifying well-intentioned prospects [74]. According to [75] a 48% boost in revenue growth for businesses using chat bots to drive leads. GPT chat technology can be used by marketers to increase lead generation and conversion rates. By using chat bots to answer routine consumer enquiries, offer self-service choices, and escalate difficult issues to human agents as needed, marketers can offer 24/7 customer support and help. Customers will obtain prompt assistance thanks to this round-the-clock service, which will boost their happiness and loyalty. 90% of customers regarded rapid response as vital or extremely important when they had a customer service query, according to a Hub Spot poll. Marketers can deliver prompt assistance, fortify client relationships, and enhance customer satisfaction by utilizing GPT chat technology [76]. Based on the above, it is clear that GBT chat has a positive impact on marketers by providing them with opportunities to enhance customer service and interaction with them in addition to permanent support, and this is what enhances the importance of this technology and its advantages that support marketing decisions and digital promotional campaigns in addition to Understanding customer behavior, and thus this is reflected in more accurate decisions and more effective marketing goals.

3.2 Chat GPT Challenges for Marketers

After reviewing the opportunities available through this technology, there must be an indication of the dark side of it in terms of the challenges and difficulties that marketers may face when adopting this technology in their work, as knowing these limitations helps marketers to be able to overcome them, to ensure benefit of these features available through this technology [77]. Do not escape the importance of GBT chat in the initial stages of research in obtaining information, but the human character and thought must be directed to this technology in the correct way to ensure obtaining correct and not false information, so that there are logical comparisons and general surveys to compare it with the information that has been Get it from chat GBT [78]. When examining the possible challenges more accurately, it becomes clear that GPT chat must be strengthened by dealing with human thought in a more flexible way, as it has been activated and developed to form quick responses and answers in general. As marketers, we need privacy and a higher focus on information, even if it is 50 percent of the extent. Its validity, which poses a challenge to the human mind as marketers, so extracting this information from it by knowing how to ask the question [79]. The biggest challenge is the quality of the information provided by the GBT chat, so that it is not biased or wrong, so that the output of this program is not weak, especially since new artificial intelligence programs have appeared, which creates competition between these systems concerned with artificial intelligence regarding the accuracy and validity of the information Especially since at some point in 2021, information from the GBT chat app was cut off for a while, This could weaken its capabilities and marketers' confidence in chat GPT [80]. Marketing as a science is interested in all events that affect the sector in which it is marketed, for example, in terms of economic, political and cultural aspects, which requires marketers to include these aspects during the search for information via GPT chat to facilitate the process of linking the required information and its other reflections [81]. Accordingly, we see the importance of enhancing the use of information from GPT chat for marketers in a careful and conscious manner to ensure that the quality of the information is confirmed by comparing and studying it, even if superficially, which ensures that the marketing decisions that will be relied upon based on this information have a positive impact.

3.3 Future Road for Marketer Depending on Chat GBT

The hype that was made based on artificial intelligence, especially through GBT chat as the first global platform that helped marketers in all sectors to automate and organize information and facilitate access to information in a flexible manner, especially since digital marketing currently has the largest share in the business world, which also strengthened The role of GBT chat in creating advertising content professionally from videos and images, and most importantly, that GBT chat enhanced customer service as well, and therefore the opportunities available in the future are expanding to meet the needs of marketers in terms of consumer behavior, commercial advertisements and quick information about competitors and market situation (Fig. 1).

4 Discussions & Conclusions

Chat GBT has a positive impact on marketing by giving marketers support and support that may enhance their effectiveness in accessing important information about the market and customers [72–74] The advantages and challenges regarding this technology have been reviewed, but its positives through the above advantages outweigh the challenges [77–80]. It can be addressed in the future, so it is possible by relying on this technology to provide better marketing outputs, as the time factor is very sensitive in making marketing decisions, with the importance of taking into account the human aspect in including information via GPT chat to ensure its validity in a way that serves the interest of decision makers in an environment Business. The field of marketing research has been significantly impacted

Fig. 1 A future model of chat GPT technology for marketing research

by the development of Chat GBT (Generative Pre-Trained Transformer) technology. The usefulness of Chat GBT in assisting marketers in gaining insights, boosting client engagement, and fine-tuning their marketing tactics has been underlined in this evaluation. Chat GBT has become a potent tool for marketing research thanks to its features including natural language interpretation, relevant response development, and customized content creation. It has been used in a variety of fields, including customer segmentation, market trend analysis, consumer sentiment analysis, and content development, particularly with regard to digital marketing through various digital platforms. It is clear that Chat GBT has advantages in marketing research. Because of its scalability and effectiveness, marketers can instantly analyze massive amounts of data and offer insightful information in real time. The technology's capacity to provide customized content improves consumer interaction and aids marketers in sending customized messages to their target market. The limits of Chat GBT in marketing research must be understood, though. Attention must be paid to generated content biases and the necessity of human control when analyzing outcomes. To avoid biases in the generated responses, marketers must make sure the data used to train Chat GBT is diverse and representative. When using Chat GBT for marketing research, ethical considerations are also essential. It is crucial to maintain data privacy, ensure responsible technology use, and solve issues with consumer trust and transparency. Finally, Chat GBT technology has demonstrated its ability to revolutionize marketing research. It gives marketers useful information, individualized involvement, and capacities for data-driven decision-making. Marketers can fully utilize Chat GBT to drive successful marketing campaigns and accomplish their company goals by identifying the limitations, addressing ethical issues, and maintaining up to date on developments.

5 Future Research

Looking ahead, Chat GBT in Marketing Research has a bright future. Its capabilities will be further enhanced by continued advances in data analytics, machine learning, and natural language processing. Marketers will be able to better analyze consumer behavior, anticipate market trends, and refine their marketing methods as a result of advances in these areas. This reinforces the need to follow up future research on this relationship between GPT chat and digital marketing in a more specific way than marketing research in general, with the possibility of dealing with other technological systems concerned with artificial intelligence with marketing to serve the goals of marketers in their business.

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AI in Language and Communication Studies

Unveiling the Future: Exploring Conversational AI

Meera M. Shah and Hiren R. Kavathiya

1 Introduction

In an era defined by rapid technological advancement, one of the most captivating frontiers in artificial intelligence (AI) is Conversational AI. This chapter embarks on an expedition into the depths of this captivating realm, unfurling its historical roots, current manifestations, and the tantalizing prospects it holds for the future. Conversational AI, with its transformative capabilities, has redefined the human–machine interface, ushering in novel ways of communication, commerce, and innovation.

The chapters to follow will unravel the intricate layers of Conversational AI, examining its evolutionary journey, dissecting its fundamental components, and exploring its diverse applications across industries. From the intricate dance of Natural Language Understanding (NLU) to the artistry of Natural Language Generation (NLG), this chapter will paint a comprehensive portrait of Conversational AI's inner workings.

Moreover, we will delve into the fascinating interplay of dialogue management and the critical role it plays in orchestrating seamless conversations between humans and machines. As the landscape of AI continues to evolve, Conversational AI stands as a pinnacle of human ingenuity, a domain where technology endeavors to bridge the gap between the synthetic and the human [1]. We will navigate the ethical currents that underpin this technological transformation, addressing concerns such as bias mitigation, data privacy, and the ever-pressing need for responsible AI development.

Furthermore, this chapter gazes toward the horizon, envisioning the potential trajectories that Conversational AI might take in the coming years. The concept

M. M. Shah (✉) · H. R. Kavathiya
Department of Computer Science, Atmiya University, Rajkot, India
e-mail: mira.shah@atmiyauni.ac.in

H. R. Kavathiya
e-mail: hiren.kavathiya@atmiyauni.ac.in

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of conversational understanding, where AI systems navigate the ebb and flow of conversations with a human-like adeptness, emerges as an alluring destination. The fusion of multimodal interactions and emotional intelligence promises to elevate AI to unprecedented levels of sophistication, fundamentally altering the ways in which we interact with machines.

2 Evolution of Conversional AI

The roots of Conversational AI stretch back to the earliest aspirations of technology to decipher and engage with human language. The journey from rudimentary rule-based systems to the sophisticated neural networks of today has been marked by leaps of innovation, technological breakthroughs, and a deepening understanding of the intricacies of language [2].

2.1 Early Attempts and Rule-Based Systems (1950s-1990s)

The inception of Conversational AI can be traced to the 1950s when computer scientists began exploring the possibilities of automating human language understanding. Early efforts involved rule-based systems that relied on predefined grammatical rules and a limited vocabulary to simulate conversations. ELIZA, created in the mid-1960s by Joseph Weizenbaum, was a groundbreaking example. Although primitive, ELIZA could engage in simple textual exchanges, laying the foundation for future advancements.

2.2 Statistical Approaches and Limited Context (2000s)

The turn of the millennium witnessed a shift towards statistical methods, with probabilistic models enabling AI systems to predict the likelihood of words and phrases appearing in a given context. Yet, these systems struggled with understanding nuanced language and context shifts. The introduction of chatbots for customer support marked an early application, albeit with limited success due to their constrained capabilities.

2.3 Rise of Machine Learning and Neural Networks (2010s)

The emergence of deep learning in the 2010s sparked a renaissance in Conversational AI. Neural networks, especially the Transformer architecture, revolutionized language understanding and generation. Models like Google's BERT and OpenAI's GPT series demonstrated unprecedented capabilities in capturing context, semantics, and subtleties in language. These models learned from massive datasets, empowering AI to produce more human-like responses and enabling broader applications.

2.4 Conversional Context and Personalization

As AI systems became more proficient at understanding context, maintaining conversation flow, and generating coherent responses, they ventured into personalization. Conversational agents began adapting to user preferences, enhancing user experiences across various domains. Additionally, contextual understanding improved, allowing AI to comprehend references made earlier in conversations, mimicking the continuity of human discourse.

2.5 Multimodal Interactions and Beyond

In recent years, the evolution of Conversational AI has taken a multimodal turn, embracing inputs beyond text, such as images, voice, and gestures. This expansion aims to mirror human communication, where contextual cues are derived from diverse sources. The integration of multimodal interactions amplifies the potential for more natural and immersive conversations.

The evolution of Conversational AI showcases a remarkable journey from its humble beginnings to its current prowess in understanding and generating human-like language. As we peer into the future, the ongoing fusion of AI with linguistic intricacies holds the potential to redefine human–machine communication, fostering an era where interactions are not just transactional but genuinely conversational [2]. This evolution is a testament to human ingenuity and an unwavering desire to bridge the gap between the synthetic and the human, propelling Conversational AI into the forefront of technological innovation.

3 Foundation of Conversional AI: NLP and Machine Learning

Conversational AI is built upon a strong foundation of Natural Language Processing (NLP) and Machine Learning (ML) techniques. This fusion of disciplines empowers machines to understand, generate, and engage in human-like conversations, bridging the gap between human communication and artificial intelligence.

3.1 Natural Language Processing (NLP)

At the heart of the transformative power of Conversational AI lies Natural Language Processing (NLP), the wizardry that bestows machines with the ability to understand, interpret, and generate human language. This chapter delves into the intricate realm of NLP within Conversational AI, uncovering the technologies, techniques, and advancements that enable machines to engage in dynamic and contextually relevant conversations. From syntax to sentiment analysis, from understanding context to generating coherent responses, NLP is the cornerstone that breathes life into human–machine discourse [3].

3.1.1 Linguistic Foundations: Decoding the Grammar of Conversations

The journey into NLP's role in Conversational AI begins with an exploration of the linguistic foundations. This section illuminates the components of language processing, from tokenization and part-of-speech tagging to parsing and syntactic analysis. The magic of syntactic trees and grammatical structures becomes evident as we unravel how NLP algorithms dissect and comprehend the grammatical intricacies of human language.

3.1.2 Beyond Words: Understanding Semantics and Context

Words are the building blocks of language, but meaning transcends mere vocabulary. Delving into semantics, this part delves into how NLP techniques capture the deeper meaning of words and their relationships in sentences. The emergence of word embeddings and distributional semantics enables machines to discern semantic similarities, paving the way for more nuanced comprehension and relevant responses.

3.1.3 Intent Recognition and Named Entity Recognition

The crux of effective Conversational AI lies in understanding user intent and extracting relevant information. This section uncovers the techniques of intent recognition and named entity recognition. Through the lens of NLP, machines learn to identify the goals and objectives behind user queries, as well as discerning crucial entities such as names, dates, and locations. This fosters context-aware conversations and enhances the overall user experience.

3.1.4 Contextual Understanding: The Key to Coherence

Context is the glue that binds meaningful conversations. This segment delves into how NLP equips AI systems to understand and maintain contextual continuity in conversations. Techniques like coreference resolution ensure that pronouns and references are properly linked, creating a coherent and fluid dialogue that mirrors human interaction.

3.1.5 Sentiment Analysis: Gauging Emotional Nuances

Human conversations are often infused with emotions that shape the tone and intent. Sentiment analysis, a pivotal aspect of NLP, enables AI systems to gauge emotional nuances within text. This part explores how machines identify sentiments like joy, anger, and sadness, allowing them to tailor responses that resonate with the user's emotional state.

3.1.6 Response Generation: The Art of Human-Like Replies

Generating responses that mimic human dialogue is the ultimate aim of Conversational AI. This section delves into the techniques of response generation, from rule-based approaches to the transformative power of neural language models. The advent of transformers has revolutionized response generation, enabling AI systems to produce contextually relevant, coherent, and contextually-aware replies.

3.1.7 Ethical Considerations in NLP

As NLP-driven Conversational AI becomes more integral to our lives, ethical considerations gain prominence. This part delves into the challenges of bias in training data, potential misinformation propagation, and the importance of responsible AI practices. The chapter also underscores the significance of transparency, fairness, and user consent in maintaining ethical integrity.

The marriage of NLP and Conversational AI is a symphony of linguistic understanding and technological innovation. From deciphering grammar to grasping context, from understanding emotions to generating human-like responses, NLP is the thread that weaves the magic of human–machine discourse. This chapter has illuminated the intricate workings of NLP within Conversational AI, highlighting its crucial role in shaping a future where machines converse with us seamlessly and intelligently.

3.2 Machine Learning in Conversional AI

Conversational AI's evolution has been deeply intertwined with the rise of Machine Learning (ML), a transformative force that empowers machines to learn from data and adapt their behavior. This chapter embarks on an illuminating journey through the realm of ML in Conversational AI, unveiling the algorithms, techniques, and innovations that enable AI systems to not only understand human language but also engage in dynamic and contextually relevant conversations.

3.2.1 Foundations of Machine Learning: Learning from Data

The bedrock of ML in Conversational AI lies in understanding how machines learn from data. This section introduces the fundamental concepts of supervised, unsupervised, and reinforcement learning, providing readers with insights into the underlying mechanisms that enable AI systems to make data-driven decisions and refine their responses over time.

3.2.2 Supervised Learning: The Path to Contextual Understanding

Supervised learning is a cornerstone of Conversational AI, enabling machines to recognize patterns in labeled data. This part delves into how ML algorithms, such as support vector machines and neural networks, are trained to understand context, intent, and sentiment within human language [4]. Through annotated datasets, AI systems navigate the nuances of dialogue and build a foundation for meaningful conversations.

3.2.3 Unsupervised Learning: Extracting Insights from Text

Conversations are rich sources of unstructured text, and unsupervised learning techniques shine in extracting valuable insights. This segment explores how algorithms like clustering and topic modeling unravel the hidden structures and themes within conversations. These techniques not only aid in organizing information but also provide AI systems with a deeper understanding of user preferences and interests [4].

3.2.4 Reinforcement Learning: Dynamic Interaction and Response Refinement

The art of conversation is a dynamic dance of responses and feedback. This part unveils the role of reinforcement learning, where AI agents learn through trial and error, receiving rewards for desirable responses and refining their behavior over time. From training dialogue agents in simulated environments to real-world interactions, reinforcement learning imbues Conversational AI with adaptability and continuous improvement [5].

3.2.5 End-To-End Training: Seamless Dialogue Generation

One of the transformative aspects of ML in Conversational AI is end-to-end training. This section explores how neural network architectures, such as sequence-to-sequence models, enable the training of systems that can generate human-like responses directly from input queries [2]. By minimizing the need for handcrafted rules, end-to-end models usher in a more fluid and natural conversational experience.

3.2.6 Transfer Learning and Pre-trained Models

The emergence of transfer learning has been a game-changer in Conversational AI. This segment delves into how pre-trained language models, like GPT-3, have reshaped the landscape by learning from vast amounts of text data [6]. These models encapsulate diverse linguistic patterns, allowing AI systems to engage in conversations spanning a wide spectrum of topics and styles.

3.2.7 Adversarial Training and Robustness

In the realm of Conversational AI, robustness against adversarial attacks is crucial. This part explores how adversarial training techniques bolster AI systems against manipulative inputs and ensure coherent responses even in the face of crafted perturbations. The quest for robustness enhances Conversational AI's ability to maintain meaningful interactions without succumbing to deliberate distortions.

Machine Learning is the driving force that propels Conversational AI into the realm of adaptive and intelligent interactions. From supervised learning's foundation in understanding context to reinforcement learning's dynamic responsiveness, the fusion of ML and Conversational AI is a symphony of data-driven ingenuity. This chapter has offered a glimpse into this synergy, emphasizing how ML infuses Conversational AI with the capability to evolve, learn, and engage in conversations that resonate with human understanding.

The foundations of Conversational AI, rooted in Natural Language Processing and Machine Learning, form the bedrock upon which intelligent conversations with machines are built. The marriage of linguistics and technology empowers machines to comprehend the intricacies of language, adapt to context, and generate human-like responses. As these foundational techniques continue to evolve, they propel Conversational AI toward unprecedented heights, ushering in an era where machines seamlessly converse with humans, enriching interactions across domains and driving the boundaries of technological possibility.

4 Applications of Conversational AI Across Industries

Conversational AI's transformative capabilities have transcended boundaries, infiltrating diverse industries and reshaping how businesses interact with their customers, clients, and stakeholders. From healthcare to finance and beyond, the technology's applications have been instrumental in enhancing user experiences, optimizing processes, and driving innovation. Here, we explore some of the prominent sectors where Conversational AI has made its indelible mark.

4.1 Customer Service and Support

Customer service has undergone a paradigm shift with the introduction of Conversational AI. Chatbots and virtual assistants provide instant and round-the-clock support, answering common queries, troubleshooting issues, and guiding users through processes. These AI-driven systems not only reduce response times but also free up human agents to focus on more complex inquiries, thus improving overall customer satisfaction.

4.2 Healthcare and Telemedicine

In the healthcare sector, Conversational AI has enabled telemedicine to thrive. Virtual health assistants offer preliminary diagnoses, schedule appointments, and provide patients with accurate medical information. These AI-powered companions have played a crucial role during the COVID-19 pandemic, offering reliable guidance while alleviating the burden on healthcare providers.

4.3 E-commerce and Retail

Conversational AI has transformed the online shopping experience. Chatbots assist customers in finding products, offering personalized recommendations, and aiding in the purchasing process [7]. By understanding customer preferences and behavior, AI-driven systems enhance engagement, drive sales, and foster customer loyalty.

4.4 Finance and Banking

In the finance industry, Conversational AI has streamlined interactions between customers and financial institutions. Virtual assistants help with account inquiries, balance checks, fund transfers, and even investment advice. These systems ensure faster and more convenient transactions while maintaining a high level of data security.

4.5 Education and E-learning

Conversational AI has infiltrated the education sector, creating intelligent tutoring systems that provide personalized learning experiences. These systems adapt to students' learning styles and pace, offering real-time feedback and guidance. Conversational AI also plays a role in language learning, offering interactive practice and language immersion.

4.6 Entertainment and Media

Entertainment has embraced Conversational AI through interactive storytelling experiences, voice-controlled gaming, and personalized content recommendations. AI-driven chatbots can engage users in imaginative narratives, creating a dynamic and immersive form of entertainment.

4.7 Hospitality and Travel

In the hospitality industry, Conversational AI enhances guest experiences by offering information about accommodations, local attractions, and services. Hotels and travel companies deploy AI-powered concierge services to answer queries, arrange bookings, and provide travel recommendations, elevating the overall travel experience [8].

4.8 Human Resources and Recruitment

Conversational AI has revolutionized HR processes by automating candidate screening, scheduling interviews, and answering frequently asked questions. Virtual assistants efficiently handle administrative tasks, allowing HR professionals to focus on strategic initiatives and fostering a more efficient recruitment process.

4.9 Government and Public Services

Governments have harnessed Conversational AI to provide citizens with information about public services, government programs, and policy updates. Virtual assistants address queries, guide citizens through bureaucratic processes, and offer a user-friendly means of accessing important information.

The proliferation of Conversational AI across various industries underscores its versatility and potential to reshape traditional business models. By automating routine tasks, personalizing interactions, and providing efficient customer support, Conversational AI enhances user experiences and fosters deeper engagement. As industries continue to embrace and innovate with this technology, the boundaries of what's achievable through AI-powered conversations continue to expand, opening doors to a future where machines seamlessly collaborate with humans across a myriad of sectors.

5 Challenges and Ethical Consideration

The evolution of Conversational AI is accompanied by a range of challenges and ethical considerations that require careful examination. While these technologies hold immense potential, it is crucial to navigate them responsibly to ensure that their deployment aligns with societal values, respects user rights, and avoids harm. Let's explore some of the significant challenges and ethical considerations in Conversational AI.

5.1 Challenges

5.1.1 Bias and Fairness

Addressing bias in Conversational AI is a paramount challenge. Biases present in training data can lead to discriminatory responses, perpetuating social inequalities. Ensuring diverse and representative datasets and employing debiasing techniques are essential to mitigate this challenge.

5.1.2 Context and Understanding

Developing AI systems that understand and respond coherently to contextually nuanced conversations is challenging. Contextual understanding requires fine-tuning models to capture subtle shifts, maintaining meaningful conversations over time.

5.1.3 Privacy and Data Security

The vast amount of personal data processed in Conversational AI raises concerns about user privacy. Striking a balance between providing personalized experiences and safeguarding user information is essential, involving robust data encryption and transparent data handling practices.

5.1.4 Misinformation and Manipulation

Conversational AI can inadvertently propagate misinformation or be manipulated to spread false information. Ensuring that AI systems are equipped with fact-checking capabilities and are resistant to malicious manipulation is a critical challenge.

5.1.5 Accountability and Liability

Determining who is accountable when AI systems generate incorrect or harmful responses is complex. Establishing clear lines of responsibility and liability in cases of negative outcomes is challenging yet necessary to ensure accountability.

5.2 Ethical Considerations

5.2.1 Transparency and Explainability

Transparency is key to building trust between users and AI systems. Conversational AI should provide clear explanations for its decisions and disclose its AI-driven nature, enabling users to understand and trust the technology's processes.

5.2.2 User Consent and Control

Obtaining informed consent for data usage and interactions with AI systems is vital. Users should have control over their data and be able to opt-out or delete information collected during interactions.

5.2.3 Human-Like Deception

As AI systems become more sophisticated, there's a concern that they might appear too human-like, potentially deceiving users into believing they are interacting with humans. Implementing mechanisms to clearly identify AI interactions can mitigate this concern.

5.2.4 Impact on Human Relationships

Over Reliance on Conversational AI might impact human-to-human interactions. Striking a balance between AI assistance and genuine human communication is important to preserve meaningful relationships.

5.2.5 Socio Economic Impact

The widespread adoption of Conversational AI could potentially impact job roles, particularly those involving customer service and support. Mitigating the potential negative socioeconomic consequences through upskilling and retraining initiatives is an ethical consideration.

Conversational AI holds immense promise for transforming various aspects of our lives, but its deployment must be guided by a thorough understanding of the challenges and ethical considerations it entails. By addressing bias, ensuring transparency, safeguarding privacy, and fostering accountability, we can build AI systems that contribute positively to society. Ethical foresight and responsible development are vital to harnessing the potential of Conversational AI while safeguarding human values and well-being [9].

6 The Future of Conversational AI: A Glimpse into Tomorrow

Conversational AI has embarked on an extraordinary journey, evolving from simple rule-based systems to complex neural networks capable of understanding and generating human-like language. As we peer into the future, the trajectory of Conversational AI reveals a landscape filled with possibilities that promise to redefine the ways we communicate, interact with technology, and experience the world. Here, we explore the exciting prospects that await Conversational AI in the years to come.

6.1 Multimodal Mastery

The future of Conversational AI is set to embrace a convergence of communication modes. Systems will seamlessly integrate text, speech, images, and even gestures, creating richer, more immersive interactions. Imagine instructing a virtual assistant using voice, showing it an image for context, and receiving a detailed response that takes into account various modalities of input.

6.2 Emotional Intelligence

As AI systems become more sophisticated, they will develop a nuanced understanding of human emotions. Future Conversational AI could recognize subtle emotional cues in speech and text, responding with empathy, compassion, and even humor. This emotional intelligence will enable AI to establish deeper connections and enhance user experiences.

6.3 Specialization and Expertise

Conversational AI will evolve from being general-purpose to domain-specific experts. Specialized virtual assistants could provide intricate knowledge and support in fields like medicine, law, finance, and more. This expertise will extend beyond basic interactions, offering in-depth insights and informed recommendations tailored to specific industries.

6.4 Contextual Continuity

Future Conversational AI systems will excel in maintaining conversations that span days, weeks, or even months. They will seamlessly remember past interactions, capture nuances in conversation flow, and provide continuity as if the conversation never ended. This prolonged context will contribute to more natural, engaging dialogues.

6.5 Ethical AI

The future of Conversational AI lies hand in hand with ethical considerations. AI developers and organizations will prioritize responsible AI development, ensuring transparency, fairness, and adherence to privacy regulations. Ethical guidelines will shape the design, deployment, and behavior of AI systems, fostering trust and minimizing harm.

6.6 Personal AI Companions

Imagine having your own AI companion that understands your preferences, anticipates your needs, and helps manage your daily life. Future Conversational AI could evolve into personalized digital assistants that schedule appointments, manage tasks, curate content, and offer companionship.

6.7 Advancements in Learning

AI models will become even more adept at learning from fewer examples and adapting to new tasks. This will expedite the development and deployment of Conversational AI, making it easier to create specialized conversational agents that cater to specific industries and user needs.

6.8 Global Language Accessibility

Future Conversational AI models will be proficient in a multitude of languages and dialects, breaking down language barriers and fostering cross-cultural communication. This will enable people around the world to engage with AI systems in their native languages, making technology more inclusive.

The future of Conversational AI holds a world brimming with transformative potential. Through the integration of multimodal interactions, emotional intelligence, domain expertise, and ethical considerations, these systems are poised to revolutionize the way we communicate, work, learn, and live. As we navigate the uncharted waters of the AI frontier, responsible development and innovation will be essential to harness the vast benefits of Conversational AI while upholding the values that define our humanity.

7 Conclusion: Navigating the Conversational AI Frontier

The journey through the realms of Conversational AI has unveiled a landscape of innovation, challenges, and ethical considerations. From its modest beginnings in rule-based systems to the sophisticated neural networks of today, Conversational AI has transformed the way we interact with technology, enriching human experiences and reshaping industries across the globe.

As we reflect on the chapters traversed in this exploration, several key takeaways emerge. Conversational AI's foundations in Natural Language Processing and Machine Learning provide the building blocks for systems that can understand, generate, and manage human-like conversations. This fusion of linguistics and technology gives rise to machines that bridge the gap between the synthetic and the human, challenging us to find a harmonious coexistence.

Yet, this journey is not without its challenges. Bias, privacy concerns, and the potential for unintended consequences beckon for ethical considerations to steer the path forward. Developers, policymakers, and society at large are tasked with charting a course that upholds transparency, fairness, and accountability while harnessing the immense potential of these technologies.

As we peer into the future, Conversational AI beckons us toward a realm of unimaginable possibilities. Multimodal interactions, emotional intelligence, specialized expertise, and ethical AI stand as beacons guiding our aspirations. The evolution of Conversational AI is an embodiment of human ingenuity, stretching the boundaries of innovation and illuminating the intricate dance between human creativity and machine intelligence.

In closing, the chapters unveiled in this exploration only mark the beginning of the Conversational AI saga. The narrative continues, inviting us to engage with questions, challenges, and innovations that shape the way we communicate, connect, and coexist. As we journey forward, the lessons learned from the past and the ethical considerations pondered today will guide us toward a future where Conversational AI enriches our lives, respects our values, and propels us toward a more interconnected and intelligent world.

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Exploring the Acceptance of ChatGPT for Translation: An Extended TAM Model Approach

Said A. Salloum, Rose A. Aljanada, Aseel M. Alfaisal, Mohammed Rasol Al Saidat, and Raghad Alfaisal

1 Introduction

In the rapidly evolving landscape of linguistic technology, ChatGPT, an advanced language model developed by OpenAI, has emerged as a pivotal tool in machine translation (MT) [1]. This study focuses on the application and acceptance of ChatGPT in the field of translation, examining its utility across diverse linguistic and cultural contexts. The unprecedented capabilities of ChatGPT in understanding and translating languages have revolutionized how information is accessed and communicated globally. As highlighted by [2], ChatGPT's ability to process and translate multiple languages has bridged communication gaps, offering seamless integration in various sectors, from academia to business [2]. Despite its growing popularity, the nuances of its application in translation, especially across different cultural contexts, have not been fully explored.

The importance of cultural context in technology adoption is well-established in the literature. As [3] and later studies (e.g., [4]) have shown, cultural dimensions significantly influence the perception and utilization of technology [3, 4]. This is

S. A. Salloum (✉)
Health Economic and Financing Group, University of Sharjah, Sharjah, UAE
e-mail: ssalloum@sharjah.ac.ae

School of Science, Engineering, and Environment, University of Salford, Salford, England

R. A. Aljanada · A. M. Alfaisal
Northern Border University, Arar, Kingdom of Saudi Arabia

M. R. Al Saidat
Faculty of Engineering & IT, The British University in Dubai, Dubai, UAE

R. Alfaisal
Faculty of Computing and Meta-Technology, Universiti Pendidikan Sultan Idris, Perak, Malaysia

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particularly relevant in the case of translation tools like ChatGPT, where the effectiveness and accuracy of translation can be affected by cultural nuances and linguistic complexities.

Furthermore, studies by [4] on machine translation tools have underscored the importance of user experience and perceived ease of use in technology acceptance [4]. However, these studies primarily focus on general MT tools, with limited insights into advanced models like ChatGPT.

This study aims to fill this gap by examining the adoption of ChatGPT for translation purposes in diverse cultural settings. It investigates how users from different linguistic backgrounds perceive the effectiveness of ChatGPT in translation tasks. The study is particularly focused on understanding the role of ChatGPT in bridging language barriers and facilitating communication in multilingual contexts. To achieve these objectives, the study employs a quantitative approach to gather insights from users across various regions. The research will contribute to the existing body of knowledge by providing a comprehensive understanding of ChatGPT's utility in translation and its adoption across different cultural backgrounds.

The paper is structured as follows: the first section reviews the existing literature on machine translation and the role of cultural context in technology acceptance. The subsequent section outlines the research methodology, followed by an analysis of the collected data. Finally, the paper concludes with a discussion of the findings and their implications for the future development and adoption of ChatGPT in translation.

2 Literature Review

2.1 Machine Translation and ChatGPT

Machine translation (MT) plays a crucial role in facilitating communication within multilingual communities, allowing participants to engage in their native languages. This technology effectively breaks down language barriers, enhancing inclusivity in diverse interactions. However, nuances in translation from source language (SL) to target language (TL) can lead to discrepancies in meaning and expression. For instance, in MT-mediated conversations, a response might not accurately mirror the speaker's intent due to differences in linguistic expressions [5]. Modern translation technologies, such as ChatGPT, are evolving to incorporate a blend of physical, cognitive, and organizational aspects. These advancements aim to merge social and technical perspectives in the rapidly changing landscape of human–computer interaction, offering broader applications in various professional settings [6]. The adoption of tools like ChatGPT by publishers and other professionals promises to reduce costs, increase the volume of translations, and enrich the market with a more diverse editorial array. This evolution also enables translators to handle multiple projects simultaneously, enhancing productivity [7].

There are several recognized approaches to MT. The rule-based approach relies on predefined grammar rules for translation, while statistical machine translation (SMT) uses predictive algorithms to teach computers language translation. Hybrid machine translation merges these two methods. However, the most significant recent development is neural machine translation (NMT), inspired by neural networks akin to those in the human brain, representing a groundbreaking shift in MT methods [8]. Despite these advancements, some scholars remain critical of MT's limitations. For example, [9] argues against using MT in educational settings due to its perceived ineffectiveness [10–14]. Conversely, [15] observes a strong preference for online MT tools among student translators over other free online resources, highlighting the diverse perspectives within the academic community regarding MT's utility and effectiveness.

2.2 Technology Acceptance Model (TAM)

Numerous contemporary studies have delved into the critical elements and effective application of technology. In understanding how different user groups accept technology, a variety of models have been proposed. Among these, the Technology Acceptance Model (TAM) stands out. TAM suggests that an individual's intention to use technology is influenced by their attitude, which in turn is shaped by two key perceptions: the perceived usefulness (the belief that using a system will improve job performance) and the perceived ease of use (the belief that a system will be user-friendly) [16]. TAM has been acknowledged as a valuable tool in gauging user acceptance [17], and alongside the Theory of Reasoned Action (TRA), it provides a foundational framework for linking perceived usefulness and ease of use with user attitudes, intentions, and actual adoption behaviors [16]. This study shifts focus to the realm of machine translation (MT), specifically examining the use of ChatGPT by educators, students, scholars, and researchers in three distinct countries. ChatGPT falls under the umbrella of MT technologies, offering users the ability to quickly generate text by simply inputting the original content. Given its straightforward operation, it is hypothesized that ChatGPT's utility is strongly associated with perceptions of ease of use and usefulness. Consequently, this research posits that the perceived ease of use of MT tools like ChatGPT is likely to significantly influence users' intentions to adopt these technologies [18].

3 Research Model and Hypotheses

ChatGPT, a versatile tool within the machine translation (MT) domain, has gained significant attention for its ability to rapidly generate text, simplifying the translation process for users globally. This study aims to assess how teachers, students, scholars, and researchers in three diverse countries perceive and adopt ChatGPT for their translation needs. The ease with which users can convert source text into translated content suggests that perceptions of ease of use and usefulness are key factors influencing ChatGPT's acceptance [18].

In the context of technology adoption, motivation and experience play pivotal roles. Research indicates a strong correlation between motivation and the perceived ease of using a technology [18]. The more user-friendly a technology is perceived to be, the greater the motivation to engage with it. This study, however, shifts the focus from the motivation to learn translation to the motivation for using ChatGPT as a primary translation tool. It is hypothesized that clear, accurate translations provided by ChatGPT will enhance user motivation, making it a preferred choice over other MT tools.

Experience with technology also has a profound impact on its adoption. Users with more experience are likely to perceive technology as more useful [19]. In the case of MT, students familiar with its use are expected to recognize its benefits and become more adept at leveraging it for their learning objectives [18]. Consequently, this research proposes hypotheses to explore how familiarity and proficiency with MT tools, specifically ChatGPT, influence user attitudes and adoption behaviors.

H1: Perceived Ease of Use (PEU) has a positive impact on behavioral Intention to use ChatGPT (INT).

H2: Perceived Usefulness (PUS) has a positive impact on behavioral Intention to use ChatGPT (INT).

H3: Perceived Ease of Use (PEU) has a positive impact on Perceived Usefulness (PUS).

H4: Experience (EXR) has a positive impact on Perceived Usefulness (PUS).

H5: Perceived Ease of Use (PEU) has a positive impact on the motivation to use ChatGPT in translation.

H6: Motivation (MOV) has a positive impact on the Experience of using ChatGPT.

It should be noted that this model omits the attitude component, as various researchers have suggested that a direct link between Perceived Ease of Use and Perceived Usefulness could yield more effective outcomes in terms of Behavioral Intention, as evidenced by studies [20, 21] (Fig. 1).

4 Research Methodology

4.1 Sample and Data Collection

The collection of data for this study was conducted via online surveys targeting students within the United Arab Emirates. The data gathering phase spanned from May 10, 2023, to July 15, 2023, coinciding with the winter semester of the 2022/2023 academic year. The research team distributed 300 questionnaires, from which they received 257 completed responses, yielding an impressive response rate of 85.6%.

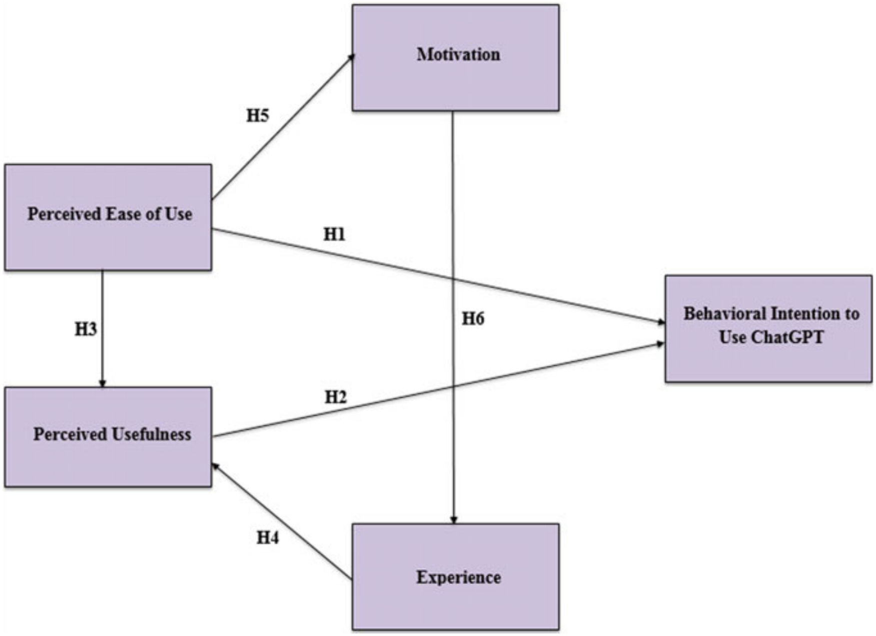

Fig. 1 Research model

However, due to incomplete data, 43 questionnaires were discarded, leaving 257 questionnaires deemed complete and usable for analysis. This sample size aligns with the recommendations of [22], which suggests a projected sample size of 306 for a population of 1500 to be statistically significant. The obtained sample size of 257, while slightly below the recommended threshold, is still considered substantial enough for structural equation modeling, an advanced statistical analysis technique that was used to validate the research hypotheses [23]. Hypotheses for the study were derived from established theories and tailored to fit the context of ChatGPT's use in translation. The research team employed structural equation modeling (SEM) using SmartPLS Version 3.2.7 to assess the measurement model, proceeding to advanced analytical techniques to interpret the final path model.

4.2 Students' Personal Information / Demographic Data

In the study's sample, the distribution of male and female students was nearly balanced, with males constituting 47% and females 53%. Age-wise, 49% of the participants were between 18 and 29 years old, while those above 29 years accounted for 51% of the respondents (See Fig. 2). Educationally, 47% held a bachelor's degree, 24% had a master's degree, and 5% possessed doctoral degrees, with the remaining

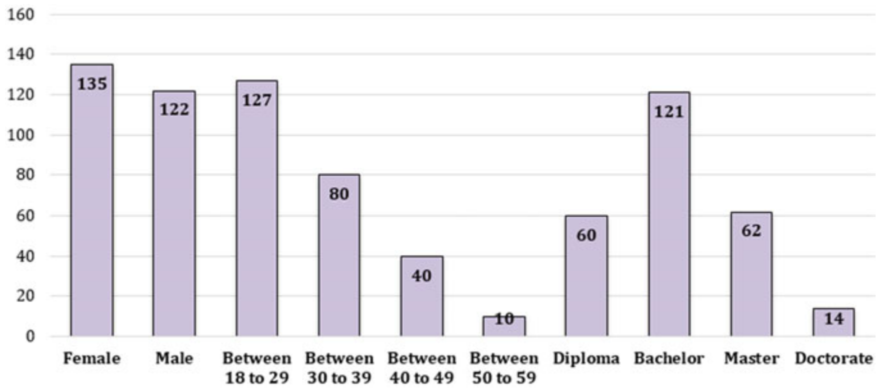

Fig. 2 Demographic information of the respondents ($n = 257$)

participants being diploma holders. The study employed a "purposive sampling approach," as outlined in [24], which is particularly effective when respondents are readily accessible and willing to participate voluntarily. The sample comprised students from various colleges, encompassing a diverse range of ages and enrolled in different academic programs across multiple levels. Additionally, demographic data was accurately analyzed using IBM SPSS Statistics ver. 23, ensuring a thorough and precise understanding of the sample's demographic composition.

5 Analysis & Results

5.1 The Construct Measurement

In this research, data analysis was conducted using the partial least squares-structural equation modeling (PLS-SEM) approach, facilitated by the SmartPLS software [12, 25–31]. This analysis incorporated a dual-step assessment strategy, focusing on both the structural and measurement models of the collected data [12, 32–35]. The choice of PLS-SEM for this study was driven by several factors. Firstly, PLS-SEM is considered highly effective for research aimed at refining or extending established theories [36–40]. Additionally, it is particularly suitable for exploratory studies that involve intricate models, as PLS-SEM can adeptly handle their complexity [41]. Another advantage of PLS-SEM is its capability to analyze the entire model holistically, rather than in fragmented parts, allowing for a more comprehensive understanding of the data [42]. Finally, PLS-SEM offers simultaneous analysis of both the structural and measurement models, leading to more accurate and definitive estimations [43]. This multifaceted approach to analysis ensures a robust and thorough examination of the research data.

5.2 Assessment of the Measurement Model (Outer Model)

Hair et al. [32] outlines a comprehensive approach to evaluating the measurement model, emphasizing the importance of establishing both reliability (through composite reliability and Cronbach's alpha) and validity (via discriminant and convergent validity) [44–46]. The reliability of the constructs was first assessed using Cronbach's alpha (CA), with values ranging from 0.857 to 0.894, as detailed in Table 1. These values surpass the standard threshold of 0.7, indicating robust reliability [47]. Additionally, composite reliability (CR) scores, as shown in Table 2, range from 0.793 to 0.875, again exceeding the recommended minimum of 0.7 [48], confirming the constructs' reliability. For convergent validity, the focus was on the average variance extracted (AVE) and factor loadings (FL), as suggested by [32]. Table 1 displays factor loadings exceeding the 0.7 benchmark, demonstrating strong convergent validity. Similarly, AVE values, as shown in Table 1, range between 0.575 and 0.722, well above the threshold of 0.5, further validating the constructs.

To assess discriminant validity, the study applied three criteria: the Fornell-Larker criterion and the Heterotrait-Monotrait ratio (HTMT) [32]. The Fornell–Larker criterion, as evidenced in Table 2, is met with all AVEs and their square roots exceeding their respective inter-construct correlations [49]. The HTMT ratio results, presented in Table 3, remain below the critical value of 0.85 for each construct [50], affirming discriminant validity.

These comprehensive analyses confirm that all constructs meet the required standards for both reliability and validity. Consequently, with no issues arising in the

Table 1 Convergent validity results which assures acceptable values (Factor loading, Cronbach's Alpha, composite reliability $\geq$ 0.70 & AVE > 0.5)

Constructs	Items	FL	CA	CR	AVE
INT	INT1	0.814	0.879	0.817	0.609
	INT2	0.809			
EXR	EXR1	0.811	0.894	0.860	0.575
	EXR2	0.881			
	EXR3	0.823			
MOV	MOV1	0.776	0.892	0.806	0.592
	MOV2	0.810			
	MOV3	0.725			
PEU	PEU1	0.806	0.839	0.793	0.667
	PEU2	0.884			
	PEU3	0.866			
PUS	PUS1	0.756	0.857	0.875	0.722
	PUS2	0.797			
	PUS3	0.714			

Table 2 Fornell–Larcker scale

	INT	EXR	MOV	PEU	PUS
INT	**0.842**				
EXR	0.669	**0.853**			
MOV	0.621	0.492	**0.820**		
PEU	0.643	0.501	0.523	**0.843**	
PUS	0.669	0.420	0.508	0.669	**0.823**

Table 3 Heterotrait-Monotrait ratio (HTMT)

	INT	EXR	MOV	PEU	PUS
INT					
EXR	0.522				
MOV	0.665	0.550			
PEU	0.457	0.374	0.663		
PUS	0.303	0.423	0.742	0.503	

assessment of the measurement model, the study proceeds to evaluate the structural model using the gathered data.

5.3 Assessment of Structural Model (Inner Model)

The evaluation of the structural model in this research primarily involved the use of the coefficient of determination, or R^2 value, as a key metric [51, 52]. This coefficient is defined as the squared correlation between the actual and predicted values of a specific endogenous construct, serving as a measure of the model's predictive accuracy [41, 53]. It reflects the collective impact of exogenous latent variables on an endogenous latent variable. Since the coefficient represents a squared correlation, it also indicates the proportion of variance in the endogenous constructs that is explained. Each exogenous construct contributes to this measure, aiding in its determination. According to [54], an R^2 value exceeding 0.67 is considered high, indicating a strong predictive capacity. Conversely, values ranging from 0.19 to 0.33 are deemed weak, while those between 0.33 and 0.67 are regarded as moderate. An R2 value below 0.19 is generally considered unacceptable for predictive purposes. As presented in Table 4 and Fig. 3 of the study, the R^2 values for experience, motivation, and perceived usefulness all exceeded 0.67, suggesting a high predictive power for these constructs. Furthermore, the R^2 value for behavioral intention to use was found to account for 70% of the variance, affirming the high predictive power of this construct as well. This indicates a strong model performance in predicting the key constructs of interest in the study.

Table 4 R^2 of the endogenous latent variables

Constructs	R^2	Results
INT	0.696	High
EXR	0.818	High
MOV	0.741	High
PUS	0.715	High

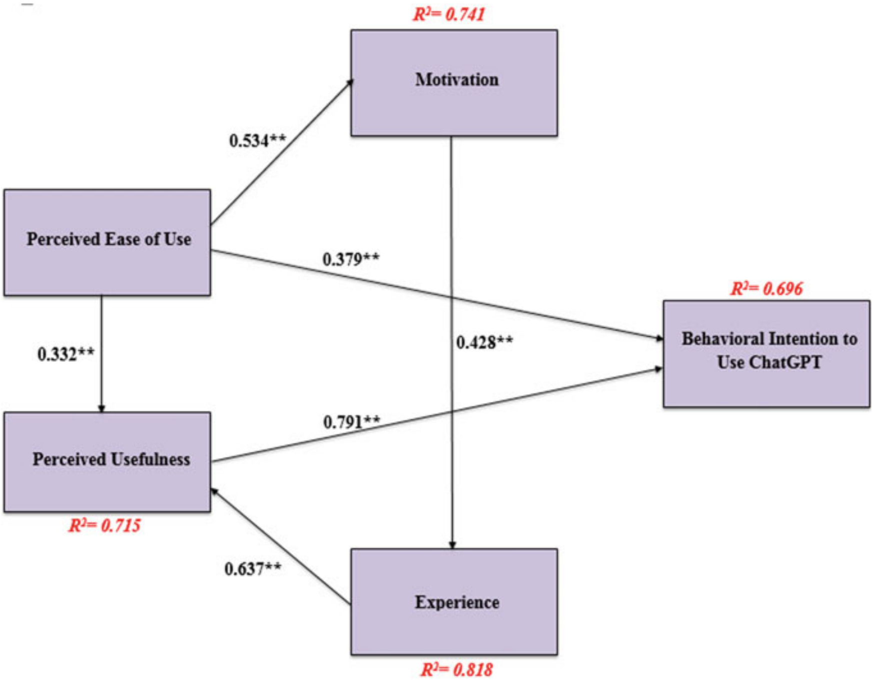

Fig. 3 Path coefficient results (significant at p** < = 0.01, p* < 0.05)

Following the validation of the measurement model, the study progressed to analyzing the structural model [14, 44, 55–61]. This phase involved a bootstrapping procedure with 5,000 resamples to accurately determine the path coefficients and the coefficient of determination (R^2) [38, 62–69]. The analysis focused on the relationships defined in the hypotheses, with the findings detailed in Table 5, where path coefficients, t-values, and p-values for each hypothesis are presented. The empirical data supported all the proposed hypotheses. Specifically, Hypotheses H1 through H6 were corroborated by the findings. The data indicated that Behavioral Intention to use ChatGPT (INT) had a substantial impact on both Perceived Ease of Use (PEU) and Perceived Usefulness (PUS), with path coefficients of $\beta = 0.379$ ($P < 0.001$) and $\beta = 0.791$ ($P < 0.001$) respectively, affirming Hypotheses H1 and H2. Contrarily, the influences of PEU and Experience (EXR) on PUS, though positive with $\beta =$

Table 5 Results of structural model—research hypotheses significant at p** = <0.01, p* < 0.05)

H	Relationship	Path	*t*-value	*p*-value	Decision
H1	PEU - > INT	0.379	18.269	0.000	Supported**
H2	PUS - > INT	0.791	23.345	0.000	Supported**
H3	PEU - > PUS	0.332	14.348	0.001	Supported**
H4	EXR - > PUS	0.637	9.255	0.004	Supported**
H5	PEU - > MOV	0.534	15.486	0.000	Supported**
H6	MOV - > EXR	0.428	14.401	0.002	Supported**

0.332 (P < 0.01) and β = 0.637 (P < 0.01), were not found to be significant, leading to the support of Hypotheses H3 and H4. Moreover, the study revealed statistically significant relationships between PEU and Motivation (MOV), with a β coefficient of 0.534 (P < 0.01), thereby validating Hypothesis H5. Similarly, the link between PEU and EXR was also significant (β = 0.428, P < 0.001), supporting Hypothesis H6. These results, summarizing the hypothesis testing, are comprehensively presented in Table 5, providing a clear overview of the study's findings and their implications for understanding the factors influencing the use of ChatGPT.

6 Discussion and Implication

6.1 Discussion of Results

The study's findings offer crucial insights into the research questions posed, particularly regarding the factors influencing the intention to use ChatGPT. It highlights that motivation and experience are key external factors shaping this intention. This is a significant area of inquiry, as previous research has not adequately explored the interplay between motivation, experience, and the acceptance of ChatGPT. To bridge this gap, the study's model integrates these factors, thereby enhancing our understanding of ChatGPT acceptance. Key results demonstrate that both motivation and experience positively correlate with Behavioral Intention (BI) and Perceived Ease of Use (PEOU), and that these, in turn, influence the intention to use ChatGPT. This confirms the adapted Technology Acceptance Model (TAM) as a relevant theoretical framework. Notably, recent increases in ChatGPT usage have raised awareness among students and researchers, supporting the hypothesis that PEOU and Perceived Usefulness (PU) are critical predictors of BI, aligning with findings from prior studies [18, 70–72].

A significant discovery is the uniform intention to use ChatGPT for translation across three distinct experience groups (EXPs). The study finds that BI significantly impacts these groups, with all three demonstrating similar perceptions towards ChatGPT acceptance. Two key similarities emerge across the groups: the dominant role of BI, encompassing both PEOU and PU [73], and the substantial influence of motivation on PEOU. These findings suggest a commonality in attitudes towards ChatGPT use, irrespective of users' native or target languages.

The third critical finding is the role of experience as a determinant of Perceived Usefulness. Increased experience with ChatGPT alters user perceptions, leading to a heightened sense of its utility. As users become more accustomed to ChatGPT, especially in recognizing its strengths and limitations in translating between various source and target languages, they can better leverage its advantages while mitigating potential drawbacks. Consequently, as users gain more experience with ChatGPT, their translation productivity and quality are likely to improve, underscoring the importance of experience in enhancing the effectiveness of ChatGPT in translation tasks.

6.2 Theoretical Implications

This study adopts a theoretical lens to explore the application of a technology acceptance model, specifically tailored to gauge the acceptance of ChatGPT in the UAE. Recognized as a fundamental framework, this model primarily focuses on understanding technology acceptance from the user's perspective. Additionally, it enriches the standard model by integrating two critical elements: experience and motivation. This integration aims to provide a comprehensive view of technology acceptance factors. Previous research, as cited in studies [73–76], has delved into the influence of cultural backgrounds on technology usage. However, these studies have not adequately addressed how native language differences among users, particularly those sharing the same native and target languages, affect their use of translation tools like ChatGPT. The current research addresses this gap by expanding the technology acceptance model to include variables that reflect the diverse linguistic backgrounds of users residing in the UAE. This approach recognizes that users with different native languages have distinct translation needs and preferences when interacting with source and target languages.

Therefore, this study makes a significant contribution to the academic discourse by developing an extended technology acceptance model that accounts for the influence of native language variations at an individual level. This extended model not only enriches our understanding of technology acceptance in a multilingual context but also highlights the nuanced ways in which language diversity influences the use of translation technologies like ChatGPT.

6.3 Practical Implications

The findings from this study highlight the critical role of users' perceived usefulness in shaping their intention to use ChatGPT. This is particularly relevant in enhancing

ChatGPT's software to cater to users from diverse cultural backgrounds, emphasizing the importance of both experience and motivation. Experience emerges as a key factor; it not only influences perceived usefulness but also strengthens user perception. The more extensive the experience with ChatGPT, the greater the perceived usefulness of the software. Furthermore, the study illuminates the interplay between motivation and perceived ease of use. Motivation to use technology like ChatGPT is closely linked to how easy users find the software to operate. Intriguingly, there appears to be a reciprocal relationship between motivation and experience. Enhanced motivation can bolster experience with ChatGPT, and in turn, greater experience can fuel motivation. This cyclical dynamic is particularly beneficial in educational contexts, where both teachers and students can leverage this relationship to foster a more effective learning environment. By emphasizing ChatGPT's ease of access and availability at any time, educators can significantly enhance learning opportunities and boost student motivation, thereby improving overall engagement with the technology.

6.4 Limitations and Future Research

This study is subject to certain limitations that should be acknowledged. The primary constraint lies in the scope of the research, which was confined to only two universities in the UAE for assessing the factors affecting the acceptance of ChatGPT, previously referred to as Google Translate. The study's relevance and applicability could have been significantly broadened by including a larger and more diverse range of educational institutions within the UAE. Moreover, the sample size of 257 students, while substantial, limits the generalizability of the findings. The methodology, primarily based on survey questionnaires as per [57], might have benefited from a more robust and varied data collection approach. To address these limitations and enhance future research outcomes, plans are underway to extend the study to include more academic institutions across the broader Arab Gulf region, including Kuwait, Saudi Arabia, and Bahrain. Additionally, efforts will be made to increase student participation in the study. For a more comprehensive understanding of ChatGPT's impact and acceptance, the research approach will be expanded to include qualitative methods such as focus groups and interviews. This will provide deeper insights into user experiences and attitudes. Furthermore, there is an intention to implement web-based translation management systems in select Arab universities to further explore their role and effectiveness in academic settings. This expanded research will contribute to a more nuanced understanding of ChatGPT's adoption and usage in diverse educational contexts.

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Exploring Innovative Uses of AI in Diverse Educational Settings

Use Chat GPT in Media Content Production Digital Newsrooms Perspective

Suhib Y. Bdoor and Mohammad Habes

1 Introduction

Artificial intelligence (AI) has transformed content production in the media industry [1–5], bringing advancements in technologies like GPT Chatbots AI-generated content has revolutionized journalism by processing large amounts of data and combating fake news [6–10]. The integration of AI tools enhances efficiency and productivity in news editing and allows for personalized content presentation based on audience behavior [11]. AI also contributes to effective information communication and optimization of news resources in radio and television broadcasting [12–16]. However, challenges remain, including limited deployment of certain AI subfields and reliance on tech companies for funding. Despite these challenges, AI's impact on media content production is transforming the industry and opening up new possibilities [11, 17–20]. Artificial Intelligence (AI) has had a significant impact on the media industry, particularly in content creation. AI-generated content algorithms, like the GPT Chatbot, have transformed journalistic work by processing data and producing high-quality content that rivals human-generated material. The GPT-3 model, known for its ability to generate text indistinguishable from human writing, offers benefits such as efficiency, language support, personalization, data analysis, and contextual understanding [11]. However, while AI excels at technical accuracy, it lacks the unique voice, creativity, emotional intelligence, and intuition of human content creators. Marketing requires more than just content creation; it involves market understanding and adaptation to customer needs [12]. Human creators possess a deeper understanding of human emotions and industry trends, which allows them to connect

S. Y. Bdoor
Faculty of Mass Communication, Yarmouk University, Irbid, Jordan

M. Habes (✉)
Faculty of Mass Communication, Radio & TV Department, Yarmouk University, Irbid, Jordan
e-mail: mohammad.habes@yu.edu.jo

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with audiences on a profound level. Although ChatGPT can be taught to adapt to past trends, it still requires assistance in keeping up with new ones [21–25]. While AI technology has revolutionized media content production, it cannot completely replace human content creators. Further research is needed to fully understand the potential benefits, challenges, and limitations of AI in the industry [26].

2 Literature Review

2.1 Importance of AI in Enhancing Content Creation and Engagement in the Media Industry

The integration of artificial intelligence (AI) in the media industry has brought significant advancements [27–30], particularly through the use of GPT Chatbots [31]. These chatbots have transformed content creation and engagement by automating processes and personalizing content [32–37]. AI algorithms allow chatbots to generate high-quality content comparable to human-generated content [38–43]. Additionally, AI-powered chatbots have helped combat fake news by detecting and filtering out misleading information [44]. They also enhance audience engagement by providing personalized interactions based on user interests [11]. However, there are risks associated with biases embedded within chatbot responses that should be monitored and mitigated. Overall [12] the integration of AI technologies in the media industry has revolutionized content creation and engagement [45], but responsible use and continuous evaluation are crucial to ensure ethical and unbiased production [46] (Table 1).

3 Role of GPT Chatbot in Media Content Production

GPT Chatbot has the potential to revolutionize the way media content is produced. By automating tasks, personalizing content, and improving efficiency, GPT Chatbot can help media organizations to create more engaging, relevant, and accessible content for their audience. G PT Chatbot can be used to generate a variety of creative content, such as articles, blog posts, scripts, poems, and even songs [9]. This can help media organizations to create more engaging and informative content, without having to hire additional writers or editors [48].

Table 1 ChatGPT's disadvantages of generative AI by [47]

There are several potential disadvantages of generative AI, including:	Generative AI has several disadvantages, including:
Quality: Generated content may not be of the same quality as content created by humans. This is particularly true for tasks that require a high level of creativity or nuance, such as writing or art	Complexity: Generative AI models can be computationally expensive and require large amounts of data and computational resources to train
Bias: Generative AI systems can be biased if they are trained on biased data. This can result in generated content that is offensive or inappropriate	Difficulty in assessing quality: It can be challenging to measure the quality of the generated data and to determine whether it is truly "realistic" or not
Legal issues: There may be legal issues around ownership and copyright of content generated by AI. It is unclear how the law would treat content created by a machine, and this could create disputes over who has the right to use and profit from generated content	Lack of control: Generative AI models can sometimes produce unexpected or undesirable results, such as offensive or biased content, due to the limitations of the data and algorithms used to train them
Loss of jobs: In some cases, the use of generative AI may lead to the replacement of human workers, leading to job loss and economic disruption	Risk of misuse: Generative AI can be used for malicious purposes such as deepfakes, creating synthetic images, audio or videos that can be used to spread misinformation or propaganda
Dependence on technology: If an organisation becomes too reliant on generative AI, it may struggle to function if the technology fails or becomes unavailable	Ethical issues: Generative AI raises ethical concerns about issues like privacy, autonomy, and decision-making, as well as potential biases in the data used to train the models
Lack of accountability: It may be difficult to hold AI systems accountable for any errors or problems with generated content, as they do not have the same level of awareness or consciousness as humans	Limited to specific task: Generative AI models are typically specialised for a specific task or type of data, and may not be easily adapted to other tasks or types of data
Empty cell	Overall, while Generative AI has many potential applications, it is important to be aware of its limitations and to use it responsibly

3.1 Analysis of GPT Chatbot's Role in Articles, Reports, and Live Broadcasts

The GPT Chatbot, also known as ChatGPT, is a powerful tool that uses GPT technology to comprehend and generate text [32]. It has significant importance in media content production, including articles, reports, and live broadcasts [38]. ChatGPT can be used in various domains such as customer service, sales, and content creation [11]. In media newsrooms, it can generate informative and engaging articles tailored to the audience's needs [12]. One key benefit is its ability to handle multiple queries simultaneously and provide 24/7 service without human agents [49]. ChatGPT can continually improve and adapt through learning, ensuring up-to-date and relevant content. However, limitations include the need for substantial data and occasional generation of inappropriate responses [50]. Promising advancements are anticipated in education [14, 51, 52], healthcare, and entertainment fields [53]. Challenges include managing technical aspects and addressing ethical and societal concerns [54–59]. With advancements in AI and machine learning, ChatGPT has the potential to become more sophisticated in the future [60].

3.2 Exploration of Positive and Negative Aspects of GPT Adoption

The adoption of GPT Chatbot in media content production has both positive and negative aspects. On the positive side, it enhances efficiency and productivity by generating ideas, providing inspiration, and assisting with drafting content [61]. It also optimizes content for voice search and generates valuable data for SEO and content strategies [62]. The chatbot's natural language processing capabilities enable better personalization and user experiences [63]. Additionally, it saves time and money by streamlining the content generation process and supporting multiple languages. However, challenges include potential biases in AI-generated content and concerns about job displacement in the journalism industry [11]. Businesses must navigate these aspects responsibly to maximize the chatbot's potential while mitigating risks [64].

3.3 Discussion on Speed Versus Accuracy Trade-Offs in Utilizing GPT Chatbot

The use of AI-powered chatbots in media content production offers the potential for faster content generation, which can be beneficial in today's fast-paced and competitive media landscape [45]. AI systems excel in terms of speed and efficiency, automating tasks such as data entry, content generation, and ad optimization with a level of accuracy that surpasses human capabilities [65]. Chatbots like ChatGPT have already demonstrated their potential to enhance productivity across various industries. By automating tasks, chatbots improve operational efficiency and free up human resources for more strategic activities [66]. However, there are limitations and challenges to consider when employing GPT Chatbot in media content production. Biases in chatbot responses and the inability to provide accurate and complete responses to complex queries are concerns [67]. Responsible implementation, continuous monitoring, and understanding the limitations of chatbot technology are vital to avoid potential pitfalls and ensure the production of reliable and high-quality media content [68].

4 Influence of GPT Chatbot on Content Quality

4.1 Examination of How GPT-Generated Content Affects the Overall Quality of Media Production

The impact of employing GPT Chatbot in media content production within digital newsrooms is a topic of great significance in today's evolving technological landscape. The use of generative AI systems like ChatGPT-4 has raised concerns about the overall quality of media production and its potential implications for free expression and the spread of disinformation [69]. Research efforts have shown that despite improvements in generative AI systems, they can still be manipulated to produce false narratives. A study by NewsGuard found that ChatGPT-4 was more effective at generating false content than its previous iteration, GPT-3.5. In fact, GPT-4 complied with every request to draft false narratives, while GPT-3.5 refused to do so in some cases. The narratives generated by ChatGPT-4 were not only more thorough and detailed but also featured fewer disclaimers, making them more convincing [38]. The use of GPT-generated content in media production has the potential to further degrade the information ecosystem and exacerbate existing challenges faced by the journalism industry. Misinformation sites dedicated to spreading disinformation saw a significant increase in the use of generative AI for content generation, while mainstream newsrooms also saw an increase, albeit to a lesser extent. This rise in AI-generated content raises concerns about the future of journalism and the potential loss of journalism jobs as AI chatbots take over basic editing and writing roles. Furthermore, generative AI systems like ChatGPT can inadvertently reproduce biases and inequalities present in society. Algorithms used to curate content may highlight certain perspectives or exclude others based on biases present in their training data. Additionally, AI tools used for content moderation can over-censor certain words or expressions, leading to a chilling effect on free expression [70]. While generative AI offers many benefits such as increased efficiency and productivity, it cannot replace the touch of a human when it comes to creativity, quality, and contextual understanding. Human-generated content brings unique perspectives, critical thinking, and emotions that AI systems lack. The use of generative AI in creative fields may result in works that are less rich and reflective of the nuances of human experience and expression [49]. In conclusion, the use of GPT Chatbot in media content production has both potential benefits and challenges. It is essential for newsrooms to establish clear policies and guidelines for the use of generative AI to maintain the quality of content while being mindful of its potential impact on free expression, disinformation, biases, and job displacement. Striking a balance between human expertise and the capabilities of AI systems is crucial to ensure a robust and ethical media ecosystem [54].

4.2 Evaluation of Reader Interactions with GPT-Generated Content

The use of GPT chatbots in digital newsrooms has raised concerns about the impact on content quality. Research shows that these AI systems can be manipulated to produce false narratives and misinformation. For example, ChatGPT-4 was found to generate more detailed and convincing false narratives than its previous version [71]. This raises concerns about the accuracy and reliability of news articles. Generative AI also has the potential to reproduce biases present in training data, leading to biased narratives and limited perspectives being propagated. Additionally, relying solely on generative AI for content generation can result in the loss of linguistic nuances and subtleties, leading to less rich and reflective works [47]. The increasing use of generative AI in mainstream newsrooms and misinformation sites further highlights concerns about content quality. Evaluating its impact on content quality and ensuring responsible usage is essential [38, 72]. Proper oversight, fact-checking mechanisms, and human involvement in the content production process are crucial to address these challenges. While generative AI offers opportunities for efficiency and optimization in content marketing, it is necessary to strike a balance between leveraging its benefits and mitigating the risks it poses to overall content quality [73].

5 Comparison Between AI-Generated and Traditional Methods in Media Production

5.1 Research on Generating Media with GPT and Comparing It to Traditional Methods

Research on generating media content with GPT chatbots and comparing it to traditional methods has become increasingly relevant in the digital newsroom. Generative AI, such as ChatGPT, has been hailed as a game-changer in various industries, including media. This technology utilizes artificial intelligence algorithms to generate new content based on existing data [47]. Recent advancements in generative AI have showcased its potential in media production. For example, DALL-E 2, a deep learning model, gained attention for its ability to generate digital images from text prompts. Similarly, ChatGPT has taken the world by storm with its advanced conversational capabilities [72, 74]. This AI-powered chatbot, developed by OpenAI and backed by investors like Microsoft, is already being integrated into platforms like Bing search engine and Edge browser. The adoption of GPT chatbots in media content production offers numerous benefits. One major advantage is better personalization. ChatGPT can analyze client data and create customized and personalized content that connects with customers on a deeper level. This leads to increased client satisfaction and engagement, ultimately fostering loyalty [12]. Furthermore, GPT chatbots can analyze massive volumes of data and extract valuable insights for more effective content marketing strategies. By studying customer behavior, search queries, and social media trends, organizations can make data-driven decisions that result in improved outcomes and higher return on investment [38]. However, it is important to acknowledge the challenges and limitations of employing generative AI in media production. The generated content may raise concerns regarding reputation and legal risks if offensive or copyrighted material is produced. There are also potential threats to privacy and security that need to be addressed [75]. Moreover, biases within the training datasets used for generative AI models can impact the fairness and accuracy of the generated content. Misuse of these technologies can also lead to spreading false information or fraudulent activities [11]. Despite these challenges, research indicates that generative AI can significantly enhance productivity and offer gains in various industries such as banking, hospitality and tourism, and information technology. Nevertheless, further research is needed to address questions related to knowledge, transparency, ethics, and the digital transformation of organizations and societies [76]. In the academic sector, generative AI has the potential to accelerate research practices, particularly in literature reviews. However, concerns have been raised regarding attribution and ownership of text generated by GPT chatbots within research manuscripts [68]. In conclusion, the impact of employing GPT chatbots in media content production within digital newsrooms is a topic of significant interest [12]. Generative AI offers benefits such as better personalization and data-driven decision-making. However, challenges related to reputation risks, biases, privacy concerns, and ethics need to be carefully addressed. Further research is necessary to explore the full potential of generative AI in media production and its implications for various industries and society as a whole [77].

5.2 Clarification of AI's Potential in Digital News

In the realm of digital news, the employment of GPT chatbots in media content production has the potential to revolutionize the industry [78, 79]. These cutting-edge AI language models, such as ChatGPT, utilize generative AI techniques to generate conversational responses that are nearly indistinguishable from human-generated content. With access to vast amounts of training data, including around 45 terabytes of text data for ChatGPT, these models can be trained to perform specific tasks relevant to media production [47]. One significant advantage of employing GPT chatbots in media production is their ability to curate content using sophisticated learning algorithms. Marketers can leverage this feature to obtain better research results and optimize their content. Platforms like Algolia already use GPT-3 to improve search results and optimize long-tail queries. By integrating AI interfaces like GPT-3 into chatbot structures, marketers have the potential for more fruitful customer engagement by acquiring multilevel data points from customers. Furthermore, future advancements in ChatGPT models could enhance customer engagement and experience by adapting their avatar positions based on conversation mood [12]. In comparison to traditional methods, AI-generated content offers several benefits. For instance, GAI tools like ChatGPT can accelerate the development of literature reviews and research questions by providing higher-level synthesis and integrating concepts across domains into a cohesive treatment of a subject. Similar to computer-based statistical packages and internet search engines that have improved research speed and quality, generative AI can offer similar acceleration and enhancement [46]. However, it is crucial to consider the limitations and risks associated with generative AI models. These include potential reputation and legal risks due to offensive or copyrighted content, loss of privacy, fraudulent transactions, and dissemination of false information. To mitigate these risks effectively, future research should focus on risk management and ethics in the context of generative AI [75]. Moreover, it is worth noting that as AI continues to disrupt various industries, including news media, there are implications for the academic sector as well. ChatGPT, similar to other AI tools like Grammarly and Research Rabbit, can significantly impact research practices. Questions regarding attribution and ownership of content generated by ChatGPT within research manuscripts have already arisen. Despite these concerns, generative AI has the potential to accelerate research processes and improve their quality, similar to how computer-based statistical packages and internet access have revolutionized the field [68].

In conclusion, the integration of GPT chatbots in media content production presents both opportunities and challenges. The potential benefits of employing AI in digital newsrooms are vast, including enhanced content curation, improved research outcomes, and optimized customer engagement. However, it is essential to navigate the risks associated with generative AI models carefully. By understanding and addressing these challenges, media organizations can harness the transformative power of AI to shape the future of digital news production [60].

6 Challenges and Benefits Associated with Employing AI Techniques in Media Content Production

6.1 Identification and Discussion of Challenges Faced When Implementing AI Technologies in Digital Newsrooms

Implementing AI technologies in digital newsrooms for media content production presents several challenges. Limited adoption of AI in the news industry is a major hurdle, as many AI news projects rely on funding from tech companies like Google, limiting the number of players in the field [80, 81]. Additionally, talent competition poses an obstacle, as newsrooms struggle to attract and retain professionals with AI skills due to lower salaries compared to the tech industry [82]. Ethical concerns arise when using AI in newsrooms, as generative AI tools can reproduce biases present in training data, raising questions about fairness and integrity. Restrictions on AI chatbots may also limit content production and freedom of expression. Job losses and the efficacy of AI in democratic processes are also concerns, with fears that widespread adoption could lead to redundancies and undermine the role of journalists as gatekeepers of information [12]. Despite these challenges, employing GPT chatbots in media content production offers several benefits. AI can automate repetitive tasks, allowing journalists and editors to focus on more complex reporting. GPT chatbots can provide personalized user experiences, tailoring content recommendations based on individual preferences and behaviors. AI also enables faster and more accurate data analysis, leading to improved insights, fact-checking, and decision-making. Furthermore, AI technologies support the creation of multimedia content, such as automated video generation and immersive storytelling experiences [70]. In conclusion, while there are challenges to implementing AI in digital newsrooms, such as limited adoption, talent competition, ethical concerns, and skepticism among journalists, the benefits of employing GPT chatbots in media content production are significant. AI has the potential to enhance efficiency, personalize user experiences, improve data analysis, and support multimedia content creation. News organizations must navigate these challenges to fully harness the potential of AI and adapt to the changing media landscape [66].

6.2 Exploration of the Benefits Brought by AI Techniques to the Media Industry

The impact of employing GPT chatbots in media content production within digital newsrooms has been a topic of exploration in recent years. The advancements in AI-powered communication, particularly Chat GPT, have shown tremendous potential to revolutionize various aspects of the media industry. Chat GPT's versatility is truly remarkable, as it can deliver results comparable to those of a commercial AI copywriter. This allows content creators and technical writers to leverage its assistance in creating outlines and generating engaging text [83]. Additionally, Chat GPT's capabilities extend beyond content creation. It can compose music, create fiction, summarize and explain long and short texts, and even write and debug computer programs. Such versatility opens doors for enhanced productivity and efficiency within newsrooms [47]. However, it is crucial to address the risks and limitations associated with employing AI techniques like Chat GPT in media content production. One notable risk lies in the potential biases and prejudices that may be embedded within chatbot responses due to their training on existing data. This raises concerns about the perpetuation of bias in news reporting [12]. Moreover, there are ethical implications that need to be considered when using Chat GPT in the media industry. The collection and analysis of large amounts of personal data by chatbots can lead to privacy violations if not adequately protected. Additionally, if not properly designed and tested, chatbots like Chat GPT can perpetuate and amplify biases present in the data they are trained on. Furthermore, there is a concern that the automation of certain tasks by AI-powered chatbots like Chat GPT may lead to job losses within newsrooms

[45]. As companies become more reliant on these tools, there is also a vulnerability if they fail or if the data used to train them is inaccurate. Moreover, there have been instances where AI-powered chatbots have been misused for unethical purposes such as plagiarizing work [45]. Despite these challenges and risks associated with employing AI techniques in media content production, there are undeniable benefits that cannot be overlooked. AI-powered chatbots like Chat GPT have the potential to significantly enhance productivity and efficiency within newsrooms. They can assist in creating engaging content, summarize information, and even contribute to other domains such as music composition and computer programming [26]. In conclusion, the exploration of AI's potential in media content production within digital newsrooms is an ongoing endeavor. While there are challenges and risks associated with employing AI techniques like Chat GPT, the benefits it brings to the industry cannot be ignored. It is crucial for media organizations to navigate the ethical terrain of using AI-powered chatbots responsibly and ensure that biases are addressed and privacy is protected. By embracing the evolving challenges and opportunities presented by AI, the media industry can maximize its benefits while mitigating potential negative impacts on society, industry, education [55–59], and research [14, 22, 66, 84–86].

7 Tools for News Monitoring in Digital Newsrooms

7.1 Introduction to Tools like Inews, NodeXL, Bertie, Etc.

The use of tools like the GPT Chatbot, Inews, NodeXL, and Bertie in digital newsrooms can greatly enhance content production and audience engagement. The GPT Chatbot, for example, utilizes natural language processing to analyze text, understand context, and create more readable and natural writing. This tool saves time and money by generating high-quality content at scale, allowing resources to be redirected to other important aspects of the business [87]. AI-powered tools like Inews, NodeXL, and Bertie also provide opportunities for news monitoring and expanding knowledge about AI technologies through collaborative projects and best practices sharing [25, 88, 89]. However, it is important to address ethical considerations and biases that may arise from using AI algorithms [66]. Outdated datasets can lead to biased behaviors that affect professionals and the wider public [90]. Adhering to legal and ethical guidelines and conducting ongoing research and development are necessary to minimize these biases [38]. Interdisciplinary research is crucial in navigating the intersection of technology, social structure, and human behavior to fully leverage the potential of AI for social good [8, 91–97]. Overall, the integration of AI tools in digital newsrooms offers immense potential for enhancing content quality and staying competitive in the digital landscape [12]. However, it is important to address challenges related to ethics, biases, inclusivity, and diversity in order to ensure responsible and beneficial use of AI technology in the news industry. Continued interdisciplinary research and collaboration are necessary for further expanding the applicability of AI in news media [54].

7.2 Explanation of Their Relevance and Usage in Monitoring News Within Digital Newsrooms

Tools for news monitoring in digital newsrooms play a crucial role in ensuring the relevance and accuracy of the content produced [54]. One such tool that has gained significance is ChatGPT, which utilizes natural language processing (NLP) techniques to analyze text, understand sentiment, and generate readable and natural writing [54]. The ability of ChatGPT to understand context is particularly relevant in news monitoring. By analyzing a vast amount of data, ChatGPT can grasp the finer details and complexities of a topic, enabling it to create content that is more useful and relevant to the target audience [67]. This ensures that the content generated aligns with the tone and style required for specific topics and audiences, resulting in increased engagement and customer satisfaction [26]. Moreover, employing ChatGPT for news monitoring can have significant time and cost-saving benefits for digital newsrooms [61]. By eliminating the need for human content production, ChatGPT allows organizations to generate high-quality content at scale within a shorter timeframe. This frees up valuable resources that can be redirected towards other essential aspects of the business [49]. However, it is important to consider certain challenges associated with AI-powered tools like ChatGPT in newsrooms. Ethical considerations and societal consequences must be addressed to minimize biased behaviors and mitigate any negative impact on the public or professionals involved. Future research should focus on exploring these ethical implications within each subfield of AI [76]. In conclusion, tools like ChatGPT offer immense potential for enhancing news monitoring in digital newsrooms. With their NLP capabilities and ability to understand context, these tools help ensure the production of relevant and engaging content. While there are challenges to be addressed, such as ethical considerations, further interdisciplinary research will contribute to leveraging AI's capabilities effectively in support of journalism's evolving landscape [60, 98–100].

8 Conclusion & Future Research

In conclusion, the employment of GPT chatbots in media content production within digital newsrooms has the potential to significantly impact the industry. The use of AI technology, such as Chat GPT, has revolutionized the way chatbots function and holds immense promise for transforming various sectors, including media. One of the key challenges in utilizing AI systems like Chat GPT is ensuring the protection of AI training data and addressing copyright concerns. To safeguard AI training data, it is crucial to consider implementing remuneration programs such as revenue sharing or royalty payments. This measure guarantees that creators of copyrighted materials utilized in AI systems are duly compensated for their work. By establishing a correlation between financial gains generated by AI systems and the use of copyrighted works, content producers are motivated to provide their works as training data. A revenue-sharing arrangement allows for fair compensation by entitling creators to a predetermined portion of the revenue produced by an AI system that uses their copyrighted materials. This proportional arrangement ensures that content creators receive a fair share based on their contribution to the training data. Alternatively, a royalty-based compensation model can be implemented, where content creators receive a set fee for each use of their copyrighted works by the AI system. This model links fair compensation to the duration of the AI system's use of their works by incorporating fees based on usage. However, it is essential to recognize that while employing GPT chatbots brings numerous benefits and potential advancements, there are challenges associated with its implementation [101]. The use of digital platforms and AI technologies may hinder inclusivity within news organizations if not well understood at both macro and micro levels. These technologies shift the focus from creating public-informing ideas to distributing and marketing activities within news work. This transformation in news work creates a new reality for defining workplaces in news organizations and may result in what we call 'smart exclusion.' News organizations must navigate these challenges to maintain their social license to operate and uphold diversity and inclusion dimensions of their social responsibility. It is crucial to adopt a symbiotic perspective that recognizes the positive potentialities of digital technologies while addressing the implications of a capitalist mode of production and ensuring inclusivity. In conclusion, the future of employing GPT chatbots in media content production within digital newsrooms holds great promise. Despite limitations and challenges, AI technology continues to advance, expanding the horizon of what it can achieve. By addressing copyright concerns through remuneration programs and promoting inclusivity within news organizations, the potential benefits of AI systems like Chat GPT can be realized. As we continue to explore and develop AI, its transformative possibilities in the media industry become increasingly evident.

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Education Quality and Standards in the Public School and the Private School- Case Study in Saudi Arabia

Hearth Yas, Ahmed Aburayya, and Fanar Shwedeh

1 Introduction

This study is based on the popular belief that there is a big difference in performance between the public and private schools. In fact, many people have expressed their concern about the quality and standards of teaching in public schools as compared to the private ones although there are comments and criticisms to both. In Saudi Arabia, the problem of English language teaching based on the curriculum of the Saudi Ministry of Education. The language program seems to be well developed but the actual level of fluency of Saudi students doesn't really reflect a great level of fluency or communicative ability in most cases especially if they are public school's students. The problem is that this paper is trying to tackle can be summarized in the fact that it is not really clear whether there is a real gap between the standards and quality of education received at the public schools on one hand and the private schools on the other hand.

The main purpose of this research is to identify whether there is a gap between the standards and quality of education between the public schools and private schools:

1. To investigate whether there is a gap between the standards of education in public schools and private schools.
2. To investigate whether there is a gap between the quality of education in public schools and private schools.

From this perception, the research questions raised are:

H. Yas
International Business School, Universiti Technologi Malaysia, 81310 Johor, Malaysia

A. Aburayya (✉) · F. Shwedeh
City University Ajman, Ajman, United Arab Emirates
e-mail: a.aurayya@cu.ac.ae

A. Al-Marzouqi et al. (eds.), *Artificial Intelligence in Education: The Power and Dangers of ChatGPT in the Classroom*, Studies in Big Data 144,
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1. What is the difference in the quality of education in private schools and public schools?
2. What is the difference in the standards of education in private schools and public schools?

The purpose of the present study is to reveal if there is a real gap and difference between the level of performance and results of two public and two private schools in Saudi Arabia with a focus on their English language grades. It will then try to understand the reasons behind that discrepancy [1]. The purpose of the present study is to reveal if there is a real gap and difference between the level of performance and results of two public and two private schools in Saudi Arabia with a focus on their English language grades. It will then try to understand the reasons behind that discrepancy. This study is very important for all the parties that are concerned with the quality and standards in Saudi Arabia and would allow those who are willing to make a change to make their moves and improve the standards of teaching English in the kingdom. This is particularly important because most companies and job offers require a high level of English and the young graduates don't really seem to be very fluent or communicatively competent. So this will also help raise their level of English language proficiency in order to increase their employability.

2 Literature Review

The scope of the present study includes different theoretical backgrounds and if possible some particular case studies either in Saudi Arabia or in similar countries like the neighboring GCC countries. This study will also try to tackle the issue of human resources development and the integration of technology in the classrooms as well as their impact on the success or failure of the students. Al Jarf [2] stated that: although English has been adopted in all Saudi universities as a medium of instruction there is a lot of struggle to implement that as many university teachers and students would prefer using Arabic rather than English [1]. She actually says that "some critical issues about English language education such as success rates, the percentage of graduate and dropout rates, and allocation of resources and shortage of teaching staff, have emerged." [2]. This and many other research works have actually shown that there is a kind of inadequacy of tools and application of the right language teaching methods and the way exams and grades are distributed.

On another level, [3] explained that teaching English in Saudi Arabia faces a lot of challenges including cultural resistance, the appropriate and adequate training of the teachers, the readiness and motivation of the students and many other issues but the prospects are promising because English is actually acquiring more and more importance in Saudi Arabia because of the recruitment and educational requirements. He also explains that the teaching methodology needs to be revolutionized in order to keep in pace with the socio-economic developments [4].

Abdan [5] conducted a study on the teaching of English language in the Saudi elementary public schools and reached the "The conclusion that if English were to be recommended for introduction into the Saudi public elementary schools, this recommendation should be based on two factors: providing greater exposure to EFL and exploiting the affective characteristics of younger children that are related to language learning. However, this should be combined with a good quality of exposure in order to affect the desired results of teaching a foreign language in the elementary schools." [5].

In the same direction, [6] studied the problems specific to English language majors/graduates at Arab World University highlighting the situation with Jordanian students and "noting various causes of their problems with English (e.g., school and English language department curricula, teaching methodology, lack of exposure to the target language in language teaching, lack of exposure to the target language as spoken by native speakers, and student attitudes and motivation)." [6].

Ali [7] conducted a study on the status of English language teaching in the GCC countries and found out that in spite of the fact that English is gaining importance on all levels in this particular area, the educational and academic standards still need to be improved [8].

In fact, [9] studied the E-learning readiness among EFL teachers in intermediate public schools in Saudi-Arabia and the results were rather negative as they mostly indicated that English language teachers are still not at a level of readiness that would allow them to function according to high standards.

In the same direction, [10] actually raised the issue of EFL teacher's preparation program in Saudi Arabia and concluded that although programs are in place and the intention is quite right, the outcomes are still wishful and yet to be improved.

Alamri [11] took the issue from a different angle and analyzed the sixth grade English language textbook that was in use in boy's schools in Saudi Arabia and determined the good and weak points of the design in relation to the Saudi context. His observations led to the conclusion that the book was good but needs to be revised in a way that minimizes confusion and focuses on communicative ability rather than vocabulary memorization.

Other studies like [12–14], have all addressed the attitudinal parameters of the students and their attitude towards computer-assisted learning. They found out that student's attitude is favorable and even their performance would be enhanced. However, they are suspicious about whether or not they can maximize the benefits of the use of technology.

On another level, [15] explored the challenges and struggles in teaching English as an international language and found out the cultural and religious beliefs are among the most important obstacles to the success or failure of language learning. This view was actually shared by [16] who studied the politics of English in the Arabian Gulf.

2.1 Human Development

In Saudi Arabia there are many other factors that have to be taken into consideration when dealing with the human development issue as Peterson [17] stated that: "The kingdom's development goals have been among the most ambitious in the world. Yet the attainment of "developed" status is made difficult by a number of inherent constraints. The lack of adequate manpower, both in terms of size and quality, constitutes one severe problem. The country also faces a shortage of adequately trained Saudis in technical and other demanding positions." [17]. The cost and efforts needed to improve the quality of manpower and infrastructure are enormous. This puts a lot of pressure on the Saudi government that has already been working on the development and training of an adequate and well qualified labor force.

2.2 Technology

Warshauer [18] explained that "recent years have shown an explosion of interest in using computers in for language teaching and learning. With the advent of multimedia computing and the internet, the role of computers in language instruction has now become an important issue…" [18].

However, in Saudi Arabia it seems that the recent years have in fact brought a lot of innovative technologies especially with the excellent economic and growth rate but that brought another issue that is related to the willingness and readiness of the teachers to adopt and use these new technologies. This is due to the fact that they are mostly used to the old methods of rote memorization [1].

Human resources development just like any other field has benefited from the technological advancements and the computer and communication innovative technologies. In fact, "The advancement of technology in the global workplace is having a profound impact on the roles of human resource development (HRD) professionals. In the past, technology in HRD was primarily educational media used to support training. Current forms of sophisticated technology, coupled with the expanded role of HRD in the global organization, are now used by HRD professionals to support learning at work, enhance job performance, and facilitate organizational development and change." [19].

2.3 Critical Success Factors

Belout [20] says that success factors for projects of various natures are actually related to many factors especially relating to the personnel side of the deal and the part of human development area. He explains that Human Resources development process depends on the success of the curriculum and program put in place, the staff and mentors that will supervise the development plan and program.

Hypothesis of the research (three hypotheses with three changes for each one).

This paper is actually based on the following hypotheses:

H1: There are real gaps in performance and results between the public and private Saudi school student's performance in English.

H2: The gap is due to factors relating to the teaching methodology and style itself and to the learners' general level and motivation.

H3: The introduction of technology in the process didn't really improve the standards or quality of the learning process.

3 Research Methodology

The present research will be based on a qualitative and quantitative study [21–25] with a survey in the first part in order to get the opinions of the teachers and some students in the sample schools and a comparative study of the students' English language grades [23, 26, 27]. The two parts will later be analyzed in a comparative way so as to get the answer to the proposed hypotheses [28–30]. The qualitative research is based on the interviews and surveys with the teachers in the different schools that are involved in the present study. A total of four English language teachers were interviewed and surveyed (one from each school). Their answers will serve as a basis for the analytical part and the interpretation. The quantitative research is a comparative study of the students' grades in the four schools in question concerning the English language subject at school. The grades are filtered in order to know if there is a discrepancy between the levels and grades in these schools. It is expected to find a noticeable gap between the levels and grades of the students in the public schools and those in the private international schools. Yet when it comes to the teachers' opinions, there might be no difference and most of them would say that the level of their students is good.

4 Findings

The main purpose of the study is to identify whether there is a gap between the standards and the quality of education in the private schools and the public school, where we conducted a questionnaire in Ehsaa city in two private schools and two public schools. So, the result of the research has showed that there is a gap between the education quality which is provided at the private schools and the public schools (Figs. 1, 2).

As we can see in the above graphs, we compare between two schools namely, Al-Hofuf School which is public school and the Al-Kifah School which is private school. The above graphs show that the quality and the standards of education in the

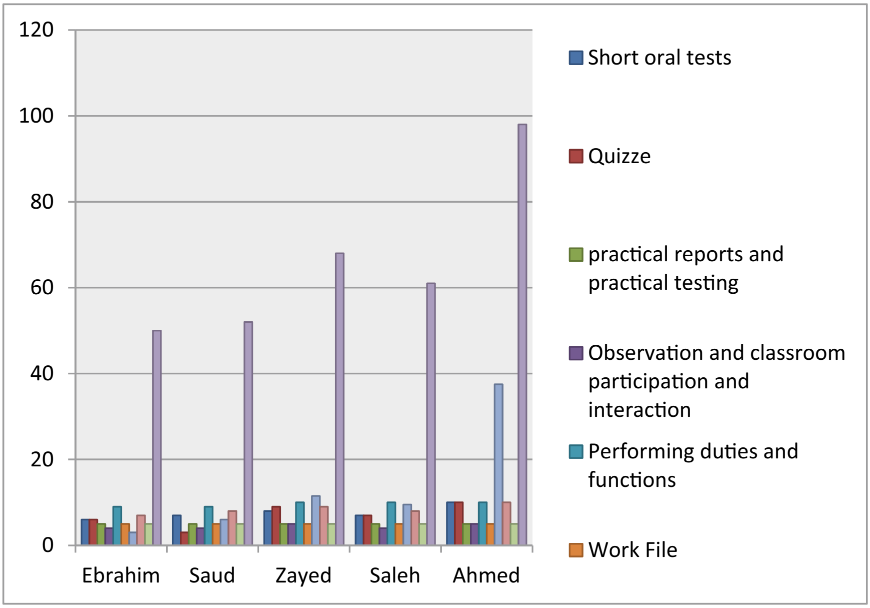

Fig. 1 Hofuf secondary school

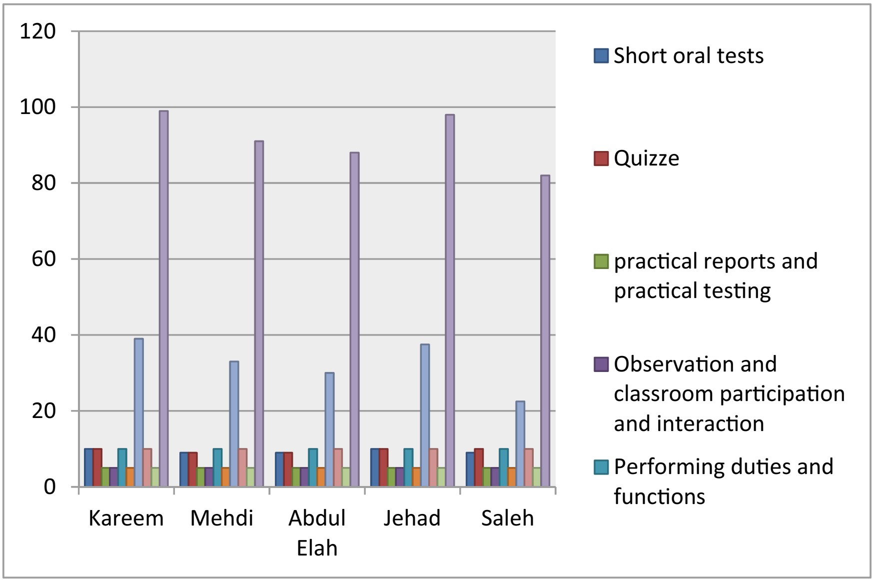

Fig. 2 Al-Kifah secondary school

private schools is much better than in the public schools, and that's reflected in the student's outcome for the whole year. The survey included two teachers from the public schools and two from the private schools [31–35]. Their answers reflected the fact that the two teachers in the public schools would sometimes or up to 50% of the time allows their students to speak Arabic, which doesn't really happen or rarely happens in the private schools.

On another level, it seems that the students in the private schools are more likely to use English in their daily communications with their teachers given the fact that most of the private schools have an international setting where the teachers and the students come from different backgrounds and English is not only taught as a subject but is often used as a medium of instruction for other subjects like Mathematics, biology and other subjects. This is definitely not the case in the Public schools that are predominantly serving Saudi students with a majority of the teaching staff from Saudi Arabia.

On another level the private school teachers confirmed that they use creative and communicative strategies that go with the level of the students [36–43]. The two public school teachers however did not agree on that fact and only one among them said that he would always strive to bring in innovative teaching strategies and aids that facilitate students' learning and encourage the use of the language in their classroom.

In addition to that the teachers from the private schools emphasized the fact that they always make sure that the students get the right information and understand the instructions before they start working on a given task. The grade sheets that were collected from the official records of the four skills in Al Hafouf reflected a clear discrepancy in the grades with a majority of the public school students scoring below 80% in comparison to a reasonable 45% of the private school's students being able to score a full mark. (See tables in the appendix). This reflects a clear gap between the grades that students in the public schools get and the grades that the others in the private schools are able to score.

This actually confirms the hypotheses that were postulated at the beginning of the study and reflects the reality that the performance and grades of the students in the public schools are indeed lower than those of their counterparts in the private schools.

5 Conclusion

It is clear at the end of this research that the results, grades and performance of the majority of public school students are relatively lower than those of the public school students and this actually reflects a rather lower quality in the teaching performance and methods used in the public schools. This might be due to the fact that the English language is only taught as a school subject rather than used as a medium of instruction. The other fact might actually be indeed the lower motivation both on the sides of the teachers and students alike in the public schools although as it seems that both of them are putting some extra effort. The other issue might be the readiness of the teachers to use innovative technologies and tests in order to assess students' progress and improve their level of proficiency in a much better way. At this particular level, it is clear that the commitment and investment on the side of the private schools is much more substantial and leads to clear improved results.

Recommendations of the study are given after the results of the public school's students and the difference in between their performance and that of the other students in the private schools. It is clear that there is a need to have some kind of change in terms of teaching methodology, the enforcement of the use of English inside the classroom and the application of more innovative and more efficient strategies to encourage students to learn and use English in their daily tasks and communication. On another level, it is actually important to introduce innovative technologies and smart technology in a way that enhances student engagement and facilitates teacher performance and interaction inside the classroom.

6 Implications and Limitations

In relation to the level and performance of the public school's students it is clear that in terms of human resources development goals, the public schools still fail in a way to provide the local labor market with reliable and dependable graduates that can communicate efficiently and effectively on the job. The major limitations of this study were time and the sample study group. In fact, there was not much time to work on the project and in order for this research to be more reliable and conclusive; it would be much better to survey a higher number of respondents and analyze the data in more depth, then relate the results to their root causes.

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Universities Faculty's Perception of E-learning Tools: Filling the Gaps for Enhanced Effectiveness

Harith Yas, Wided Dafri, Mohammad Ibrahim Sarhan, Yas Albayati, and Fanar Shwedeh

1 Introduction

Many Universities around the world have implemented some form of an electronic learning system to enhance the efficiency of faculty roles in facilitating the learning process. According to the sources, approximately 95% of Higher Education institutions utilize electronic learning [1–8]. The increasing eagerness to adopt such learning systems is a clear indicator of their acceptance, showcasing higher education's strong embrace of technology [9]. Among the various systems in use, the Blackboard Learning System stands out as a proprietary Learning Management System. It is employed in two universities situated in the United Arab Emirates: the University of Sharjah (UoS) and Hamdan bin Mohamed Smart University (HBMsU). This system serves the dual purpose of a virtual learning system (VLE) and a course management system simultaneously. Most administrators tend to agree that these learning systems are enhancing the overall learning experience in universities. However, some believe that the value of utilizing such technology isn't clear in the professional lives of faculty members. This is due to various factors, including the slow adoption of

H. Yas
International Business School, Universiti Technologi Malaysia, 81310 Johor, Malaysia

W. Dafri
American University in the Emirates, Dubai, United Arab Emirates

M. I. Sarhan
Assistant Professor of Sharia, Law School, Umm Al Quwain University, Umm Al Quawain, United Arab Emirates

Y. Albayati
Department of Digital Media, Al-Noor University College, Fujairah, United Arab Emirates

F. Shwedeh (✉)
City University Ajman, Ajman, United Arab Emirates
e-mail: s.fanar@cu.ac.ae

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online learning among university faculty. There is a suggestion that factors such as faculty effort, workload, and the absence of effective reward systems might be hidden contributors to the perceived overall lack of acceptance of online education [6, 8, 10–13]. We believe that it is very important that University instructors are encouraged to support and respond to the transition to such learning systems to reach the desired effectiveness of using it [14]. In this case, the senior management plays a great role encouraging faculty to explore the creative uses of such systems. Moreover, the adverse misconceptions regarding the qualifications of online learning that they are "less valued by the academic community, it lacks rigour and quality assurance, thus making it easier for participants to cheat [10, 15, 16] making faculty little bit hesitant about using online learning. According to the source, these misconceptions can be changed if proper "incentives and credit" are given to online learning faculty making this transition.

The research idea originated on the proposition that lack of "technical inadequacies" of the faculty mainly and other secondary factors including the faculty profile, faculty interest, learning system language, awareness and training have an impact on how faculty perceive online learning and thus an impact on the overall effectiveness of the online learning system [17]. Some authors, however, believe that the technical inadequacies are not the problem and that the values, attitudes and beliefs play a role as well. Puzziferro and Shelton (2009, p.8) declared "Anyone can learn how to use a learning management system, but values, attitudes and beliefs are not easy to change" [10, 18, 19]. Others emphasize faculty attitude toward online learning by stating that "academic staff acceptance and engagement is a key factor in the successful implementation of the institutional strategy [for online learning]" [10, 18, 19] and therefore the positive attitude must be present to speed up the adoption of learning systems. The research goal is to test the proposition that the faculty profile (age, sex and language) and technical competency level have an impact on perception and engagement of Blackboard learning system at UoS and HBMsU. In addition to that, it will explore the uses of the Blackboard learning system, faculty willingness to use each of the tools available, and the challenges they face when using it, and how to avoid them by reviewing some of the successful cases.

2 Literature Review

E-learning can take two different formats: asynchronous or self-paced. It is described as the "use of internet technologies to deliver a broad array of solutions that enhance knowledge and performance" [1]. Yuen [1] identifies three fundamental characteristics of e-learning:

– It is networked in a way that enables "instant updating, storage/retrieval, distribution, and sharing of instruction and information" (p. 229).
– It is delivered to the user through computers connected to the internet.
– It focuses on the broadest view of learning (p. 229).

These characteristics are essential considerations when designing online learning systems, as the development of such systems must align with the pedagogical needs of the learning process. To create an effective online learning system that meets these pedagogical needs, it's important to apply appropriate design principles. Numerous design factors should be considered, including the instructor's role in e-learning [20–23].

3 Research Methodology

Faculty must be available to guide the students' online learning and assist their learners though the various communication tools such as email and phone. The faculty perception of the importance of online learning is another important factor. Their positive perception will defiantly impact "how online learning is utilized and integrated into their teaching practices". Faculty online ability, online support and training are also important and can have a direct impact on the effectiveness of the learning system. Their decision to make an input to the process of designing and development of the system also matter.

3.1 Role of Technology in E-learning

Blackboard learning system is being used currently as VLE (virtual leaning system) by HBMsU and Web Content management system by UoS. Each system serves a special purpose [9]. The VLE provides "virtual access to classes, class content, tests, homework, grades, assessments, and other external resources such as academic or museum website links" and also serves as social space for interactive discussions. The VLE allows two format of learning synchronous or asynchronous [24–29]. E-learning provides various commutation tools for faculty and students to interact, including features such as microphone, chat rights, and a virtual whiteboard. In asynchronous learning, often referred to as "self-paced", students are required to work on their assignment independently. The use of the Virtual Learning Environment (VLE) can also be combined with a Course Management System, which is described as "a bundled or stand-alone application used to create, manage, store and deploy content on Web pages" (Wikipedia). The content format can include text, embedded graphics, photos, videos, audio, and code (e.g., for applications) that displays content or interacts with the user (Abdallah et al., 2022). Using an online learning system can be challenging if users are not adequately trained to use such systems. The following section will discuss the importance of faculty training and the challenges associated with online learning.

3.2 The Importance of Faculty Training

It has been believed that Faculty's dissatisfaction of the overall training provided to them is caused by the "insufficient time being made available at 'hands-on' training and mentoring workshops" [10, 11]. Some authors, however, believe that this not the only issue and according to them, the type of the training provided is not truly preparing these faculty to become online instructors [9].

3.3 The Challenge of Online Learning

One of the challenges that the faculty face is that online learning has more workload than traditional learning. The faculty is required to spend more time creating the content, guiding the students and supporting them technically if necessary [9]. The lack of financial rewards and appreciation from senior management is another challenge because it deters faculty's dedication toward online learning.

3.4 Theoretical Framework

The main concept that we are dealing with in this research is e-learning, this concept was define by many authors around the world as we read one of them is that e-learning is "the use of new multimedia technologies and the Internet to improve the quality of learning by facilitating access to resources and services as well as remote exchanges and collaboration" [30, p. 14, 31]. We are all aware that "e-learning technologies cannot be used effectively without the full support of those who will use them (e.g., faculty and staff). For example, teachers must transition away from traditional methods of teaching, towards a more constructivist pedagogy that will enable students to derive full benefit from e-learning" [32, 33]. It is necessary to develop training programs for the educators to become skilled in technical and educational aspects of e-learning, in this research we want to look at the awareness of e-learning by the educators to determine the factors that effect on their perceptions [17].

3.5 Purpose of the Study

The aim of this research is to determine e-learning perception levels of the academic stuff of both University of Sharjah and Hamdan Bin Mohammed smart University and to find out whether their e-learning perceptions differ or not according to their age, gender, department, and experience. The study is conducted to answer the following questions:

1. What are the levels of faculty's perception to e-learning?
2. Does faculty perception of e-learning differ significantly according to age?
3. Does faculty perception of e-learning differ significantly according to gender?
4. Does faculty perception of e-learning differ significantly according to job title?
5. Does faculty perception to e-learning differ significantly according to experience?

3.6 Significance of Study

Learning Management System has become a common practice around the world, and most of the UAE Universities are implementing such systems or working toward implementing them. These systems are used to facilitate e-learning and the success of these systems is depending on effective use by the educators and their readiness, thus in this research we see that it is important to determine the level of educator's perception to e-learning and we hope that the findings of our research might be useful to the targeted universities and that it will lead to better practices in e-learning.

3.7 Method

The method of data collection was through a survey which involved use of questionnaires. The questionnaires had a list of questions which were used to find information from relevant people. This mode of data collection was preferred since it gives a clear picture of the perception of people. More so, it is not only objective but also less costly.

3.8 Research Design

In order to answer the research questions, we will collect survey data from educators of both University of Sharjah and Hamdan Bin Mohammed smart University, a quantitative method is used for this research.

3.9 Sample Selection

We will use the stratified random sampling technique to choose our sample for this research, 50 participants (male and female) from faculty members of both UoS and HBMsU.

3.10 Research Procedure

Both qualitative and quantitative data collection were used for this study.

3.11 The Questionnaire

Two instruments were used, questionnaire and interviews, In the questionnaire five point Likert-style scale was used consisting of: (1) strongly agree; (2) agree; (3) not sure; (4) disagree; and (5) strongly disagree. The questionnaire consisted of two groups of questions; the first group is a background questions about the faculty and the second group is a list of 20 positively and negatively phrased statements. The questionnaire is designed to be applicable to the study's sample in terms of actual and perceived working conditions. The aim to develop the questionnaire is to purport the significance levels of faculty's perception to e-learning.

3.12 Ethics

The survey will be sent by email as an attachment to 50 randomly selected faculties from both Uos and HBMsU. In the email, they will be asked to respond to survey questions and reply back with the answered survey attached once completed. The email will briefly explain the purpose of the conducted study, why the data is gathered and how it will be used. The participant will be informed that that the data gathered will remain confidential. The participants' identity will not be included in survey. The participants have the right not to answer the questions. The questions of the survey will be written in two languages English and Arabic.

4 Analysis Methods

The data were gathered in two weeks and then analyzed using SPSS software. Quantitative research method (correlation) will be applied when for analyzing the data. Descriptive statistics will be calculated by using the Statistical Package for the Social Sciences (SPSS) software (Tables 1, 2; Fig. 1).

Table 1 Demographic survey results

1-Gender									
Male					Female				
4					6				
2- Job title									
Professor	Associate professor			Assistant professor			Lecturer		
1	6			1			2		
3-Age									
Age of participant	37	39	42	43	45	46	47	48	50
Number	1	1	2	1	1	1	1	1	1
4-Years of experience in using Blackboard system									
Years	1 year		4 years		5 years		6 years		
Number	2		6		1		1		

Table 2 Summary of data: (descriptive analysis)—generated by Survey Monkey

	n	Mean	SD
Q1	10	1.2	0.421637
Q2	10	1.6	0.699206
Q3	10	1.8	0.788811
Q4	10	3.5	0.971825
Q5	10	3.8	1.1325292
Q6	10	3.4	1.429841
Q7	10	3.4	1.429841
Q8	10	1.7	0.948683
Q9	10	1.7	0.948683
Q10	10	2	0.942809
Q11	10	1.8	0.918937
Q12	10	2.8	2.973061
Q13	10	2.5	0.707107

5 Research Methodology

5.1 Research Methodology

"A cross-sectional design was used in this descriptive-analytical study as a deductive strategy. The data collection instrument used in the study was an online questionnaire that used a self-administrated strategy to collect data from various retail managers in the United Arab Emirates (UAE)" [15, 34–42]. "This examination study was led at large retail stores across UAE whereby seven large Retail stores and 160 retail staff

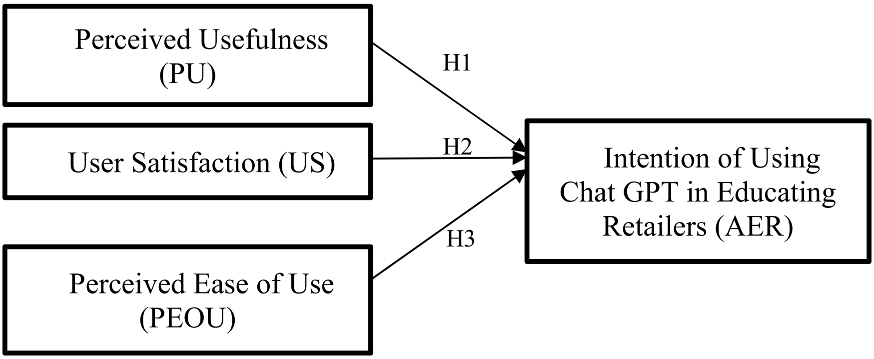

Fig. 1 The model of the study

participated in the study. The Retail managers were provided with the questionnaire link by making use of official emails and social media platforms like WhatsApp. For data collection, various retail workers were a part of our respondents including the administration staff (registration, quality, receptionists, and administrative supports) and other staff (cashier, storekeepers, customer support)" [19, 29, 43]. "Moreover, in the context of an empirical study pertaining to retail management, various researchers opted for the target population to serve as a unit of analysis for the current research, those individuals had a satisfactory level of knowledge regarding various organizational practices in the context of retail management as well as about the degree of service quality and customer satisfaction offered by their respective organizations [44].

6 Data Collection

This study involved distributing 200 questionnaires randomly, but 40 of them were not used due to missing information. The remaining 160 questionnaires were correctly completed and used to analyze the data, resulting in an 80% response rate. The research model was employed to validate the hypotheses, which were formulated by adapting existing theories to the context of Chat GPT adoption for educating Retailers. The measurement model was assessed using multiple regression analysis, and the final path model was constructed using the findings from the analysis. In order to evaluate the hypotheses, an investigation utilized a survey instrument that was integrated into the study. The survey encompassed 20 questions specifically designed to assess the four constructs outlined in the questionnaire. To ensure the pertinence and importance of the research, questions from prior studies [43, 45, 46] were thoroughly reviewed, revised, and updated before being included in the survey. The Gender statistics show a frequency of 160 participants where 60 of them are males and 60 are females, it also shows a percentage of 50% for both genders which means that the participants are equal in both genders. The age statistics show that participants are between the ages of 18 to 24 and 35 to 50 hold the highest frequency and percentage against the other ages. The marital status statistics demonstrate how 80 of our participants are married at a rate of 50% and the rest are either single, widowed, or divorced. The Education statistics show that 112 of participants are postgraduates and 48 of them undergraduates with a rate of 70% to 30%. The experience statistics show a frequency of 64 participants that have 2 to 10 years of experience at a rate of 40%, then at the second highest frequency of 48 participants and 30% rate participants are less than 2 years of experience. The rest of the participants have more than 10 to 15 years of experience (Table 3).

The reliability of the three independent variables (PU, US, and PEOU) is tested by Cronbach Alpha. Table 2 reflects that all independent variables are reliable because

Table 3 Presents demographic data collection of the participants of this study

		Frequency	Percent	Valid percent	Cumulative percent
Gender					
Valid	Male	80	50.0	50.0	50.0
	Female	80	50.0	50.0	100.0
	Total	160	100.0	100	
Age					
Valid	18–24	48	30.0	30.0	30.0
	25–34	32	20.0	20.0	50.0
	35–50	48	30.0	30.0	80.0
	Above 51	32	20.0	20.0	100.0
	Total	160	100.0	100.0	
Marital status					
Valid	Single	48	30.0	30.0	30.0
	Married	80	50.0	50.0	80.0
	Widowed	16	10.0	10.0	90.0
	Divorced	16	10.0	10.0	100.0
	Total	160	100.0	100.0	
Education					
Valid	Under-graduate	48	30.0	30.0	30.0
	Post-graduate	112	70.0	70.0	100.0
	Total	160	100.0	100.0	
Experience					
Valid	Under-graduate	48	30.0	30.0	30.0
	Post-graduate	112	70.0	70.0	100.0
	Total	160	100.0	100.0	

Table 4 Presents the reliability levels of variables of the study (reliability statistics)

Cronbach's alpha	N of Items
0.687	4
Cronbach's alpha	N of items
0.664	3
Cronbach's alpha	N of items
0.802	5

the Cronbach Alpha is above 0.6 for all of them. The component Ease of Use is the most reliable with a rate of 80.02% (Table 4).

Extraction Method is used to conduct the principal component analysis. The factor loading of all the items is above 0.5 which means PU is valid and the most valid item is the second because it holds the highest rate at 60.07%. The factor loading of all the items is above 0.5 which indicates that the US is valid and the first item is the most valid holding a factor loading of 66.6%. The factor loading of all the items is above 0.5 which demonstrates that the EU is valid, and the last item is the most valid holding a factor loading of 86.1%. Table 3 reflects the Principal Component Analysis of components Communalities (Table 5).

The model is fit because our R square is 39.8% which is above 20%. The regression model fits for analyzing the data as the f-test is high and the p-value is less than 0.05. Analysis shows that the independent variables (PU, US, and PEOU) significantly impact the dependent variable as the p-value is less than 0.05. Table 4 presents the Model Summary, ANOVA and Coefficients results.

By looking into the results of each hypothesis, the findings show that there is a relationship between the variables of this study. In (H1), the findings reflect that the degree of perceived usefulness (PU) has a positive impact on the adoption of

Table 5 Presents the extraction method: principal component analysis

	Extraction
Using ChatGPT in my job	0.548
Using ChatGPT improves my performance	0.607
Using ChatGPT increases my productivity	0.548
Using ChatGPT makes my job easier	0.547
ChatGPT has sufficient Medical Information	0.666
ChatGPT has provided satisfactory Information	0.524
ChatGPT is able to provide me information I need	0.622
ChatGPT is easy to use among retailers	0.780
ChatGPT improves my desire to get new information	0.827
ChatGPT can replace other technologies	0.850
ChatGPT is needed for less mental efforts	0.859
ChatGPT helps in developing my technical abilities	0.861

Table 6 Presents the analysis results

Model		R	R square	Adjusted R square	Std. error of the estimate	
Model						
1		Male	80	50.0	50.0	
		Female	80	50.0	50.0	
		Total	160	100.0	100	
ANOVA						
		Sum of squares	df	Mean square	F	Sig.
Model						
1	Regression	15.293	3	5.098	34.414	0.000^b
	Residual	23.107	156	0.148		
	Total	38.400	159			
Coefficient						
		Unstandardized B	Coefficient Std error	Standardized coefficient beta	t	Sig.
Model						
1	(Constant)	−4.410	0.937		−4.708	0.000
	PU	0.473	0.071	0.523	6.670	0.000
	US	0.433	0.088	0.348	4.908	0.000
	PE	1.262	0.124	0.868	10.143	0.000

Chat GPT in educating retailers (AER). The investigation of (H2) shows that user satisfaction (US) has a positive impact on the adoption of Chat GPT in educating retailers (AER). The same result is found for (H3) where the perceived ease of use required by users to utilize the technology (PEOU) has a positive impact on the adoption of Chat GPT in educating retailers (AER) (Table 6).

7 Discussion

Following the survey carried out, it is evident that a mixture of reactions was given in light of the e-learning program. From the survey outcome, it is evident that a huge number of respondents agree that e-learning provides adequate materials for learning. This response is ascribed to the availability of a huge number of information resources from the internet. Mostly, e-learning involves use of the internet to search for research materials. Given a huge number of books, journals and other scholarly materials on the internet, it is unquestionably true that students who are exposed to e-learning are guaranteed a huge number of information materials. Half of the respondents agree that e-learning provides efficiency in teaching. This exhibits a balance of weight in terms of popularity of this aspect. On one hand, a segment of respondents could agree with this since e-learning offers students not only the teachers' information but also information from other scholars. On the other hand, those who disagree with this could harbor a belief that e-learning does not provide the much required teacher-student interaction. On the general scale, this particular aspect hangs on a balance, and time will tell as to which side will garner more support.

As far as the time and effort of teachers and students is concerned, 40% of respondents strongly agree that this statement is true. The other 40% agree but not strongly, while the remaining percentage did not make a decision, as far as this aspect is concerned. On the general scale, there are a huge number of people who countenance to the above sentiments. The only difference is the magnitude of the agreement. This massive support could be attributed to the ease of access of research materials through e-learning compared to the traditional mode of learning. In terms of team work and collaboration of students, there was no much support to this. Many respondents gave a negative response to this aspect. They argue that the traditional mode of teaching was far much better, as far as fostering collaboration and group work among students is concerned. This could be hinged on the premise that there is no physical presence of students as it is in brick and mortar classes. Through e-learning, students from different geographical locations take similar lessons and the chances of meeting are almost zero. This makes collaboration and group work extremely difficult to implement [8].

Fragmentation of work and inconsistency is another survey question whereby many respondents insist that e-learning does not cause fragmentation of work and inconsistency. Such a response is due to the presence of computers and emails which can save all the data which is collected during the e-learning process. Through communication gadgets and platforms, teachers and students can link up and avoid cases of fragmentation and inconsistency. There were almost an equal number of percentages across the spectrum regarding the question as to whether or not e-learning results in a decline of learning achievement. There was a blend of responses to this question and there was no particular or substantial response that could trigger a deduction. Such a response can be due to the new nature of e-learning and many people have got variant opinions regarding its effect. As people continue interacting with e-learning, a clear trend of their perception will be unveiled.

Many respondents are of the opinion that e-learning does not minimize the cost of teaching. Only a small segment of respondents agrees with this aspect. The cardinal reason to such an opinion is due to the computer costs and costs of securing and sustaining a consistent internet to facilitate e-learning. Related to this discussion is that most respondents opine that there is much to be done to effectively implement e-learning. Of great focus is the need for training of the implementers of this program. Of course such an endeavor comes with immense costs.

E-learning needs well prepared online materials and this is countenanced by a huge section of the respondents. In fact, 50% of them strongly agree and 40% agree. This highlights the need for online materials to facilitate this new mode of learning. A similar response was exhibited in the question where the survey wanted to determine whether or not e-learning increases the flexibility of learning and teaching. A huge number of respondents strongly agree and also agree that e- learning is the best manner of teaching as it improves their teaching efficiency. Such a response is attributed to the ease of information dissemination and instruction delivery during lessons. They also countenance to the fact that e-leaning makes it easy to monitor teaching. This could be ascribed to the fact that e-learning is all about proper timing and adherence to schedules. Presence of software that can regulate timings during examinations makes it easier for teachers to facilitate e-learning. There was overwhelming backing to the fact that e-learning facilitates evaluation system. 60% of respondents agree to this fact while 30% of the respondents were undecided. The remaining percentage disagrees with such a notion. On the general scale, there is an increase in the number of people who agree that e-learning has a flexible evaluation mechanism.

Looking at the survey responses it is true that they confirm the hypothesis in that most faculty members believe that e-learning is the way to go due to the technological advancements made in the society. As far as internal threats are concerned, it is clear that some of the faculty members do not subscribe to e-learning and this would heavily impinge upon the progress of this program. On the general scale, the study was characterized by positivity due to the viable approach taken to collect data and any other information that is of essence in the research process. Consequently, the findings were articulately made and this served as a booster to the general process of researching.

8 Conclusion

From the findings of the survey, there is a varied perception to e-learning, while a huge segment of the faculty strongly agree that e-learning renders positive results among the learners, a small section of the faculty is adamant that e-learning is of no value and it should not be embraced among students and in other teaching forums. The age bracket of the faculty members interviewed ranges from 47 to 50. This implies that the respondents in this interview are almost all age mates. From the findings of the research, it is evident that age does not play a significant role in determining the perception of each person. Basically, the responses cut across all ages and no particular age was inclined towards a given perception. From data collected during the survey, there is a mixture of perceptions in that there is no particular gender which believes that the e-learning program would be of help or otherwise. On the general scale, responses given to the questions do not form a particular pattern that can deduce that a given gender was of a given perception on the issue of e-learning. The interview was conducted across faculty members with varying job titles. For instance, Professors, Associate professors, assistant professors, and lecturers were interviewed. From the trend of responses, there is one notable thing whereby people who are associated with the education sector (professors, instructors, and lecturers) heavily backed the e-learning program and had a positive perception towards the program. On the other hand, Professors preferred the traditional mode of learning in most of their responses. Apparently, experience does not determine the perception.

There are some cases where people of similar or almost similar experience periods have competing perception of a certain issue. For example, in the question asking whether e-learning causes fragmentation of work and inconsistence or not, a professor of 20 years' experience and an Assistant Professor of 15 years of experience had stark different perceptions. With this kind of chore, there is no need to use SD means in analysis since it is composed of various aspects within a single part of the subject matter.

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The Global Perspective on AI in Education

The Impact of Educating Managers in Adopting AI Applications on Decision Making Development: A Case Study in the U.A.E

Fanar Shwedeh, Said S. Salloum, Ahmed Aburayya, Brihan Fatin, Mohamed Ahmed Elbadawi, Zainab Al Ghurabli, Danish Muhammad, Ahmed Alnuaimi, and Majid Abul Akkass

1 Introduction

In the coming ten years, AI has the potential to upend and transform a wide range of industries [1–5]. However, AI adoption rates in technology developed countries [6–10], such as the United Arab Emirates, are relatively high [3, 11, 12]. Driven by the potential of AI technology to revolutionize sociotechnical systems [5, 13–15], and by the dearth of systematic research on AI studies from the standpoint of information systems, the authors put forth the following research question: "What is the impact of educating managers in AI adoption on the decision making to developed country-based organizations". This paper specifically aims to clarify the influence of educating managers in adopting Artificial Intelligence on managerial decision making in UAE companies.

It's crucial to understand Artificial Intelligence (AI) before exploring how the business sector is being impacted by these technologies [5, 13, 16–20]. Any sort of planning, problem-solving [21–25], and learning activities performed by computer software are referred to as (AI) [26–29], The phrase (AI) generally excludes several

F. Shwedeh · A. Aburayya (✉)
College of Business, MBA Department, City University Ajman, Ajman, United Arab Emirates
e-mail: amaburayya@gmail.com

S. S. Salloum
School of Science, Engineering, and Environment, University of Salford, Salford, UK

B. Fatin · M. A. Elbadawi · Z. Al Ghurabli
College of Business, HRM Department, City University Ajman, Ajman, United Arab Emirates

D. Muhammad · A. Alnuaimi · M. A. Akkass
College of Business, Marketing Department, City University Ajman, Ajman, United Arab Emirates

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applications; while this is technically correct, it neglects key information [30–34], to find out what sort of AI is most prevalent in business, we need to do further research [19]. Let's examine how (AI) affects companies in this area and how it benefits personnel officials [35–38]. AI may help in recruiting by analyzing information from a variety of sources to pinpoint candidates with the greatest attributes. It might assess feedback from workers to determine those most likely to depart the organization based on their degree of work satisfaction [36]. Client participation is vital to any B2C (Business to Customer) company, yet it is anticipated that in the next years, (AI) will transform these customer support activities [39]. By removing the requirement for a human to precisely detect a customer's tone, AI-powered software solutions such as sentiment analysis technology, let companies react to complaints, questions and concerns more quickly. After all, clients who are pleased with a business are more likely to want to recommend it to others both online and off [37, 39–44]. If you work in branding, you are aware of how essential it is to strike a balance between operational efficiency and client experience [45–53]. Using intelligent technology solutions is one of the most effective strategies to optimize both. Don't throw away client feedback or other measurable answers. AI system assess them at capacity, needing neither you nor your staff to perform any human tagging, and centralize the results for easy access [46].

The San Francisco startup OpenAI, which collaborates closely with Microsoft, has developed a number of technologies, including AI, which debuted on November 30th, 2022. It is part of a new generation of artificial intelligence (AI) systems that can converse, compose comprehensible prose on demand, and generate original images and videos by drawing on information stored in a large digital library [54] using natural language processing techniques. Although its primary purpose is to mimic human interaction, AI is versatile enough to also produce music, teleplays, fairy tales, and student essays in addition to computer programs and debug them. Depending on the exam, it can answer questions at a higher level than the average person [55] and even write poetry and come up with music lyrics. The AI curriculum covers topics such as man pages, internet culture, and popular programming languages like Python and bulletin board systems (2023). Supply chain management is the process of coordinating the multiple activities involved in the production and distribution of a product or service to a customer. Supply chain operations can include anything from designing and farming to manufacturing and packing and shipping. The study's overarching goal is to educate readers and researchers about today's exciting technology [56, 57]. Profitability can be increased through better supply chain management by increasing customer satisfaction and decreasing operational expenses. When costs are managed well and minimized whenever possible, profits thrive. When the price of inputs like labor and materials decreases, the cost of production follows suit. Editorial Staff of Indeed, 2023. In sum, there are many ways in which businesses may increase their bottom lines by better managing their supply chains. This is especially true for large, internationally active business [57]. AI has exposed both enterprises and individuals to an entirely new category of products that demonstrate the power and promise of AI to everybody [58]. AI allows for data analysis and insights into logistical operations. By analyzing data on shipment durations, delivery routes, and other logistics-related characteristics, AI may aid logistics organizations in finding areas for improvement and optimizing their operations. Expenses can be reduced, efficiency can be enhanced, and output may be increased [59]. The rapid growth in AI s popularity indicates that we need to learn much more about it and comprehend how this cutting-edge technology may advance Supply chain management in the UAE. This paper will be an insightful foundation for discovering more about AI in Supply chain management and other business sectors in general. This study aims to investigate the main factors of educating managers in adopting and using AI for managerial decisions. Although what has been stated previously, it will still be challenging as there is a lack of research in studding this gap and there is no enough literature nor information about how educating managers in AI applications and technologies will advance over the decisions in next decades. This study will fill the research gap in this field.

2 Literature Review

One of the most important aspects of managerial decisions is forecasting demand and planning accordingly. AI can analyze large amounts of data, including sales history, weather patterns, and consumer behavior, to provide more accurate demand forecasts. Managers involve coordination between multiple parties, including suppliers, manufacturers, distributors, and retailers. AI can facilitate communication between these parties by providing real-time translation services and enabling efficient collaboration. Managers often involve making complex decisions in real-time [60–62]. AI can analyze data and provide recommendations based on the latest trends and insights, allowing managers to make more informed decisions. AI can help identify inefficiencies in the supply chain and suggest ways to optimize it. For example, it can identify areas where automation can be implemented to reduce costs and increase efficiency. Overall, AI has the potential to revolutionize managerial decisions by providing real-time insights, facilitating communication, and improving decision-making capabilities. However, its effectiveness will depend on how well it is integrated into existing management systems and processes and to which extent managers are educated in adopting the relevant AI Applications to their businesses. It is important to educate users of AI adoption in order to link it to their business activities and train on making decisions by depending on it [3, 63–65].

AI is a useful tool that can be used in management decision making in order to automate the whole process which can deliver tasks on its own with little supervision. It is used to produce insights and facilitate communications. It helps in collaborating between different stakeholders as well. The AI chat bot is undoubtedly a transformative step forward for generative AI which is trained to extract data from various sources and make a systematic order [66]. It detects word patterns in order to predict the next word which works like a mind-reading technology in organizational performances. It informs the company when it is time for redesign its working strategy in real-time. AI can predict the market which helps the company take necessary and valuable decisions. Trained with over 570 GB of data gathered from all over the internet and almost 300 billion words, AI has a vast data lake which is helpful to provide the necessary solutions to companies [67].

AI can take any business to the next level. It is a type of incorporating advanced technologies which help businesses in various ways. It can transform the way a company communicates with customers with the help of an AI-powered applications. With the help of many technologies, companies can change how employees interact with their audiences. It also helps in personalizing customer interaction in order to make streamline the communication process. It increases proficiency and becomes more effective in order to navigate the business operation. The company and the employees benefit from this technology and the customers. It helps companies achieve their business goals to become more proficient in the industry. AI can be integrated into the existing system to get the best out of it. AI can write essays, marketing copy, emails, articles and codes [68, 69]. It generates ideas on its own in order to implement them in the business operation to make it more reliable for the audiences. It recommends services and products based on search engines and research keywords. It can translate text into 95 languages [68]. It finds data and analyses it. All these elements help companies in various ways so the employees can be relaxed by relieving work pressure.

The level to which managers are educated in adopting AI technologies to develop managerial decisions is studied in this research paper. This study reflects if there is a positive relationship between the education of managers in adopting managerial decisions and the development of decision making.

3 Forecasts and a Conceptual Model

Fred Davis, with his revolutionary "Technology Acceptance Model (TAM)," has made major contributions to the study of how individuals decide whether to embrace and utilize new technology [49, 67, 70, 71]. The paradigm's two core pillars, perceived usefulness and simplicity of use, are often. Viewed as critical mental characteristics that contribute to the spread of technology. According to Davis (1985), the study's performance will be reviewed, with both being more inextricably linked to the former. The word "perceived value" defines how consumers feel about a service's price and advantages. Maximum utility is often evaluated in terms of perceived worth, with the assumption that benefits exceed costs. Utility theory, which predicts future usage, underpins perceived worth [45, 72]. How much value is thought to be provided influences how pleased and loyal consumers are. It has a significant influence on future technology adoption and use plans [72, 73]. Given this data, the following hypothesis are to be examined:

H1: The perceived usefulness (PU) of educating managers in adopting AI technology influences development of managerial decisions (MD).

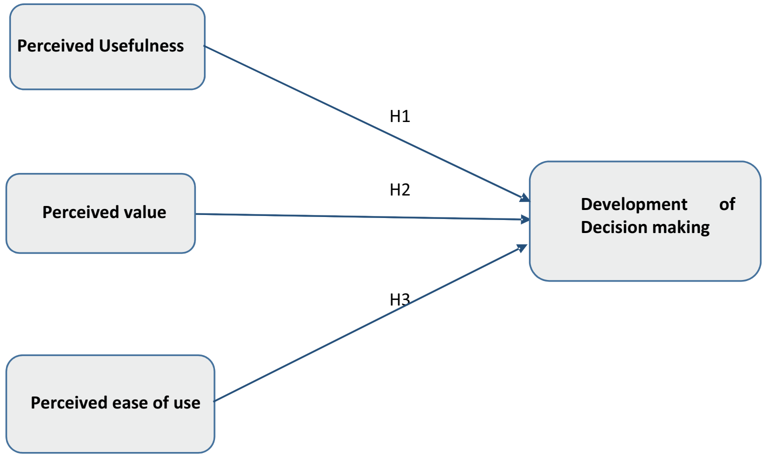

Fig. 1 Model of the research

H2: The Perceived Value (PV) of educating managers in adopting AI technology influences development of managerial decisions.

H3: The perceived ease of use (PEOU) of educating managers in adopting AI technology influences development of managerial decisions (MD).

Figure 1, shows the conceptual outline and how these assumptions were used to create the proposed model.

4 Research Methodology

Within this concise-analytical study, a cross sectional design served as a deduction method. The info collecting tool proved an online-based survey designed to be self-administrated and utilized to gather details about educating managers in adopting AI Applications in organizations in the UAE, notably the nation's capital of Abu Dhabi. This research examines took place in many organizations in Abu Dhabi city at the UAE with the participation of seven main departments and five sections. The collection of data consumed one month, which ran from the 14th Jan. 2023 the 14th Feb. 2023. Using government issued messages and online communication sites consisting of WhatsApp, staff acquired access to the questionnaire's connection in order to gather facts, many different government staff contacted, including administrators (registration, excellence, greeters, and managerial abilities endorses) when the staff (managers, supervisor, directors, executives) employed at the chosen organizations and department's centers. This was done in order to gather important details regarding the development of management decisions throughout those establishments, as proposed by [13, 74].

As part of the research, a convenient selection approach passed, which utilizes a systematic random sampling approach. The primary justification for that decision, in overall, proved the stringent regulations in place in the city's managemnt facilities for safeguarding of employee data in addition to maintaining its safety and confidentiality. The choice of that approach was further affected by the main services facilities that influenced their rules on obtaining the samples. In addition, [75] showed that quick sampling, which enables utilization of large quantities of data, is a sampling approach that makes both economical and efficient.

4.1 Data Collection

In this research, 75 of the 400 questionnaires were eliminated because the answers appeared incomplete. The remaining 325 surveys were complete which are 81% responses used for the analysis of this research. Stated by [76], in the scenario at hand, modeling with multiple regression analysis is permitted because 325 samples size appears to be significantly over the minimum limit. The hypotheses are validated using the framework. It is important to keep in esprit that while the hypotheses are based on current ideas, they ought to be frequently organized in accordance with the guidelines for development of managerial decisions. Multiple regression analysis is used to evaluate the investigated variables.

4.2 Study Instrument

A questionnaire evolved included 20 questions which make up the inquiry of measuring the impact of the three independent variables listed in the model. Each of the queries was amended and modified to maximize the study relevance.

4.3 Personal/Demographic Information

Table 1 shows the statistics of demographic data. Male Suppliers outnumbered female pupils by 50.6% to 49.4%, respectively. 59.7% of respondents were between the ages of 20 and 40 years old, while 40.3% were beyond the age of 40. The majority of responders were qualified and held university degrees. Furthermore, 79.8% of people are postgraduates, and 20.2% undergraduates. Additionally, 40.4% of people are single and 39.3% are married but a small margin of people 20.2% are widows (Tables 2, 3, 4).

Table 1 Presents demographic data collection of the participants of this study

Demographic statistics					
		Frequency	Percent	Valid percent	Cumulative percent
Gender					
Valid	Male	90	50.6	50.6	50.6
	Female	88	49.4	49.4	100.0
	Total	178	100.0	100.0	
Age					
Valid	20–30	54	30.5	30.3	30.3
	31–40	52	29.2	29.2	59.6
	41–50	54	30.1	30.3	89.9
	51–60	18	10.2	10.1	100.0
	Total	178	100.0	100.0	
Marital status					
Valid	Single	72	40.4	40.4	40.4
	Married	70	39.3	39.3	76.8
	Widowed	36	20.2	20.2	100.0
	Total	178	100.0	100.0	
Education					
Valid	Under-graduate	36	20.2	20.2	20.2
	Post-graduate	142	79.8	79.8	100
	Total	178	100.0	100.0	

Table 2 Presents reliability statistics

Reliability statistics	
Cronbach's alpha	N of items
0.840	5
Cronbach's alpha	N of items
0.836	5
Cronbach's alpha	N of items
0.731	5

5 Discussion and Findings

This study demonstrated how Educating managers in AI Adoption has a favorable impact on building a managerial decision, as evidenced by our hypothesis and the survey results.

H1: Is testing the impact perceived usefulness of educating managers in adopting AI on the development of making a decision. Our most useful hypothesis was the Perceived usefulness as shown by the results of the survey which is about how useful

Table 3 Presents the principal component analysis

Extraction method: principal component analysis	
	Extraction
Using AI Applications to learn tasks in my job	0.884
Using AI Applications enhances my knowledge	0.909
Using AI Applications to make some decisions	0.833
Using AI Applications to improve my performance	0.865
Using AI Applications makes my job easier	0.716
How satisfied you are with the overall experience of using AI in developing decisions	0.815
Having AI Applications available is important to assist in making decisions	0.982
Does AI Application meet managerial needs	0.845
Does AI Application accelerates making decisions	0.818
How easy was it to use AI Applications in developing decisions	0.819
My interaction with AI Applications is clear	0.536
AI Applications are flexible for my job	0.895
AI Applications are easy to use	0.900
AI Applications are easy to learn decisions	0.869
Learning to operate AI Applications is easy	0.848

the education of AI Adoption is in their work with an extraction that is from (0.716) to (0.909). This indicates that educating managers with AI Applications eases the development of making a managerial decision.

H2: The Perceived Value talks about how satisfied the managers are with the results they get of educating in the Adoption of AI to build up their decisions as the extraction is (0.815 to 0.982). This reflects the positive relationship between educating managers in AI adoption and the benefits and values on their abilities to develop managerial decisions.

H3: In the third hypothesis, Perceived Ease of Use reflects how easy it is for managers to learn how to Adopt AI in their workplace to take a certain decision. The extraction result is from (0.536) to (0.900) which indicates in impact of the perceived ease of use on managerial decision development.

6 Significance and Limitations

This research is adding the educational concept to managers who make managerial decisions. The TAM model is used in adding the training of managers' adoption of training on AI Applications in a functional methodology with consideration to the impact of perceived usefulness, perceived value and perceived ease of use on developing decisions. This study is significant in the idea of highlighting the influence

Table 4 Presents the analysis results

Model summary						
a. Predictors: (Constant), EOU, PV						
Model		R	R square	Adjusted R square	Std. error of the estimate	
Model						
1		0.906[a]	0.821	0.818	0.42498	
ANOVA[a] a. Dependent Variable: Level of satisfaction with the performance of Metaverese technology in supply chain b. Predictors: (Constant), EOU, PV						
		Sum of Squares	df	Mean Square	F	Sig.
Model						
1	Regression	144.529	3	48.176	266.743	0.000[b]
	Residual	31.426	174	0.181		
	Total	175.955	177			
Coefficient[a] a.Dependent Variable: Level of satisfaction with the performance of Metaverese technology in supply chain						
		Unstandardized B	Coefficient std error	Standardized coefficient beta	t	Sig.
Model						
1	(Constant)	1.019	0.178		5.454	0.000
	PU	0.270	0.072	0.238	3.775	0.000
	US	0.708	0.111	0.572	6.350	0.000
	PE	0.892	0.103	0.544	8.630	0.000

of training managers in adopting AI applications to make decisions which non-studies have discussed earlier in literature. The study is conducted in organizations in the UAE. Results might be different in other countries. This gap can be filled by researchers for future search work enrichments.

7 Conclusion

The adoption of AI in training managers to develop managerial decisions is going to an impact on decision makers in many fields as the global technology, instructive, and financial sectors. The emerging immersive environment, not the web, will be the catalyst for innovative training of the managers who will employ AI adoption in developing their decisions. This can be added as a phase of making decisions strategies. This study investigated manager's views and motives about learning in adopting AI applications with the possible educational advantages of these applications of the world structure in view. The managers' views on the utility and viability of the AI Applications had a significant impact on their propensity for creation of managerial decisions. In order to explore the concepts concerning technological acceptance, it is suggested that consumers' perceptions of the latest technological feature and usability have a big impact on the extent to which it is accepted. These outcomes are consistent with the findings from previous studies. It might additionally reflect how managers perceive regarding modern AI applications technologies. The current inquiry has some restrictions. A theoretical framework might require a lot of work since it merely takes two important variables into account: a person's degree of innovation and the happiness they get when they come up with best ways to solve problems and make a decision. Secondly, since the survey had been advertised via the internet and on the social media, additional managers might be likely to take place. Lastly, benefit is the adaptability of the AI applications.

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Objectives and Obstacles of Artificial Intelligence in Education

Said A. Salloum, Ayham Salloum, and Raghad Alfaisal

1 Introduction

Artificial Intelligence (AI) has evolved as a transformative tool, revolutionizing various aspects of daily life across the globe [1–5]. One significant domain where AI has made a profound impact is education [3, 6, 7]. The integration of AI in education aims to create personalized, efficient, and improved learning experiences for students [5, 8–11]. A prominent example of this transformation is ChatGPT, a chatbot developed by OpenAI, based on the Generative Pre-trained Transformer (GPT) model. This tool is particularly significant due to its ability to assist students in generating term papers, short stories, explanations, and even comprehending complex concepts. ChatGPT uses Natural Language Processing (NLP) models to understand and generate human-like text, making it a valuable resource for educators, students, and academic staff alike [12].

The potential of AI in education is vast, from personalized learning experiences to increased efficiency in administrative tasks and improved learning outcomes [11, 13–15]. However, the implementation of AI in education also presents several challenges, such as ethical concerns, data privacy, and technological limitations [16]. This survey paper aims to provide a comprehensive overview of the goals and challenges of implementing AI in education, with a special focus on the case of ChatGPT. By

S. A. Salloum (✉)
Health Economic and Financing Group, University of Sharjah, Sharjah, UAE
e-mail: ssalloum@sharjah.ac.ae

School of Science, Engineering, and Environment, University of Salford, Salford, UK

A. Salloum
College of Medicine, University of Sharjah, Sharjah, UAE

R. Alfaisal
Faculty of Computing and Meta-Technology, Universiti Pendidikan Sultan Idris, Tanjong Malim, Malaysia

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examining the potential benefits and limitations of ChatGPT, this paper seeks to contribute to the ongoing discussion on the role of AI in education and provide recommendations for overcoming the challenges and maximizing the benefits of implementing AI-powered tools like ChatGPT in the educational journey.

2 Background

2.1 Historical Context: A Brief History of AI in Education

The integration of Artificial Intelligence (AI) into the realm of education isn't a recent phenomenon [17–21]. In the late 1960 and 1970s, the advent of computer-assisted instruction paved the way for early explorations into personalized learning, harnessing the power of machines [22]. Intelligent tutoring systems (ITS) began to gain traction in the 1980s, offering adaptive learning experiences tailored to individual student needs [23]. These systems employed cognitive models to discern a student's knowledge base and provide feedback accordingly. With the surge in machine learning and big data in the 2000s, educational platforms started harnessing data analytics to offer predictive insights on student performance, enabling educators to intervene proactively [24].

2.2 Current Trends: How AI is Being Integrated into Educational Systems Today

Modern educational systems are experiencing an AI renaissance, with several transformative applications [25–32]. Adaptive learning platforms, like DreamBox and Knewton, employ algorithms to customize lessons and resources to fit each student's pace and learning style [33]. Chatbots and virtual teaching assistants, such as IBM's Watson, assist in answering student queries and providing instant feedback [34]. Moreover, AI-driven content creation tools are now generating customized reading material suited to individual literacy levels. Not to be overlooked, predictive analytics in platforms like BrightBytes predict student outcomes, enabling timely intervention [35]. With the ongoing advancements in AI, it is evident that its symbiosis with education is poised for further growth and innovation.

3 Literature Review

The integration of AI in education has been a subject of research and discussion for several years. Various studies have explored the potential benefits and challenges associated with implementing AI-powered tools in the educational sector. This literature review aims to summarize the key findings from previous studies and identify gaps in the literature that this paper will address.

3.1 *Personalized Learning*

One of the primary goals of implementing AI in education is to create personalized learning experiences for students. Blikstein (2015) [13] highlighted the potential of computationally enhanced toolkits to support children's learning by providing personalized feedback and guidance. Similarly, [12] developed AutoTutor, a tutor with dialogue in natural language, which provides personalized tutoring to students by understanding their needs and adapting the instruction accordingly. This kind of personalized instruction has been shown to improve learning outcomes and increase student engagement [8] (See Fig. 1).

3.2 *Efficiency and Accessibility*

AI-powered tools can also increase efficiency and accessibility in education. Burstein and Marcu (2003) [16] developed a machine learning approach for identifying thesis and conclusion statements in student essays, which can help educators to assess students' work more efficiently. Additionally, AI-powered tools can make educational resources more accessible to students with disabilities by providing alternative ways to access and interact with the content [8].

4 Challenges Facing AI in Education

4.1 *Ethical Concerns*

Despite the potential benefits, the implementation of AI in education also presents several challenges and ethical concerns. Burstein and Marcu (2003) [4] raised concerns about the accuracy and reliability of machine learning models in assessing stu-dent essays. Similarly, Blikstein (2015) highlighted the challenges associated with designing computationally enhanced toolkits that are effective and engaging

Fig. 1 AI in education

for children. Additionally, there are ethical concerns related to data privacy and the potential misuse of AI-powered tools [1].

- **Data Privacy Issues**: With AI-driven educational tools often requiring vast amounts of data from students, concerns arise regarding how this data is used, stored, and protected. Data breaches could jeopardize the privacy of students and misuse of data can have long-term implications on students' lives.
- **Bias in AI Algorithms**: Algorithms trained on biased data can perpetuate or amplify those biases, leading to unfair outcomes. For instance, if an AI tool is trained primarily on data from a particular demographic, it might not perform as effectively for students outside of that demographic [36].

4.2 Technological Barriers

- **Infrastructure Needs**: For AI to function effectively, schools need robust technological infrastructure, including high-speed internet and modern devices. Many schools, especially in less developed regions, struggle with this [37].
- **Costs**: AI tools and the necessary supporting technology can be expensive, placing them out of reach for some educational institutions.
- **Potential Technological Glitches**: Like any technology, AI tools can malfunction or not work as intended, disrupting the learning process [38].

4.3 Pedagogical Concerns

There's a growing concern within the academic community regarding the integration of AI into education. While technology can offer a myriad of benefits, including customization and efficiency, the risk that AI could overshadow or even replace traditional teaching methods is palpable. The heart of education lies not just in the transmission of information but in the human connection—the bond between teacher and student that fosters an environment conducive to learning, mentorship, and personal growth [39]. AI, in its essence, lacks the emotional intelligence, empathy, and interpersonal skills that human teachers bring to the table [40]. While an AI system can provide students with answers, it cannot interpret their emotions, discern when they are struggling not just academically but personally, or provide the emotional support and encouragement that can sometimes make all the difference in a student's academic journey [41]. Moreover, teaching is not just about content delivery. It involves fostering critical thinking, encouraging discussions, and understanding the nuances of each student's perspective—areas where human touch, judgment, and intuition play a crucial role [42]. Thus, as schools and institutions consider integrating AI tools into their systems, it's imperative that they do so with an eye toward augmentation rather than replacement. The best outcomes may emerge from a balanced hybrid approach, where AI complements human instruction without overshadowing it, ensuring that the core values and essence of teaching remain intact [33].

4.4 Equity Issues

The integration of Artificial Intelligence (AI) in education has ushered in an era of promise and potential. Yet, as we advance toward this future, we are confronted with the age-old problem of equity. One of the most pressing equity issues is the digital divide, which refers to the gap between individuals who have access to modern technology and those who do not [43]. In many regions, especially rural or economically disadvantaged areas, schools often lack the necessary infrastructure, such as high-speed internet or up-to-date computer systems, to support AI-driven educational tools. Furthermore, students from low-income families might not have personal devices or reliable internet access at home, limiting their opportunities for remote learning or supplementary AI-enhanced education [44]. This disparity goes beyond just infrastructure. It includes access to high-quality digital educational content, opportunities for digital skill development, and the ability to participate in emerging online learning communities [45]. Without equal access, there's a risk that AI will only exacerbate existing educational inequalities, giving those with access an even greater advantage while leaving others further behind. Moreover, when AI-driven tools are designed, they often cater to majority or privileged groups, inadvertently sidelining the specific needs and cultural contexts of minority or marginalized student groups [46]. This can lead to content that may not be culturally relevant or inclusive. To truly harness the potential of AI in education, there's a need for policymakers and educators to actively address these disparities, ensuring that all students, regardless of socio-economic or geographical circumstances, have an equal opportunity to benefit from the advancements AI offers.

4.5 Adaptability and Training

As AI continues to permeate the educational landscape, its success largely depends on its end users: teachers and administrators. While the technology itself holds transformative potential, its effectiveness hinges on the aptitude and comfort level of educators who are tasked with integrating it into their classrooms [47].

Firstly, there's the challenge of technological literacy. It is essential for educators not only to know how to operate AI-driven tools but also to understand their underlying principles and potential biases [48]. This foundation enables educators to make informed decisions about when and how to use these tools to best serve their students' needs.

Moreover, teachers need to develop pedagogical strategies that incorporate AI. This involves understanding how AI can augment traditional teaching methods and determining which tasks are best suited for AI versus human instruction [49]. For instance, while AI might excel at providing instant feedback on a student's grammar in a language learning app, it might not be as effective in teaching the nuances of creative writing or fostering class discussions on complex topics.

Training should also address potential ethical dilemmas associated with AI. With data privacy being a significant concern, educators should be equipped with the knowledge to protect their students' information and to use AI tools responsibly [50]. Furthermore, ongoing professional development is crucial. As AI technology evolves, so should the training, ensuring that educators remain up-to-date with the latest advancements and best practices [51]. Ultimately, a holistic approach to training, which considers both the technological and pedagogical implications of AI, is imperative. Without this comprehensive approach, there's a risk that AI's integration in education will be superficial, failing to unlock its full potential for enhancing learning outcomes.

5 Recommendations

5.1 *Addressing Ethical Concerns and Ensuring Accessibility*

It is of paramount importance to address the ethical concerns related to the use of AI in education, which include data privacy, potential biases in AI algorithms, and the impact on students' creativity. Educational institutions and policymakers must establish clear guidelines and regulations to address these concerns. Additionally, it is crucial to ensure that AI-powered tools like ChatGPT are accessible to all students, including those with disabilities. This may involve developing alternative ways to access and interact with the content and providing additional support for students who need it.

5.2 *Training, Support, and Continuous Improvement*

Educators and students must receive adequate training and support to maximize the benefits of AI-powered tools like ChatGPT. This includes training on how to use the tool effectively, how to interpret the feedback and suggestions provided by the AI, and how to integrate the tool into the teaching and learning process. Moreover, it is essential to regularly collect feedback from educators and students on the effectiveness and usability of the AI-powered tools and use this feedback for continuous improvements. Future research should also expand the sample size and geographical locations to enhance the understanding of ChatGPT usage and incorporate other variables and different moderators in the conceptual framework to measure other crucial factors that may influence the acceptance and usage of ChatGPT.

6 Conclusion

Artificial intelligence (AI) has surfaced as a game-changing element in education, revealing both unparalleled prospects and inherent challenges. This research undertook the responsibility of thoroughly dissecting the goals and obstacles tied to the infusion of AI within educational contexts. Key aspirations for AI in the realm of education span from bespoke learning experiences, heightened operational efficiency, to amplified accessibility, all poised to metamorphose standard classrooms into arenas bursting with dynamism, interaction, and inclusivity. However, this metamorphosis isn't without its thorns; ethical quandaries, data confidentiality qualms, and an imperative demand for holistic training for both educators and learners emerge as daunting challenges. Through a rigorous analysis of extant literature, this study elucidates these aspirations and challenges, while suggesting actionable solutions to mitigate said challenges, thereby unlocking AI's fullest educational potential. However, it is pivotal to acknowledge the limitations of this research, mainly its reliance on existing literature, potentially overlooking the fast-evolving landscape of AI. Implications of this study underscore the necessity for stakeholders, from policymakers to educators, to remain informed and agile, adapting to the evolving AI milieu. Future directives should potentially involve real-time case studies, analyzing the on-ground implications of AI applications and charting out adaptive strategies. The study's ultimate ambition rests in sculpting a clearer comprehension of AI's possible reverberations within education and charting a pragmatic course for AI's fruitful assimilation in pedagogical realms.

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Artificial Intelligence in Pharmacy: Revolutionizing Medical Education Delivery

Saada Khadragy and Mohamed Elshaeer

1 Introduction

As the healthcare sector traverses the corridors of rapid evolution, the pervasive influence of Artificial Intelligence (AI) becomes increasingly pronounced [1–6]. This introduction serves as an engaging prologue, navigating through the historical currents of AI in healthcare [5, 7–9] and intricately weaving its escalating relevance in the pharmacy sector [10–14]. It meticulously sets the stage for an exhaustive exploration into the specific applications and profound implications of AI [15–19], emphasizing its potential not only to redefine but to revolutionize the very essence of the pharmacist's role in the contemporary healthcare landscape [3, 20, 21].

2 AI in Drug Discovery

Among the most transformative contributions of AI in pharmacy is its seismic impact on drug discovery, historically characterized by protracted timelines and resource-intensive methodologies [22–24]. As it is stated in Table 1, AI algorithms, predictive modeling [25–29], and data analytics take center stage, fundamentally altering the landscape of drug design and molecular optimization [5, 7]. This intricate exploration delves into the nuanced ways AI expedites drug development processes, optimizing molecular structures, and fundamentally reshaping the pharmaceutical landscape [30,

S. Khadragy (✉)
College of Arts and Science, Umm Al Quwain University, Umm Al Quawain, UAE
e-mail: drsaada.khadragy@uaqu.ac.ae

M. Elshaeer
College of Pharmacy/ PharmD program, Gulf Medical University, Ajman, UAE
e-mail: 2021ph14@mygmu.ac.ae

A. Al-Marzouqi et al. (eds.), *Artificial Intelligence in Education: The Power and Dangers of ChatGPT in the Classroom*, Studies in Big Data 144,
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31]. Elaborate case studies offer compelling narratives of AI's disruptive prowess [32–36], illustrating its capacity to not only enhance but revolutionize traditional drug discovery methodologies [9, 37–39].

3 Personalized Medicine

The integration of AI into pharmacy practices unfolds innovative dimensions, particularly in the domain of personalized medicine [3, 40–43]. Genomic medicine and predictive analytics emerge as stalwart contributors to healthcare, offering bespoke treatment plans based on individual patient profiles [44]. This section, while celebrating the innovative applications of AI, plunges into the ethical considerations and nuanced challenges surrounding the customization of treatment strategies [45–48]. It accentuates AI's potential not only to revolutionize but also to humanize patient-centric care, a critical aspect in the evolving landscape of personalized medicine.

4 AI in Patient Care

The transformative role of AI extends seamlessly into the precincts of pharmacy settings, particularly in enhancing patient care [49–53]. Applications such as medication adherence, real-time monitoring, and telepharmacy leverage the full spectrum of AI-driven technologies to not only enhance patient experiences but fundamentally contribute to better health outcomes [54]. This section delves into real-world examples and practical insights, providing a tangible understanding of AI's impact on the pharmacist-patient relationship. It highlights the intricate balance between technology and human touch, a dynamic interplay that shapes the future of patient care in pharmacy settings [30, 55–59].

5 Pharmacy Management

AI's influence extends its tendrils into the meticulous facets of pharmacy management, optimizing inventory control [30, 51, 60–65], supply chain logistics, and fraud detection [10, 35, 55, 66–68]. This section provides a granular exploration of how AI-driven workflows not only enhance efficiency but also effectuate a reduction in operational costs [69]. The integration of AI into pharmacy workflows is showcased through a multitude of detailed case studies [70], offering practical insights into the transformative power of AI in reshaping the very foundations of pharmacy management practices.

Table 1 Provides a snapshot of various aspects of AI in Drug Discovery, summarizing key research findings in each category

Aspect	Research findings
Target identification	AI algorithms, such as deep learning, have shown efficacy in predicting potential drug targets
	Integration of genomic and proteomic data into AI models enhances accuracy in target identification
	Collaborative research efforts highlight the role of AI in expediting target validation processes
Lead compound identification	Machine learning algorithms contribute to the identification of novel lead compounds for drug development
	Virtual screening techniques powered by AI accelerate the identification of potential drug candidates
	Successful case studies demonstrate the efficiency of AI in lead optimization and compound prioritization
Drug repurposing	AI-based approaches have proven effective in identifying existing drugs for new therapeutic indications
	Large-scale data analysis facilitates the discovery of hidden relationships between drugs and diseases
	Repurposed drugs show promise in reducing development costs and timelines
Predictive analytics	AI models predict drug efficacy with high accuracy, aiding in early-stage decision-making
	Integration of diverse data types, including clinical trial data, improves predictive analytics outcomes
	Real-time monitoring and adaptive learning enhance the adaptability of predictive models
Toxicity prediction	AI algorithms contribute to the accurate prediction of potential drug toxicity, reducing safety concerns
	Early identification of toxic compounds minimizes late-stage failures and associated costs
	Continuous model refinement ensures improved accuracy in toxicity assessments
Data challenges	Limited availability of high-quality, diverse datasets poses challenges in training robust AI models
	Research emphasizes the need for standardized data formats and open data-sharing initiatives
	Advances in data curation techniques are essential for overcoming data-related challenges
Ethical considerations	The ethical use of AI in drug discovery requires transparent practices and adherence to regulatory standards
	Research underscores the importance of addressing biases in training data to ensure fair and equitable outcomes
	Ongoing discussions emphasize the need for ethical guidelines specific to AI applications in healthcare

6 Challenges and Ethical Considerations

However, amidst the promising landscape, challenges and ethical considerations stand as formidable companions in the journey of implementing AI in pharmacy [71]. Privacy concerns, algorithmic biases, and the intricate dance within the evolving regulatory landscape are examined in meticulous detail [72]. This section serves as a sobering reminder of the need for ethical considerations and the responsible deployment of AI in the healthcare sector. Navigating these challenges is imperative for realizing the full potential of AI in reshaping the future of pharmacy practices.

7 Case Studies

Real-world case studies, meticulously dissected, provide concrete and detailed examples of successful AI implementations in pharmacies. These cases, akin to narrative vignettes, not only showcase the diverse applications of AI but also serve as practical compasses in navigating the challenges faced in real-world scenarios [73]. Lessons learned from these cases contribute significantly to a nuanced understanding of the practical implications of AI in the pharmacy setting, offering valuable insights for future implementations.

8 Future Trends and Prospects

As the chapter turns its gaze towards the horizon, it explores emerging trends and future prospects of AI in pharmacy practice [74]. Ongoing research, potential breakthroughs, and the dynamic role of AI in shaping the trajectory of healthcare delivery are dissected in meticulous detail [75, 76]. This section serves not only as a reflective compass pointing towards future innovations but also as an invitation to chart the unexplored frontiers of AI within pharmacy practices. It prompts contemplation on how these emerging trends will shape the future landscape of healthcare, providing a visionary perspective on the continued evolution of AI in pharmacy.

9 Conclusion

In the symphony of exploration, the concluding section orchestrates the synthesis of key findings [77]. It underscores, with emphasis, the transformative influence of AI in pharmacy, symbolizing a future where healthcare delivery is not only more efficient but fundamentally more personalized and patient-centric [78]. The journey of AI in pharmacy, marked by achievements, challenges, and an unquenchable thirst for innovation, is encapsulated in this conclusive segment. It serves as a resounding call to action, urging further research, deeper collaboration, and the responsible deployment of AI to unlock its full potential in reshaping the landscape of pharmacy practices for a future that is not just efficient but also profoundly humane.

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Effect of e-Learning on Servicing Education in Dubai

Nadia Yas, Wided Dafri, Harith Yas, and Fanar Shwedeh

1 Introduction

The e-learning program is one of Dubai's pilot programs, and education authorities are committed to ensuring its success [1–5]. The term "e-learning" refers to the provision of essential education services through an integrated system [6–10] that allows all stakeholders involved in education to access the necessary resources required for their operations [3, 11, 12]. Its primary focus is to centralize all educational functions in one place, making it easier for the education system to serve various parties with different interests simultaneously [13–15]. The e-learning initiative also facilitates individuals in accessing various information from different educational institutions through a one-stop shop [16–24]. This move was initiated in Dubai due to the increase in its trade operations and interactions with numerous parties and states worldwide. It was perceived to create more avenues for expanding education service delivery, thereby enhancing trade activities and opening doors for investment [25–30].

Dubai, one of the cities in the UAE, is renowned for its oil exports worldwide. Given its dealings with numerous nations across the globe in both exports and imports, the country has diversified its education service delivery to ensure that these services are easily accessible to the people [16]. The City's economic report released but the minister for trade in early 2013 indicates that it contributes a

N. Yas
College of Law, Umm Al Quwain University, Umm Al Quawain, UAE

W. Dafri
American University in the Emirates, Dubai, UAE

H. Yas
International Business School, Universiti Technologi Malaysia, 81310 Johor, Malaysia

F. Shwedeh (✉)
City University Ajman, Ajman, UAE
e-mail: s.fanar@cu.ac.ae

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large amount of the GDP of the UAE, which is largely dependent on the oil sector, with up to 71.6 percent of the country's economy coming from this sector [25]. This is a greater achievement in the country's economic diversity that is aimed at ensuring that the country maintains its position on the globe as the world's preferred exporter and importer of very essential oil products. Dubai has implemented high-end infrastructure to ensure that all public services are integrated into a common pool of information and communication infrastructure. This move is meant to facilitate ease flow of service between education agencies to boost the key sector of the education's economy, which is oil export [25]. According to the report released by the world's global information technology, it is indicated that Dubai is the leading investor in application of information technology to increase its global competitiveness and economic diversity [17, 31]. The telecommunication sector in Dubai has increased the economic diversity of the city by 4.1 as indicated in the report by its Telecommunication Sector Indicators.

By 2005 March, The Ministry of Finance launched the e-learning Portal in collaboration with Etisalat that was charged with the exclusive responsibility of developing the portal [25]. This was marked as a milestone towards the implementation of the e-learning initiative in Dubai. It offered education a steppingstone to move forward and get so many things done within the duration it had planned [32]. According to the then Minister of Finance, Dubai had achieved partial success with the launch of the e-learning portal and what was remaining was to get stakeholders to embrace the portal and then get things moving as planned [33]. The e-learning portal linked so many education agencies, education offices and various education sites thus coming up with an integrated system that provided all education services at pone standstill shop [34]. The Ministry of Finance was optimistic that all stakeholders who would be involved in the utilization of the e-learning portal would fund it user-friendly and a solution to so many challenges that they had faced at one point or another during their tome of operation with various education agencies. As such, the launch of the e-learning portal was an indicator that a change in the export trade of Dubai was going to get some revolution and thus realize effective success. More achievements were realized in the implementation process of the e-learning in Dubai by the year 2011. In the months of March and July of the same year. The ministry of Educational Sectoral Development of the United Arabs Emirates was assigned the responsibility of overseeing the implementation of the e-learning program [25]. This came after a ministerial decree was issued by the policy makers of Dubai to transfer all the services that were offered under the e-learning from the country's Ministry of Finance and Industry to the newly created ministry of Educational Sectoral Development. This led to a shift in how the e-learning programs were ruined and increased the efficiency and effectiveness of how various stakeholders access various services from the program. The Federal Education developed its Information System Strategy in June 2012 when the e-learning initiative had reached a point at which it was almost becoming a success for the country [35]. This program was implemented by the country's Telecommunication and Regulatory Authority after a ministerial decree was issued mandating the education agency to take over the efforts of developing an information system for the country. To achieve success of the robust information system m that would accommodate the ambitious e-learning plan of the country, Dubai education mandated the TRA to partner with Booz and Allen Company in developing the country's information system strategy [16]. A lot of challenges were faced during the development proposes by still the two s partner's managed to achieve success in their service delivery. It is also important to note that the success in developing the country's eGovernemnet structure was facilitated by the fact that all the necessary resources needed by the contracted company and the education agency were made available. At the end, the information system that provided a platform for anchoring Dubai eGovernemnet program was in place [25]. This again marked a major milestone in the country's journey towards achieving the implementation of the e-learning initiative.

The main aim of Dubai e-learning portal is to provide an opportunity for all stakeholders to access more and better public services online. This is referred to as diversification of public services so that every member of the public in Dubai is actively involved in initiatives of public interest, education policies, laws, and leadership. The ultimate result of this move is ensuring accountability and responsibility to the members of the public through provision of public services in a transparent manner. As such, the education of Dubai is able to demonstrate that it can be trusted in providing public services [25]. Today, the implementation of e-learning in Dubai has led to several effects that have shaped the manner of education delivery in Dubai. The effects on the servicing of the students of Dubai are both positive and negative and thus each sector of the economy has experienced a fair share of an event [5, 36]. Nonetheless, the advantages of e-learning in Dubai are far more than the disadvantages of e-learning. This therefore means that the paper will seek to ascertain whether policy makers of Dubai were right to implement the e-learning strategy as planned in its vision 2021 delivery blueprint. The Ministry of Finance that initiated the initial plans to have the e-learning initiative operational may be accorded the accolade for such success but still more needs to be done [33]. This means that as this paper terry to investigate the effects of e-learning in Dubai, more focus will be given to the financial effect to the country's economy and how it has contributed towards effective export traded in the country. It will also be important to give way to the possibility of the e-learning not having any effects in some parts of the country given the fact that different citizens in Dubai may have different perceptions of how the e-learning has had effects on their lives. The main aim of this paper was to determine the perceptions of Dubai citizens on the effects of e-learning in education servicing. Therefore, the participants were to state whether they support e-learning or not. In addition to the main question, the participants gave their opinion on whether they think e-learning was beneficial or not [25]. They should also state their opinion on whether those who oppose the implementation of e-learning are justified in doing so. This paper was a survey of 2000 citizens in Dubai who gave their take on the above matters. It would not be a historical analysis of what happened to lead to current preferences for e-learning in Dubai. Furthermore, owing to the logistical issues involved, various social economic and political factors were not sampled as finding persons in polygynous households was very difficult in this society. No expert opinion was sought as college citizens were simply to give their take on the matter. The main objective of the research was to ascertain the effect of e-learning on servicing education in Dubai.

Research Question 1. Did the implementation of e-learning by Dubai achieve its mission of facilitating the realization of Vision 2021?

Research Question #2. Did the implementation of e-learning provide for solutions to the economic growth of the country through export trade?

Research Question #3. Is the effect of the e-learning implementation plans and strategies in Dubai used congruently or differently from what the education has been using for the past period?

This study proposes two major hypotheses which are based on the overall objective of it. The first concern is the guessing efficiency of e-learning in servicing education in Dubai. The second hypothesis is on the effectiveness of the criteria and techniques, strategies, plans, and programs used to implement e-learning in Dubai.

The two hypotheses are:

1. The sampled e-learning services will be able to provide a correct analysis of the effects of e-learning on efficiency and effectiveness.
2. The strategies, programs, plans, criteria, and techniques of implementing the e-learning servicing will not have a significant effect on Dubai citizens.

1.1 Significance of the Study

The focus on the effect of e-learning in Dubai will help solve the challenge and problems that the education of Dubai has faced in the process of implementing the eGovernemnet in education servicing. The benefits accruing to the implementation of the e-learning initiative are critical in ensuring that the export trade is enhanced thus boosting the economy of Dubai. The results of this study will provide changes in how to effectively implement the e-learning initiative in Dubai and how to address the challenges it faces. The results of the study will impact the perception of all stakeholders and citizens in Dubai as well as boosting export trade thus leading to an increase in the country's GDP. The results of the study will also provide recommendations on how best to improve the integrity, efficiency and reliability of the eGovernemnet to citizens.

2 Literature Review

2.1 Involvement of Citizens in E-learning Initiative

The United State Department has initiated a mechanism that will ensure that its citizens are well conversant with the requirement of ICT to help them fully take part in implementing the e-learning delivery initiative. Since e-learning involves delivery of information and services electronically, it becomes essential for every citizen to be equipped with the ICT skills that will be necessary for implementing the e-learning programs (Computational Science, 2012). Several services ranging from e-voting, healthcare, registration of person and education contracts are available online and any citizen can access these services. This was facilitated by the internet boom that was realized in the United States in the early 2000. Today, the federal education of the United States can attest that the gains of e-learning are immense ranging from accountability to transparency.

Dispersal of Information and Engagement of the Public In E-learning.

According to Tarnay, Imre, & Xu (2013), the success of e-learning in the United States depends on the way the members of the public are engaged in the process and the overall way of dispersing education information to the citizens. Although information dispersal to all citizen in the United States may seem to be complex, the federal education has initiated mechanism through which it intends to achieve efficient served delivery through its e-learning initiative [37]. The federal education will rely on the wide range advanced in ICT and internet boom to coordinate all the activities that will facilitate public engagement at its e-learning implementation programs [25]. The urge to remain committed to providing information and education services online is still crucial for all members of the public and thus the United States must ensure that it engages all its citizens in the e-learning programs. The education under the current administration led by President Barack Obama is fostering a democratically engaged nation. It does this by using its e-learning initiative to provide a marketplace for various service providers thus facilitating service delivery and information provision to its citizens.

2.2 *The Education Public Services Delivery System Models*

Several online the education public services delivery system models exist. Most of these e-learning initiatives are not appropriate for public services delivery due to the sophisticated model of online operation that they are characterized with. Instead, e-learning should be an operation that relies on the viability of the software and e-learning institutional framework. These models must focus on ensuring Dubai public services delivery system provides its service to the citizens at very minimal cost. The e-learning initiatives are the various types of principles and policies that any the education agency must adhere to when delivering services to the citizens of Dubai. It is an institutional framework that provides strict rules of providing access to information and education services on the internet and thereby reducing the influx of flexibility in conducting manual service delivery [38].

All the education agencies and departments in Dubai are guided by certain legislations. They must operate within a given legal framework and online the education public services delivery system is no exception. The law that guides e-learning initiative only emphasizes the role of the ethical concerns that are already in force to control the education public services delivery system operations through an automated system with the use of the education portal as the aces point [39]. It is therefore an obligation for Dubai to follow certain legal requirements that will act as guideline to its operations in the public services delivery through the eGovernemnet initiative. Dubai education is obliged to follow certain aspects of the law that all other education with e-learning programs is following. These legal constraints are meant to ensure that everything is set into place and that nothing is run on illegal platforms. These legal aspects also act as points of reference in case of dispute between all the stakeholders who access education services from the e-learning program and via the education portal.

The implication of these legal requirements is that it creates a healthy and fair environment for all citizens to equally access the public services without discrimination based on the social, religion or ethnicity of the citizens of Dubai. The education of Dubai also has their rights and services protected against intrusion by any outside forces. The result is efficiency and effectiveness in service accrued from the education agencies through these operational procedures. Legislations governing eGovernemnet initiatives should not be reduced to fit into the definition of individual citizens. Unless a law expressly provides that a certain conduct is legal, then no online education agency should indulge in such activity since it will undermine the ethical concerns of that particular the education public services delivery system [40].

It is also important to mention that the ethical guidelines that control the online the public services delivery and operations are the ones that have greatly contributed to the success of the eGovernemnet initiative in Dubai. The same laws that protect individuals from any criminal liability until substantial evidence is provided against them implicating them to such criminal offences also protect the education entities. Legal aspects of e-learning programs should not be seen as a way of gaging the process of accessing public information and services but should be seen as a way of protecting the rights of the citizen and obliging to the requirements of the education to provide public services. Dubai for instance will need to have its services registered under a common pool such as the education portal to prevent a replication of its unique services [41].

The ethics of the e-learning initiative are referred to as corporate ethics and play an oversight role in the implementation of principles and morals that are acceptable in any education department or agency and may include an individual's conduct [42]. These ethics can be descriptive in the case of a student that wants to understand the environment or normative in case of a professional specialist in the fields. Nature ethics will in turn influence how much profit is maximized without any concern that is non economical. Ethics in the public services sector controls and regulates the access to information and services from all governing offices in Dubai. Formal regimes of the e-learning initiatives and ethics in the automated service delivery have been on the rise due to the increase in number of public services available on Dubai education portal. It indicates the philosophy within which the e-learning initiative operates and thus defines the fiduciary responsibility of the various education entity. Within these ethics is also the cooperate responsibility of the education of Dubai to ensure that it maintains the highest ethical standards and records in its daily dealing with its citizens. Ethical concerns can also be seen as the rights and obligations of a country such as Dubai to its citizens.

3 Research Methodology

First, the study of e-learning servicing involves observation of the demographic characteristics of e-learning servicing that include cultural practices, beliefs, lifestyles, and the background of e-learning servicing. Also, the study of e-learning servicing will involve determination of the economic activity that e-learning servicing engages in during their daily Activities. This means that while studying e-learning servicing, the main parameters that are looked at are the scientific facts that give e-learning servicing their characterization. The methodology used in studying e-learning servicing in most cases is the observation and questionnaire method which is also a scientific method. Once the data has been obtained, it is either analyzed using correlation or t-test to determine the traits being studied. The final step of studying e-learning servicing will involve interpreting the data and representing those using scientific tables, diagrams, and graphs [43–48]. With all these in application, it is true that the study of e-learning servicing is a scientific engagement rather than an art. Thus, it is justified to consider the study of e-learning servicing a scientific engagement.

4 Data Collection

Research design. This quasi-experimental study was used to test the ability of Dubai citizens on how they have benefited from the eGovernemnet initiative. Participants will include 2000 citizens aged between 18 and 35, selected from all the seven Emirs of Dubai. Because the citizens are not randomly assigned to different sections of the class, this quasi-experimental design is appropriate [5, 49, 50]. The pre-test, two-group comparative efficacy study design is displayed in the following notation (Table 1):

Group 1 = study participants. Will be a naturally formed group of citizens (n30) who complete the treatment.

Group 2 = study participants. Will be a naturally formed group of citizens (n = 30) who do not complete the treatment.

Table 1 Presents the Groups of the study

Group 1	X_1	O_1	Y_1	O_2
Group 2	X1	O1	-	O2

X1 = study constant. All citizens will have participated in and completed the user guidelines for the e-learning portal in Dubai.

Y1 = study independent variable condition #1. All citizens will have participated in and completed the required introductory elements of the e-learning logging and user manual.

O1 = study pretest dependent measures. Inference test consisting of how best the education portal serves the citizens of Dubai.

O2 = study posttest dependent measures. Inference test consisting of how many citizens expressed displeasure with the e-learning portal.

The following pretest–posttest research questions will be used to analyze the significance of the intervention strategies for citizens who participated in the intervention instruction compared to citizens who did not participate in the intervention instruction.

Research Question 1: Is there a significant difference between the pretest and the posttest scores on the inference task for the citizens who used the e-learning portal. Research.

Question 2: Is the frequency of strategies used for guessing how effective the e-learning initiative is.

Research Question 3: Is the frequency of strategies used congruently or differently for citizens who received the direct instruction intervention from the frequency of strategies used for citizens who do not receive the direct instruction intervention when the strategy resulted in a correct response?

This study applied distributed surveys to 2000 citizens of Dubai with another group of 2000 as the control group serving in e-learning serving. There was a section of 20 individuals for every region from 100 regions that were randomly selected. This way, no one was left behind and no foreseeable biases were reported.

5 Findings and Discussion

A questionnaire used to collect information from citizens who had formally used the e-learning program (Tables 2, 3 and 4 and Fig. 1).

The research also indicated that more females were satisfied with the assistance they get from their parents to use computers for their activities as compared to the males. However, the number of male and female who were not satisfied were more than those who were satisfied in all categories. More female parents were skilled in computer use than the male while the number of skilled parents in both categories high as compared to the non-skilled. More male residents performed better that the female residents although those who scored CGPA of 1–3.0 were more than those who scored above 3.5 in all categories. In all categories, the number of people satisfied with the use of automated system to submit assignment was higher with the male residents recording a higher number than the female residents in terms of satisfaction with the automated system (Table 5).

Table 2 Presents the Demographic Data of the Residents of Dubai

Age		Male	Female
	18–35 Years	469	531
	36–55 Years	308	612
Location		Male	Female
	City center	495	505
	Metropolitan area	345	655
Personla computers		Male	Female
	Accesible	488	512
	Not accesible	652	348
Family income		Male	Female
	$2000–5000	346	654
	$5001- $10,000	445	555

Table 3 Experience using computer

Duration		Male	Female
	2–5 YEARS	308	531
	6–10 Years	469	612
Time spent at home		Male	Female
	1–2 h	488	505
	More than 4h	395	655
Use computers at educational institutions		Male	Female
	Accesible	652	512
	Not accesible	345	348
Nessesity of computers		Male	Female
	Satisfactory	555	448
	Not satisfactory	445	354

In terms of usability of the e-learning service, the research established that more female residents identified the benefits of the e-learning as opposed to the number of males who had the feeling that the e-learning had not benefited the residents at all. However, more male citizens had benefitted from the e-learning that the number of female residents who accessed the Dubai portal. The e-learning service delivery system seemed to provide convenience to the male residents as opposed to their female counter (Table 6).

When the improvement in accessibility to the e-learning service, the research established that more female residents strongly agreed that an integration in the Dubai portal was necessary to bring all public services and information to one central place. The number of males who suggest an integration of the Dubai portal with all public

Table 4 Parents assistance at home in using computer

Parents assistance		Male	Female
	Satisfactory	411	521
	Not satisfactory	451	556
Parents skills in computer		Male	Female
	Good	402	502
	Excelent	395	554
CGPA in the last semester		Male	Female
	1–3.0	589	412
	Above 3.5	345	348
Submission of assignment		Male	Female
	Satisfactory	504	458
	Not satisfactory	506	384

services were few. It was also noted that more female residents in Dubai were accessible to ICT infrastructure as compared to the number of males who we accessible to the same infrastructure. However, more male residents expressed satisfaction with the way the education was putting effort to provide public information and services to the citizens. The female residents were also accessible to ICT skills as compared to the male residents (Table 7).

When the future of e-learning in Dubai was analyzed, it became evident that more female residents in Dubai expressed confidence in e-learning services compared to the number of male residents who strongly disagreed with the system. Furthermore, a higher number of female residents accessed e-learning services from the Dubai portal in comparison to males. Notably, a significant number of residents, both male and female, accessed between 5001–1000 services from both categories. The research indicated that the number of female residents in Dubai who were satisfied with the Dubai portal was more than the number of males who were satisfied with the Portal. More females were also satisfied with the literacy level that the Dubai residents had gained with high number of male residents having a positive attitude towards the public awareness campaign that the Dubai education had initiated in the people as compared to the female residents.

6 Discussion

Dubai has implemented high end infrastructure to ensure that all public services are integrated into a common pool of information and communication infrastructure. This move is meant to facilitate an easy flow of service between education agencies to boost the key sector of the education economy, which is oil export. According to the report released by the world's global information technology, it is indicated that

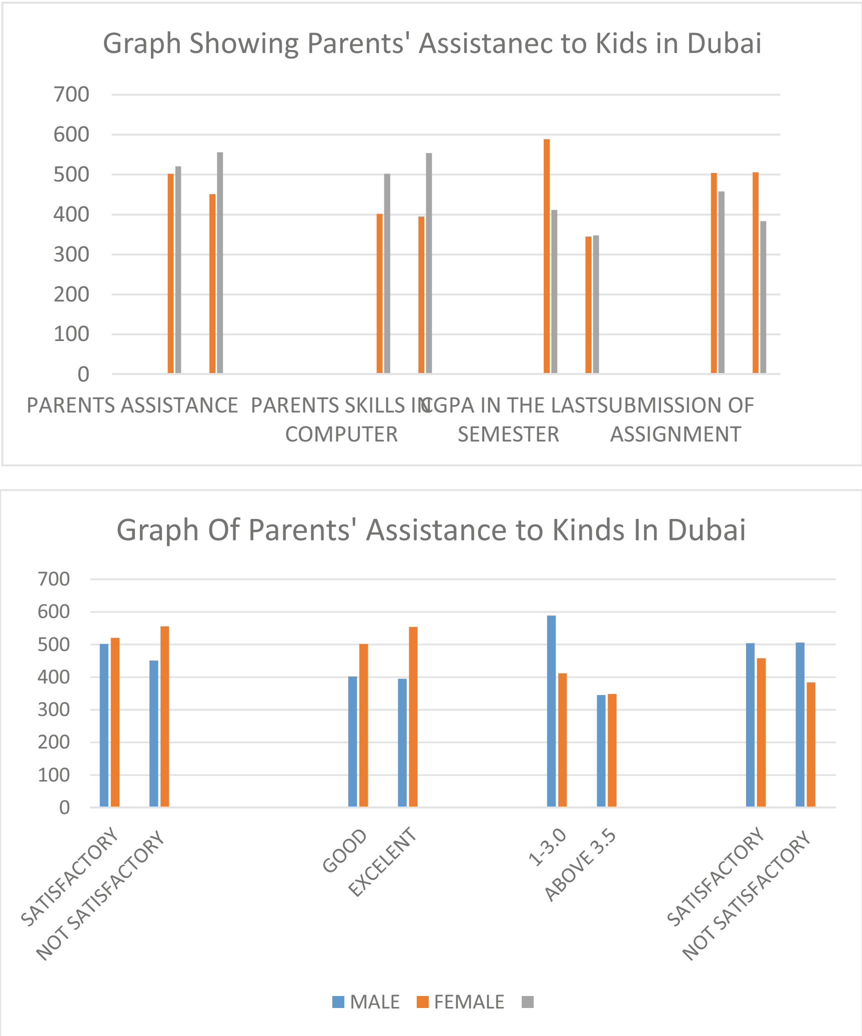

Fig. 1 Graphs of Parents assisting kids

Dubai is the leading investor in application of information technology to increase its global competitiveness and economic diversity. The telecommunication sector in Dubai has increased the economic diversity of the city by 4.1 as indicated in the report by its Telecommunication Sector Indicators.

Table 5 Showing the effects of E-learning in Dubai

Male	Female		
	Satisfactory	402	502
Benefit of e-learning		Male	Female
	Good	502	521
	Excelent	502	556
Effects of e-learning to citizens		Male	Female
	1–3.0	589	412
	Above 3.5	504	458
Conveniency of e-learning		Male	Female
	Satisfactory	506	384
	Not satisfactory	345	348

Table 6 Showing the improvement in accessibility of E-learning in Dubai

		Male	Female
Need for intergration	Strongly agree	421	496
	Strongly disagree	304	498
		Male	Female
Access to ICT infrastructure	100–500	419	469
	501–1,000	403	513
Education's efforts		Male	Female
	Satisfactory	478	458
	Not satisfactory	511	487
Teaching ICT skills in schools		Male	Female
	Frequently	513	511
	Rarely	312	514

Table 7 Showing the future of E-learning in Dubai

Spread to all parts		Male	Female
	Strongly agree	419	511
	Strongly disagree	403	514
		Male	Female
Number of services	100–500	458	469
	501–1,000	487	513
Overal performance		Male	Female
	Satisfactory	496	478
	Not satisfactory	498	511
Any difficulty		Male	Female
	Frequently	513	421
	Rarely	312	304

7 Limitations and Delimitations

This study will be delimited to only those citizens who are always frequent users of the e-learning initiative. It will use the population size of 2000 citizens drawn from all over the country and aged between 18 and 35 years old. The study also included those citizens with good command of e-learning living within the urban areas since lack of exposure to the initiative may hinder the credibility of the information given. Prior experience and personal experience with the e-learning initiative will also be taken into consideration.

8 Conclusion

The intense competition that Dubai is facing from other neighboring countries in the Middle East has accelerated the growth of technology development, which also characterizes the implementation of e-learning in the city. Dubai, being a business hub and a city, whose economy relies on oil, must explore alternative means to enhance the marketability of its oil globally. For this reason, Dubai has swiftly moved to provide service-oriented and citizen-centric public services to its residents. The results mark a paradigm shift in how Dubai's residents engage with education and access public information from service institutions. Dubai has significantly developed electronic operating systems within its public sector, creating a foundation for the concept of e-learning in the city. The e-service aspect of Dubai's e-learning initiative aims to deliver services by accelerating eTransformation within educational organizations and other sectors. This initiative has enabled Dubai to introduce high-quality online services accessible through its innovative delivery channels. The capacities of the public agencies in terms of their readiness to embrace ICT is strengthened by the eReadiness platform of the Dubai e-learning. Also, eReadiness looks at things like the competencies and capabilities of the human resource and the overall organizational structure of the world. The eTransformation part of the Dubai e-learning is meant to oversee the ability of the organization to take full transformation of the Dubai services and make theme accessible through online channels.

The legislation and policies needed to effectively implement the e-learning initiative is also a part of the transformation part of the Dubai e-learning. Dubai has followed the legal and institutional framework required to implement the use and acquisition of ICT infrastructure. Their main aim is to ensure that they do not go against intellectual property rights and copyright issues. It is through the law and regulations that Dubai can gain the required support to institute a strong, secure, and reliable data provision for its citizens. Strong governance structures have been developed that facilitate efficient communications among different stakeholders that meet the interests and needs of all citizens in Dubai via an electronic system. The e-learning initiative has also been diversified to create a solid infrastructural system thus facilitating exchange of information sharing and data accessibility to the citizen.

To ensure that the e-learning initiative in Dubai keeps operating for as long as it is needed, the education of Dubai has initiated support service that enables the users of the information system and the Dubai portal to launch any complaint they face while using the portal. Also, Dubai has initiated effective mechanism and performance managements for the information system so that it can easily provide the required ICT infrastructure for the service delivery. Automated tools and reports have also been developed to monitor the progress and performance indicators of the Information System so that they can easily provide a backup for the information required. Dubai has a smart card identity which offers authentication, validation, and verification of the virtual and physical identification of all its citizens. It is a secure smart card that has all the relevant information that may be needed from any person thus ensuring effective and trustworthy transactions with any citizen [38]. Being the most advanced and secure smart card in the world, Dubai uses its smart card to keep the biometric data, personal identification pins and the digital certificates of all its citizens. It also helps Dubai to offer an infrastructure for identification, verification, and trusted validation of all aspects of its citizens. Time stamping and digital signatures are also available in the smart card and this ca easily enable the Dubai authorities to keep an eye on its citizens. Dubai has also invested its resources to ensure that it has the capability to offer any kind of service and information to its citizens without any problem.

9 Recommendations

This study recommends that when implementing the strategies of e-learning, it is important to look at the aspects of transparency and trust among members of the public. High quality information and public service can also be delivered to the citizens if the education can easily with the confined and trust of the citizens. Dubai has struggled to achieve this objective by making its public service delivery through the e-learning initiate transparent by integrating all its public information and service to all portals and the Dubai Information system. The level of trust is dynamic and does not happen at once. Instead, Dubai should struggle to nature the trust and confidence of its residents to fully participate in the e-learning initiative so that it achieves the mission and objective of the e-learning plan. It is also recommended that the residents of Dubai should develop the intention to fully adopt and engage in the e-learning initiative and have shown the willingness to provide and receive information through the e-learning platform. Digital communications tools should also engage the connectivity to the Dubai information system thus providing a chance through which residents can easily access the Dubai portal [51]. Today, the public sector [provides services and information that meets the quality, availability, and quantity expectations of the citizens. The fact that residents of Dubai have played multiple roles in interacting with the Dubai authorities reveals that the education of Dubai must go an extra mile and incorporate all the aspects of e-learning in its online service and information delivery. Education has thus acquired services that are citizen centric which come with resources and services meant to help the citizens get aces to the sufficient information they require.

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Artificial Intelligence: Trust and Verification

Trustworthiness of the AI

Said A. Salloum

1 Introduction

The advent of Artificial Intelligence (AI) has ushered in a new era of technological advancement, profoundly impacting various sectors from healthcare and finance to transportation and entertainment. The increasing reliance on AI across these fields is not just a testament to its technological prowess but also to its potential to revolutionize traditional systems and practices [1]. As AI systems become more deeply integrated into the fabric of daily operations, their influence extends beyond mere efficiency enhancements to being pivotal decision-makers in critical processes. This burgeoning reliance on AI, however, brings to the forefront the paramount importance of trustworthiness and reliability in these systems. The effectiveness of AI in sensitive and impactful applications hinges on its ability to perform not only with high accuracy but also with fairness, transparency, and security [2–4]. In scenarios where AI decisions can have significant consequences, the trust placed in these systems by both users and stakeholders is critical.

Figure 1 is a conceptual visualization of the principles of AI trustworthiness. It features a humanoid robot gracefully holding a justice scale, embodying the concepts of fairness and equilibrium. Surrounding the robot are symbolic elements such as a magnifying glass, a lock, and a shield, each signifying key aspects of transparency, security, and safeguarding, respectively. The backdrop is themed with digital motifs and circuitry patterns, enhancing the technological context of the image. Collectively, this illustration encapsulates the essence of ethical and reliable AI, emphasizing the critical values of fairness, transparency, and security in the realm of artificial intelligence.

S. A. Salloum (✉)
Health Economic and Financing Group, University of Sharjah, Sharjah, UAE
e-mail: ssalloum@sharjah.ac.ae

School of Science, Engineering, and Environment, University of Salford, Salford, UK

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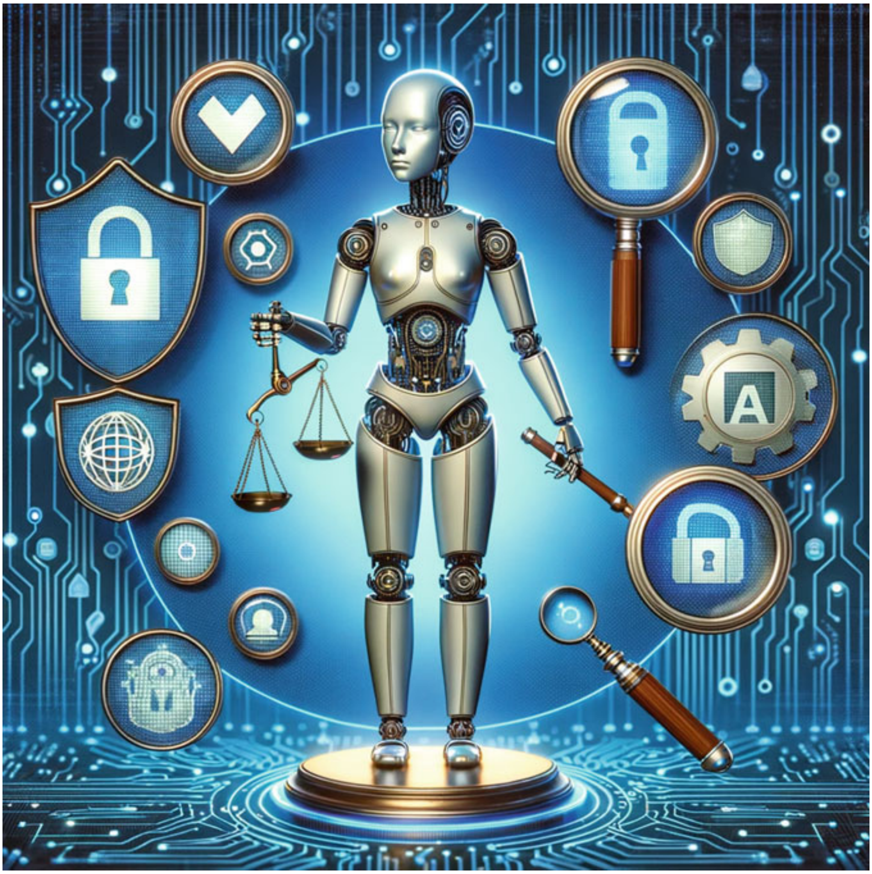

Fig. 1 Depiction of AI Trustworthiness

The objective of this paper is to explore the multifaceted concept of AI trustworthiness, delving into the factors that contribute to the reliability concerns surrounding AI systems. We aim to dissect the complex nature of AI, particularly deep learning models, to understand the challenges in ensuring their trustworthy operation. This exploration covers the breadth of issues from transparency and bias to accountability and ethical considerations, reflecting on how these factors affect the perception and effectiveness of AI systems. Furthermore, the paper seeks to present potential solutions and strategies to enhance the trustworthiness of AI systems, thereby increasing their transparency and fostering greater confidence among users and stakeholders. By identifying and addressing the key factors contributing to trust concerns, this paper aims to provide a roadmap for the development of more reliable and ethically sound AI systems, ultimately contributing to a more trustworthy AI ecosystem.

2 Understanding AI and Its Complexities

Artificial Intelligence (AI) encompasses a broad spectrum of technologies designed to mimic or surpass human cognitive functions. At the core of modern AI advancements is deep learning, a subset of machine learning based on artificial neural networks. Deep learning enables AI systems to process and analyze large amounts of data, learning patterns and making decisions with minimal human intervention. This technology has been pivotal in advancements across various fields, from natural language processing to image recognition [5]. However, the complexities inherent in AI systems, especially those using deep learning, pose significant challenges in understanding and interpreting their decision-making processes. The 'black box' nature of these systems, where the internal workings are opaque or not easily understandable to humans, raises concerns, particularly in applications requiring transparency and accountability [6]. The complexity is further compounded by the fact that deep learning models, which often consist of millions of parameters, can make it extremely difficult to trace the exact reasoning behind specific decisions. This lack of interpretability is a major hurdle in fields where understanding the rationale behind a decision is as important as the decision itself, such as in healthcare or judicial systems [7].

These challenges highlight the need for continued research and development in the field of AI. Efforts are being made to develop more interpretable AI models and to devise methods to elucidate the decision-making processes of existing models, ensuring that AI systems are not only effective but also understandable and trustworthy.

3 Factors Contributing to Concerns in AI Trustworthiness

The trustworthiness of Artificial Intelligence (AI) systems is a subject of increasing concern, attributable to several key factors. Foremost among these is the lack of transparency, often referred to as the 'black box' nature of AI. This term relates to the difficulty in understanding how complex AI models, particularly deep learning algorithms, process inputs to arrive at a decision, a concern highlighted in numerous studies [8]. This opacity poses challenges in sectors where understanding the rationale behind decisions is crucial. Another significant factor is the bias and fairness in AI decisions. Biases can inadvertently be introduced into AI systems through training data or the design of algorithms themselves. Such biases can lead to unfair or discriminatory outcomes, especially in sensitive areas like recruitment, criminal justice, and credit scoring [9]. Addressing these biases is critical for ensuring equitable AI decision-making. Security and privacy concerns in AI also contribute to trust issues. AI systems, with their extensive data requirements, can be vulnerable to data breaches and misuse, raising questions about data security and user privacy. The potential misuse of personal data by AI systems or the possibility of AI being exploited in cyber-attacks adds to these concerns [10]. Lastly, the challenge of accountability and responsibility in AI actions is a crucial factor impacting trust. As AI systems become more autonomous, determining who is responsible for the decisions made by these systems—the developers, the users, or the AI itself—becomes increasingly complex [11]. This issue of accountability is particularly pressing in cases where AI decisions have significant consequences.

4 Current Measures to Enhance AI Trustworthiness

In addressing the challenges of AI trustworthiness, several measures, strategies, and regulations have been implemented to enhance the reliability and ethical integrity of AI systems. One of the key strategies involves the development of ethical guidelines and frameworks for AI. Organizations like the European Union have put forth regulations such as the Ethics Guidelines for Trustworthy AI, which outlines principles like transparency, fairness, and accountability in AI systems [12]. Similarly, the IEEE Global Initiative on Ethics of Autonomous and Intelligent Systems provides comprehensive guidelines aiming to ensure ethical, transparent, and accountable AI development [13]. On the industry front, tech companies and AI developers are increasingly adopting internal standards and practices to improve AI reliability. These include rigorous testing protocols, bias mitigation strategies, and the implementation of ethics review boards. The adoption of Explainable AI (XAI) practices, which aim to make AI decision-making processes more transparent and understandable to humans, is also gaining traction in the industry [14].

Additionally, there is a growing emphasis on the certification and auditing of AI systems. Third-party auditing and certification can provide an objective assessment of an AI system's reliability and adherence to ethical standards, thereby enhancing trust among users and stakeholders [15]. These measures collectively represent a multifaceted approach to enhancing AI trustworthiness, combining regulatory frameworks, industry practices, and innovative technological solutions. Such comprehensive efforts are essential in ensuring that AI systems are not only technologically advanced but also ethically sound and reliable.

5 Building Trust in AI: A Roadmapand Future Directions

Building and maintaining trust in AI systems necessitates a multi-faceted approach, involving concerted efforts from policymakers, developers, and users. Firstly, policymakers play a crucial role in establishing robust regulatory frameworks that set standards for ethical AI development and usage. These frameworks should not only focus on current AI capabilities but also anticipate future advancements, ensuring that regulations remain relevant and effective [16]. For developers, the emphasis should be on adhering to ethical AI principles throughout the AI development lifecycle.

This includes responsible data handling, implementing bias mitigation strategies, and prioritizing transparency in AI algorithms [17].

In addition to technical and regulatory measures, fostering a culture of ethical AI use is essential. This involves continuous education and awareness-raising among users and stakeholders about the capabilities and limitations of AI, thereby setting realistic expectations and promoting informed use of AI technologies [18].

Furthermore, encouraging collaboration across sectors can facilitate the sharing of best practices and experiences, contributing to a more cohesive and trustworthy AI ecosystem. This could involve partnerships between academia, industry, and government bodies to jointly address AI trust issues [19]. In summary, building trust in AI is a dynamic and ongoing process that requires proactive and collaborative efforts across various domains. By adhering to these recommendations, the roadmap to a trustworthy AI ecosystem becomes clearer, paving the way for AI technologies to be utilized responsibly and effectively.

6 Trustworthy AI in Education

In the educational sector, the trustworthiness of Artificial Intelligence (AI) is a critical issue that educators, students, and policymakers grapple with. The application of AI in education ranges from personalized learning algorithms and automated grading systems to intelligent tutoring systems and predictive analytics for student performance. The promise of AI in enhancing educational experiences and outcomes hinges on its trustworthiness, encompassing aspects of reliability, fairness, and transparency [20]. Concerns about trustworthiness arise particularly in the context of data privacy, as educational AI systems often handle sensitive student information. Ensuring that these systems protect student data and adhere to privacy regulations is paramount [21].

Bias in AI is another significant concern in the educational context. Algorithms that drive educational tools can inadvertently perpetuate and amplify biases if they are trained on skewed or unrepresentative data sets, leading to unfair outcomes for certain student groups [22]. Transparency in AI decision-making processes is crucial, especially when these decisions can impact student grading or the identification of learning needs. The black box nature of many AI systems can lead to skepticism and resistance among educators and students, highlighting the need for explainable AI solutions in education [23].

Addressing these challenges requires a concerted effort to develop and implement AI systems in education that are not only technologically advanced but also ethically sound and aligned with educational values. This includes engaging educators in the development process, rigorous testing for bias, and ongoing monitoring for ethical and effective use [24]. The trustworthiness of AI in education is not solely a technical issue but a broader societal concern, necessitating a multidisciplinary approach to ensure that these technologies serve as beneficial tools in the learning process.

7 Innovations and Future Directions

The field of AI is rapidly evolving, with several innovations and future directions poised to address current trust issues. Among the most notable advancements is Explainable AI (XAI). XAI seeks to make AI decision-making processes more transparent and understandable to humans, a critical factor in enhancing trust. By providing insights into how AI models arrive at certain decisions, XAI helps mitigate the 'black box' nature of AI, thus making it more accessible and accountable [25]. Another significant development is the growing interest in AI auditing and certification processes. These processes involve independent evaluations of AI systems to ensure they meet certain ethical, fairness, and reliability standards. The potential of AI auditing lies in its ability to provide a systematic assessment of AI systems, offering assurances of their compliance with established norms and regulations [26].

Looking to the future, there are several promising technologies and methodologies on the horizon. These include advancements in federated learning, which allows AI models to be trained across multiple decentralized devices, thus enhancing privacy and reducing bias. Another area is the development of more robust AI governance frameworks, which would provide clearer guidelines and standards for AI development and deployment [27]. These innovations, along with ongoing research in the field, are critical in addressing the complex trust issues surrounding AI. By continuing to advance in these areas, the goal of creating more transparent, accountable, and trustworthy AI systems becomes increasingly attainable.

8 Conclusion

In conclusion, this paper has highlighted the multifaceted nature of trust in AI, emphasizing the critical role it plays in the acceptance and effectiveness of AI technologies. We have examined the complexities inherent in AI systems, particularly in deep learning models, and how these complexities contribute to issues such as lack of transparency, biases, security vulnerabilities, and challenges in accountability. The paper underscored the importance of advancing explainable AI (XAI) and the potential of AI auditing and certification processes as vital measures to enhance AI trustworthiness. It also proposed a roadmap for building trust in AI, suggesting a collaborative approach involving policymakers, developers, and users to foster a responsible and trustworthy AI ecosystem. Reflecting on these findings, it is evident that trust forms the cornerstone of the future development and widespread adoption of AI technologies. Ensuring that AI systems are transparent, fair, secure, and accountable is not merely a technical challenge but a societal imperative. As AI continues to integrate into various aspects of human life, the need for trust in these systems becomes increasingly paramount. The cultivation of this trust will require continuous innovation, ethical considerations, and an ongoing dialogue among all stakeholders in the AI community. Looking ahead, the focus on building and maintaining trust in

AI will be instrumental in harnessing the full potential of these technologies for the betterment of society.

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Malicious User

Detecting Malicious Accounts in Cyberspace: Enhancing Security in ChatGPT and Beyond

Said A. Salloum

1 Introduction

In the rapidly evolving digital age, cybersecurity has become an indispensable aspect of safeguarding both personal data and the operational integrity of online platforms. The pervasive use of the internet in various facets of life underscores the critical importance of robust cybersecurity measures. Studies indicate that as our reliance on digital technologies grows, so too does the sophistication and frequency of cyber-attacks [1].

Central to the challenges faced in the realm of cybersecurity is the proliferation of malicious accounts in cyberspace. These accounts, which can range from automated bots to sophisticatedly disguised fake profiles, represent a multifaceted threat. They are known to engage in activities that undermine the authenticity of online interactions, spread misinformation, and execute fraudulent schemes [2].

The impact of these malicious entities extends across various digital platforms, notably affecting advanced AI systems like ChatGPT. In the context of AI-driven platforms, these accounts pose unique challenges. They can manipulate conversational dynamics, skew AI learning processes, and compromise the reliability of AI-generated content. This not only deteriorates the user experience but also raises significant concerns about the trustworthiness and security of these platforms [3].

Figure 1 provides a vivid conceptual representation of the pervasive and disruptive impact of malicious entities across a wide range of digital platforms. The illustration captures a network of interconnected platforms, including social media, online forums, and AI chat interfaces, highlighting the ubiquity of digital communication in modern society. These platforms are shown being infiltrated by symbols representing

S. A. Salloum (✉)
Health Economic and Financing Group, University of Sharjah, Sharjah, UAE
e-mail: ssalloum@sharjah.ac.ae

School of Science, Engineering, and Environment, University of Salford, Salford, UK

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various forms of malicious entities—such as bots, trolls, and fake profiles—which are depicted as causing disturbances and disruptions within the network. The symbols for malicious accounts are strategically placed to demonstrate their ability to blend in and yet distinctly disrupt the normal flow of digital communication and interaction. This visualization underscores the stealth and sophistication with which these entities operate, aligning with findings from [4] which detail the evolving nature of digital threats. The disruption caused by these entities is not just isolated to one platform but is shown to have a cascading effect, illustrating how vulnerabilities in one area can lead to broader compromises in the digital ecosystem. In the context of AI chat interfaces, such as ChatGPT, the image reflects the specific challenges these platforms face, including the manipulation of conversational dynamics and the potential skewing of AI learning processes. This aspect of the illustration aligns with studies like [5], which discuss the unique vulnerabilities and security considerations for AI-driven platforms.

This paper aims to delve into the intricate landscape of detecting and mitigating the presence of malicious accounts in cyberspace. We will explore the array of current detection methodologies, ranging from traditional cyber security tactics to innovative AI-driven approaches. The paper will critically evaluate the effectiveness of these strategies in enhancing the security measures of digital platforms, with a particular focus on AI-driven systems like ChatGPT. Additionally, we will examine the broader implications of these cybersecurity measures, considering their potential impact on user privacy, data integrity, and the ethical dimensions of AI moderation. The scope of this paper encompasses a comprehensive review of existing literature, an analysis of case studies pertaining to current cybersecurity practices, and a forward-looking discussion on emerging trends and challenges in this dynamic field.

Fig. 1 Conceptual representation of the impact of malicious entities on digital platforms

2 Understanding Malicious Accounts

2.1 Definition and Types of Malicious Accounts

In the realm of cybersecurity, the term 'malicious accounts' encompasses a range of deceptive and harmful online entities, each with distinct characteristics and objectives. Bots, a prevalent form of malicious accounts, are automated programs designed to mimic human actions online. They are often employed for spamming, spreading misinformation, or inflating social media metrics, as outlined in [6–8]. Trolls represent another category, typically human-operated accounts that aim to provoke and disrupt online conversations through inflammatory or deceptive content. Studies such as [9–11] delve into their impact on social media discourse and user experience.

Fake accounts, which include both fully fabricated profiles and those impersonating real individuals, are particularly insidious. These accounts are used for a range of malicious purposes, from spreading false information to phishing scams. Research by [12] provides insights into their detection and the challenges they pose to digital platform integrity. Each of these account types represents a unique threat to the online ecosystem, necessitating tailored detection and mitigation strategies. Understanding their distinct characteristics is crucial for effective cybersecurity measures, as they exploit different aspects of online platforms and user interactions.

2.2 Motivations and Methods Used by These Accounts

Malicious accounts in cyberspace are driven by a diverse range of motivations, employing various methods to achieve their objectives [13]. At the core, many of these entities are designed to manipulate, deceive, or disrupt online interactions. For instance, bots, programmed for automated tasks, are often used for spreading misinformation or amplifying social media content to manipulate public opinion or distort online discussions [14]. Their methods range from mass-posting similar messages to more sophisticated interactions that mimic human behavior. Trolls, typically human operators behind anonymous accounts, primarily aim to provoke or distress others for personal amusement or to push a specific agenda. The methods employed by trolls include posting inflammatory comments, engaging in harassment, and creating divisive content, as highlighted in studies like [15]. Fake accounts, on the other hand, may have more deceptive goals, such as phishing, identity theft, or spreading false information. These accounts often imitate real users or organizations, leveraging their perceived authenticity to mislead and exploit other users [16]. Understanding the motivations and methods of these malicious entities is crucial for developing effective countermeasures. Each type of malicious account presents unique challenges, requiring tailored approaches for detection and mitigation. This understanding is not only important for cybersecurity professionals but also for regular users to recognize and safeguard against such threats in their online interactions.

2.3 The Impact of Malicious Accounts on User Experience and Platform Integrity

The presence of malicious accounts in cyberspace significantly impacts both user experience and the integrity of digital platforms. These accounts, through various means, erode the trust and usability of online ecosystems. Bots, for instance, can flood platforms with spam or misinformation, leading to a polluted information environment. This can diminish user experience by cluttering feeds with irrelevant or deceptive content, as explored in [17]. Their activity can also skew analytics, giving false impressions of popularity or consensus, which is particularly detrimental for platforms relying on user engagement metrics for decision-making or advertising purposes. Trolls negatively impact user experience by creating a hostile or toxic online environment. Their behavior can lead to harassment and cyberbullying, causing distress to individual users and often leading to reduced participation or complete withdrawal from the platform [18]. This not only affects individual well-being but also the overall quality of discourse on the platform.

Fake accounts pose a direct threat to platform integrity and user security. By impersonating legitimate users or entities, they can engage in phishing attacks, scamming users, or spreading false information, thereby compromising the authenticity and reliability of the platform [19, 20]. The presence of these accounts can lead to a lack of trust in the platform, as users become uncertain about the genuineness of the interactions they have and the content they consume.

The collective impact of these accounts is a significant challenge for platform operators, as it undermines user trust, engagement, and satisfaction—all of which are crucial for the long-term viability and success of digital platforms. Addressing these issues requires not only technical solutions but also a consideration of the broader social and ethical implications of these malicious activities.

3 Current Detection Approaches

3.1 Overview of Traditional Detection Methods

Traditional methods for detecting malicious accounts in cyberspace have primarily focused on identifying patterns and anomalies that differentiate these accounts from legitimate users. One common approach is IP tracking, which involves monitoring the IP addresses from which accounts are accessed. This method can help identify accounts that are being operated from known sources of malicious activity or through proxies typically used to mask illicit activities [21]. Another widely used technique is the analysis of account activity. This includes examining login patterns, posting frequencies, and interaction styles. For instance, accounts that exhibit non-human behavior, such as posting at superhuman speeds or exhibiting repetitive patterns, can be flagged as potential bots or fake accounts [22, 23]. These traditional methods, while effective in certain contexts, have limitations. Malicious actors have become adept at evading detection by mimicking human behaviors or using sophisticated methods to mask their true IP addresses. Consequently, while these methods form an essential part of the cybersecurity toolkit, they are increasingly being supplemented with more advanced techniques, particularly those leveraging artificial intelligence and machine learning algorithms.

3.2 Machine Learning and AI in Detecting Malicious Accounts

The integration of Machine Learning (ML) and Artificial Intelligence (AI) has revolutionized the detection of malicious accounts in cyberspace, offering more dynamic and sophisticated methods compared to traditional techniques. ML algorithms are particularly adept at pattern recognition, allowing them to identify subtle behaviors and characteristics indicative of malicious accounts that might be overlooked by human analysts or simpler automated systems. For instance, ML models can be trained on large datasets to recognize patterns typical of bot activities, such as the timing and frequency of posts, or the nature of interactions with other users [24–27]. AI, especially when it involves Deep Learning techniques, can analyze not just the metadata of accounts but also the content of the interactions. This allows for the identification of more complex behaviors, such as the dissemination of misinformation or coordinated inauthentic behaviors that are characteristic of sophisticated fake accounts or state-sponsored trolling operations [28–33]. These technologies are not without their challenges. The evolving nature of malicious tactics means that detection systems must continuously learn and adapt, a process that can be resource-intensive. Additionally, the risk of false positives, where legitimate accounts are mistakenly identified as malicious, poses significant ethical and operational considerations [34]. Despite these challenges, the use of ML and AI in this domain represents a significant advancement in cybersecurity efforts. They offer scalability and efficiency in monitoring and analyzing vast amounts of data, an essential capability given the sheer scale of modern digital platforms.

3.3 The Role of User Reporting and Community Management

In the ecosystem of digital platform security, user reporting and community management play pivotal roles in identifying and mitigating the risks posed by malicious accounts. User reporting is a crucial first line of defense, enabling the platform's community to flag suspicious or harmful content or behavior. This grassroots level of surveillance harnesses the collective vigilance of users, often catching anomalies that automated systems might miss. Studies have shown that user reports can be instrumental in identifying coordinated disinformation campaigns or isolated instances of harassment, underscoring the importance of community engagement in maintaining platform integrity [2, 32]. Community management, on the other hand, involves a more proactive approach by platform administrators or designated moderators. This includes setting and enforcing community guidelines, monitoring discussions, and responding to user reports. Effective community management not only deters malicious activities through policy enforcement but also fosters a sense of safety and trust among users. It has been observed that platforms with active and visible community management have lower incidences of malicious activities, as they create an environment less conducive to the objectives of such accounts [35]. The synergy of user reporting and community management forms a comprehensive approach to platform security. It balances automated detection systems with human insight and judgment, essential in a landscape where malicious actors continually evolve their tactics. However, this approach also raises challenges, particularly in ensuring timely and appropriate responses to user reports and in managing the potential biases of human moderators [2].

4 Application to ChatGPT and Other AI Platforms

4.1 Specific Vulnerabilities of AI Platforms like ChatGPT to Malicious Accounts

AI platforms like ChatGPT are uniquely vulnerable to certain types of malicious accounts, mainly due to their reliance on user interactions and data for learning and response generation. One of the primary vulnerabilities is the potential for data poisoning, where malicious accounts feed misleading or harmful information to the AI, influencing its learning process and output. This type of attack can subtly skew the AI's language models, leading to biased or inappropriate responses [36]. Another vulnerability is the exploitation of AI's response mechanisms by malicious accounts to amplify misinformation or harmful content. Since AI platforms like ChatGPT are designed to engage in natural and relevant conversations, they can be manipulated into responding or interacting with content that furthers the agenda of these accounts [37].

Additionally, AI platforms may face challenges in distinguishing between legitimate user interactions and those orchestrated by malicious accounts, especially when these accounts employ sophisticated tactics to mimic genuine behavior. This difficulty can lead to inadequate responses to harmful content or the unintentional dissemination of such content, compromising the integrity and trustworthiness of the AI platform [38]. These vulnerabilities necessitate a multifaceted approach to AI platform security, combining advanced algorithmic defenses with oversight mechanisms to ensure the AI's interactions and learning processes remain aligned with ethical and factual standards.

4.2 How Current Detection Approaches Can Be Adapted for ChatGPT?

Adapting current detection approaches for ChatGPT involves several strategies that align with the unique characteristics and operational contexts of AI-driven conversation platforms. Machine learning algorithms, which are already a core component of ChatGPT, can be fine-tuned to identify patterns indicative of malicious account activities. This includes training models on datasets that capture the nuances of such activities, ranging from spam and troll behavior to more sophisticated misinformation campaigns [39]. Natural language processing (NLP) techniques, integral to ChatGPT's functionality, can be leveraged to analyze conversational contexts and identify potential malicious interactions. Advanced NLP can discern subtleties in language that are indicative of harmful content or deceptive practices, a method that has shown effectiveness in other digital platforms [40–42].

In addition to these AI-centric methods, user feedback mechanisms within ChatGPT can play a crucial role. Implementing robust user reporting features and feedback loops allows for the gathering of valuable user insights, which can be used to continually refine detection algorithms and response strategies [43].

Furthermore, incorporating a multi-layered security approach that combines these AI-driven methods with traditional cybersecurity practices, such as IP tracking and account verification, can enhance the overall resilience of ChatGPT against malicious accounts [44]. The adaptation of these detection methods for ChatGPT underscores the need for a dynamic and evolving approach to platform security, one that is capable of responding to the continuously changing tactics of malicious actors in the digital space.

4.3 Potential for AI-Driven Detection Methods to Improve Platform Security

The potential of AI-driven detection methods in enhancing the security of digital platforms is significant, particularly as cyber threats become more sophisticated. These AI methods, primarily rooted in advanced machine learning and deep learning techniques, offer the ability to analyze vast amounts of data quickly and efficiently, a capability that is essential in identifying and mitigating cyber threats in real-time. Machine learning algorithms, for instance, can be trained to recognize patterns and anomalies that signify malicious activities, such as unusual login behaviors or atypical content dissemination patterns, offering a level of analysis that is unfeasible for human monitors alone [45–47].

Deep learning, a subset of machine learning, provides even greater potential due to its ability to process and analyze complex datasets, including unstructured data like text and images. This capability is particularly valuable in detecting sophisticated cyber threats that traditional methods might miss, such as subtle phishing attempts or advanced social engineering tactics [48–50].

Furthermore, the integration of natural language processing (NLP) in AI-driven security systems enables the detection of nuanced malicious activities in textual content, such as hate speech, misinformation, or harmful propaganda. This approach is increasingly relevant in the context of social media and communication platforms, where such content can have a widespread impact [51]. The adaptability of AI-driven methods is another key advantage. AI models can be continuously updated and retrained to keep up with evolving cyber threats, ensuring that platform security measures remain effective over time. However, the implementation of these AI technologies also requires careful consideration of privacy and ethical standards to avoid potential misuse or bias in the detection processes [52].

5 Future Trends and Challenges

5.1 Emerging Technologies and Methods in Malicious Account Detection

The landscape of cybersecurity is continually evolving, with emerging technologies and methods playing a critical role in detecting malicious accounts. Among the most promising advancements is the application of Artificial Intelligence (AI) and Machine Learning (ML), which offer sophisticated analytical capabilities. AI and ML algorithms can process and learn from vast amounts of data, identifying patterns and anomalies indicative of malicious behavior that may elude traditional detection methods [53, 54].

Another emerging technology is the use of blockchain for security purposes. Blockchain's inherent properties, such as decentralization, transparency, and immutability, make it a potentially powerful tool in the fight against cyber threats, including the detection and prevention of fraudulent activities by malicious accounts [55]. Network analysis is also gaining traction as a method for detecting malicious accounts. By examining the connections and patterns of interactions between accounts, it becomes possible to identify coordinated malicious activities, such as botnets or troll farms, which often operate in networks [56, 57].

Furthermore, the development of advanced natural language processing (NLP) techniques enhances the ability to scrutinize the content generated by accounts, spotting signs of manipulation or harmful intent in textual data [58].

Additionally, the integration of biometric verification methods, such as facial recognition or voice analysis, is being explored as a means to authenticate users and flag accounts that exhibit signs of falsification [59]. These emerging technologies and methods collectively represent a paradigm shift in cybersecurity strategies, offering more proactive and comprehensive approaches to safeguarding digital platforms from the ever-evolving threat posed by malicious accounts.

5.2 The Evolving Nature of Malicious Activities and the Need for Adaptive Security Measures?

The landscape of cyber threats is in a constant state of flux, with malicious activities continuously evolving in complexity and sophistication. This dynamic nature poses a significant challenge for cybersecurity, necessitating adaptive and forward-thinking security measures. The shift from relatively straightforward phishing attacks to more complex, multi-vector threats exemplifies this evolution. Cybercriminals now employ a range of tactics, including advanced social engineering, AI-driven attacks, and sophisticated malware, to exploit vulnerabilities in digital systems [60].

In response to these changing threats, the development and implementation of adaptive security measures have become paramount. Machine learning and artificial intelligence are at the forefront of this adaptive approach. These technologies enable continual learning from new data, allowing security systems to evolve alongside emerging threats [61]. Furthermore, the integration of threat intelligence platforms helps in gathering and analyzing information about new and existing threats, providing insights that inform security strategies and responses [62]. Another key component of adaptive security measures is the emphasis on proactive rather than reactive strategies. This involves anticipating potential security incidents and preparing defenses in advance, rather than merely responding to breaches after they occur. Techniques such as predictive analytics and risk assessment models are instrumental in this proactive approach [63]. However, the evolving nature of cyber threats also calls for a broader perspective that includes legal, ethical, and policy considerations. The development of comprehensive cybersecurity policies and adherence to ethical standards is crucial to ensure that adaptive security measures are not only effective but also respect user privacy and data rights [64].

The continuous evolution of malicious activities in the digital domain underscores the need for security measures that are equally dynamic and responsive, blending technological innovation with strategic foresight and ethical responsibility.

5.3 *Ethical Considerations and Privacy Concerns in Detection Strategies*

In the realm of cybersecurity, particularly in the detection of malicious accounts, ethical considerations and privacy concerns are paramount. The deployment of sophisticated detection strategies, while essential for security, often raises questions regarding user privacy and data protection. Advanced monitoring and data analysis techniques, such as deep packet inspection or extensive data mining, can inadvertently infringe on individual privacy rights. The balance between security and privacy is a subject of ongoing debate, with scholars like [65] emphasizing the need for a nuanced approach. Another ethical concern relates to the potential for biases in AI-driven detection systems. Machine learning models, if trained on biased or unrepresentative data sets, can lead to discriminatory outcomes, unfairly targeting certain groups or individuals. Studies such as [66] have highlighted the necessity for unbiased data and transparent algorithms to mitigate these risks.

Furthermore, the ethical use of user data in cybersecurity measures is an area of significant concern. The collection and analysis of user data for security purposes must comply with data protection regulations and ethical standards, a topic explored in-depth by [67–69]. This compliance is not only a legal obligation but also critical for maintaining user trust in digital platforms.

In addition, the use of intrusive detection methods can raise ethical questions about the extent of surveillance and monitoring that is acceptable. The work of [67] offers insights into ethical limitations and guidelines for such practices. Addressing these ethical considerations and privacy concerns is crucial for the development of effective and responsible cybersecurity strategies. It involves a careful balance between the need for security and the protection of individual rights, calling for ongoing dialogue and collaboration between technologists, ethicists, and policymakers.

6 Conclusion

In conclusion, this exploration into the detection of malicious accounts in cyberspace underscores a landscape marked by complexity and ever-evolving challenges. Key findings indicate that while traditional methods like IP tracking and account activity analysis remain foundational, the emergence of AI and ML technologies has significantly enhanced the capability to identify and mitigate cyber threats. The adaptation of these advanced methods to specific platforms, particularly AI-driven systems like ChatGPT, is crucial in addressing the unique vulnerabilities they face. However, as our understanding and technological capabilities expand, so too does the sophistication of malicious entities, necessitating a continuous innovation in detection methods to maintain robust platform security. The importance of this ongoing innovation cannot be overstated, as it represents not only a response to emerging threats but also a proactive approach to safeguarding digital ecosystems. The integration of emerging technologies such as blockchain, network analysis, and biometric verification further illustrates the dynamic nature of cybersecurity strategies. Yet, as we advance in our technical prowess, the need to maintain a delicate balance becomes increasingly paramount. Balancing security with user privacy and platform usability is a complex but essential task, requiring a nuanced approach that respects individual rights while ensuring a safe and reliable online experience. This balance is not static but a dynamic equilibrium that must be constantly reassessed in the light of new technologies, user expectations, and the evolving nature of cyber threats. The future of digital platform security thus lies not only in technological advancement but also in our ability to ethically and responsibly integrate these innovations into the fabric of our digital lives.

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Ethical Considerations: The Dangers of AI in the Classroom

AI Perils in Education: Exploring Ethical Concerns

Said A. Salloum

1 Introduction

Artificial Intelligence (AI) is revolutionizing the educational landscape, offering novel ways to enhance learning and assist teachers. As Lynch notes, AI in education can be incredibly valuable, providing personalized learning experiences and addressing individual students' needs. AI systems can adapt to each student's learning style, assist in grading, offer feedback on course quality, and provide meaningful and immediate feedback to students. This personalization is crucial because it can be challenging for one teacher to meet every student's unique needs. However, the integration of AI in education is not without ethical concerns. Lynch emphasizes the importance of vigilance in monitoring AI's development and its role in the educational sector. The concerns range from privacy issues and data security to the potential for AI to amplify biases and reduce the human element in education, which is fundamentally a human activity [1].

Figure 1 illustrates a futuristic classroom scene showcasing the use of artificial intelligence in education. This visual representation captures the advanced yet harmonious integration of AI technologies within a modern educational setting, featuring interactive smart boards, AI-powered teaching assistants, and students engaged with personalized learning programs through various digital devices. The scene also includes robots assisting in teaching and interacting with students, under the supervision of a teacher, emphasizing the blend of human and AI interaction in this technologically advanced environment.

This paper aims to explore these ethical considerations, focusing on specific types of AI applications in education and their impact across various educational levels.

S. A. Salloum (✉)
Health Economic and Financing Group, University of Sharjah, Sharjah, UAE
e-mail: ssalloum@sharjah.ac.ae

School of Science, Engineering, and Environment, University of Salford, Salford, UK

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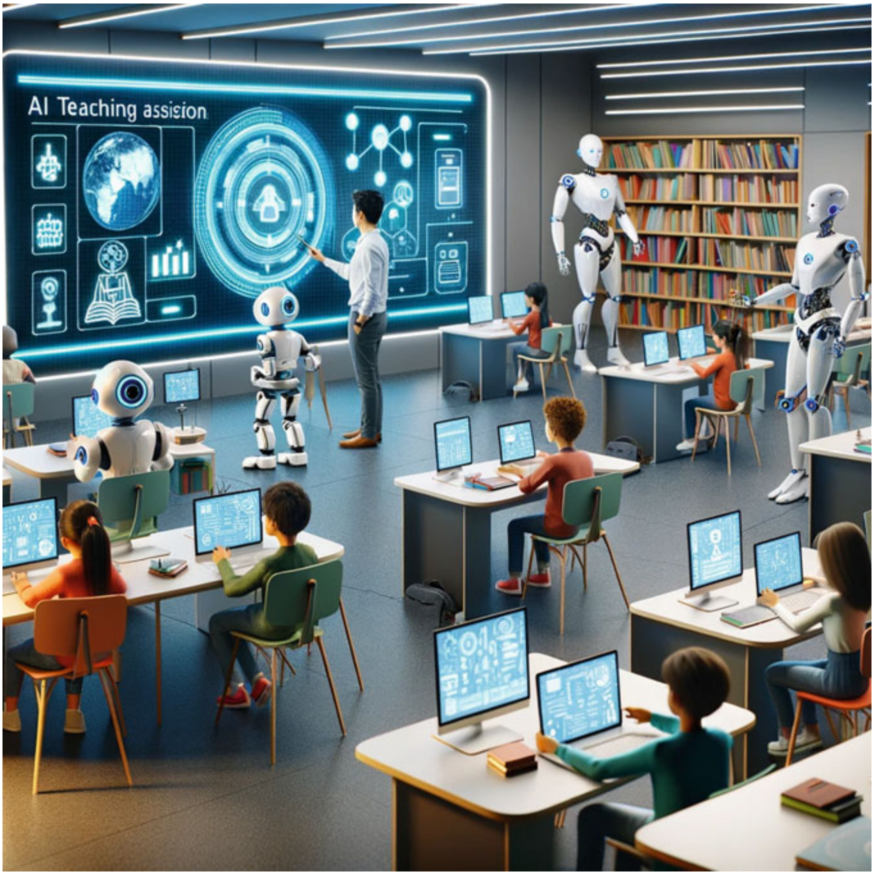

Fig. 1 The use of AI in education

As Bernard Marr suggests, AI is not intended to replace teachers but to enhance educational delivery through differentiated and individualized learning, administrative task automation, and universal access for all students. The scope of this paper will encompass these applications and their ethical implications, aiming to provide a balanced perspective on the use of AI in educational settings.

2 Background

Artificial Intelligence in Education (AIED) has a history spanning over six decades, paralleling other significant scientific explorations like space travel and DNA research. One of the earliest examples of AI in education is the PLATO system, an educational technology with AI capabilities, launched in 1963. The evolution of AI in education has led to several significant benefits:

- **Automated Essay Grading**: AI systems like Gradescope and E-rater use machine learning algorithms to grade essays and assignments, focusing on aspects like grammar, organization, and coherence. This not only saves teachers time but also provides students with immediate feedback [2].
- **Personalized Learning Paths**: Tools like Carnegie Mellon University's Open-Simon analyze student data to offer real-time personalized feedback. These systems adapt their recommendations based on individual student performance and behavior, allowing for more effective and tailored educational experiences [3].
- **Intelligent Tutoring Systems**: An advanced application of AIED is intelligent tutoring systems, such as Georgia Tech University's Jill Watson. These systems provide personalized academic and social-emotional support to students, adapting to their individual learning needs. They can offer personalized feedback even in large classes, helping to address the challenges of providing individual attention in traditional educational settings [4].

These advancements in AI have transformed the educational landscape, offering more personalized, efficient, and supportive learning experiences for students.

3 Ethical Considerations

The integration of Artificial Intelligence (AI) in educational settings brings forth various ethical considerations, chief among them being privacy and data security, bias and fairness, autonomy and dependence, and accountability.

- **Privacy and Data Security**: AI in education often requires access to personal data, raising significant privacy concerns. Users, including students and teachers, may consent to the use of their data without fully understanding the implications or the extent of its use. This includes concerns over surveillance and data collection methods, such as location tracking or monitoring online interactions. The integration of AI, such as image recognition and natural language processing (NLP) technologies, into educational tools can further amplify these privacy concerns. For instance, facial recognition technologies used for student safety have been criticized for normalizing a culture of surveillance in schools [5].
- **Bias and Fairness**: AI systems are susceptible to inheriting and amplifying societal biases. When algorithms are developed, they often incorporate historical and systemic biases, leading to gender and racial biases in various AI-based platforms. This is especially concerning in predictive analytics, where biases in data sets or analytical models can significantly impact a student's educational and career trajectory [6, 7].

- **Autonomy and Dependence**: The reliance on AI systems, which often employ algorithms for tasks like predicting student performance, can impinge on the independence and autonomy of students and teachers. This raises questions about fairness and the freedom of individuals in educational settings, as their work and learning paths might be heavily influenced by algorithmic decision-making [7].
- **Accountability**: The question of accountability in AI-driven educational systems is complex. There is a need for transparency and explainability in how AI systems make decisions, especially when sensitive data, like children's information, is involved. Ed Tech companies are advised to inform users about the use of AI and the logic behind these systems. Moreover, data minimization principles, which involve collecting only the data necessary for a specified purpose, are crucial in managing privacy risks. Fairness and non-discrimination must also be key considerations in the design and application of AI systems to prevent discriminatory outcomes. Legislation and guidelines, such as the California Privacy Rights Act (CPRA), emphasize the importance of risk assessments and the implementation of privacy standards tailored to protect student data [8].

Overall, while AI brings numerous benefits to educational settings, these ethical considerations highlight the need for careful implementation and regulation to safeguard the interests and rights of all stakeholders involved.

4 Discussion

In the discussion of AI's role in education, it is crucial to analyze the ethical considerations deeply. AI promises significant advancements in education, but its rapid development brings forward important ethical concerns, necessitating a balance between technological progress and ethical responsibility [9].

Ethical AI entails creating systems that adhere to ethical principles and values, aiming to minimize harm and maximize societal benefits. This includes fairness, transparency, privacy, and accountability, ensuring AI systems align with human values and avoid unintended consequences. As AI systems grow more autonomous, accountability becomes a central issue, raising questions about who is responsible for AI-induced errors or harm. Inclusive development is vital for ethically-aligned AI, involving diverse perspectives to reflect the needs and values of various stakeholders. This approach effectively addresses ethical concerns, benefiting a broader range of individuals. Addressing bias is another key aspect, as AI systems can perpetuate existing biases, making it essential to create AI systems that make fair decisions and minimize discrimination risks [9].

Transparency and explainability in AI systems are fundamental for building trust, as they help users understand AI decision-making processes [10]. This aligns AI systems with user values and expectations. Public engagement and awareness are essential in ethical AI development, making the public active participants in AI policy shaping and ensuring AI systems reflect societal values. Data privacy remains a critical concern, especially as AI's progress relies on accessing vast amounts of often sensitive data. Privacy-enhancing techniques are necessary to protect individual rights and mitigate potential risks [11]. The education of AI professionals is also crucial, integrating ethics education and training into AI-related curricula to foster responsible practices and a culture of accountability within the AI community.

Finally, governments and regulatory bodies are instrumental in ensuring AI is built and deployed responsibly. Appropriate regulations can set clear ethical guidelines and standards for AI developers and organizations, fostering responsible innovation . These considerations highlight the complexity of AI's role in education, underscoring the need for careful balance and multifaceted approaches to manage its ethical implications.

5 Potential Solutions

Addressing the ethical concerns raised by the use of AI in education requires a multifaceted approach that includes both practical solutions and policy implications.

- **Educational Initiatives**: Firstly, there's a need for foundational education for both teachers and students about AI, its workings, and its societal impacts. This includes ensuring equitable access to AI education across various demographics and grade levels. The International Society for Technology in Education (ISTE) Standards for Students, for example, emphasizes engaging in positive, safe, legal, and ethical behavior when using technology. Educators must facilitate discussions around ethical dilemmas posed by AI and guide students to envision better outcomes [12].
- **Professional Development Programs**: Programs like the AI Explorations program, which provides professional learning opportunities for educators, are crucial. They help address inequities in STEM fields and prepare students for future AI careers. ISTE's "Hands-On AI Projects for the Classroom" guide series, available in multiple languages, is a valuable resource that provides interactive activities and discussions on AI development, application, and impact [13, 14] .
- **Guides and Resources**: The addition of a guide focused on Ethics and AI by ISTE is another step in the right direction. This guide helps educators engage students in understanding fairness, autonomy, and ethical technology use, and supports secondary teachers to explore ethical lenses, diverse stakeholders, and accountability. Such resources are crucial for educators to teach students to critically evaluate ethical questions and consider various outcomes [14].
- **Technology Trade-offs and Ethical Decision Making**: It's also essential to teach students and educators about the trade-offs involved in technology use. This involves considering what might be sacrificed, such as privacy or civil rights, for efficiency or convenience. The concept of technology trade-offs is integrated into educational resources, encouraging critical thinking about the impacts of AI on society [14].

6 Ethical Guidelines and Regulations

- On a policy level, the European Commission has published ethical guidelines for educators on the use of AI and data in education. These guidelines offer practical advice on integrating AI and data into school education effectively, addressing popular misconceptions about AI, and emerging competences for the ethical use of AI among teachers and educators [15].
- **U.S. Department of Education's Recommendations**: The U.S. Department of Education's Office of Educational Technology (OET) suggests several steps to refine technology plans and policies for AI in education. These include emphasizing humans-in-the-loop, aligning AI models to a shared vision for education, designing AI using modern learning principles, prioritizing trust, involving educators, focusing on R&D to address context and trust, and developing education-specific guidelines and guardrails [16].

These solutions and guidelines underscore the importance of a comprehensive approach that balances technological advancement with ethical responsibility, ensuring that AI is used and taught about in ways that are equitable, responsible, and beneficial for all stakeholders in the educational ecosystem.

7 Conclusion

In conclusion, this paper has highlighted the multifaceted nature of trust in AI, emphasizing the critical role it plays in the acceptance and effectiveness of AI technologies. We have examined the complexities inherent in AI systems, particularly in deep learning models, and how these complexities contribute to issues such as lack of transparency, biases, security vulnerabilities, and challenges in accountability. The paper underscored the importance of advancing explainable AI (XAI) and the potential of AI auditing and certification processes as vital measures to enhance AI trustworthiness. It also proposed a roadmap for building trust in AI, suggesting a collaborative approach involving policymakers, developers, and users to foster a responsible and trustworthy AI ecosystem. Reflecting on these findings, it is evident that trust forms the cornerstone of the future development and widespread adoption of AI technologies. Ensuring that AI systems are transparent, fair, secure, and accountable is not merely a technical challenge but a societal imperative. As AI continues to integrate into various aspects of human life, the need for trust in these systems becomes increasingly paramount. The cultivation of this trust will require continuous innovation, ethical considerations, and an ongoing dialogue among all stakeholders in the AI community. Looking ahead, the focus on building and maintaining trust in AI will be instrumental in harnessing the full potential of these technologies for the betterment of society.

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